



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 10:54:24 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 562.85um = .83phi

Surface Weighted Mean:

D[3,2] = 476.54um = 1.07phi

Span :

1.135

Uniformity:

0.353

Specific Surface Area*:

0.0126 m²/g

Percentiles:

d(0.05) = 262.67um = 1.93phi

d(0.10) = 300.32um = 1.74phi

d(0.16) = 336.93um = 1.57phi

d(0.50) = 519.55um = .94phi

d(0.84) = 796.38um = .33phi

d(0.90) = 889.85um = .17phi

d(0.95) = 1010.92um = -.02phi

Distribution Properties:

Kurtosis = .43

Skewness = .86

Standard Deviation = 232.28 um = 2.11phi

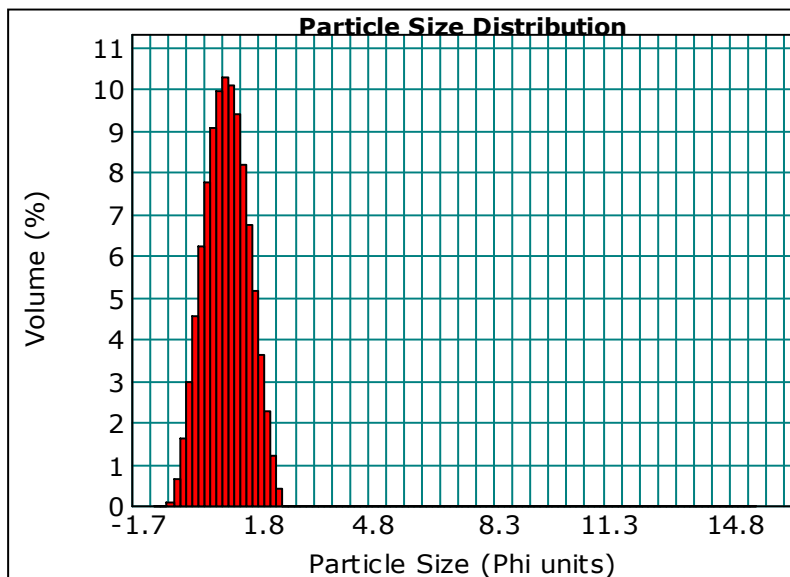
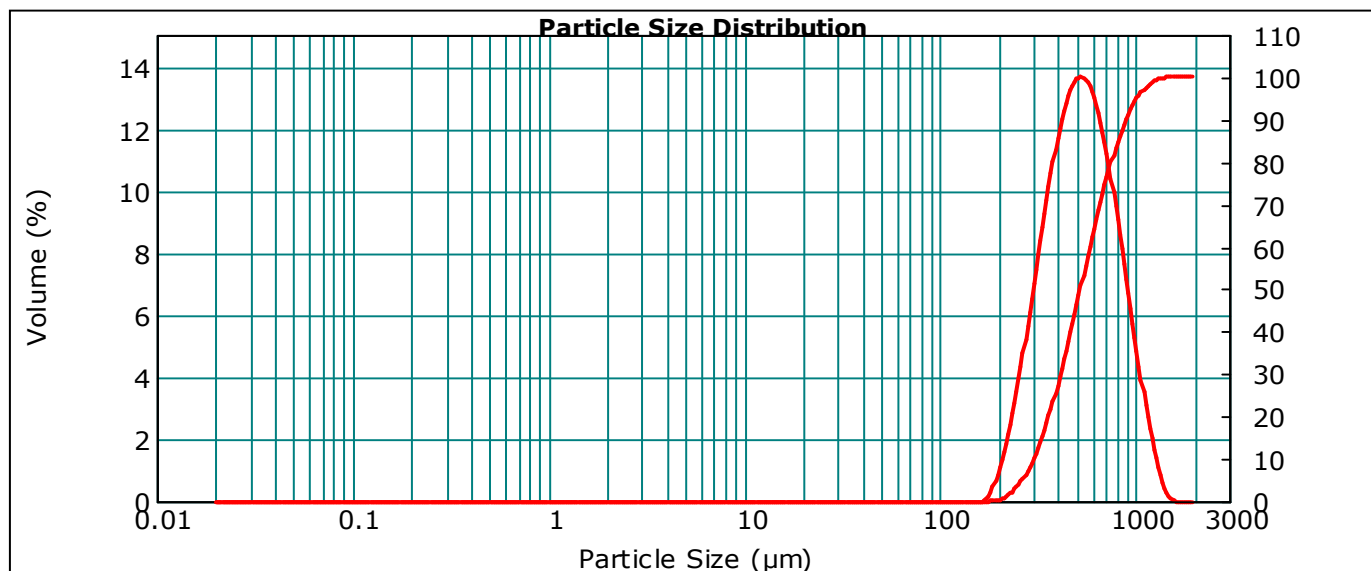
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.05
-0.5	5.30
0.0	18.67
0.5	29.41
1.0	27.60
1.5	15.31
2.0	3.66
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



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Sample Name: C_1_1_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 10:55:32 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 569.44um = .81phi

Surface Weighted Mean:

D[3,2] = 480.13um = 1.06phi

Span :

1.151

Uniformity:

0.358

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 263.04um = 1.93phi

d(0.10) = 301.24um = 1.73phi

d(0.16) = 338.39um = 1.56phi

d(0.50) = 524.41um = .93phi

d(0.84) = 808.25um = .31phi

d(0.90) = 904.9um = .14phi

d(0.95) = 1030.3um = -0.04phi

Distribution Properties:

Kurtosis = .44

Skewness = .87

Standard Deviation = 238.07 um = 2.07phi

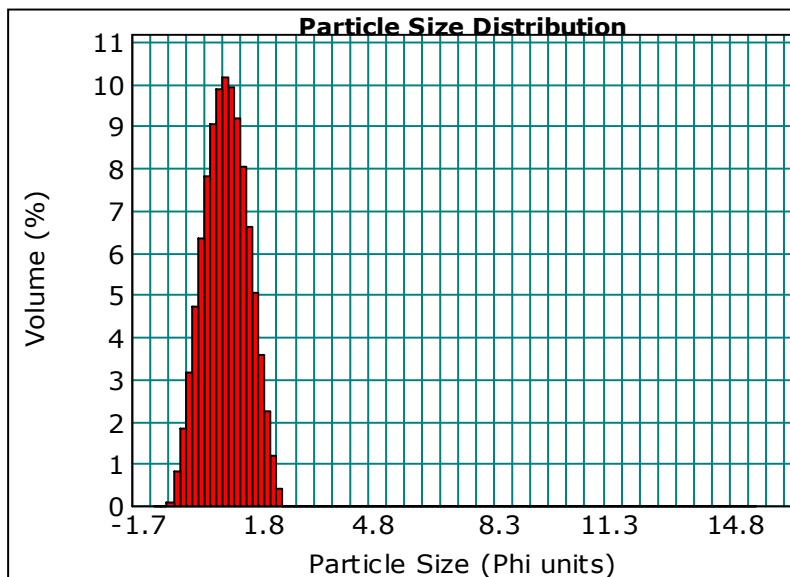
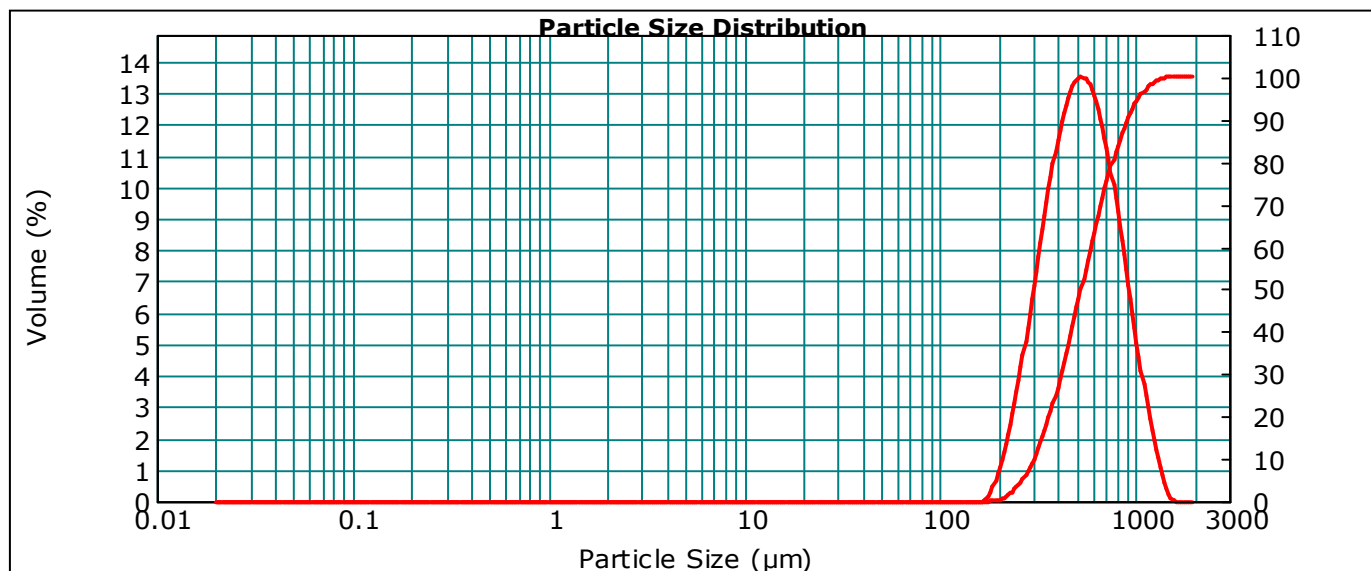
Inclusive SD (Sorting Coeff.) = .61 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .93 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.08
-0.5	5.89
0.0	19.02
0.5	29.22
1.0	27.13
1.5	15.02
2.0	3.64
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



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Sample Name: C_1_1_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 10:56:39 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 570.6um = .81phi

Surface Weighted Mean:

D[3,2] = 482.8um = 1.05phi

Span :

1.136

Uniformity:

0.353

Specific Surface Area*:

0.0124 m²/g

Percentiles:

d(0.05) = 265.66um = 1.91phi

d(0.10) = 303.9um = 1.72phi

d(0.16) = 341.23um = 1.55phi

d(0.50) = 526.96um = .92phi

d(0.84) = 807.68um = .31phi

d(0.90) = 902.49um = .15phi

d(0.95) = 1024.96um = -.04phi

Distribution Properties:

Kurtosis = .4

Skewness = .85

Standard Deviation = 235.54 um = 2.09phi

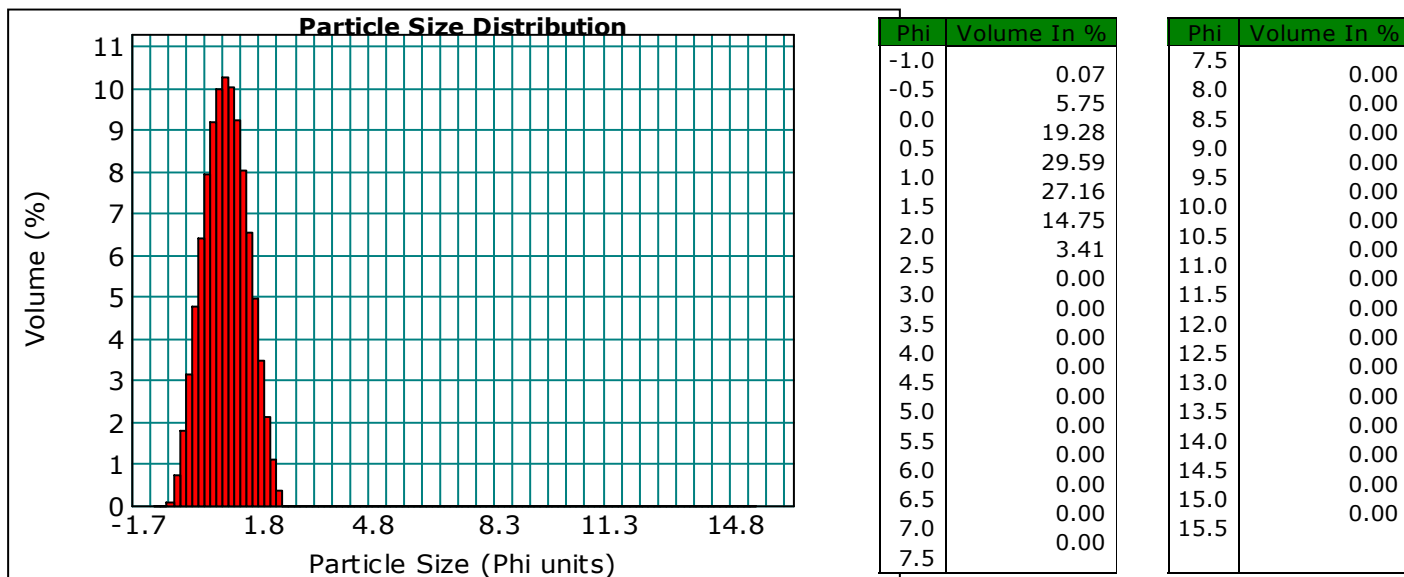
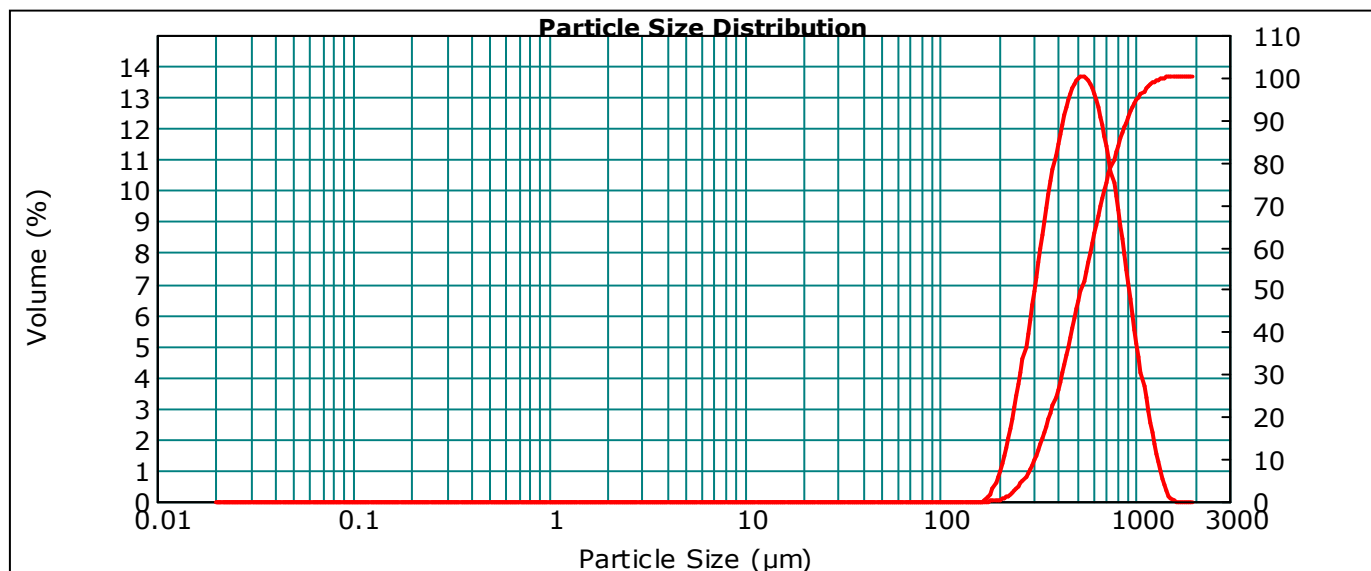
Inclusive SD (Sorting Coeff.) = .61 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .93 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





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Sample Name: C_1_1_21 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 10:54:24 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 567.63um = .82phi

Surface Weighted Mean:

D[3,2] = 479.81um = 1.06phi

Span :

1.141

Uniformity:

0.354

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 263.76um = 1.92phi

d(0.10) = 301.82um = 1.73phi

d(0.16) = 338.82um = 1.56phi

d(0.50) = 523.63um = .93phi

d(0.84) = 804.09um = .31phi

d(0.90) = 899.11um = .15phi

d(0.95) = 1022.19um = -0.03phi

Distribution Properties:

Kurtosis = .43

Skewness = .86

Standard Deviation = 235.33 um = 2.09phi

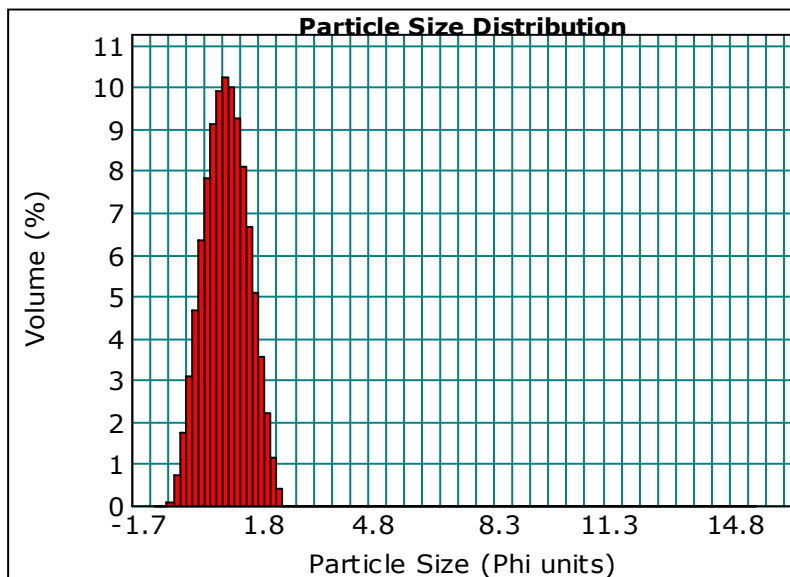
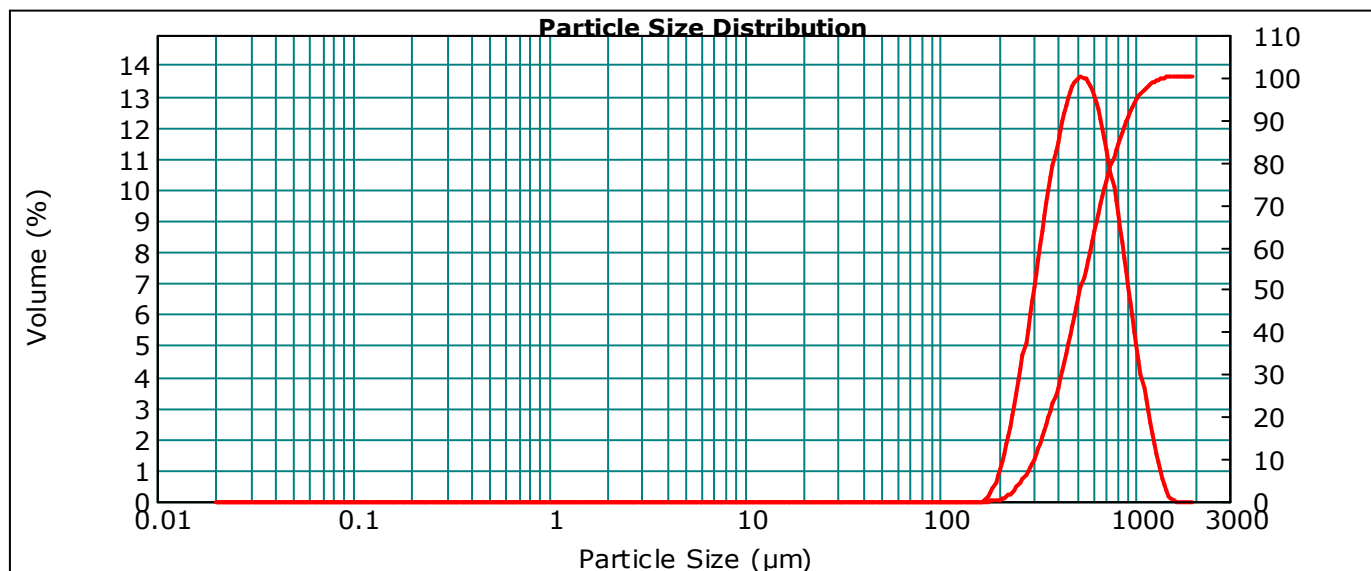
Inclusive SD (Sorting Coeff.) = .61 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .94 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.07
-0.5	5.65
0.0	18.99
0.5	29.41
1.0	27.29
1.5	15.03
2.0	3.57
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



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Sample Name: C_1_1_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:06:09 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 600.92um = .73phi

Surface Weighted Mean:

D[3,2] = 502.59um = .99phi

Span :

1.179

Uniformity:

0.364

Specific Surface Area*:

0.0119 m²/g

Percentiles:

d(0.05) = 273.1um = 1.87phi

d(0.10) = 313.33um = 1.67phi

d(0.16) = 352.62um = 1.5phi

d(0.50) = 549.83um = .86phi

d(0.84) = 855.54um = .23phi

d(0.90) = 961.74um = .06phi

d(0.95) = 1103.53um = -.14phi

Distribution Properties:

Kurtosis = .83

Skewness = .97

Standard Deviation = 259.56 um = 1.95phi

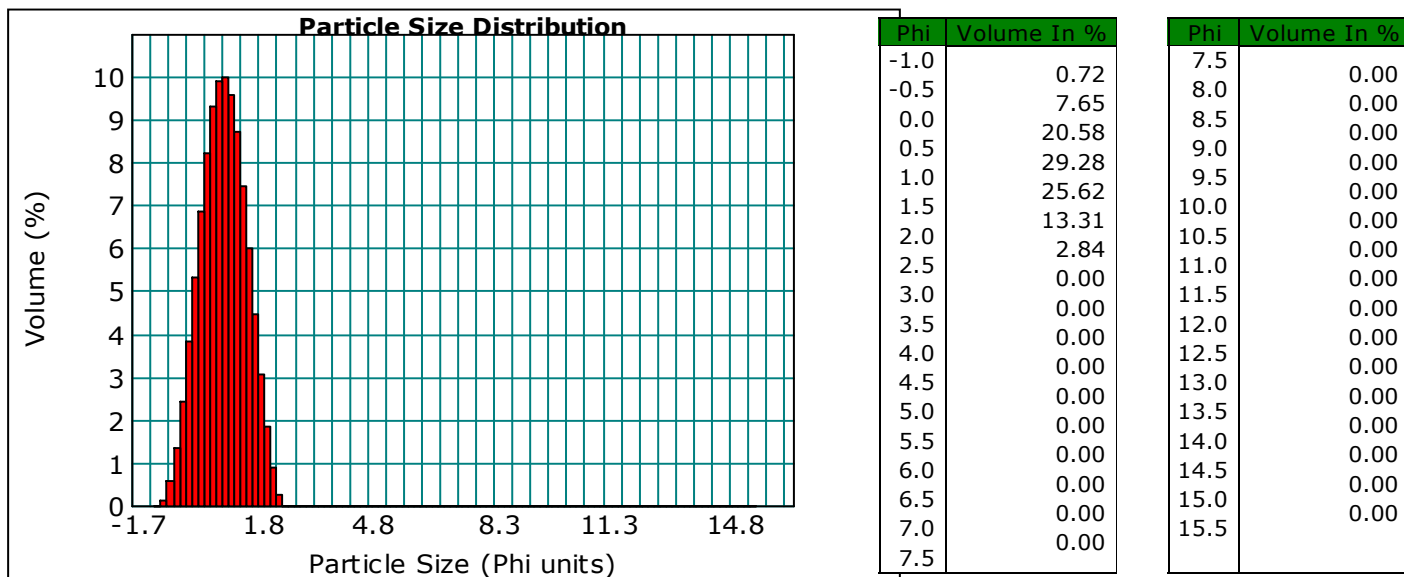
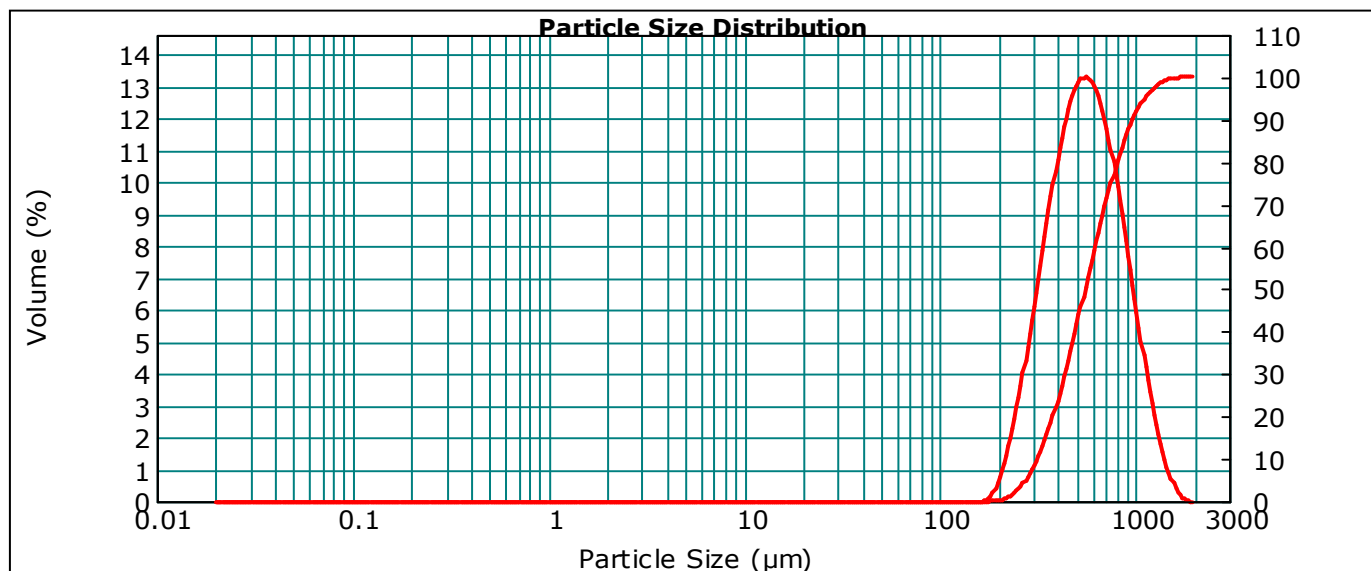
Inclusive SD (Sorting Coeff.) = .62 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .86 phi (Coarse Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





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Sample Name: C_1_1_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:07:16 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 603.88um = .73phi

Surface Weighted Mean:

D[3,2] = 502.08um = .99phi

Span :

1.204

Uniformity:

0.372

Specific Surface Area*:

0.012 m²/g

Percentiles:

d(0.05) = 271.62um = 1.88phi

d(0.10) = 311.73um = 1.68phi

d(0.16) = 350.99um = 1.51phi

d(0.50) = 549.21um = .86phi

d(0.84) = 862.15um = .21phi

d(0.90) = 972.98um = .04phi

d(0.95) = 1122.26um = -.017phi

Distribution Properties:

Kurtosis = 1.16

Skewness = 1.06

Standard Deviation = 267.53 um = 1.9phi

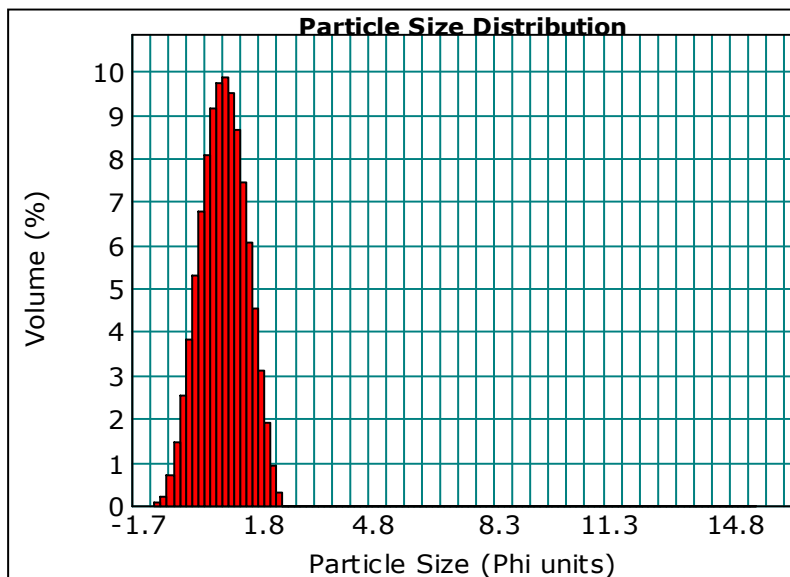
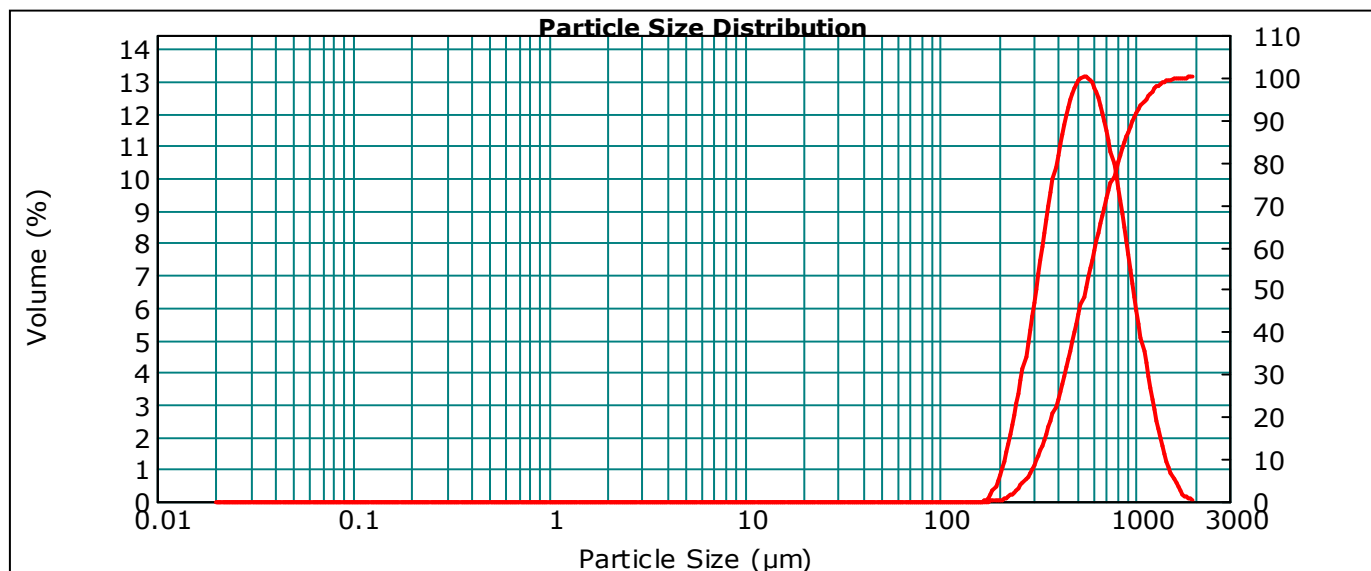
Inclusive SD (Sorting Coeff.) = .63 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .86 phi (Coarse Sand)

Inclusive Skewness = -0.01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.00
-0.5	7.87
0.0	20.30
0.5	28.86
1.0	25.54
1.5	13.47
2.0	2.95
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



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Sample Name: C_1_1_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:08:23 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 615.21um = .7phi

Surface Weighted Mean:

D[3,2] = 509.28um = .97phi

Span :

1.219

Uniformity:

0.378

Specific Surface Area*:

0.0118 m²/g

Percentiles:

d(0.05) = 273.96um = 1.87phi

d(0.10) = 314.94um = 1.67phi

d(0.16) = 355um = 1.49phi

d(0.50) = 558.07um = .84phi

d(0.84) = 880.22um = .18phi

d(0.90) = 995.03um = .01phi

d(0.95) = 1150.82um = -.02phi

Distribution Properties:

Kurtosis = 1.28

Skewness = 1.09

Standard Deviation = 276.41 um = 1.86phi

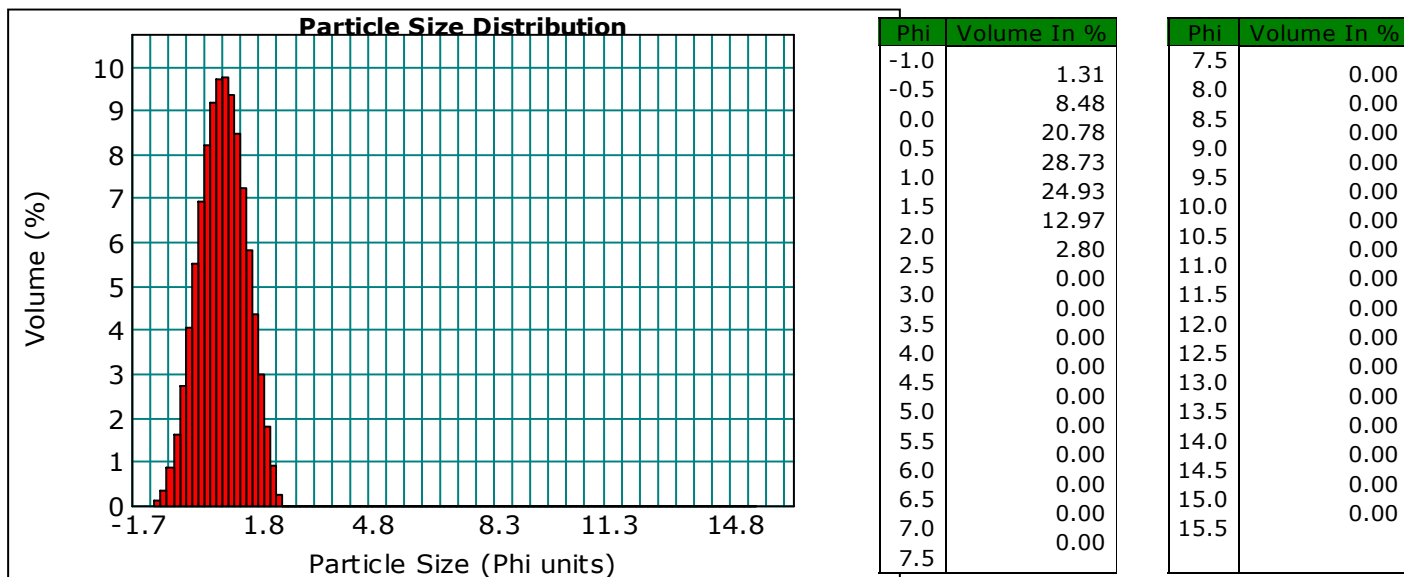
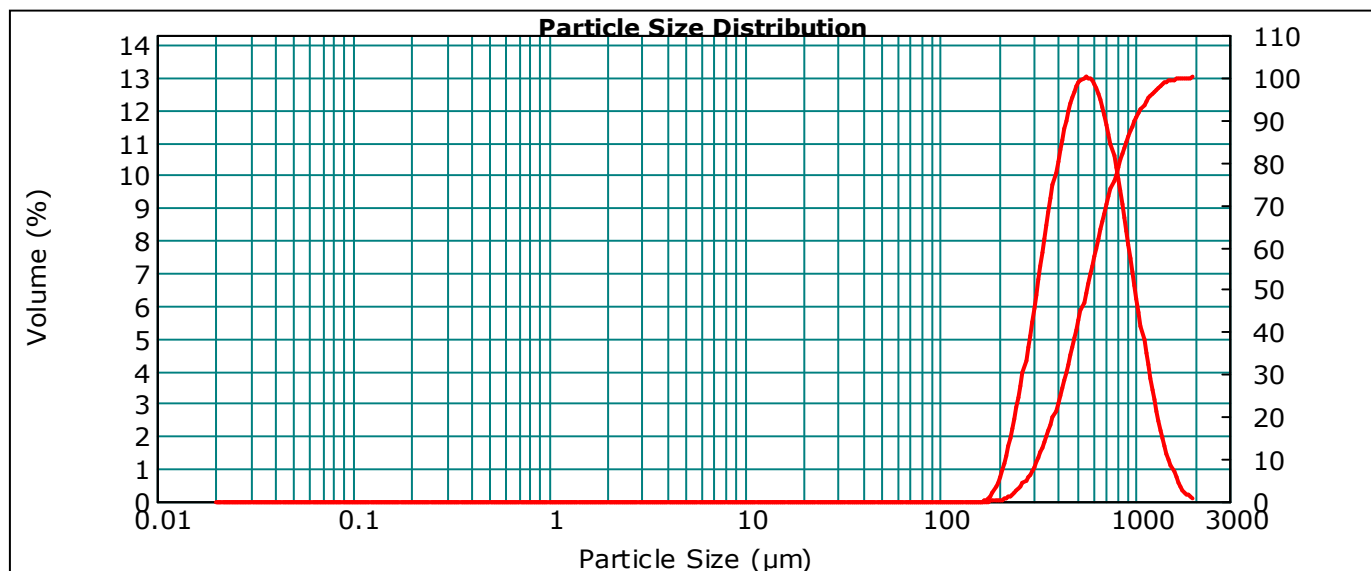
Inclusive SD (Sorting Coeff.) = .64 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .84 phi (Coarse Sand)

Inclusive Skewness = -0.01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





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Sample Name: C_1_1_20 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:06:09 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 606.67um = .72phi

Surface Weighted Mean:

D[3,2] = 504.63um = .99phi

Span :

1.201

Uniformity:

0.371

Specific Surface Area*:

0.0119 m²/g

Percentiles:

d(0.05) = 272.9um = 1.87phi

d(0.10) = 313.31um = 1.67phi

d(0.16) = 352.84um = 1.5phi

d(0.50) = 552.33um = .86phi

d(0.84) = 865.83um = .21phi

d(0.90) = 976.54um = .03phi

d(0.95) = 1125.49um = -.017phi

Distribution Properties:

Kurtosis = 1.13

Skewness = 1.04

Standard Deviation = 267.99 um = 1.9phi

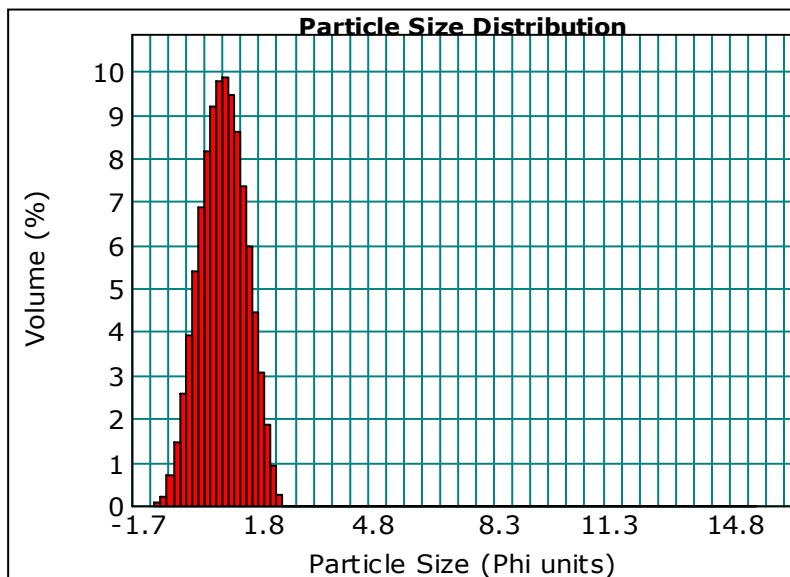
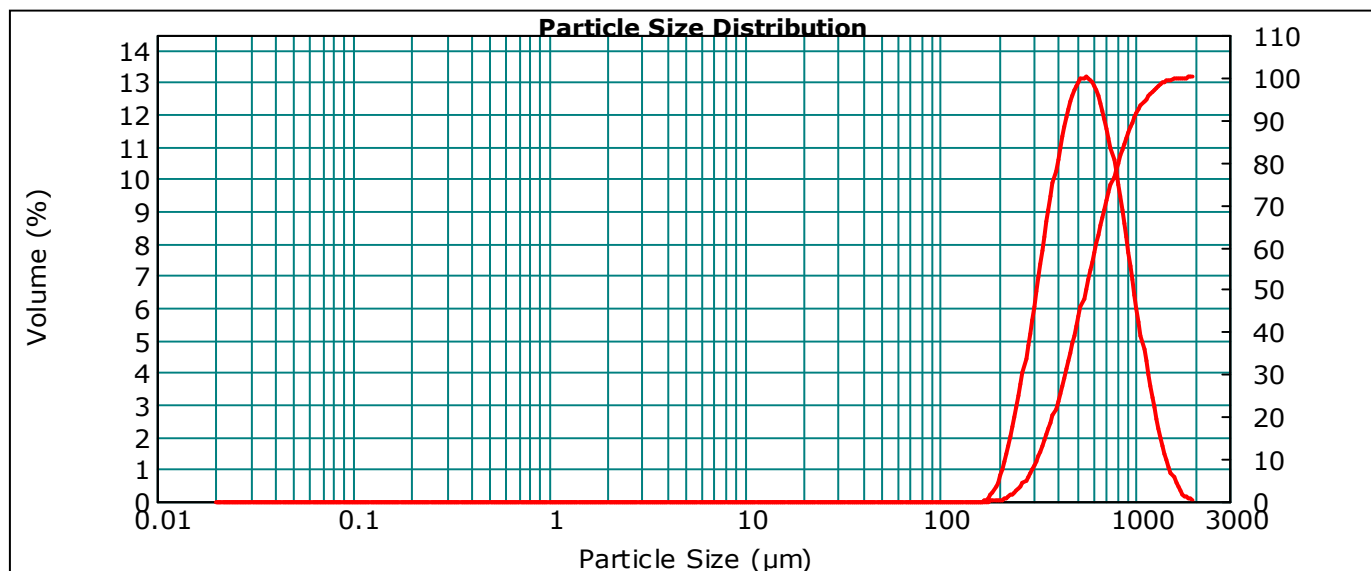
Inclusive SD (Sorting Coeff.) = .63 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .86 phi (Coarse Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.01
-0.5	8.00
0.0	20.55
0.5	28.96
1.0	25.36
1.5	13.25
2.0	2.87
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:21:31 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 571.63um = .81phi

Surface Weighted Mean:

D[3,2] = 491.9um = 1.02phi

Span :

1.065

Uniformity:

0.331

Specific Surface Area*:

0.0122 m²/g

Percentiles:

d(0.05) = 278.39um = 1.84phi

d(0.10) = 317.12um = 1.66phi

d(0.16) = 353.95um = 1.5phi

d(0.50) = 532.09um = .91phi

d(0.84) = 794.87um = .33phi

d(0.90) = 883.68um = .18phi

d(0.95) = 999.86um = 0phi

Distribution Properties:

Kurtosis = .48

Skewness = .84

Standard Deviation = 224.02 um = 2.16phi

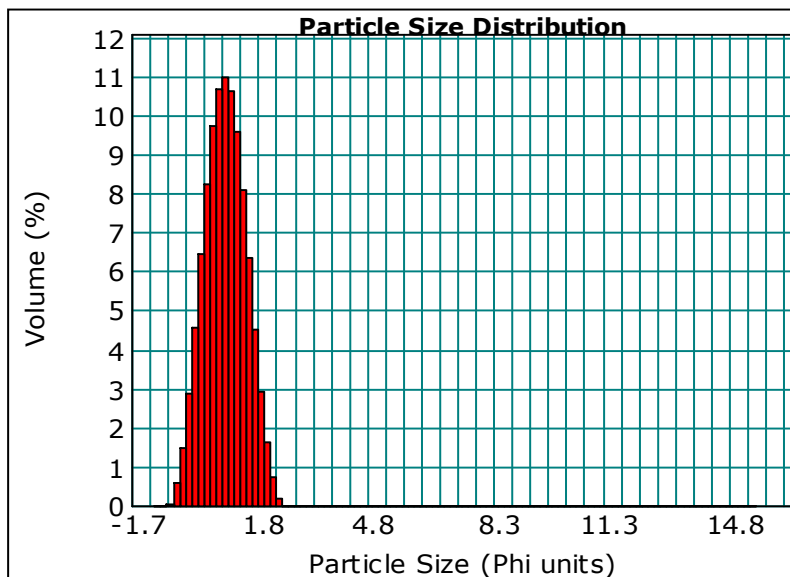
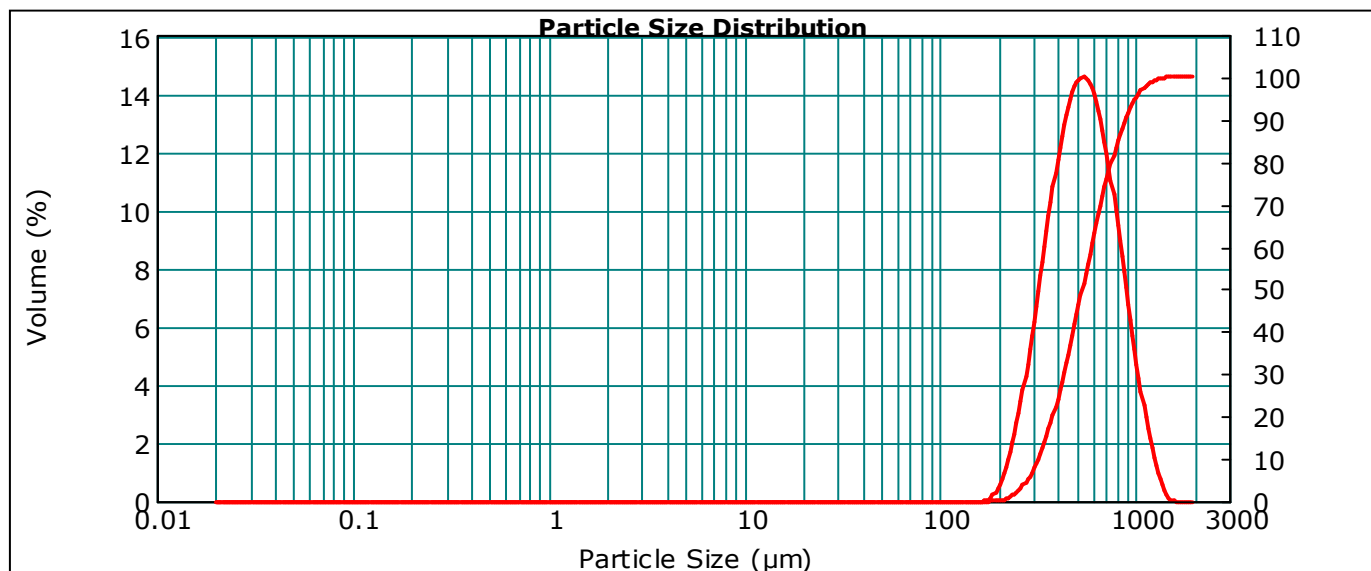
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .91 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.05
-0.5	4.95
0.0	19.39
0.5	31.54
1.0	28.14
1.5	13.53
2.0	2.40
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:22:38 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 568.74um = .81phi

Surface Weighted Mean:

D[3,2] = 489.25um = 1.03phi

Span :

1.067

Uniformity:

0.332

Specific Surface Area*:

0.0123 m²/g

Percentiles:

d(0.05) = 276.85um = 1.85phi

d(0.10) = 315.31um = 1.67phi

d(0.16) = 351.99um = 1.51phi

d(0.50) = 529.09um = .92phi

d(0.84) = 790.94um = .34phi

d(0.90) = 879.73um = .18phi

d(0.95) = 995.99um = .01phi

Distribution Properties:

Kurtosis = .51

Skewness = .85

Standard Deviation = 223.36 um = 2.16phi

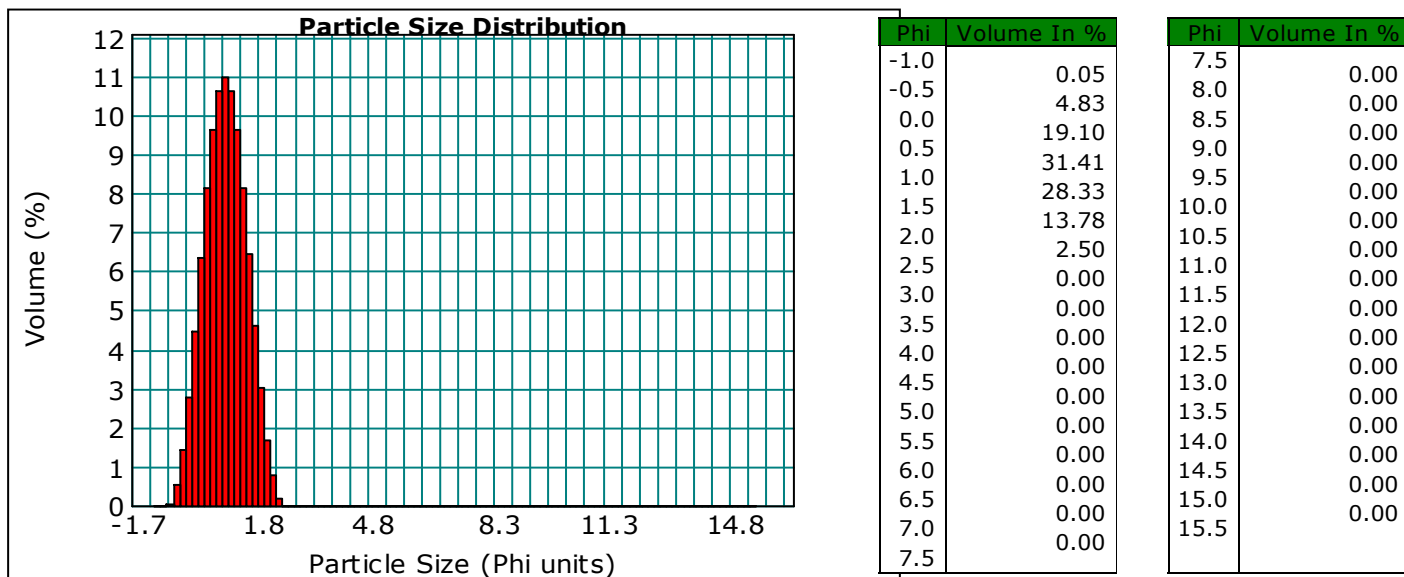
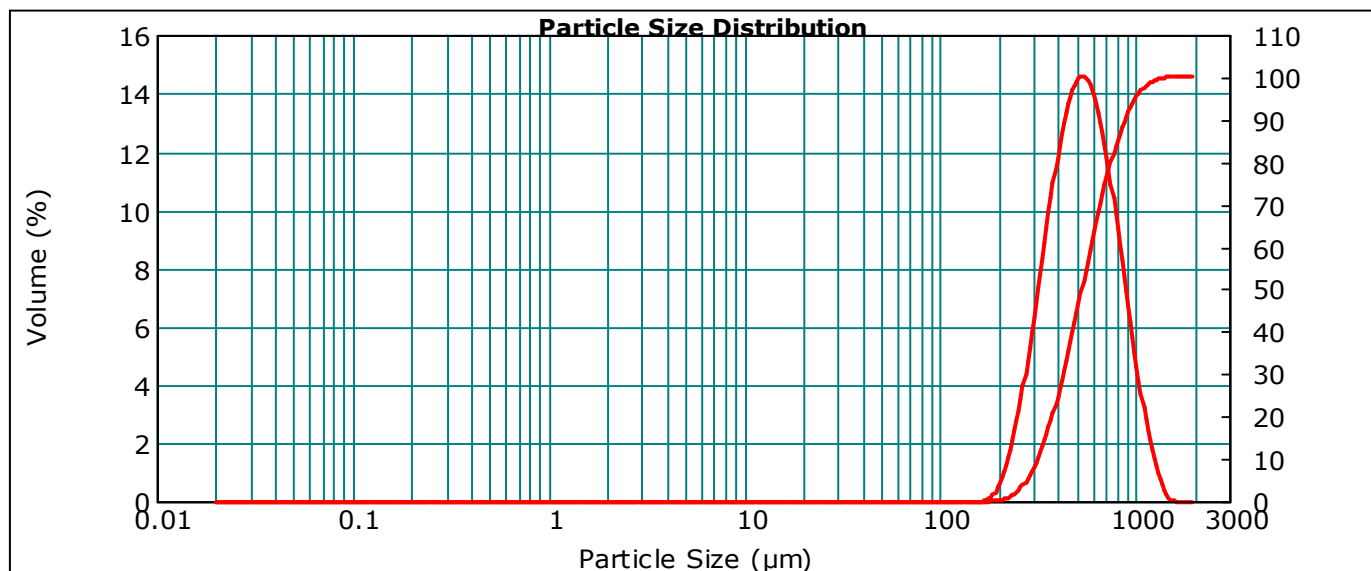
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .92 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:23:46 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 571.81um = .81phi

Surface Weighted Mean:

D[3,2] = 491.53um = 1.02phi

Span :

1.070

Uniformity:

0.333

Specific Surface Area*:

0.0122 m²/g

Percentiles:

d(0.05) = 277.89um = 1.85phi

d(0.10) = 316.59um = 1.66phi

d(0.16) = 353.42um = 1.5phi

d(0.50) = 531.67um = .91phi

d(0.84) = 795.64um = .33phi

d(0.90) = 885.26um = .18phi

d(0.95) = 1002.92um = 0phi

Distribution Properties:

Kurtosis = .51

Skewness = .86

Standard Deviation = 225.2 um = 2.15phi

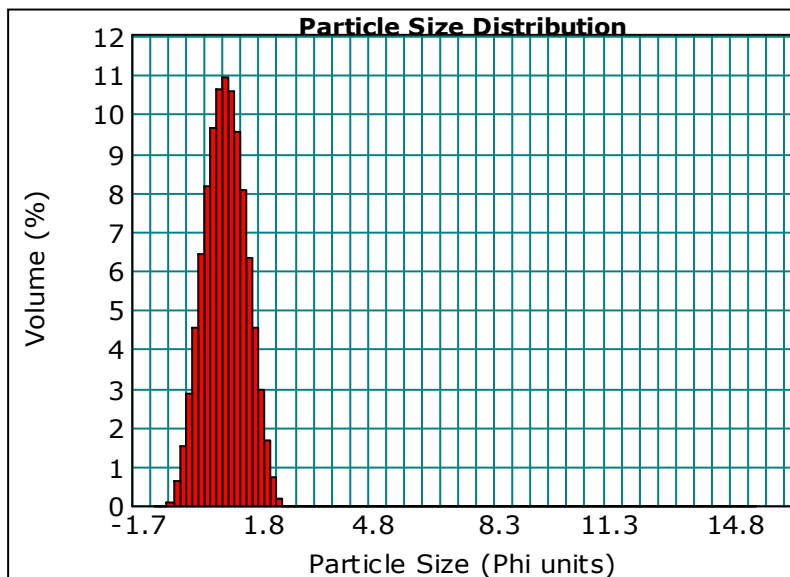
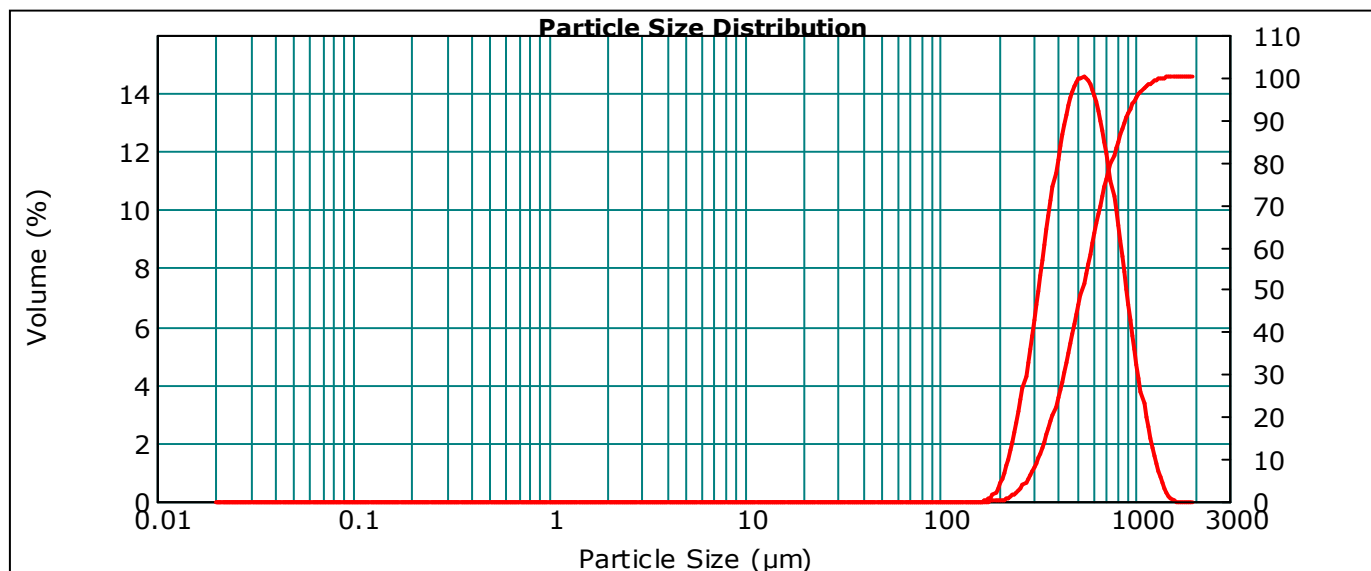
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .91 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.06
-0.5	5.04
0.0	19.31
0.5	31.43
1.0	28.14
1.5	13.59
2.0	2.44
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_19 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:21:31 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 570.73um = .81phi

Surface Weighted Mean:

D[3,2] = 490.89um = 1.03phi

Span :

1.067

Uniformity:

0.332

Specific Surface Area*:

0.0122 m²/g

Percentiles:

d(0.05) = 277.7um = 1.85phi

d(0.10) = 316.33um = 1.66phi

d(0.16) = 353.12um = 1.5phi

d(0.50) = 530.95um = .91phi

d(0.84) = 793.83um = .33phi

d(0.90) = 882.9um = .18phi

d(0.95) = 999.61um = 0phi

Distribution Properties:

Kurtosis = .5

Skewness = .85

Standard Deviation = 224.2 um = 2.16phi

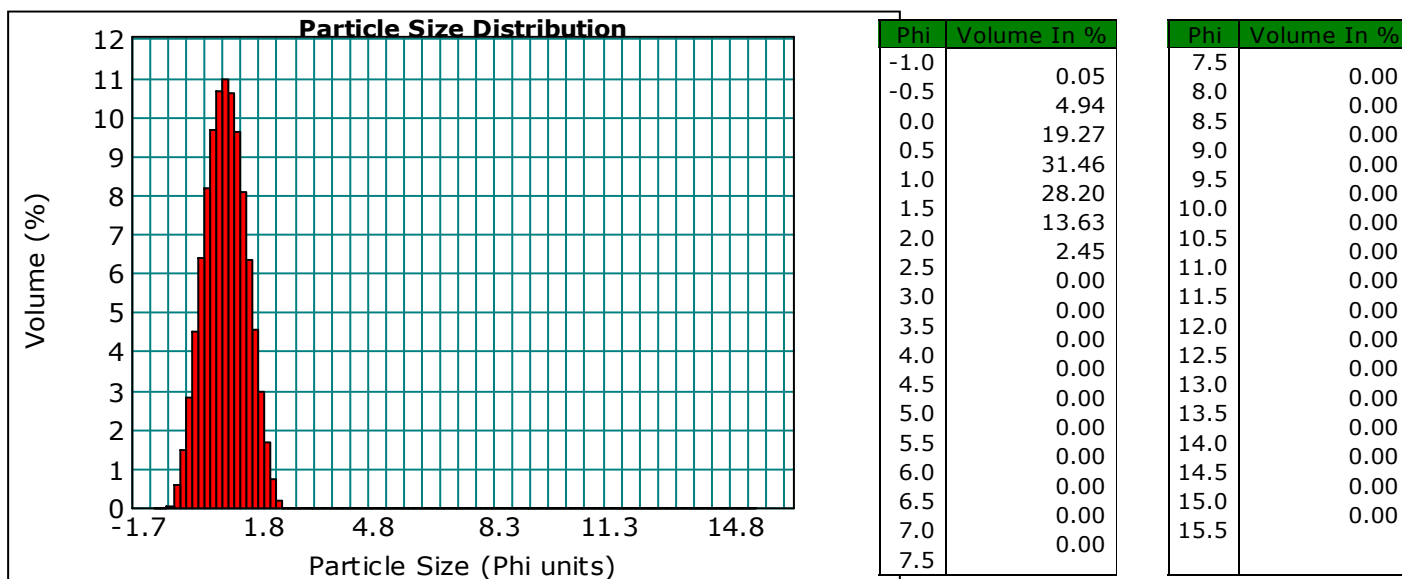
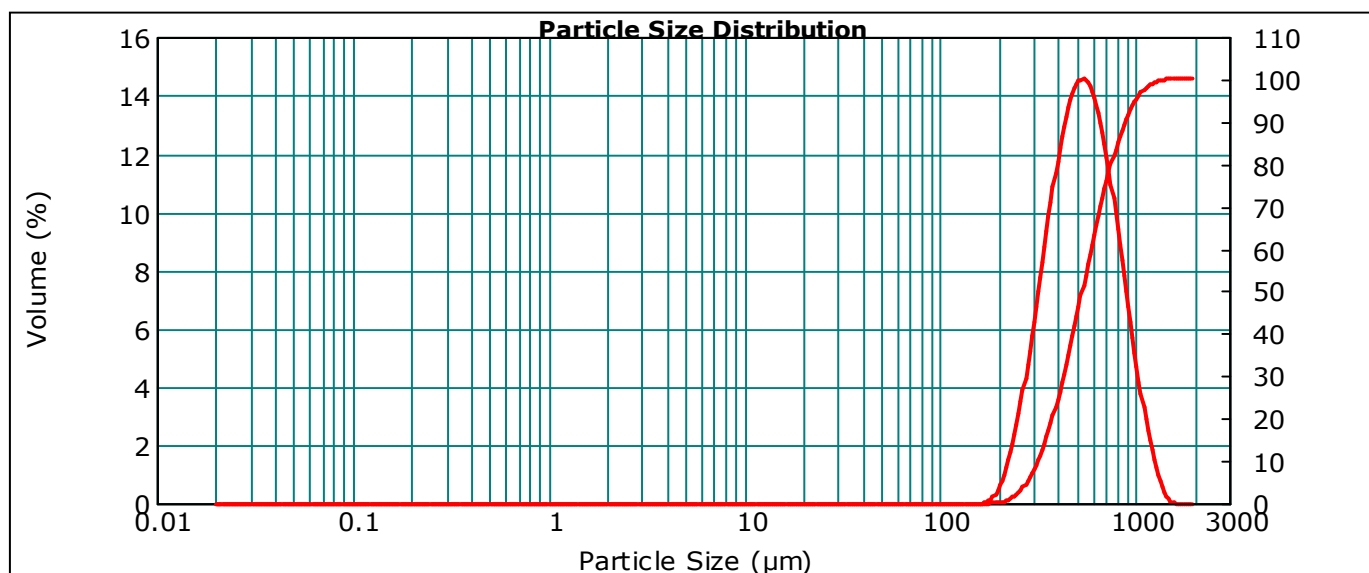
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .92 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:32:43 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 585.61um = .77phi

Surface Weighted Mean:

D[3,2] = 505.11um = .99phi

Span :

1.056

Uniformity:

0.326

Specific Surface Area*:

0.0119 m²/g

Percentiles:

d(0.05) = 286.96um = 1.8phi

d(0.10) = 326.39um = 1.62phi

d(0.16) = 364.03um = 1.46phi

d(0.50) = 545.8um = .87phi

d(0.84) = 812.8um = .3phi

d(0.90) = 902.79um = .15phi

d(0.95) = 1020.27um = -.03phi

Distribution Properties:

Kurtosis = .44

Skewness = .83

Standard Deviation = 227.56 um = 2.14phi

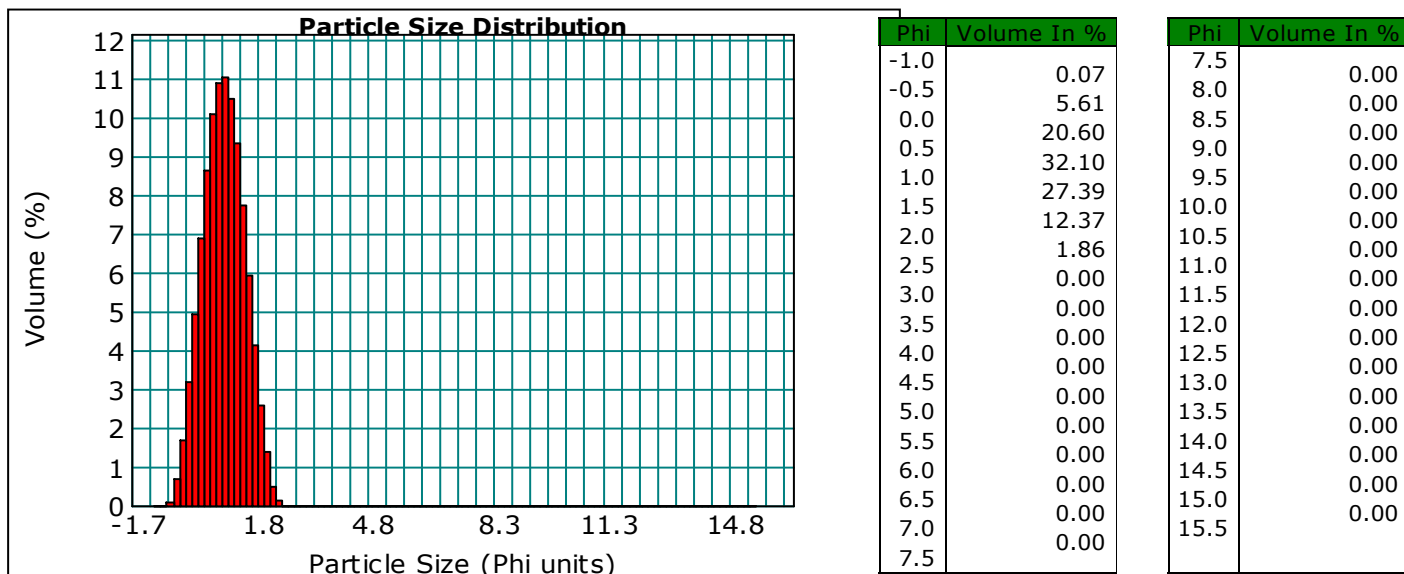
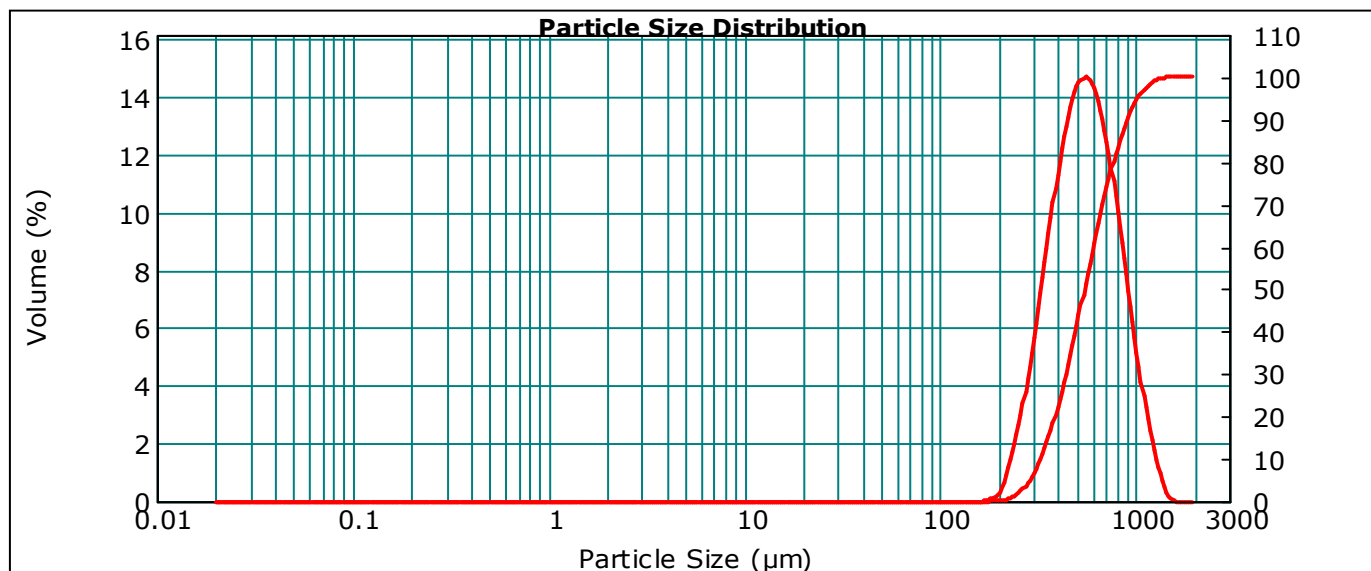
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .88 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:33:50 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 590.44um = .76phi

Surface Weighted Mean:

D[3,2] = 508.22um = .98phi

Span :

1.063

Uniformity:

0.327

Specific Surface Area*:

0.0118 m²/g

Percentiles:

d(0.05) = 287.75um = 1.8phi

d(0.10) = 327.62um = 1.61phi

d(0.16) = 365.72um = 1.45phi

d(0.50) = 549.69um = .86phi

d(0.84) = 820.58um = .29phi

d(0.90) = 912um = .13phi

d(0.95) = 1032.01um = -0.05phi

Distribution Properties:

Kurtosis = .51

Skewness = .85

Standard Deviation = 231.31 um = 2.11phi

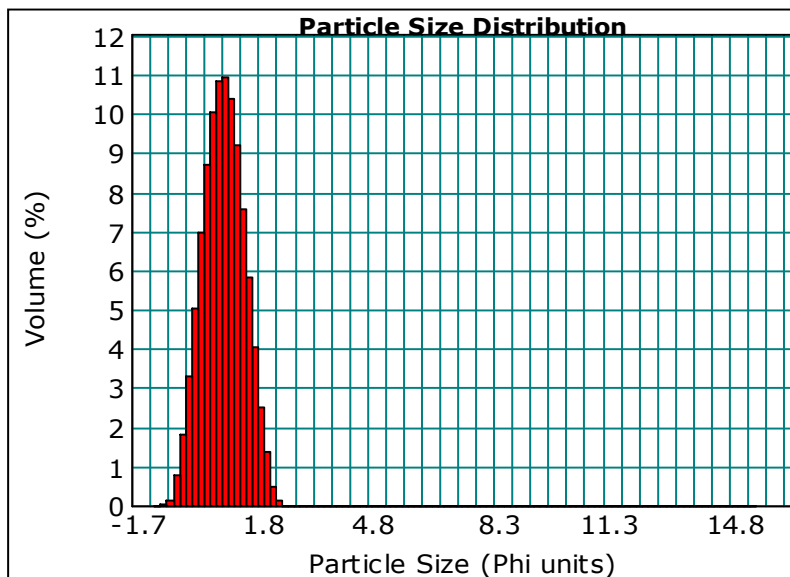
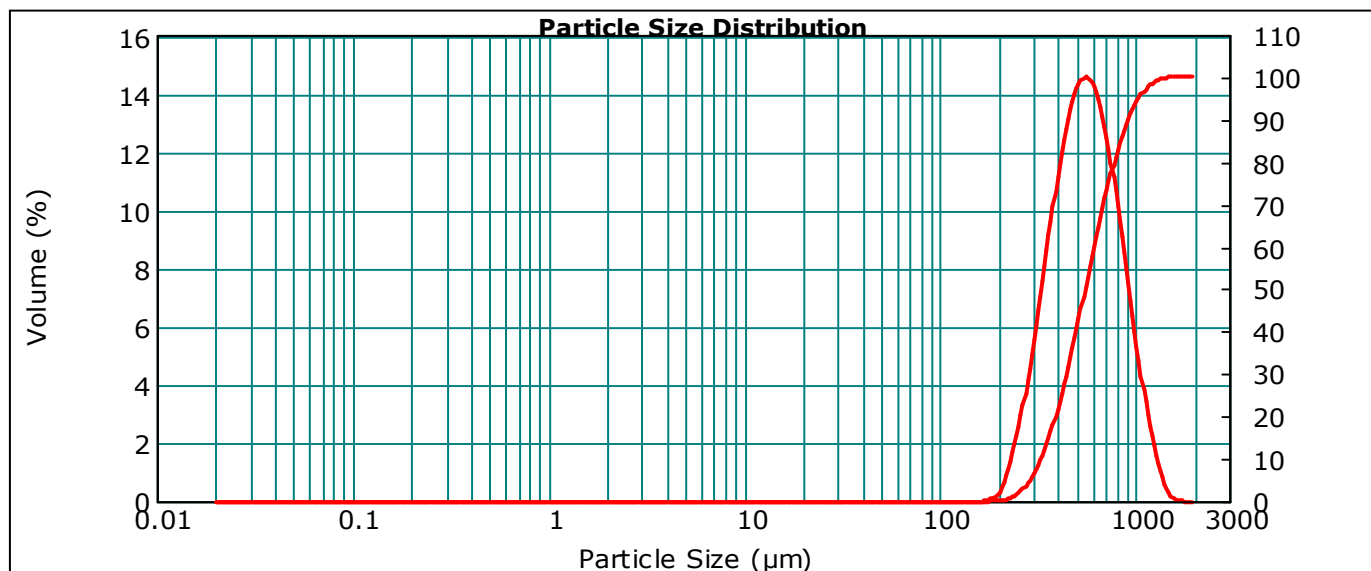
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .87 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.14
-0.5	5.94
0.0	20.90
0.5	32.01
1.0	27.03
1.5	12.15
2.0	1.83
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:34:58 AM

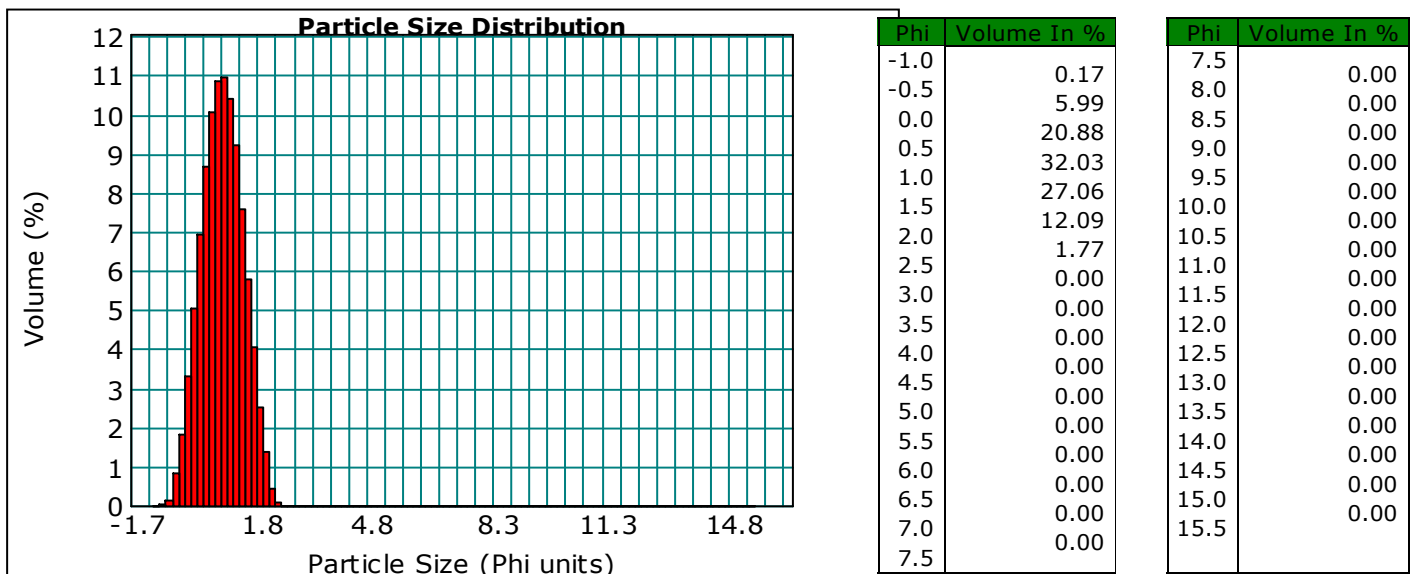
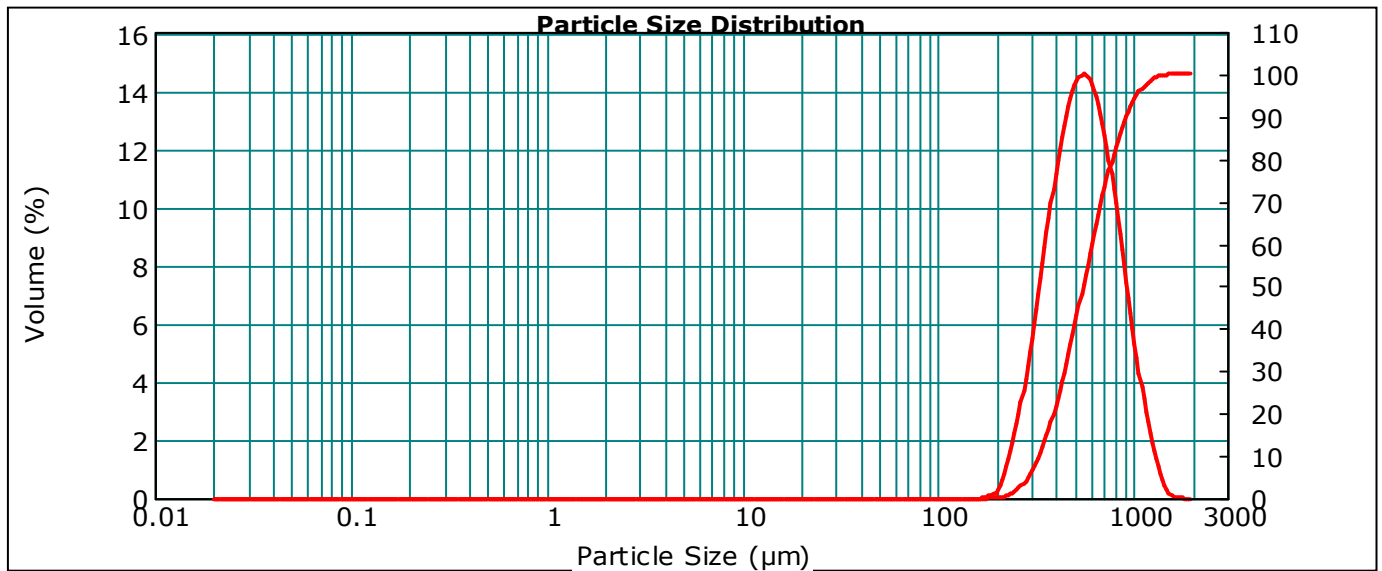
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 591.35um = .76phi
Surface Weighted Mean:
D[3,2] = 509.08um = .97phi
Span :
1.063
Uniformity:
0.328
Specific Surface Area*:
0.0118 m²/g

Percentiles:
d(0.05) = 288.64um = 1.79phi
d(0.10) = 328.41um = 1.61phi
d(0.16) = 366.42um = 1.45phi
d(0.50) = 550.09um = .86phi
d(0.84) = 821.47um = .28phi
d(0.90) = 913.42um = .13phi
d(0.95) = 1034.49um = -.05phi

Distribution Properties:
Kurtosis = .57
Skewness = .87
Standard Deviation = 232.06 um = 2.11phi
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = .86 phi (Coarse Sand)
Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_18 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:32:43 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 589.13um = .76phi

Surface Weighted Mean:

D[3,2] = 507.47um = .98phi

Span :

1.061

Uniformity:

0.327

Specific Surface Area*:

0.0118 m²/g

Percentiles:

d(0.05) = 287.78um = 1.8phi

d(0.10) = 327.47um = 1.61phi

d(0.16) = 365.38um = 1.45phi

d(0.50) = 548.52um = .87phi

d(0.84) = 818.3um = .29phi

d(0.90) = 909.43um = .14phi

d(0.95) = 1029.02um = -.04phi

Distribution Properties:

Kurtosis = .51

Skewness = .85

Standard Deviation = 230.33 um = 2.12phi

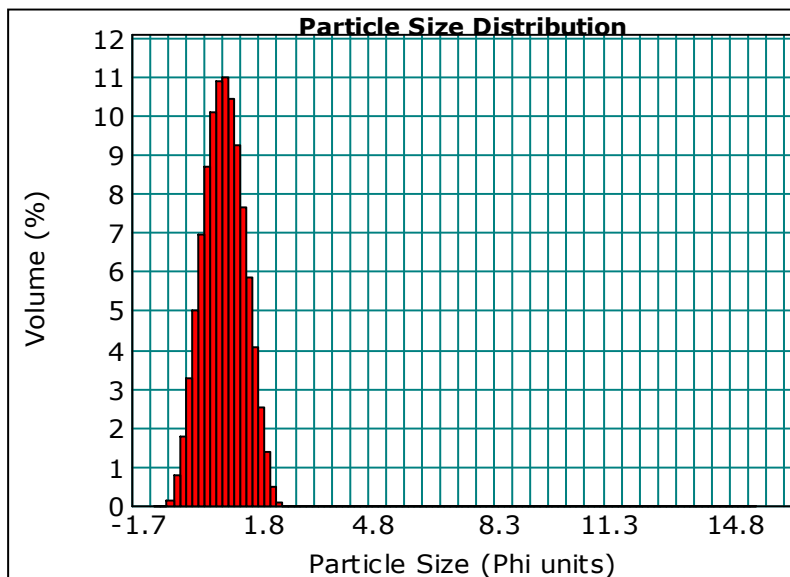
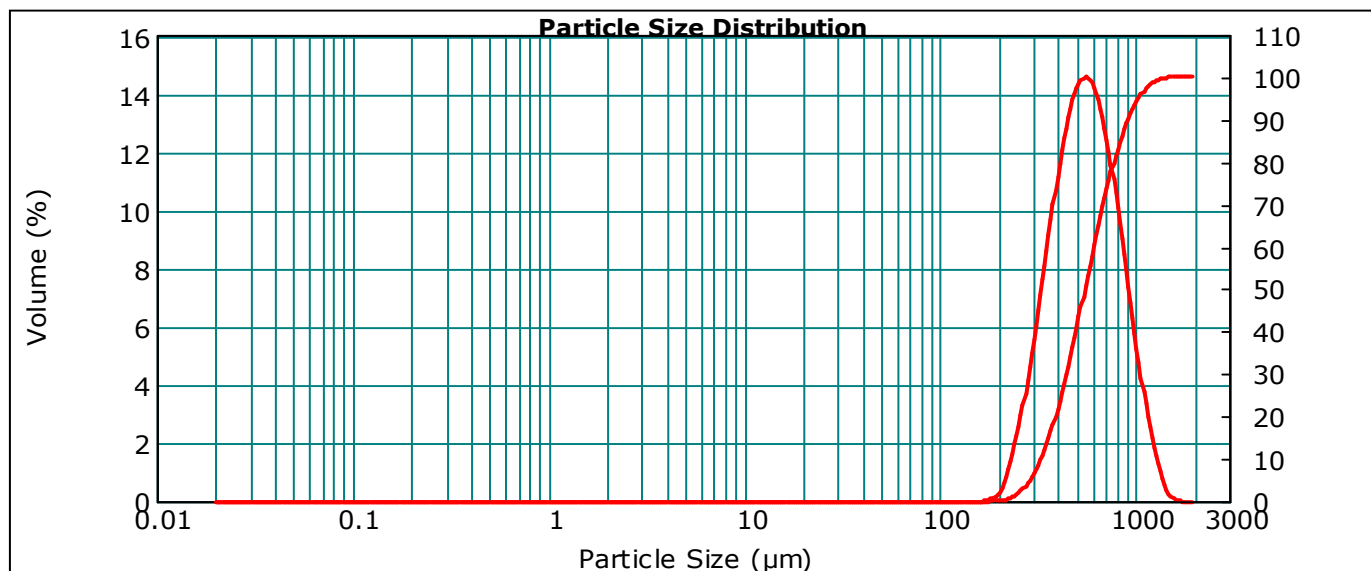
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .87 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.13
-0.5	5.84
0.0	20.79
0.5	32.05
1.0	27.16
1.5	12.21
2.0	1.82
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_17
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:43:40 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 580.94um = .78phi

Surface Weighted Mean:

D[3,2] = 500.39um = 1phi

Span :

1.062

Uniformity:

0.329

Specific Surface Area*:

0.012 m²/g

Percentiles:

d(0.05) = 283.64um = 1.82phi

d(0.10) = 322.88um = 1.63phi

d(0.16) = 360.33um = 1.47phi

d(0.50) = 540.95um = .89phi

d(0.84) = 807.07um = .31phi

d(0.90) = 897.1um = .16phi

d(0.95) = 1014.67um = -.02phi

Distribution Properties:

Kurtosis = .46

Skewness = .84

Standard Deviation = 226.95 um = 2.14phi

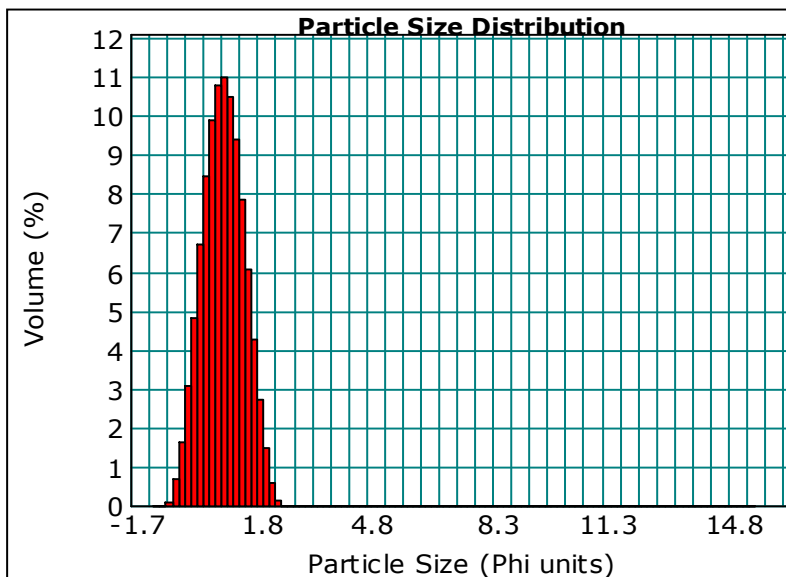
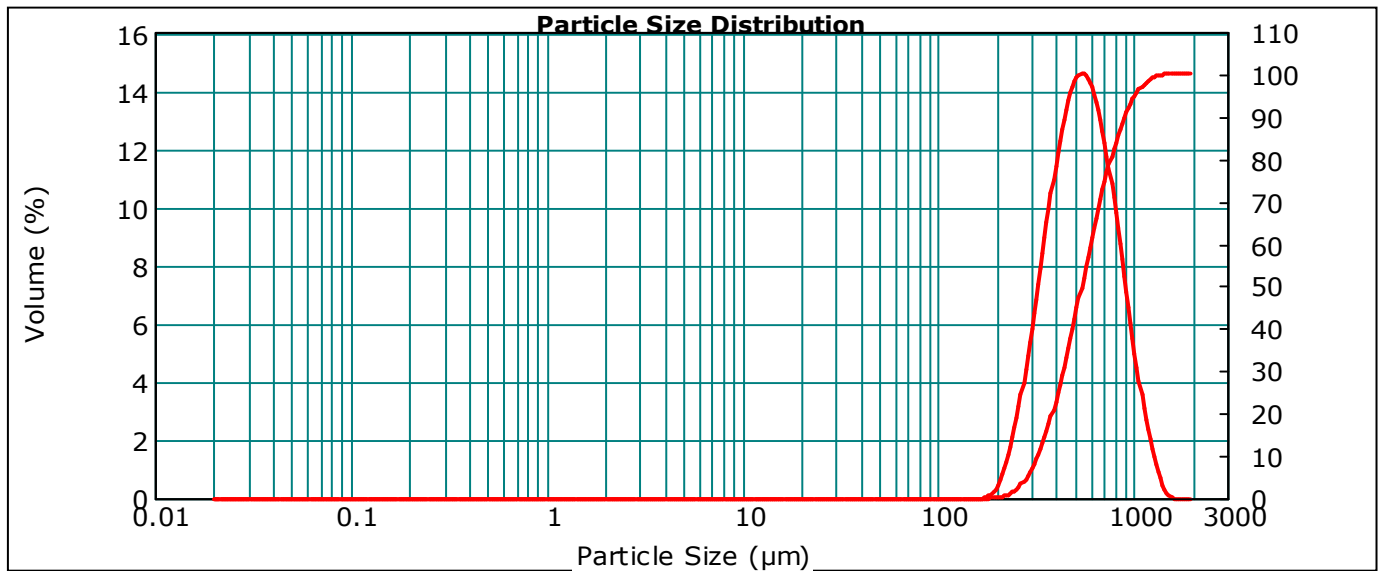
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .89 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.06
-0.5	5.43
0.0	20.16
0.5	31.86
1.0	27.64
1.5	12.80
2.0	2.05
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_17
Measured by: WATERCHEM
Particle Name: Marine Sediment
SOP Name: Unknown Sample (RI=1.55 AI=0.1) - quick & no
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:27:20 PM

Dispersant: Water
Sonication (seconds): 0
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 575.42um = .8phi

Surface Weighted Mean:

D[3,2] = 495.94um = 1.01phi

Span :

1.059

Uniformity:

0.329

Specific Surface Area*:

0.0121 m²/g

Percentiles:

d(0.05) = 281.5um = 1.83phi

d(0.10) = 320.32um = 1.64phi

d(0.16) = 357.36um = 1.48phi

d(0.50) = 535.89um = .9phi

d(0.84) = 798.89um = .32phi

d(0.90) = 887.79um = .17phi

d(0.95) = 1004.32um = -.01phi

Distribution Properties:

Kurtosis = .48

Skewness = .84

Standard Deviation = 224.43 um = 2.16phi

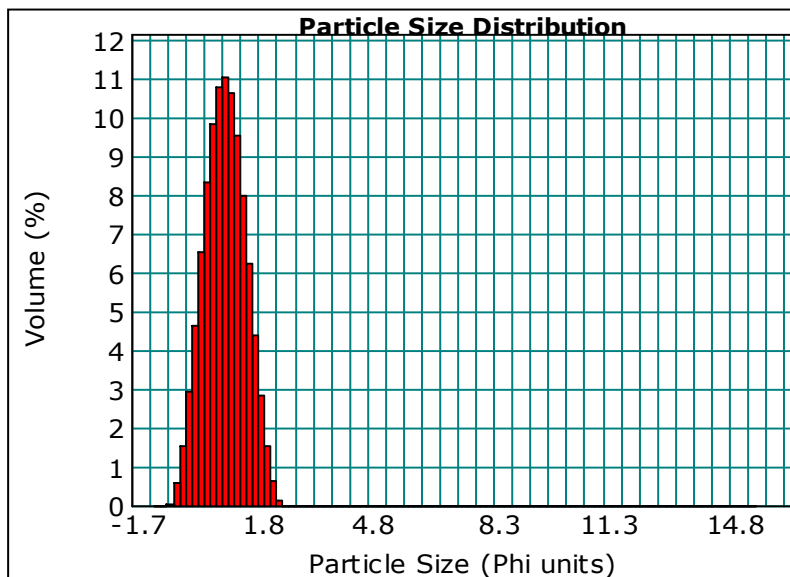
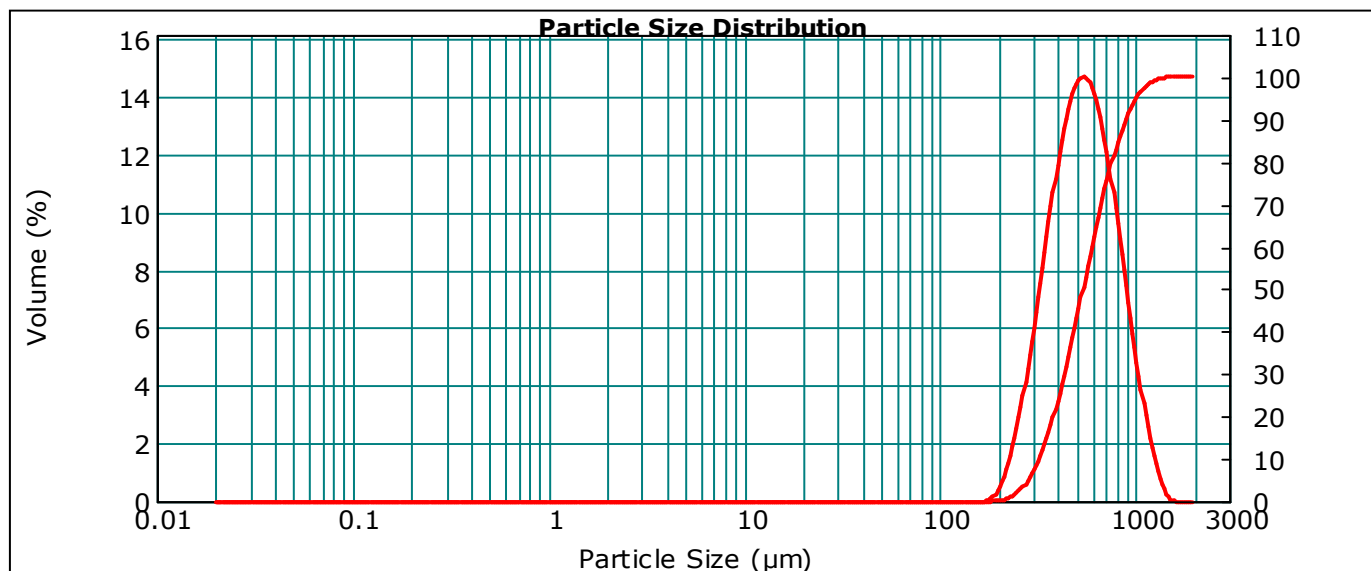
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .9 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.05
-0.5	5.09
0.0	19.69
0.5	31.81
1.0	28.02
1.5	13.16
2.0	2.18
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_17
Measured by: WATERCHEM
Particle Name: Marine Sediment
SOP Name: Unknown Sample (RI=1.55 AI=0.1) - quick & no
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:28:08 PM

Dispersant: Water
Sonication (seconds): 0
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 577.17um = .79phi

Surface Weighted Mean:

D[3,2] = 496.5um = 1.01phi

Span :

1.068

Uniformity:

0.331

Specific Surface Area*:

0.0121 m²/g

Percentiles:

d(0.05) = 281.28um = 1.83phi

d(0.10) = 320.11um = 1.64phi

d(0.16) = 357.19um = 1.49phi

d(0.50) = 536.52um = .9phi

d(0.84) = 802.51um = .32phi

d(0.90) = 892.96um = .16phi

d(0.95) = 1011.91um = -.02phi

Distribution Properties:

Kurtosis = .51

Skewness = .86

Standard Deviation = 227 um = 2.14phi

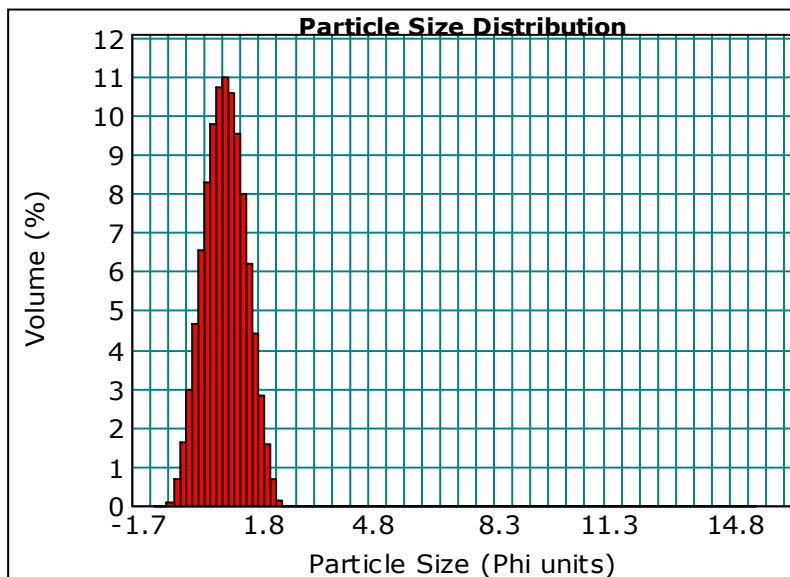
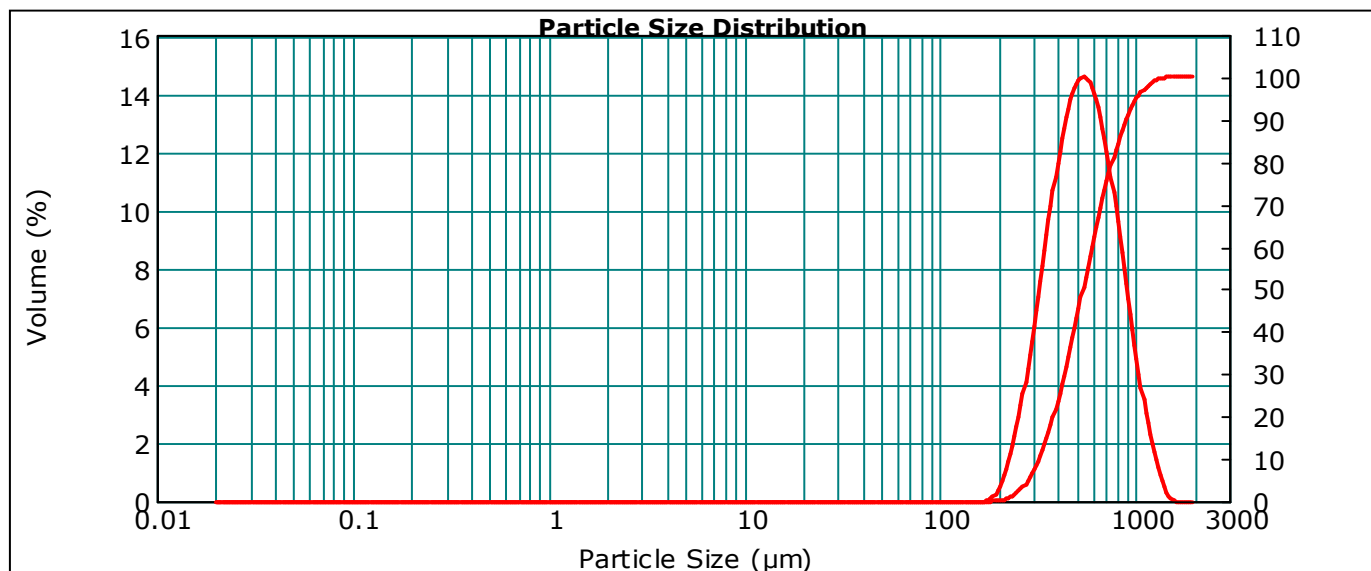
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .9 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.07
-0.5	5.33
0.0	19.70
0.5	31.62
1.0	27.92
1.5	13.18
2.0	2.19
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_17
Measured by: WATERCHEM
Particle Name: Marine Sediment
SOP Name: Unknown Sample (RI=1.55 AI=0.1) - quick & no
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:28:56 PM

Dispersant: Water
Sonication (seconds): 0
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 583.67um = .78phi

Surface Weighted Mean:

D[3,2] = 501.05um = 1phi

Span :

1.076

Uniformity:

0.333

Specific Surface Area*:

0.012 m²/g

Percentiles:

d(0.05) = 283.17um = 1.82phi

d(0.10) = 322.44um = 1.63phi

d(0.16) = 359.97um = 1.47phi

d(0.50) = 541.61um = .88phi

d(0.84) = 812.44um = .3phi

d(0.90) = 905.1um = .14phi

d(0.95) = 1027.53um = -.04phi

Distribution Properties:

Kurtosis = .66

Skewness = .89

Standard Deviation = 231.83 um = 2.11phi

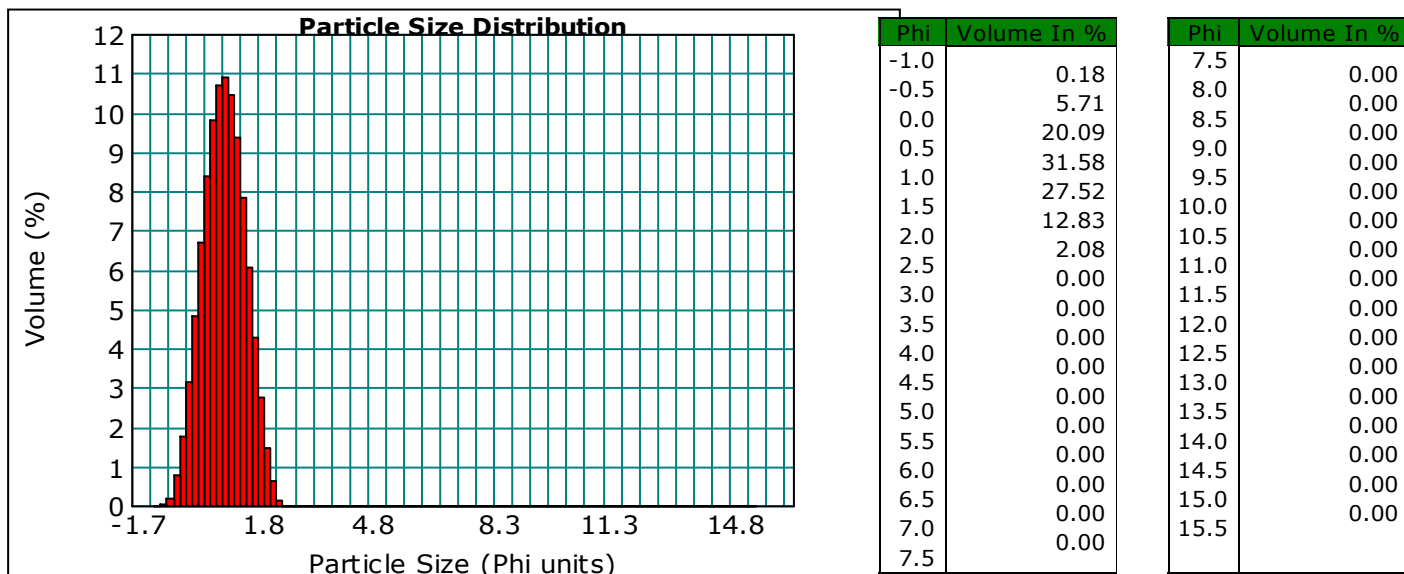
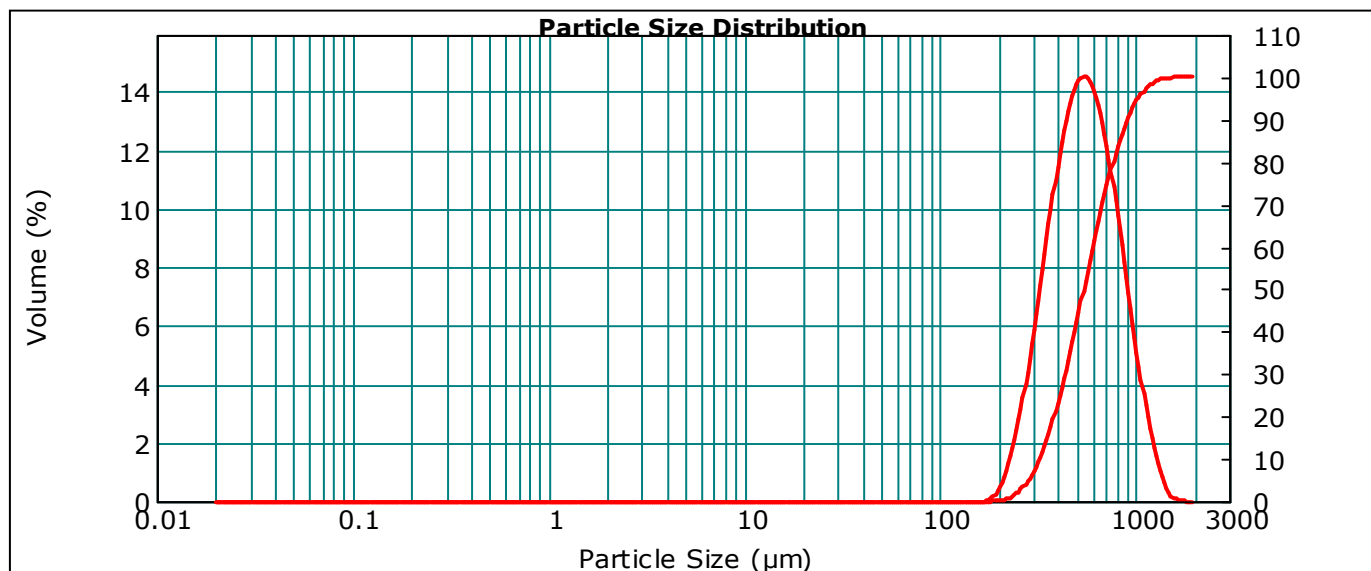
Inclusive SD (Sorting Coeff.) = .58 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .89 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_17 - Average
Measured by: WATERCHEM
Particle Name: Marine Sediment
SOP Name: Unknown Sample (RI=1.55 AI=0.1) - quick & no
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:27:20 PM

Dispersant: Water
Sonication (seconds): 0
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 578.75um = .79phi

Surface Weighted Mean:

D[3,2] = 497.82um = 1.01phi

Span :

1.068

Uniformity:

0.331

Specific Surface Area*:

0.0121 m²/g

Percentiles:

d(0.05) = 281.97um = 1.83phi

d(0.10) = 320.94um = 1.64phi

d(0.16) = 358.16um = 1.48phi

d(0.50) = 537.99um = .89phi

d(0.84) = 804.53um = .31phi

d(0.90) = 895.25um = .16phi

d(0.95) = 1014.53um = -.02phi

Distribution Properties:

Kurtosis = .56

Skewness = .87

Standard Deviation = 227.8 um = 2.13phi

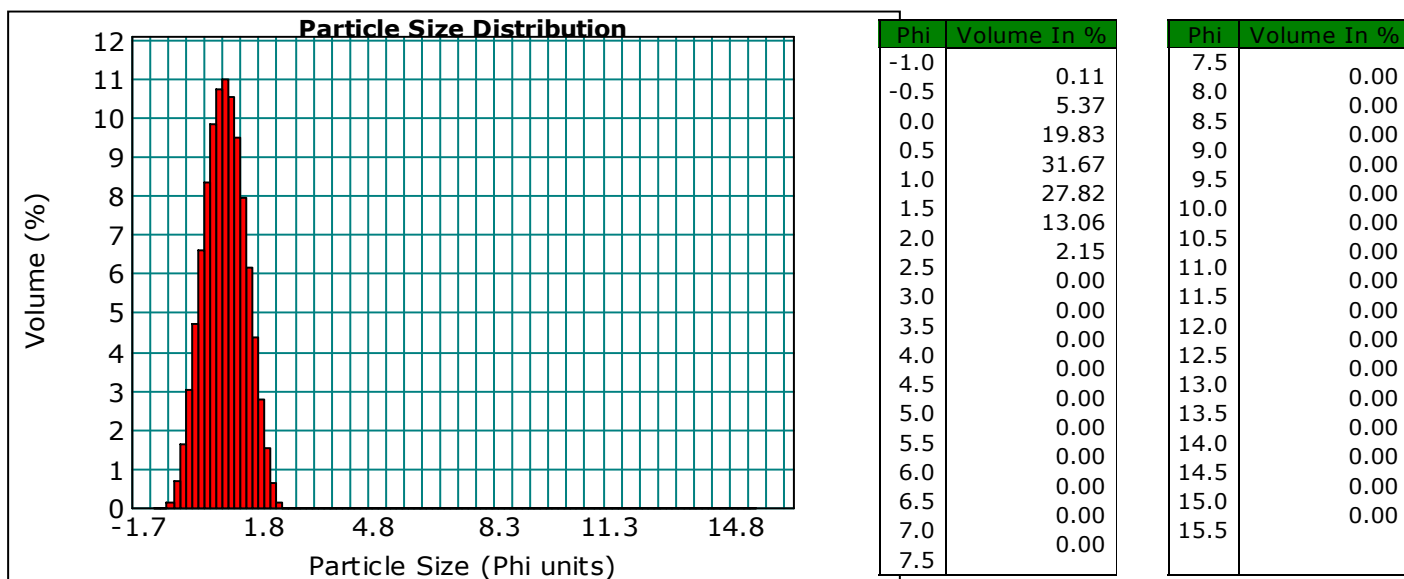
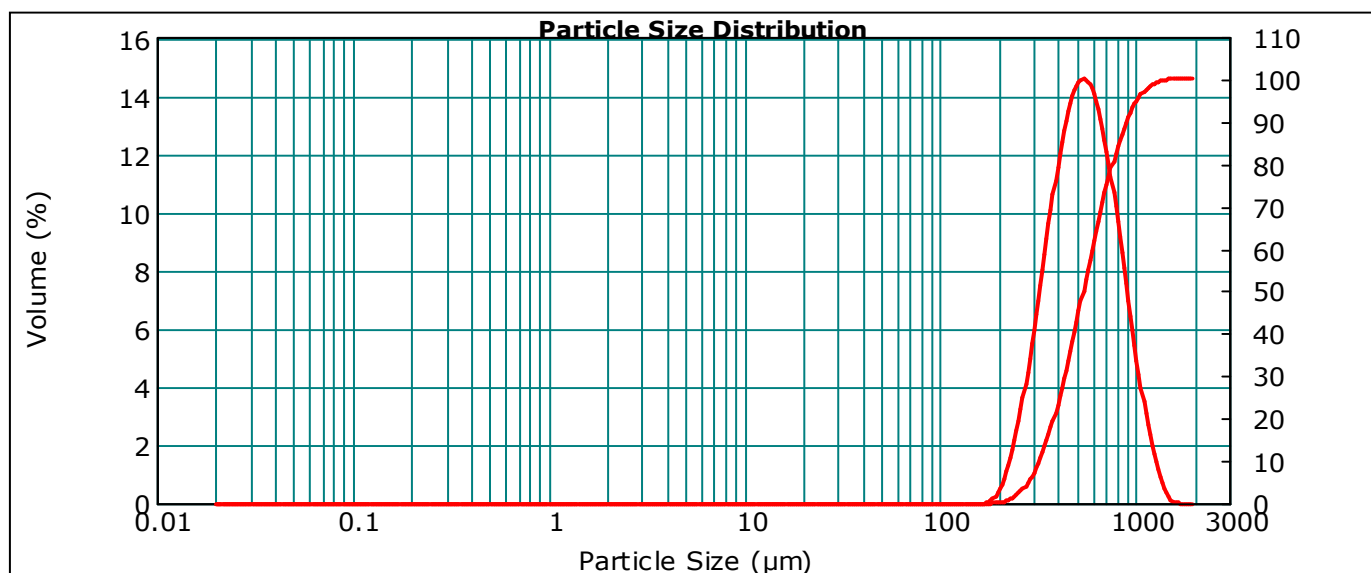
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .9 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:45:06 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 565.72um = .82phi

Surface Weighted Mean:

D[3,2] = 480.99um = 1.06phi

Span :

1.115

Uniformity:

0.346

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 265.92um = 1.91phi

d(0.10) = 303.95um = 1.72phi

d(0.16) = 341.04um = 1.55phi

d(0.50) = 524.85um = .93phi

d(0.84) = 798.49um = .32phi

d(0.90) = 889.06um = .17phi

d(0.95) = 1004.9um = -.01phi

Distribution Properties:

Kurtosis = .28

Skewness = .8

Standard Deviation = 228.91 um = 2.13phi

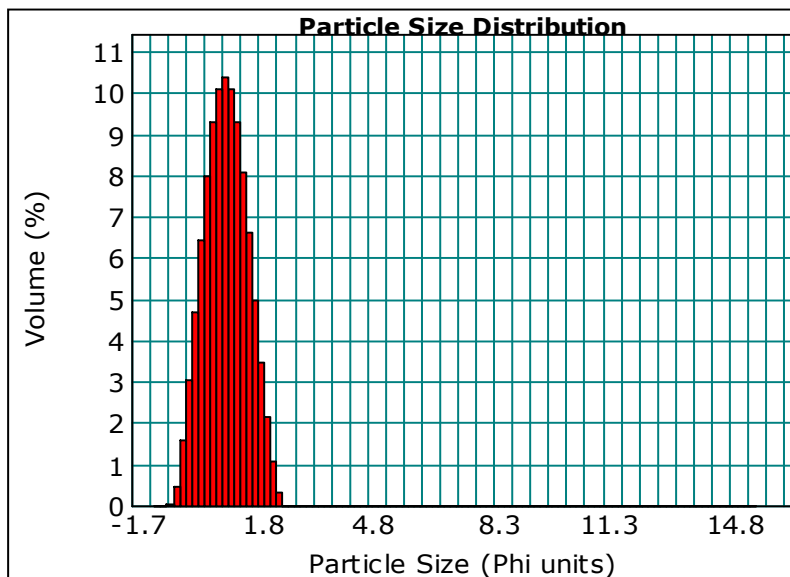
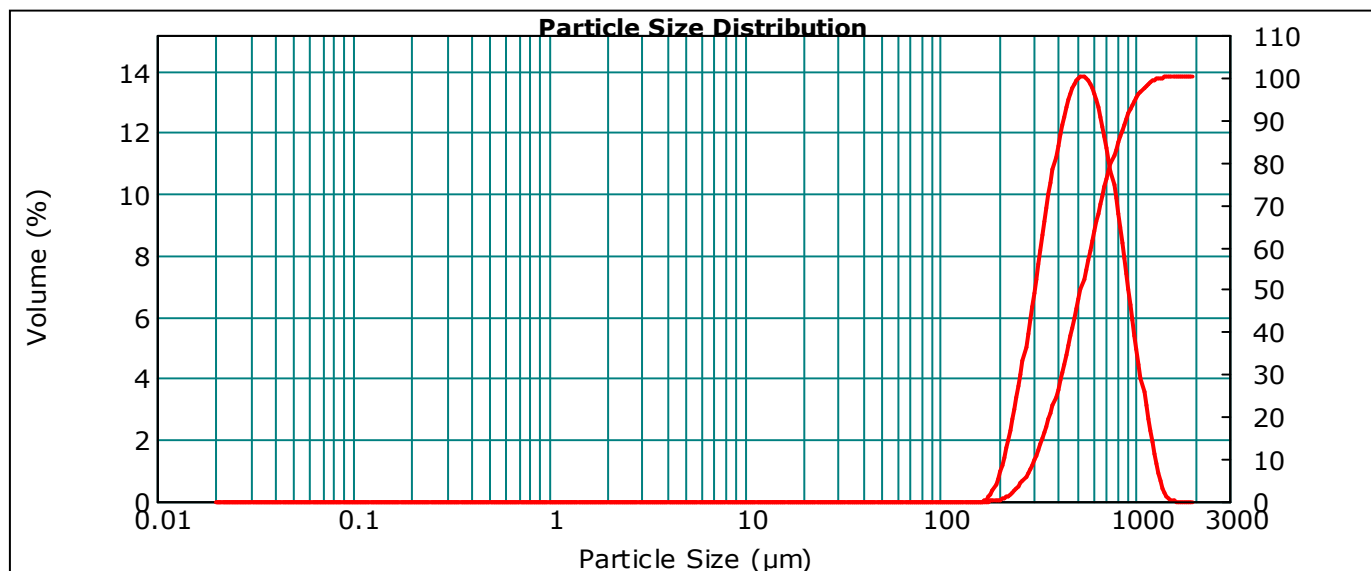
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .94 phi (Coarse Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.02
-0.5	5.14
0.0	19.28
0.5	29.93
1.0	27.42
1.5	14.83
2.0	3.37
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:46:13 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 563.92um = .83phi

Surface Weighted Mean:

D[3,2] = 487.79um = 1.04phi

Span :

1.043

Uniformity:

0.325

Specific Surface Area*:

0.0123 m²/g

Percentiles:

d(0.05) = 277.94um = 1.85phi

d(0.10) = 316.11um = 1.66phi

d(0.16) = 352.43um = 1.5phi

d(0.50) = 526.99um = .92phi

d(0.84) = 781.06um = .36phi

d(0.90) = 865.77um = .21phi

d(0.95) = 975.67um = .04phi

Distribution Properties:

Kurtosis = .38

Skewness = .8

Standard Deviation = 216.34 um = 2.21phi

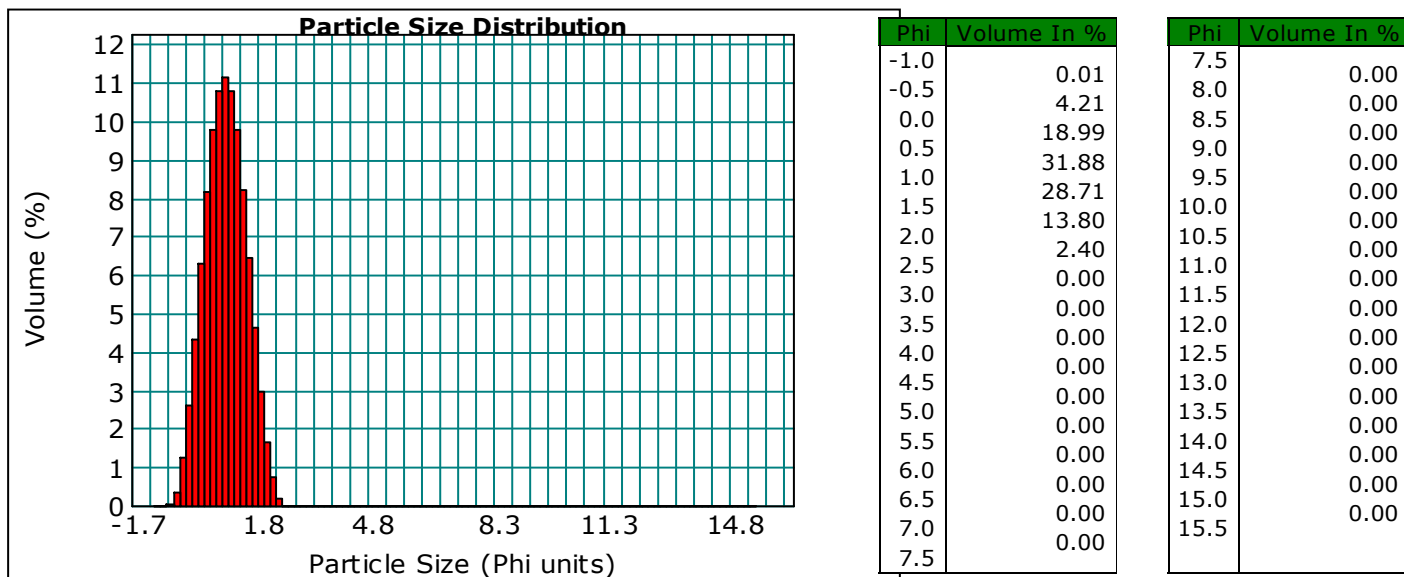
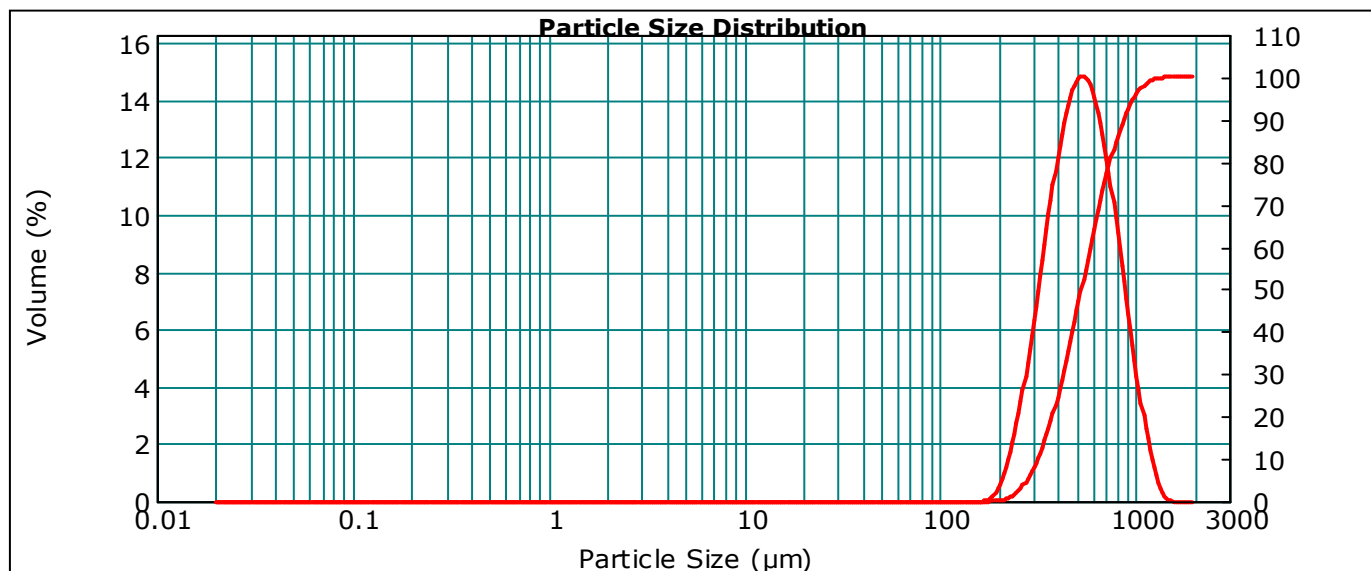
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .93 phi (Coarse Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:47:21 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 563.98um = .83phi

Surface Weighted Mean:

D[3,2] = 487.63um = 1.04phi

Span :

1.046

Uniformity:

0.326

Specific Surface Area*:

0.0123 m²/g

Percentiles:

d(0.05) = 277.88um = 1.85phi

d(0.10) = 315.92um = 1.66phi

d(0.16) = 352.16um = 1.51phi

d(0.50) = 526.62um = .93phi

d(0.84) = 781.5um = .36phi

d(0.90) = 866.66um = .21phi

d(0.95) = 977.32um = .03phi

Distribution Properties:

Kurtosis = .4

Skewness = .81

Standard Deviation = 216.95 um = 2.2phi

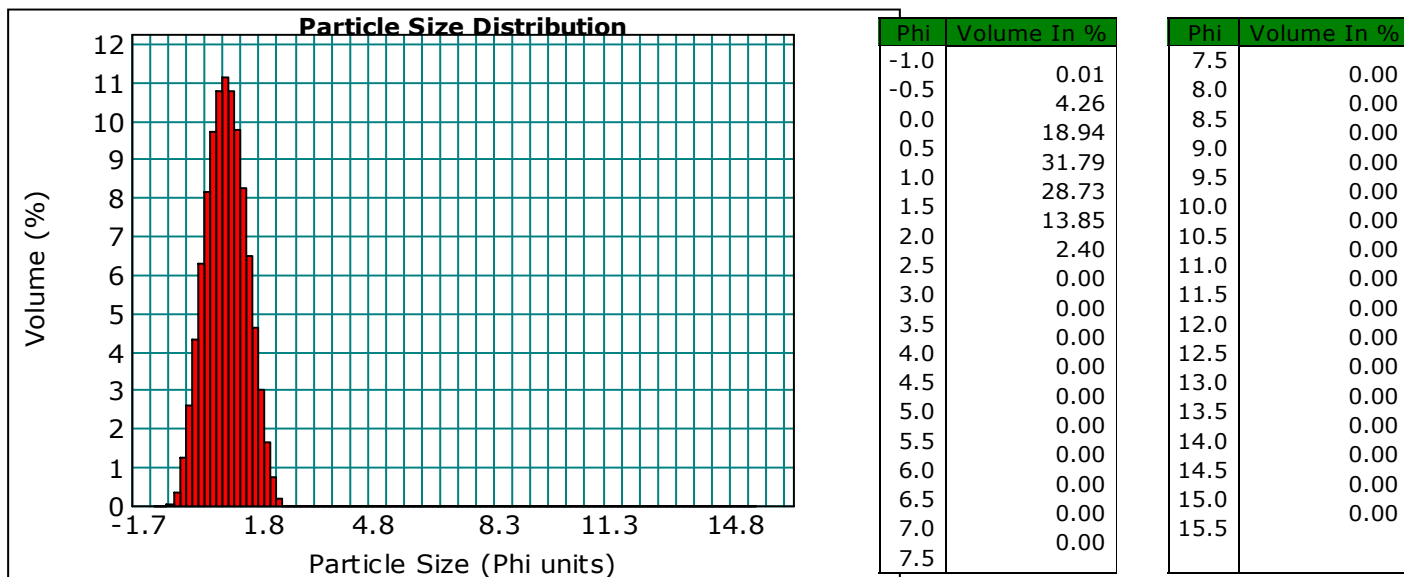
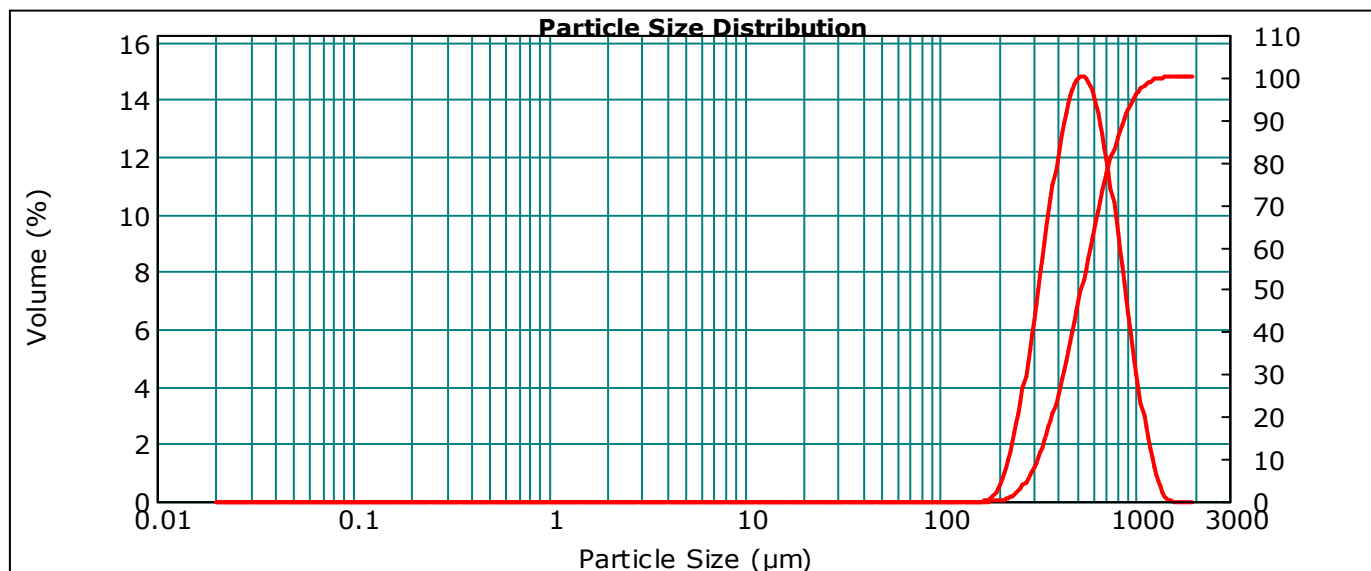
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .93 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





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THE UNIVERSITY OF
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Sample Name: C_1_1_16 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:45:06 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 564.54um = .82phi

Surface Weighted Mean:

D[3,2] = 485.45um = 1.04phi

Span :

1.068

Uniformity:

0.332

Specific Surface Area*:

0.0124 m²/g

Percentiles:

d(0.05) = 273.71um = 1.87phi

d(0.10) = 311.91um = 1.68phi

d(0.16) = 348.56um = 1.52phi

d(0.50) = 526.18um = .93phi

d(0.84) = 786.78um = .35phi

d(0.90) = 873.79um = .19phi

d(0.95) = 985.93um = .02phi

Distribution Properties:

Kurtosis = .36

Skewness = .81

Standard Deviation = 220.81 um = 2.18phi

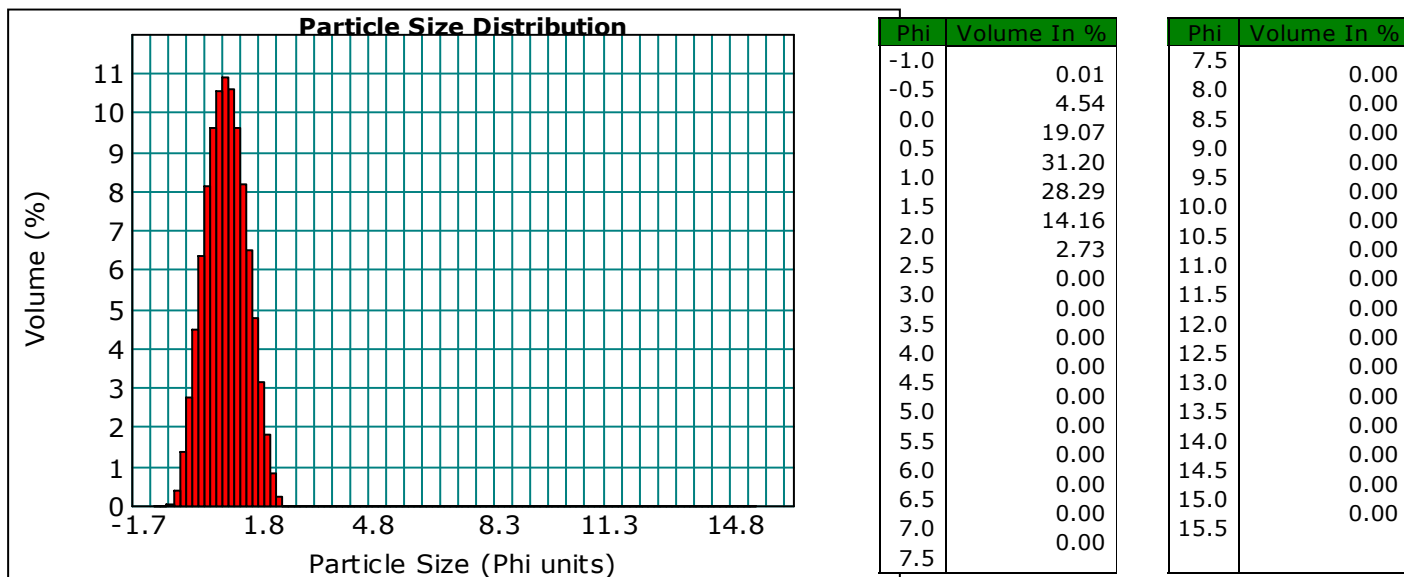
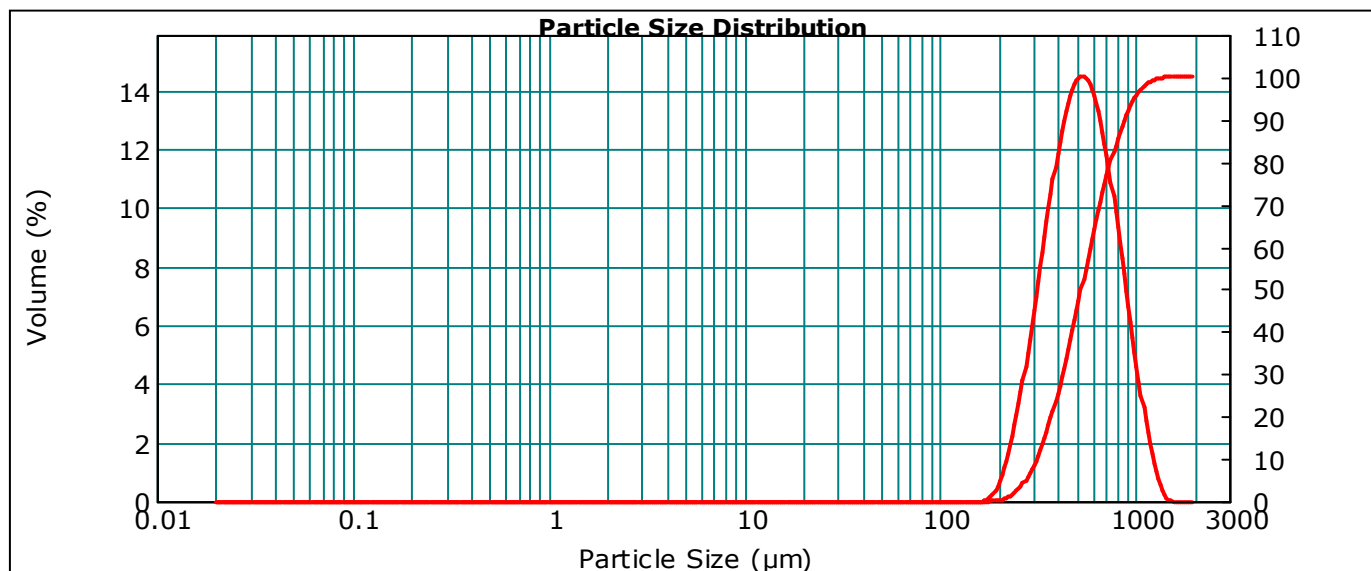
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .93 phi (Coarse Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:56:04 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 553.41um = .85phi

Surface Weighted Mean:

D[3,2] = 478.81um = 1.06phi

Span :

1.044

Uniformity:

0.326

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 273.29um = 1.87phi

d(0.10) = 310.42um = 1.69phi

d(0.16) = 345.85um = 1.53phi

d(0.50) = 516.76um = .95phi

d(0.84) = 766.66um = .38phi

d(0.90) = 850.14um = .23phi

d(0.95) = 958.42um = .06phi

Distribution Properties:

Kurtosis = .38

Skewness = .81

Standard Deviation = 212.42 um = 2.24phi

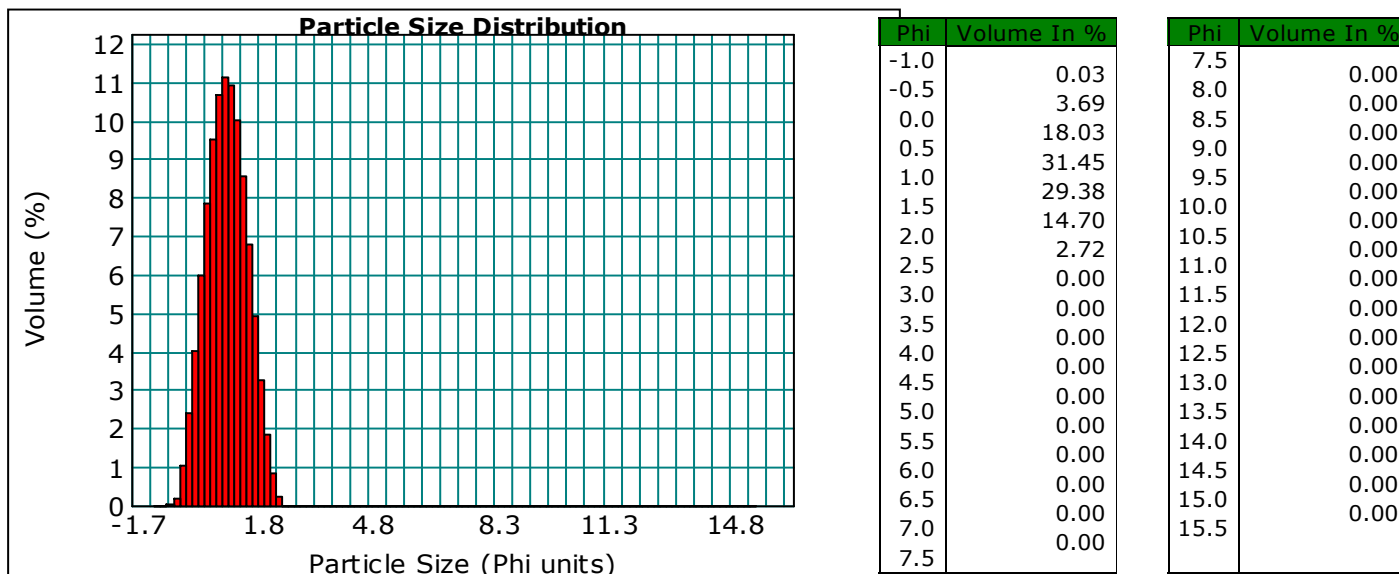
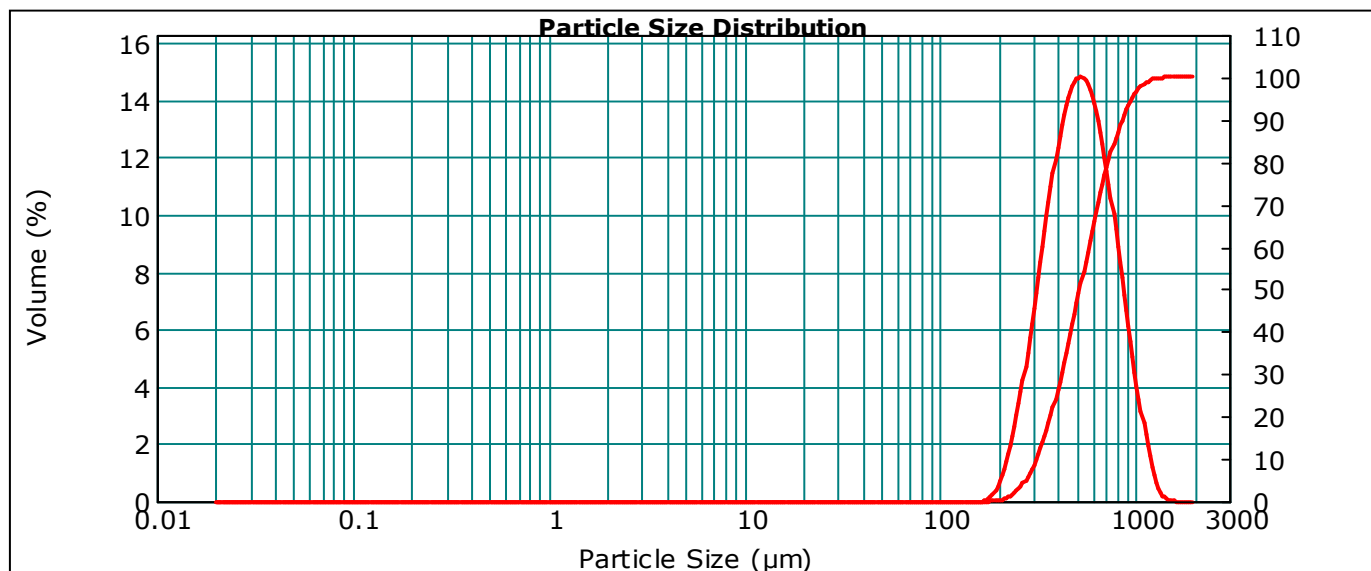
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .96 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:57:11 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 551.97um = .86phi

Surface Weighted Mean:

D[3,2] = 478.49um = 1.06phi

Span :

1.036

Uniformity:

0.323

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 273.73um = 1.87phi

d(0.10) = 310.8um = 1.69phi

d(0.16) = 346.14um = 1.53phi

d(0.50) = 516.26um = .95phi

d(0.84) = 763.74um = .39phi

d(0.90) = 845.88um = .24phi

d(0.95) = 952.14um = .07phi

Distribution Properties:

Kurtosis = .31

Skewness = .79

Standard Deviation = 210.02 um = 2.25phi

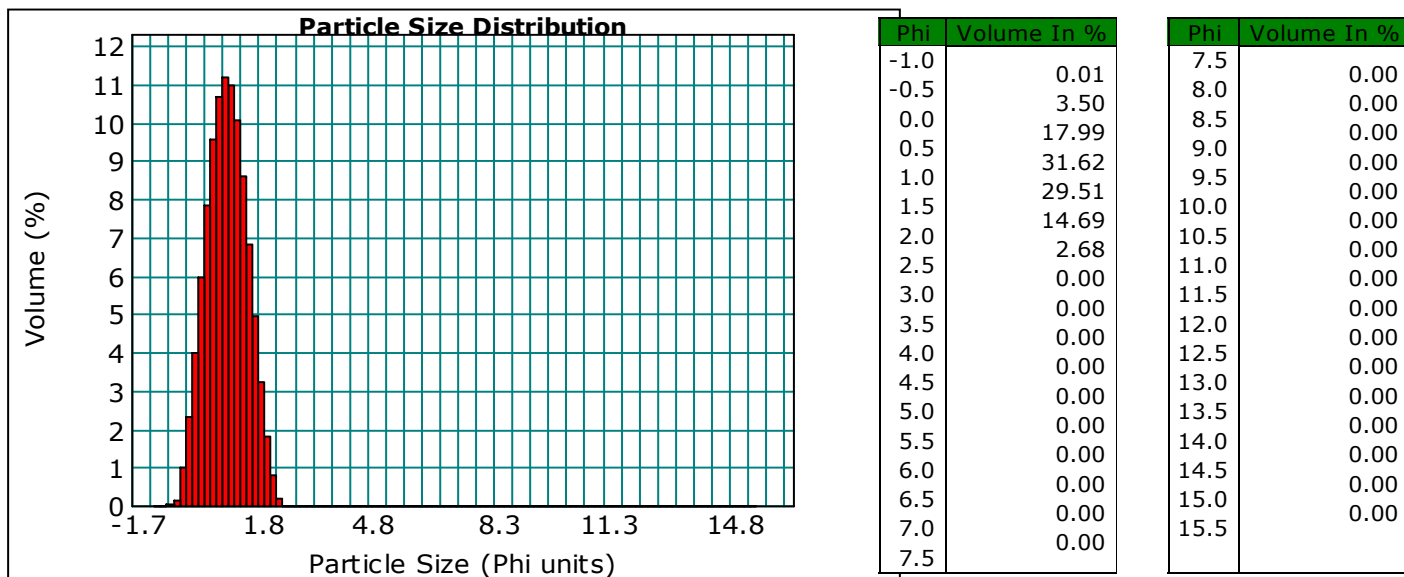
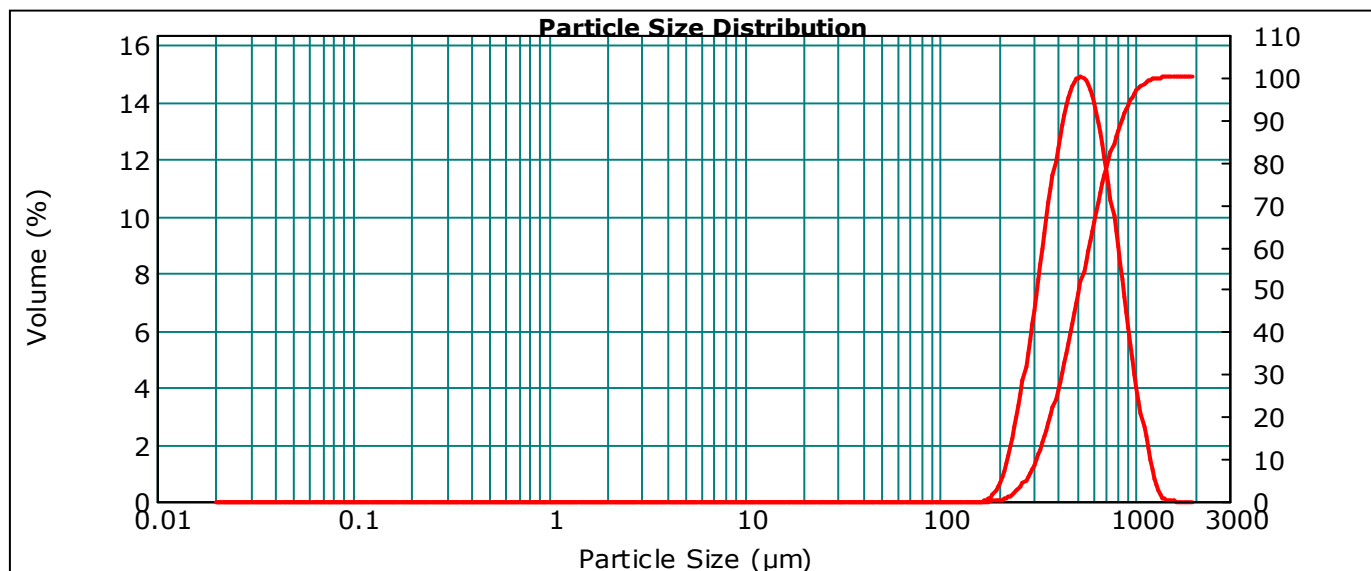
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .96 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:58:18 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 557.21um = .84phi

Surface Weighted Mean:

D[3,2] = 481.32um = 1.05phi

Span :

1.052

Uniformity:

0.328

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 274.27um = 1.87phi

d(0.10) = 311.61um = 1.68phi

d(0.16) = 347.26um = 1.53phi

d(0.50) = 519.46um = .94phi

d(0.84) = 772.75um = .37phi

d(0.90) = 857.94um = .22phi

d(0.95) = 968.75um = .05phi

Distribution Properties:

Kurtosis = .45

Skewness = .83

Standard Deviation = 215.57 um = 2.21phi

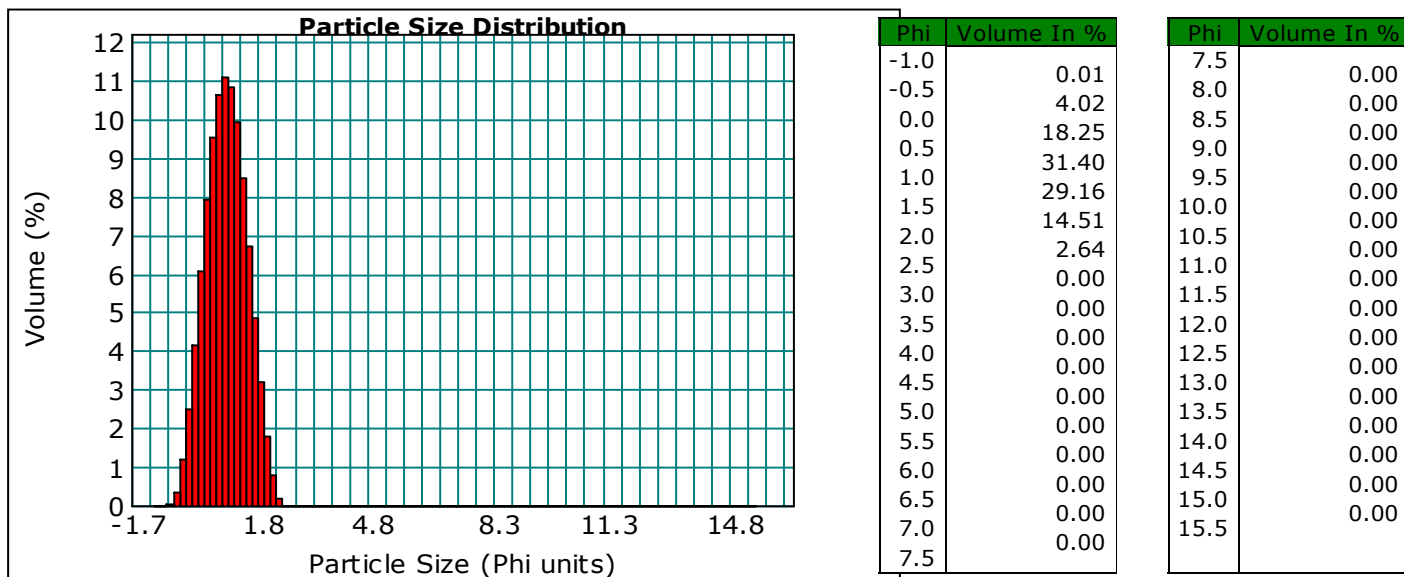
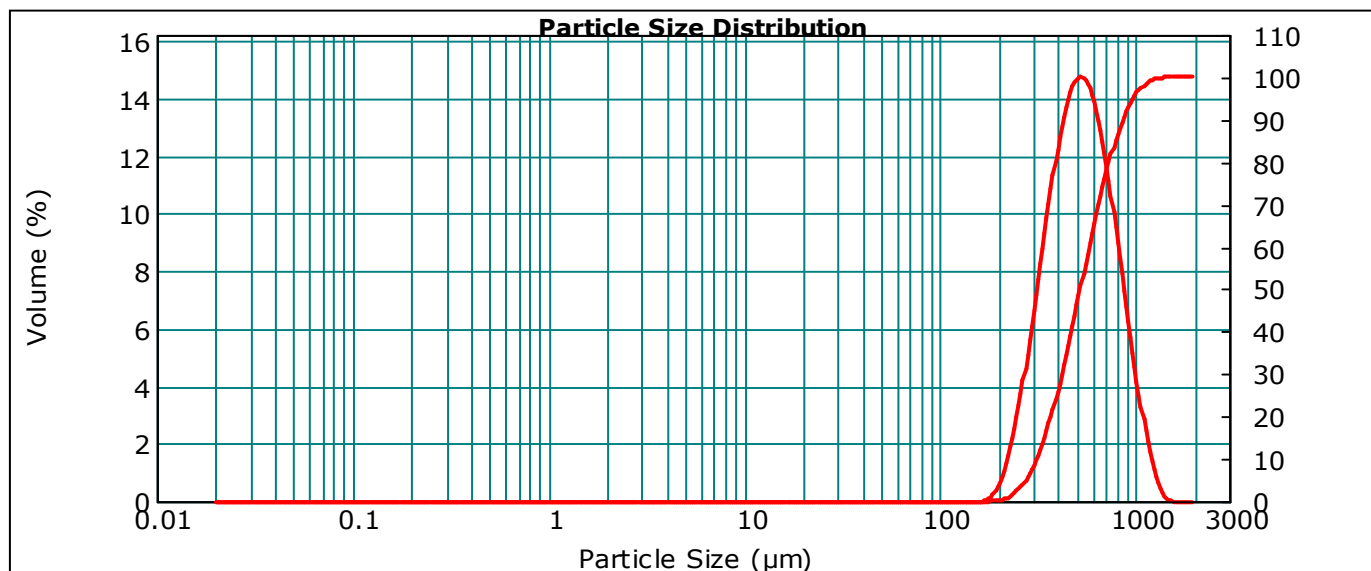
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_15 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 1:56:04 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 554.2um = .85phi

Surface Weighted Mean:

D[3,2] = 479.54um = 1.06phi

Span :

1.044

Uniformity:

0.326

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 273.76um = 1.87phi

d(0.10) = 310.94um = 1.69phi

d(0.16) = 346.41um = 1.53phi

d(0.50) = 517.49um = .95phi

d(0.84) = 767.65um = .38phi

d(0.90) = 851.3um = .23phi

d(0.95) = 959.59um = .06phi

Distribution Properties:

Kurtosis = .39

Skewness = .81

Standard Deviation = 212.69 um = 2.23phi

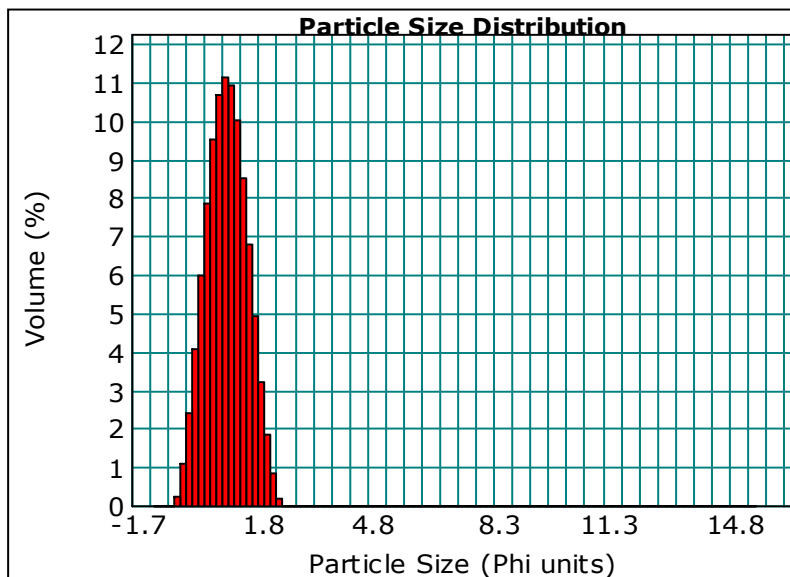
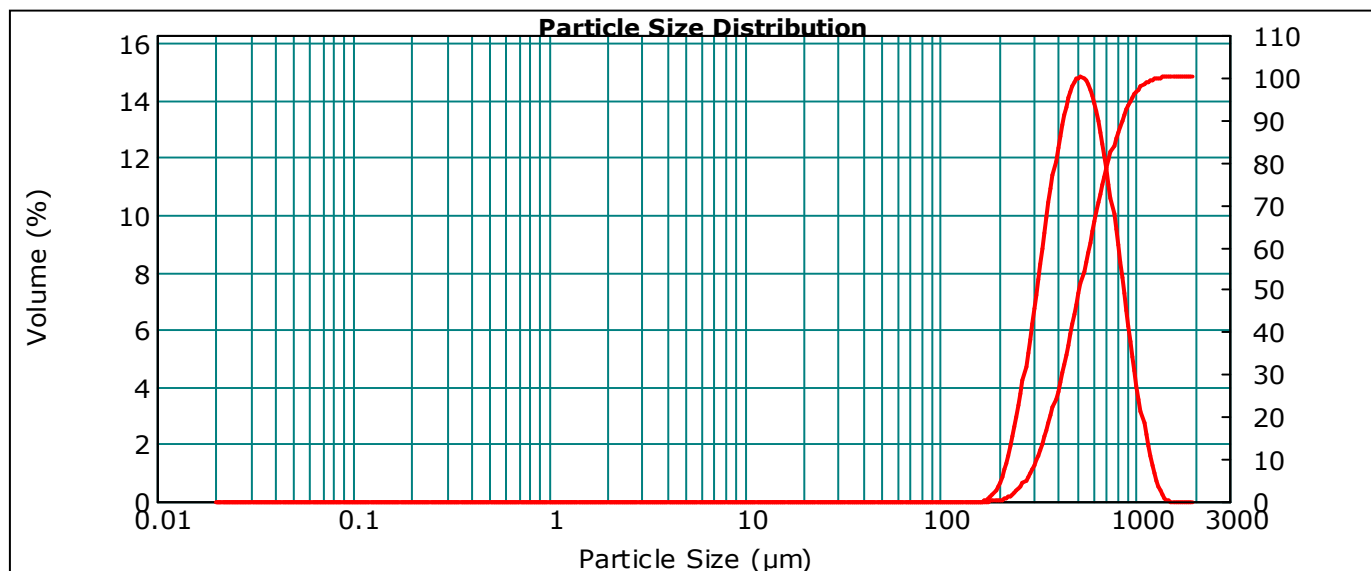
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	3.75
0.0	18.09
0.5	31.49
1.0	29.35
1.5	14.63
2.0	2.68
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:07:25 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 562.58um = .83phi

Surface Weighted Mean:

D[3,2] = 474.52um = 1.08phi

Span :

1.150

Uniformity:

0.357

Specific Surface Area*:

0.0126 m²/g

Percentiles:

d(0.05) = 259.95um = 1.94phi

d(0.10) = 297.54um = 1.75phi

d(0.16) = 334.33um = 1.58phi

d(0.50) = 518.47um = .95phi

d(0.84) = 798.91um = .32phi

d(0.90) = 893.54um = .16phi

d(0.95) = 1015.99um = -.02phi

Distribution Properties:

Kurtosis = .42

Skewness = .86

Standard Deviation = 234.67 um = 2.09phi

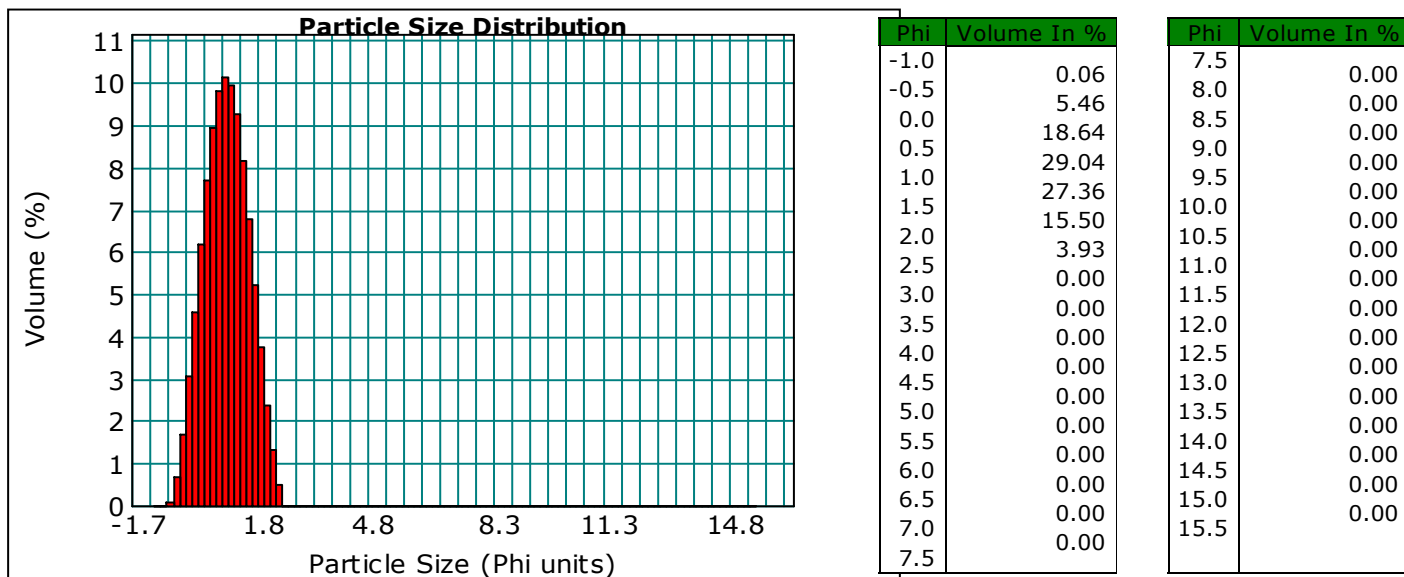
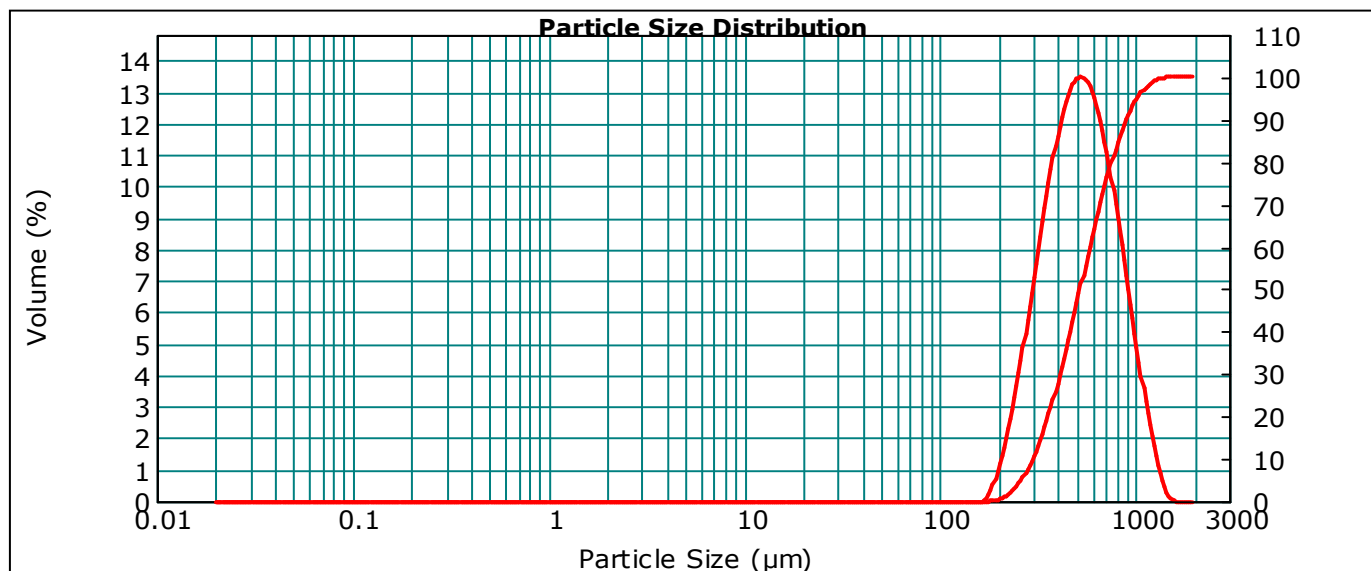
Inclusive SD (Sorting Coeff.) = .61 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:08:33 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 555.48um = .85phi

Surface Weighted Mean:

D[3,2] = 478.9um = 1.06phi

Span :

1.058

Uniformity:

0.33

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 271.72um = 1.88phi

d(0.10) = 309.22um = 1.69phi

d(0.16) = 344.95um = 1.54phi

d(0.50) = 517.74um = .95phi

d(0.84) = 771.75um = .37phi

d(0.90) = 857.06um = .22phi

d(0.95) = 967.82um = .05phi

Distribution Properties:

Kurtosis = .44

Skewness = .83

Standard Deviation = 216.02 um = 2.21phi

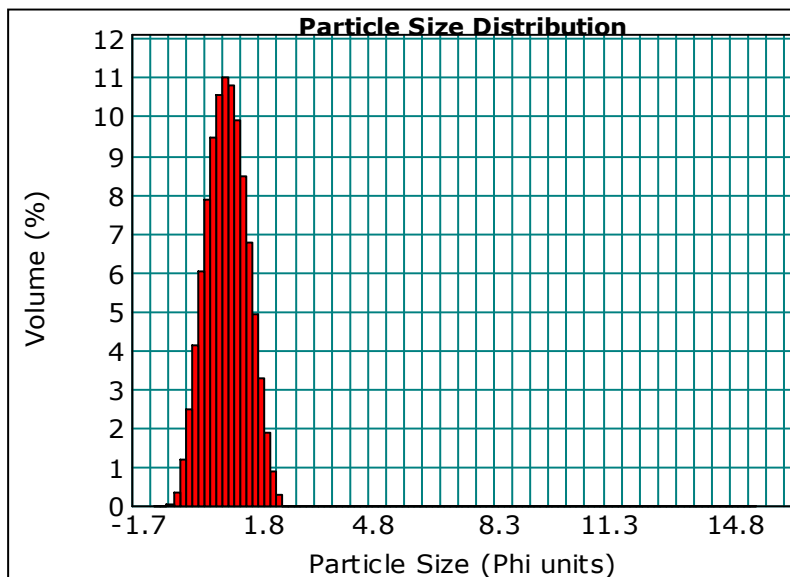
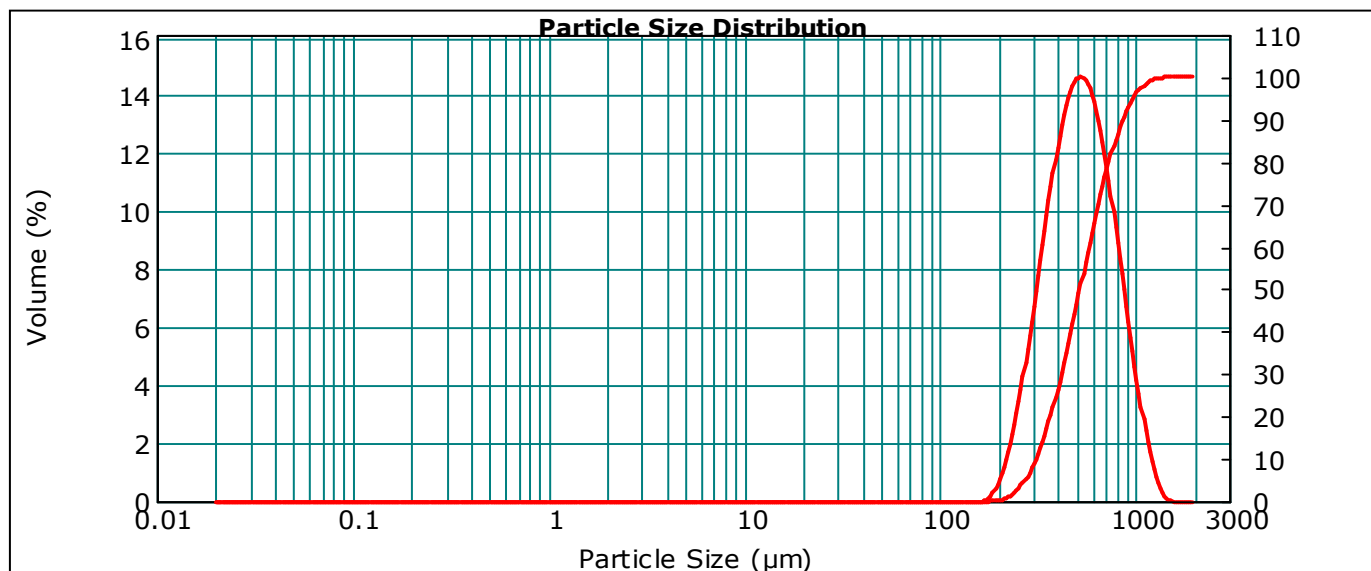
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.01
-0.5	3.99
0.0	18.16
0.5	31.18
1.0	29.08
1.5	14.71
2.0	2.87
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:09:40 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 557.41um = .84phi

Surface Weighted Mean:

D[3,2] = 480.38um = 1.06phi

Span :

1.059

Uniformity:

0.33

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 272.39um = 1.88phi

d(0.10) = 310.02um = 1.69phi

d(0.16) = 345.92um = 1.53phi

d(0.50) = 519.48um = .94phi

d(0.84) = 774.63um = .37phi

d(0.90) = 860.31um = .22phi

d(0.95) = 971.69um = .04phi

Distribution Properties:

Kurtosis = .44

Skewness = .83

Standard Deviation = 217.02 um = 2.2phi

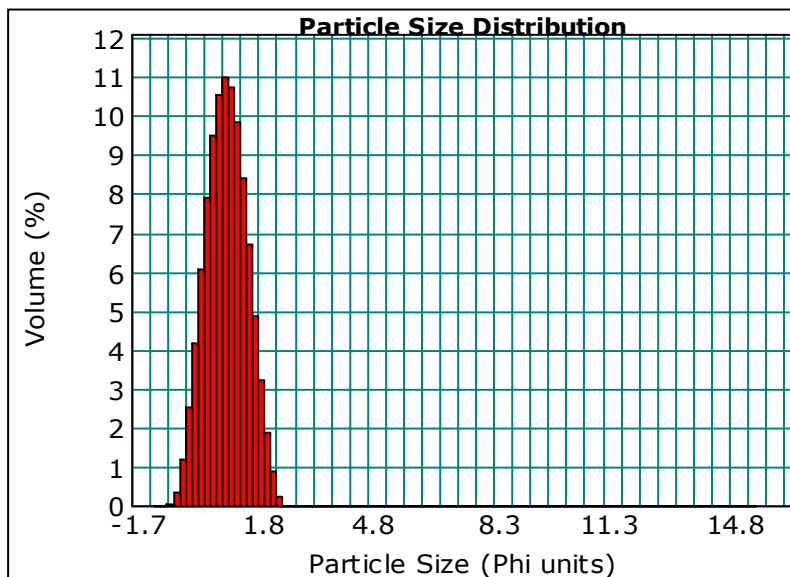
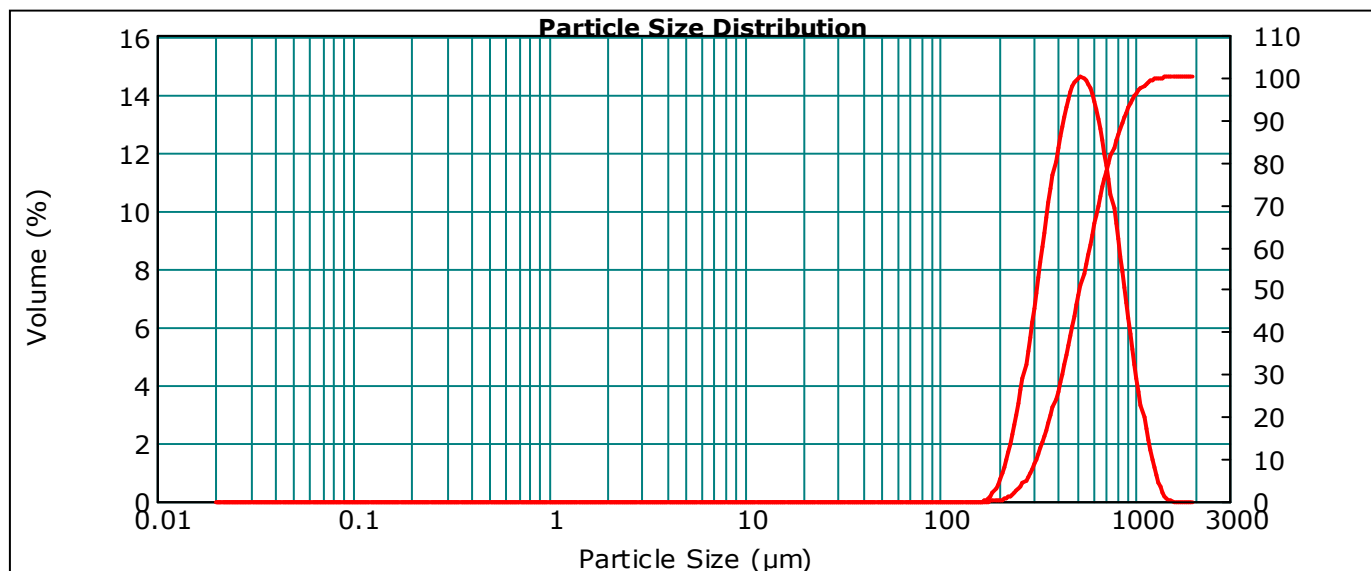
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.01
-0.5	4.10
0.0	18.32
0.5	31.22
1.0	28.95
1.5	14.57
2.0	2.82
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_14 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:07:25 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 558.49um = .84phi

Surface Weighted Mean:

D[3,2] = 477.92um = 1.07phi

Span :

1.088

Uniformity:

0.339

Specific Surface Area*:

0.0126 m²/g

Percentiles:

d(0.05) = 267.77um = 1.9phi

d(0.10) = 305.5um = 1.71phi

d(0.16) = 341.81um = 1.55phi

d(0.50) = 518.57um = .95phi

d(0.84) = 781.43um = .36phi

d(0.90) = 869.95um = .2phi

d(0.95) = 985.25um = .02phi

Distribution Properties:

Kurtosis = .46

Skewness = .84

Standard Deviation = 222.76 um = 2.17phi

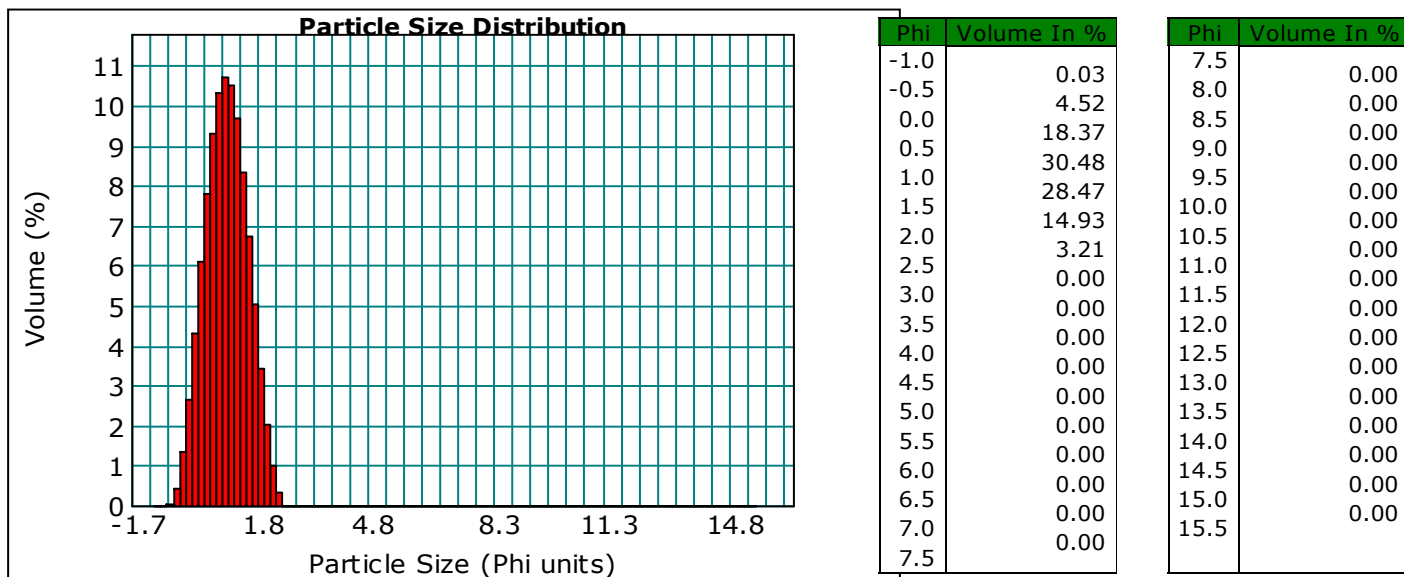
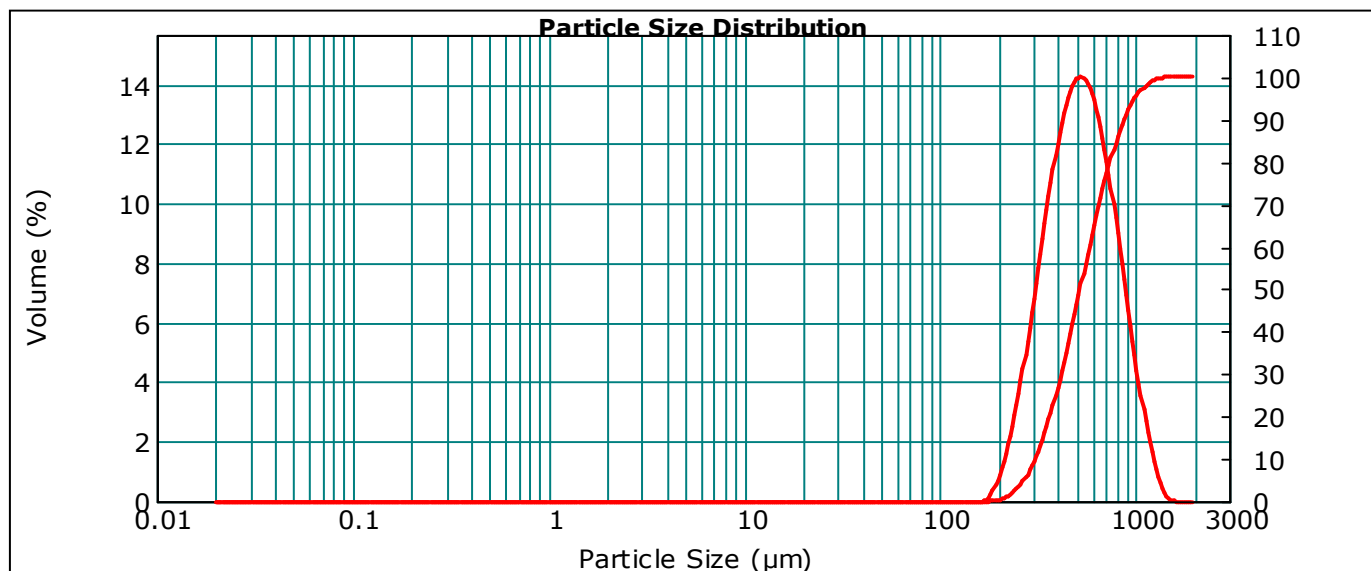
Inclusive SD (Sorting Coeff.) = .58 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:19:01 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 554.05um = .85phi

Surface Weighted Mean:

D[3,2] = 479.77um = 1.06phi

Span :

1.040

Uniformity:

0.325

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 274.12um = 1.87phi

d(0.10) = 311.38um = 1.68phi

d(0.16) = 346.89um = 1.53phi

d(0.50) = 517.67um = .95phi

d(0.84) = 766.78um = .38phi

d(0.90) = 849.99um = .23phi

d(0.95) = 957.99um = .06phi

Distribution Properties:

Kurtosis = .38

Skewness = .81

Standard Deviation = 211.97 um = 2.24phi

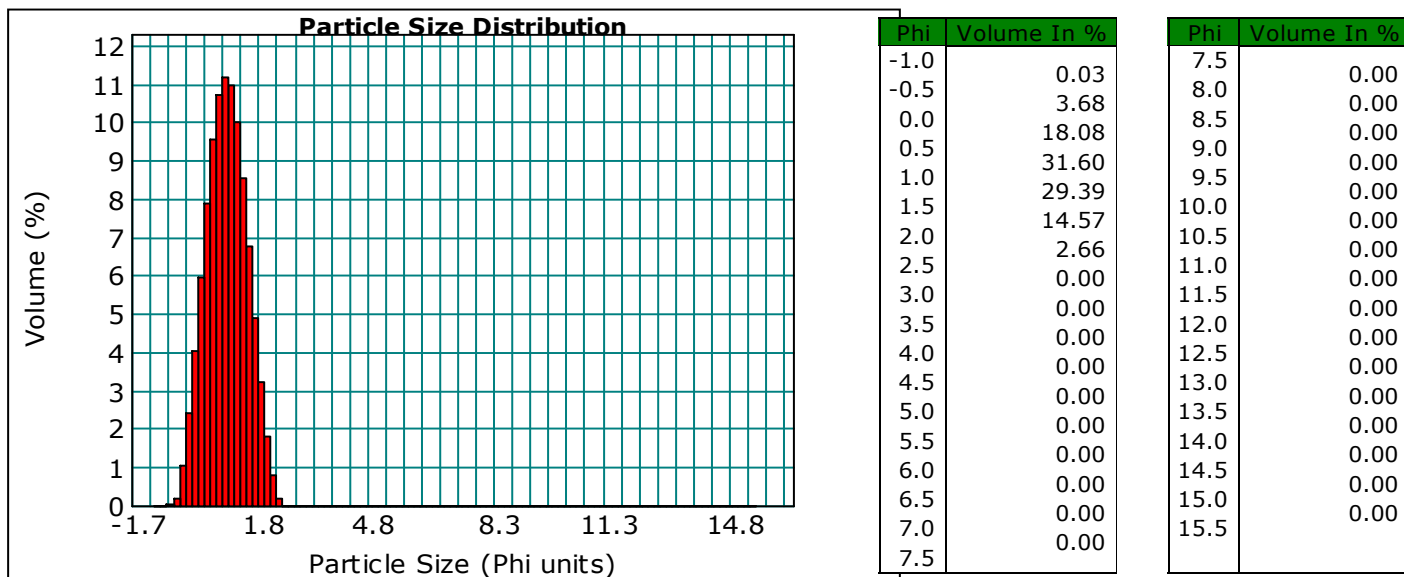
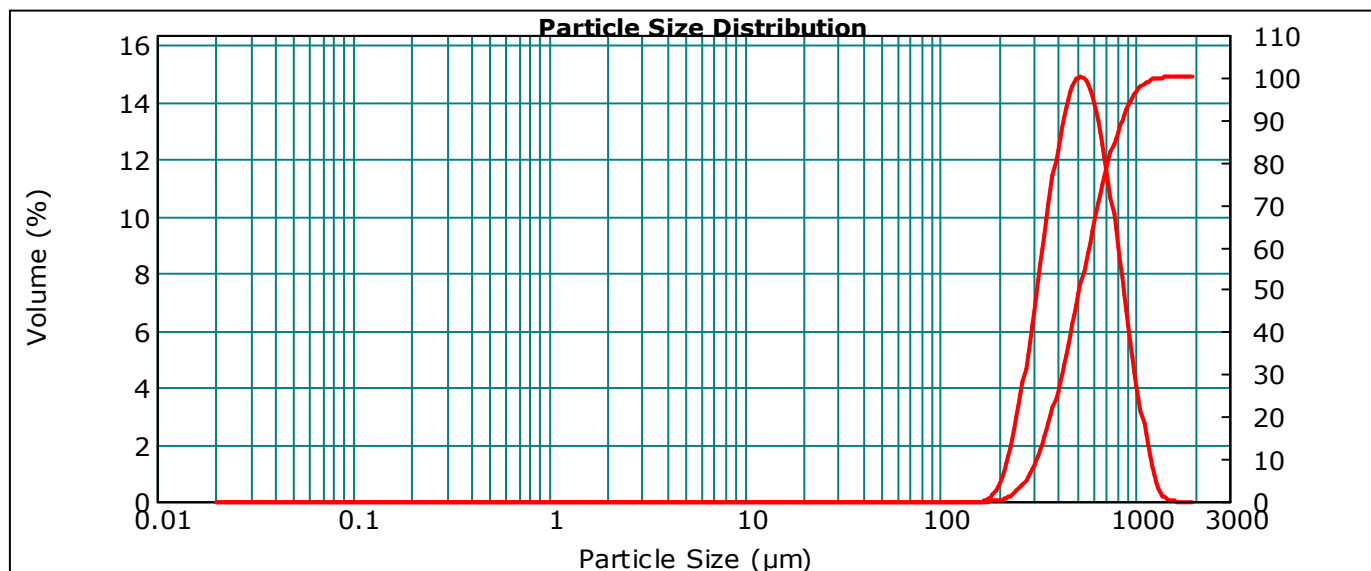
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:20:08 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 559.34um = .84phi

Surface Weighted Mean:

D[3,2] = 482.52um = 1.05phi

Span :

1.057

Uniformity:

0.33

Specific Surface Area*:

0.0124 m²/g

Percentiles:

d(0.05) = 274.46um = 1.87phi

d(0.10) = 311.98um = 1.68phi

d(0.16) = 347.82um = 1.52phi

d(0.50) = 520.94um = .94phi

d(0.84) = 776.38um = .37phi

d(0.90) = 862.58um = .21phi

d(0.95) = 975.16um = .04phi

Distribution Properties:

Kurtosis = .49

Skewness = .84

Standard Deviation = 217.61 um = 2.2phi

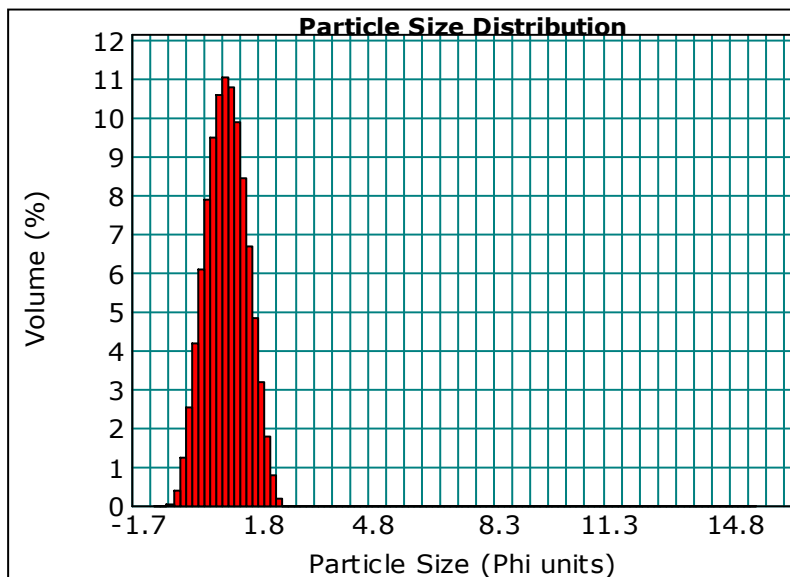
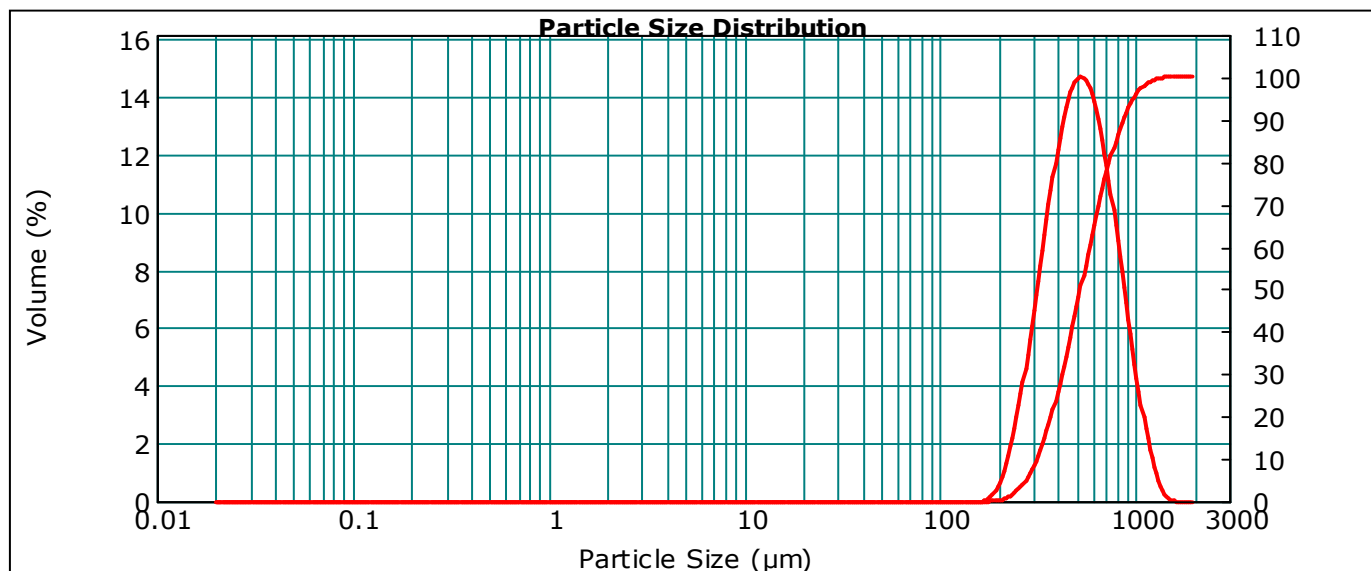
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .94 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.02
-0.5	4.21
0.0	18.37
0.5	31.34
1.0	29.01
1.5	14.40
2.0	2.64
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:21:16 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 551.64um = .86phi

Surface Weighted Mean:

D[3,2] = 477.11um = 1.07phi

Span :

1.046

Uniformity:

0.327

Specific Surface Area*:

0.0126 m²/g

Percentiles:

d(0.05) = 272.4um = 1.88phi

d(0.10) = 309.37um = 1.69phi

d(0.16) = 344.61um = 1.54phi

d(0.50) = 514.73um = .96phi

d(0.84) = 764.27um = .39phi

d(0.90) = 847.89um = .24phi

d(0.95) = 956.79um = .06phi

Distribution Properties:

Kurtosis = .42

Skewness = .82

Standard Deviation = 212.31 um = 2.24phi

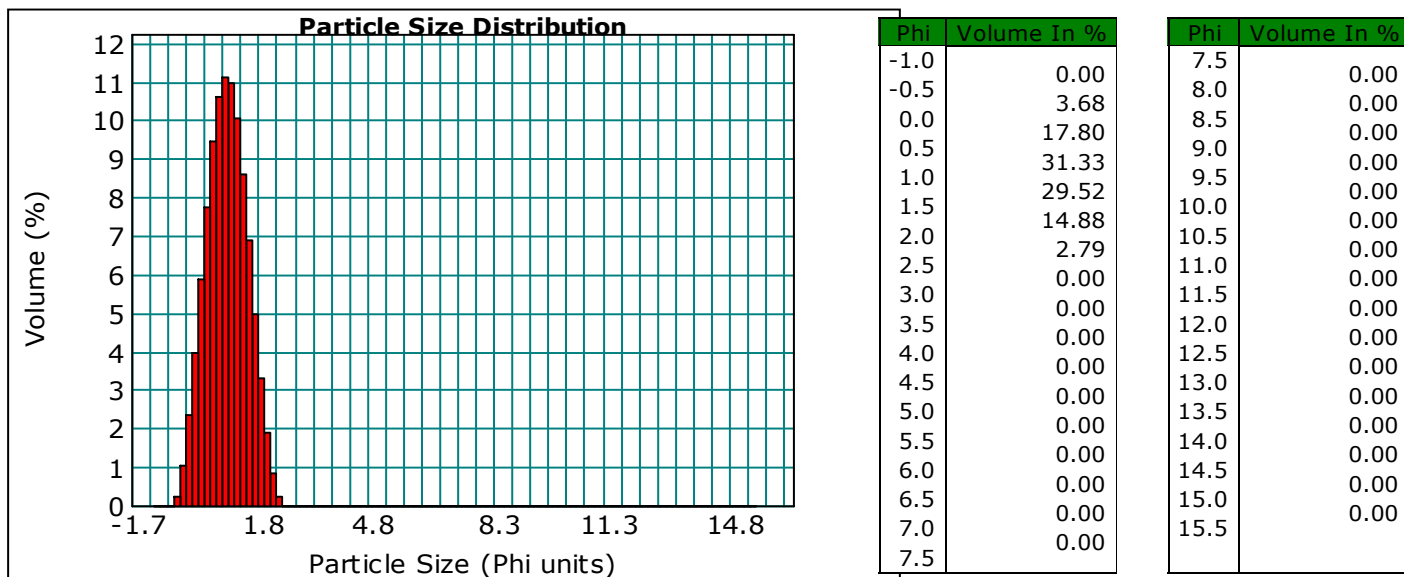
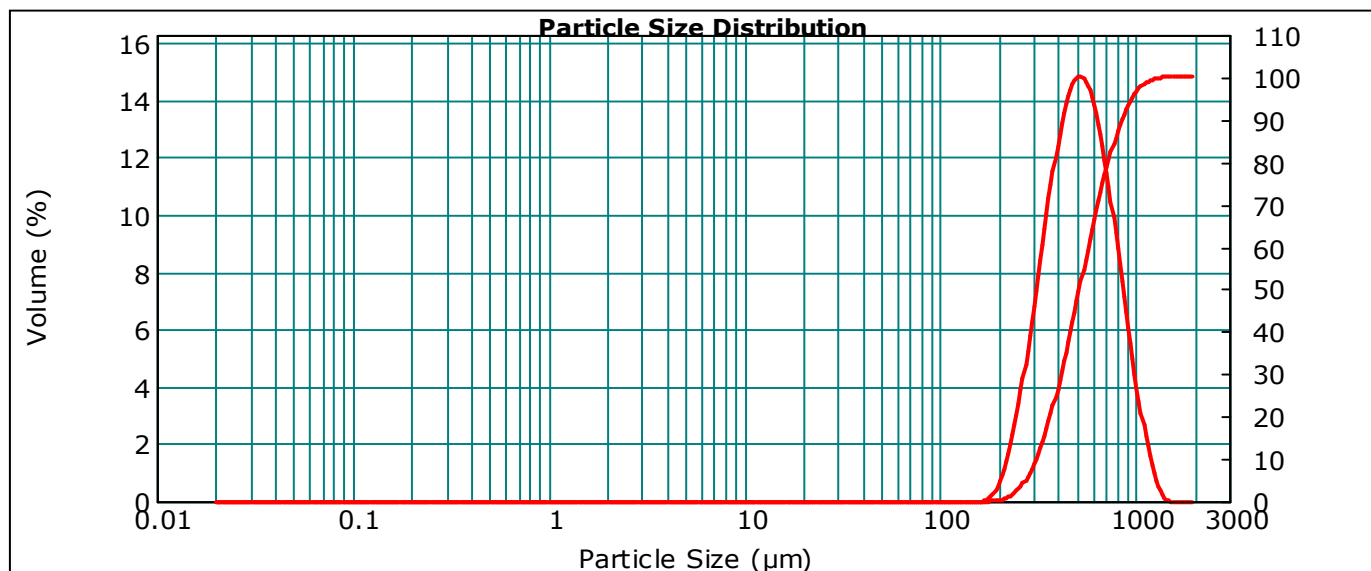
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .96 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_13 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:19:01 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 555.01um = .85phi

Surface Weighted Mean:

D[3,2] = 479.79um = 1.06phi

Span :

1.048

Uniformity:

0.327

Specific Surface Area*:

0.0125 m²/g

Percentiles:

d(0.05) = 273.65um = 1.87phi

d(0.10) = 310.89um = 1.69phi

d(0.16) = 346.42um = 1.53phi

d(0.50) = 517.76um = .95phi

d(0.84) = 769.12um = .38phi

d(0.90) = 853.56um = .23phi

d(0.95) = 963.11um = .05phi

Distribution Properties:

Kurtosis = .44

Skewness = .82

Standard Deviation = 214 um = 2.22phi

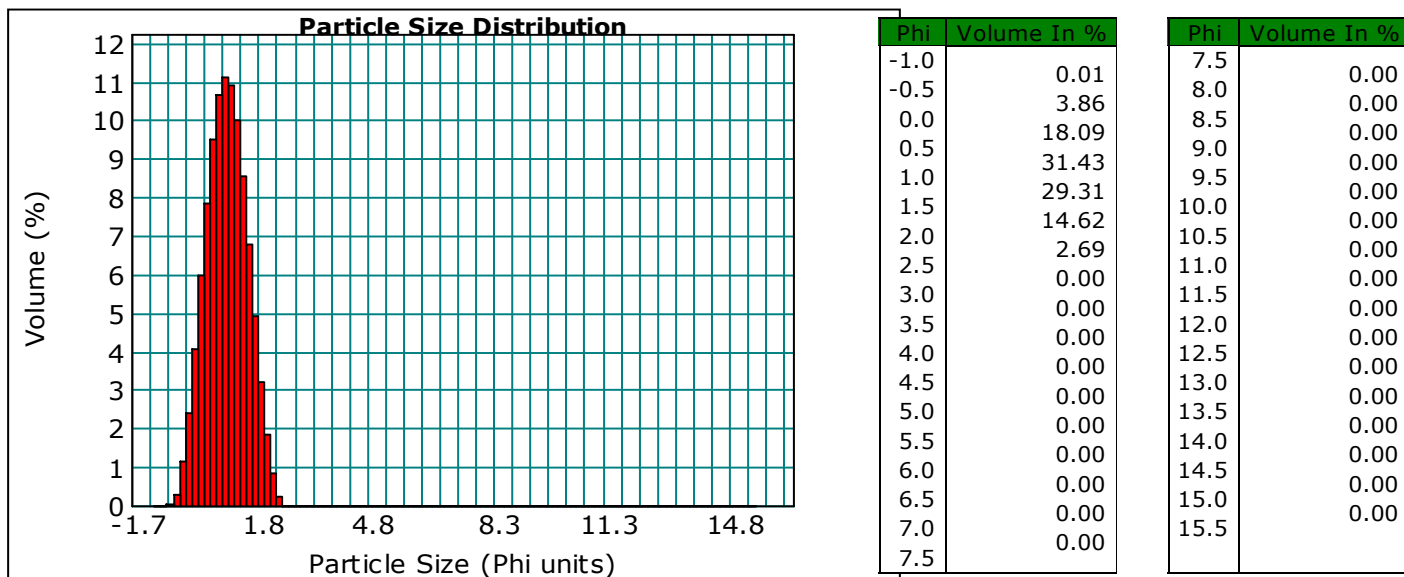
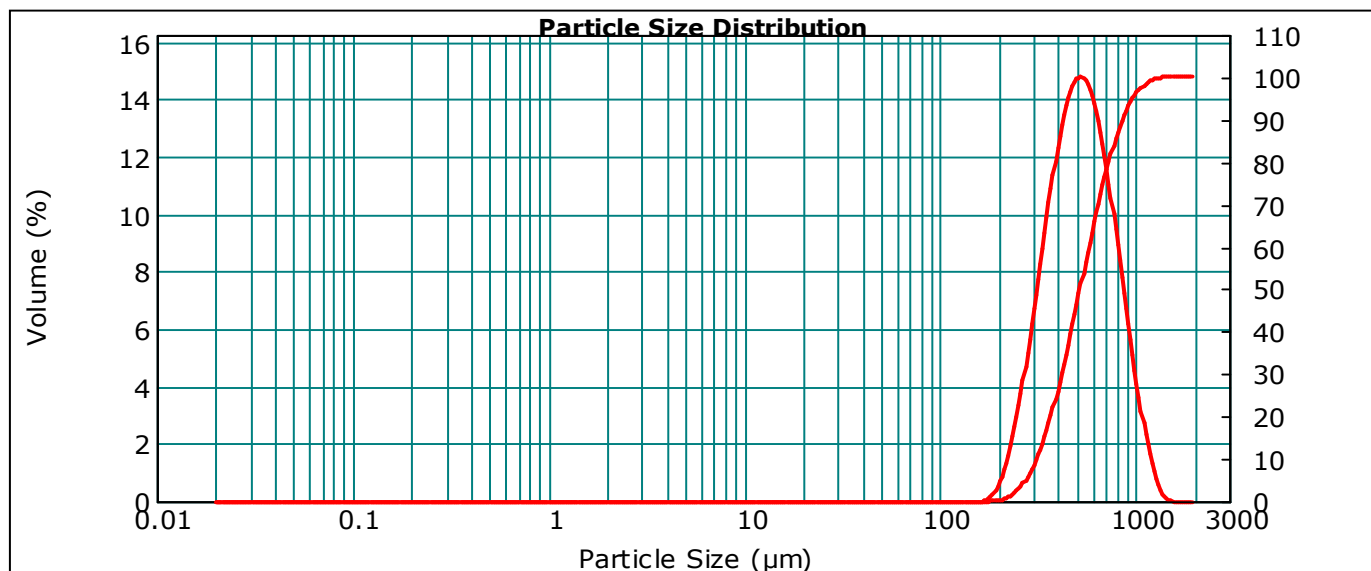
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = .95 phi (Coarse Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:29:44 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 540.49um = .89phi

Surface Weighted Mean:

D[3,2] = 460um = 1.12phi

Span :

1.115

Uniformity:

0.346

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 254.3um = 1.98phi

d(0.10) = 290.57um = 1.78phi

d(0.16) = 326.01um = 1.62phi

d(0.50) = 501.9um = .99phi

d(0.84) = 763.91um = .39phi

d(0.90) = 850.08um = .23phi

d(0.95) = 959.42um = .06phi

Distribution Properties:

Kurtosis = .09

Skewness = .76

Standard Deviation = 217.3 um = 2.2phi

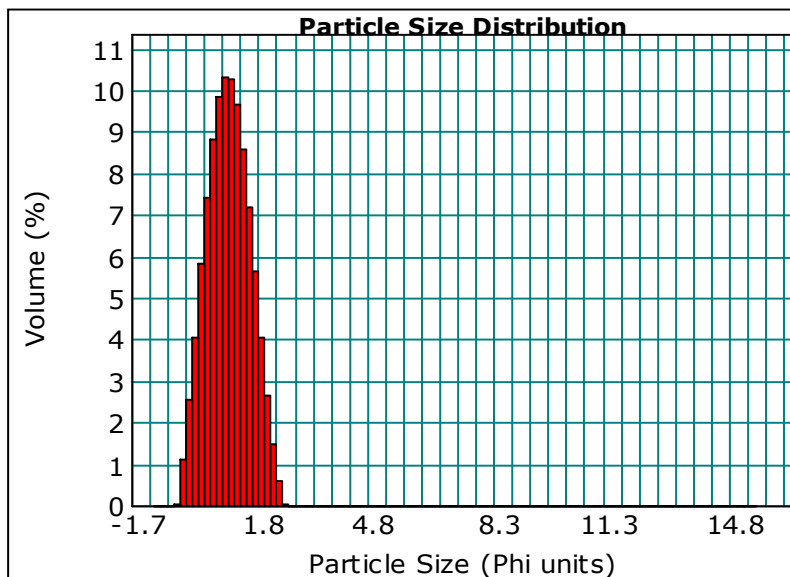
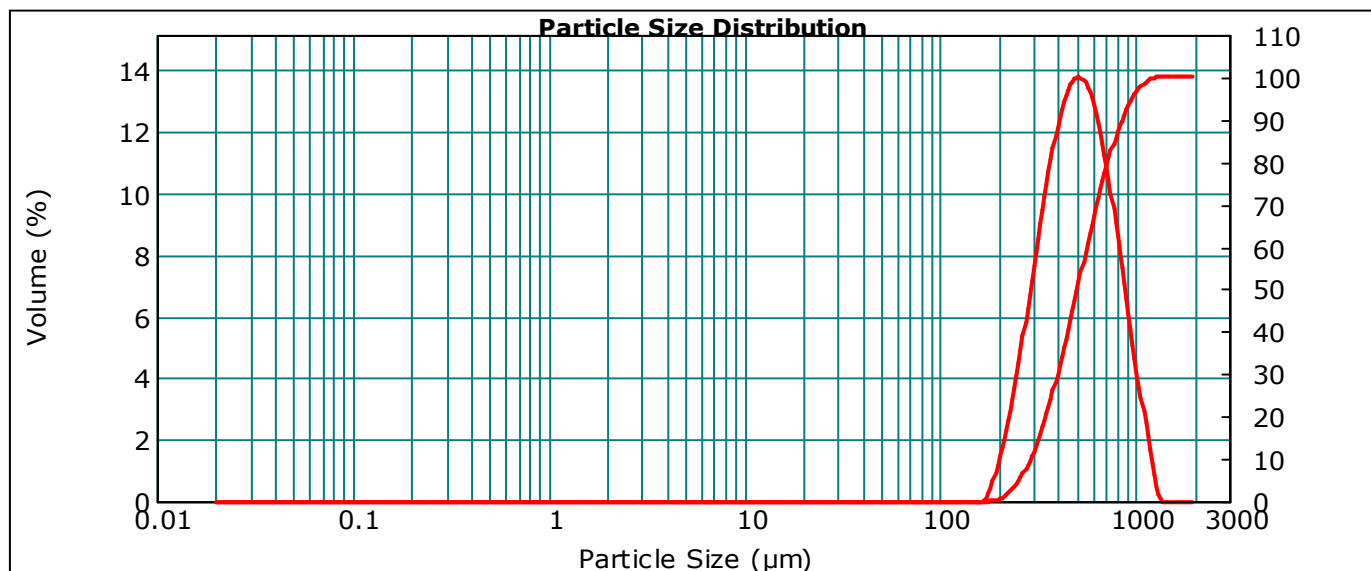
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	3.73
0.0	17.45
0.5	29.15
1.0	28.48
1.5	16.67
2.0	4.51
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:30:51 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 540.28um = .89phi

Surface Weighted Mean:

D[3,2] = 460.65um = 1.12phi

Span :

1.106

Uniformity:

0.343

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 255.18um = 1.97phi

d(0.10) = 291.53um = 1.78phi

d(0.16) = 326.96um = 1.61phi

d(0.50) = 502.51um = .99phi

d(0.84) = 762.32um = .39phi

d(0.90) = 847.26um = .24phi

d(0.95) = 954.87um = .07phi

Distribution Properties:

Kurtosis = .08

Skewness = .75

Standard Deviation = 215.75 um = 2.21phi

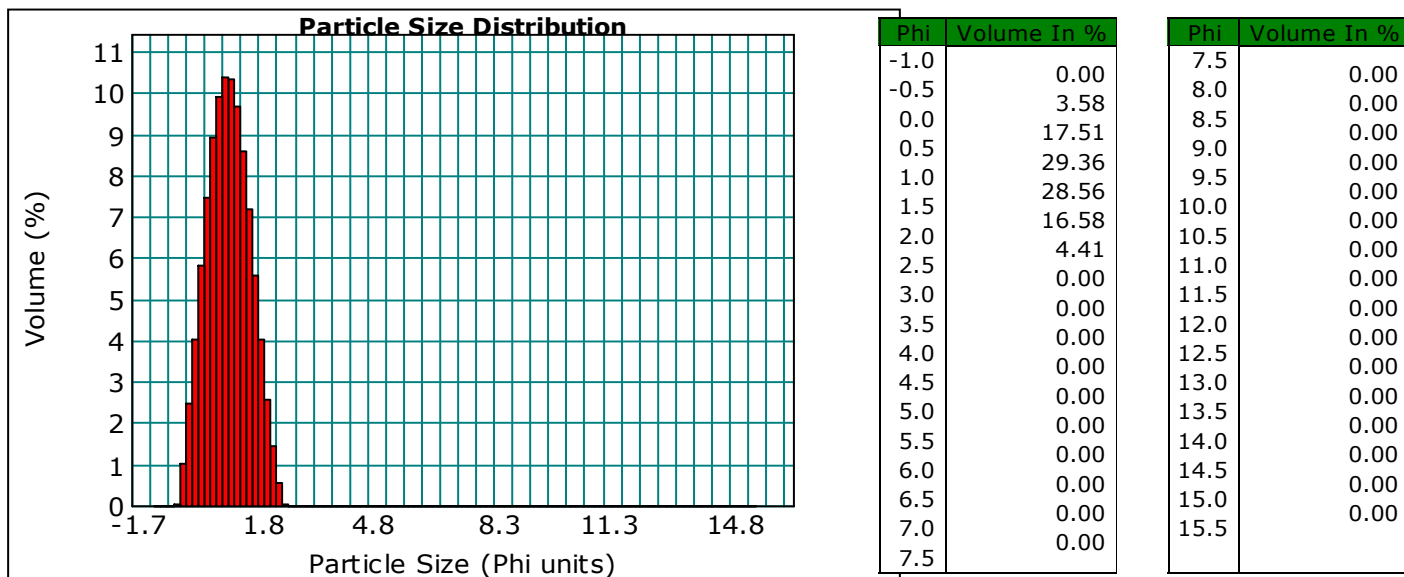
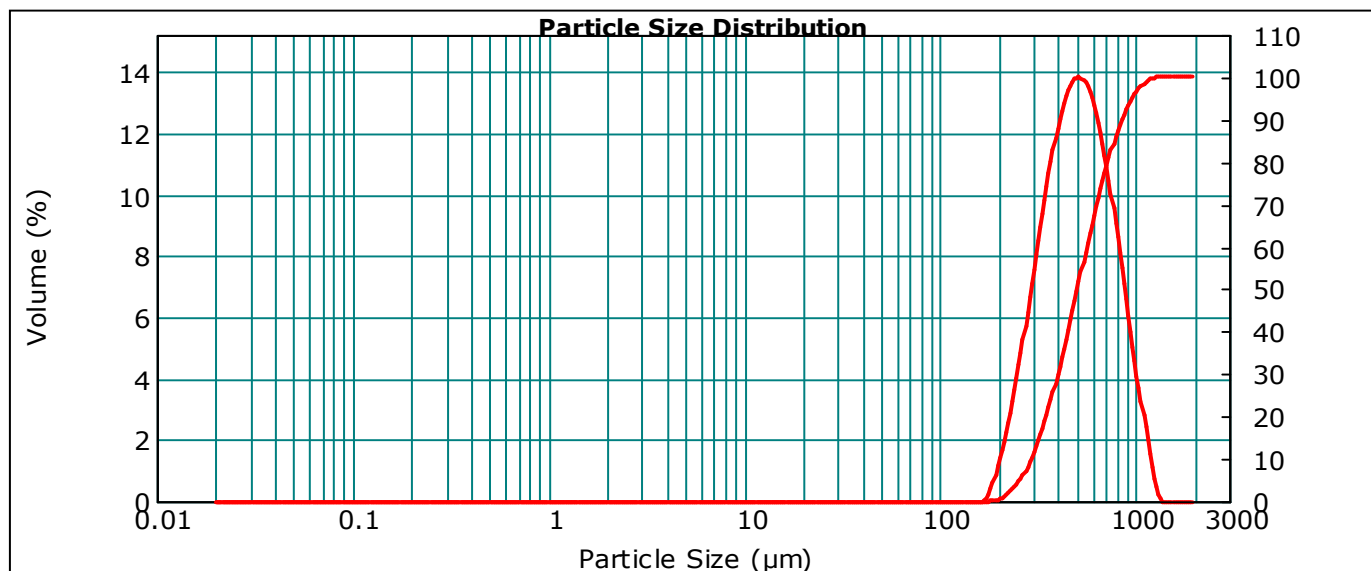
Inclusive SD (Sorting Coeff.) = .59 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1 phi (Coarse Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:31:59 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 539.48um = .89phi

Surface Weighted Mean:

D[3,2] = 458.98um = 1.12phi

Span :

1.117

Uniformity:

0.346

Specific Surface Area*:

0.0131 m²/g

Percentiles:

d(0.05) = 253.76um = 1.98phi

d(0.10) = 289.86um = 1.79phi

d(0.16) = 325.19um = 1.62phi

d(0.50) = 500.66um = 1phi

d(0.84) = 762.82um = .39phi

d(0.90) = 849.13um = .24phi

d(0.95) = 958.8um = .06phi

Distribution Properties:

Kurtosis = .1

Skewness = .76

Standard Deviation = 217.3 um = 2.2phi

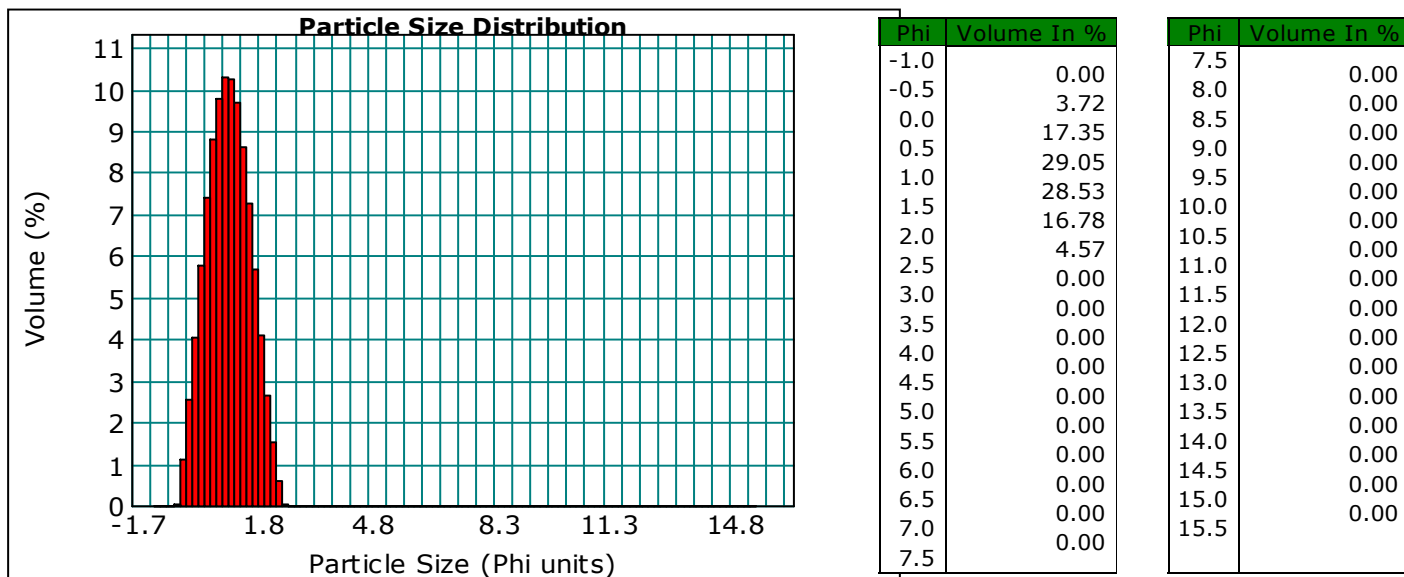
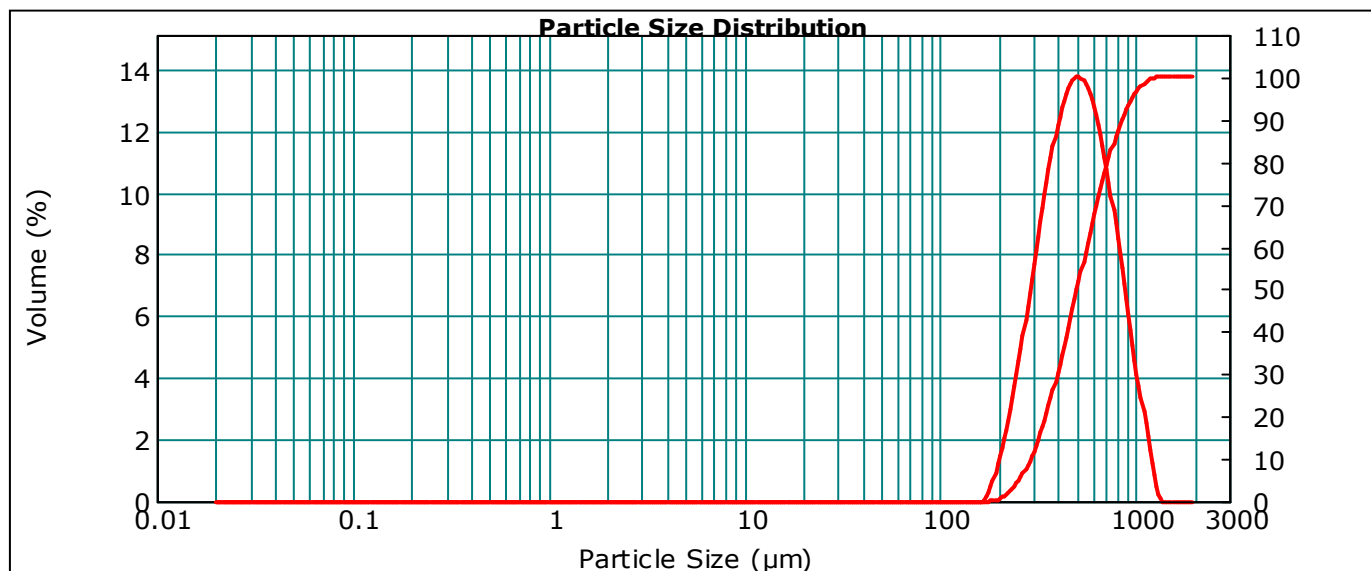
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_12 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:29:44 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 540.09um = .89phi

Surface Weighted Mean:

D[3,2] = 459.88um = 1.12phi

Span :

1.113

Uniformity:

0.345

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 254.4um = 1.97phi

d(0.10) = 290.65um = 1.78phi

d(0.16) = 326.05um = 1.62phi

d(0.50) = 501.69um = 1phi

d(0.84) = 763.02um = .39phi

d(0.90) = 848.8um = .24phi

d(0.95) = 957.69um = .06phi

Distribution Properties:

Kurtosis = .09

Skewness = .76

Standard Deviation = 216.78 um = 2.21phi

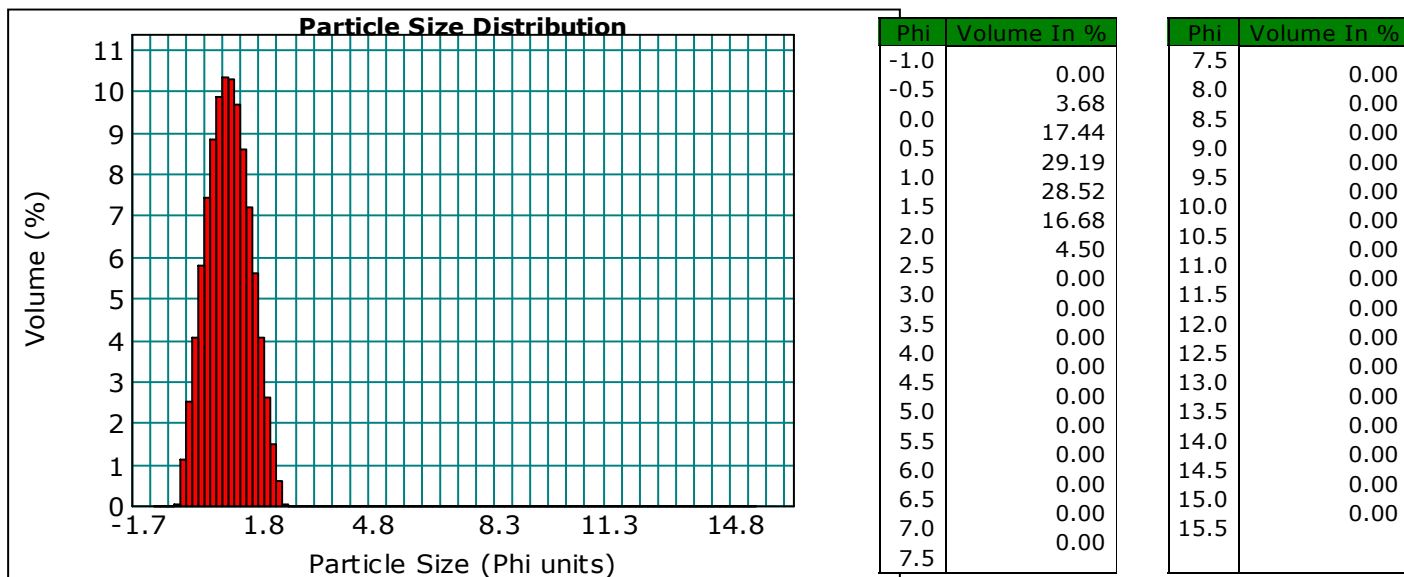
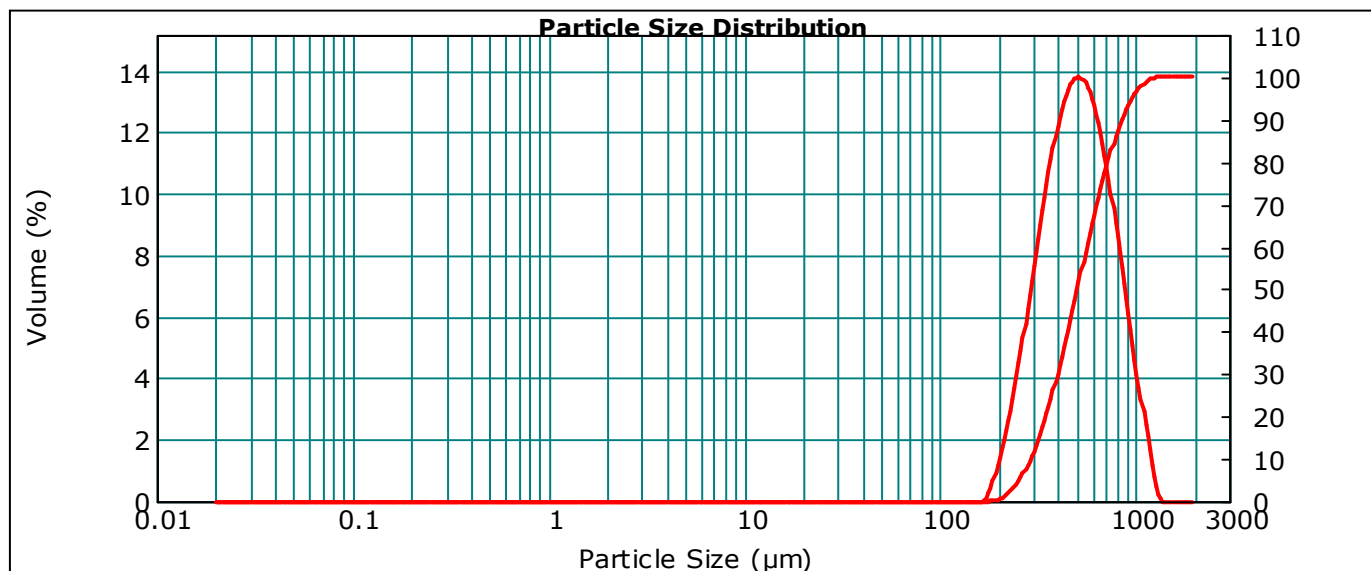
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:41:10 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 522.5um = .94phi

Surface Weighted Mean:

D[3,2] = 451.94um = 1.15phi

Span :

1.048

Uniformity:

0.324

Specific Surface Area*:

0.0133 m²/g

Percentiles:

d(0.05) = 257.54um = 1.96phi

d(0.10) = 292.62um = 1.77phi

d(0.16) = 326.16um = 1.62phi

d(0.50) = 487.87um = 1.04phi

d(0.84) = 724.79um = .46phi

d(0.90) = 803.75um = .32phi

d(0.95) = 905.78um = .14phi

Distribution Properties:

Kurtosis = .29

Skewness = .79

Standard Deviation = 200.54 um = 2.32phi

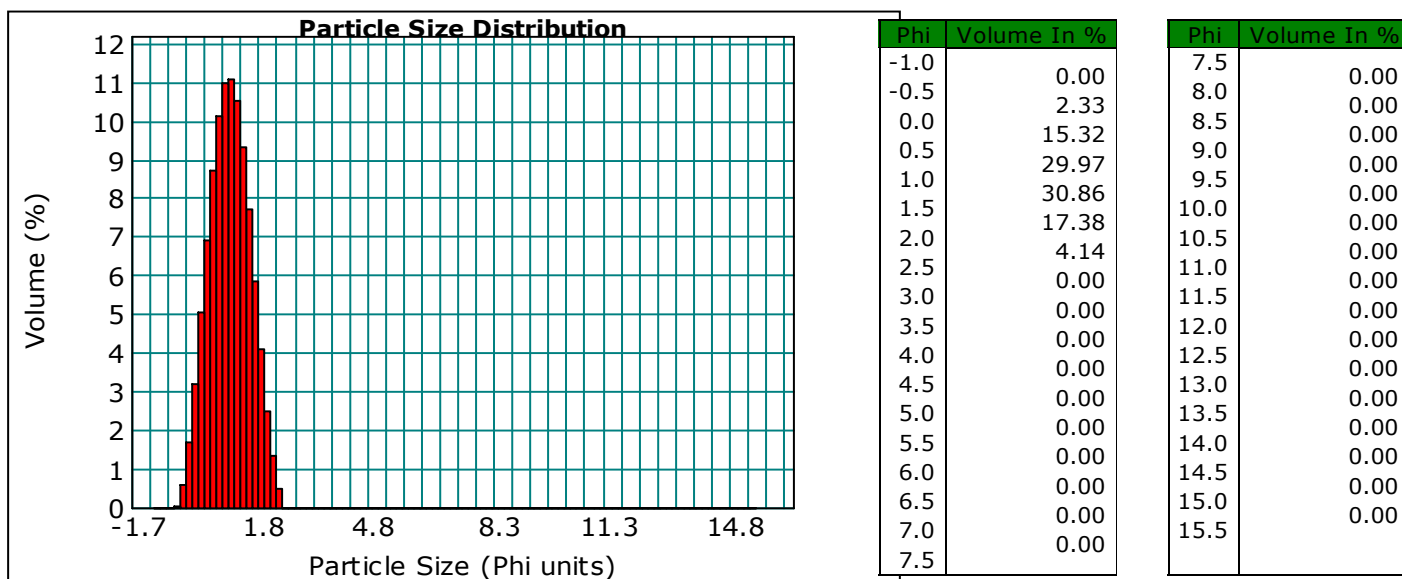
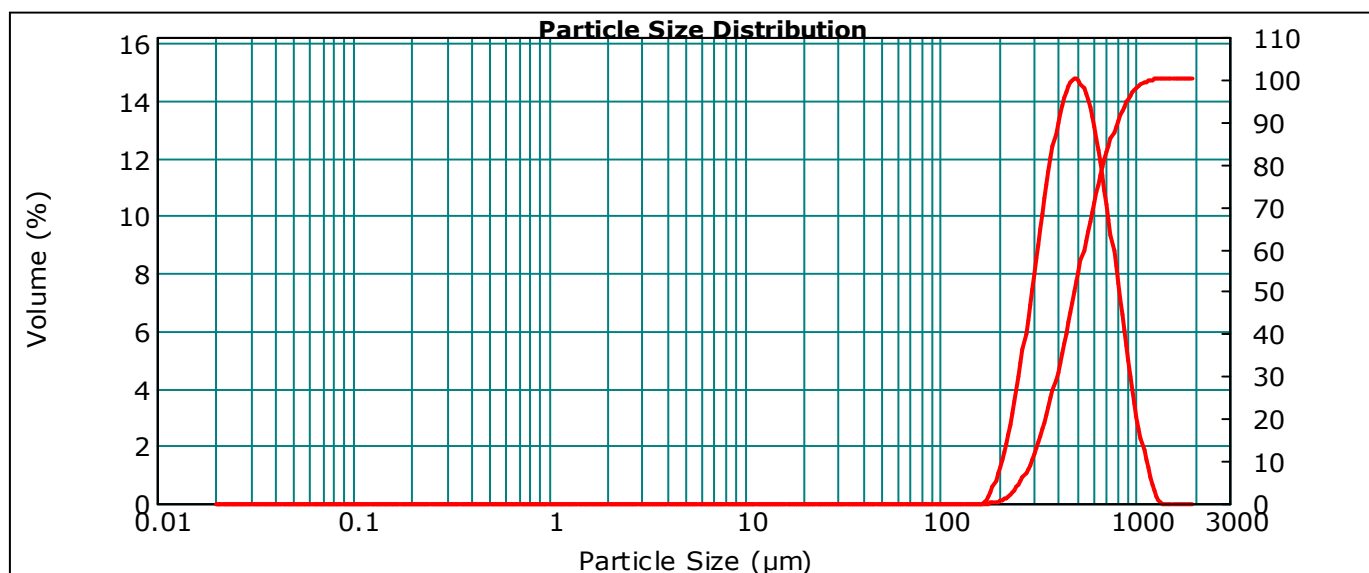
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.04 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:42:18 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 524.28um = .93phi

Surface Weighted Mean:

D[3,2] = 453.37um = 1.14phi

Span :

1.048

Uniformity:

0.325

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 258.22um = 1.95phi

d(0.10) = 293.45um = 1.77phi

d(0.16) = 327.1um = 1.61phi

d(0.50) = 489.56um = 1.03phi

d(0.84) = 727.44um = .46phi

d(0.90) = 806.66um = .31phi

d(0.95) = 909um = .14phi

Distribution Properties:

Kurtosis = .28

Skewness = .79

Standard Deviation = 201.31 um = 2.31phi

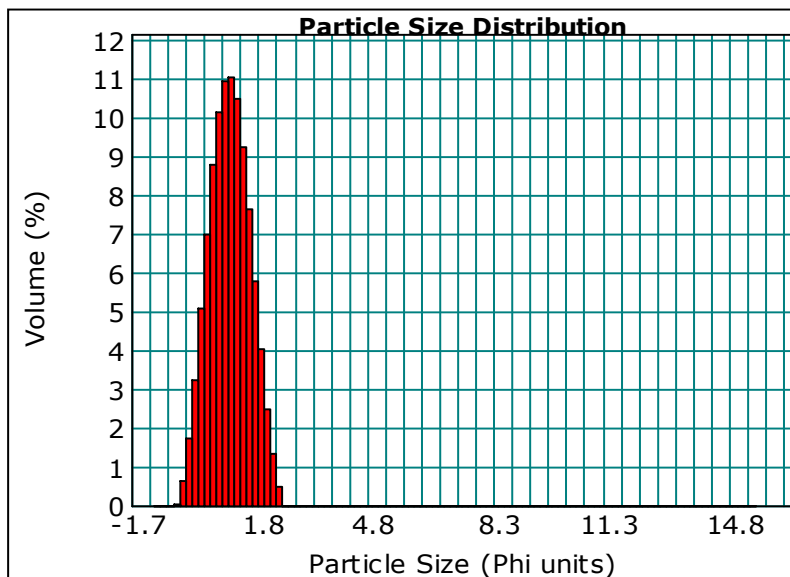
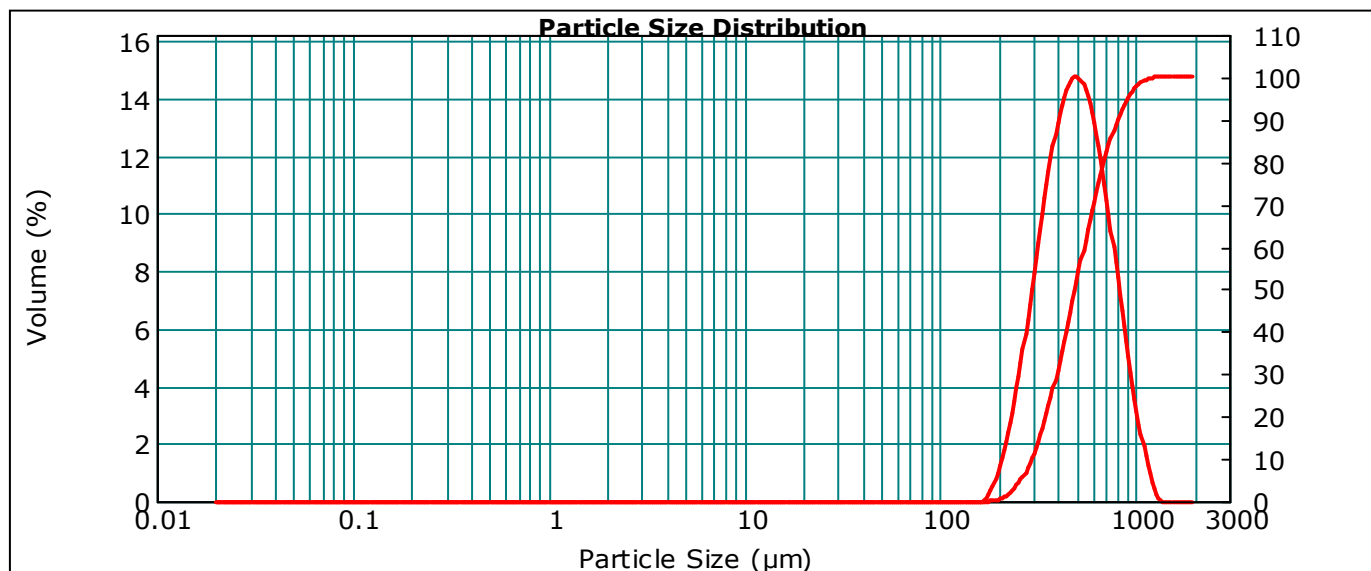
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.41
0.0	15.50
0.5	30.05
1.0	30.75
1.5	17.22
2.0	4.07
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:43:25 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 524.51um = .93phi

Surface Weighted Mean:

D[3,2] = 453.33um = 1.14phi

Span :

1.052

Uniformity:

0.326

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 258.23um = 1.95phi

d(0.10) = 293.37um = 1.77phi

d(0.16) = 326.95um = 1.61phi

d(0.50) = 489.28um = 1.03phi

d(0.84) = 728.08um = .46phi

d(0.90) = 808um = .31phi

d(0.95) = 911.3um = .13phi

Distribution Properties:

Kurtosis = .3

Skewness = .8

Standard Deviation = 202.03 um = 2.31phi

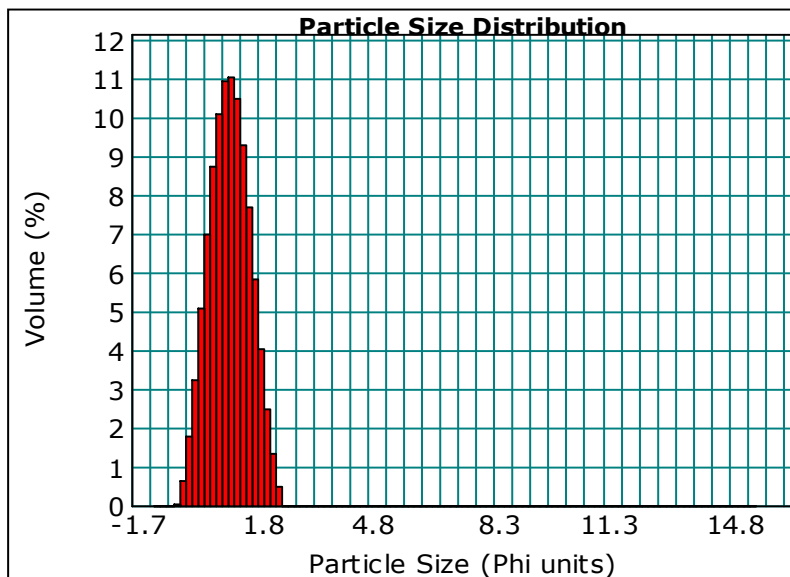
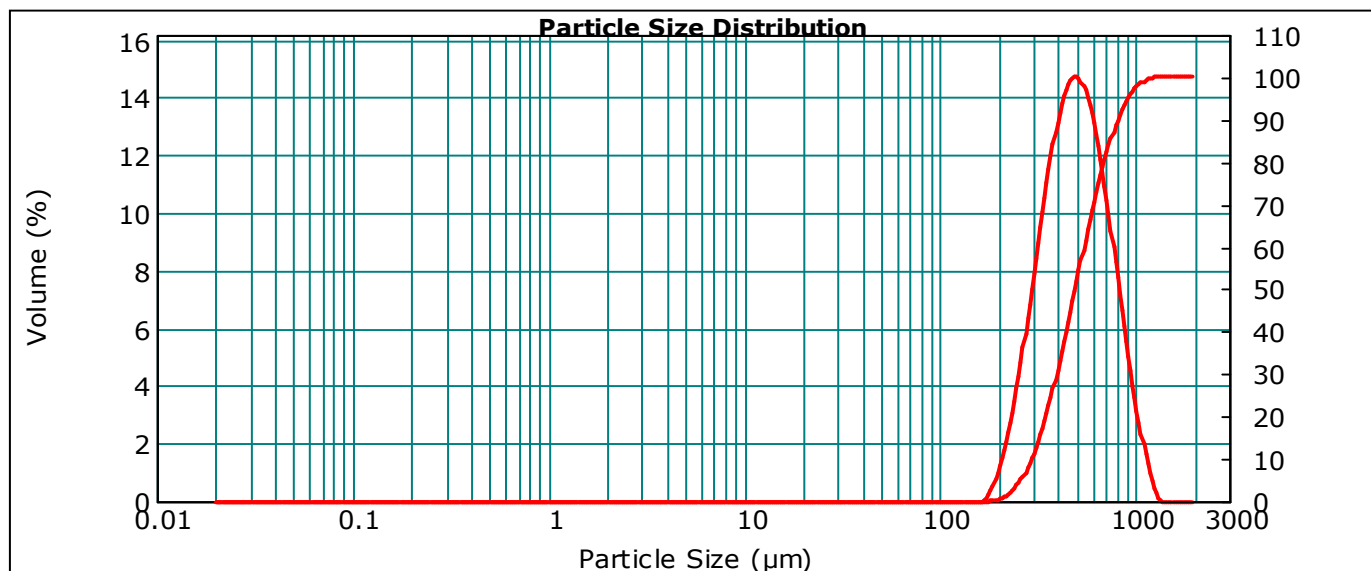
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.49
0.0	15.48
0.5	29.95
1.0	30.76
1.5	17.27
2.0	4.07
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_11 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:41:10 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 523.76um = .93phi

Surface Weighted Mean:

D[3,2] = 452.88um = 1.14phi

Span :

1.049

Uniformity:

0.325

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 258um = 1.95phi

d(0.10) = 293.14um = 1.77phi

d(0.16) = 326.73um = 1.61phi

d(0.50) = 488.9um = 1.03phi

d(0.84) = 726.77um = .46phi

d(0.90) = 806.1um = .31phi

d(0.95) = 908.71um = .14phi

Distribution Properties:

Kurtosis = .29

Skewness = .79

Standard Deviation = 201.3 um = 2.31phi

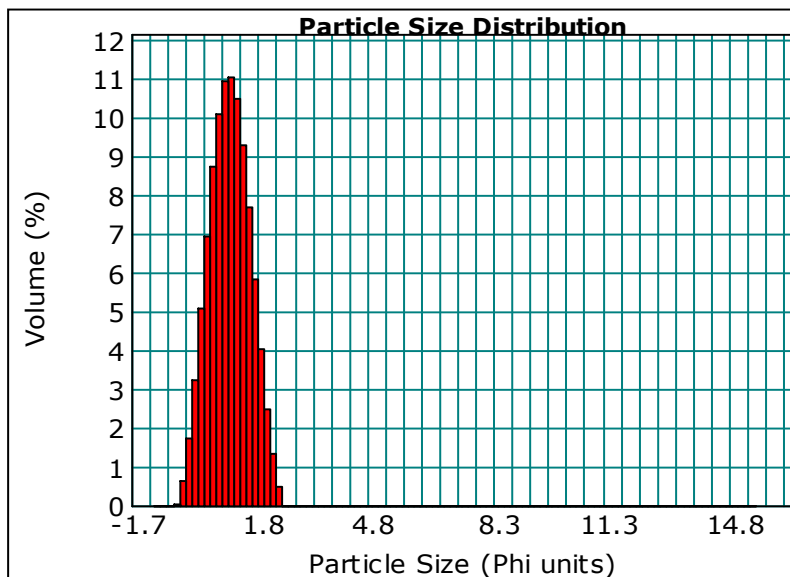
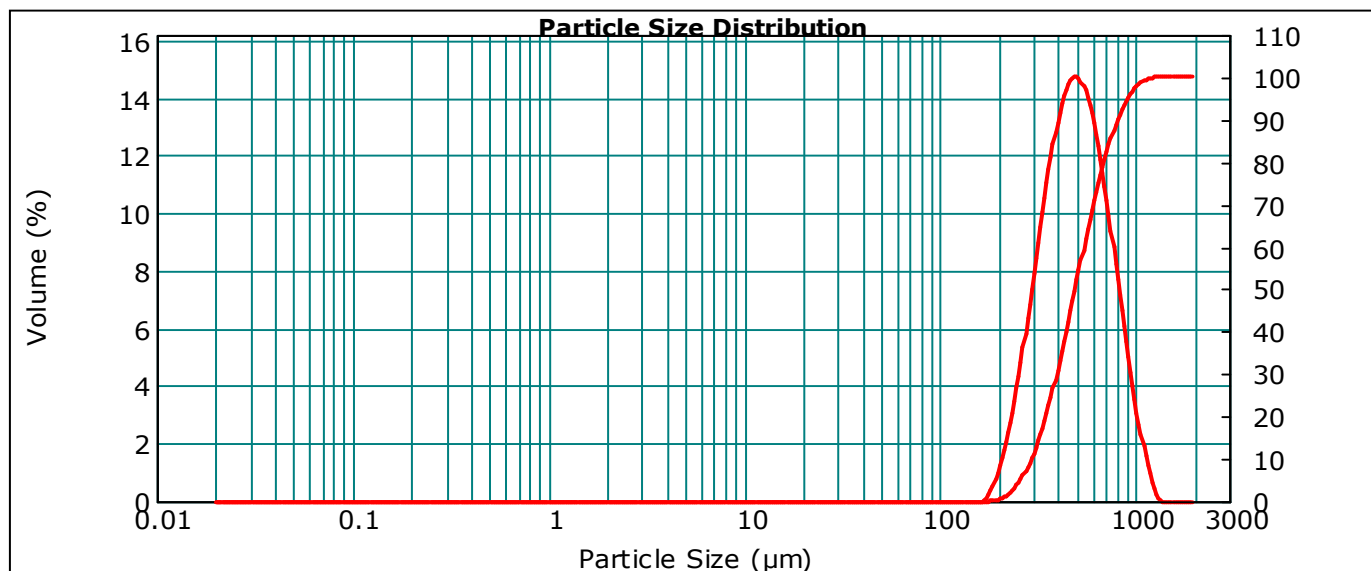
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.04 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.41
0.0	15.43
0.5	29.99
1.0	30.79
1.5	17.29
2.0	4.09
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:54:15 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 530.59um = .91phi

Surface Weighted Mean:

D[3,2] = 451.45um = 1.15phi

Span :

1.117

Uniformity:

0.345

Specific Surface Area*:

0.0133 m²/g

Percentiles:

d(0.05) = 249.72um = 2phi

d(0.10) = 285.14um = 1.81phi

d(0.16) = 319.72um = 1.65phi

d(0.50) = 492.17um = 1.02phi

d(0.84) = 749.97um = .42phi

d(0.90) = 835.13um = .26phi

d(0.95) = 942.94um = .08phi

Distribution Properties:

Kurtosis = .15

Skewness = .77

Standard Deviation = 214.02 um = 2.22phi

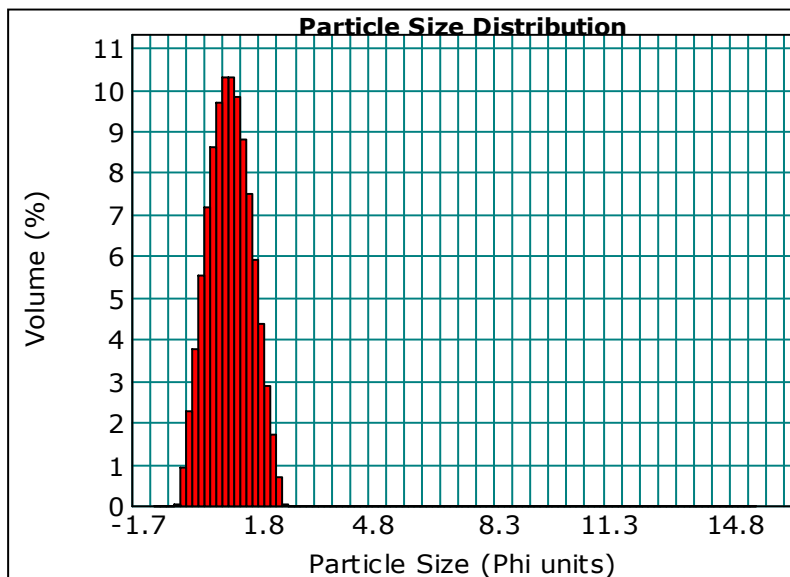
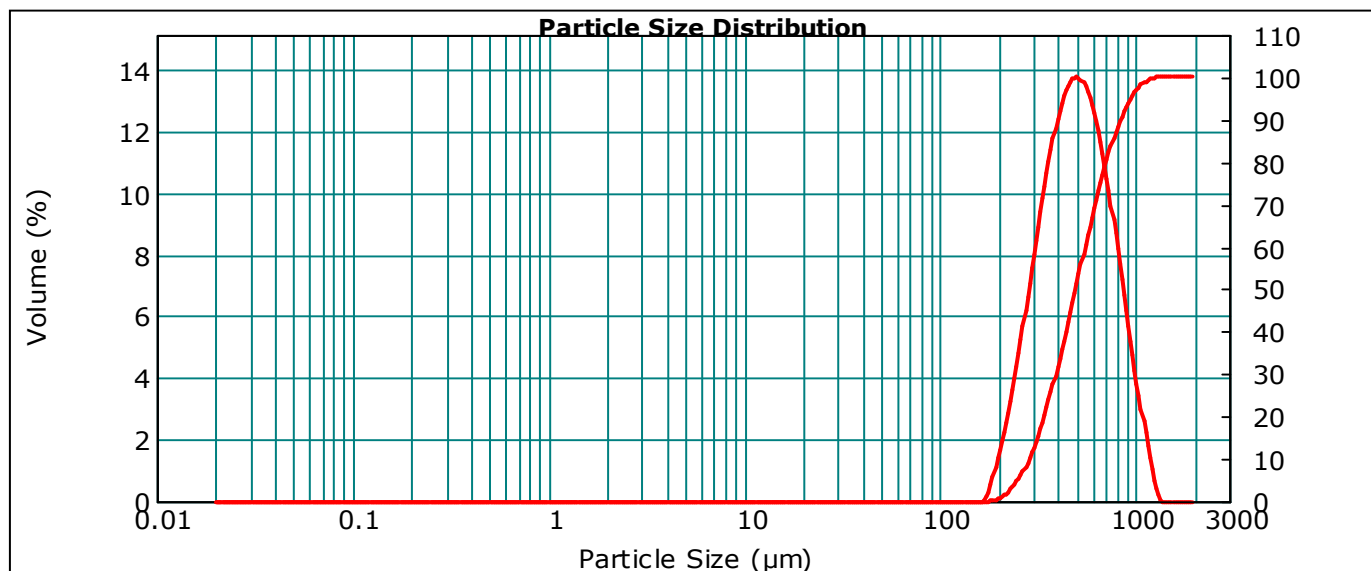
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	3.27
0.0	16.61
0.5	28.70
1.0	28.89
1.5	17.50
2.0	5.03
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:55:22 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 532.88um = .91phi

Surface Weighted Mean:

D[3,2] = 452.87um = 1.14phi

Span :

1.123

Uniformity:

0.347

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 250.11um = 2phi

d(0.10) = 285.69um = 1.81phi

d(0.16) = 320.43um = 1.64phi

d(0.50) = 493.9um = 1.02phi

d(0.84) = 754.15um = .41phi

d(0.90) = 840.24um = .25phi

d(0.95) = 949.6um = .07phi

Distribution Properties:

Kurtosis = .14

Skewness = .78

Standard Deviation = 215.76 um = 2.21phi

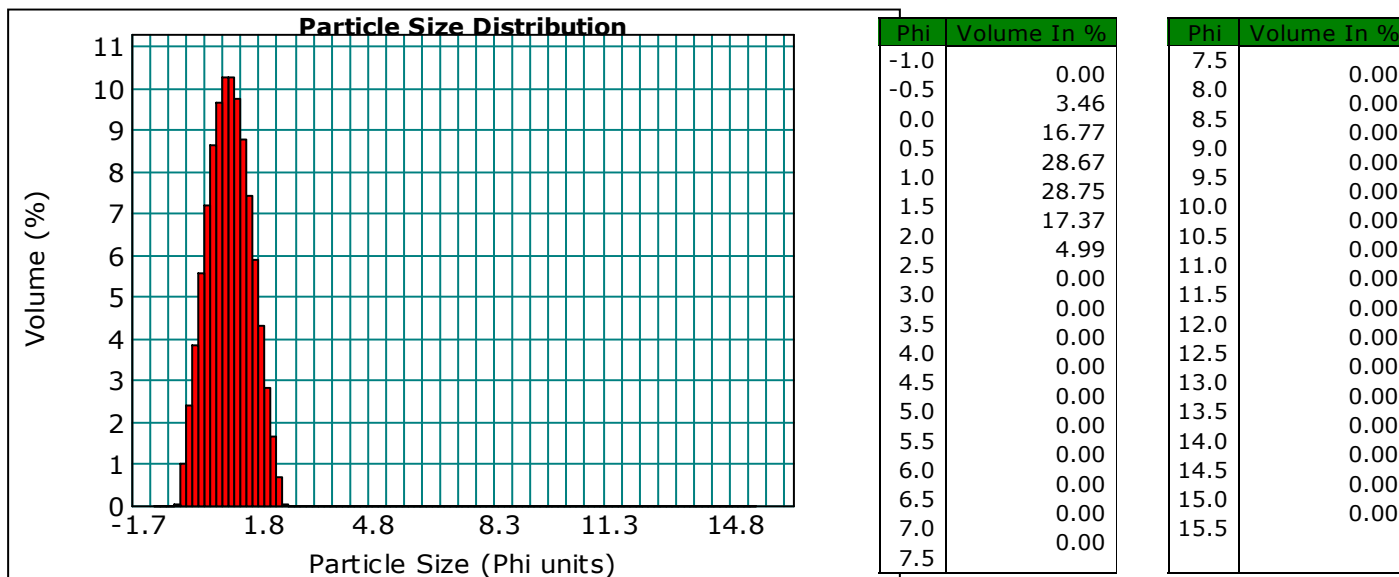
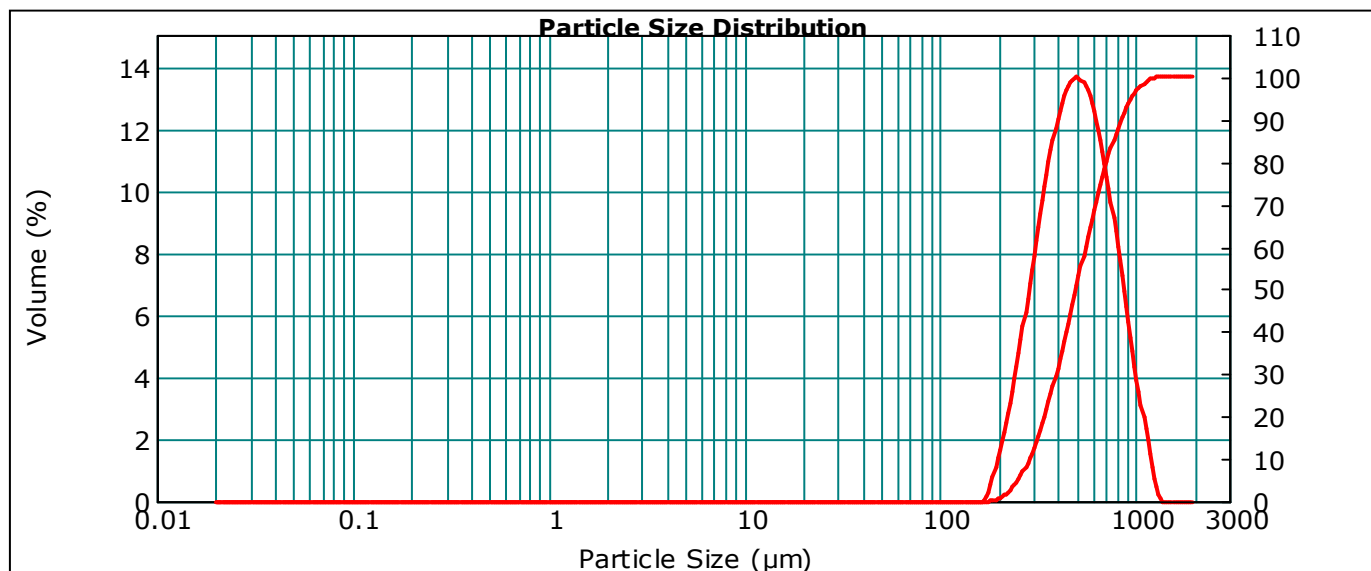
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.02 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:56:31 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 527.51um = .92phi

Surface Weighted Mean:

D[3,2] = 448.9um = 1.16phi

Span :

1.118

Uniformity:

0.345

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 248.65um = 2.01phi

d(0.10) = 283.69um = 1.82phi

d(0.16) = 317.98um = 1.65phi

d(0.50) = 488.93um = 1.03phi

d(0.84) = 745.47um = .42phi

d(0.90) = 830.48um = .27phi

d(0.95) = 938.24um = .09phi

Distribution Properties:

Kurtosis = .18

Skewness = .79

Standard Deviation = 213.05 um = 2.23phi

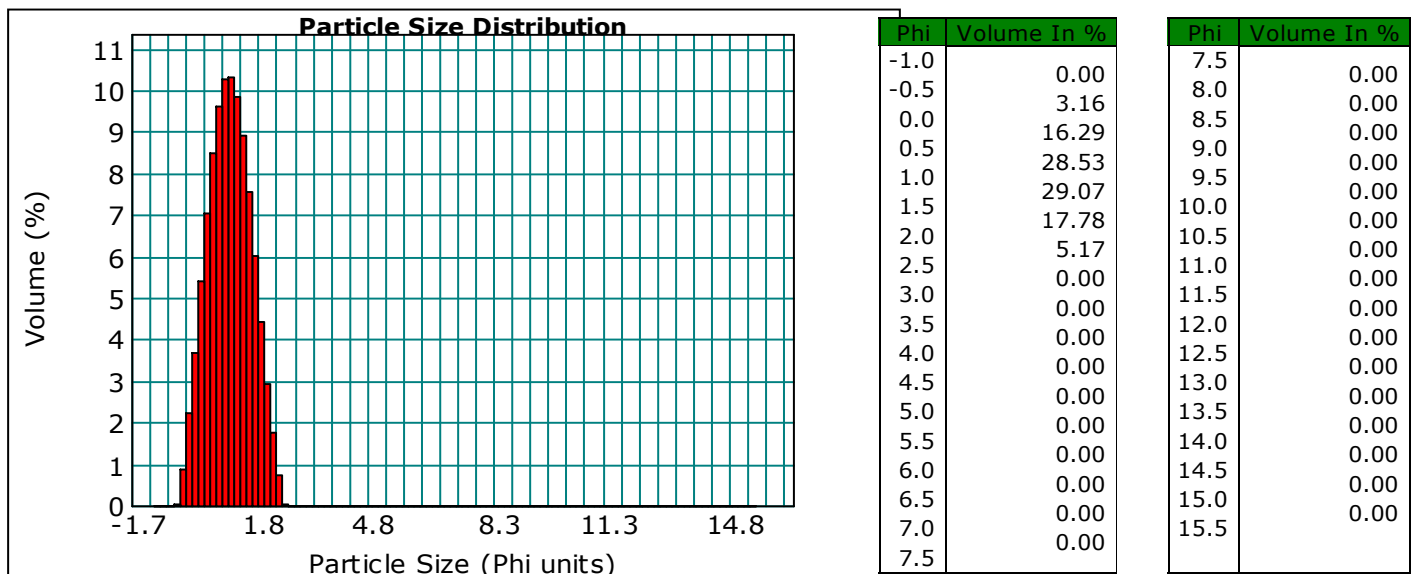
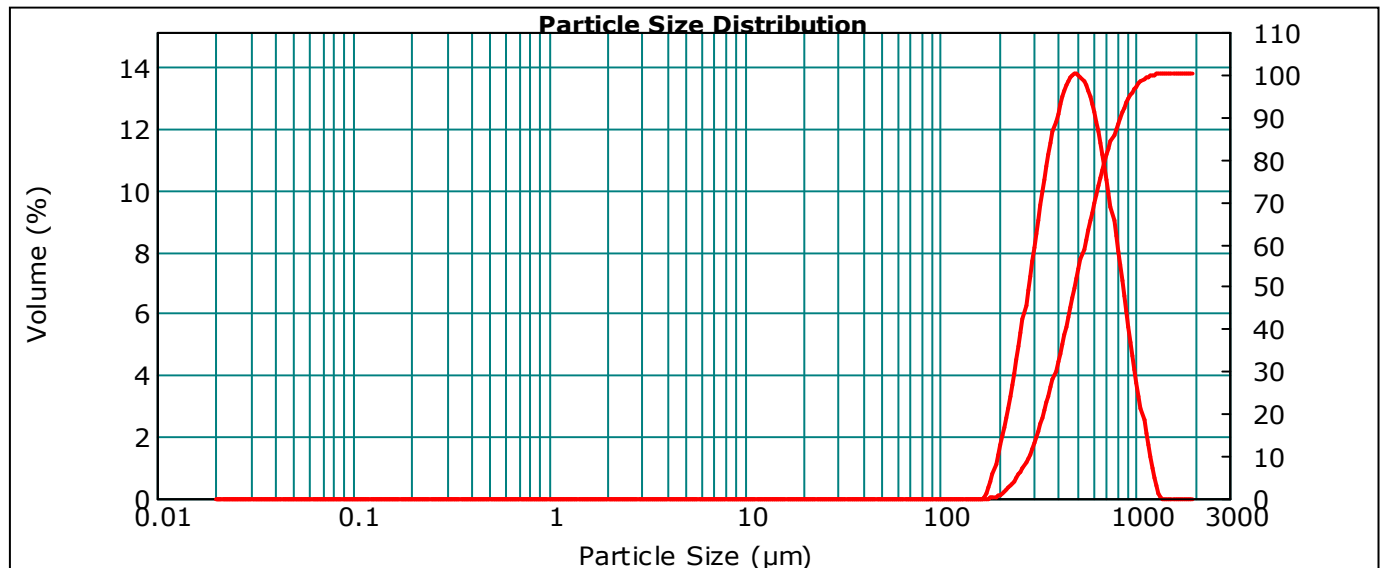
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.04 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_10 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 2:54:15 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 530.32um = .92phi

Surface Weighted Mean:

D[3,2] = 451.07um = 1.15phi

Span :

1.120

Uniformity:

0.346

Specific Surface Area*:

0.0133 m²/g

Percentiles:

d(0.05) = 249.49um = 2phi

d(0.10) = 284.83um = 1.81phi

d(0.16) = 319.37um = 1.65phi

d(0.50) = 491.66um = 1.02phi

d(0.84) = 749.85um = .42phi

d(0.90) = 835.32um = .26phi

d(0.95) = 943.66um = .08phi

Distribution Properties:

Kurtosis = .16

Skewness = .78

Standard Deviation = 214.29 um = 2.22phi

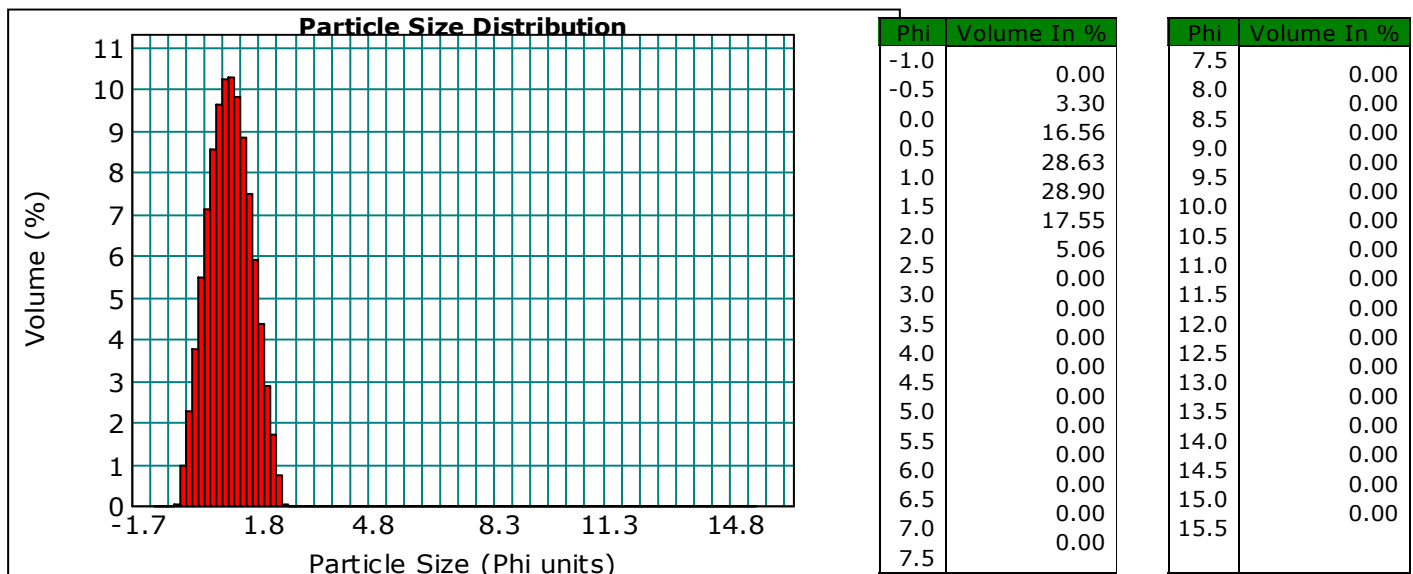
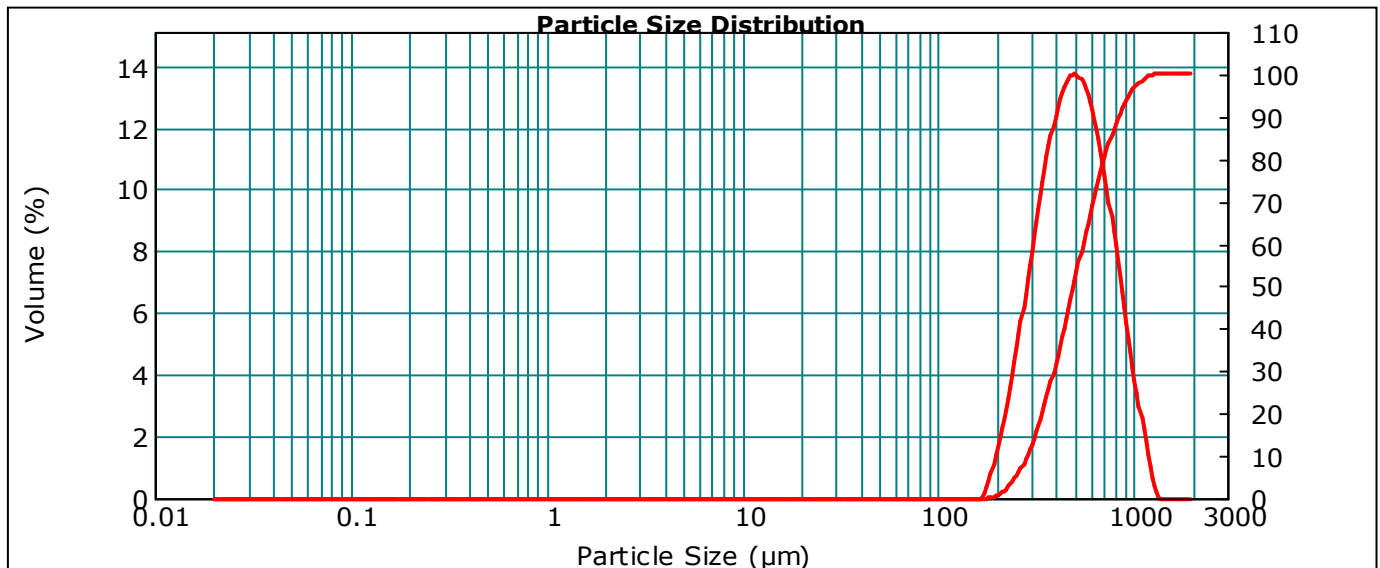
Inclusive SD (Sorting Coeff.) = .6 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



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Sample Name: C_1_1_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:05:01 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 528.47um = .92phi

Surface Weighted Mean:

D[3,2] = 458.18um = 1.13phi

Span :

1.038

Uniformity:

0.322

Specific Surface Area*:

0.0131 m²/g

Percentiles:

d(0.05) = 262.2um = 1.93phi

d(0.10) = 297.55um = 1.75phi

d(0.16) = 331.35um = 1.59phi

d(0.50) = 494.09um = 1.02phi

d(0.84) = 731.62um = .45phi

d(0.90) = 810.53um = .3phi

d(0.95) = 911.98um = .13phi

Distribution Properties:

Kurtosis = .26

Skewness = .78

Standard Deviation = 201 um = 2.31phi

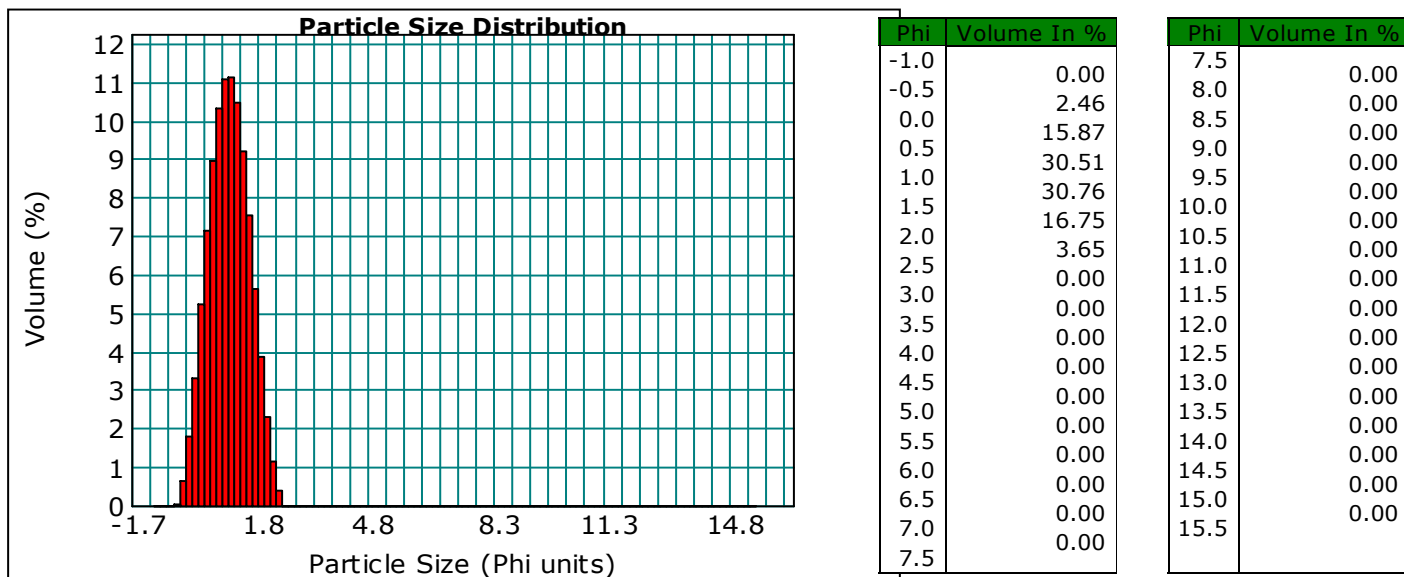
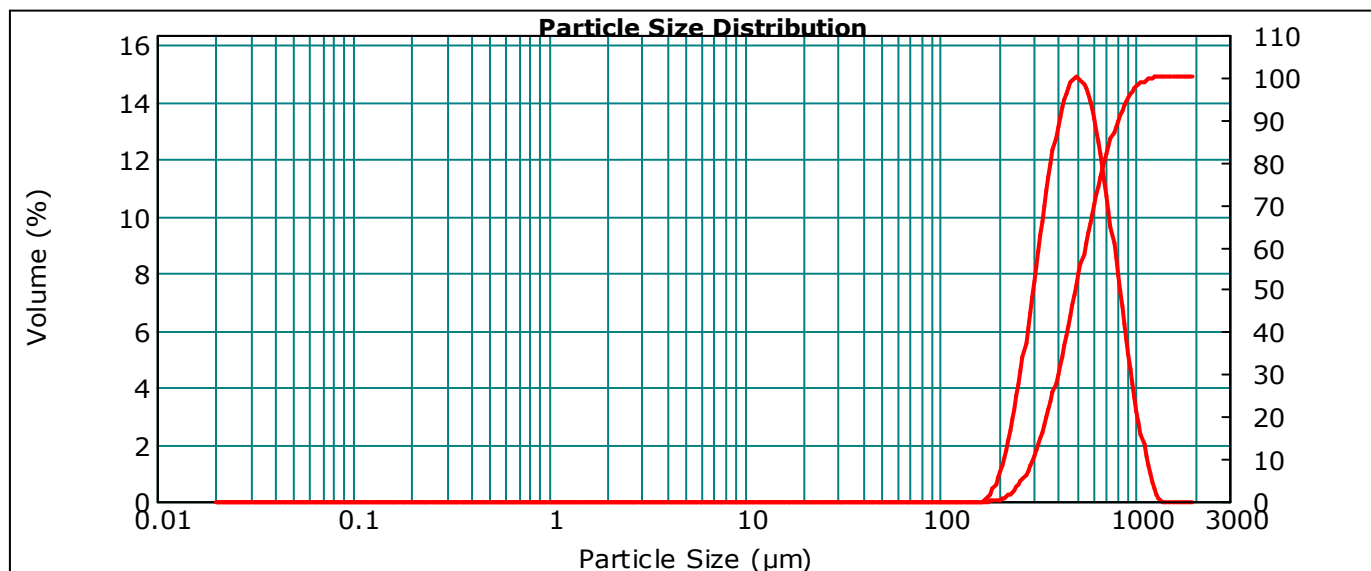
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.02 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:06:08 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 524.52um = .93phi

Surface Weighted Mean:

D[3,2] = 454.73um = 1.14phi

Span :

1.039

Uniformity:

0.322

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 260.33um = 1.94phi

d(0.10) = 295.35um = 1.76phi

d(0.16) = 328.81um = 1.6phi

d(0.50) = 490.23um = 1.03phi

d(0.84) = 726.15um = .46phi

d(0.90) = 804.57um = .31phi

d(0.95) = 905.76um = .14phi

Distribution Properties:

Kurtosis = .28

Skewness = .79

Standard Deviation = 199.7 um = 2.32phi

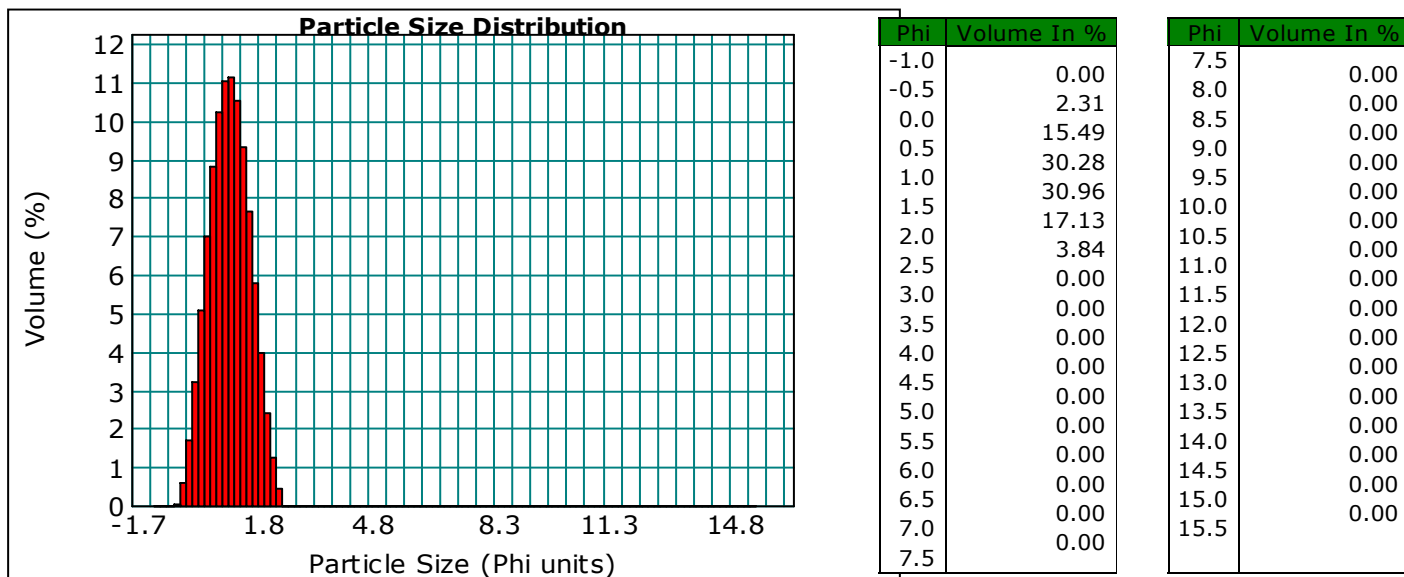
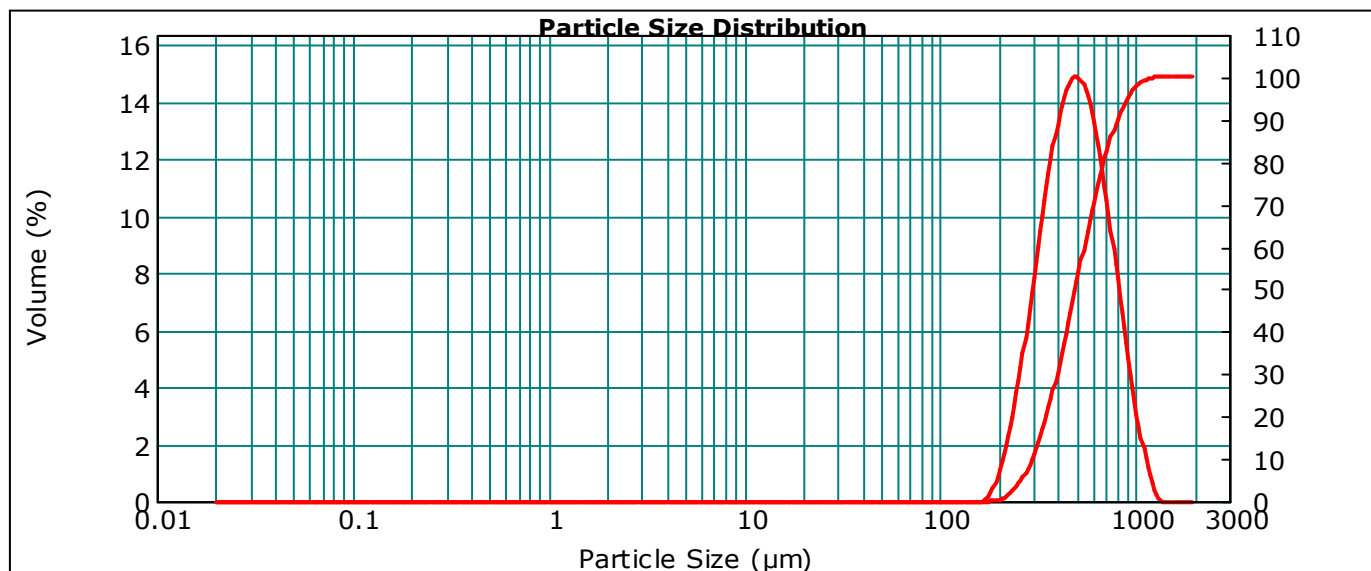
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:07:16 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 526.13um = .93phi

Surface Weighted Mean:

D[3,2] = 455.77um = 1.13phi

Span :

1.043

Uniformity:

0.323

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 260.72um = 1.94phi

d(0.10) = 295.81um = 1.76phi

d(0.16) = 329.35um = 1.6phi

d(0.50) = 491.32um = 1.03phi

d(0.84) = 728.82um = .46phi

d(0.90) = 808.13um = .31phi

d(0.95) = 910.5um = .14phi

Distribution Properties:

Kurtosis = .29

Skewness = .79

Standard Deviation = 201.04 um = 2.31phi

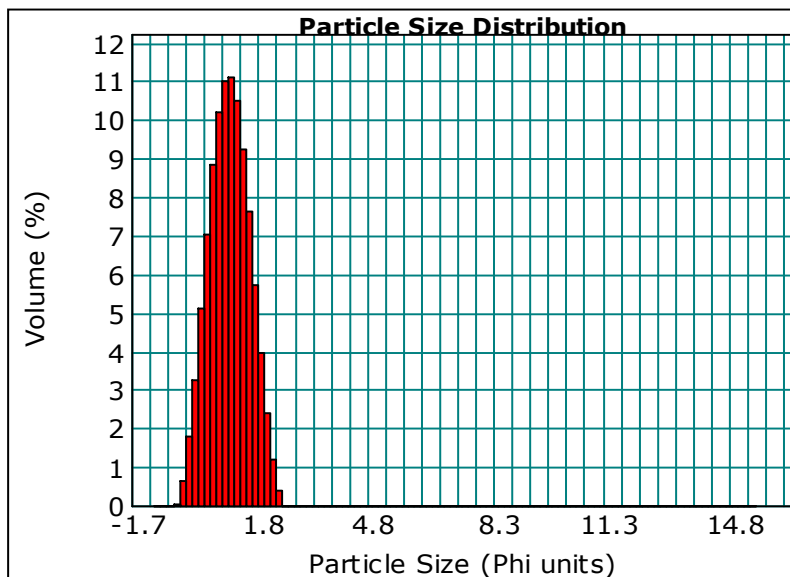
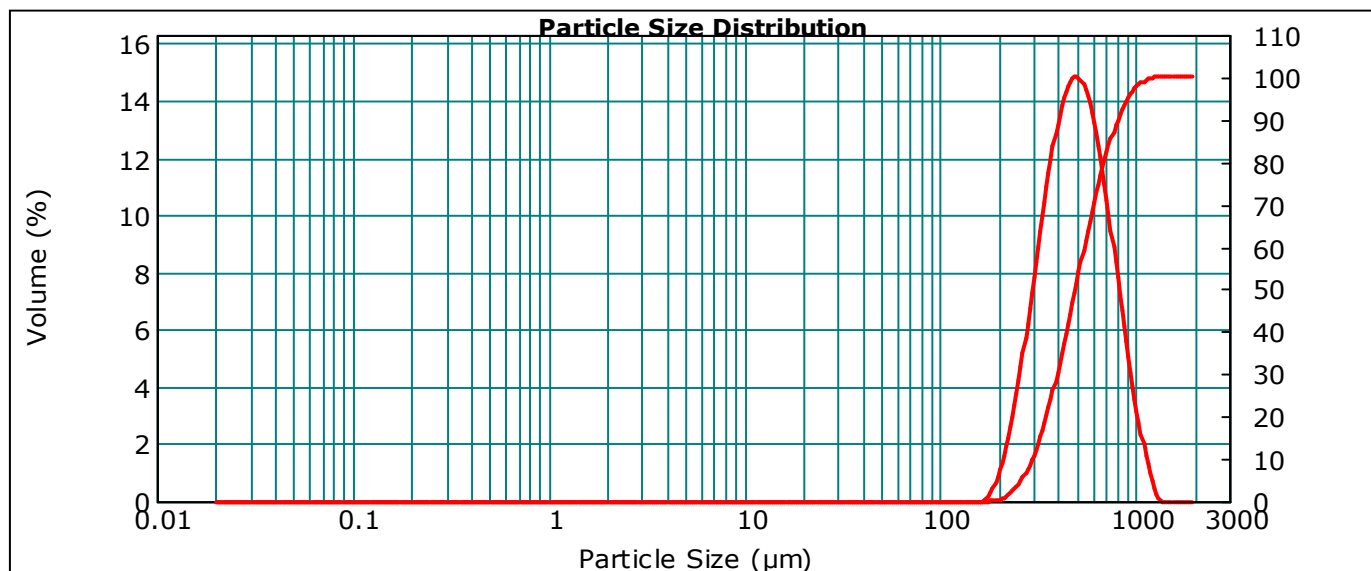
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.45
0.0	15.60
0.5	30.25
1.0	30.86
1.5	17.04
2.0	3.80
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_9 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:05:01 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 526.37um = .93phi

Surface Weighted Mean:

D[3,2] = 456.22um = 1.13phi

Span :

1.040

Uniformity:

0.323

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 261.07um = 1.94phi

d(0.10) = 296.23um = 1.76phi

d(0.16) = 329.83um = 1.6phi

d(0.50) = 491.88um = 1.02phi

d(0.84) = 728.83um = .46phi

d(0.90) = 807.72um = .31phi

d(0.95) = 909.44um = .14phi

Distribution Properties:

Kurtosis = .28

Skewness = .79

Standard Deviation = 200.59 um = 2.32phi

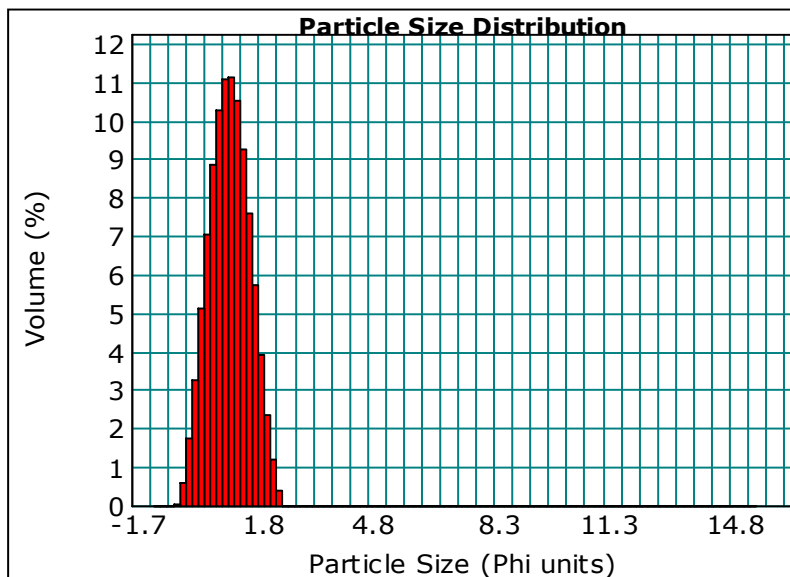
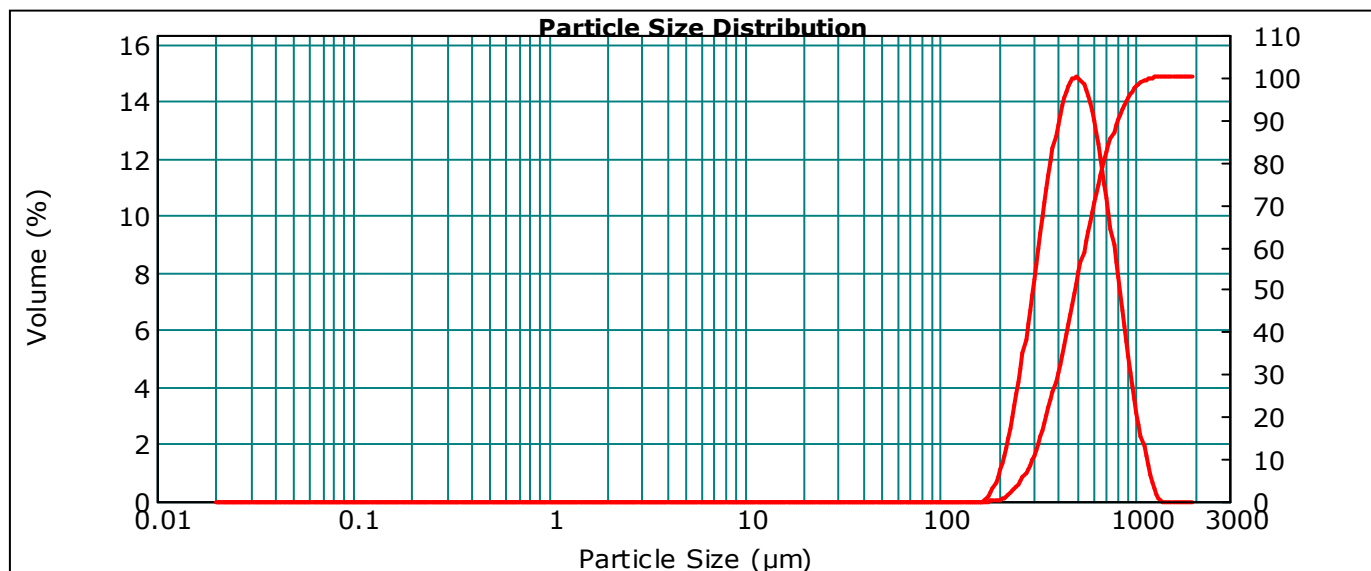
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.41
0.0	15.65
0.5	30.34
1.0	30.86
1.5	16.97
2.0	3.76
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:15:55 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 523.13um = .93phi

Surface Weighted Mean:

D[3,2] = 454.24um = 1.14phi

Span :

1.033

Uniformity:

0.32

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 260.9um = 1.94phi

d(0.10) = 295.65um = 1.76phi

d(0.16) = 328.87um = 1.6phi

d(0.50) = 489.15um = 1.03phi

d(0.84) = 723.34um = .47phi

d(0.90) = 801.13um = .32phi

d(0.95) = 901.26um = .15phi

Distribution Properties:

Kurtosis = .28

Skewness = .78

Standard Deviation = 198.17 um = 2.34phi

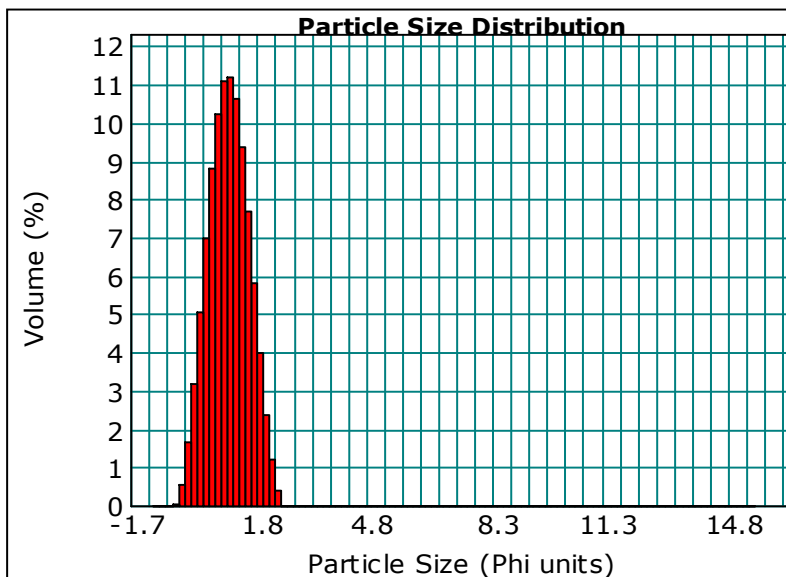
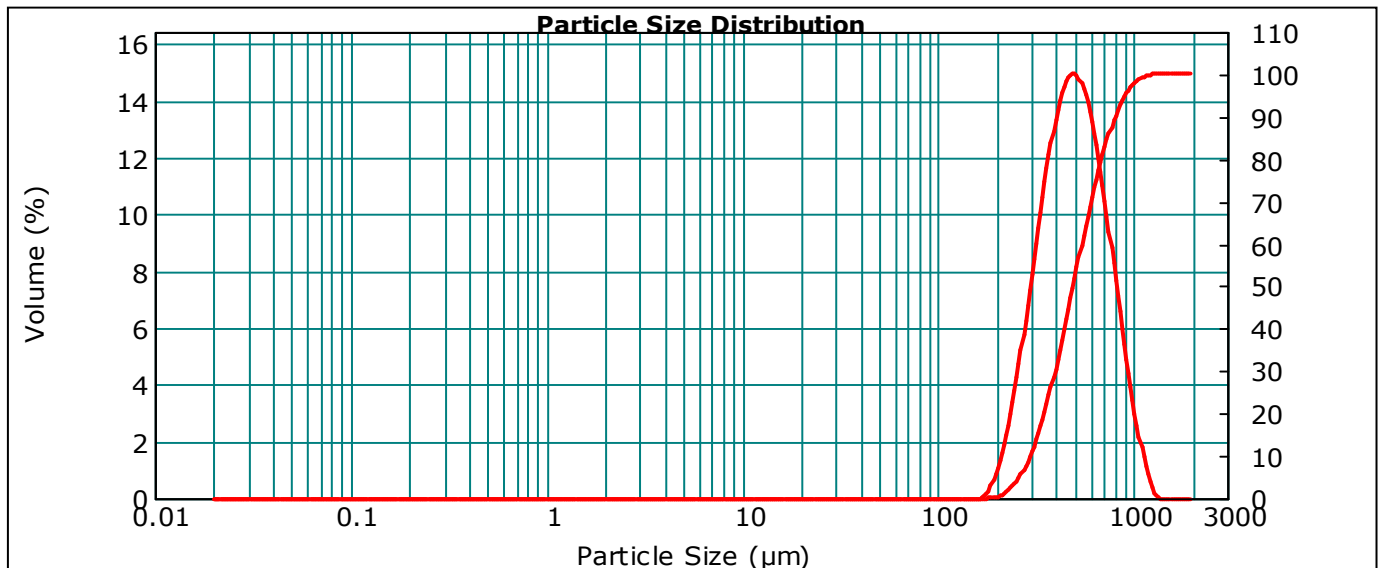
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	0.00
0.0	2.19
0.5	15.34
1.0	30.32
1.5	31.16
2.0	17.22
2.5	3.77
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:17:02 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 521.48um = .94phi

Surface Weighted Mean:

D[3,2] = 453.02um = 1.14phi

Span :

1.032

Uniformity:

0.319

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 260.41um = 1.94phi

d(0.10) = 295.02um = 1.76phi

d(0.16) = 328.09um = 1.61phi

d(0.50) = 487.74um = 1.04phi

d(0.84) = 720.89um = .47phi

d(0.90) = 798.22um = .33phi

d(0.95) = 897.45um = .16phi

Distribution Properties:

Kurtosis = .28

Skewness = .78

Standard Deviation = 197.17 um = 2.34phi

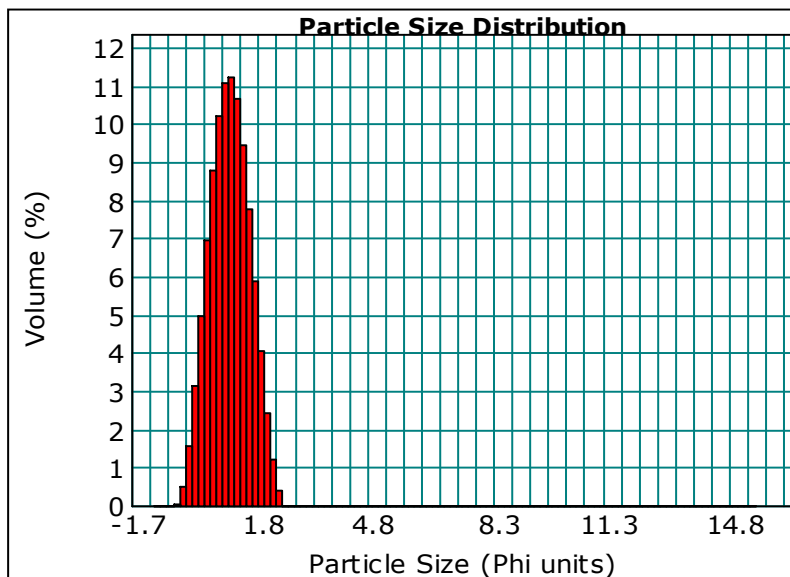
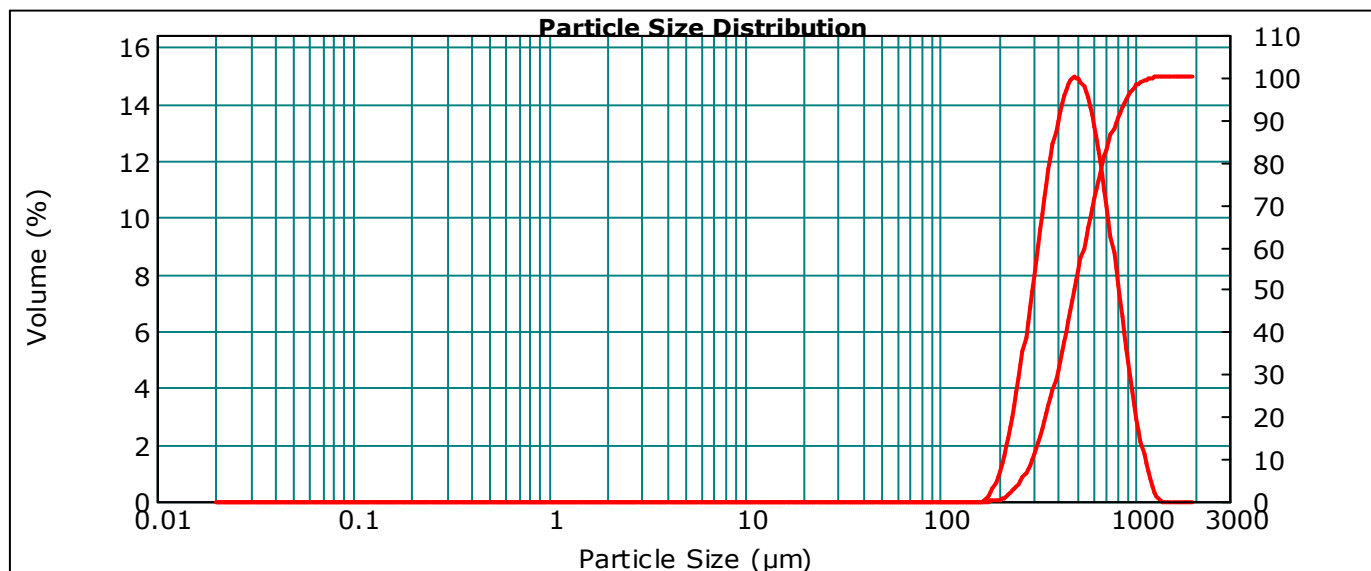
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.04 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.09
0.0	15.21
0.5	30.26
1.0	31.26
1.5	17.36
2.0	3.82
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:18:10 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 523.78um = .93phi

Surface Weighted Mean:

D[3,2] = 455.13um = 1.14phi

Span :

1.030

Uniformity:

0.319

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 261.53um = 1.93phi

d(0.10) = 296.39um = 1.75phi

d(0.16) = 329.7um = 1.6phi

d(0.50) = 490.21um = 1.03phi

d(0.84) = 723.86um = .47phi

d(0.90) = 801.14um = .32phi

d(0.95) = 900.32um = .15phi

Distribution Properties:

Kurtosis = .26

Skewness = .77

Standard Deviation = 197.65 um = 2.34phi

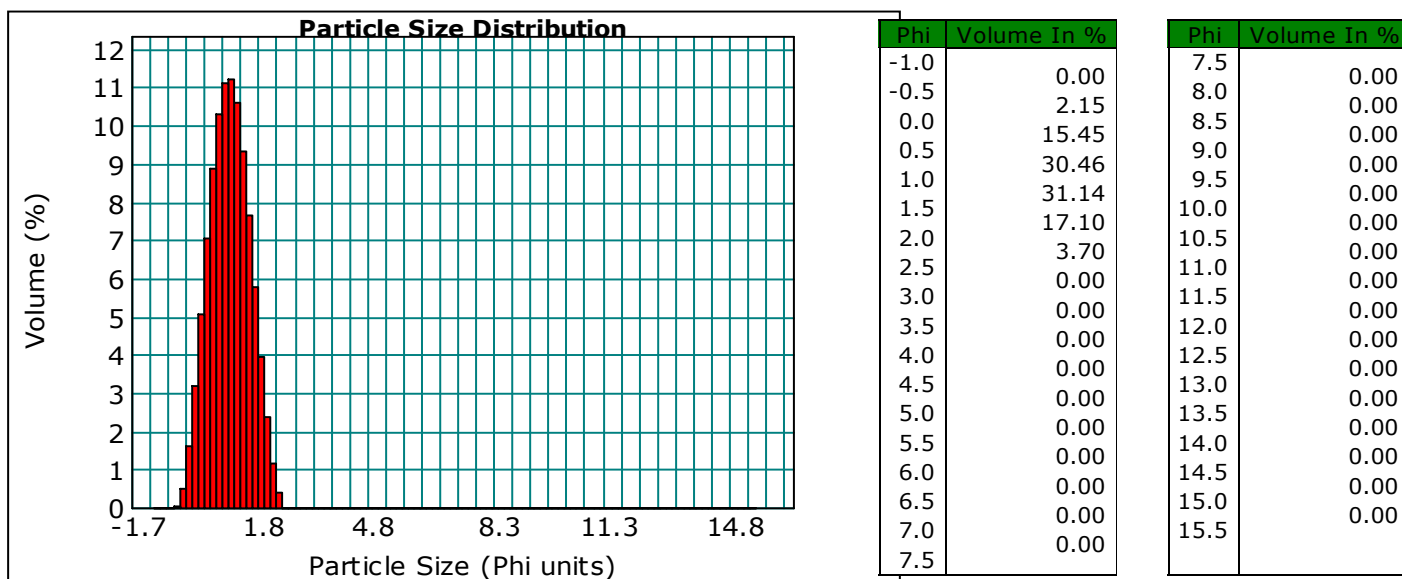
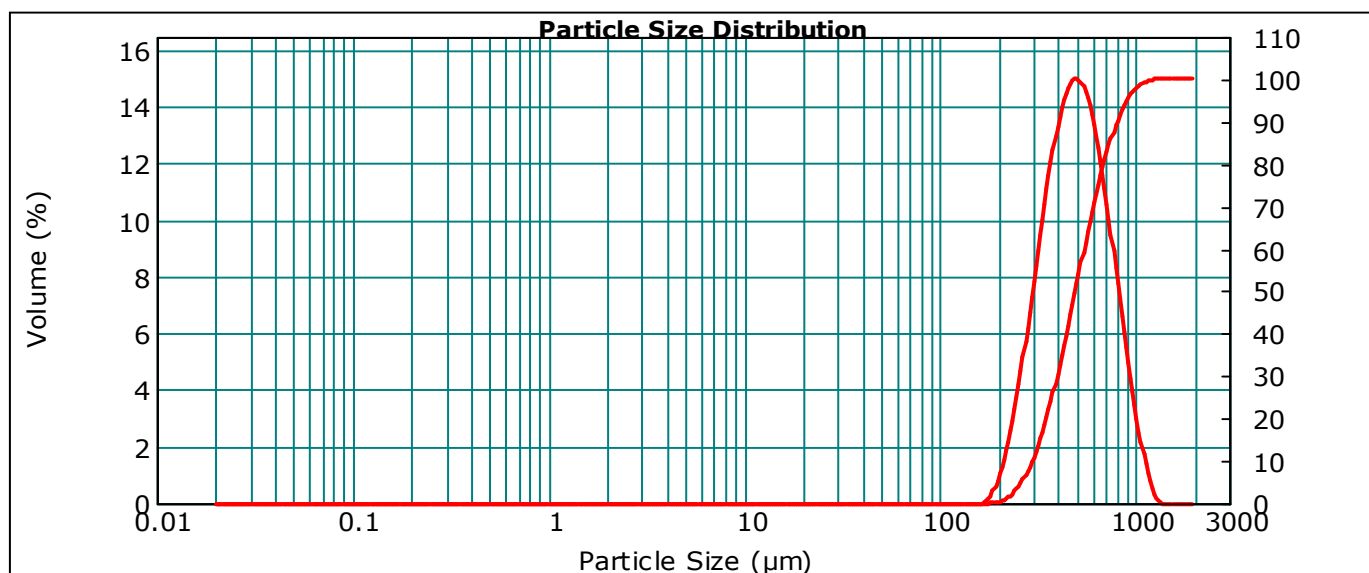
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_8 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:15:55 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 522.8um = .94phi

Surface Weighted Mean:

D[3,2] = 454.13um = 1.14phi

Span :

1.032

Uniformity:

0.32

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 260.94um = 1.94phi

d(0.10) = 295.68um = 1.76phi

d(0.16) = 328.88um = 1.6phi

d(0.50) = 489.03um = 1.03phi

d(0.84) = 722.7um = .47phi

d(0.90) = 800.17um = .32phi

d(0.95) = 899.68um = .15phi

Distribution Properties:

Kurtosis = .27

Skewness = .78

Standard Deviation = 197.67 um = 2.34phi

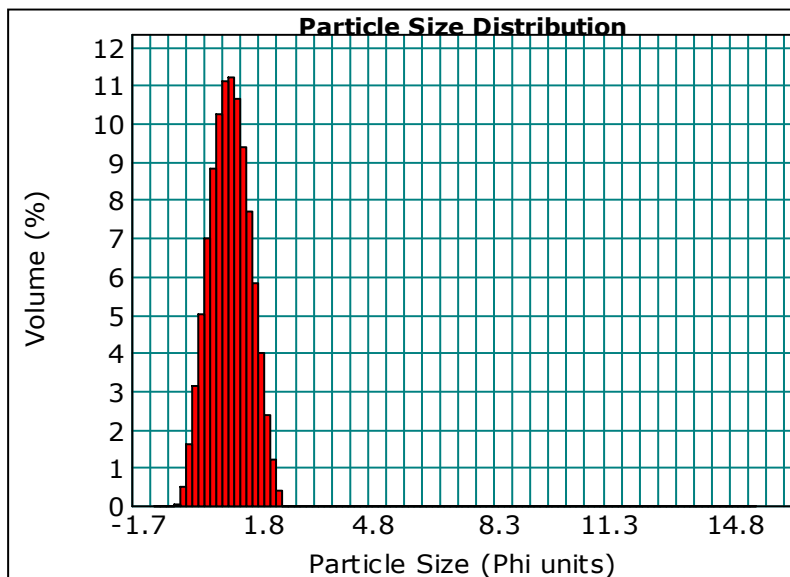
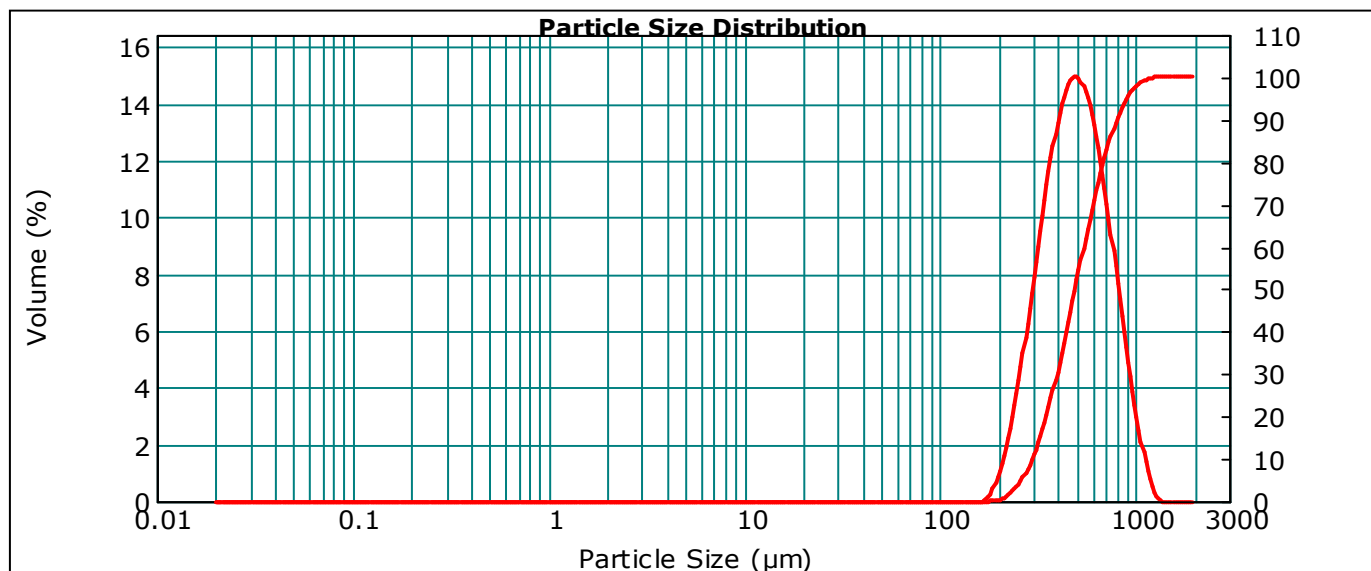
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.14
0.0	15.33
0.5	30.35
1.0	31.19
1.5	17.23
2.0	3.76
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:28:33 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 512.14um = .97phi

Surface Weighted Mean:

D[3,2] = 446.58um = 1.16phi

Span :

1.016

Uniformity:

0.313

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 258.25um = 1.95phi

d(0.10) = 292.1um = 1.78phi

d(0.16) = 324.5um = 1.62phi

d(0.50) = 480.12um = 1.06phi

d(0.84) = 705.66um = .5phi

d(0.90) = 779.98um = .36phi

d(0.95) = 874.96um = .19phi

Distribution Properties:

Kurtosis = .25

Skewness = .76

Standard Deviation = 190.81 um = 2.39phi

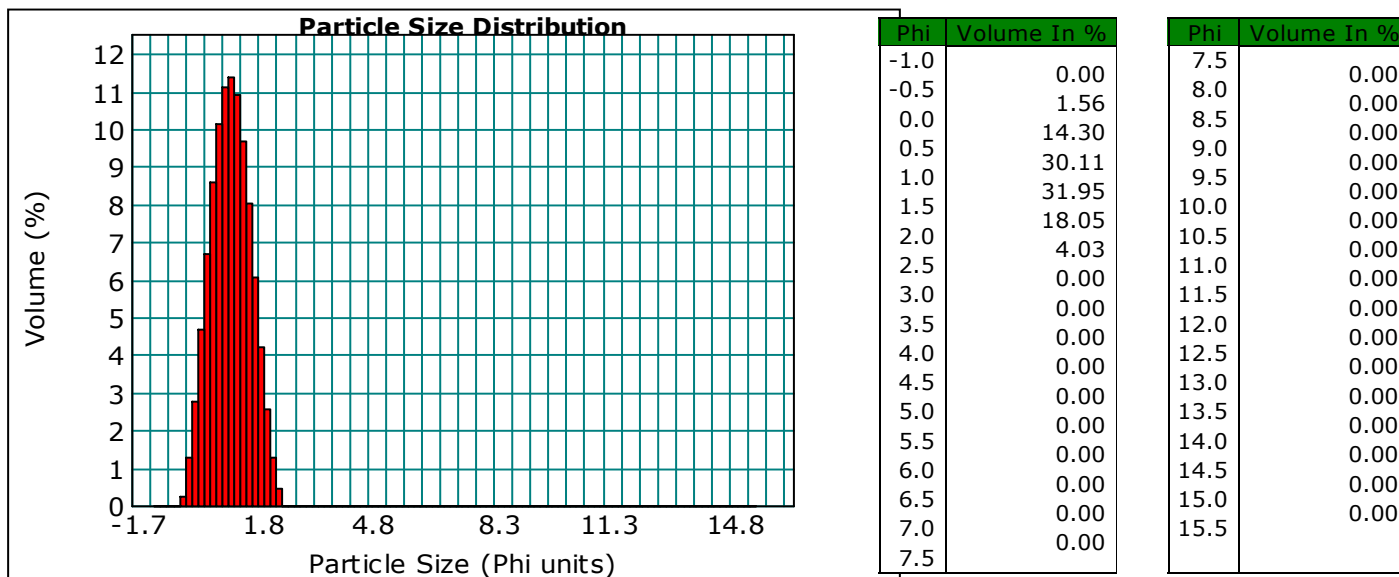
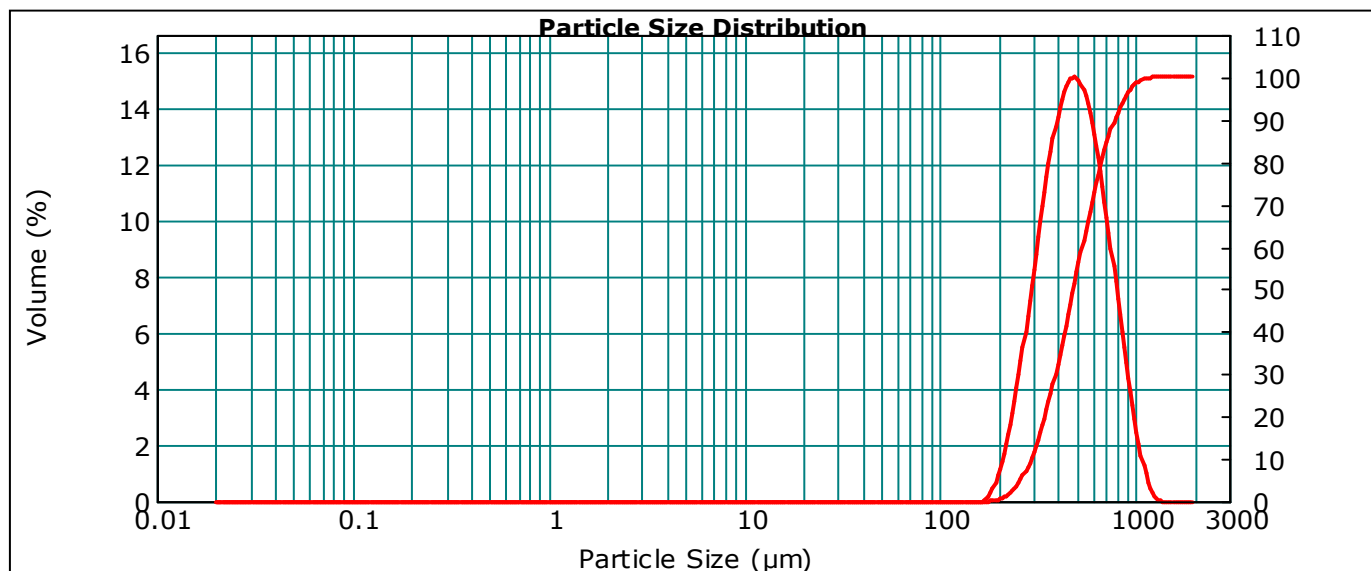
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.06 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:29:41 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 511.06um = .97phi

Surface Weighted Mean:

D[3,2] = 445.32um = 1.17phi

Span :

1.020

Uniformity:

0.314

Specific Surface Area*:

0.0135 m²/g

Percentiles:

d(0.05) = 257.5um = 1.96phi

d(0.10) = 291.18um = 1.78phi

d(0.16) = 323.4um = 1.63phi

d(0.50) = 478.5um = 1.06phi

d(0.84) = 704.48um = .51phi

d(0.90) = 779.38um = .36phi

d(0.95) = 875.49um = .19phi

Distribution Properties:

Kurtosis = .29

Skewness = .78

Standard Deviation = 191.3 um = 2.39phi

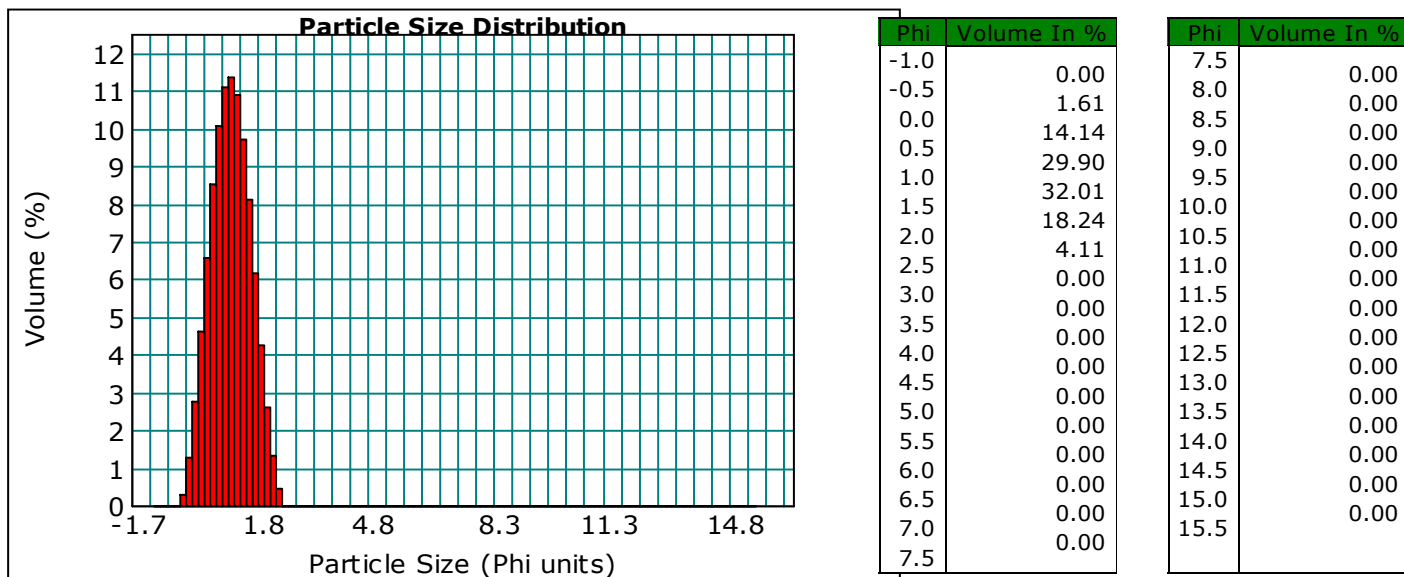
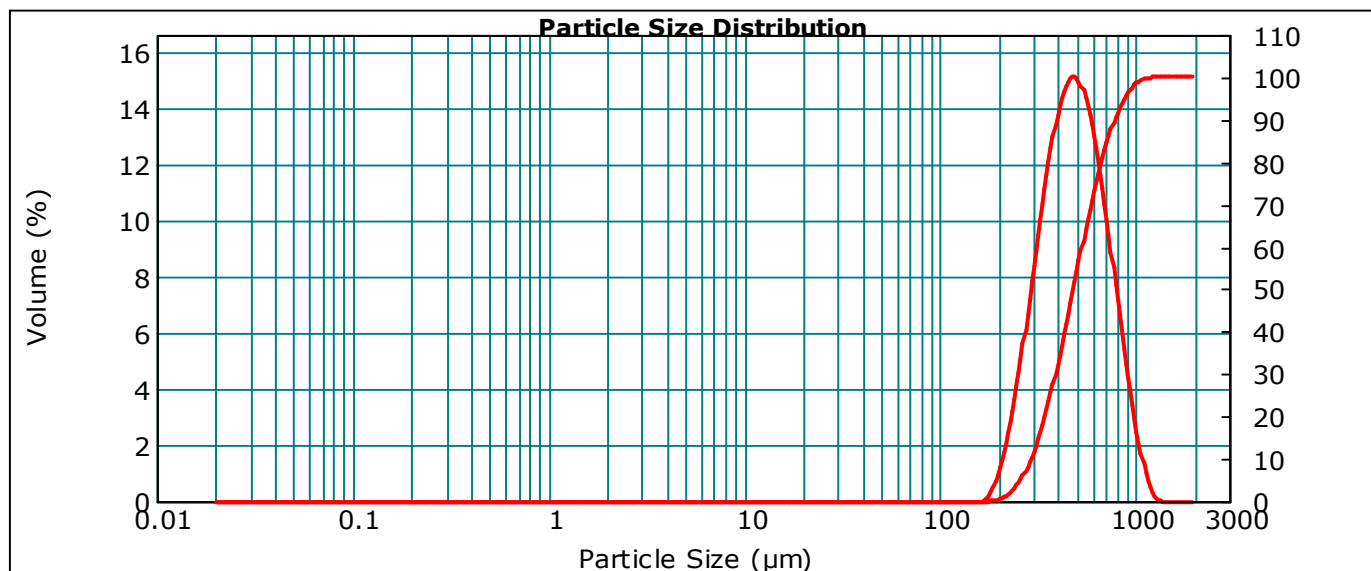
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.07 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:30:48 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 509.99um = .97phi

Surface Weighted Mean:

D[3,2] = 445.32um = 1.17phi

Span :

1.011

Uniformity:

0.311

Specific Surface Area*:

0.0135 m²/g

Percentiles:

d(0.05) = 258.17um = 1.95phi

d(0.10) = 291.79um = 1.78phi

d(0.16) = 323.96um = 1.63phi

d(0.50) = 478.42um = 1.06phi

d(0.84) = 701.93um = .51phi

d(0.90) = 775.43um = .37phi

d(0.95) = 869.02um = .2phi

Distribution Properties:

Kurtosis = .23

Skewness = .76

Standard Deviation = 188.97 um = 2.4phi

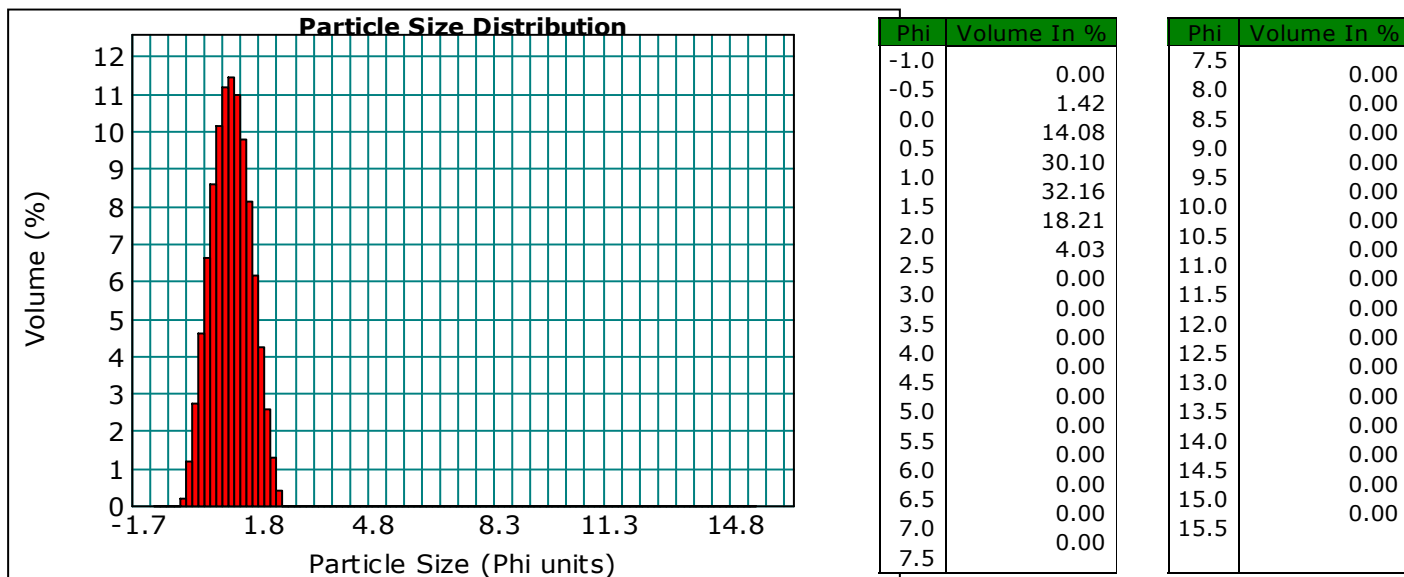
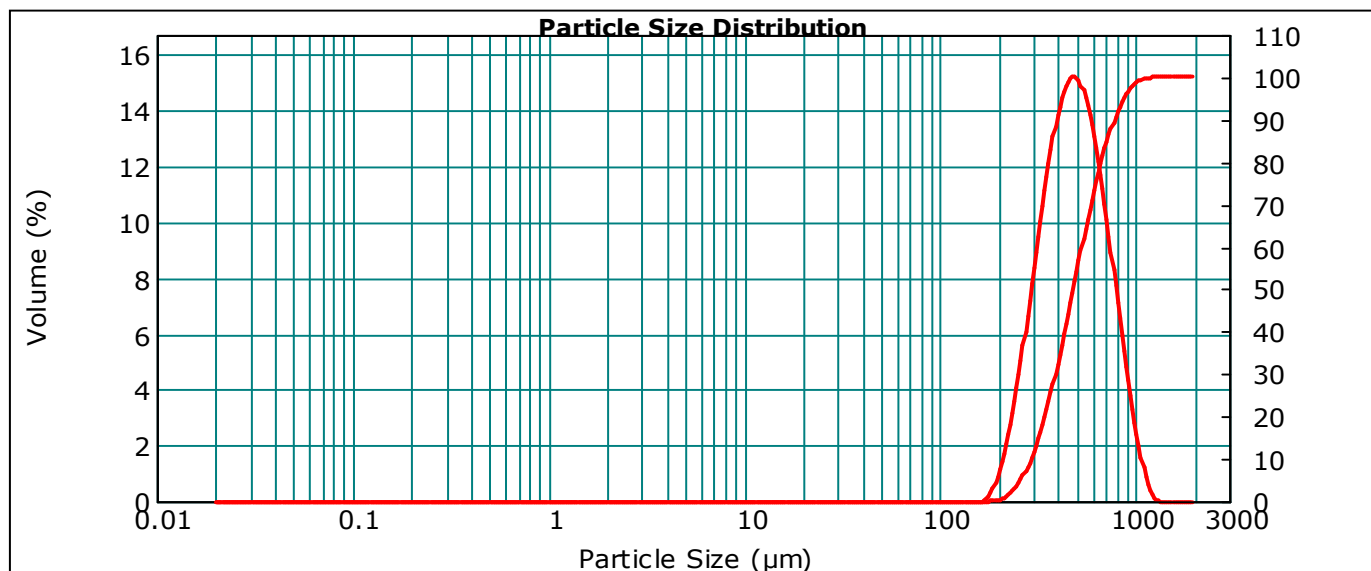
Inclusive SD (Sorting Coeff.) = .54 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.07 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_7 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:28:33 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 511.06um = .97phi

Surface Weighted Mean:

D[3,2] = 445.74um = 1.17phi

Span :

1.016

Uniformity:

0.312

Specific Surface Area*:

0.0135 m²/g

Percentiles:

d(0.05) = 257.97um = 1.95phi

d(0.10) = 291.69um = 1.78phi

d(0.16) = 323.95um = 1.63phi

d(0.50) = 479.01um = 1.06phi

d(0.84) = 704.02um = .51phi

d(0.90) = 778.27um = .36phi

d(0.95) = 873.14um = .2phi

Distribution Properties:

Kurtosis = .25

Skewness = .77

Standard Deviation = 190.37 um = 2.39phi

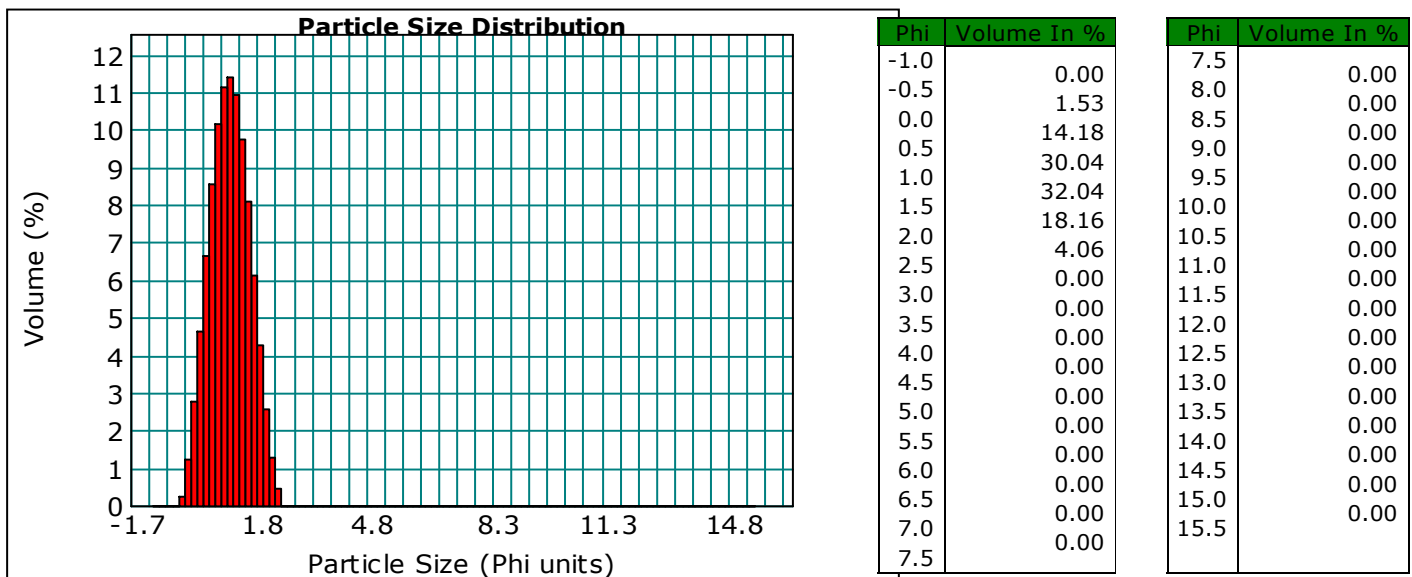
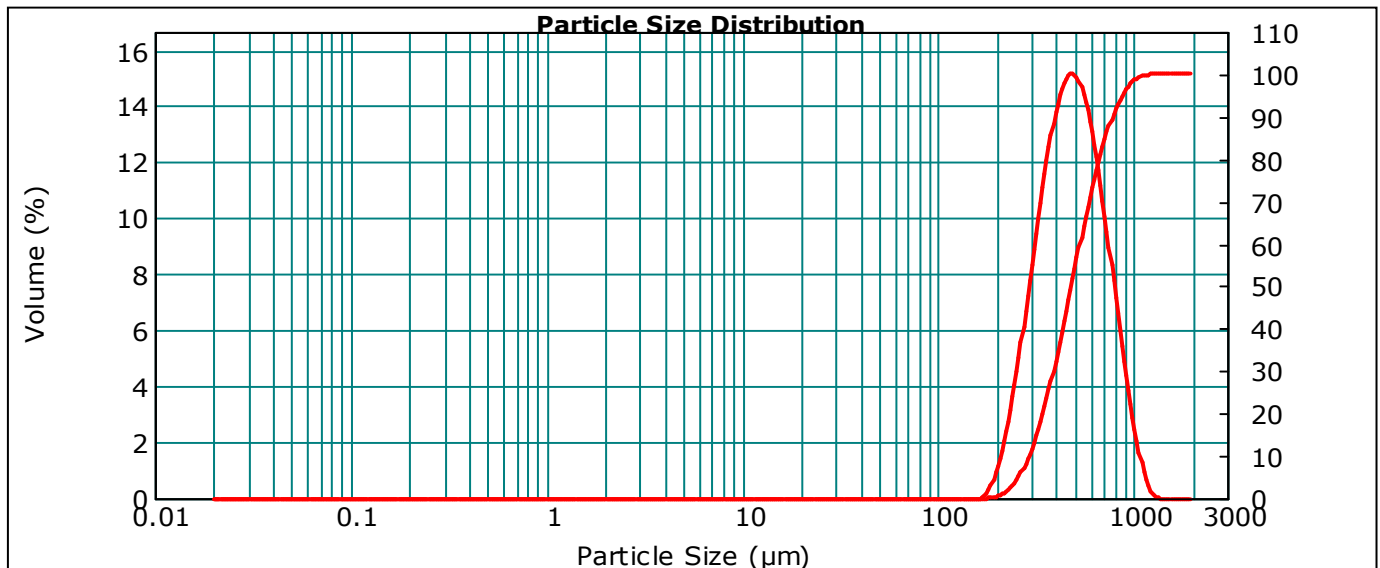
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.06 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:39:59 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 517.39um = .95phi

Surface Weighted Mean:

D[3,2] = 450.48um = 1.15phi

Span :

1.026

Uniformity:

0.317

Specific Surface Area*:

0.0133 m²/g

Percentiles:

d(0.05) = 260.76um = 1.94phi

d(0.10) = 294.61um = 1.76phi

d(0.16) = 326.96um = 1.61phi

d(0.50) = 483.47um = 1.05phi

d(0.84) = 713.56um = .49phi

d(0.90) = 790.48um = .34phi

d(0.95) = 889.63um = .17phi

Distribution Properties:

Kurtosis = .34

Skewness = .8

Standard Deviation = 194.84 um = 2.36phi

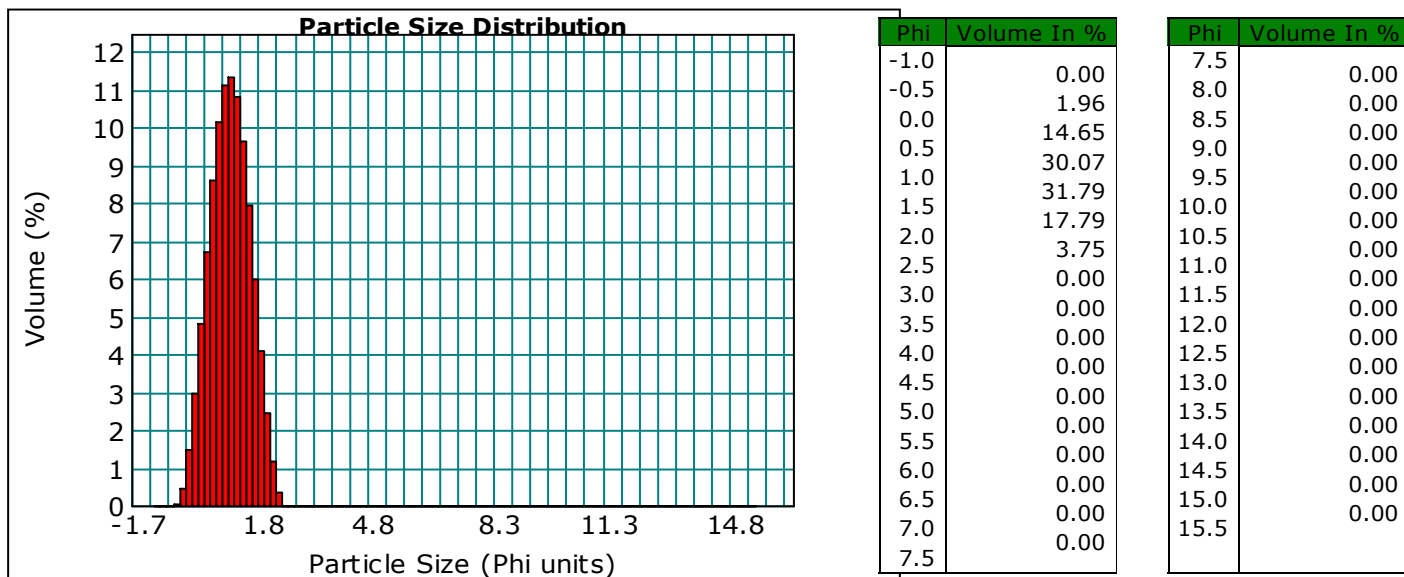
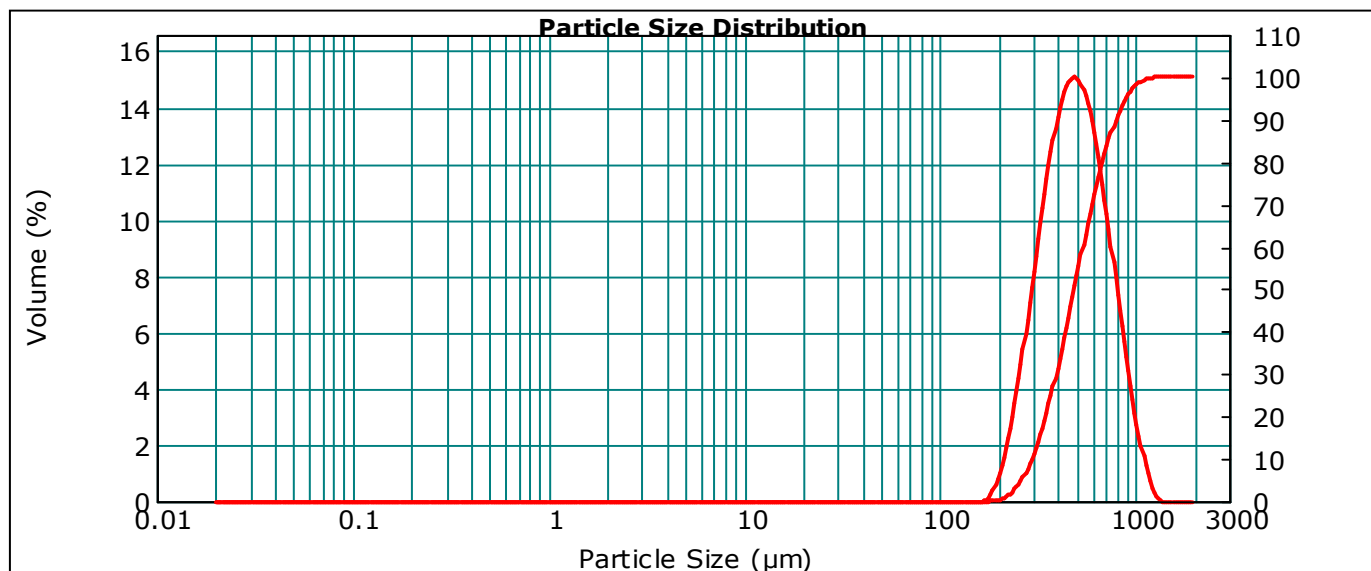
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:41:06 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 516um = .95phi

Surface Weighted Mean:

D[3,2] = 449.85um = 1.15phi

Span :

1.019

Uniformity:

0.314

Specific Surface Area*:

0.0133 m²/g

Percentiles:

d(0.05) = 260.56um = 1.94phi

d(0.10) = 294.44um = 1.76phi

d(0.16) = 326.81um = 1.61phi

d(0.50) = 483.05um = 1.05phi

d(0.84) = 710.94um = .49phi

d(0.90) = 786.45um = .35phi

d(0.95) = 883.6um = .18phi

Distribution Properties:

Kurtosis = .29

Skewness = .78

Standard Deviation = 192.89 um = 2.37phi

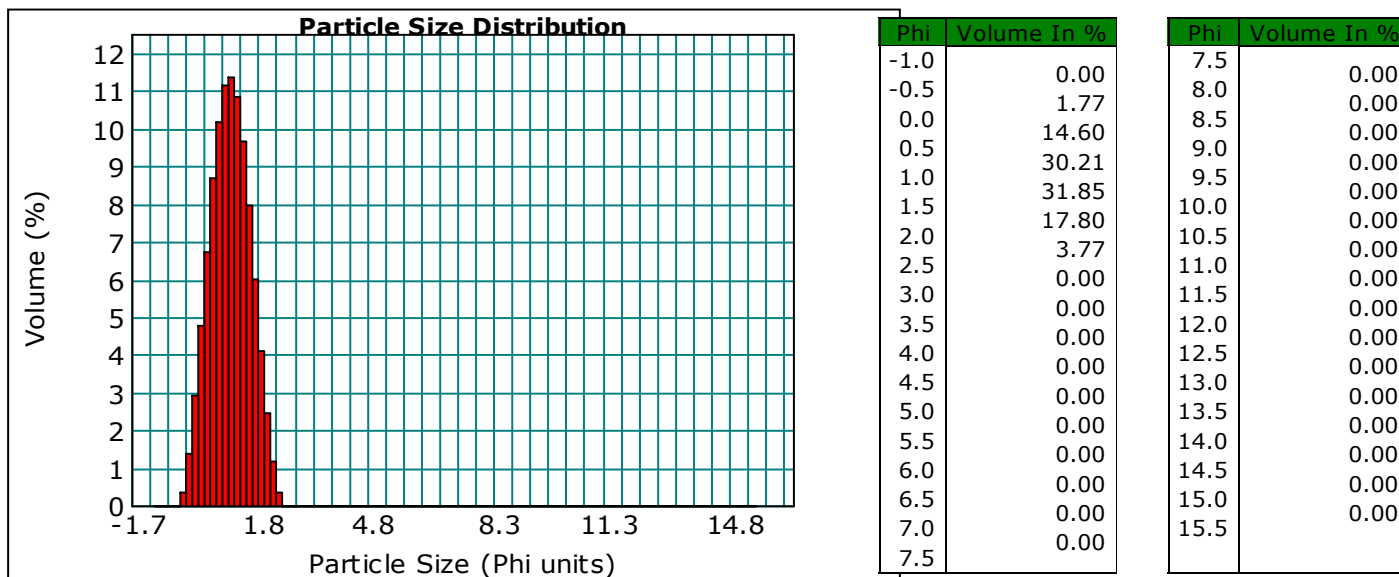
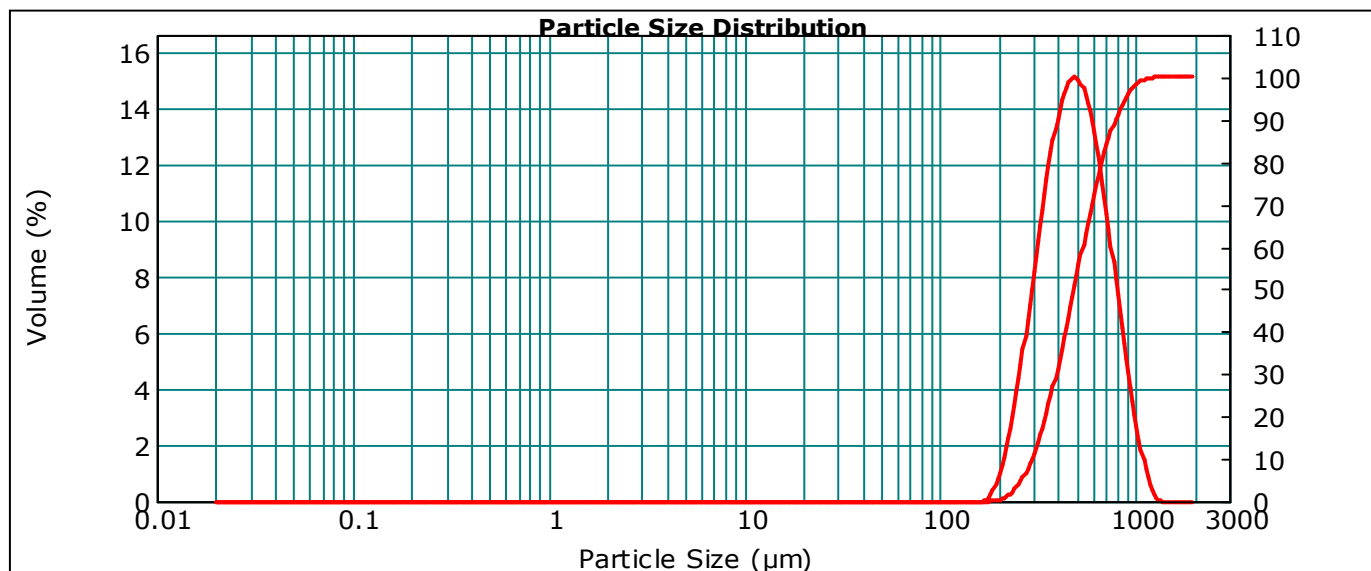
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:42:14 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 512.47um = .96phi

Surface Weighted Mean:

D[3,2] = 447.9um = 1.16phi

Span :

1.009

Uniformity:

0.311

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 260.63um = 1.94phi

d(0.10) = 294.18um = 1.77phi

d(0.16) = 326.18um = 1.62phi

d(0.50) = 480.32um = 1.06phi

d(0.84) = 704.49um = .51phi

d(0.90) = 778.67um = .36phi

d(0.95) = 873.68um = .19phi

Distribution Properties:

Kurtosis = .28

Skewness = .78

Standard Deviation = 189.78 um = 2.4phi

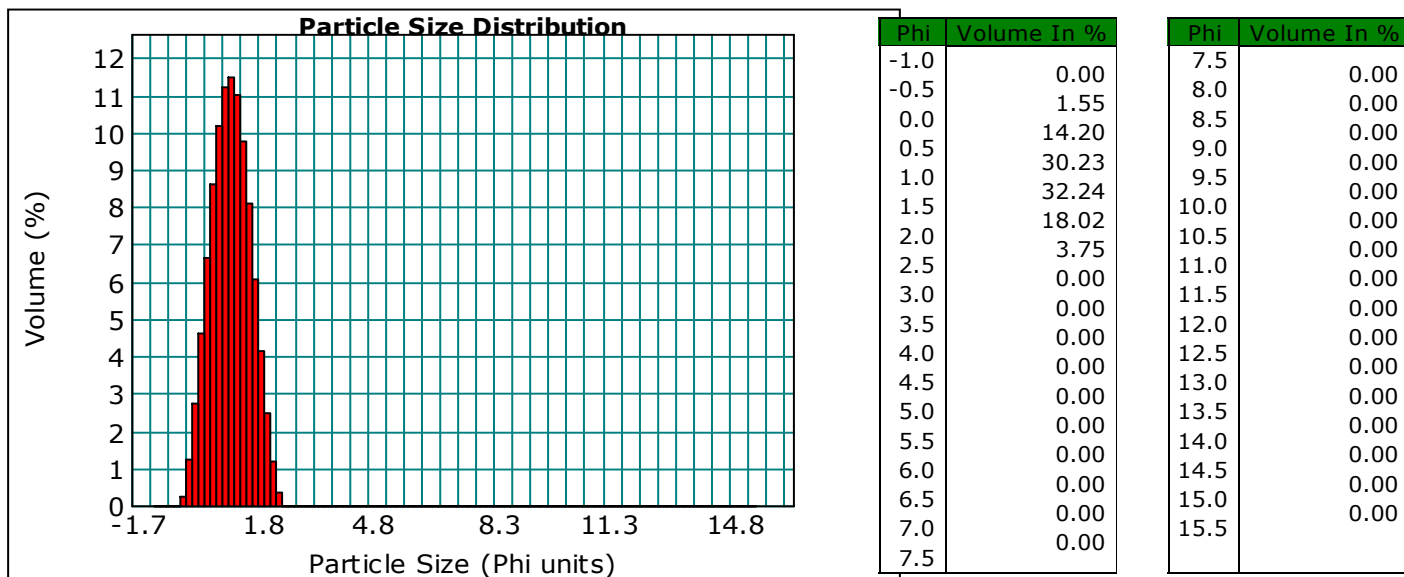
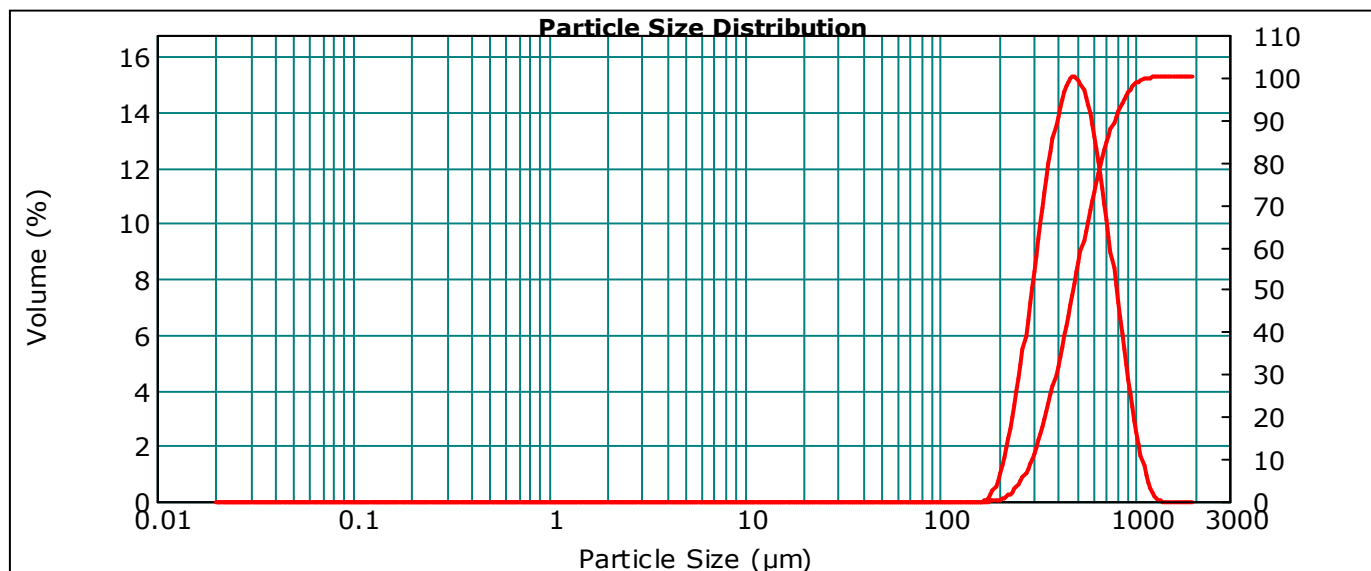
Inclusive SD (Sorting Coeff.) = .54 phi (Moderately Well)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.06 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_6 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:39:59 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 515.29um = .96phi

Surface Weighted Mean:

D[3,2] = 449.41um = 1.15phi

Span :

1.018

Uniformity:

0.314

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 260.65um = 1.94phi

d(0.10) = 294.41um = 1.76phi

d(0.16) = 326.65um = 1.61phi

d(0.50) = 482.27um = 1.05phi

d(0.84) = 709.62um = .49phi

d(0.90) = 785.13um = .35phi

d(0.95) = 882.41um = .18phi

Distribution Properties:

Kurtosis = .3

Skewness = .79

Standard Deviation = 192.53 um = 2.38phi

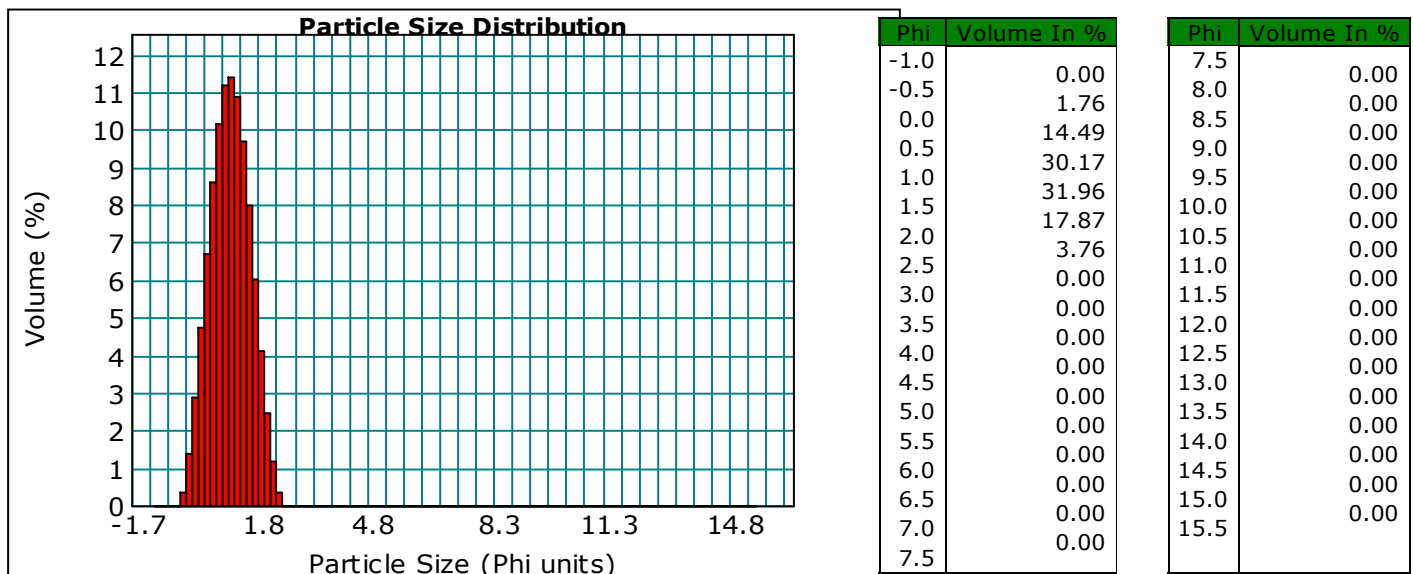
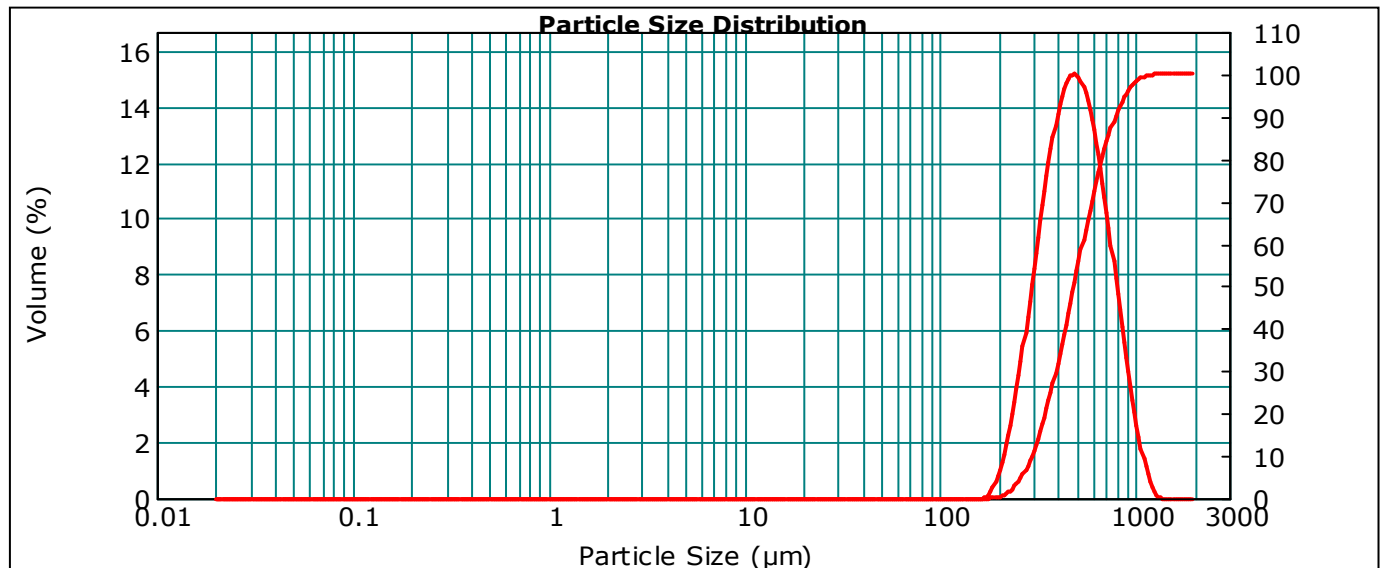
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:51:24 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 526.23um = .93phi

Surface Weighted Mean:

D[3,2] = 456.67um = 1.13phi

Span :

1.040

Uniformity:

0.323

Specific Surface Area*:

0.0131 m²/g

Percentiles:

d(0.05) = 263.11um = 1.93phi

d(0.10) = 297.53um = 1.75phi

d(0.16) = 330.51um = 1.6phi

d(0.50) = 490.51um = 1.03phi

d(0.84) = 727.86um = .46phi

d(0.90) = 807.87um = .31phi

d(0.95) = 911.77um = .13phi

Distribution Properties:

Kurtosis = .35

Skewness = .82

Standard Deviation = 200.81 um = 2.32phi

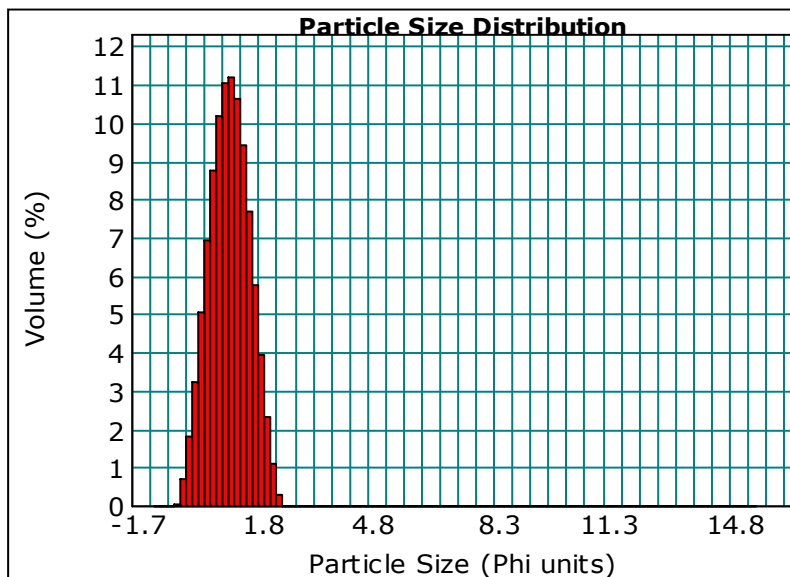
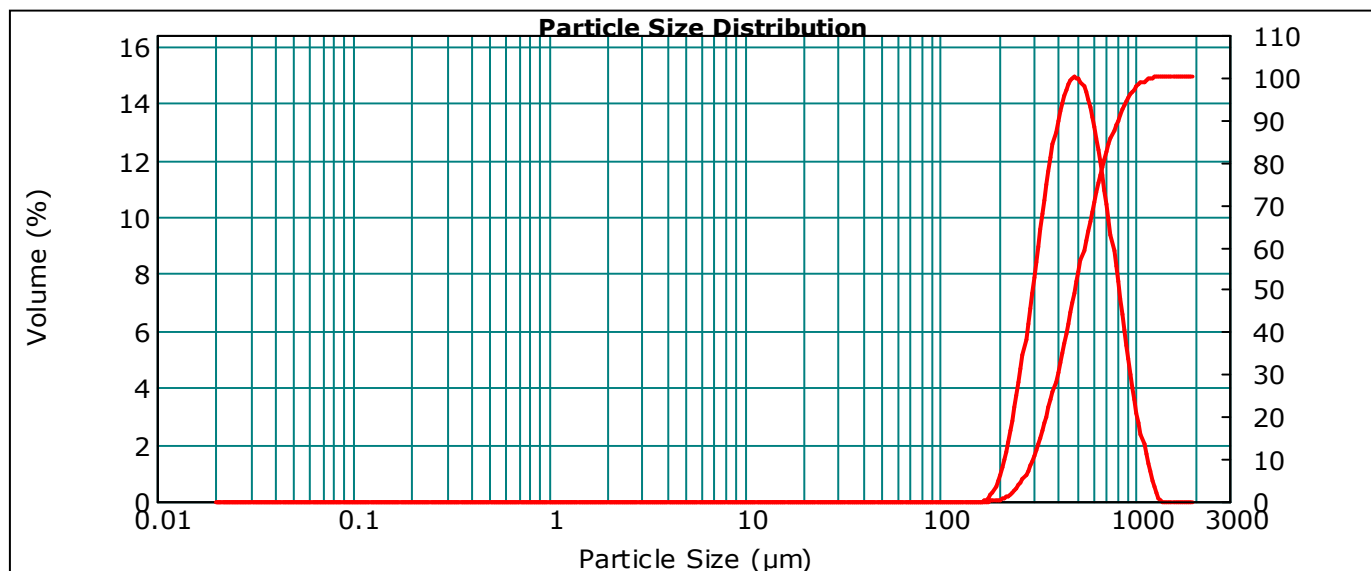
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.52
0.0	15.42
0.5	30.18
1.0	31.20
1.5	17.16
2.0	3.52
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:52:32 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 523.86um = .93phi

Surface Weighted Mean:

D[3,2] = 455.72um = 1.13phi

Span :

1.030

Uniformity:

0.319

Specific Surface Area*:

0.0132 m²/g

Percentiles:

d(0.05) = 263.63um = 1.92phi

d(0.10) = 297.82um = 1.75phi

d(0.16) = 330.56um = 1.6phi

d(0.50) = 489um = 1.03phi

d(0.84) = 722.94um = .47phi

d(0.90) = 801.53um = .32phi

d(0.95) = 903.47um = .15phi

Distribution Properties:

Kurtosis = .35

Skewness = .81

Standard Deviation = 198.13 um = 2.34phi

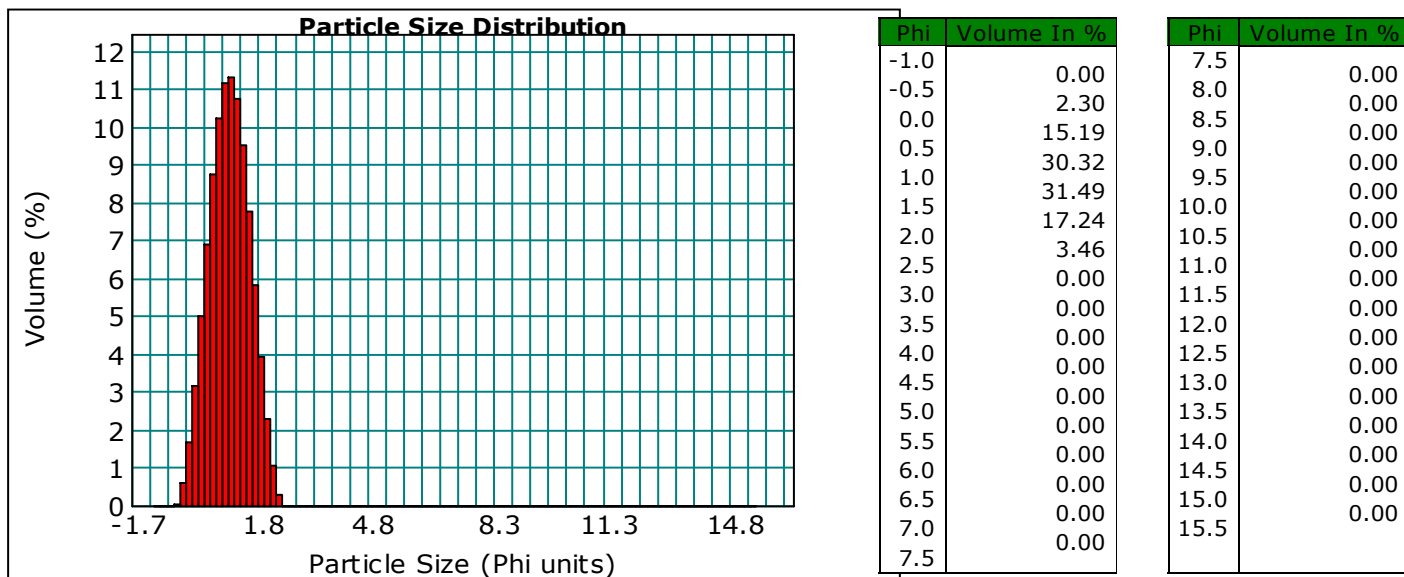
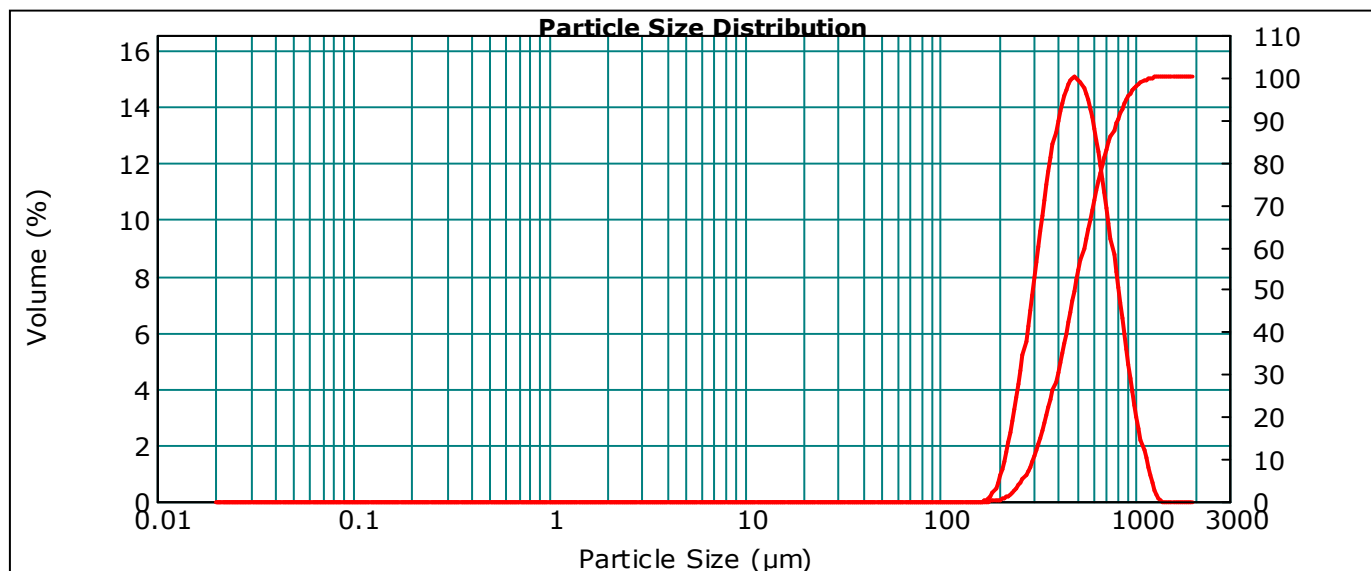
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:53:40 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 526um = .93phi

Surface Weighted Mean:

D[3,2] = 457.2um = 1.13phi

Span :

1.033

Uniformity:

0.32

Specific Surface Area*:

0.0131 m²/g

Percentiles:

d(0.05) = 264um = 1.92phi

d(0.10) = 298.4um = 1.74phi

d(0.16) = 331.37um = 1.59phi

d(0.50) = 490.94um = 1.03phi

d(0.84) = 726.51um = .46phi

d(0.90) = 805.51um = .31phi

d(0.95) = 908.03um = .14phi

Distribution Properties:

Kurtosis = .33

Skewness = .81

Standard Deviation = 199.38 um = 2.33phi

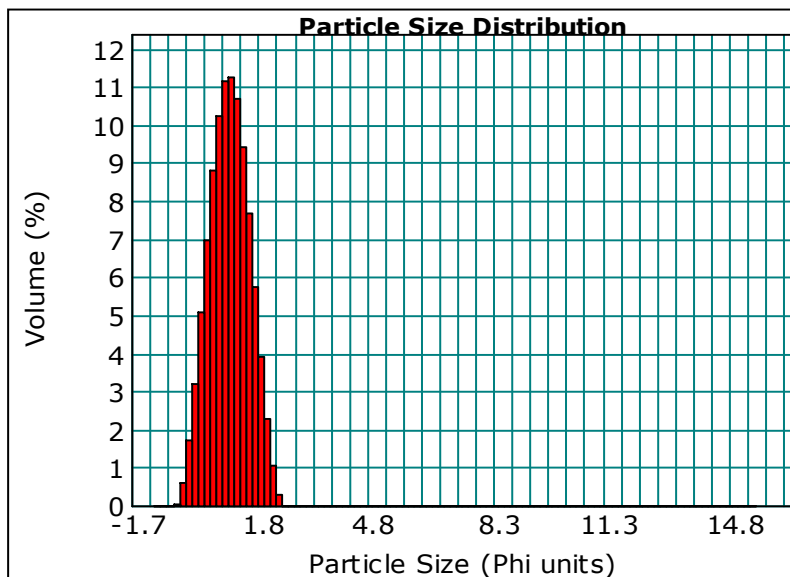
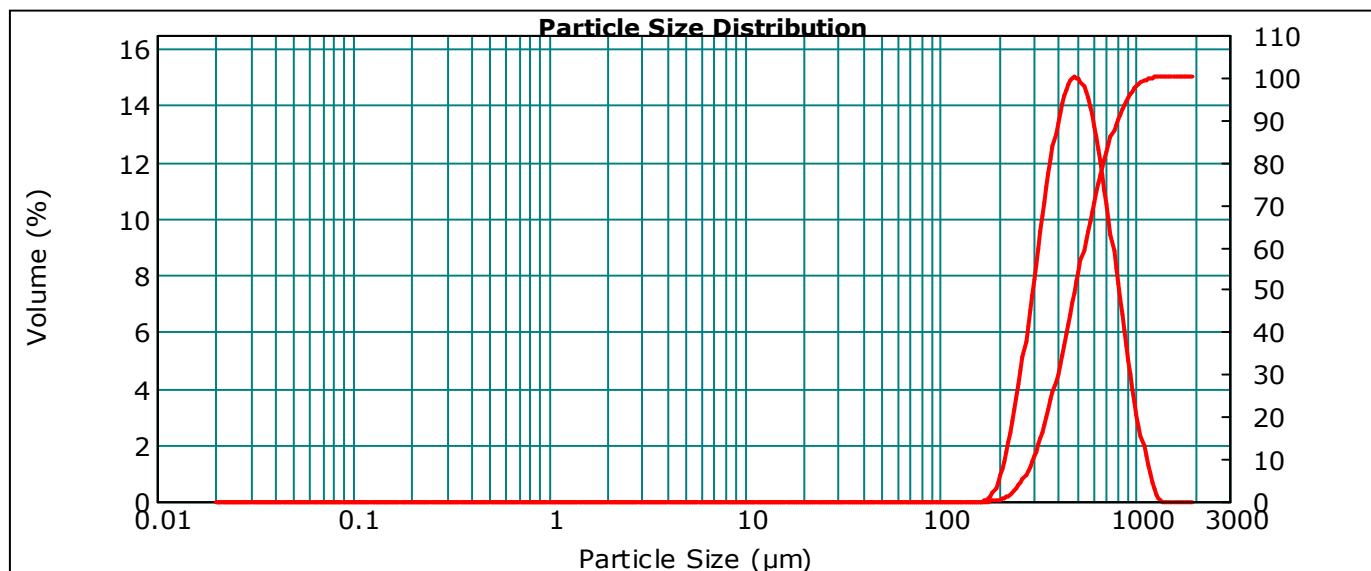
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.40
0.0	15.43
0.5	30.37
1.0	31.29
1.5	17.07
2.0	3.43
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_5 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 3:51:24 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 525.36um = .93phi

Surface Weighted Mean:

D[3,2] = 456.53um = 1.13phi

Span :

1.034

Uniformity:

0.321

Specific Surface Area*:

0.0131 m²/g

Percentiles:

d(0.05) = 263.58um = 1.92phi

d(0.10) = 297.92um = 1.75phi

d(0.16) = 330.81um = 1.6phi

d(0.50) = 490.15um = 1.03phi

d(0.84) = 725.77um = .46phi

d(0.90) = 804.95um = .31phi

d(0.95) = 907.78um = .14phi

Distribution Properties:

Kurtosis = .34

Skewness = .81

Standard Deviation = 199.45 um = 2.33phi

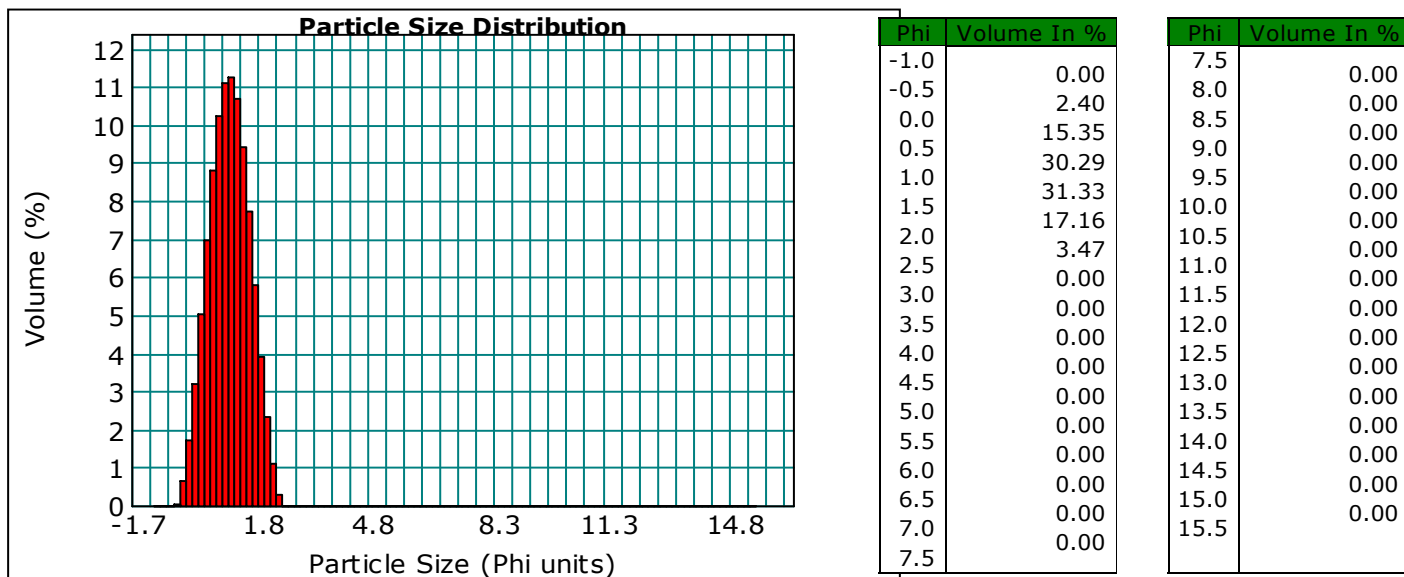
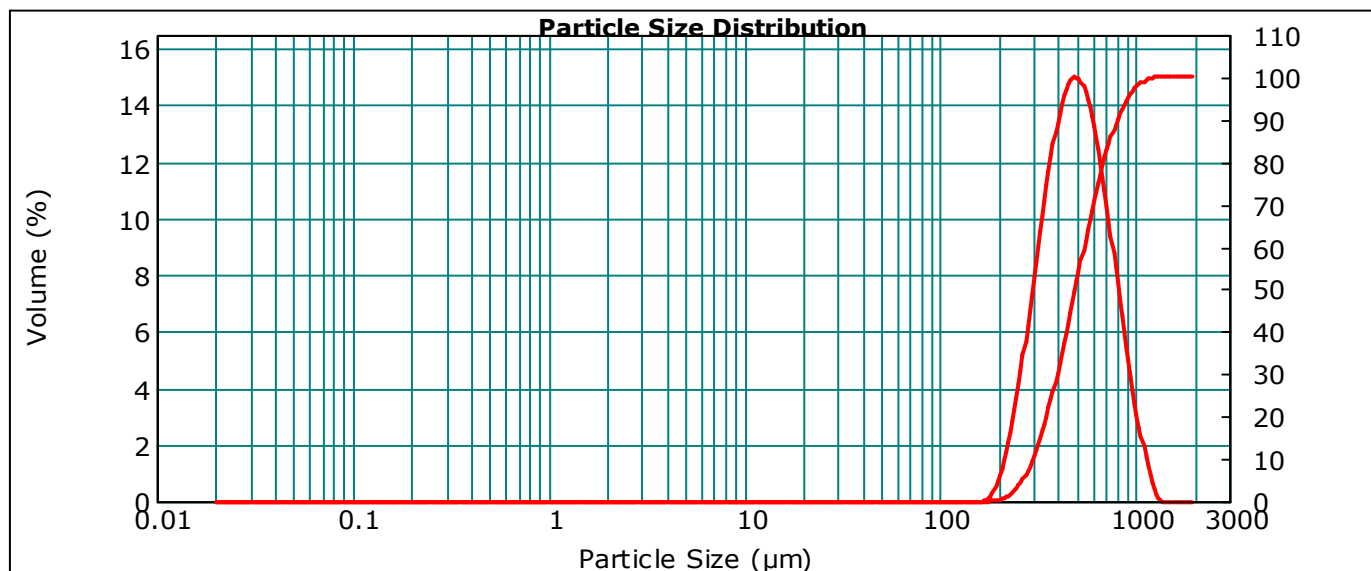
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.03 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:02:45 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 527.53um = .92phi

Surface Weighted Mean:

D[3,2] = 460.01um = 1.12phi

Span :

1.017

Uniformity:

0.316

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 266.55um = 1.91phi

d(0.10) = 301.23um = 1.73phi

d(0.16) = 334.35um = 1.58phi

d(0.50) = 493.96um = 1.02phi

d(0.84) = 726.53um = .46phi

d(0.90) = 803.68um = .32phi

d(0.95) = 903.05um = .15phi

Distribution Properties:

Kurtosis = .28

Skewness = .78

Standard Deviation = 197.01 um = 2.34phi

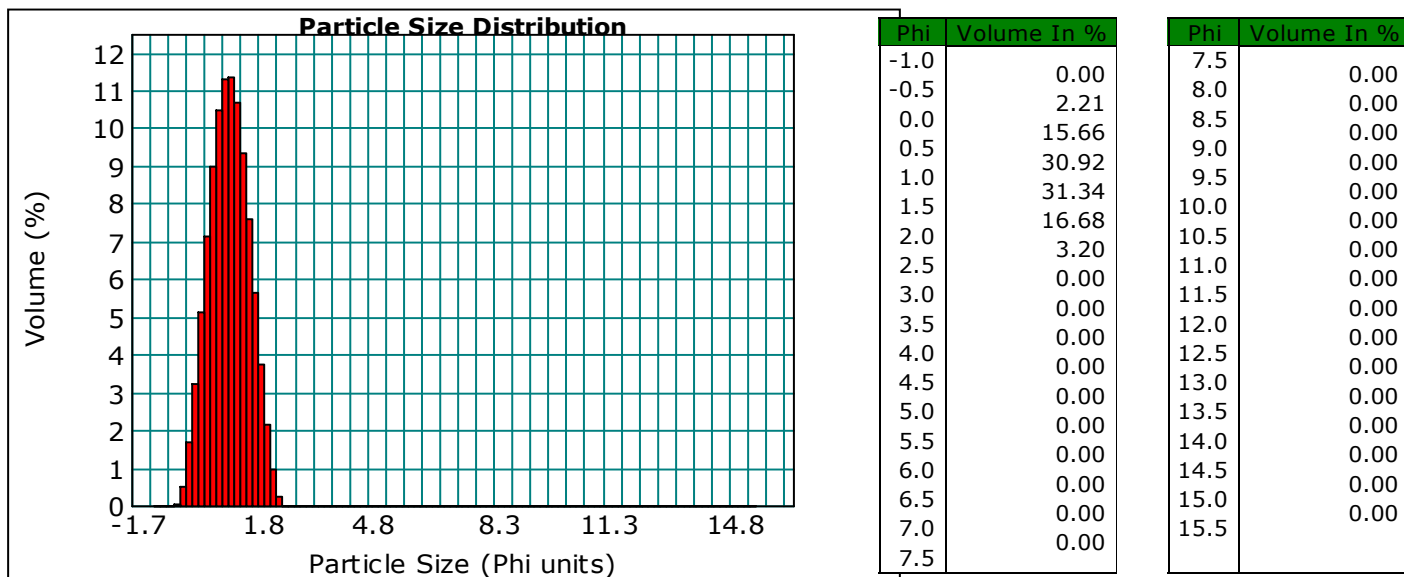
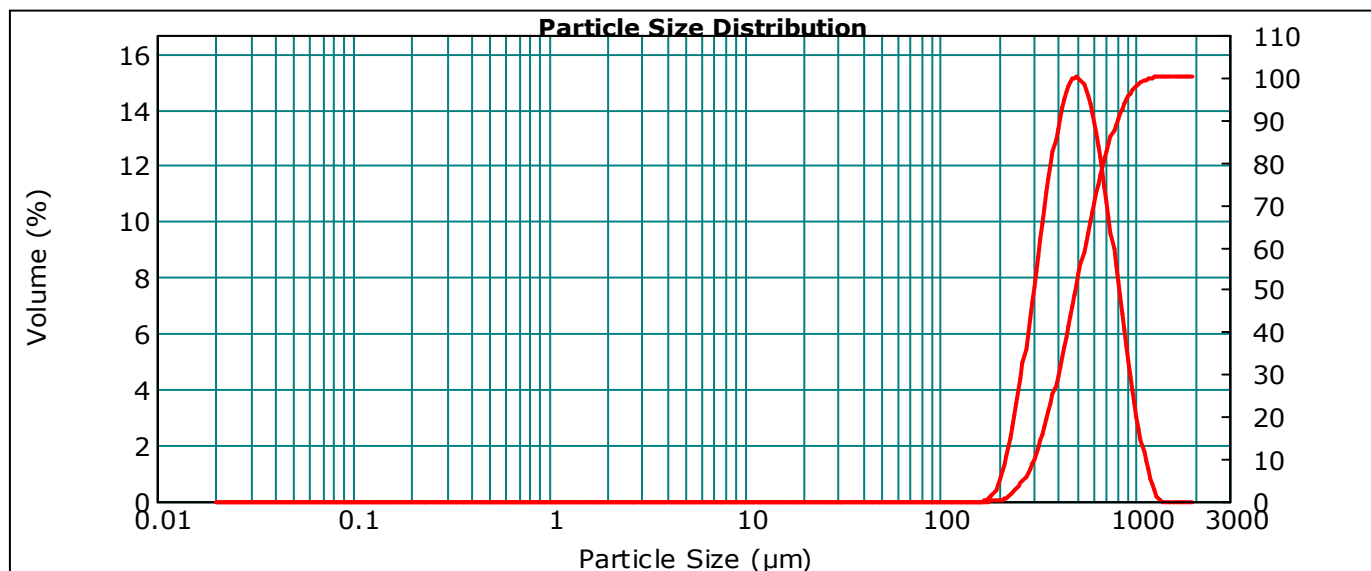
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.02 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:03:52 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 529.81um = .92phi

Surface Weighted Mean:

D[3,2] = 461.26um = 1.12phi

Span :

1.025

Uniformity:

0.319

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 266.74um = 1.91phi

d(0.10) = 301.55um = 1.73phi

d(0.16) = 334.79um = 1.58phi

d(0.50) = 495.4um = 1.01phi

d(0.84) = 730.72um = .45phi

d(0.90) = 809.19um = .31phi

d(0.95) = 910.43um = .14phi

Distribution Properties:

Kurtosis = .29

Skewness = .79

Standard Deviation = 199.23 um = 2.33phi

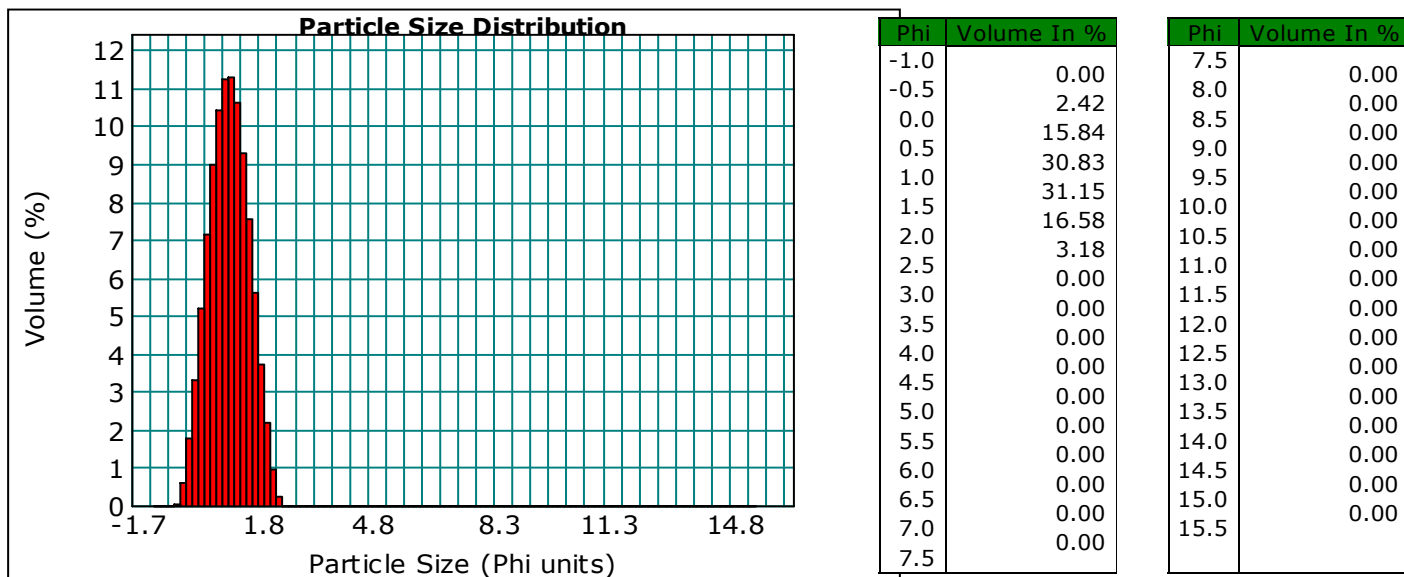
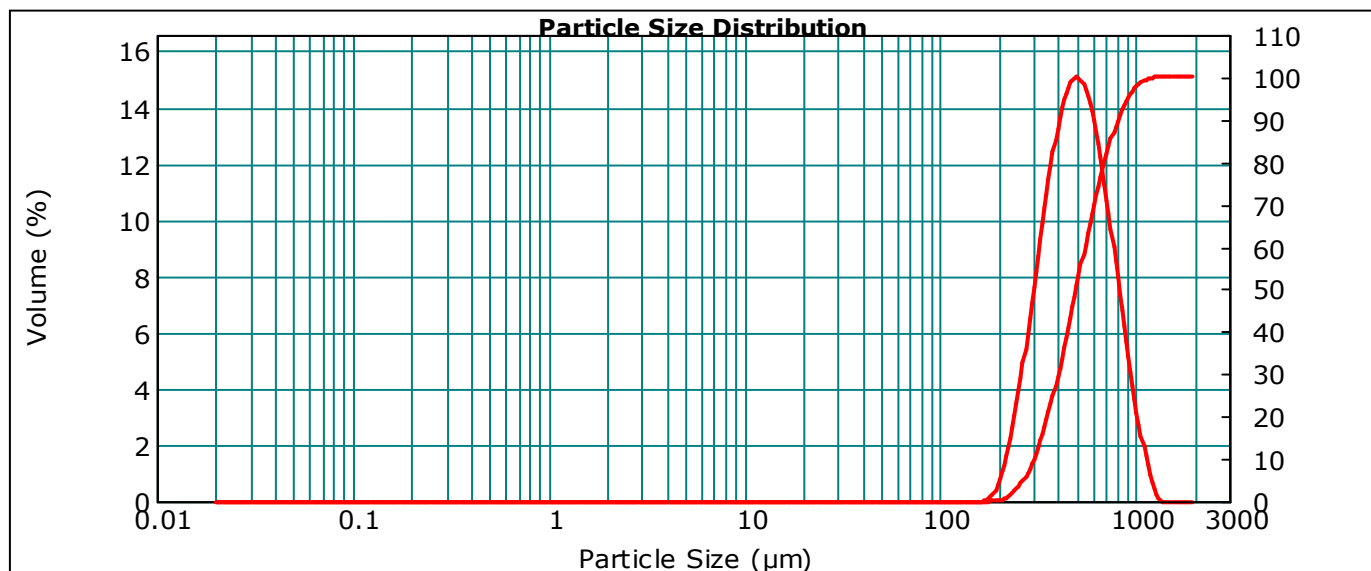
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.01 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:05:00 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 529.86um = .92phi

Surface Weighted Mean:

D[3,2] = 461.29um = 1.12phi

Span :

1.025

Uniformity:

0.319

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 266.66um = 1.91phi

d(0.10) = 301.51um = 1.73phi

d(0.16) = 334.78um = 1.58phi

d(0.50) = 495.55um = 1.01phi

d(0.84) = 730.85um = .45phi

d(0.90) = 809.22um = .31phi

d(0.95) = 910.24um = .14phi

Distribution Properties:

Kurtosis = .28

Skewness = .79

Standard Deviation = 199.19 um = 2.33phi

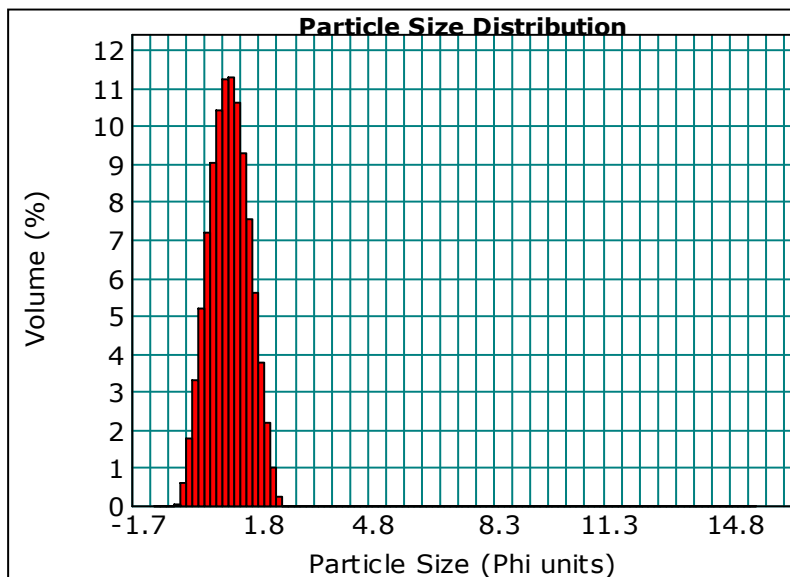
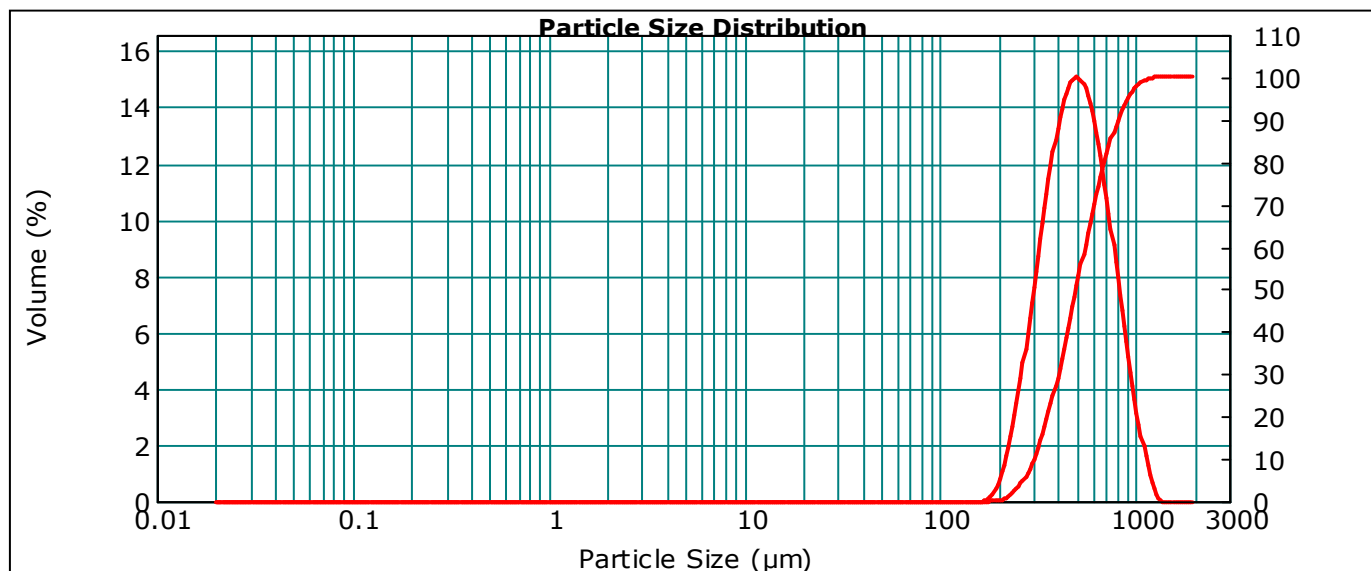
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.01 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.41
0.0	15.86
0.5	30.84
1.0	31.13
1.5	16.57
2.0	3.19
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_4 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:02:45 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 529.07um = .92phi

Surface Weighted Mean:

D[3,2] = 460.85um = 1.12phi

Span :

1.022

Uniformity:

0.318

Specific Surface Area*:

0.013 m²/g

Percentiles:

d(0.05) = 266.65um = 1.91phi

d(0.10) = 301.44um = 1.73phi

d(0.16) = 334.64um = 1.58phi

d(0.50) = 494.97um = 1.01phi

d(0.84) = 729.32um = .46phi

d(0.90) = 807.3um = .31phi

d(0.95) = 907.94um = .14phi

Distribution Properties:

Kurtosis = .28

Skewness = .79

Standard Deviation = 198.48 um = 2.33phi

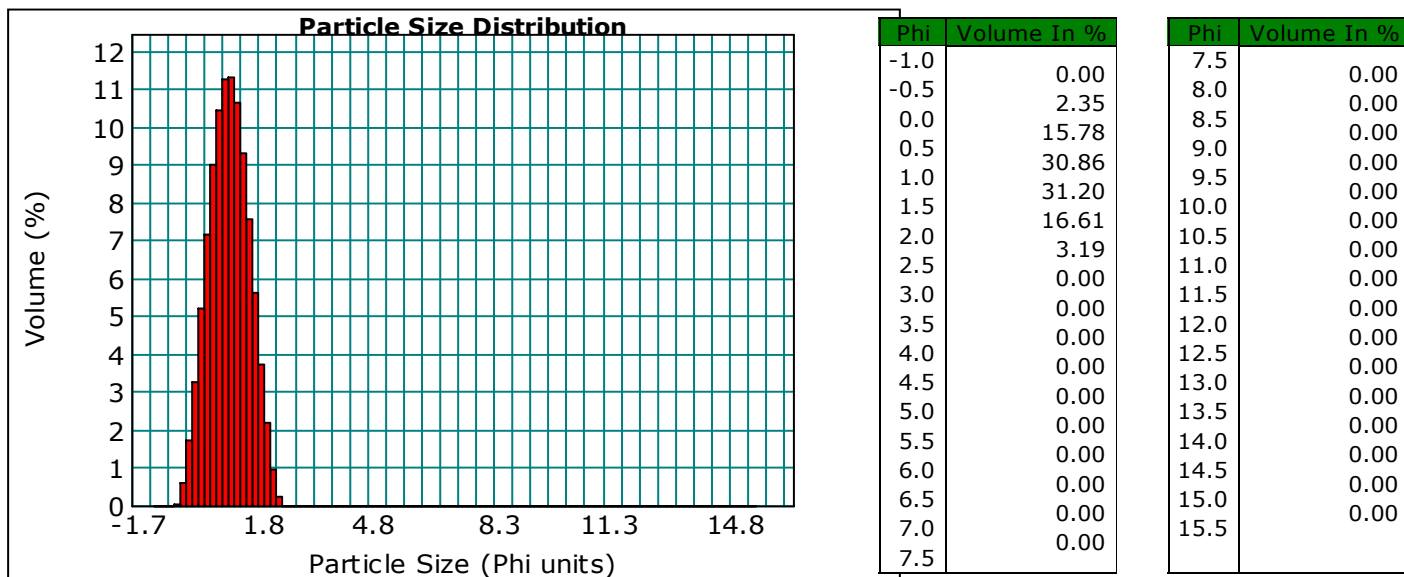
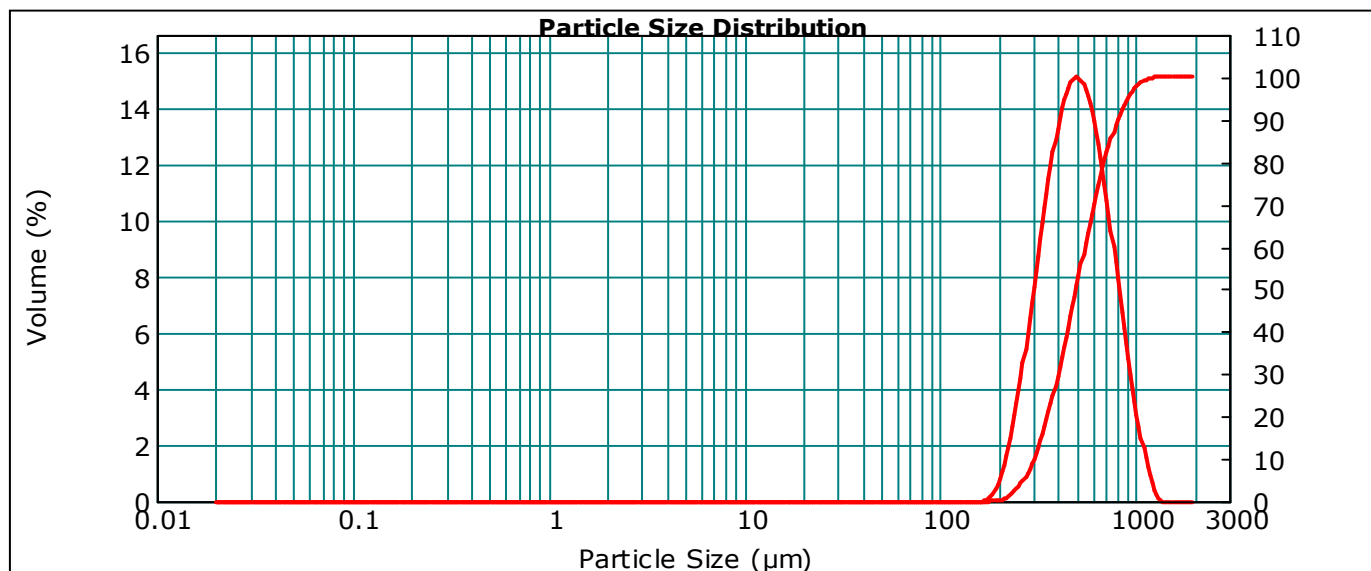
Inclusive SD (Sorting Coeff.) = .55 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.02 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:14:27 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 507.58um = .98phi

Surface Weighted Mean:

D[3,2] = 440.67um = 1.18phi

Span :

1.037

Uniformity:

0.318

Specific Surface Area*:

0.0136 m²/g

Percentiles:

d(0.05) = 253.75um = 1.98phi

d(0.10) = 287.14um = 1.8phi

d(0.16) = 319.06um = 1.65phi

d(0.50) = 473.53um = 1.08phi

d(0.84) = 701.58um = .51phi

d(0.90) = 778.28um = .36phi

d(0.95) = 877.71um = .19phi

Distribution Properties:

Kurtosis = .39

Skewness = .82

Standard Deviation = 193.36 um = 2.37phi

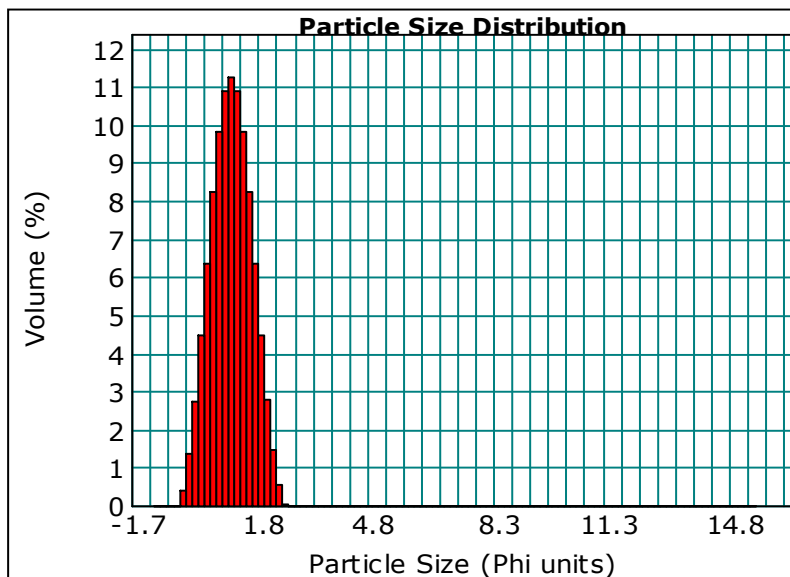
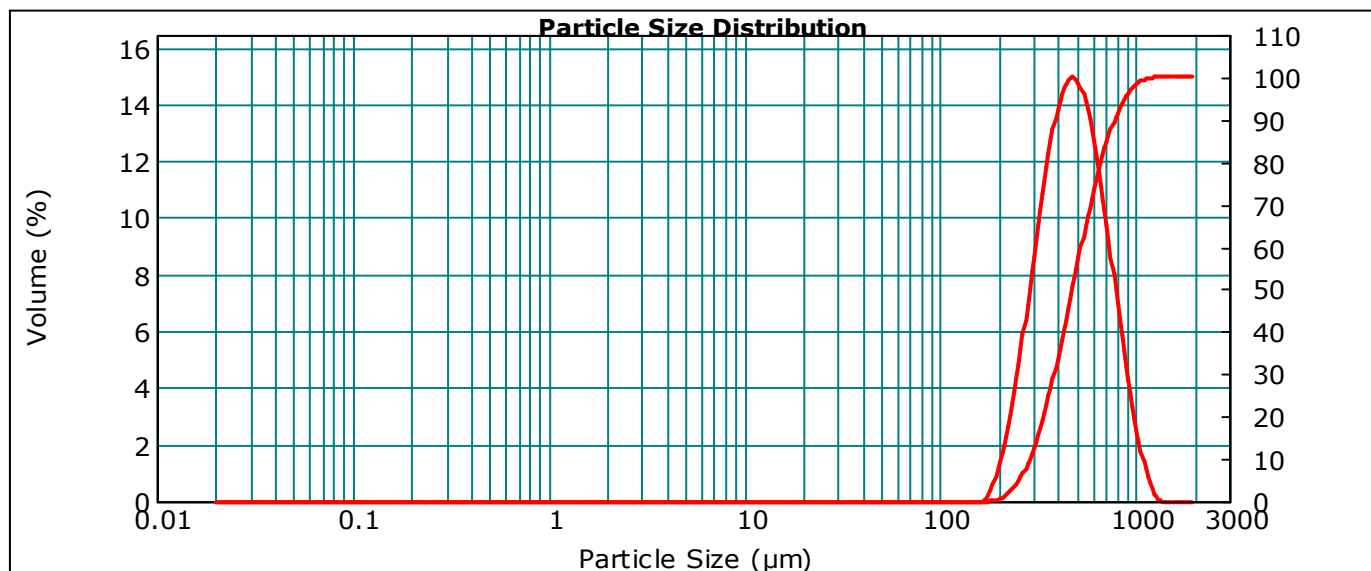
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.08 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	1.75
0.0	13.73
0.5	29.19
1.0	31.98
1.5	18.81
2.0	4.54
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:15:34 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 510.56um = .97phi

Surface Weighted Mean:

D[3,2] = 442.66um = 1.18phi

Span :

1.043

Uniformity:

0.32

Specific Surface Area*:

0.0136 m²/g

Percentiles:

d(0.05) = 254.56um = 1.97phi

d(0.10) = 288.1um = 1.8phi

d(0.16) = 320.19um = 1.64phi

d(0.50) = 475.61um = 1.07phi

d(0.84) = 706.32um = .5phi

d(0.90) = 784.38um = .35phi

d(0.95) = 886.21um = .17phi

Distribution Properties:

Kurtosis = .42

Skewness = .84

Standard Deviation = 195.78 um = 2.35phi

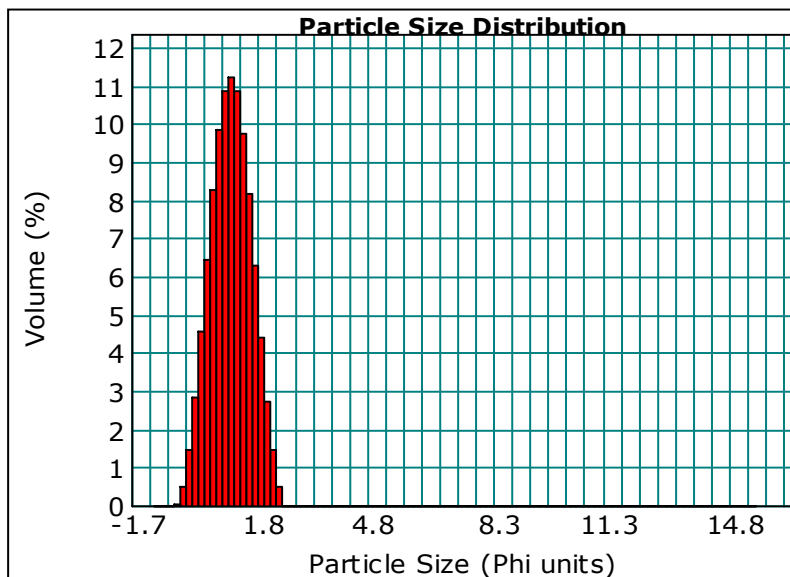
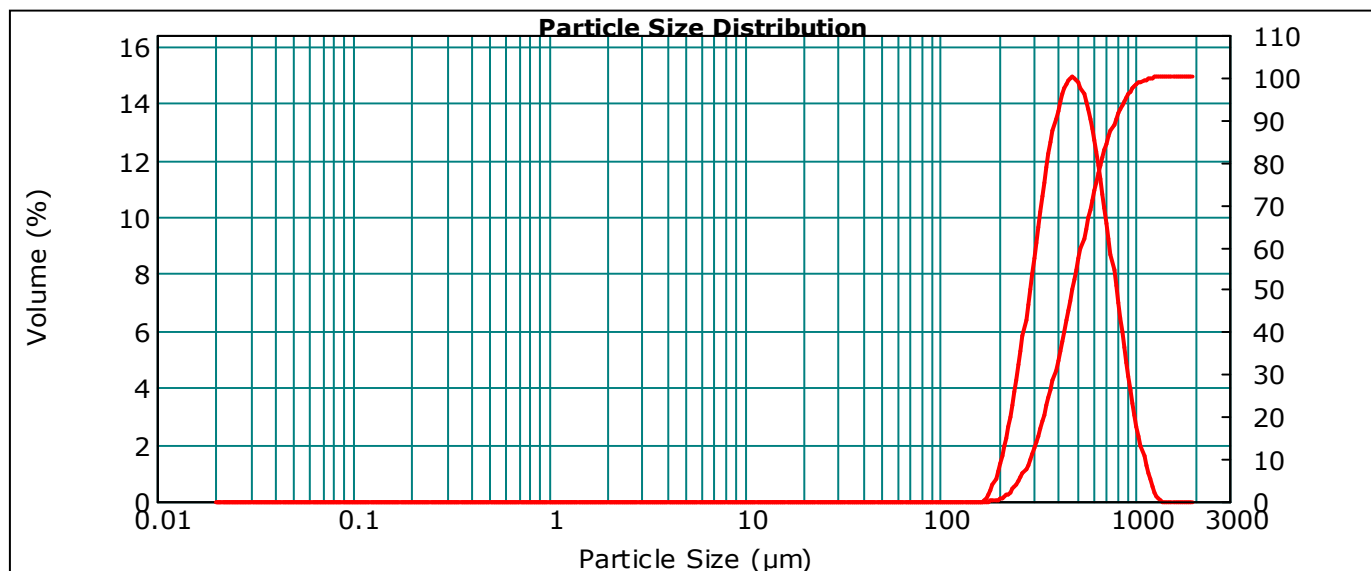
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.07 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	1.97
0.0	13.96
0.5	29.19
1.0	31.82
1.5	18.61
2.0	4.44
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:16:41 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 510.56um = .97phi

Surface Weighted Mean:

D[3,2] = 442.65um = 1.18phi

Span :

1.042

Uniformity:

0.32

Specific Surface Area*:

0.0136 m²/g

Percentiles:

d(0.05) = 254.23um = 1.98phi

d(0.10) = 287.88um = 1.8phi

d(0.16) = 320.08um = 1.64phi

d(0.50) = 476.04um = 1.07phi

d(0.84) = 706.58um = .5phi

d(0.90) = 784.11um = .35phi

d(0.95) = 884.82um = .18phi

Distribution Properties:

Kurtosis = .38

Skewness = .82

Standard Deviation = 195.39 um = 2.36phi

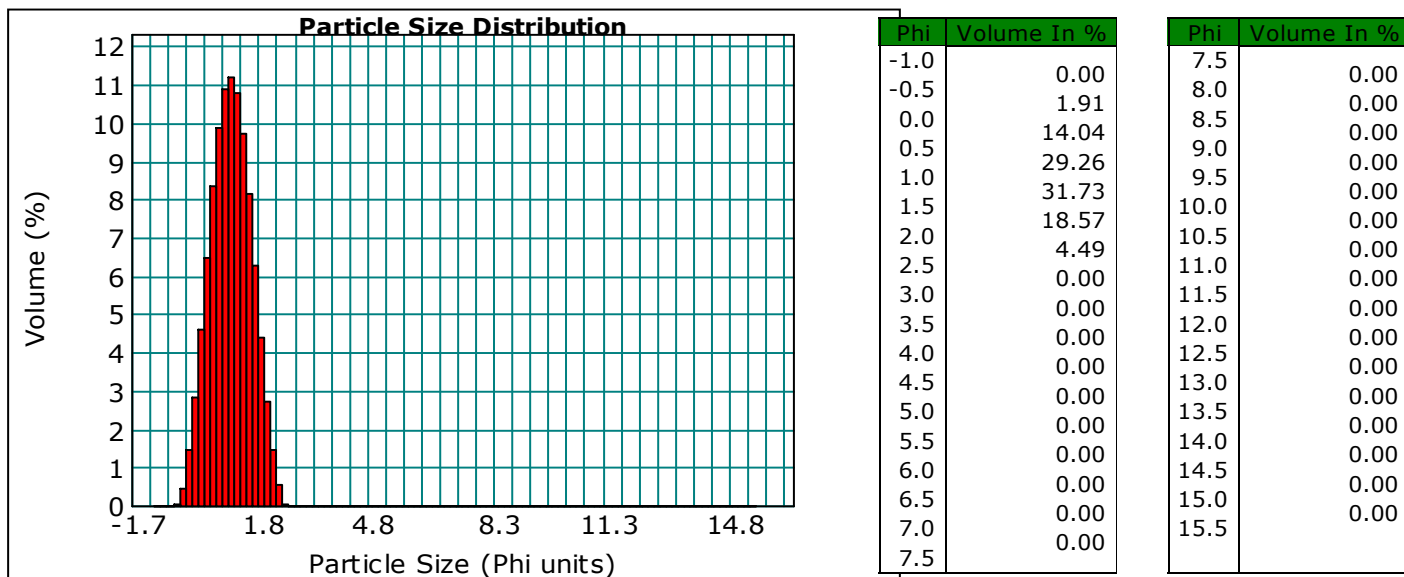
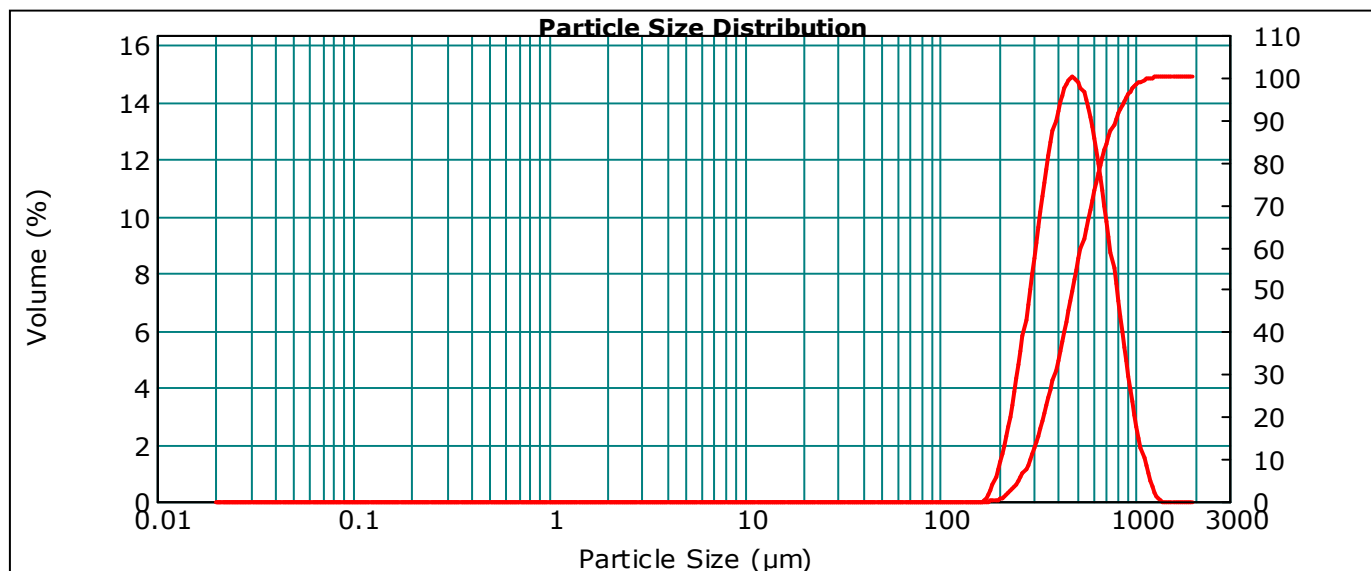
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.07 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_3 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:14:27 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 509.56um = .97phi

Surface Weighted Mean:

D[3,2] = 441.99um = 1.18phi

Span :

1.041

Uniformity:

0.319

Specific Surface Area*:

0.0136 m²/g

Percentiles:

d(0.05) = 254.18um = 1.98phi

d(0.10) = 287.71um = 1.8phi

d(0.16) = 319.77um = 1.64phi

d(0.50) = 475.06um = 1.07phi

d(0.84) = 704.83um = .5phi

d(0.90) = 782.27um = .35phi

d(0.95) = 882.95um = .18phi

Distribution Properties:

Kurtosis = .4

Skewness = .83

Standard Deviation = 194.85 um = 2.36phi

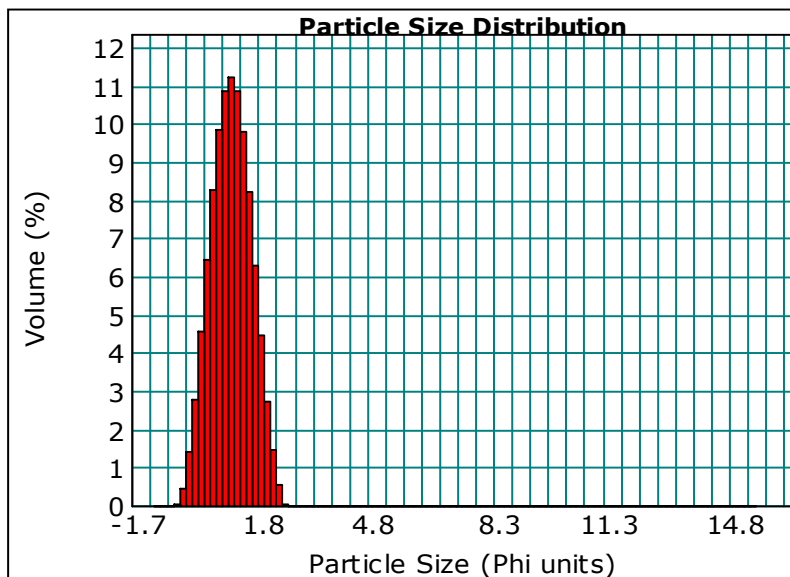
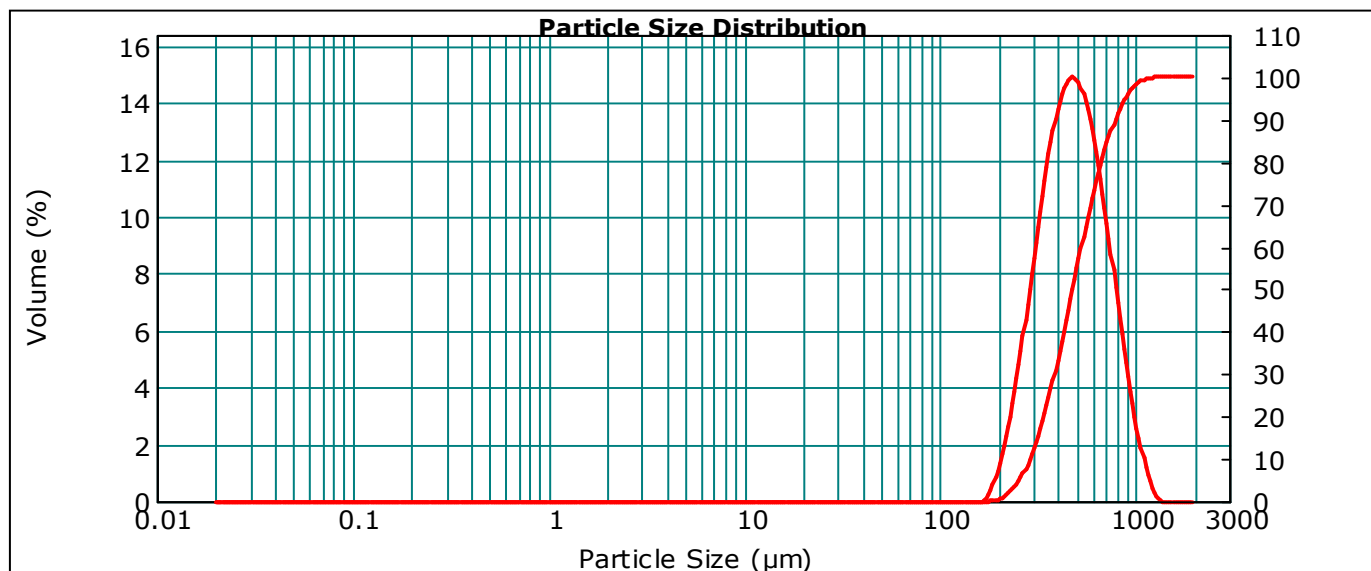
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.07 phi (Medium Sand)

Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	1.88
0.0	13.91
0.5	29.21
1.0	31.84
1.5	18.66
2.0	4.49
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:25:45 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 495.61um = 1.01phi

Surface Weighted Mean:

D[3,2] = 430.26um = 1.22phi

Span :

1.036

Uniformity:

0.32

Specific Surface Area*:

0.0139 m²/g

Percentiles:

d(0.05) = 247.41um = 2.02phi

d(0.10) = 280.23um = 1.84phi

d(0.16) = 311.55um = 1.68phi

d(0.50) = 462.89um = 1.11phi

d(0.84) = 685.21um = .55phi

d(0.90) = 759.57um = .4phi

d(0.95) = 855.52um = .23phi

Distribution Properties:

Kurtosis = .35

Skewness = .8

Standard Deviation = 188.32 um = 2.41phi

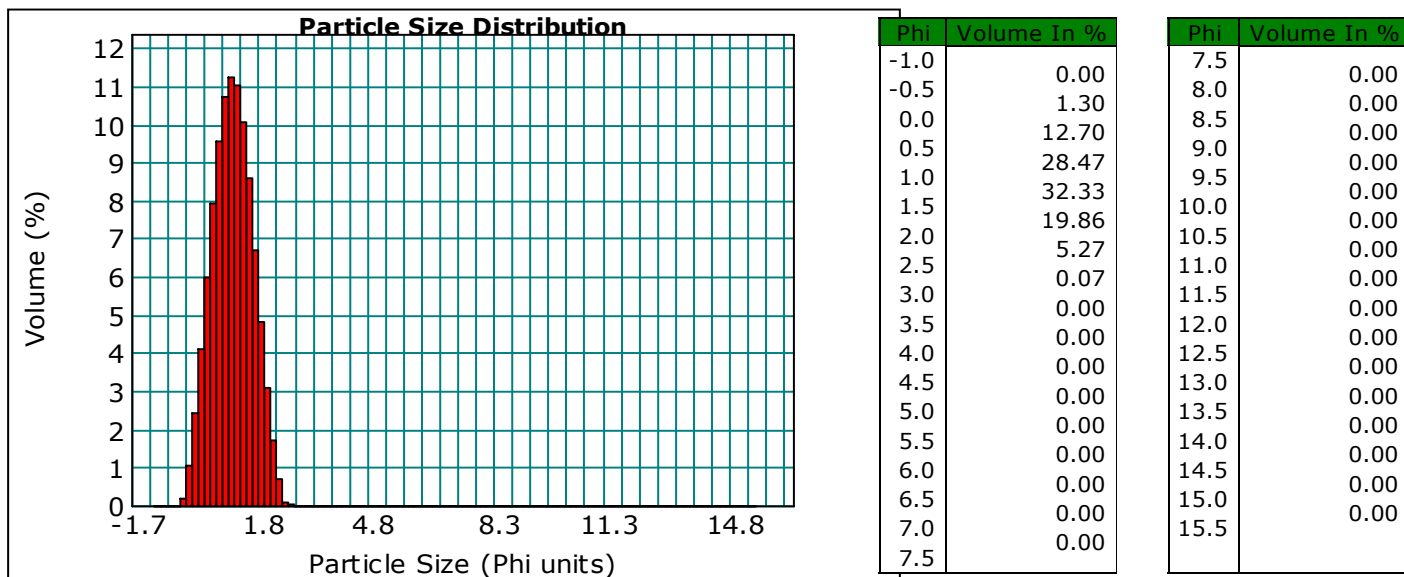
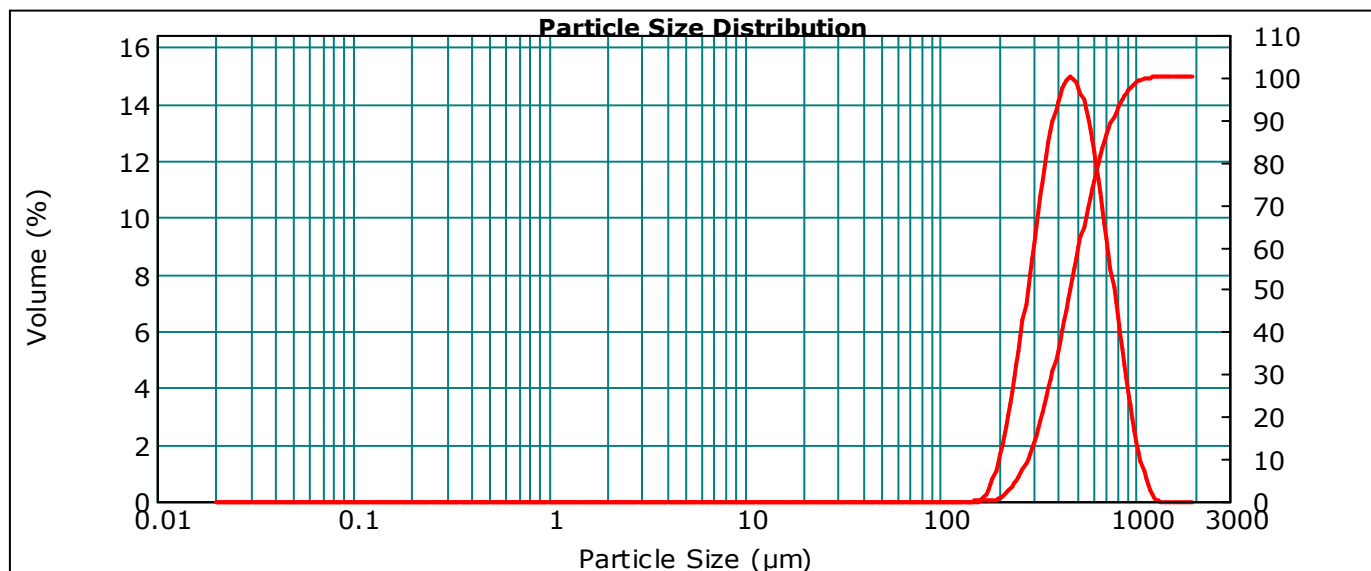
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.11 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:26:52 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 496.97um = 1.01phi

Surface Weighted Mean:

D[3,2] = 430.88um = 1.21phi

Span :

1.040

Uniformity:

0.322

Specific Surface Area*:

0.0139 m²/g

Percentiles:

d(0.05) = 247.19um = 2.02phi

d(0.10) = 280.18um = 1.84phi

d(0.16) = 311.67um = 1.68phi

d(0.50) = 463.93um = 1.11phi

d(0.84) = 687.83um = .54phi

d(0.90) = 762.76um = .39phi

d(0.95) = 859.62um = .22phi

Distribution Properties:

Kurtosis = .36

Skewness = .81

Standard Deviation = 189.69 um = 2.4phi

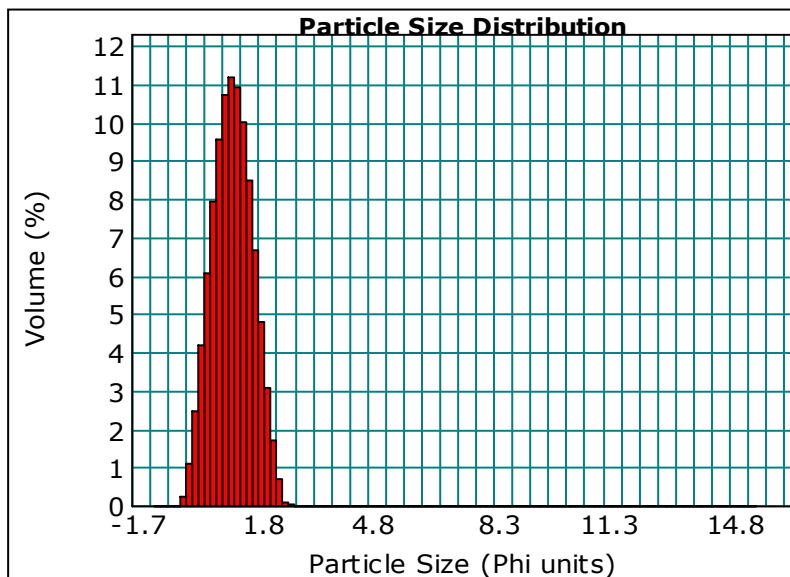
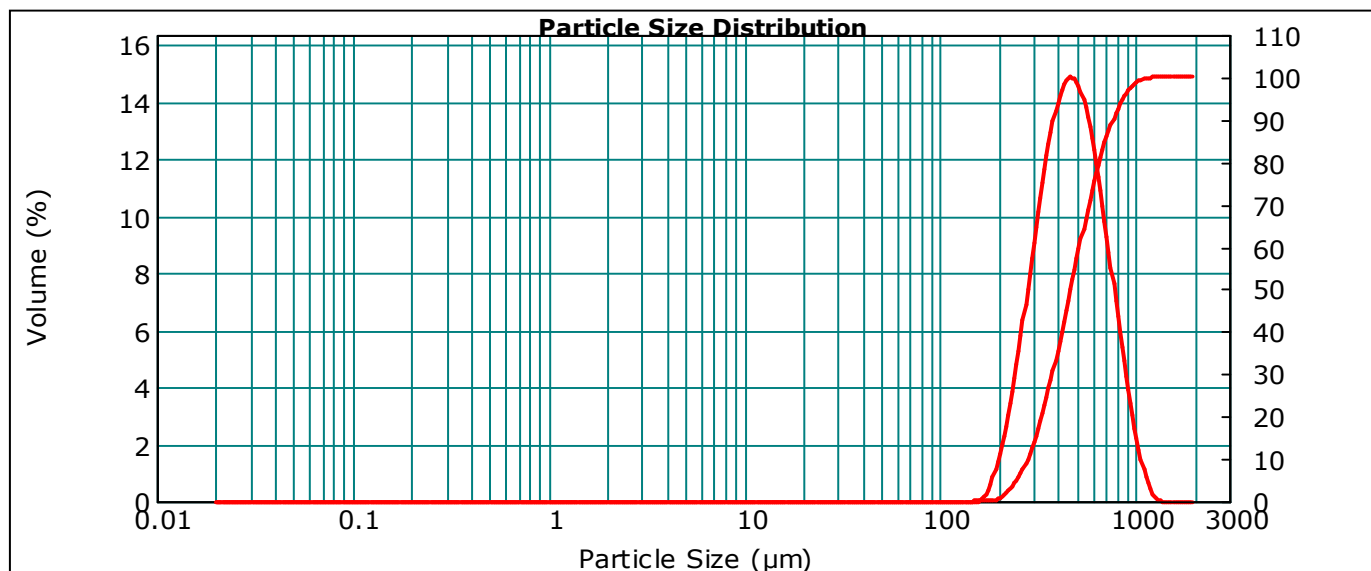
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.11 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	1.38
0.0	12.86
0.5	28.47
1.0	32.16
1.5	19.75
2.0	5.28
2.5	0.09
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_1_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:28:00 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 497.48um = 1.01phi

Surface Weighted Mean:

D[3,2] = 431.98um = 1.21phi

Span :

1.034

Uniformity:

0.319

Specific Surface Area*:

0.0139 m²/g

Percentiles:

d(0.05) = 248.29um = 2.01phi

d(0.10) = 281.3um = 1.83phi

d(0.16) = 312.82um = 1.68phi

d(0.50) = 464.92um = 1.1phi

d(0.84) = 687.71um = .54phi

d(0.90) = 762.01um = .39phi

d(0.95) = 857.82um = .22phi

Distribution Properties:

Kurtosis = .33

Skewness = .8

Standard Deviation = 188.71 um = 2.41phi

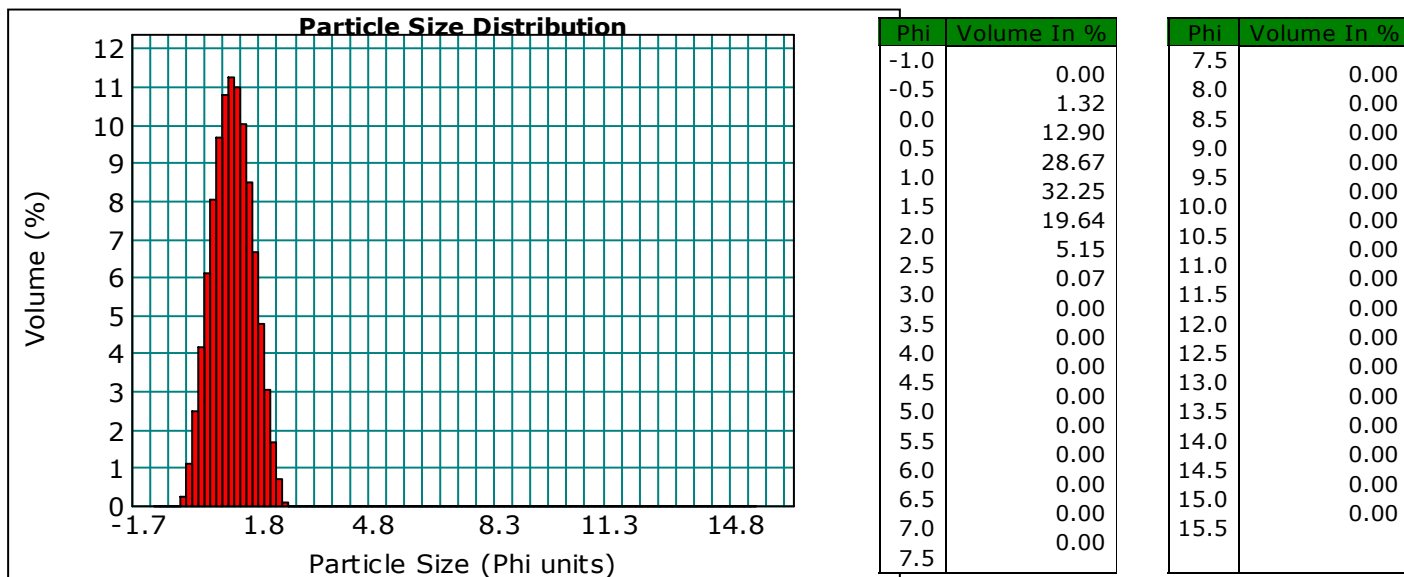
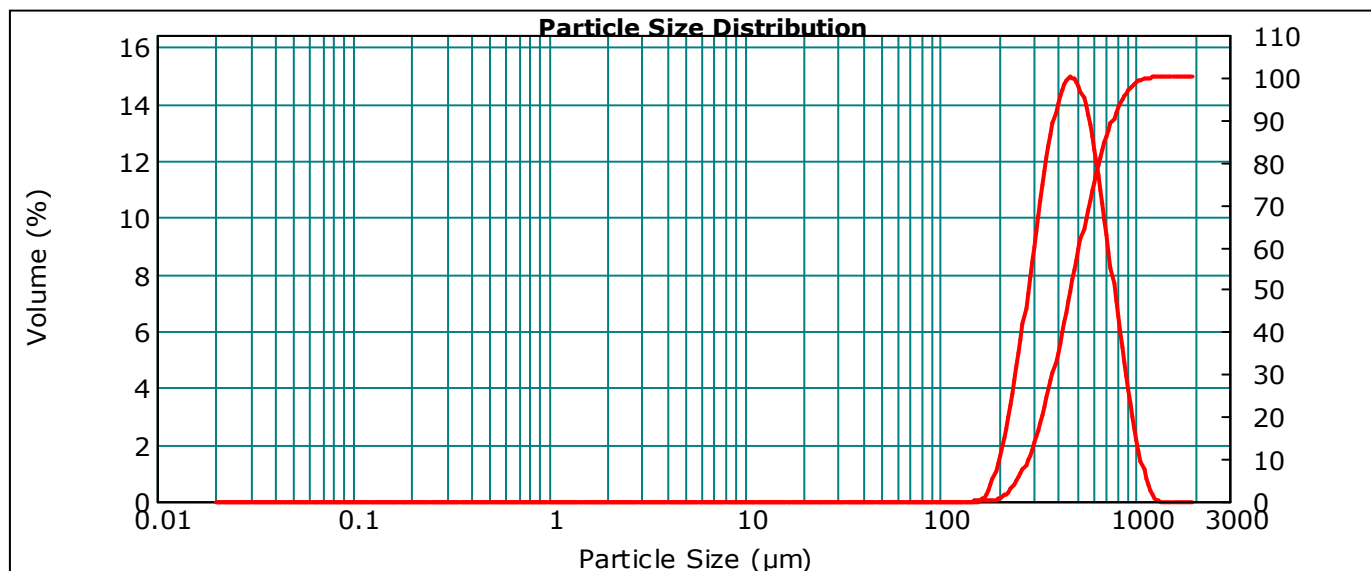
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.11 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_2 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:25:45 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 496.69um = 1.01phi

Surface Weighted Mean:

D[3,2] = 431.04um = 1.21phi

Span :

1.037

Uniformity:

0.32

Specific Surface Area*:

0.0139 m²/g

Percentiles:

d(0.05) = 247.63um = 2.01phi

d(0.10) = 280.57um = 1.83phi

d(0.16) = 312.01um = 1.68phi

d(0.50) = 463.91um = 1.11phi

d(0.84) = 686.92um = .54phi

d(0.90) = 761.45um = .39phi

d(0.95) = 857.66um = .22phi

Distribution Properties:

Kurtosis = .35

Skewness = .8

Standard Deviation = 188.91 um = 2.4phi

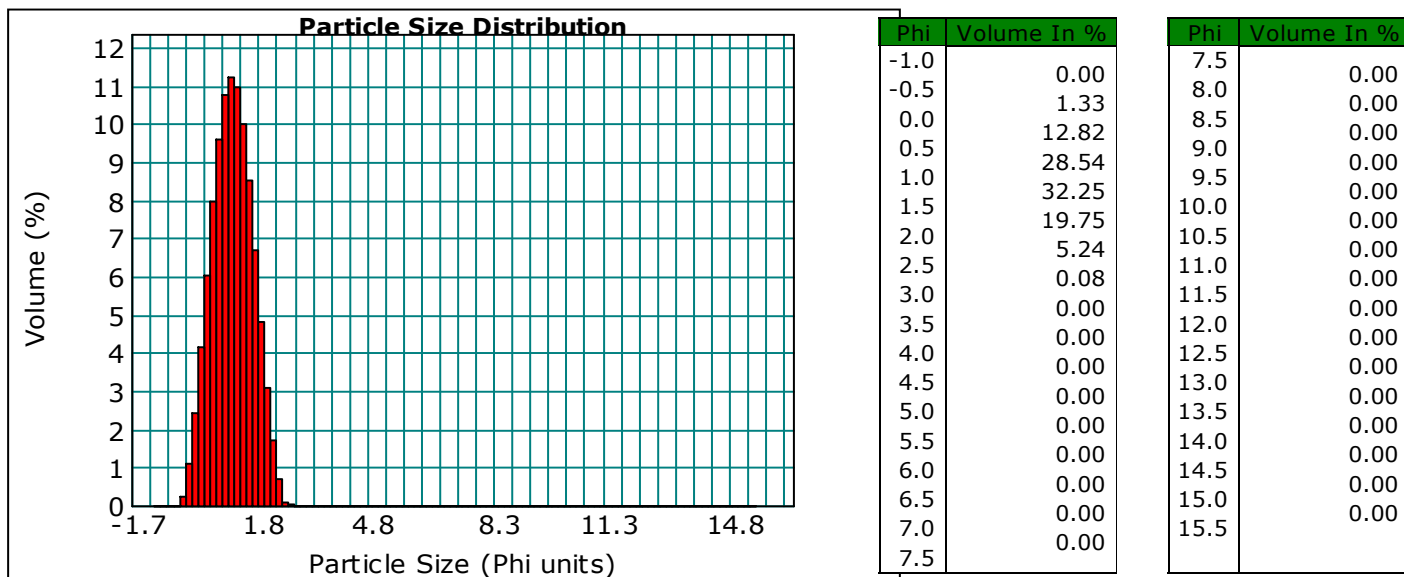
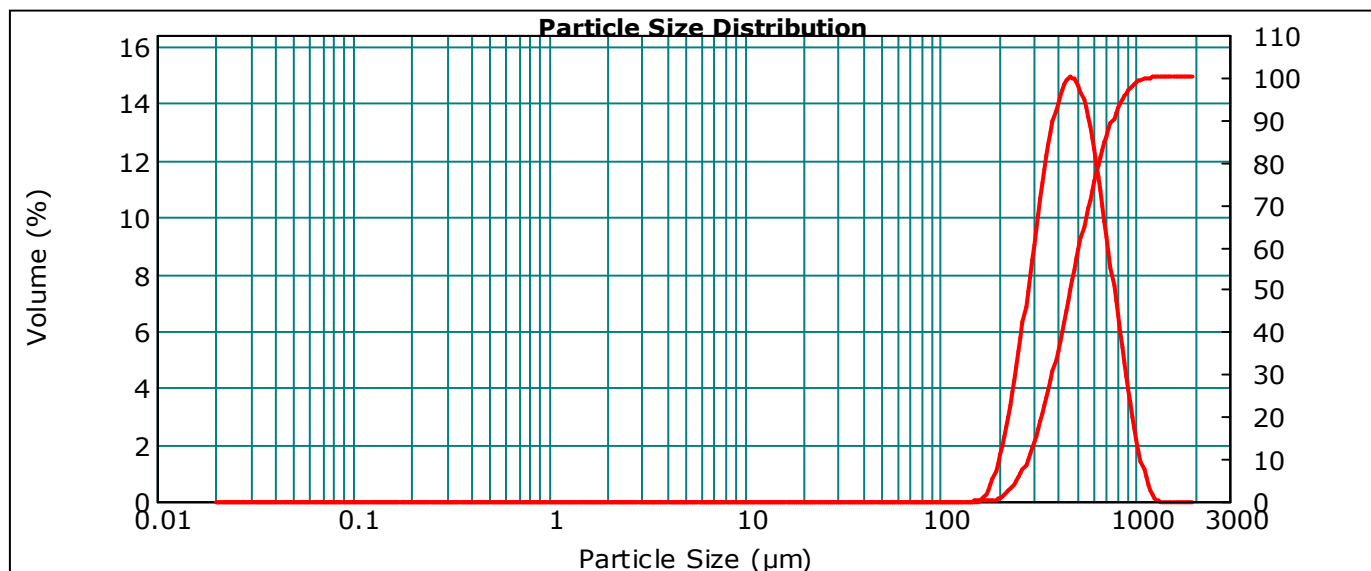
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.11 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:37:27 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 516.16um = .95phi

Surface Weighted Mean:

D[3,2] = 447.4um = 1.16phi

Span :

1.041

Uniformity:

0.321

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 256.16um = 1.96phi

d(0.10) = 290.52um = 1.78phi

d(0.16) = 323.41um = 1.63phi

d(0.50) = 482.08um = 1.05phi

d(0.84) = 714.74um = .48phi

d(0.90) = 792.36um = .34phi

d(0.95) = 892.14um = .16phi

Distribution Properties:

Kurtosis = .32

Skewness = .8

Standard Deviation = 196.95 um = 2.34phi

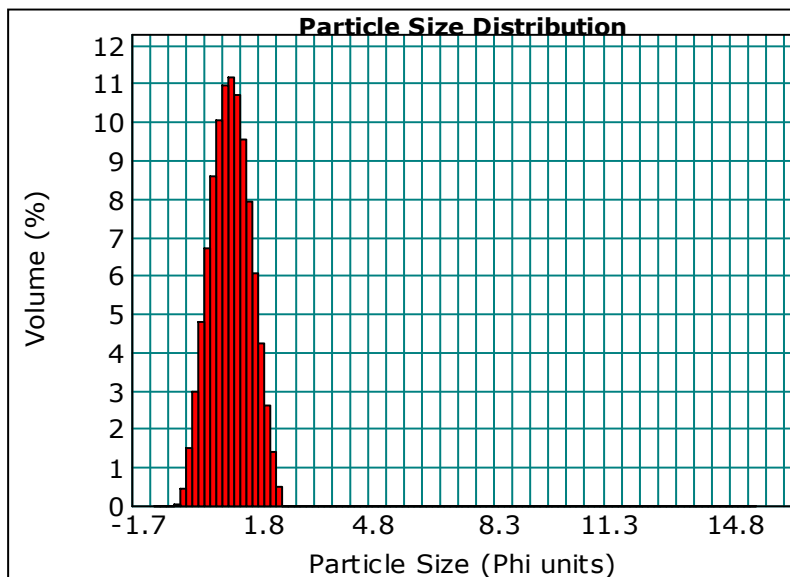
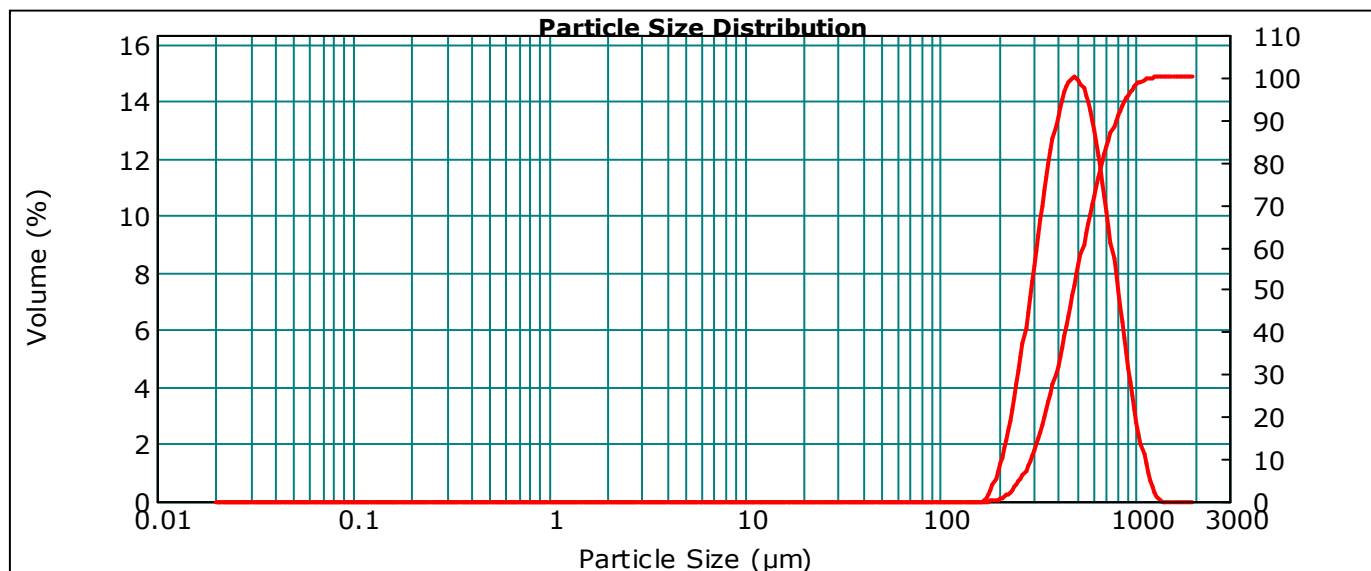
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.06 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.02
0.0	14.69
0.5	29.73
1.0	31.35
1.5	17.93
2.0	4.27
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:38:35 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 517.57um = .95phi

Surface Weighted Mean:

D[3,2] = 448.11um = 1.16phi

Span :

1.046

Uniformity:

0.323

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 256.28um = 1.96phi

d(0.10) = 290.69um = 1.78phi

d(0.16) = 323.65um = 1.63phi

d(0.50) = 482.85um = 1.05phi

d(0.84) = 717.33um = .48phi

d(0.90) = 795.91um = .33phi

d(0.95) = 897.42um = .16phi

Distribution Properties:

Kurtosis = .34

Skewness = .81

Standard Deviation = 198.5 um = 2.33phi

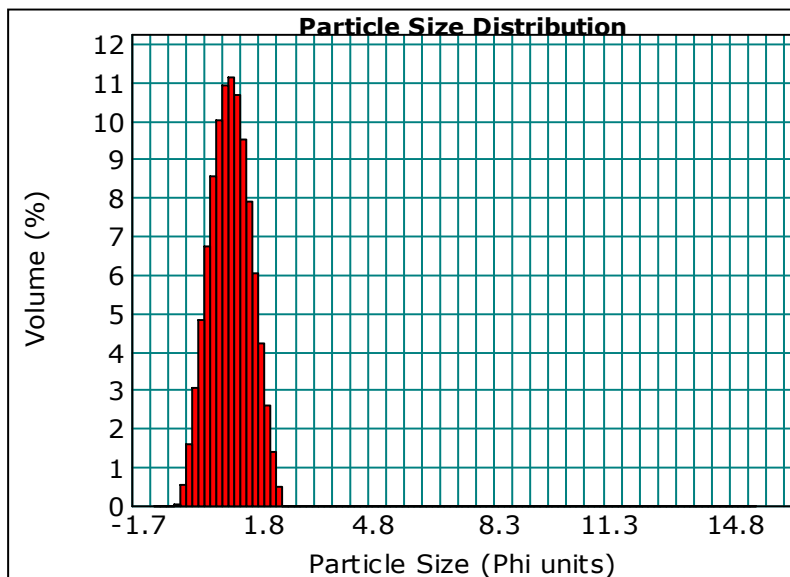
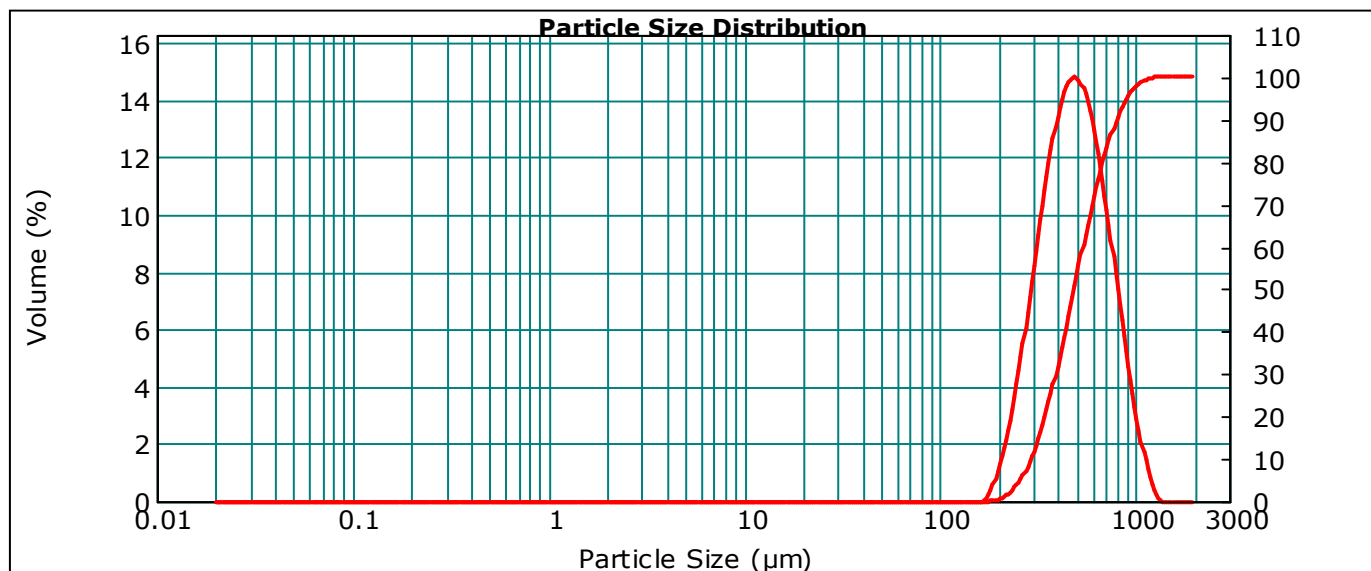
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.16
0.0	14.78
0.5	29.67
1.0	31.25
1.5	17.88
2.0	4.26
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:39:42 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 517.74um = .95phi

Surface Weighted Mean:

D[3,2] = 448.57um = 1.16phi

Span :

1.043

Uniformity:

0.322

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 256.79um = 1.96phi

d(0.10) = 291.21um = 1.78phi

d(0.16) = 324.18um = 1.63phi

d(0.50) = 483.27um = 1.05phi

d(0.84) = 717.17um = .48phi

d(0.90) = 795.39um = .33phi

d(0.95) = 896.25um = .16phi

Distribution Properties:

Kurtosis = .33

Skewness = .8

Standard Deviation = 197.99 um = 2.34phi

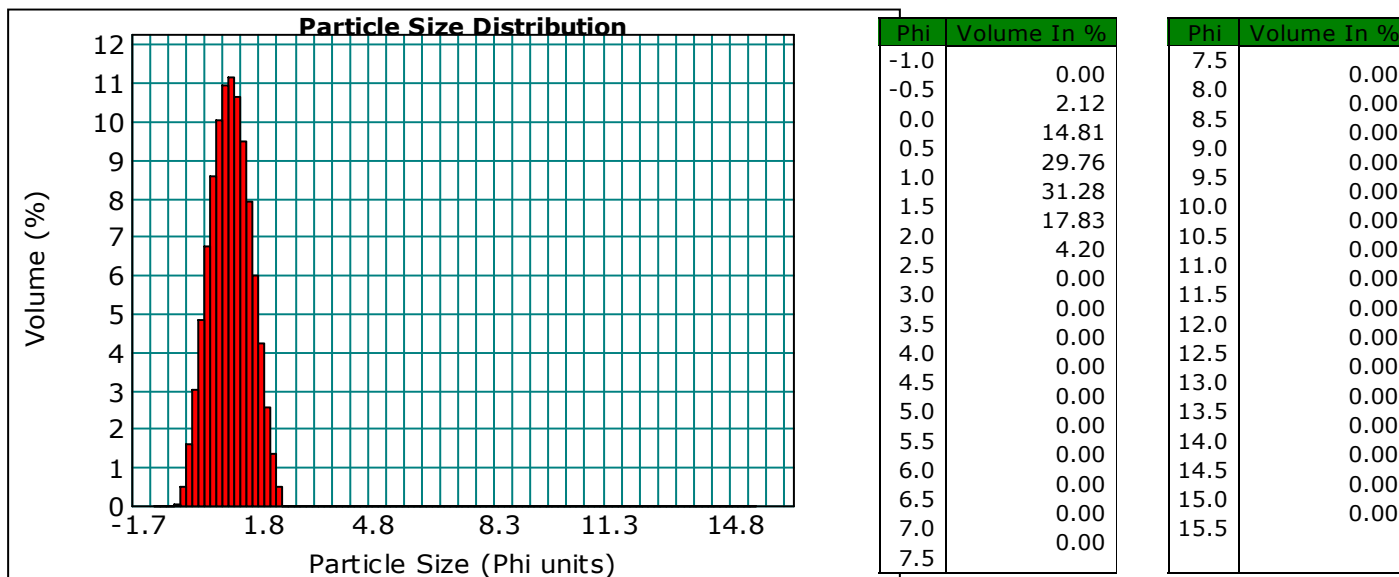
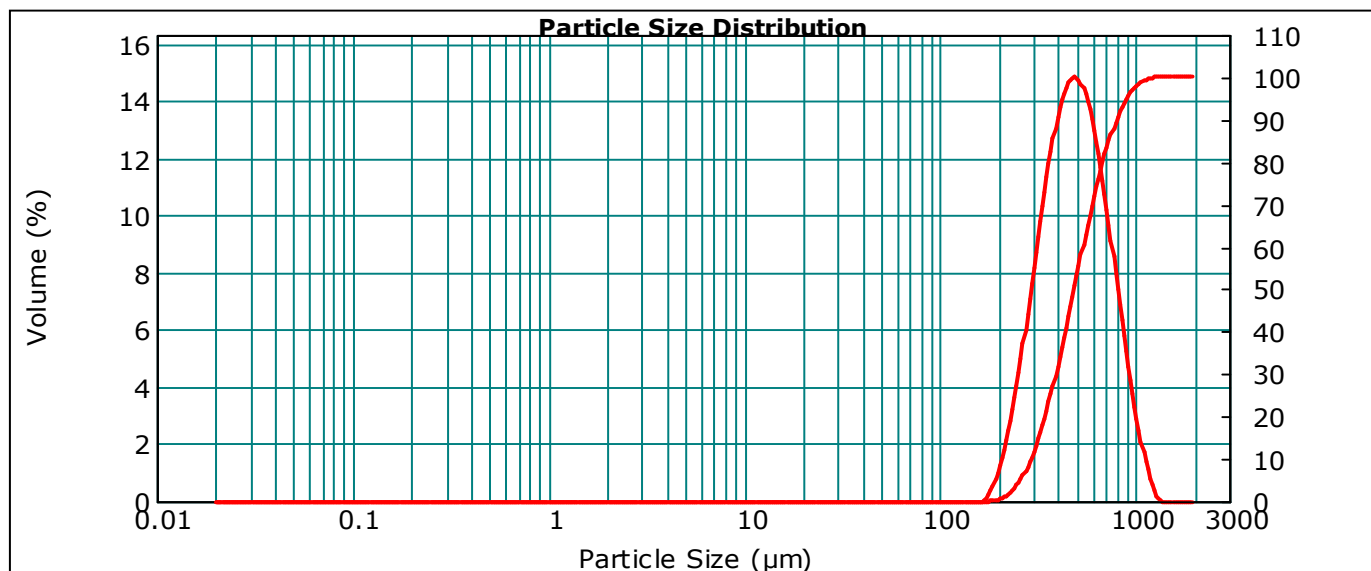
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_1_1 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 4:37:27 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 517.16um = .95phi

Surface Weighted Mean:

D[3,2] = 448.02um = 1.16phi

Span :

1.044

Uniformity:

0.322

Specific Surface Area*:

0.0134 m²/g

Percentiles:

d(0.05) = 256.41um = 1.96phi

d(0.10) = 290.81um = 1.78phi

d(0.16) = 323.75um = 1.63phi

d(0.50) = 482.74um = 1.05phi

d(0.84) = 716.42um = .48phi

d(0.90) = 794.56um = .33phi

d(0.95) = 895.28um = .16phi

Distribution Properties:

Kurtosis = .33

Skewness = .8

Standard Deviation = 197.82 um = 2.34phi

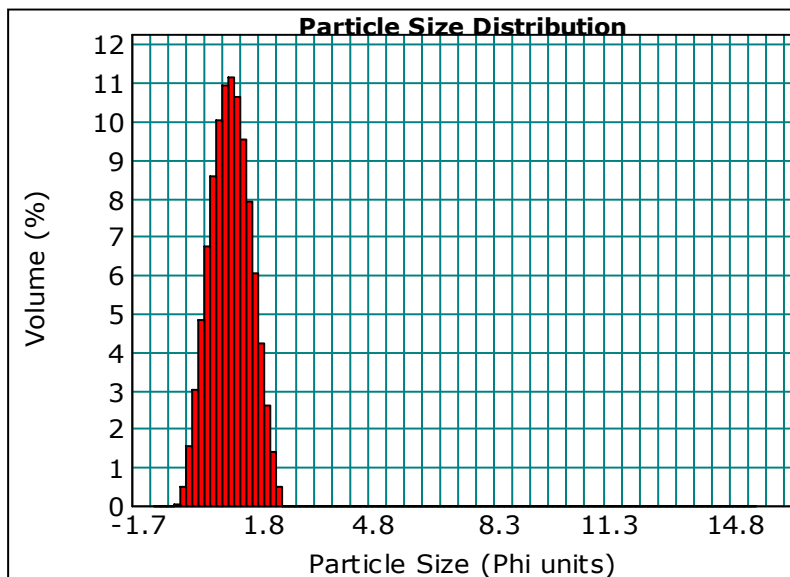
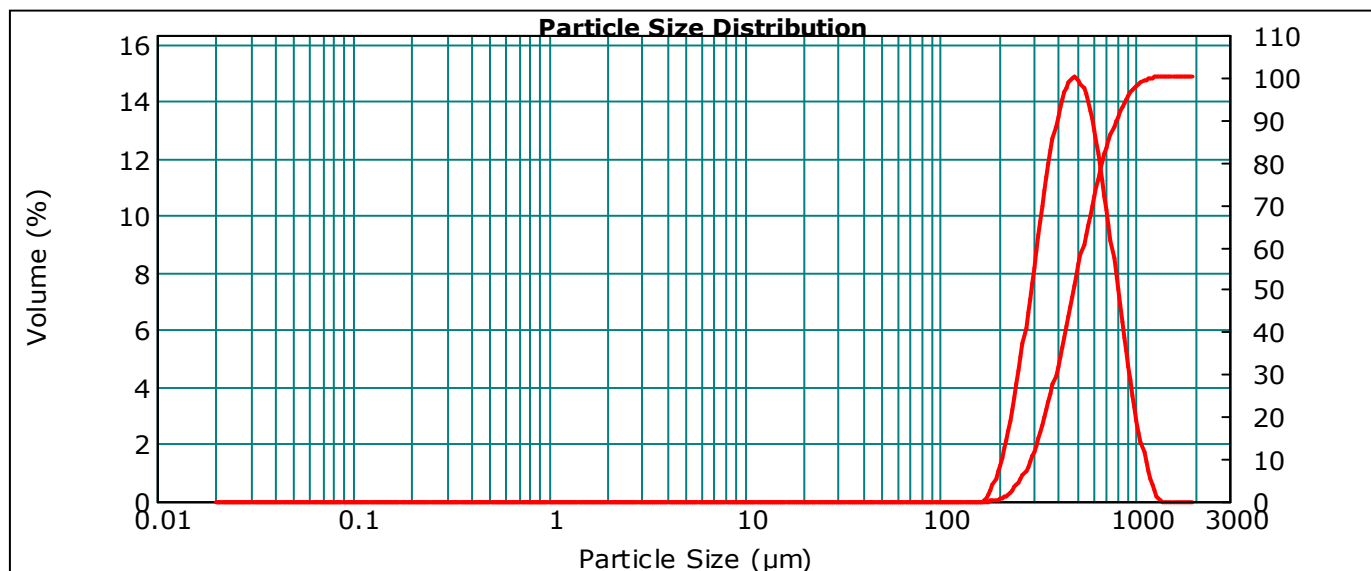
Inclusive SD (Sorting Coeff.) = .56 phi (Moderately Well)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.05 phi (Medium Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	2.10
0.0	14.76
0.5	29.72
1.0	31.29
1.5	17.88
2.0	4.25
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	0.00
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 10:55:46 AM

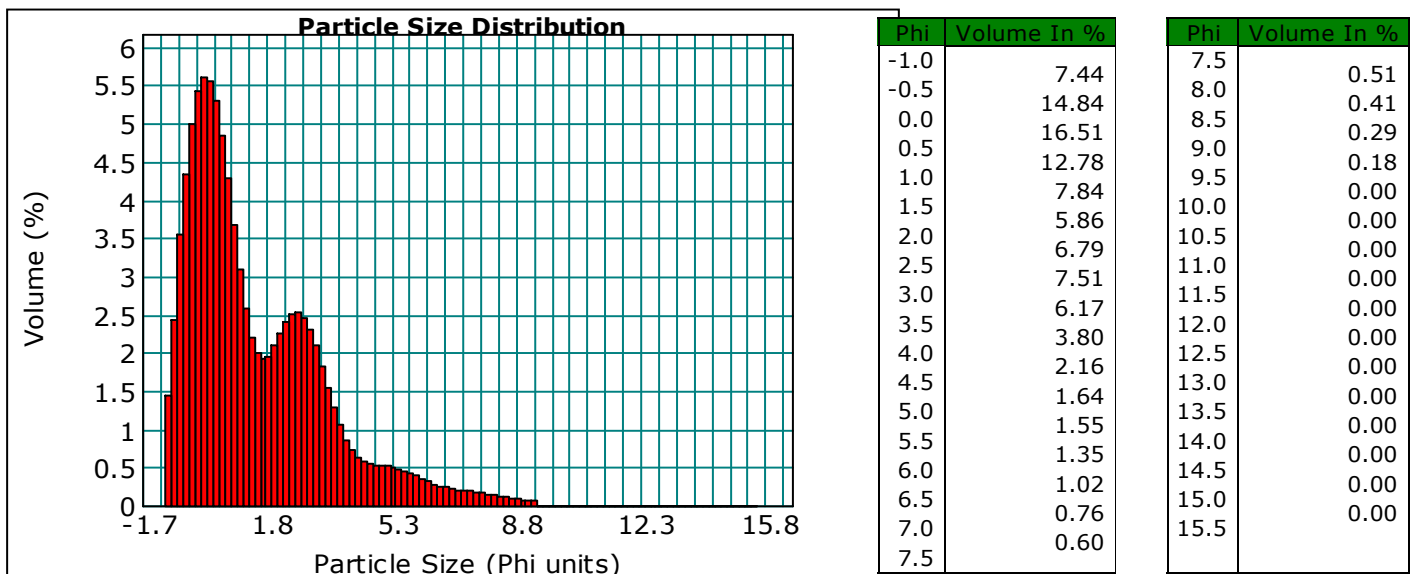
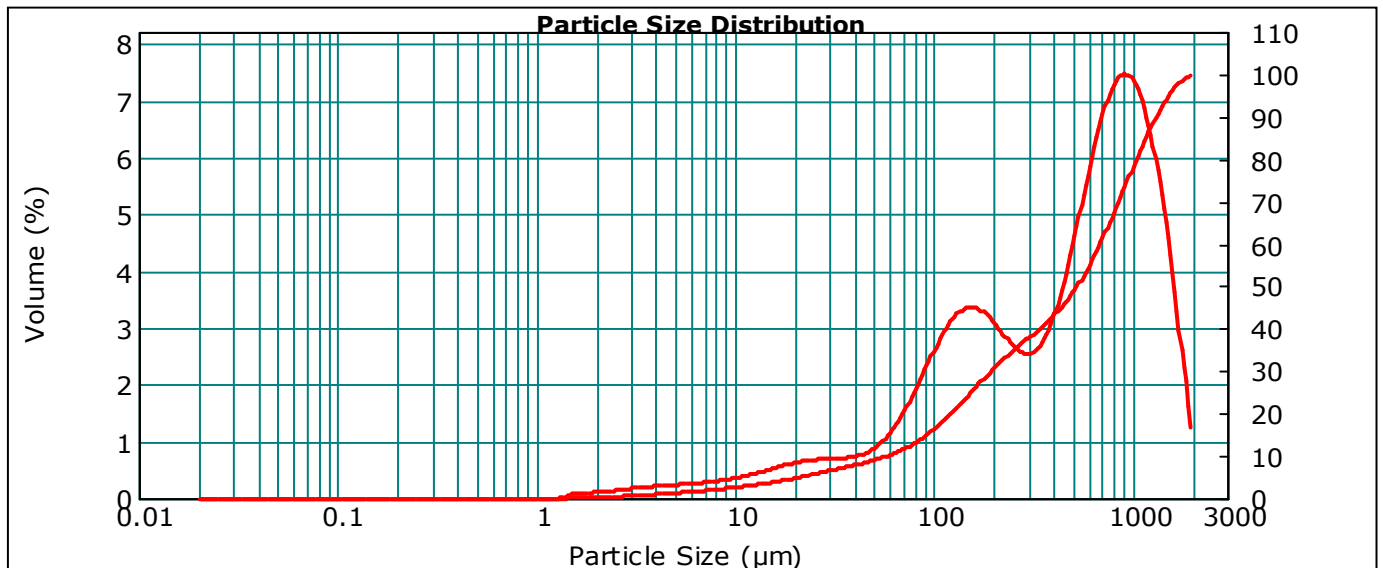
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
 alkaline calgon

Vol. Weighted Mean:
 D[4,3] = 602.98um = .73phi
Surface Weighted Mean:
 D[3,2] = 81.75um = 3.61phi
Span :
 2.392
Uniformity:
 0.778
Specific Surface Area*:
 0.0734 m²/g

Percentiles:
 d(0.05) = 21.47um = 5.54phi
 d(0.10) = 58.81um = 4.09phi
 d(0.16) = 98.85um = 3.34phi
 d(0.50) = 526.69um = .92phi
 d(0.84) = 1143.19um = -0.19phi
 d(0.90) = 1318.54um = -0.4phi
 d(0.95) = 1525.6um = -0.61phi

Distribution Properties:
 Kurtosis = -0.53
 Skewness = .65
 Standard Deviation = 488.54 um = 1.03phi
 Inclusive SD (Sorting Coeff.) = 1.81 phi (Poorly Sorted)
 Inclusive Kurtosis = .96 phi (Mesokurtic)
 Inclusive Mean = 1.36 phi (Medium Sand)
 Inclusive Skewness = .43 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 10:56:53 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 609.08um = .72phi

Surface Weighted Mean:

D[3,2] = 83.02um = 3.59phi

Span :

2.354

Uniformity:

0.763

Specific Surface Area*:

0.0723 m²/g

Percentiles:

d(0.05) = 21.81um = 5.52phi

d(0.10) = 60um = 4.06phi

d(0.16) = 100.48um = 3.31phi

d(0.50) = 536.84um = .9phi

d(0.84) = 1148.79um = -0.2phi

d(0.90) = 1323.67um = -0.4phi

d(0.95) = 1529.78um = -0.61phi

Distribution Properties:

Kurtosis = -0.55

Skewness = .63

Standard Deviation = 488.79 um = 1.03phi

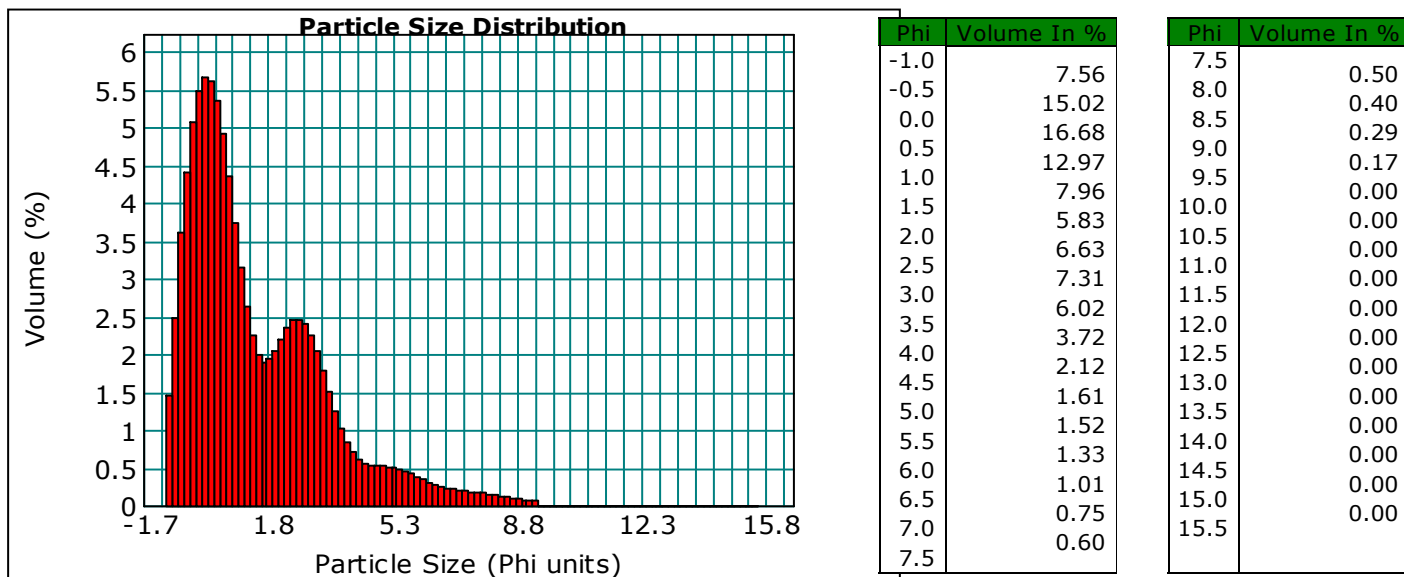
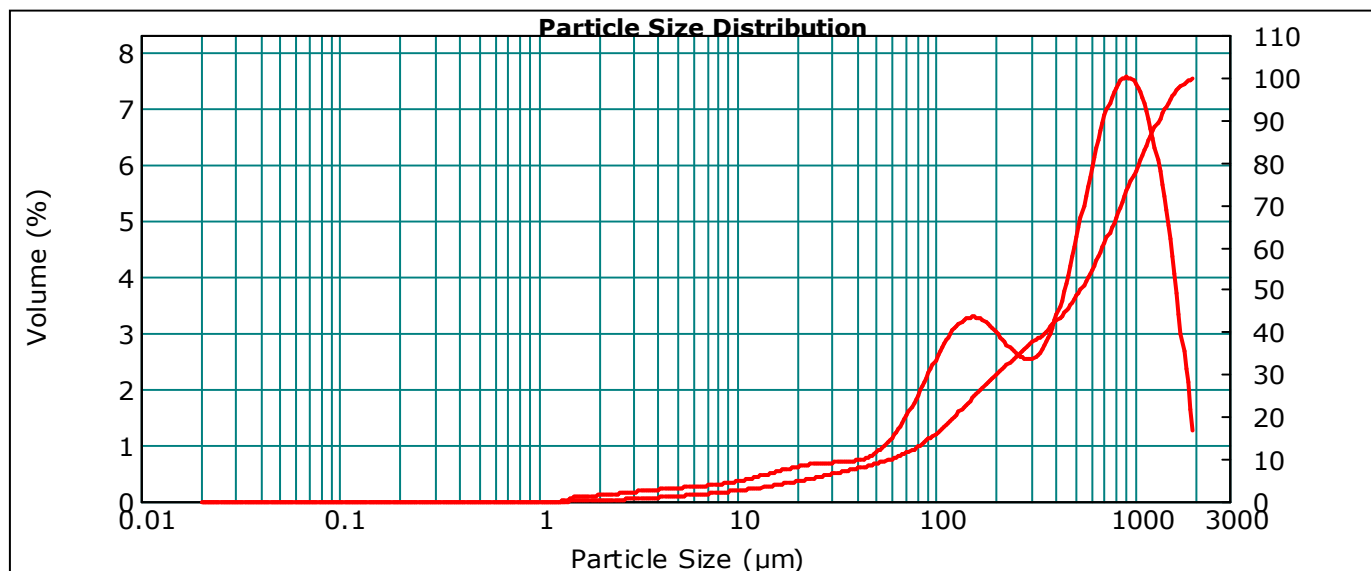
Inclusive SD (Sorting Coeff.) = 1.81 phi (Poorly Sorted)

Inclusive Kurtosis = .97 phi (Mesokurtic)

Inclusive Mean = 1.34 phi (Medium Sand)

Inclusive Skewness = .44 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_21
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 10:58:01 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
 alkaline calgon

Vol. Weighted Mean:

D[4,3] = 584.76um = .77phi

Surface Weighted Mean:

D[3,2] = 79.23um = 3.66phi

Span :

2.480

Uniformity:

0.802

Specific Surface Area*:

0.0757 m²/g

Percentiles:

d(0.05) = 20.39um = 5.62phi

d(0.10) = 55.69um = 4.17phi

d(0.16) = 95.71um = 3.39phi

d(0.50) = 498.81um = 1phi

d(0.84) = 1114.38um = -0.16phi

d(0.90) = 1292.67um = -0.37phi

d(0.95) = 1505.27um = -0.59phi

Distribution Properties:

Kurtosis = -0.42

Skewness = .7

Standard Deviation = 481.03 um = 1.06phi

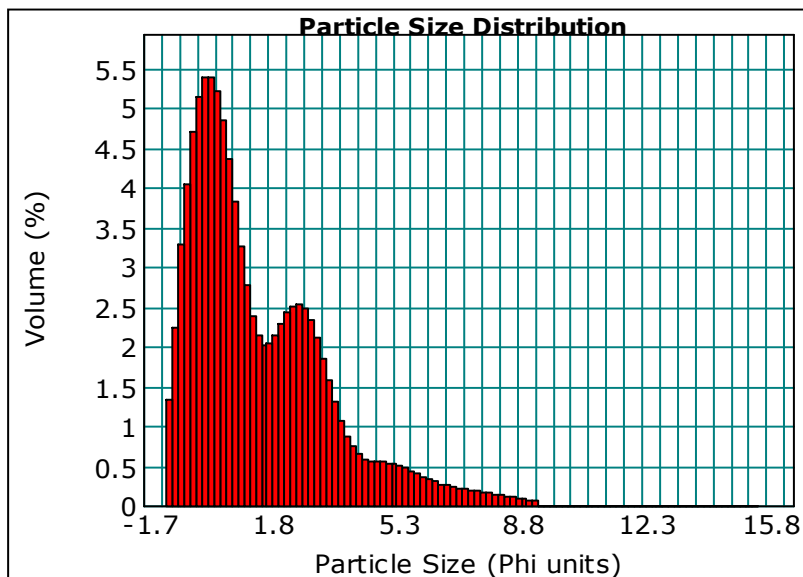
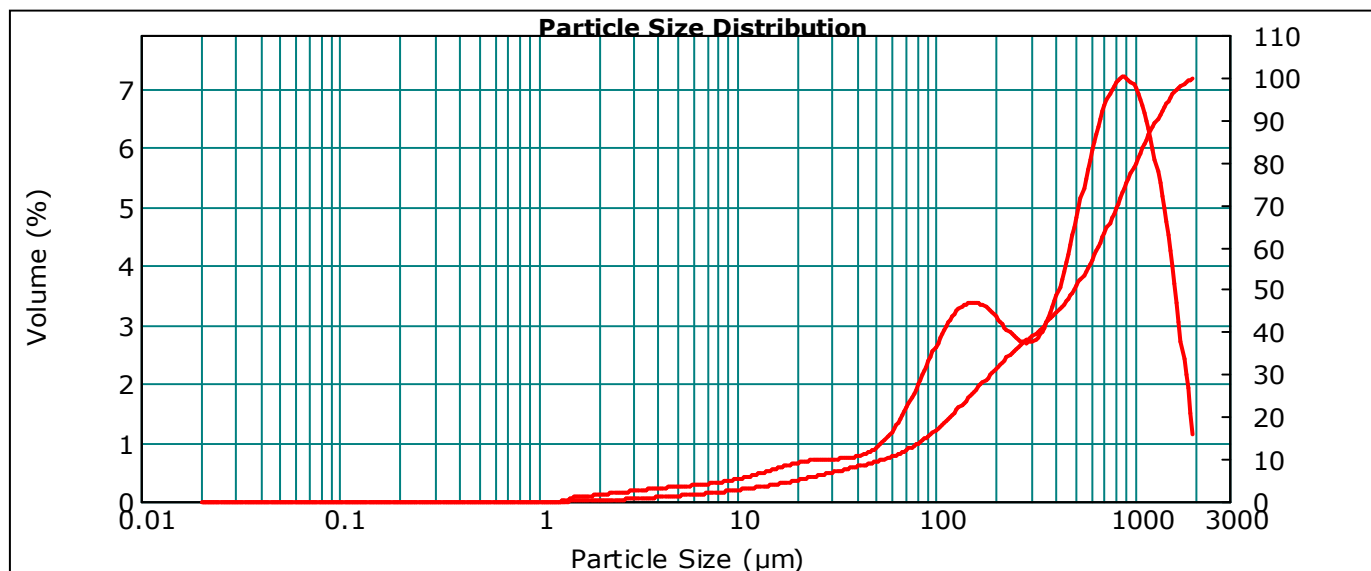
Inclusive SD (Sorting Coeff.) = 1.83 phi (Poorly Sorted)

Inclusive Kurtosis = .98 phi (Mesokurtic)

Inclusive Mean = 1.41 phi (Medium Sand)

Inclusive Skewness = .42 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.89
-0.5	13.97
0.0	16.04
0.5	13.03
1.0	8.38
1.5	6.22
2.0	6.90
2.5	7.55
3.0	6.26
3.5	3.90
4.0	2.23
4.5	1.69
5.0	1.60
5.5	1.41
6.0	1.08
6.5	0.80
7.0	0.63
7.5	

Phi	Volume In %
7.5	0.53
8.0	0.42
8.5	0.30
9.0	0.18
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_21 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 10:55:46 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 598.94um = .74phi

Surface Weighted Mean:
D[3,2] = 81.3um = 3.62phi

Span :
2.408

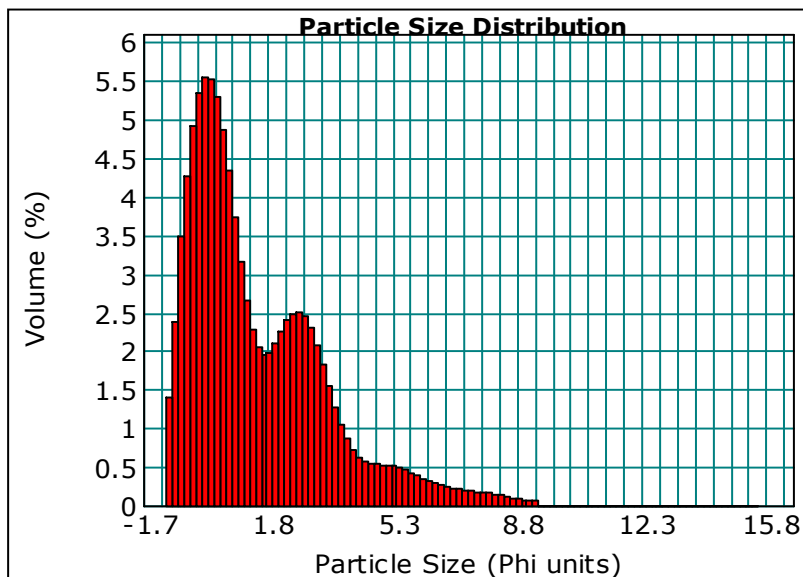
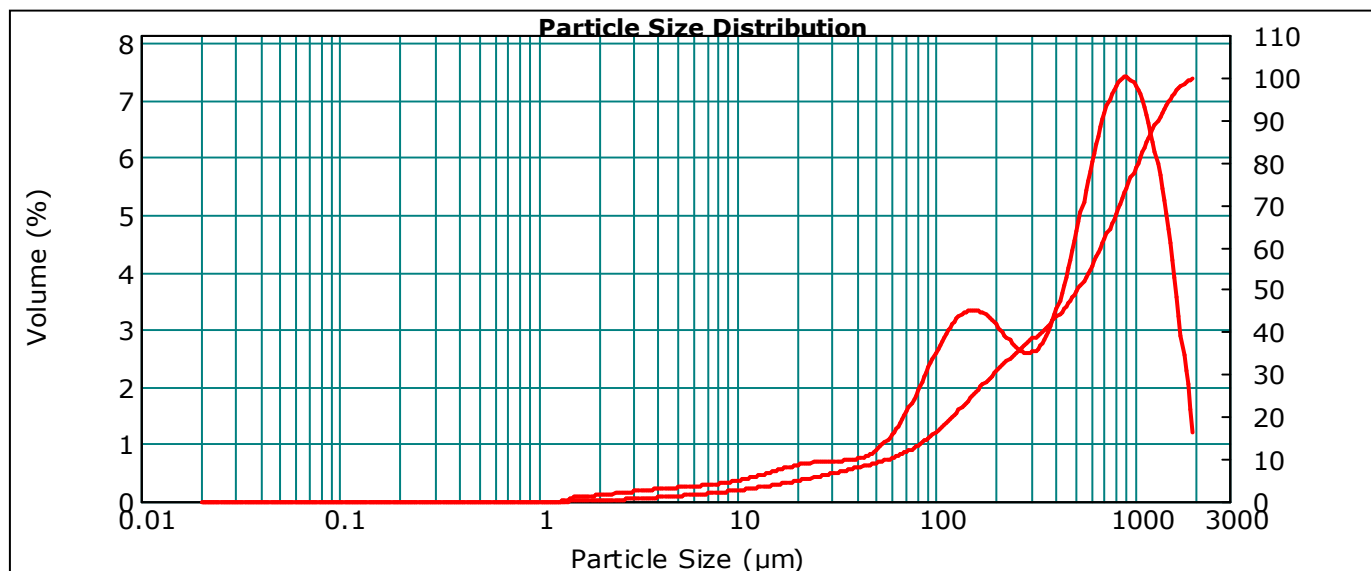
Uniformity:
0.781

Specific Surface Area*:
0.0738 m²/g

Percentiles:
d(0.05) = 21.2um = 5.56phi
d(0.10) = 58.14um = 4.1phi
d(0.16) = 98.32um = 3.35phi
d(0.50) = 520.63um = .94phi
d(0.84) = 1135.77um = -0.18phi
d(0.90) = 1311.92um = -0.39phi
d(0.95) = 1520.4um = -0.6phi

Distribution Properties:
Kurtosis = -0.5
Skewness = .66
Standard Deviation = 486.24 um = 1.04phi
Inclusive SD (Sorting Coeff.) = 1.82 phi (Poorly Sorted)
Inclusive Kurtosis = .97 phi (Mesokurtic)
Inclusive Mean = 1.37 phi (Medium Sand)
Inclusive Skewness = .43 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	7.29
-0.5	14.61
0.0	16.41
0.5	12.93
1.0	8.06
1.5	5.97
2.0	6.77
2.5	7.46
3.0	6.15
3.5	3.81
4.0	2.17
4.5	1.64
5.0	1.56
5.5	1.36
6.0	1.04
6.5	0.77
7.0	0.61
7.5	

Phi	Volume In %
7.5	0.52
8.0	0.41
8.5	0.29
9.0	0.18
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:07:14 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 534.74um = .9phi

Surface Weighted Mean:

D[3,2] = 70.01um = 3.84phi

Span :

3.127

Uniformity:

1.02

Specific Surface Area*:

0.0857 m²/g

Percentiles:

d(0.05) = 18.72um = 5.74phi

d(0.10) = 49.13um = 4.35phi

d(0.16) = 83.63um = 3.58phi

d(0.50) = 386.43um = 1.37phi

d(0.84) = 1071.75um = -0.1phi

d(0.90) = 1257.66um = -0.33phi

d(0.95) = 1479.14um = -0.56phi

Distribution Properties:

Kurtosis = -0.23

Skewness = .86

Standard Deviation = 479.44 um = 1.06phi

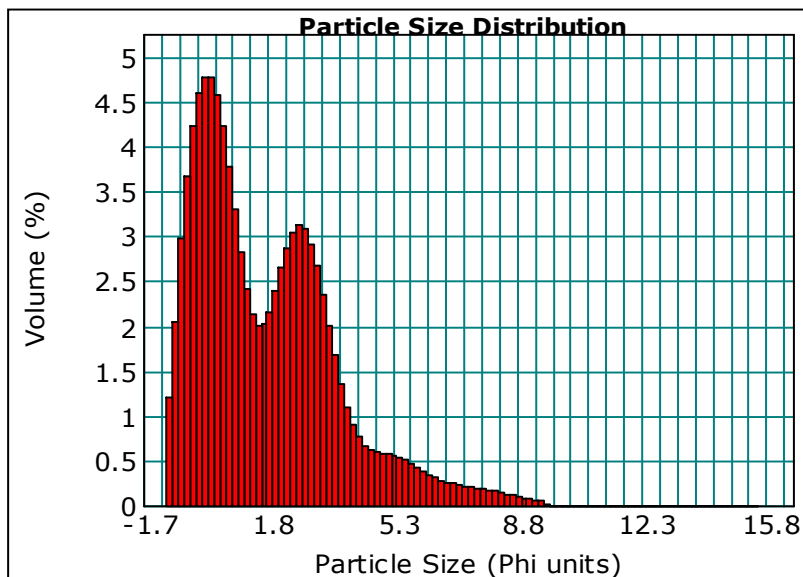
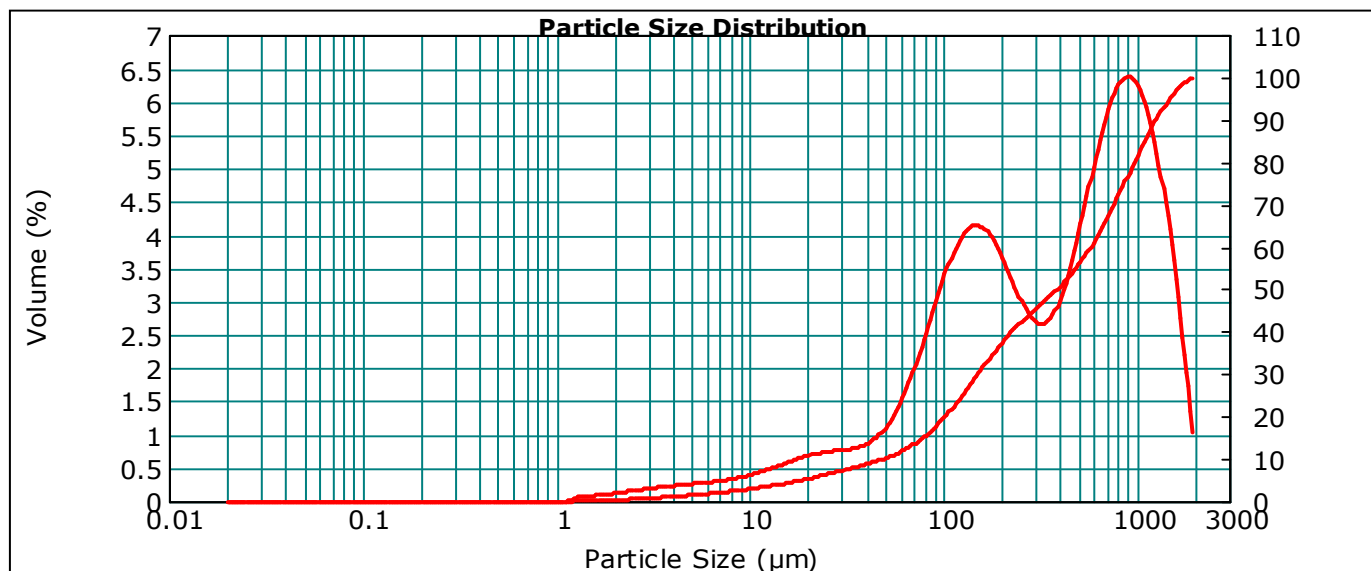
Inclusive SD (Sorting Coeff.) = 1.88 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.62 phi (Medium Sand)

Inclusive Skewness = .29 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.26
-0.5	12.55
0.0	14.16
0.5	11.30
1.0	7.36
1.5	6.23
2.0	7.98
2.5	9.28
3.0	7.89
3.5	4.96
4.0	2.72
4.5	1.87
5.0	1.70
5.5	1.48
6.0	1.14
6.5	0.84
7.0	0.67
7.5	

Phi	Volume In %
7.5	0.57
8.0	0.46
8.5	0.33
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:08:22 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 544.14um = .88phi

Surface Weighted Mean:

D[3,2] = 70.97um = 3.82phi

Span :

2.937

Uniformity:

0.96

Specific Surface Area*:

0.0845 m²/g

Percentiles:

d(0.05) = 18.87um = 5.73phi

d(0.10) = 49.83um = 4.33phi

d(0.16) = 84.74um = 3.56phi

d(0.50) = 413.48um = 1.27phi

d(0.84) = 1080.15um = -0.11phi

d(0.90) = 1264.07um = -0.34phi

d(0.95) = 1483.79um = -0.57phi

Distribution Properties:

Kurtosis = -0.29

Skewness = .82

Standard Deviation = 480.51 um = 1.06phi

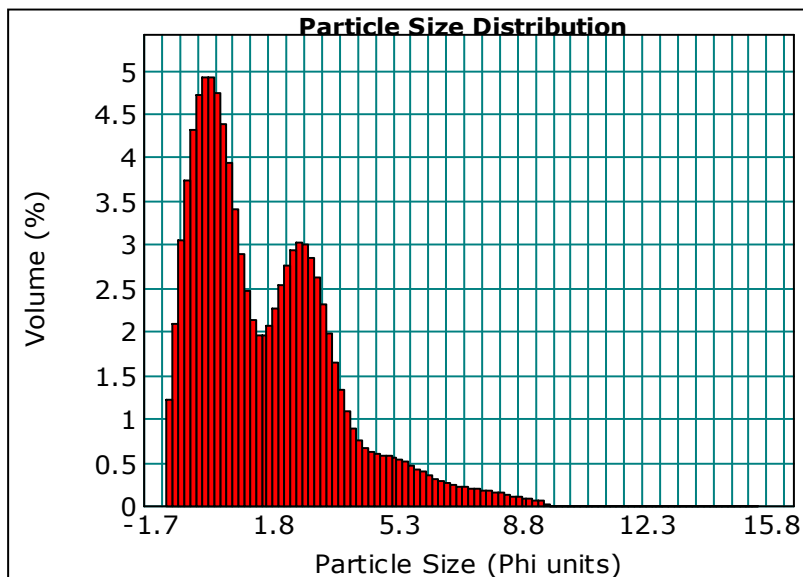
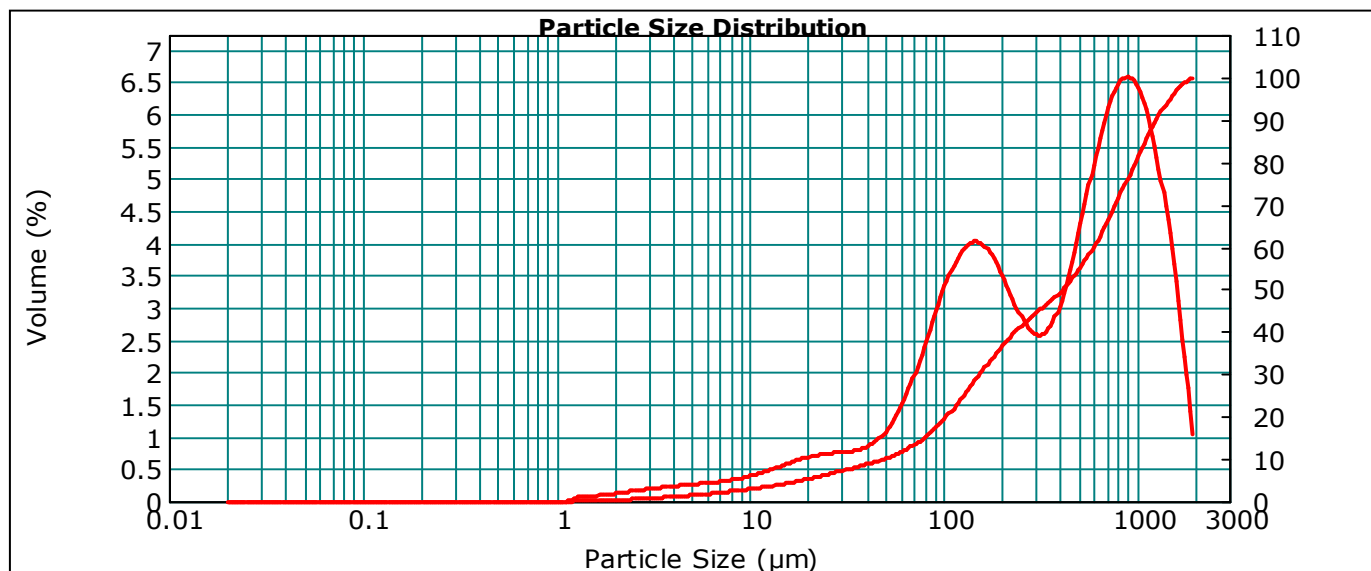
Inclusive SD (Sorting Coeff.) = 1.87 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.57 phi (Medium Sand)

Inclusive Skewness = .33 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.36
-0.5	12.84
0.0	14.61
0.5	11.70
1.0	7.44
1.5	5.98
2.0	7.61
2.5	8.99
3.0	7.72
3.5	4.86
4.0	2.66
4.5	1.84
5.0	1.69
5.5	1.48
6.0	1.13
6.5	0.83
7.0	0.66
7.5	

Phi	Volume In %
7.5	0.56
8.0	0.45
8.5	0.32
9.0	0.20
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_20
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:09:29 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 541.48um = .89phi

Surface Weighted Mean:

D[3,2] = 69.83um = 3.84phi

Span :

3.041

Uniformity:

0.995

Specific Surface Area*:

0.0859 m²/g

Percentiles:

d(0.05) = 18.38um = 5.77phi

d(0.10) = 48.26um = 4.37phi

d(0.16) = 83.37um = 3.58phi

d(0.50) = 400.6um = 1.32phi

d(0.84) = 1082.41um = -0.11phi

d(0.90) = 1266.47um = -0.34phi

d(0.95) = 1485.91um = -0.57phi

Distribution Properties:

Kurtosis = -0.3

Skewness = .83

Standard Deviation = 482.64 um = 1.05phi

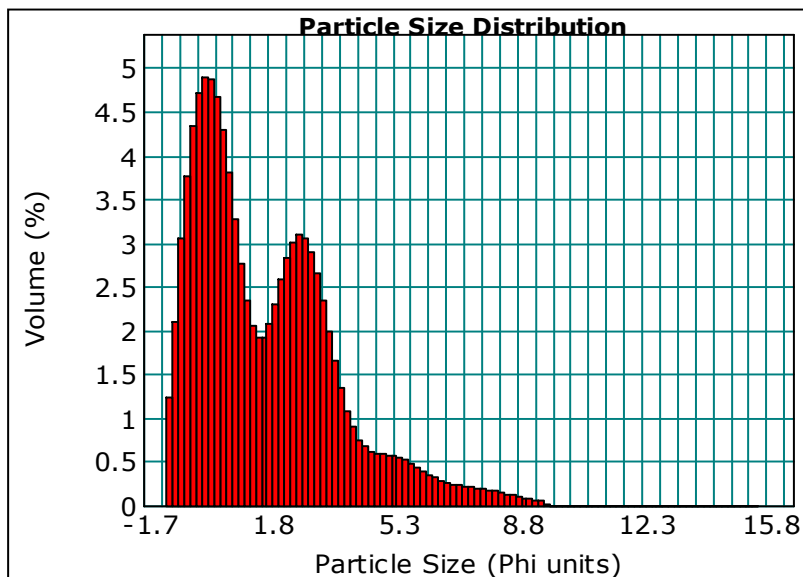
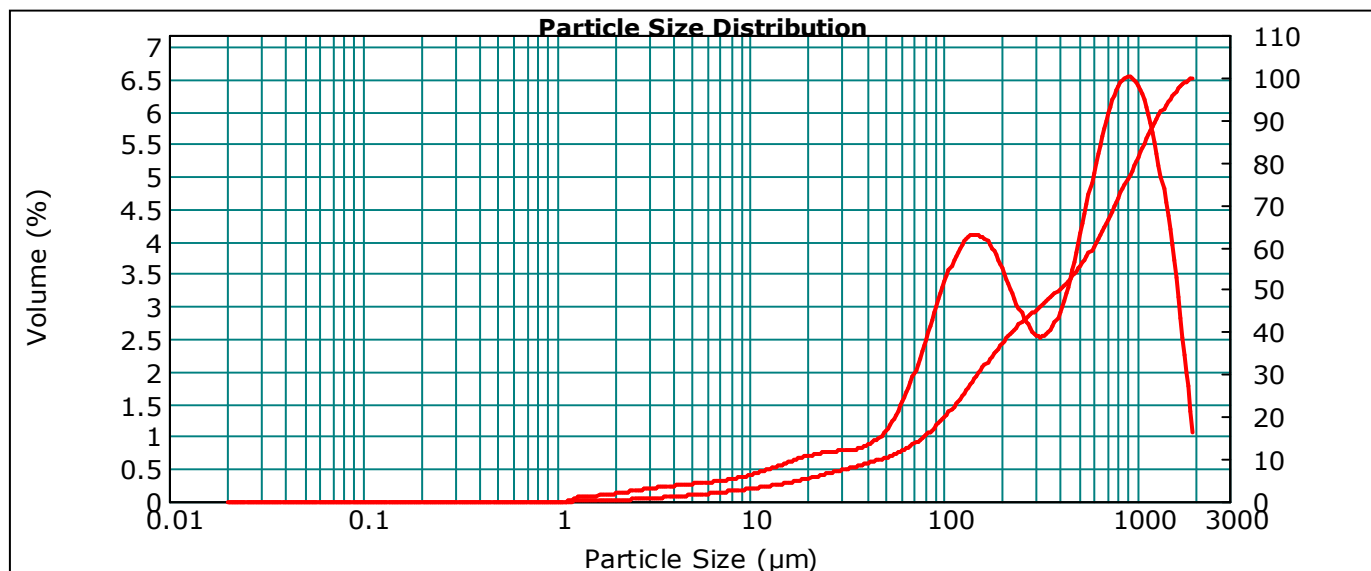
Inclusive SD (Sorting Coeff.) = 1.88 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.6 phi (Medium Sand)

Inclusive Skewness = .31 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.41
-0.5	12.87
0.0	14.47
0.5	11.34
1.0	7.13
1.5	5.94
2.0	7.78
2.5	9.18
3.0	7.84
3.5	4.91
4.0	2.69
4.5	1.87
5.0	1.73
5.5	1.52
6.0	1.16
6.5	0.86
7.0	0.68
7.5	

Phi	Volume In %
7.5	0.57
8.0	0.46
8.5	0.33
9.0	0.20
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_20 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:07:14 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 540.12um = .89phi

Surface Weighted Mean:
D[3,2] = 70.27um = 3.83phi

Span :
3.033

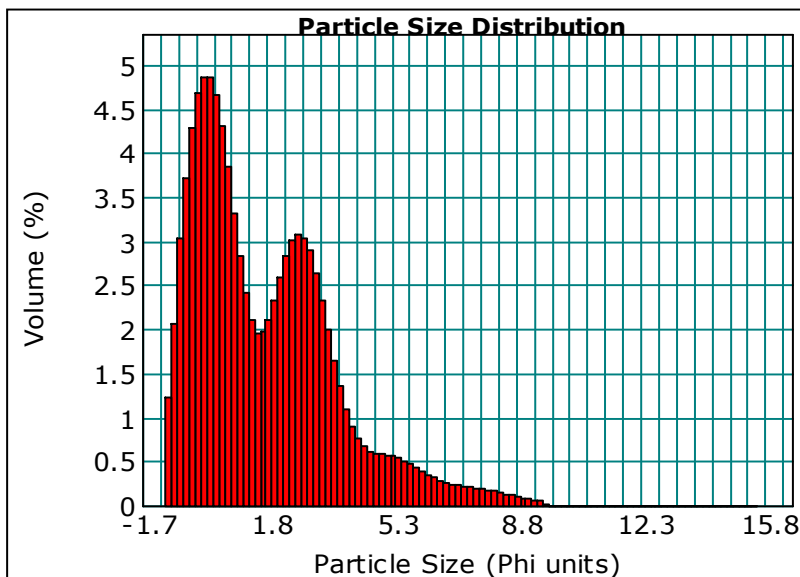
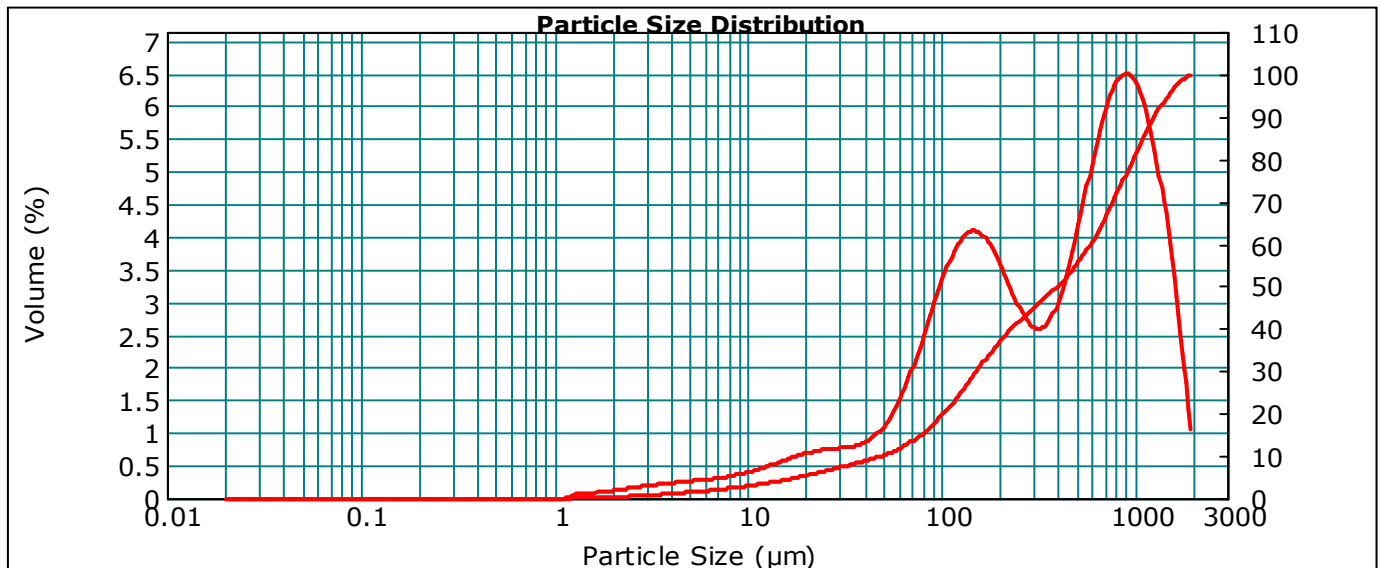
Uniformity:
0.99

Specific Surface Area*:
0.0854 m²/g

Percentiles:
d(0.05) = 18.66um = 5.74phi
d(0.10) = 49.08um = 4.35phi
d(0.16) = 83.91um = 3.58phi
d(0.50) = 400.21um = 1.32phi
d(0.84) = 1078.15um = -0.11phi
d(0.90) = 1262.77um = -0.34phi
d(0.95) = 1482.98um = -0.57phi

Distribution Properties:
Kurtosis = -0.27
Skewness = .84
Standard Deviation = 480.88 um = 1.06phi
Inclusive SD (Sorting Coeff.) = 1.88 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.6 phi (Medium Sand)
Inclusive Skewness = .31 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.34
-0.5	12.75
0.0	14.41
0.5	11.44
1.0	7.31
1.5	6.05
2.0	7.79
2.5	9.15
3.0	7.82
3.5	4.91
4.0	2.69
4.5	1.86
5.0	1.70
5.5	1.49
6.0	1.14
6.5	0.84
7.0	0.67
7.5	

Phi	Volume In %
7.5	0.57
8.0	0.46
8.5	0.33
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:18:25 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 559.55um = .84phi

Surface Weighted Mean:

D[3,2] = 73.27um = 3.77phi

Span :

2.723

Uniformity:

0.89

Specific Surface Area*:

0.0819 m²/g

Percentiles:

d(0.05) = 19.57um = 5.68phi

d(0.10) = 52.68um = 4.25phi

d(0.16) = 88.56um = 3.5phi

d(0.50) = 449.3um = 1.15phi

d(0.84) = 1094.32um = -0.13phi

d(0.90) = 1276.28um = -0.35phi

d(0.95) = 1493.34um = -0.58phi

Distribution Properties:

Kurtosis = -0.35

Skewness = .78

Standard Deviation = 481.56 um = 1.05phi

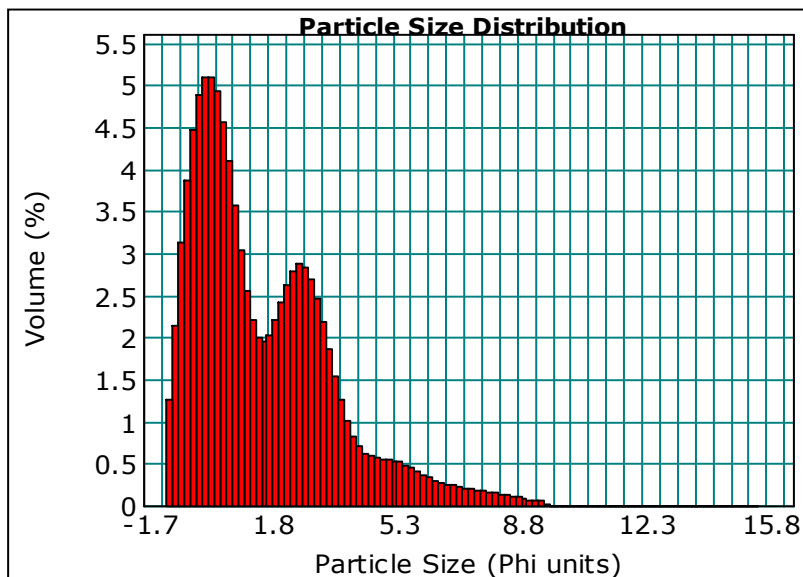
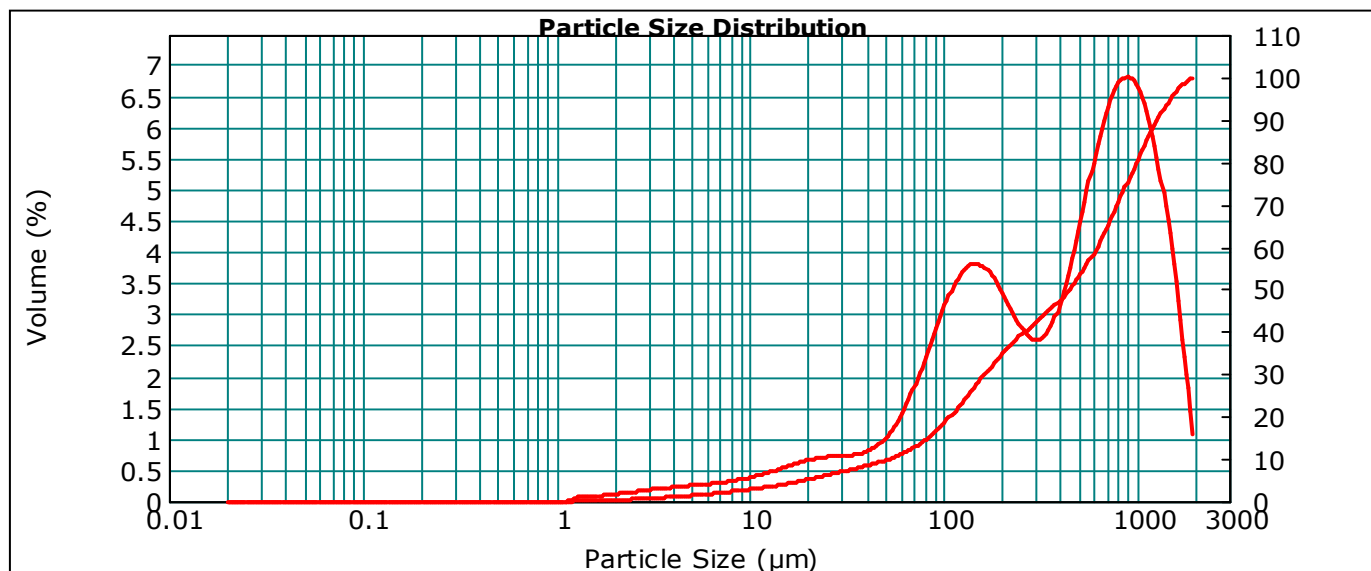
Inclusive SD (Sorting Coeff.) = 1.85 phi (Poorly Sorted)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.51 phi (Medium Sand)

Inclusive Skewness = .37 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.59
-0.5	13.27
0.0	15.14
0.5	12.20
1.0	7.76
1.5	5.98
2.0	7.30
2.5	8.52
3.0	7.29
3.5	4.57
4.0	2.50
4.5	1.75
5.0	1.62
5.5	1.43
6.0	1.10
6.5	0.81
7.0	0.65
7.5	

Phi	Volume In %
7.5	0.55
8.0	0.44
8.5	0.31
9.0	0.20
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:19:32 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 554.31um = .85phi

Surface Weighted Mean:

D[3,2] = 71.15um = 3.81phi

Span :

2.933

Uniformity:

0.958

Specific Surface Area*:

0.0843 m²/g

Percentiles:

d(0.05) = 18.7um = 5.74phi

d(0.10) = 49.82um = 4.33phi

d(0.16) = 85.83um = 3.54phi

d(0.50) = 421.61um = 1.25phi

d(0.84) = 1102.87um = -0.14phi

d(0.90) = 1286.41um = -0.36phi

d(0.95) = 1502.58um = -0.59phi

Distribution Properties:

Kurtosis = -0.36

Skewness = .8

Standard Deviation = 487.33 um = 1.04phi

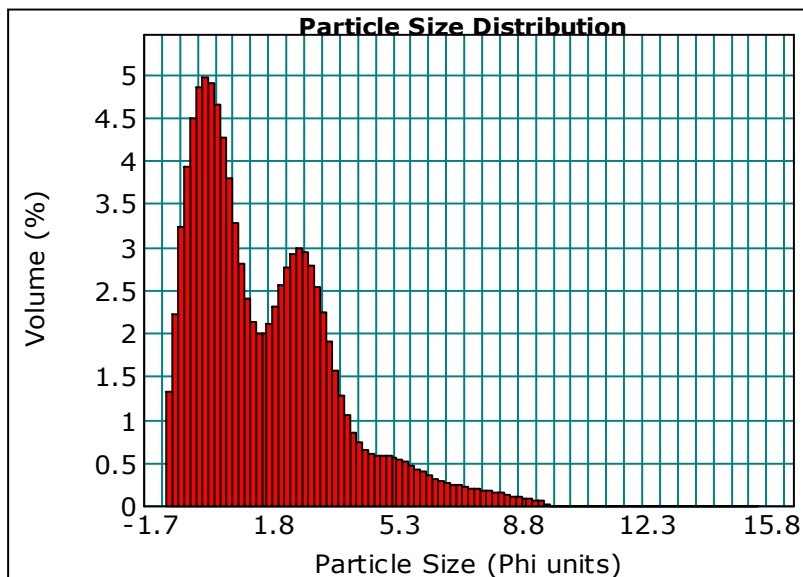
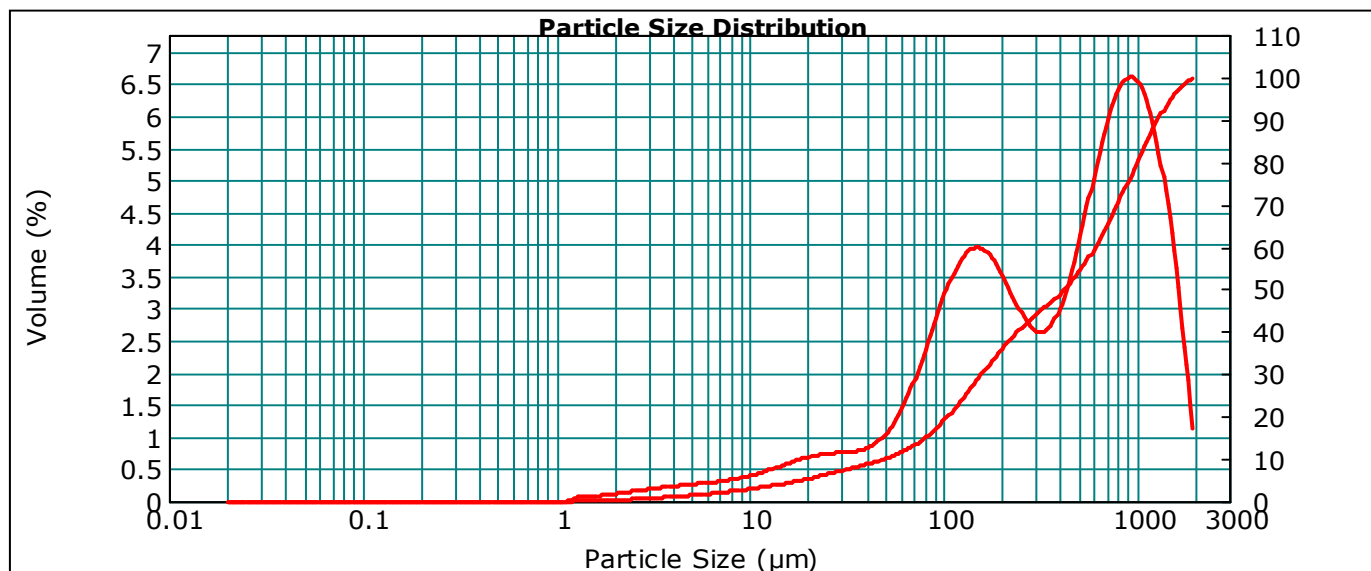
Inclusive SD (Sorting Coeff.) = 1.88 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.55 phi (Medium Sand)

Inclusive Skewness = .33 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.80
-0.5	13.33
0.0	14.56
0.5	11.33
1.0	7.31
1.5	6.13
2.0	7.69
2.5	8.85
3.0	7.49
3.5	4.68
4.0	2.57
4.5	1.82
5.0	1.69
5.5	1.49
6.0	1.14
6.5	0.85
7.0	0.67
7.5	

Phi	Volume In %
7.5	0.57
8.0	0.45
8.5	0.32
9.0	0.20
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_19
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:20:39 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 573.48um = .8phi

Surface Weighted Mean:

D[3,2] = 75.99um = 3.72phi

Span :

2.696

Uniformity:

0.884

Specific Surface Area*:

0.079 m²/g

Percentiles:

d(0.05) = 19.58um = 5.67phi

d(0.10) = 52.52um = 4.25phi

d(0.16) = 88.98um = 3.49phi

d(0.50) = 464.58um = 1.11phi

d(0.84) = 1124.64um = -0.17phi

d(0.90) = 1304.88um = -0.38phi

d(0.95) = 1516.53um = -0.6phi

Distribution Properties:

Kurtosis = -0.46

Skewness = .73

Standard Deviation = 491.27 um = 1.03phi

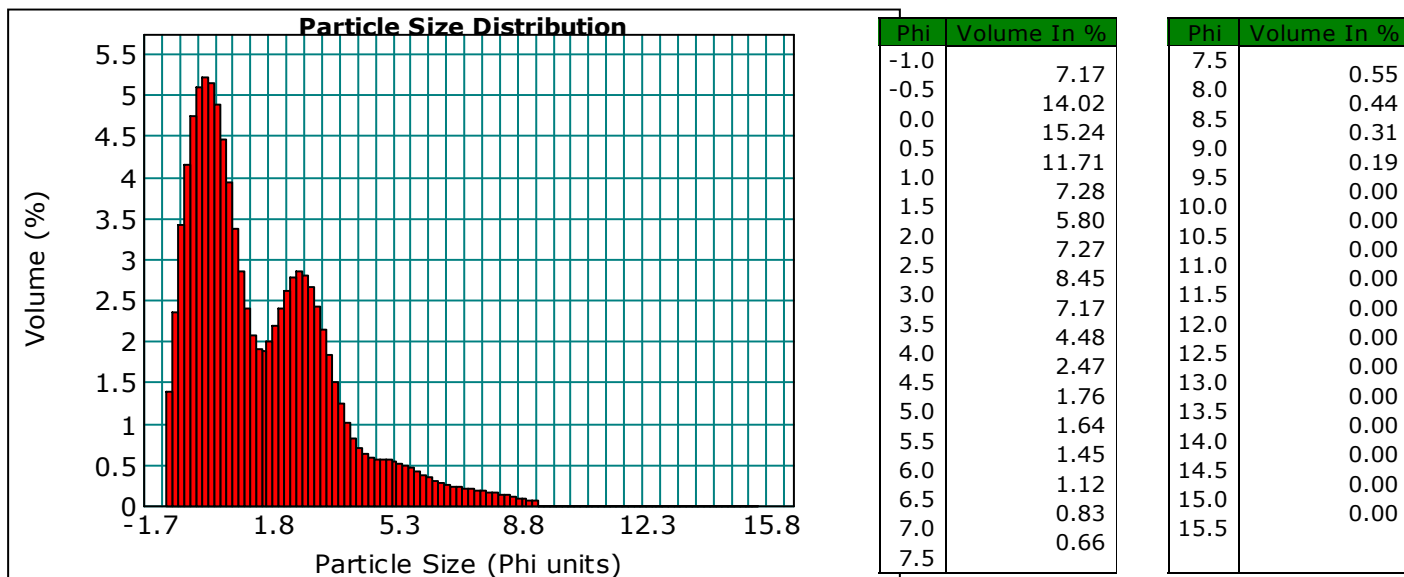
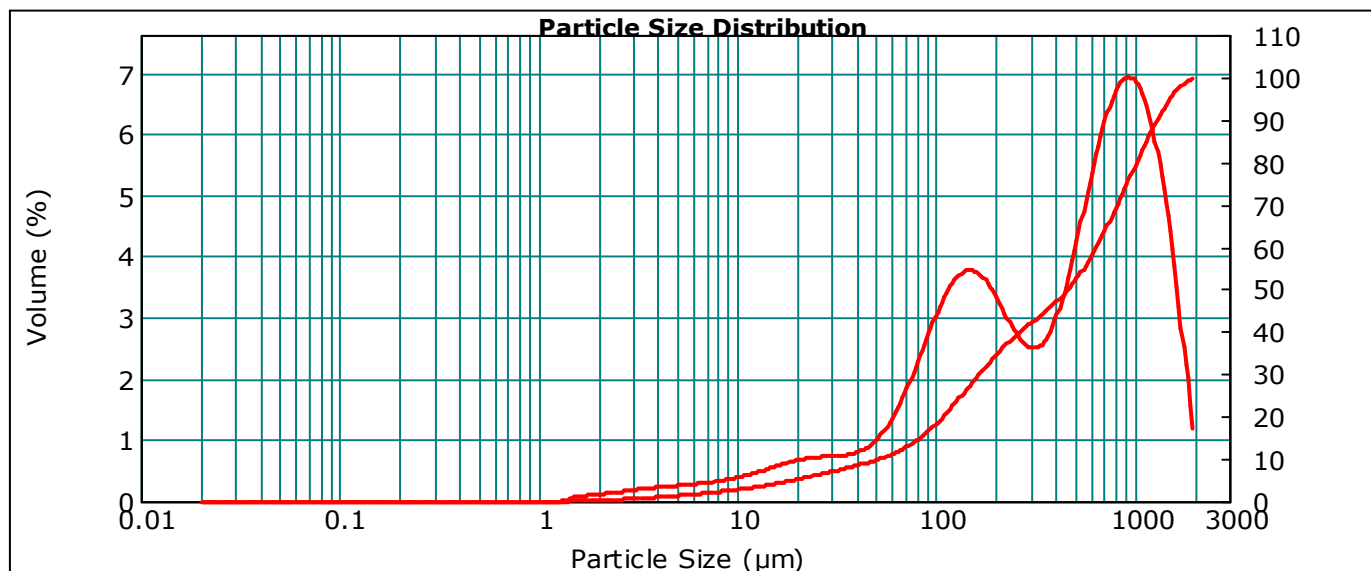
Inclusive SD (Sorting Coeff.) = 1.87 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.48 phi (Medium Sand)

Inclusive Skewness = .38 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_19 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:18:25 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 562.45um = .83phi

Surface Weighted Mean:

D[3,2] = 73.42um = 3.77phi

Span :

2.779

Uniformity:

0.909

Specific Surface Area*:

0.0817 m²/g

Percentiles:

d(0.05) = 19.27um = 5.7phi

d(0.10) = 51.66um = 4.27phi

d(0.16) = 87.78um = 3.51phi

d(0.50) = 445.4um = 1.17phi

d(0.84) = 1107.45um = -0.15phi

d(0.90) = 1289.44um = -0.37phi

d(0.95) = 1504.42um = -0.59phi

Distribution Properties:

Kurtosis = -0.39

Skewness = .77

Standard Deviation = 486.8 um = 1.04phi

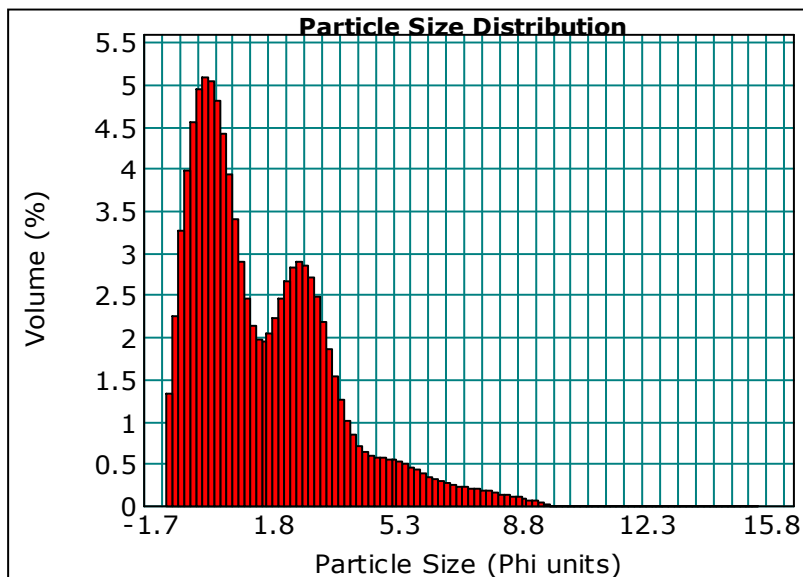
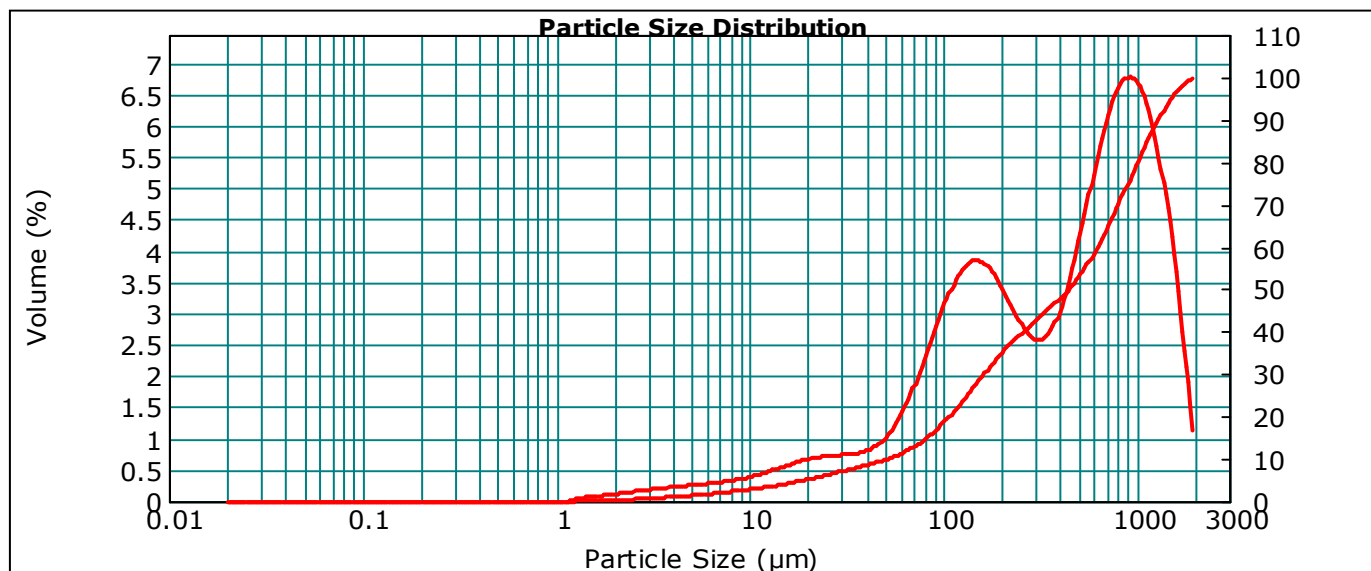
Inclusive SD (Sorting Coeff.) = 1.87 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.51 phi (Medium Sand)

Inclusive Skewness = .36 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.85
-0.5	13.54
0.0	14.98
0.5	11.75
1.0	7.45
1.5	5.97
2.0	7.42
2.5	8.60
3.0	7.32
3.5	4.58
4.0	2.51
4.5	1.77
5.0	1.65
5.5	1.46
6.0	1.12
6.5	0.83
7.0	0.66
7.5	

Phi	Volume In %
7.5	0.56
8.0	0.44
8.5	0.31
9.0	0.20
9.5	0.04
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:29:26 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 578.57um = .79phi

Surface Weighted Mean:

D[3,2] = 84.67um = 3.56phi

Span :

2.577

Uniformity:

0.84

Specific Surface Area*:

0.0709 m²/g

Percentiles:

d(0.05) = 23.15um = 5.43phi

d(0.10) = 63.23um = 3.98phi

d(0.16) = 98.61um = 3.34phi

d(0.50) = 478.53um = 1.06phi

d(0.84) = 1116.16um = -0.16phi

d(0.90) = 1296.59um = -0.37phi

d(0.95) = 1509.88um = -0.59phi

Distribution Properties:

Kurtosis = -0.41

Skewness = .74

Standard Deviation = 483.7 um = 1.05phi

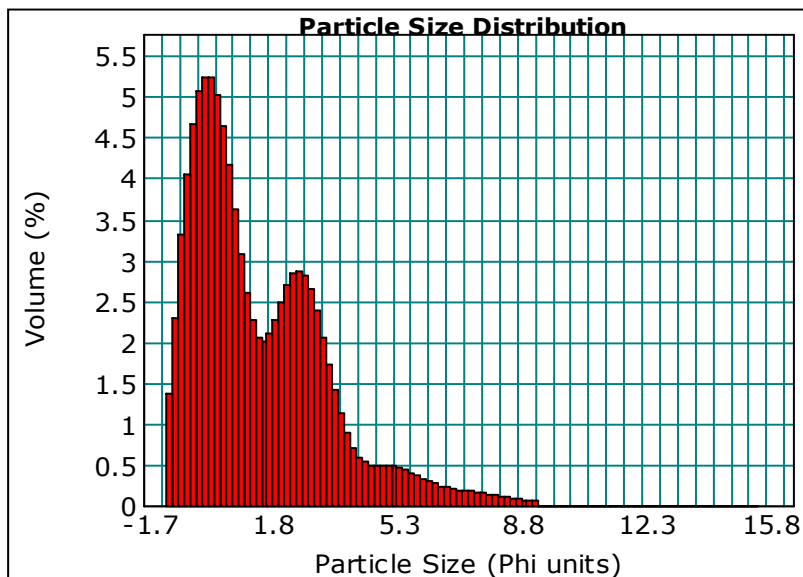
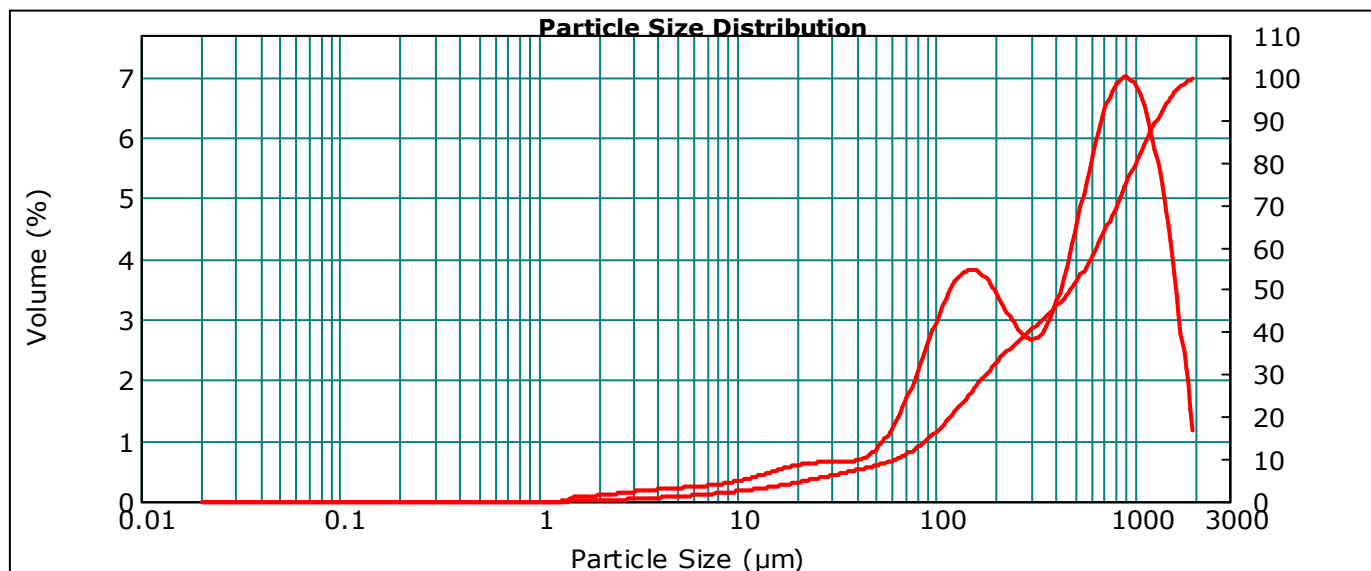
Inclusive SD (Sorting Coeff.) = 1.79 phi (Poorly Sorted)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.42 phi (Medium Sand)

Inclusive Skewness = .38 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	7.00
-0.5	13.85
0.0	15.53
0.5	12.40
1.0	7.90
1.5	6.18
2.0	7.50
2.5	8.53
3.0	7.03
3.5	4.17
4.0	2.15
4.5	1.50
5.0	1.45
5.5	1.29
6.0	0.97
6.5	0.71
7.0	0.56
7.5	

Phi	Volume In %
7.5	0.47
8.0	0.38
8.5	0.27
9.0	0.16
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:30:33 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 588.29um = .77phi

Surface Weighted Mean:

D[3,2] = 84.95um = 3.56phi

Span :

2.538

Uniformity:

0.83

Specific Surface Area*:

0.0706 m²/g

Percentiles:

d(0.05) = 23.07um = 5.44phi

d(0.10) = 63.22um = 3.98phi

d(0.16) = 99.1um = 3.33phi

d(0.50) = 492.3um = 1.02phi

d(0.84) = 1133.89um = -0.18phi

d(0.90) = 1312.71um = -0.39phi

d(0.95) = 1522.59um = -0.61phi

Distribution Properties:

Kurtosis = -0.48

Skewness = .71

Standard Deviation = 488.84 um = 1.03phi

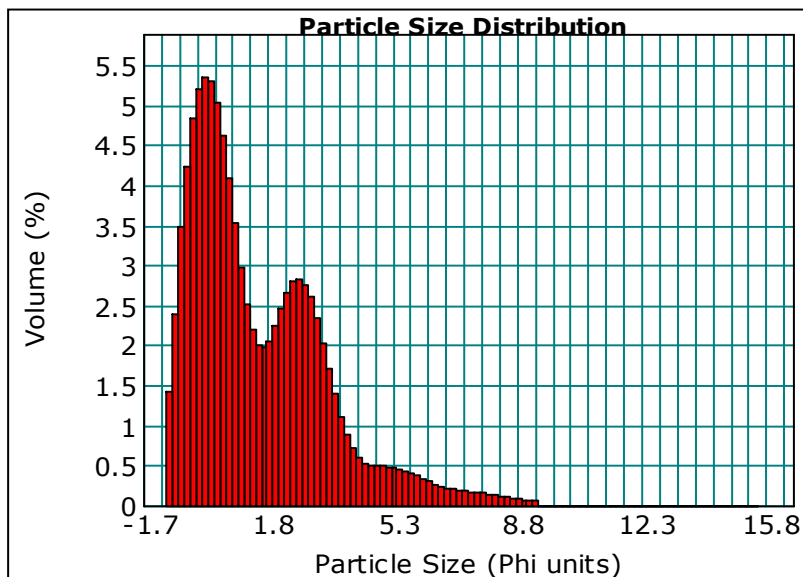
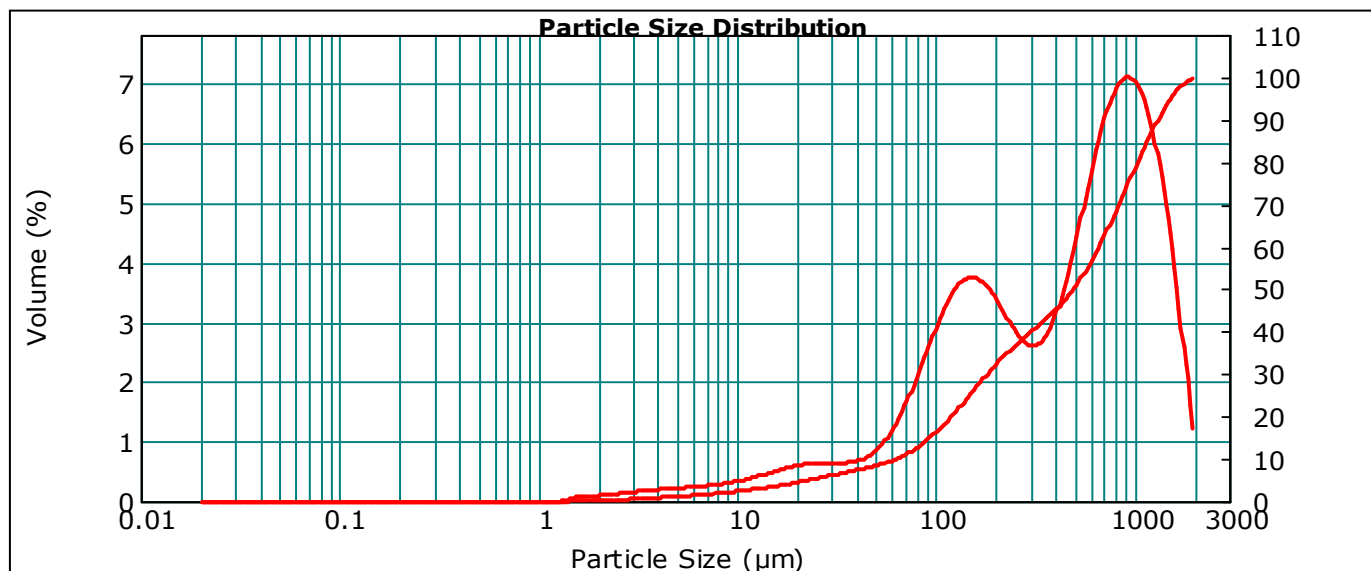
Inclusive SD (Sorting Coeff.) = 1.79 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.39 phi (Medium Sand)

Inclusive Skewness = .39 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	7.33
-0.5	14.34
0.0	15.70
0.5	12.20
1.0	7.65
1.5	6.05
2.0	7.41
2.5	8.40
3.0	6.91
3.5	4.11
4.0	2.13
4.5	1.50
5.0	1.45
5.5	1.29
6.0	0.98
6.5	0.71
7.0	0.56
7.5	

Phi	Volume In %
7.5	0.47
8.0	0.38
8.5	0.27
9.0	0.16
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_18
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:31:41 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 606.44um = .72phi

Surface Weighted Mean:
D[3,2] = 91um = 3.46phi

Span :
2.378

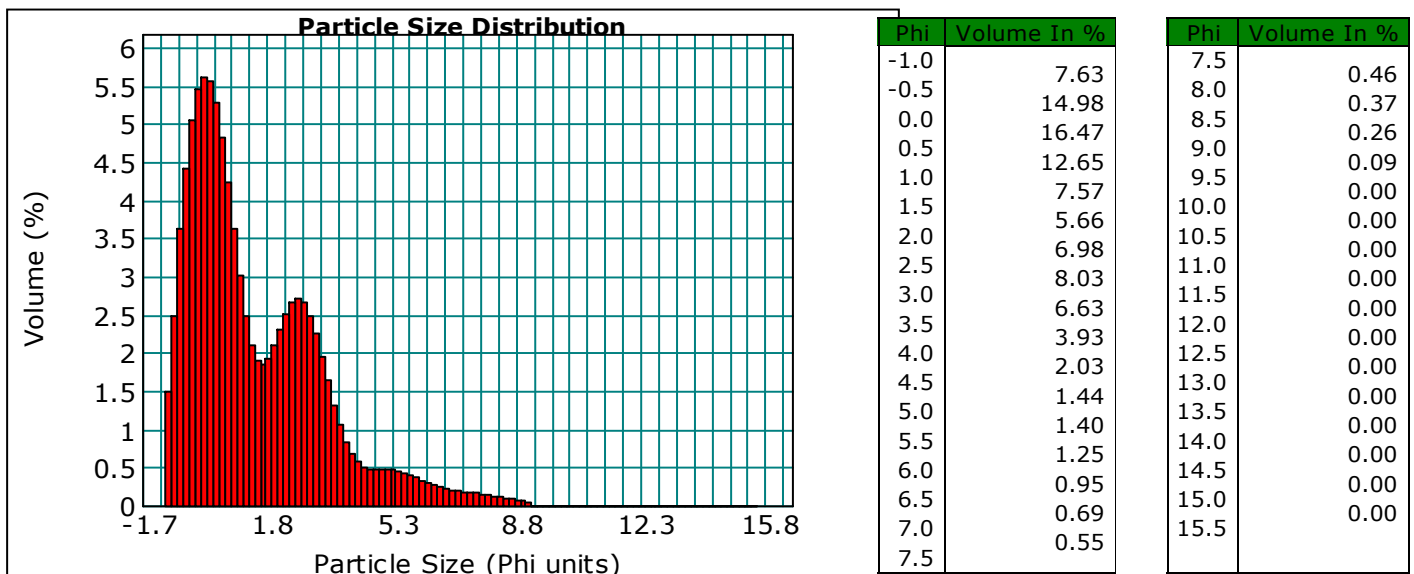
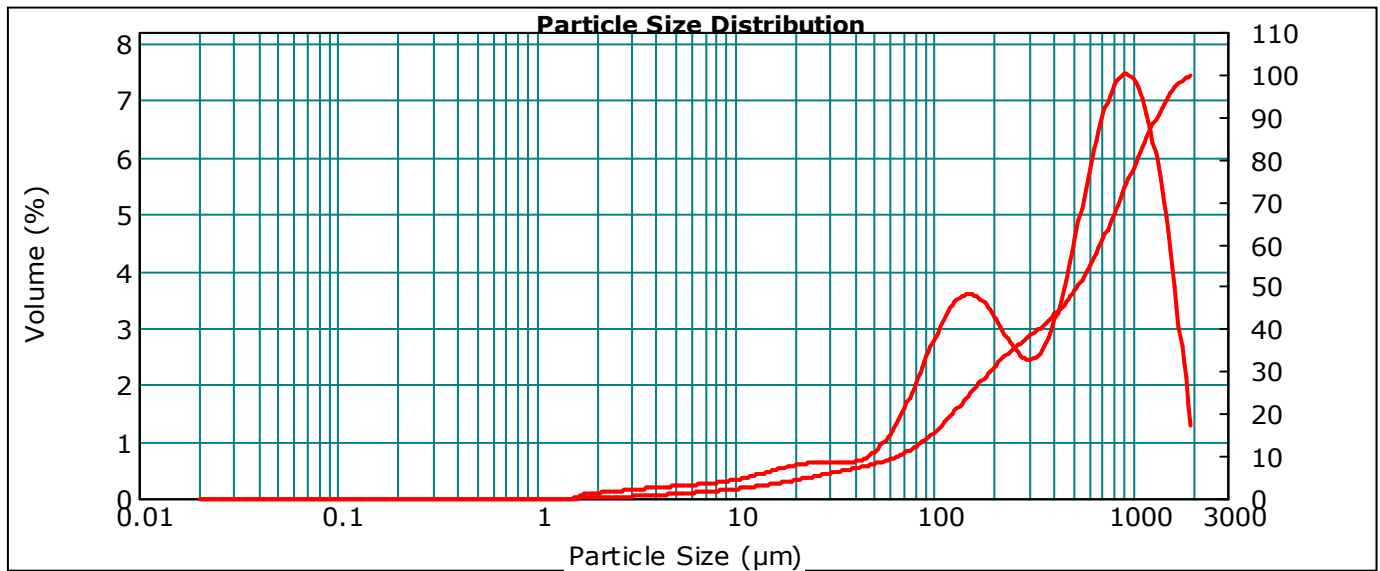
Uniformity:
0.777

Specific Surface Area*:
0.0659 m²/g

Percentiles:
d(0.05) = 24.32um = 5.36phi
d(0.10) = 66.43um = 3.91phi
d(0.16) = 102.82um = 3.28phi
d(0.50) = 529.73um = .92phi
d(0.84) = 1150.65um = -0.2phi
d(0.90) = 1326.3um = -0.41phi
d(0.95) = 1532.55um = -0.62phi

Distribution Properties:
Kurtosis = -0.55
Skewness = .65
Standard Deviation = 490.54 um = 1.03phi
Inclusive SD (Sorting Coeff.) = 1.78 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.33 phi (Medium Sand)
Inclusive Skewness = .42 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_18 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:29:26 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 591.1um = .76phi

Surface Weighted Mean:

D[3,2] = 86.78um = 3.53phi

Span :

2.494

Uniformity:

0.815

Specific Surface Area*:

0.0691 m²/g

Percentiles:

d(0.05) = 23.5um = 5.41phi

d(0.10) = 64.27um = 3.96phi

d(0.16) = 100.13um = 3.32phi

d(0.50) = 500.44um = 1phi

d(0.84) = 1133.89um = -0.18phi

d(0.90) = 1312.16um = -0.39phi

d(0.95) = 1521.87um = -0.61phi

Distribution Properties:

Kurtosis = -0.48

Skewness = .7

Standard Deviation = 487.84 um = 1.04phi

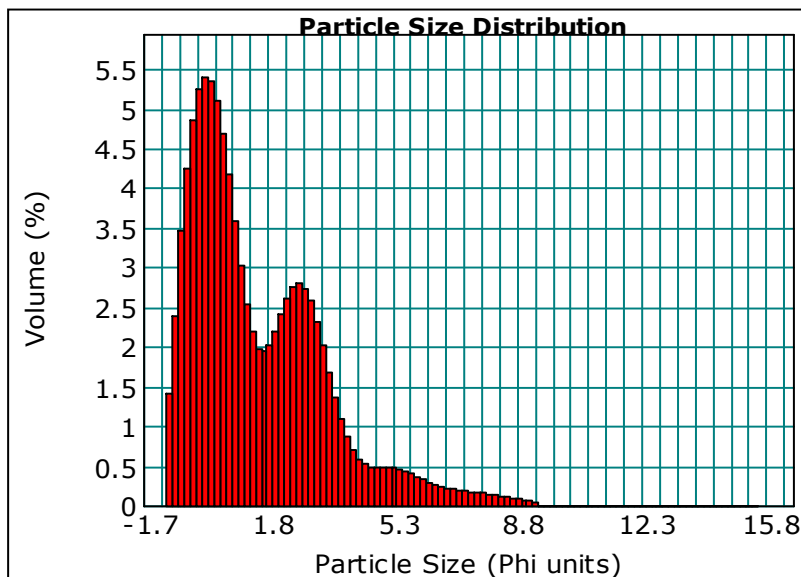
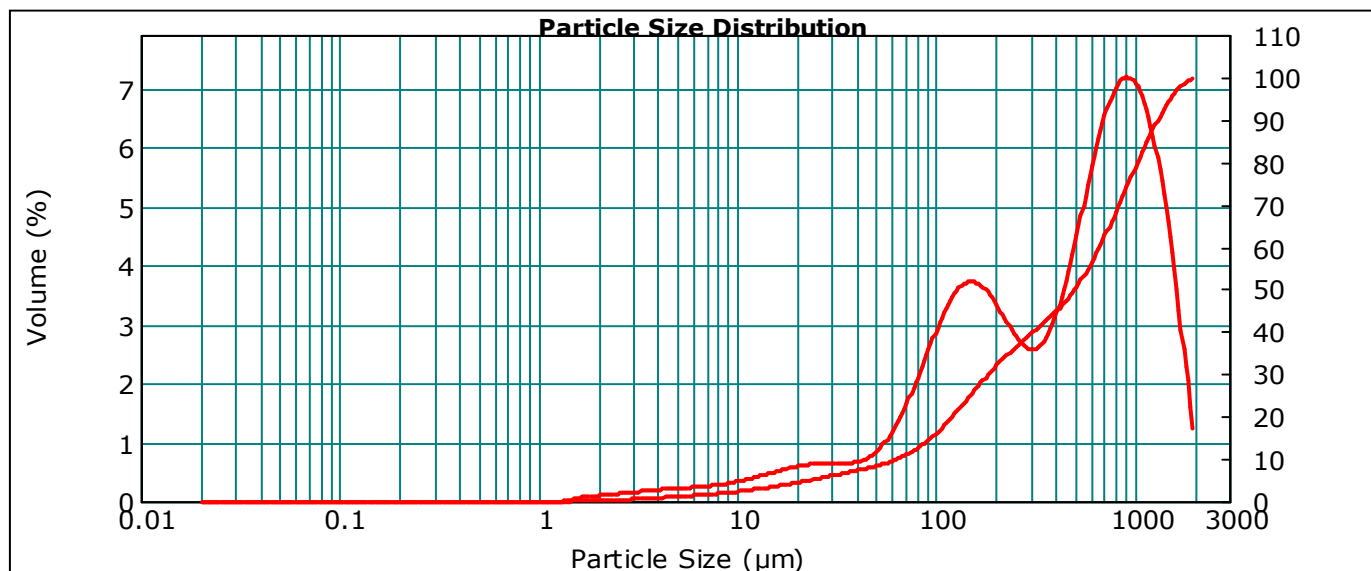
Inclusive SD (Sorting Coeff.) = 1.79 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.38 phi (Medium Sand)

Inclusive Skewness = .4 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	7.32
-0.5	14.39
0.0	15.90
0.5	12.42
1.0	7.71
1.5	5.96
2.0	7.30
2.5	8.32
3.0	6.85
3.5	4.07
4.0	2.10
4.5	1.48
5.0	1.43
5.5	1.28
6.0	0.97
6.5	0.70
7.0	0.55
7.5	

Phi	Volume In %
7.5	0.47
8.0	0.38
8.5	0.27
9.0	0.14
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_17
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:40:37 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 537.23um = .9phi

Surface Weighted Mean:

D[3,2] = 67.71um = 3.88phi

Span :

3.266

Uniformity:

1.07

Specific Surface Area*:

0.0886 m²/g

Percentiles:

d(0.05) = 17.62um = 5.83phi

d(0.10) = 45.77um = 4.45phi

d(0.16) = 80.63um = 3.63phi

d(0.50) = 376.21um = 1.41phi

d(0.84) = 1089.11um = -0.12phi

d(0.90) = 1274.33um = -0.35phi

d(0.95) = 1493.19um = -0.58phi

Distribution Properties:

Kurtosis = -0.31

Skewness = .84

Standard Deviation = 487.58 um = 1.04phi

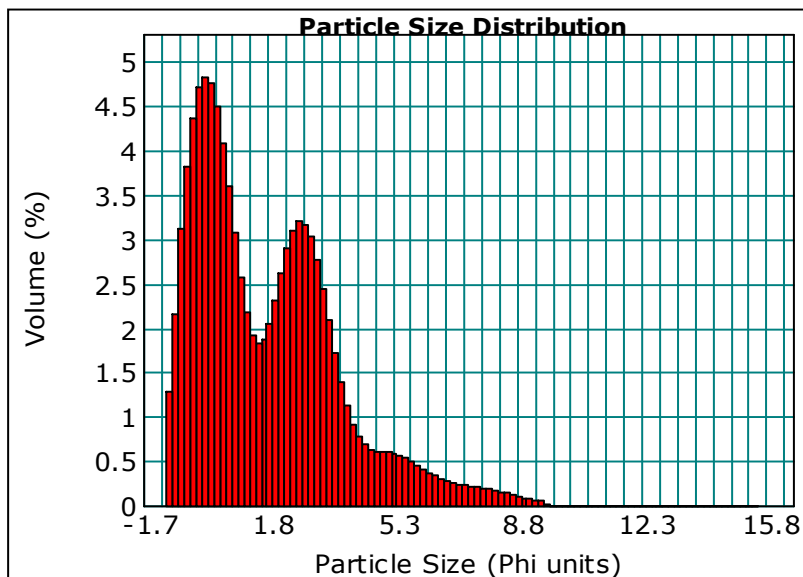
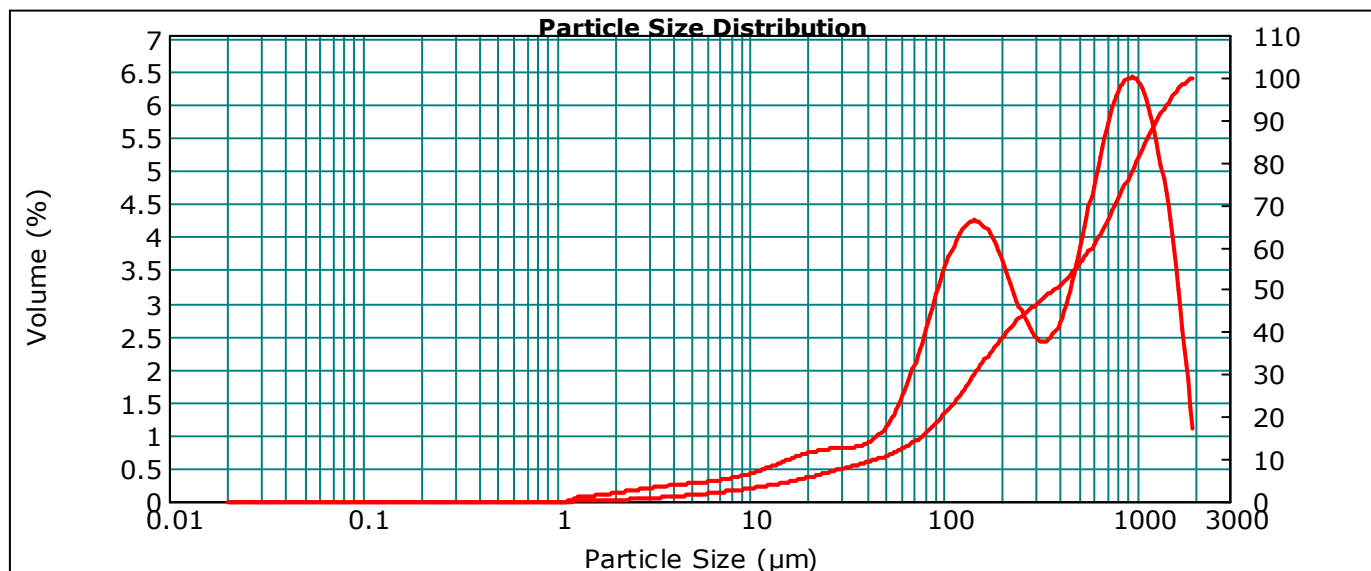
Inclusive SD (Sorting Coeff.) = 1.91 phi (Poorly Sorted)

Inclusive Kurtosis = .92 phi (Mesokurtic)

Inclusive Mean = 1.64 phi (Medium Sand)

Inclusive Skewness = .28 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.57
-0.5	12.94
0.0	14.10
0.5	10.73
1.0	6.66
1.5	5.77
2.0	7.89
2.5	9.49
3.0	8.17
3.5	5.12
4.0	2.76
4.5	1.92
5.0	1.79
5.5	1.59
6.0	1.22
6.5	0.89
7.0	0.70
7.5	

Phi	Volume In %
7.5	0.59
8.0	0.48
8.5	0.34
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_17
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:41:44 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 510.88um = .97phi

Surface Weighted Mean:
D[3,2] = 65.18um = 3.94phi

Span :
3.594

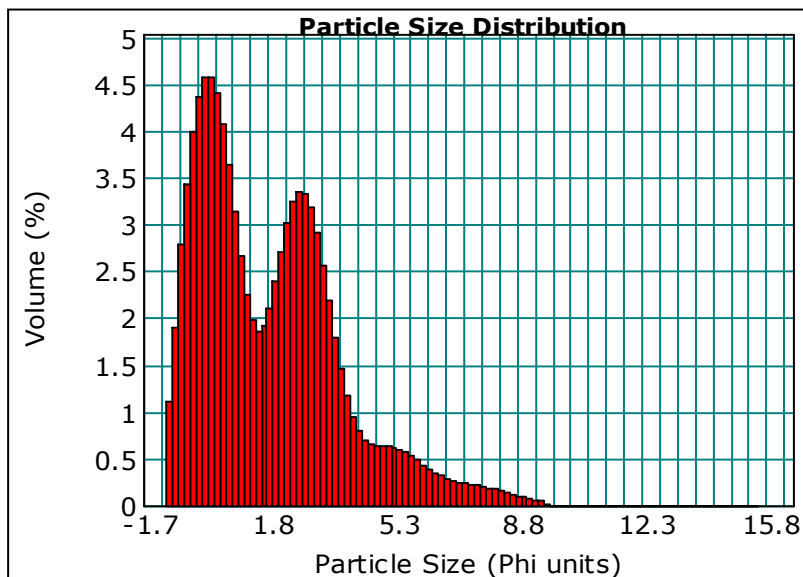
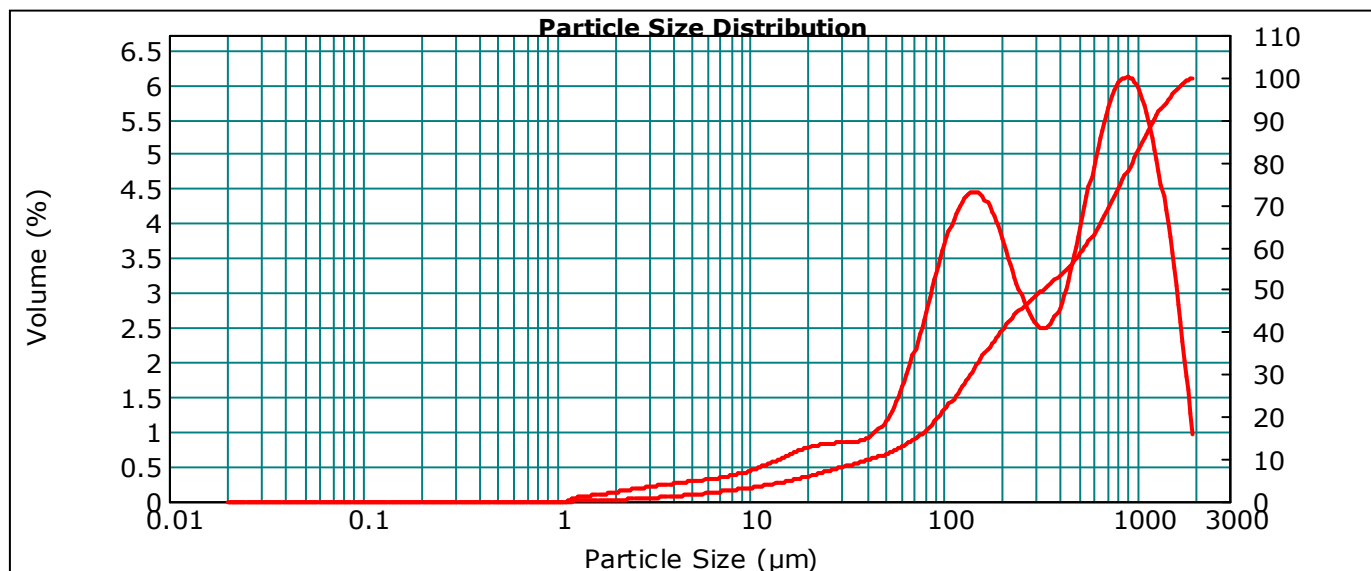
Uniformity:
1.17

Specific Surface Area*:
0.0921 m²/g

Percentiles:
d(0.05) = 16.75um = 5.9phi
d(0.10) = 42.58um = 4.55phi
d(0.16) = 77.41um = 3.69phi
d(0.50) = 330.96um = 1.6phi
d(0.84) = 1043.6um = -0.06phi
d(0.90) = 1232.08um = -0.3phi
d(0.95) = 1458.63um = -0.54phi

Distribution Properties:
Kurtosis = -0.11
Skewness = .92
Standard Deviation = 475.23 um = 1.07phi
Inclusive SD (Sorting Coeff.) = 1.91 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.74 phi (Medium Sand)
Inclusive Skewness = .23 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.82
-0.5	11.84
0.0	13.58
0.5	10.83
1.0	6.86
1.5	5.91
2.0	8.19
2.5	9.95
3.0	8.58
3.5	5.34
4.0	2.86
4.5	1.99
5.0	1.88
5.5	1.68
6.0	1.28
6.5	0.93
7.0	0.74
7.5	

Phi	Volume In %
7.5	0.62
8.0	0.50
8.5	0.35
9.0	0.22
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_17
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:42:52 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 530.11um = .92phi

Surface Weighted Mean:
D[3,2] = 65.93um = 3.92phi

Span :
3.467

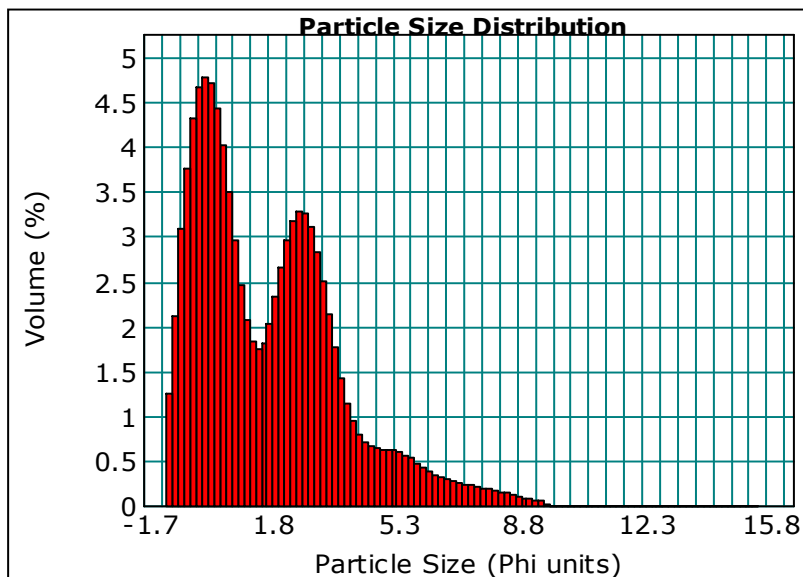
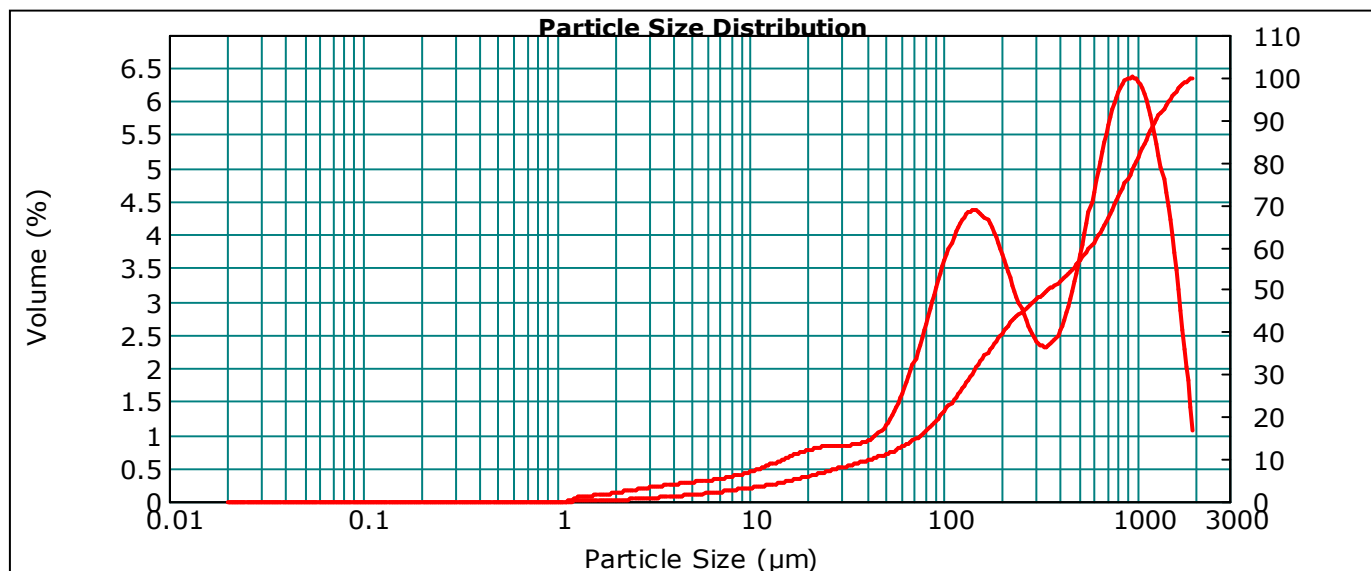
Uniformity:
1.13

Specific Surface Area*:
0.091 m²/g

Percentiles:
d(0.05) = 16.84um = 5.89phi
d(0.10) = 42.95um = 4.54phi
d(0.16) = 78.1um = 3.68phi
d(0.50) = 353.59um = 1.5phi
d(0.84) = 1083.4um = -0.12phi
d(0.90) = 1269um = -0.34phi
d(0.95) = 1488.83um = -0.57phi

Distribution Properties:
Kurtosis = -0.29
Skewness = .86
Standard Deviation = 487.7 um = 1.04phi
Inclusive SD (Sorting Coeff.) = 1.93 phi (Poorly Sorted)
Inclusive Kurtosis = .92 phi (Mesokurtic)
Inclusive Mean = 1.69 phi (Medium Sand)
Inclusive Skewness = .25 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.47
-0.5	12.80
0.0	13.95
0.5	10.45
1.0	6.33
1.5	5.62
2.0	8.02
2.5	9.74
3.0	8.38
3.5	5.23
4.0	2.83
4.5	1.98
5.0	1.87
5.5	1.66
6.0	1.28
6.5	0.93
7.0	0.74
7.5	

Phi	Volume In %
7.5	0.62
8.0	0.49
8.5	0.34
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_17 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:40:37 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 526.07um = .93phi

Surface Weighted Mean:
D[3,2] = 66.26um = 3.92phi

Span :
3.442

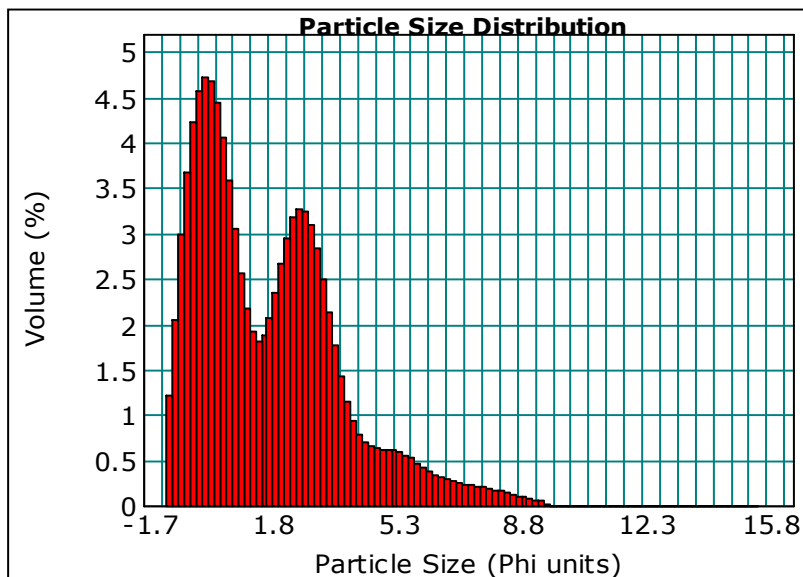
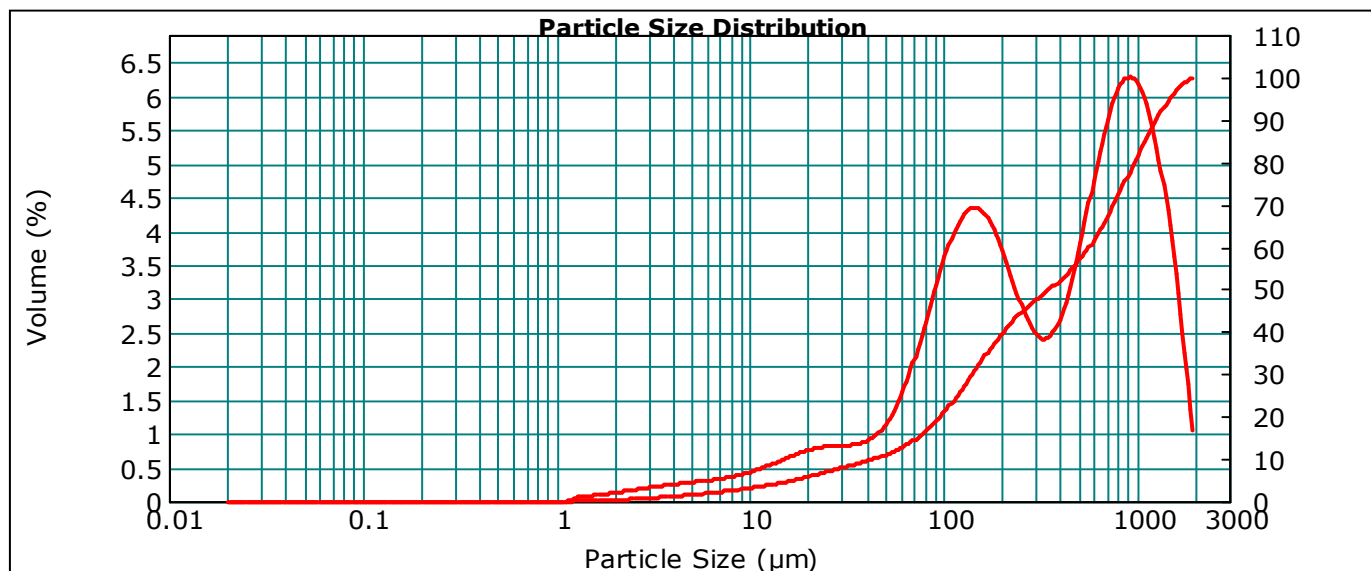
Uniformity:
1.12

Specific Surface Area*:
0.0906 m²/g

Percentiles:
d(0.05) = 17.06um = 5.87phi
d(0.10) = 43.75um = 4.51phi
d(0.16) = 78.7um = 3.67phi
d(0.50) = 353.11um = 1.5phi
d(0.84) = 1072.43um = -0.1phi
d(0.90) = 1259.04um = -0.33phi
d(0.95) = 1480.61um = -0.57phi

Distribution Properties:
Kurtosis = -0.24
Skewness = .87
Standard Deviation = 483.67 um = 1.05phi
Inclusive SD (Sorting Coeff.) = 1.92 phi (Poorly Sorted)
Inclusive Kurtosis = .92 phi (Mesokurtic)
Inclusive Mean = 1.69 phi (Medium Sand)
Inclusive Skewness = .25 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.29
-0.5	12.53
0.0	13.88
0.5	10.67
1.0	6.62
1.5	5.77
2.0	8.03
2.5	9.73
3.0	8.38
3.5	5.23
4.0	2.82
4.5	1.96
5.0	1.84
5.5	1.64
6.0	1.26
6.5	0.92
7.0	0.72
7.5	

Phi	Volume In %
7.5	0.61
8.0	0.49
8.5	0.34
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:51:32 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 505.8um = .98phi

Surface Weighted Mean:

D[3,2] = 67.41um = 3.89phi

Span :

3.664

Uniformity:

1.18

Specific Surface Area*:

0.089 m²/g

Percentiles:

d(0.05) = 17.71um = 5.82phi

d(0.10) = 45.13um = 4.47phi

d(0.16) = 79.71um = 3.65phi

d(0.50) = 321.41um = 1.64phi

d(0.84) = 1033.12um = -0.05phi

d(0.90) = 1222.68um = -0.29phi

d(0.95) = 1450.93um = -0.54phi

Distribution Properties:

Kurtosis = -0.04

Skewness = .95

Standard Deviation = 471.12 um = 1.09phi

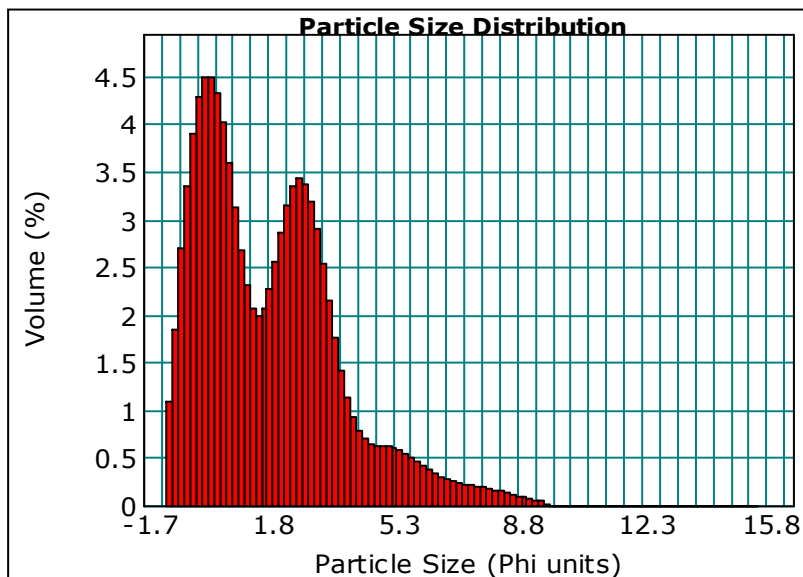
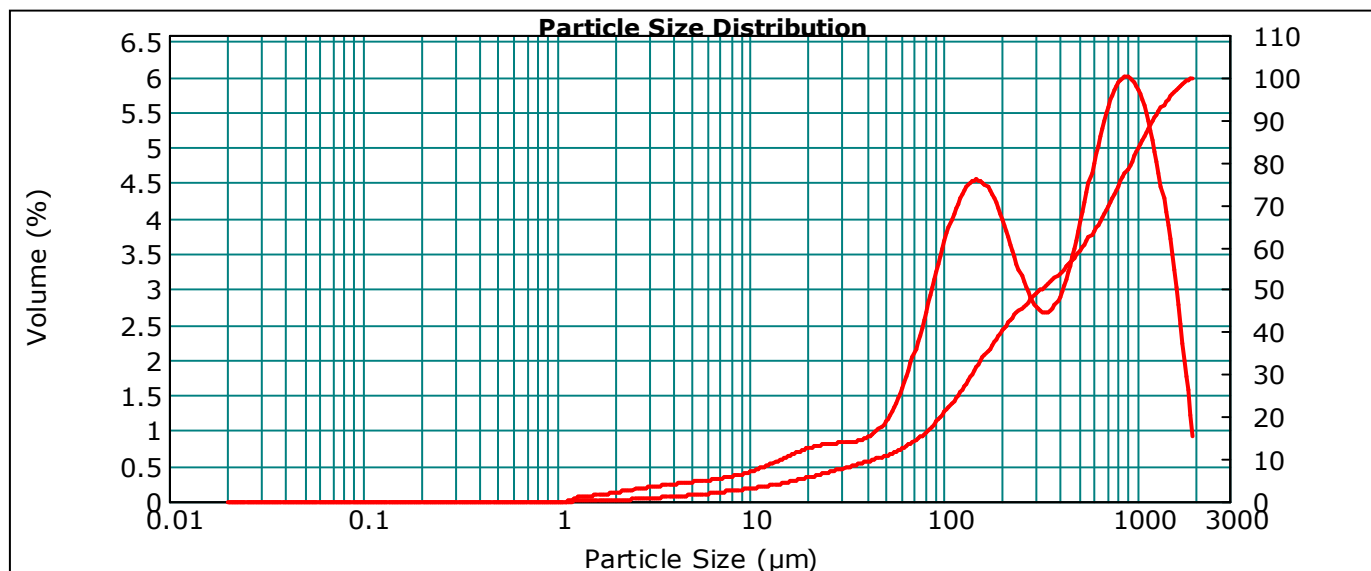
Inclusive SD (Sorting Coeff.) = 1.89 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.75 phi (Medium Sand)

Inclusive Skewness = .2 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.66
-0.5	11.58
0.0	13.35
0.5	10.72
1.0	7.04
1.5	6.38
2.0	8.65
2.5	10.16
3.0	8.55
3.5	5.24
4.0	2.80
4.5	1.96
5.0	1.83
5.5	1.61
6.0	1.22
6.5	0.88
7.0	0.70
7.5	

Phi	Volume In %
7.5	0.59
8.0	0.47
8.5	0.33
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:52:39 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 490.97um = 1.03phi

Surface Weighted Mean:
D[3,2] = 65.16um = 3.94phi

Span :
3.976

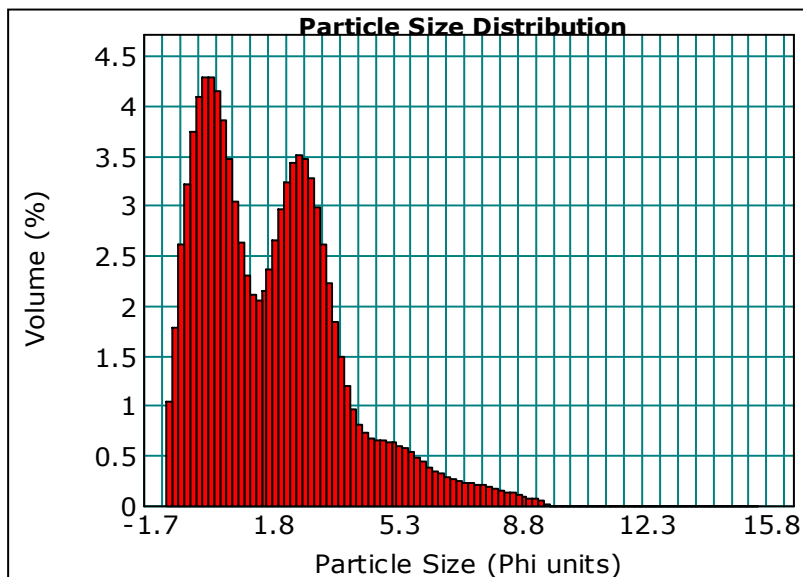
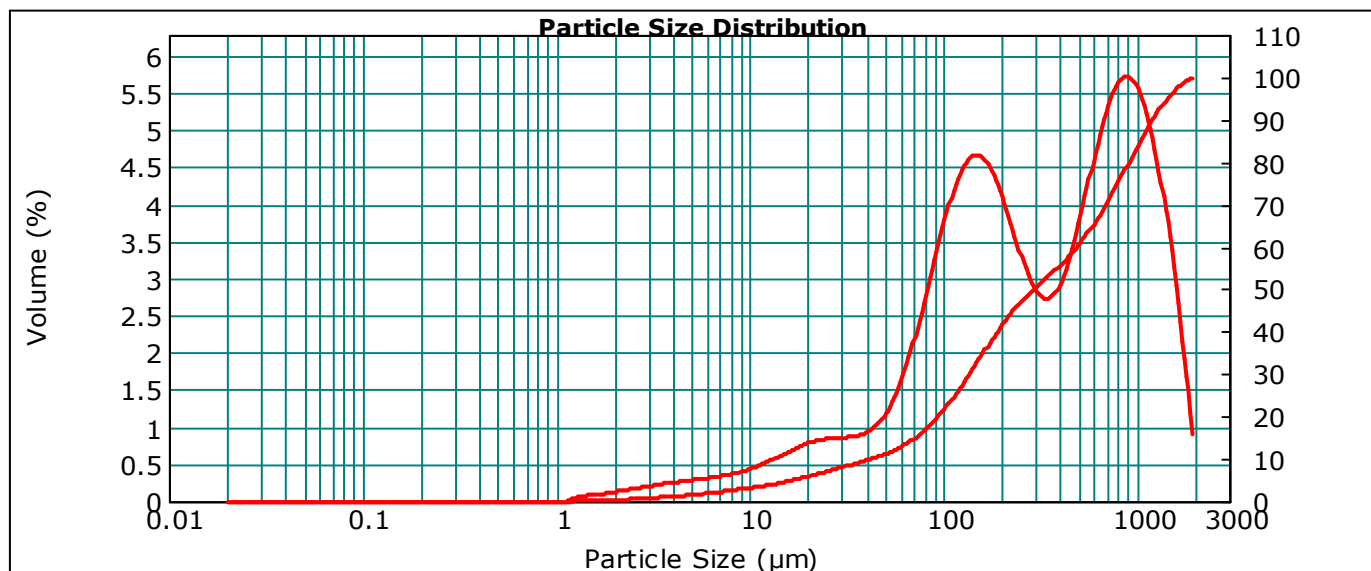
Uniformity:
1.27

Specific Surface Area*:
0.0921 m²/g

Percentiles:
d(0.05) = 16.89um = 5.89phi
d(0.10) = 42.24um = 4.57phi
d(0.16) = 76.63um = 3.71phi
d(0.50) = 293.16um = 1.77phi
d(0.84) = 1014.87um = -0.02phi
d(0.90) = 1207.91um = -0.27phi
d(0.95) = 1440um = -0.53phi

Distribution Properties:
Kurtosis = .07
Skewness = 1.01
Standard Deviation = 467.59 um = 1.1phi
Inclusive SD (Sorting Coeff.) = 1.9 phi (Poorly Sorted)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = 1.82 phi (Medium Sand)
Inclusive Skewness = .16 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.45
-0.5	11.08
0.0	12.73
0.5	10.33
1.0	7.03
1.5	6.58
2.0	8.91
2.5	10.43
3.0	8.81
3.5	5.44
4.0	2.92
4.5	2.04
5.0	1.91
5.5	1.68
6.0	1.28
6.5	0.93
7.0	0.73
7.5	

Phi	Volume In %
7.5	0.62
8.0	0.49
8.5	0.34
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_16
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:53:47 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 484.42um = 1.05phi

Surface Weighted Mean:
D[3,2] = 64.38um = 3.96phi

Span :
3.982

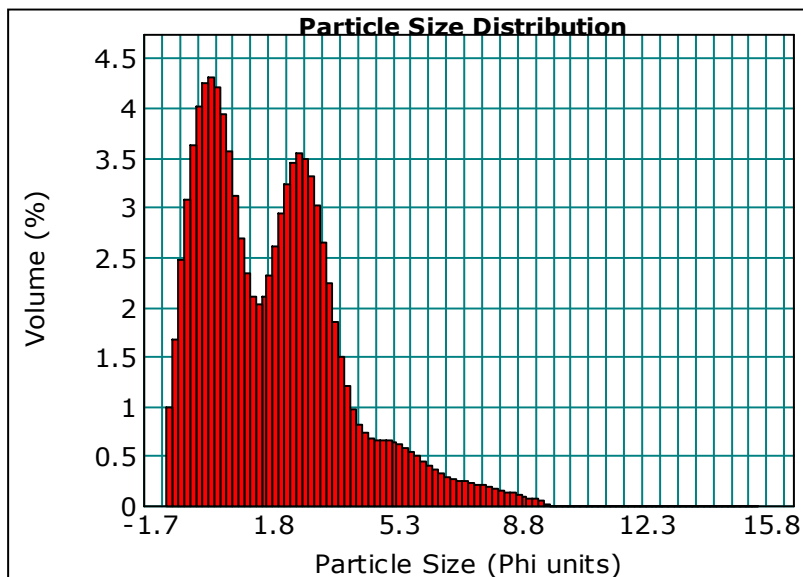
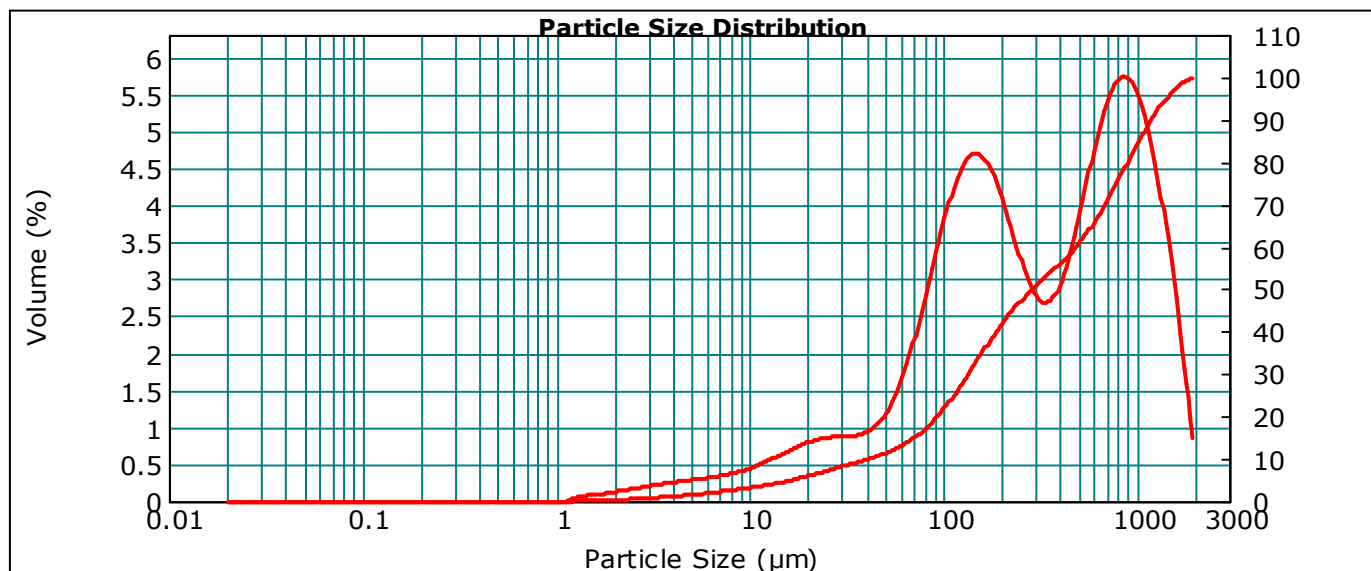
Uniformity:
1.27

Specific Surface Area*:
0.0932 m²/g

Percentiles:
d(0.05) = 16.54um = 5.92phi
d(0.10) = 41.01um = 4.61phi
d(0.16) = 75.64um = 3.72phi
d(0.50) = 288.56um = 1.79phi
d(0.84) = 998.02um = 0phi
d(0.90) = 1189.95um = -0.25phi
d(0.95) = 1423.65um = -0.51phi

Distribution Properties:
Kurtosis = .13
Skewness = 1.02
Standard Deviation = 462.17 um = 1.11phi
Inclusive SD (Sorting Coeff.) = 1.9 phi (Poorly Sorted)
Inclusive Kurtosis = .95 phi (Mesokurtic)
Inclusive Mean = 1.84 phi (Medium Sand)
Inclusive Skewness = .16 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.16
-0.5	10.77
0.0	12.80
0.5	10.59
1.0	7.08
1.5	6.46
2.0	8.87
2.5	10.51
3.0	8.89
3.5	5.47
4.0	2.93
4.5	2.06
5.0	1.95
5.5	1.72
6.0	1.31
6.5	0.95
7.0	0.75
7.5	

Phi	Volume In %
7.5	0.63
8.0	0.50
8.5	0.35
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_16 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 11:51:32 AM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 493.73um = 1.02phi

Surface Weighted Mean:
D[3,2] = 65.62um = 3.93phi

Span :
3.876

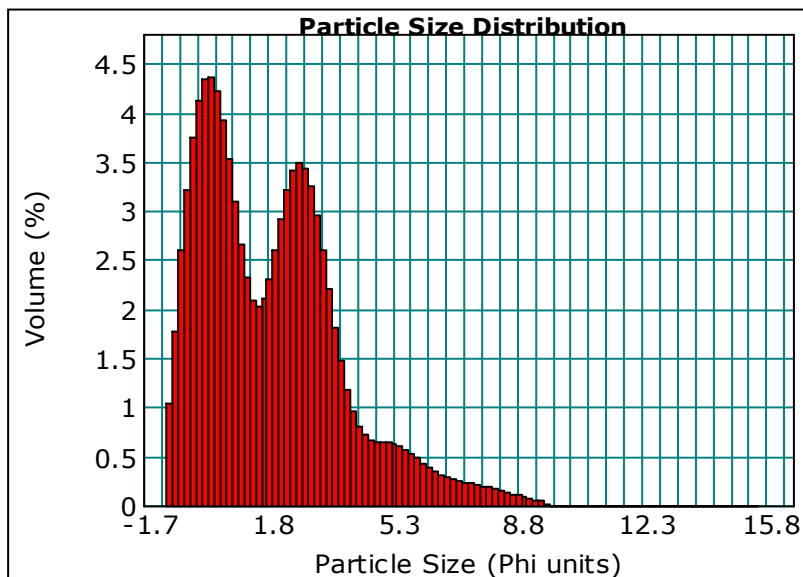
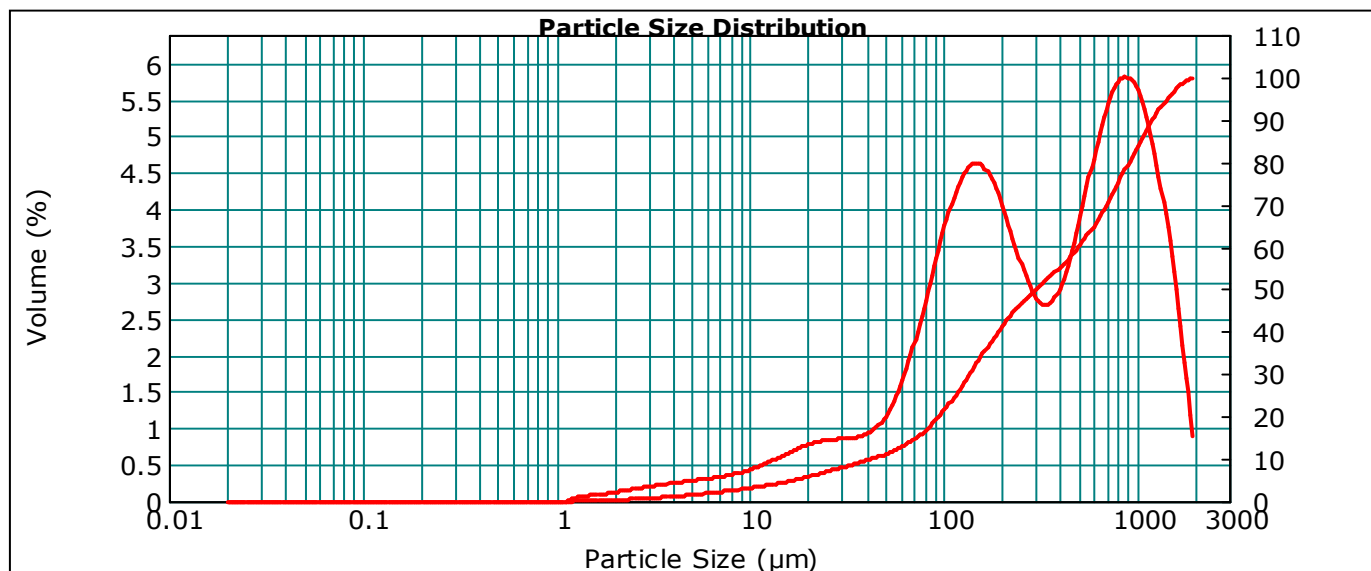
Uniformity:
1.24

Specific Surface Area*:
0.0914 m²/g

Percentiles:
d(0.05) = 17.03um = 5.88phi
d(0.10) = 42.76um = 4.55phi
d(0.16) = 77.31um = 3.69phi
d(0.50) = 300.38um = 1.74phi
d(0.84) = 1015.53um = -0.02phi
d(0.90) = 1207.03um = -0.27phi
d(0.95) = 1438.38um = -0.52phi

Distribution Properties:
Kurtosis = .05
Skewness = 1
Standard Deviation = 467.06 um = 1.1phi
Inclusive SD (Sorting Coeff.) = 1.9 phi (Poorly Sorted)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = 1.8 phi (Medium Sand)
Inclusive Skewness = .17 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.43
-0.5	11.14
0.0	12.96
0.5	10.55
1.0	7.05
1.5	6.47
2.0	8.81
2.5	10.37
3.0	8.75
3.5	5.38
4.0	2.89
4.5	2.02
5.0	1.90
5.5	1.67
6.0	1.27
6.5	0.92
7.0	0.73
7.5	

Phi	Volume In %
7.5	0.61
8.0	0.49
8.5	0.34
9.0	0.21
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:02:23 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 496.54um = 1.01phi

Surface Weighted Mean:
D[3,2] = 57.37um = 4.12phi

Span :
4.258

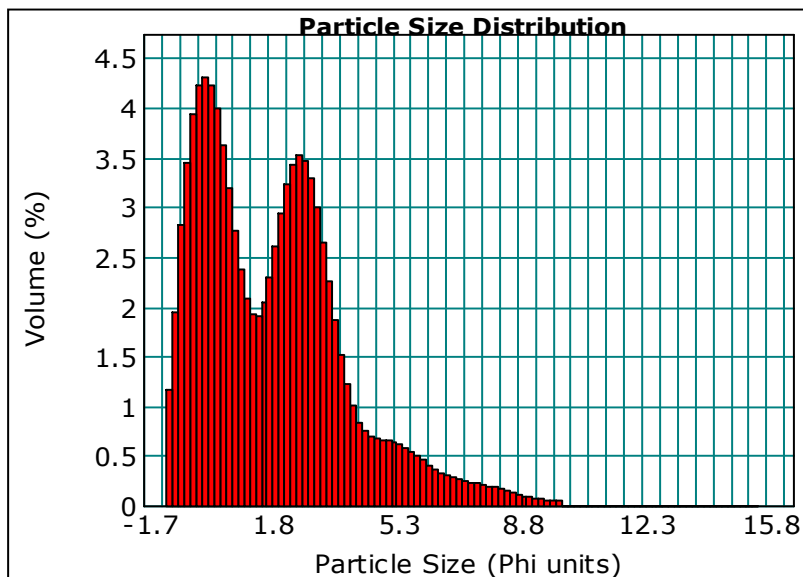
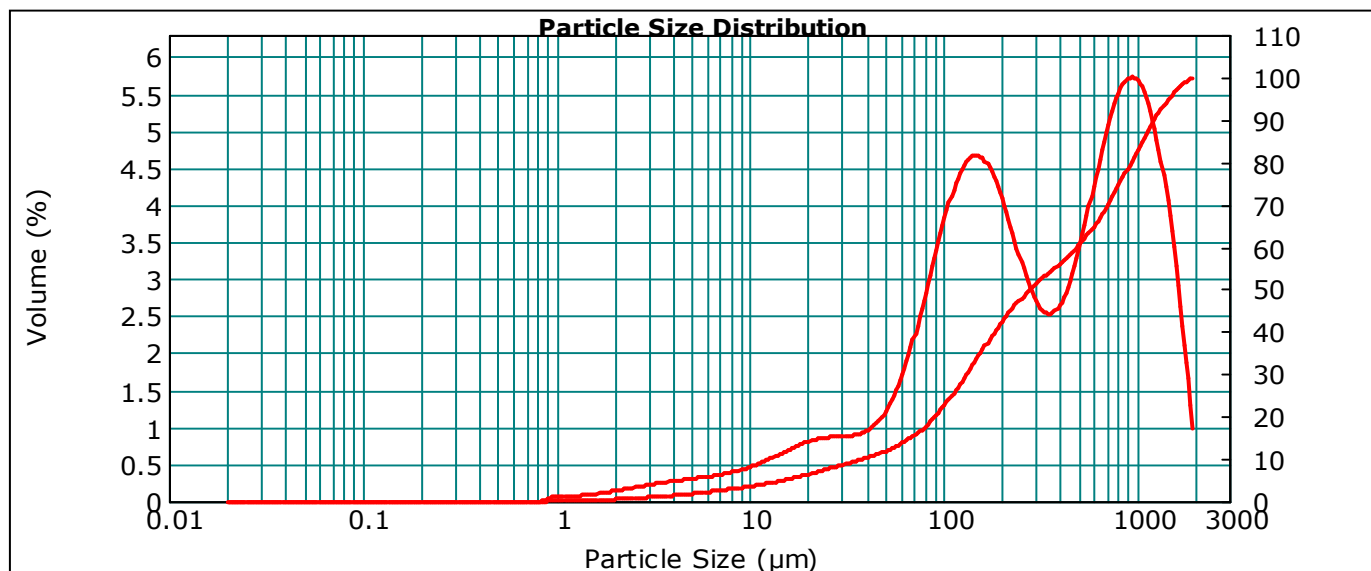
Uniformity:
1.35

Specific Surface Area*:
0.105 m²/g

Percentiles:
d(0.05) = 15.25um = 6.04phi
d(0.10) = 38.41um = 4.7phi
d(0.16) = 73.19um = 3.77phi
d(0.50) = 281.55um = 1.83phi
d(0.84) = 1043.72um = -0.06phi
d(0.90) = 1237.33um = -0.31phi
d(0.95) = 1465.76um = -0.55phi

Distribution Properties:
Kurtosis = -0.04
Skewness = .99
Standard Deviation = 479.38 um = 1.06phi
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = 1.85 phi (Medium Sand)
Inclusive Skewness = .15 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.95
-0.5	11.64
0.0	12.55
0.5	9.57
1.0	6.37
1.5	6.30
2.0	8.85
2.5	10.46
3.0	8.89
3.5	5.54
4.0	3.00
4.5	2.09
5.0	1.95
5.5	1.74
6.0	1.34
6.5	0.99
7.0	0.78
7.5	

Phi	Volume In %
7.5	0.66
8.0	0.53
8.5	0.38
9.0	0.24
9.5	0.16
10.0	0.03
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:03:30 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 500.17um = 1phi

Surface Weighted Mean:

D[3,2] = 59.59um = 4.07phi

Span :

4.204

Uniformity:

1.34

Specific Surface Area*:

0.101 m²/g

Percentiles:

d(0.05) = 15.36um = 6.02phi

d(0.10) = 38.32um = 4.71phi

d(0.16) = 73.11um = 3.77phi

d(0.50) = 286.39um = 1.8phi

d(0.84) = 1049.59um = -0.07phi

d(0.90) = 1242.42um = -0.31phi

d(0.95) = 1469.62um = -0.56phi

Distribution Properties:

Kurtosis = -0.08

Skewness = .97

Standard Deviation = 481.09 um = 1.06phi

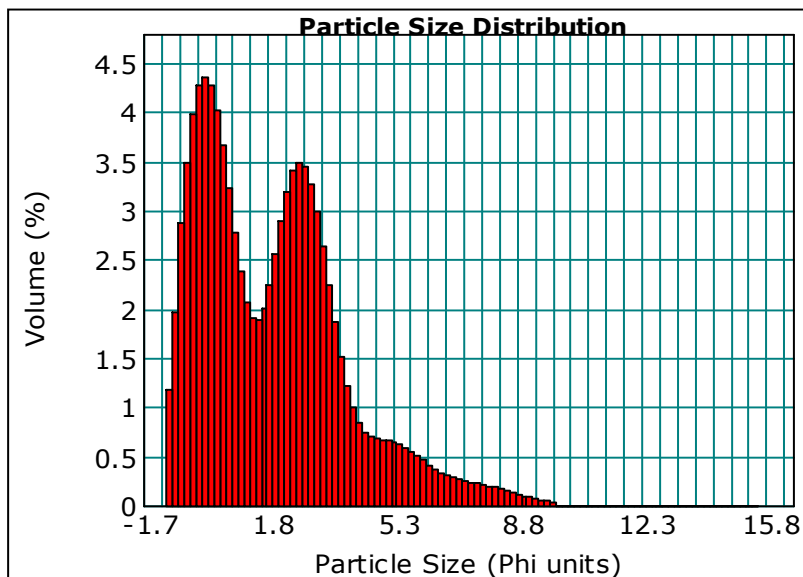
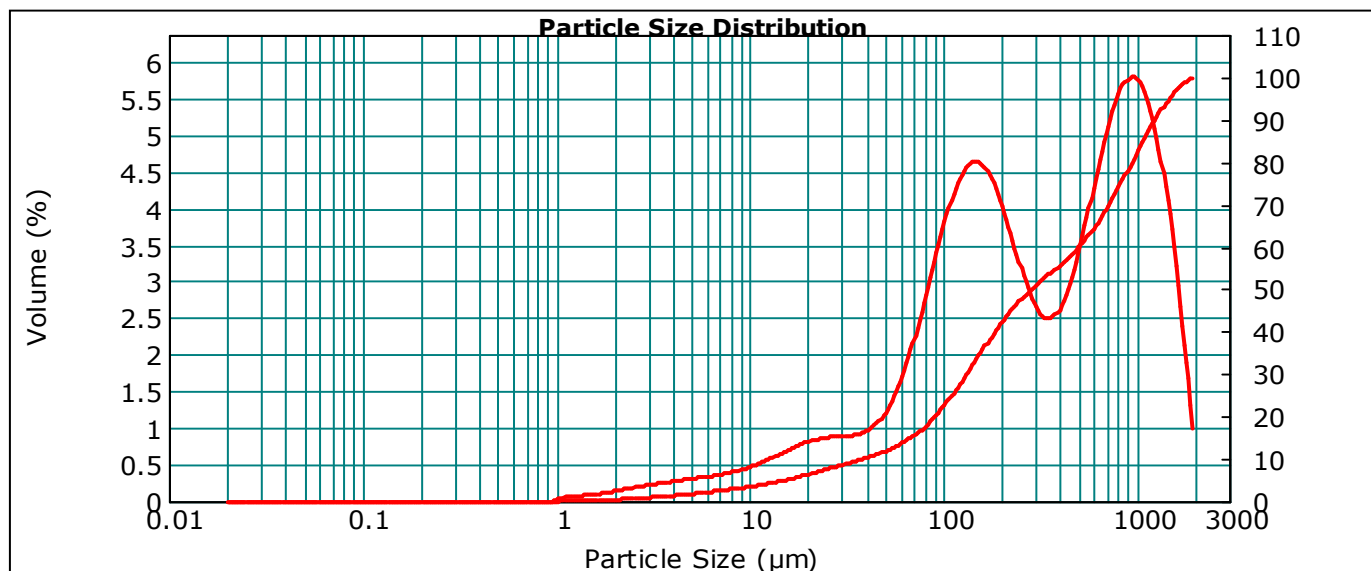
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.84 phi (Medium Sand)

Inclusive Skewness = .15 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.03
-0.5	11.78
0.0	12.68
0.5	9.64
1.0	6.33
1.5	6.17
2.0	8.72
2.5	10.37
3.0	8.84
3.5	5.52
4.0	3.00
4.5	2.10
5.0	1.97
5.5	1.75
6.0	1.35
6.5	1.00
7.0	0.79
7.5	

Phi	Volume In %
7.5	0.66
8.0	0.53
8.5	0.38
9.0	0.23
9.5	0.13
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_15
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:04:38 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 502.58um = .99phi

Surface Weighted Mean:

D[3,2] = 59.79um = 4.06phi

Span :

4.194

Uniformity:

1.34

Specific Surface Area*:

0.1 m²/g

Percentiles:

d(0.05) = 15.34um = 6.03phi

d(0.10) = 38.15um = 4.71phi

d(0.16) = 73.17um = 3.77phi

d(0.50) = 288.35um = 1.79phi

d(0.84) = 1054.8um = -0.08phi

d(0.90) = 1247.39um = -0.32phi

d(0.95) = 1473.69um = -0.56phi

Distribution Properties:

Kurtosis = -0.1

Skewness = .96

Standard Deviation = 482.75 um = 1.05phi

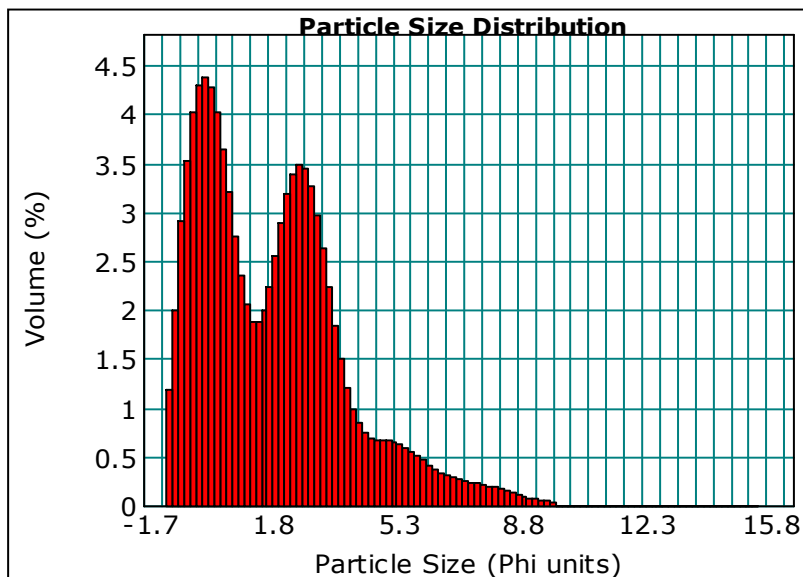
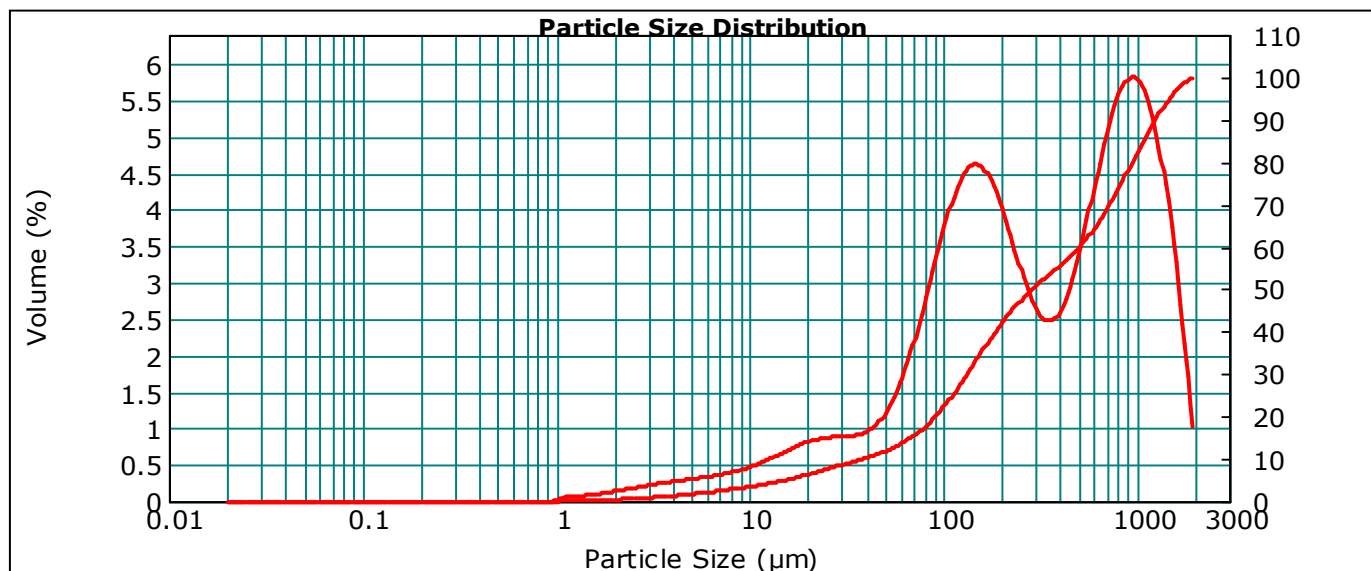
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.83 phi (Medium Sand)

Inclusive Skewness = .16 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.12
-0.5	11.89
0.0	12.72
0.5	9.59
1.0	6.28
1.5	6.15
2.0	8.71
2.5	10.34
3.0	8.80
3.5	5.48
4.0	2.98
4.5	2.11
5.0	1.98
5.5	1.76
6.0	1.36
6.5	1.00
7.0	0.79
7.5	

Phi	Volume In %
7.5	0.66
8.0	0.53
8.5	0.37
9.0	0.23
9.5	0.13
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_15 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:02:23 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 499.76um = 1phi

Surface Weighted Mean:

D[3,2] = 58.89um = 4.09phi

Span :

4.219

Uniformity:

1.34

Specific Surface Area*:

0.102 m²/g

Percentiles:

d(0.05) = 15.32um = 6.03phi

d(0.10) = 38.29um = 4.71phi

d(0.16) = 73.16um = 3.77phi

d(0.50) = 285.38um = 1.81phi

d(0.84) = 1049.4um = -0.07phi

d(0.90) = 1242.4um = -0.31phi

d(0.95) = 1469.72um = -0.56phi

Distribution Properties:

Kurtosis = -0.07

Skewness = .97

Standard Deviation = 481.08 um = 1.06phi

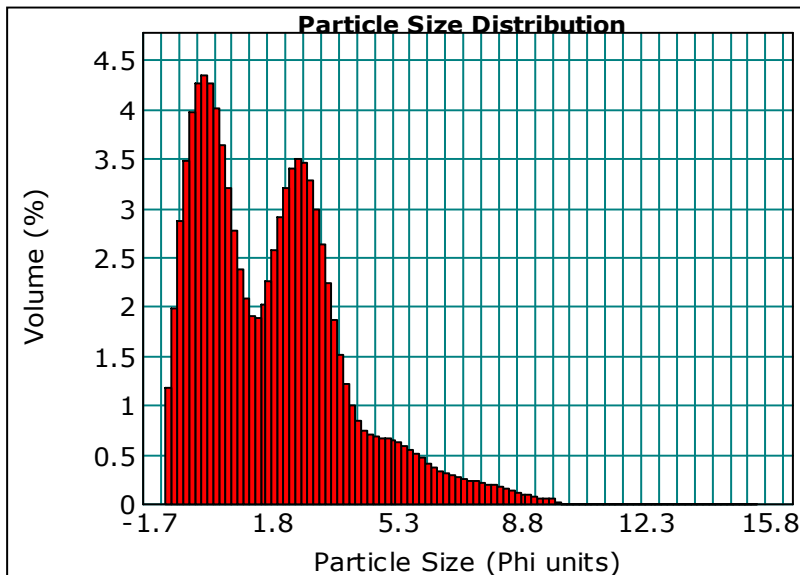
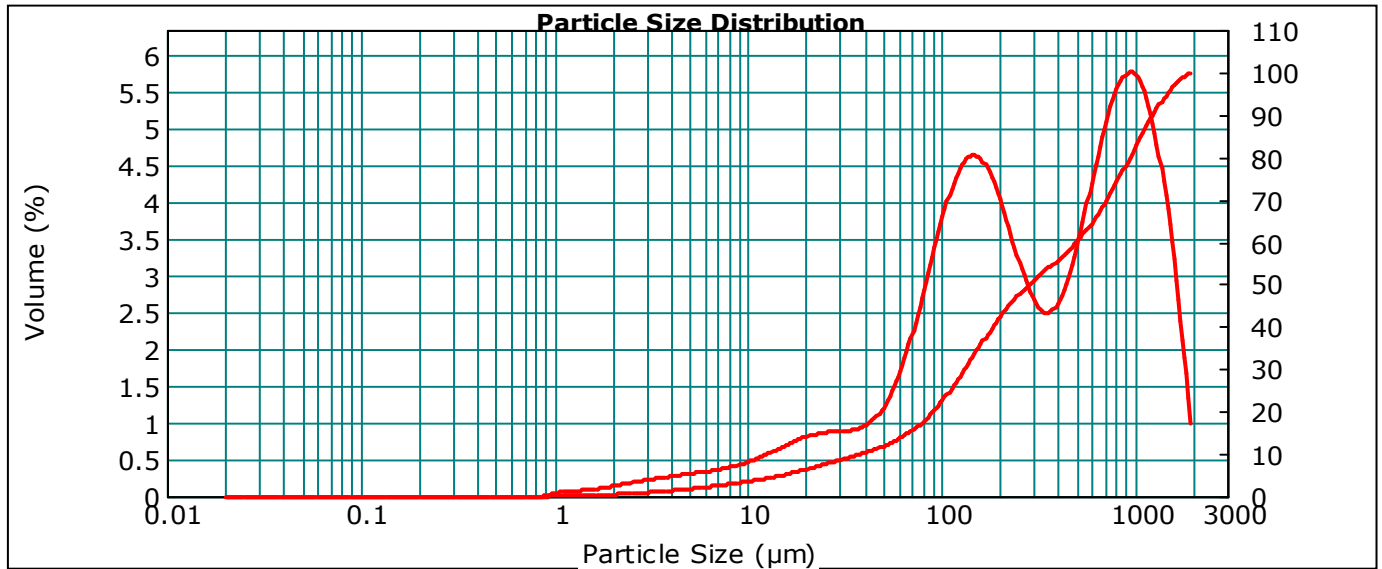
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.84 phi (Medium Sand)

Inclusive Skewness = .15 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.04
-0.5	11.77
0.0	12.65
0.5	9.60
1.0	6.33
1.5	6.21
2.0	8.76
2.5	10.39
3.0	8.84
3.5	5.52
4.0	3.00
4.5	2.10
5.0	1.97
5.5	1.75
6.0	1.35
6.5	1.00
7.0	0.79
7.5	

Phi	Volume In %
7.5	0.66
8.0	0.53
8.5	0.38
9.0	0.23
9.5	0.14
10.0	0.01
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:13:30 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 483.68um = 1.05phi

Surface Weighted Mean:
D[3,2] = 58.62um = 4.09phi

Span :
4.576

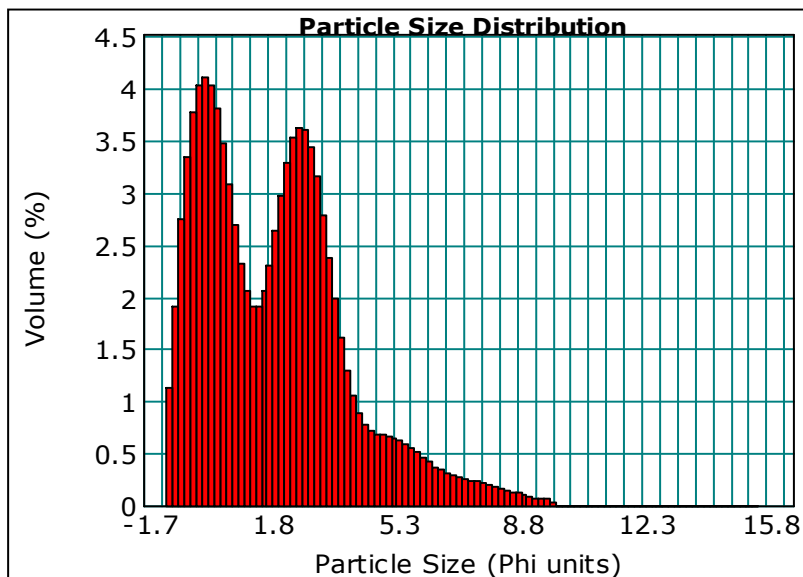
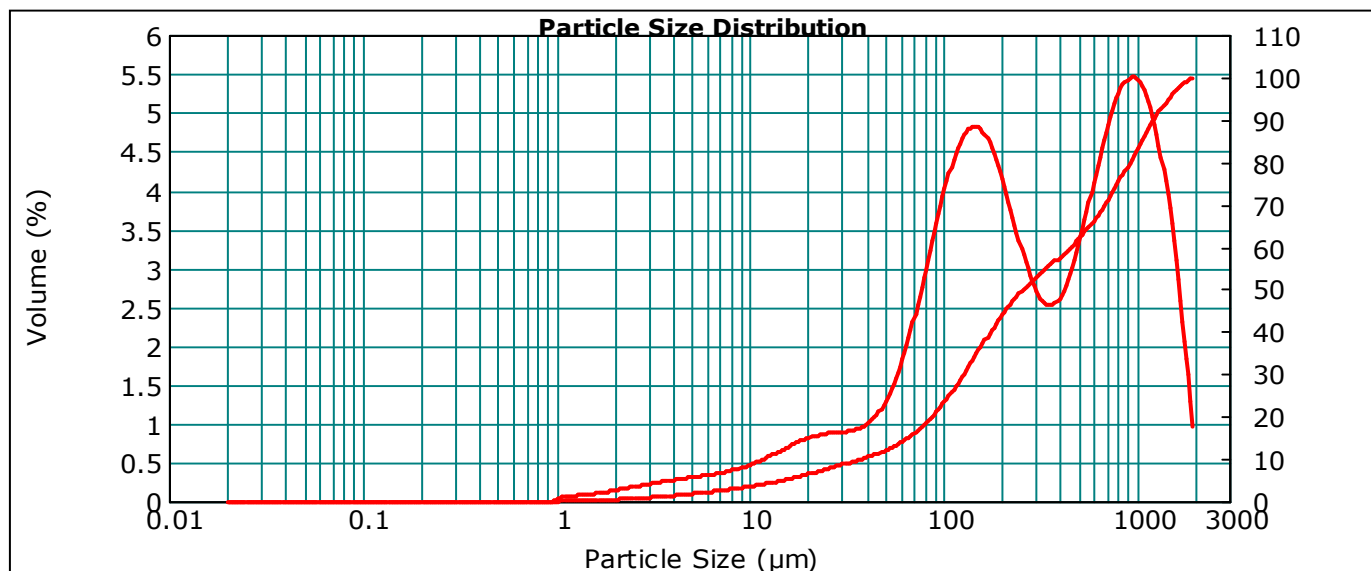
Uniformity:
1.44

Specific Surface Area*:
0.102 m²/g

Percentiles:
d(0.05) = 15.25um = 6.04phi
d(0.10) = 37.88um = 4.72phi
d(0.16) = 71.2um = 3.81phi
d(0.50) = 259.99um = 1.94phi
d(0.84) = 1029.23um = -0.04phi
d(0.90) = 1227.49um = -0.3phi
d(0.95) = 1459.79um = -0.55phi

Distribution Properties:
Kurtosis = .07
Skewness = 1.04
Standard Deviation = 477.04 um = 1.07phi
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.9 phi (Medium Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.82
-0.5	11.20
0.0	11.97
0.5	9.23
1.0	6.29
1.5	6.31
2.0	8.99
2.5	10.79
3.0	9.32
3.5	5.88
4.0	3.18
4.5	2.16
5.0	1.98
5.5	1.76
6.0	1.36
6.5	1.00
7.0	0.79
7.5	

Phi	Volume In %
7.5	0.67
8.0	0.54
8.5	0.38
9.0	0.24
9.5	0.13
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:14:38 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 451.19um = 1.15phi

Surface Weighted Mean:
D[3,2] = 52.09um = 4.26phi

Span :
4.806

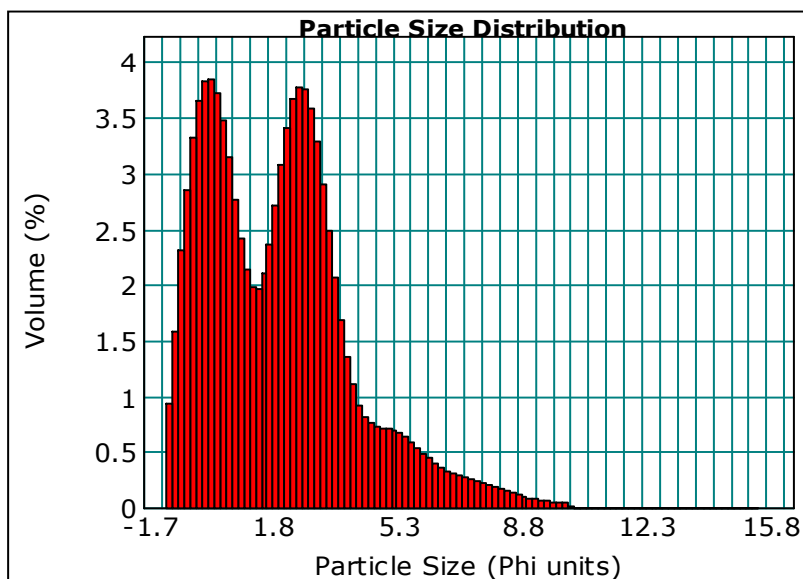
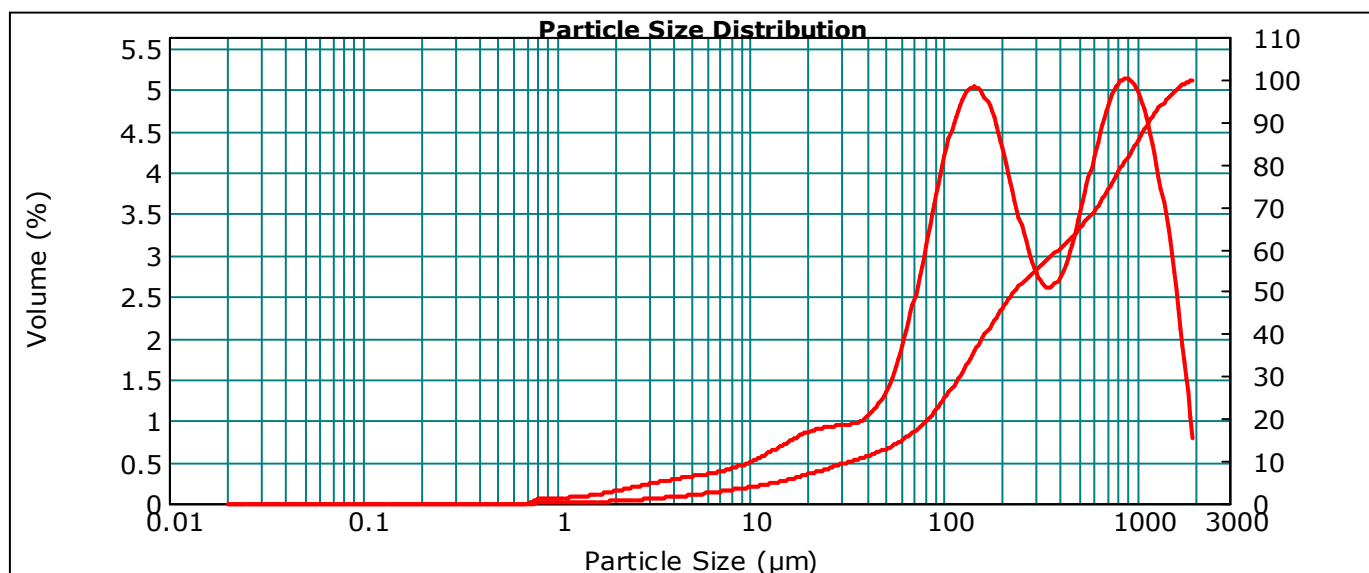
Uniformity:
1.48

Specific Surface Area*:
0.115 m²/g

Percentiles:
d(0.05) = 13.98um = 6.16phi
d(0.10) = 34.01um = 4.88phi
d(0.16) = 66.93um = 3.9phi
d(0.50) = 234.67um = 2.09phi
d(0.84) = 961.27um = .06phi
d(0.90) = 1161.82um = -0.22phi
d(0.95) = 1403.56um = -0.49phi

Distribution Properties:
Kurtosis = .41
Skewness = 1.15
Standard Deviation = 456.84 um = 1.13phi
Inclusive SD (Sorting Coeff.) = 1.97 phi (Poorly Sorted)
Inclusive Kurtosis = .96 phi (Mesokurtic)
Inclusive Mean = 2.02 phi (Fine Sand)
Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	4.83
-0.5	9.87
0.0	11.44
0.5	9.38
1.0	6.53
1.5	6.50
2.0	9.30
2.5	11.24
3.0	9.72
3.5	6.12
4.0	3.32
4.5	2.28
5.0	2.10
5.5	1.86
6.0	1.44
6.5	1.06
7.0	0.84
7.5	

Phi	Volume In %
7.5	0.71
8.0	0.56
8.5	0.40
9.0	0.24
9.5	0.16
10.0	0.10
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_14
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:15:45 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 479.03um = 1.06phi

Surface Weighted Mean:
D[3,2] = 58.15um = 4.1phi

Span :
4.571

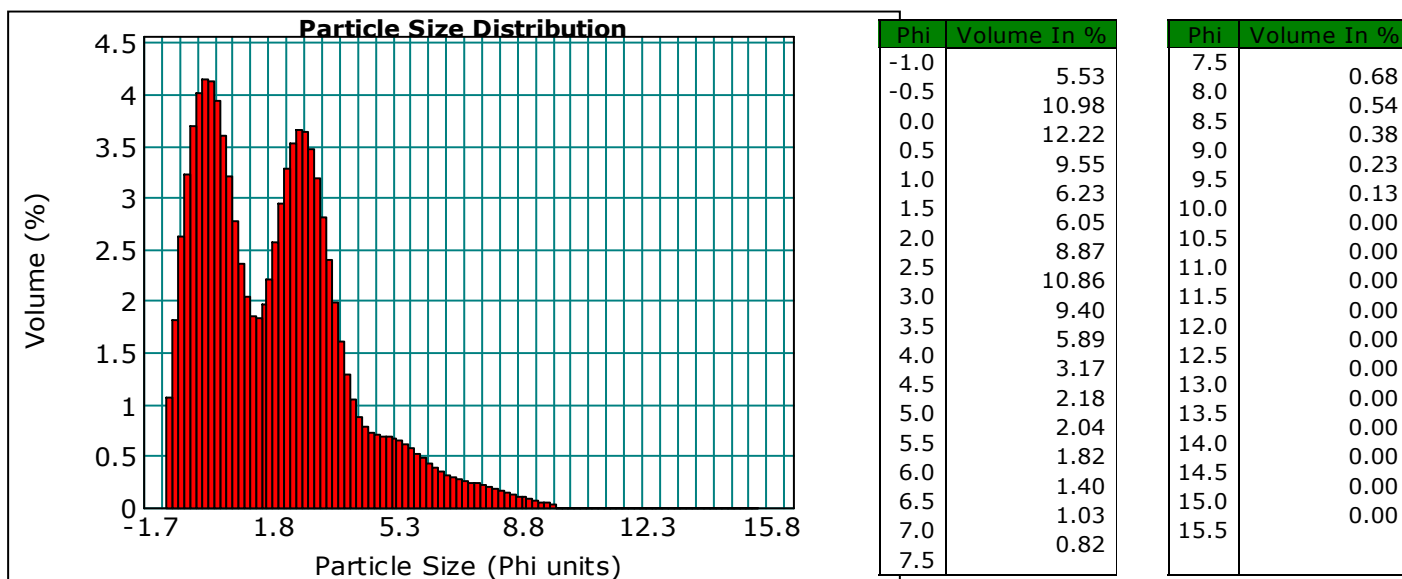
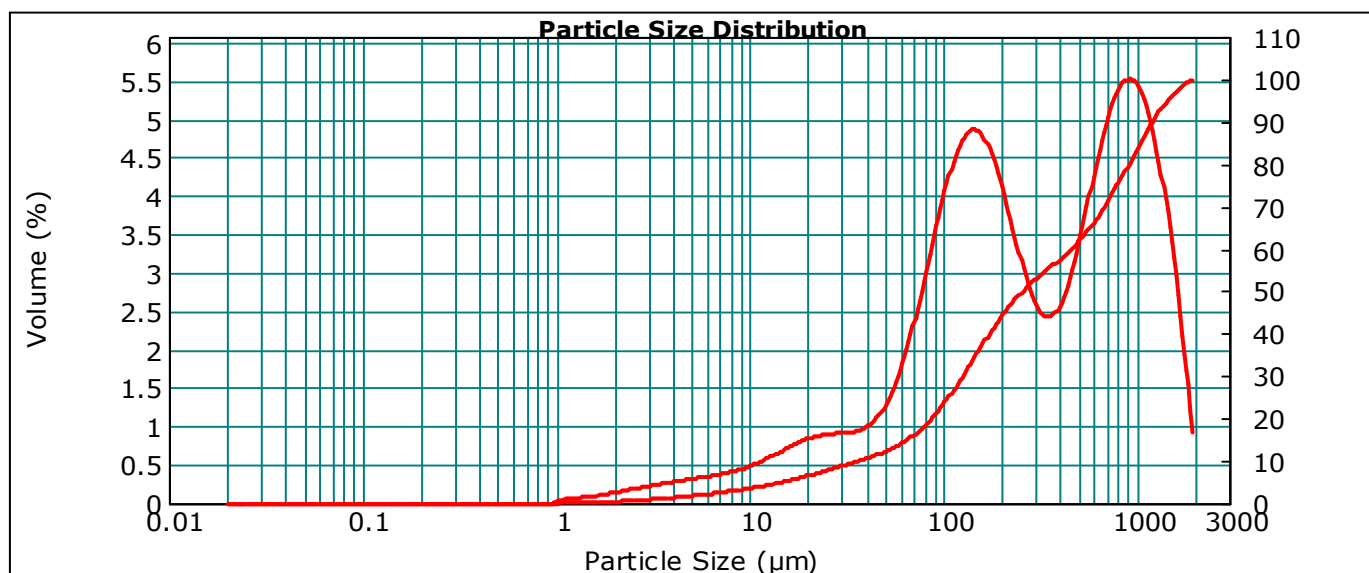
Uniformity:
1.44

Specific Surface Area*:
0.103 m²/g

Percentiles:
d(0.05) = 14.89um = 6.07phi
d(0.10) = 36.43um = 4.78phi
d(0.16) = 70.15um = 3.83phi
d(0.50) = 256.93um = 1.96phi
d(0.84) = 1014.45um = -0.02phi
d(0.90) = 1210.72um = -0.28phi
d(0.95) = 1444.23um = -0.53phi

Distribution Properties:
Kurtosis = .1
Skewness = 1.04
Standard Deviation = 472.29 um = 1.08phi
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = 1.92 phi (Medium Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_14 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:13:30 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 471.3um = 1.09phi

Surface Weighted Mean:

D[3,2] = 56.12um = 4.16phi

Span :

4.665

Uniformity:

1.46

Specific Surface Area*:

0.107 m²/g

Percentiles:

d(0.05) = 14.7um = 6.09phi

d(0.10) = 36.05um = 4.79phi

d(0.16) = 69.41um = 3.85phi

d(0.50) = 249.73um = 2phi

d(0.84) = 1002.27um = 0phi

d(0.90) = 1200.94um = -0.26phi

d(0.95) = 1436.98um = -0.52phi

Distribution Properties:

Kurtosis = .19

Skewness = 1.08

Standard Deviation = 469.02 um = 1.09phi

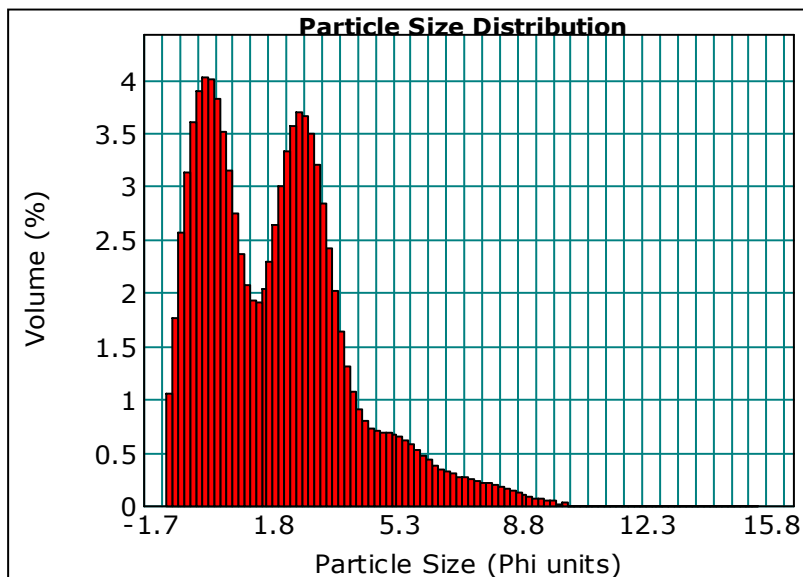
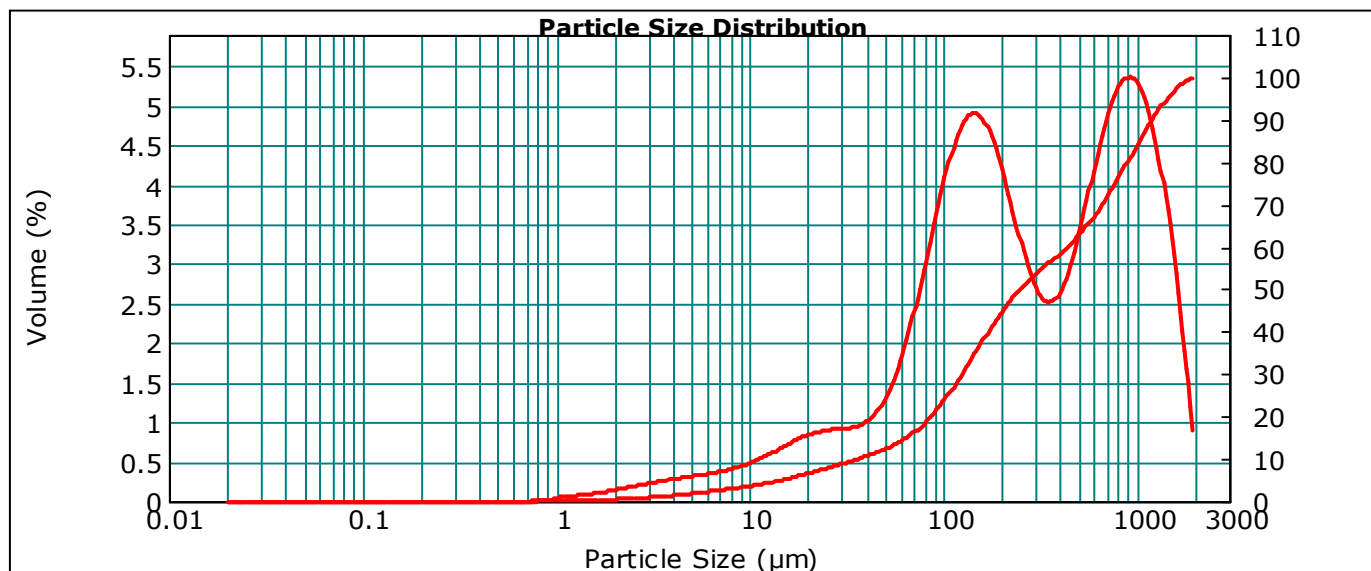
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .94 phi (Mesokurtic)

Inclusive Mean = 1.95 phi (Medium Sand)

Inclusive Skewness = .1 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.39
-0.5	10.68
0.0	11.88
0.5	9.39
1.0	6.35
1.5	6.29
2.0	9.05
2.5	10.97
3.0	9.48
3.5	5.96
4.0	3.22
4.5	2.21
5.0	2.04
5.5	1.81
6.0	1.40
6.5	1.03
7.0	0.82
7.5	

Phi	Volume In %
7.5	0.69
8.0	0.55
8.5	0.39
9.0	0.24
9.5	0.14
10.0	0.03
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:24:34 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 453.71um = 1.14phi

Surface Weighted Mean:

D[3,2] = 48.6um = 4.36phi

Span :

4.925

Uniformity:

1.52

Specific Surface Area*:

0.123 m²/g

Percentiles:

d(0.05) = 13.32um = 6.23phi

d(0.10) = 32.81um = 4.93phi

d(0.16) = 66.46um = 3.91phi

d(0.50) = 231.31um = 2.11phi

d(0.84) = 972.38um = .04phi

d(0.90) = 1172.1um = -0.23phi

d(0.95) = 1412.07um = -0.5phi

Distribution Properties:

Kurtosis = .35

Skewness = 1.14

Standard Deviation = 460.94 um = 1.12phi

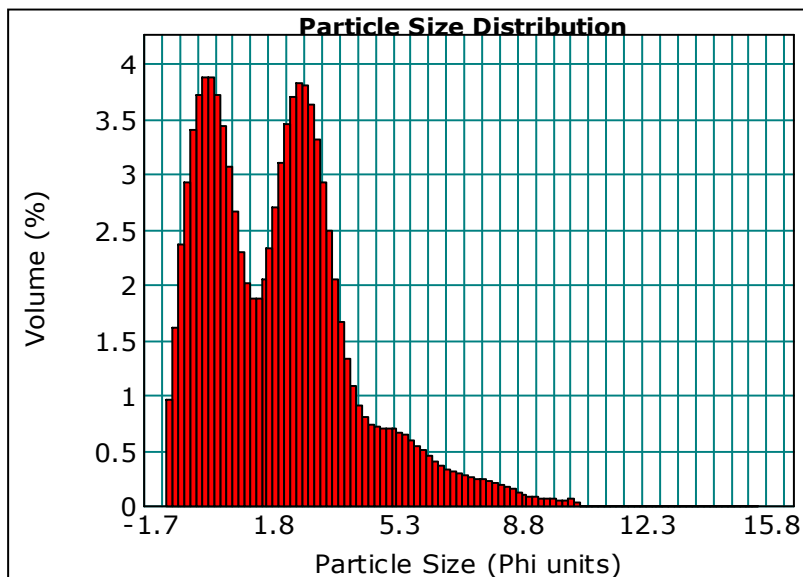
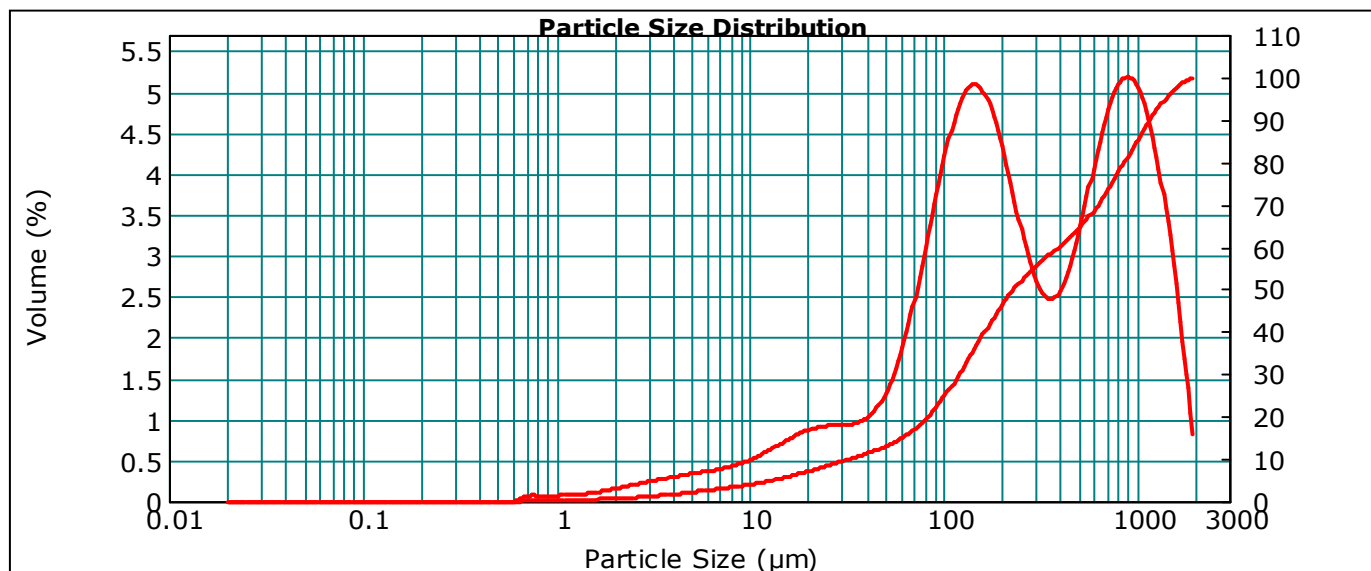
Inclusive SD (Sorting Coeff.) = 1.99 phi (Poorly Sorted)

Inclusive Kurtosis = .96 phi (Mesokurtic)

Inclusive Mean = 2.02 phi (Fine Sand)

Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	4.97
-0.5	10.10
0.0	11.51
0.5	9.16
1.0	6.17
1.5	6.30
2.0	9.34
2.5	11.38
3.0	9.79
3.5	6.10
4.0	3.25
4.5	2.23
5.0	2.09
5.5	1.88
6.0	1.46
6.5	1.08
7.0	0.86
7.5	

Phi	Volume In %
7.5	0.73
8.0	0.58
8.5	0.41
9.0	0.26
9.5	0.17
10.0	0.15
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:25:43 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 474.47um = 1.08phi

Surface Weighted Mean:
D[3,2] = 52.32um = 4.26phi

Span :
4.820

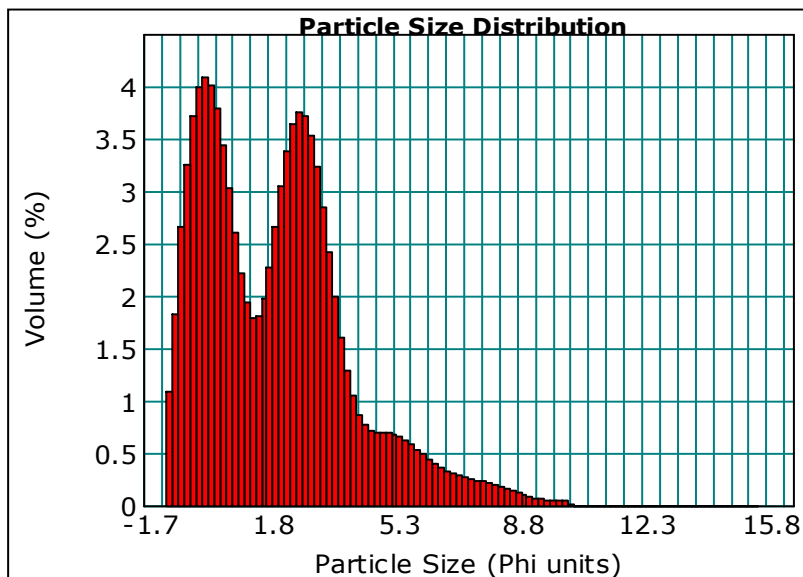
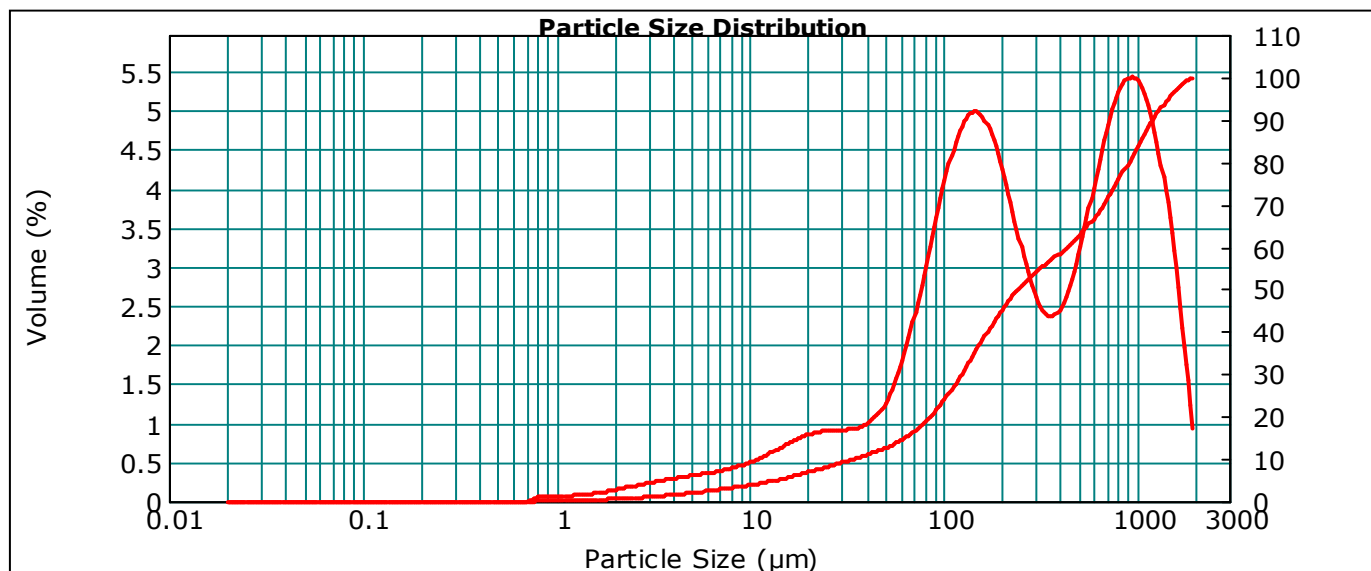
Uniformity:
1.5

Specific Surface Area*:
0.115 m²/g

Percentiles:
d(0.05) = 13.91um = 6.17phi
d(0.10) = 34.36um = 4.86phi
d(0.16) = 68.74um = 3.86phi
d(0.50) = 244.76um = 2.03phi
d(0.84) = 1016.32um = -0.02phi
d(0.90) = 1214.04um = -0.28phi
d(0.95) = 1447.66um = -0.53phi

Distribution Properties:
Kurtosis = .12
Skewness = 1.06
Standard Deviation = 474.29 um = 1.08phi
Inclusive SD (Sorting Coeff.) = 1.99 phi (Poorly Sorted)
Inclusive Kurtosis = .95 phi (Mesokurtic)
Inclusive Mean = 1.96 phi (Medium Sand)
Inclusive Skewness = .09 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.59
-0.5	10.98
0.0	11.91
0.5	9.04
1.0	5.91
1.5	6.11
2.0	9.17
2.5	11.14
3.0	9.53
3.5	5.89
4.0	3.13
4.5	2.17
5.0	2.05
5.5	1.85
6.0	1.44
6.5	1.06
7.0	0.85
7.5	

Phi	Volume In %
7.5	0.71
8.0	0.57
8.5	0.40
9.0	0.25
9.5	0.16
10.0	0.10
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_13
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:26:50 PM

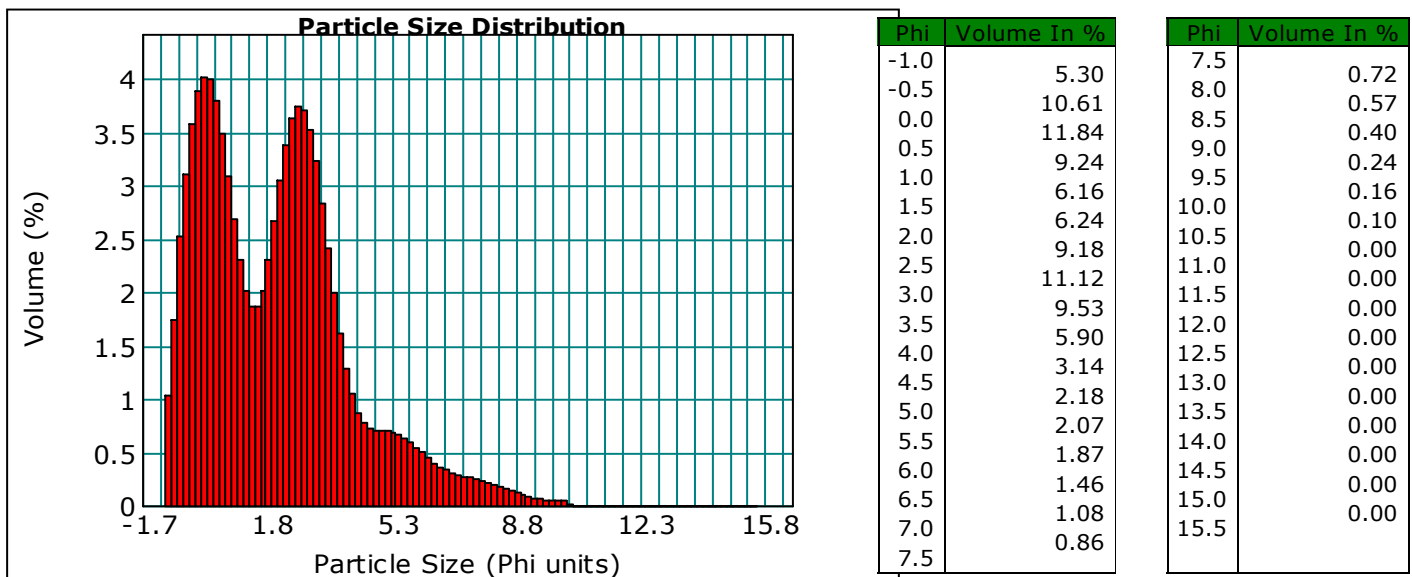
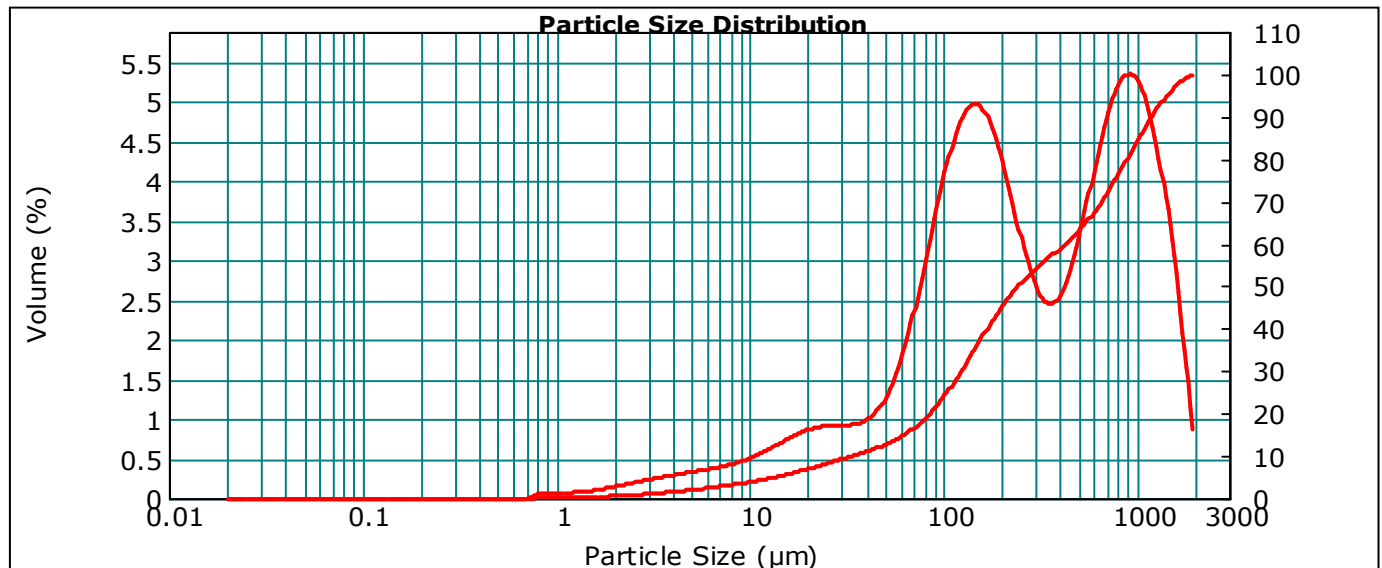
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 467.34um = 1.1phi
Surface Weighted Mean:
D[3,2] = 52.13um = 4.26phi
Span :
4.776
Uniformity:
1.49
Specific Surface Area*:
0.115 m²/g

Percentiles:
d(0.05) = 13.76um = 6.18phi
d(0.10) = 33.74um = 4.89phi
d(0.16) = 68.12um = 3.88phi
d(0.50) = 243.29um = 2.04phi
d(0.84) = 997.64um = 0phi
d(0.90) = 1195.78um = -0.26phi
d(0.95) = 1432.01um = -0.52phi

Distribution Properties:
Kurtosis = .21
Skewness = 1.09
Standard Deviation = 468.08 um = 1.1phi
Inclusive SD (Sorting Coeff.) = 1.98 phi (Poorly Sorted)
Inclusive Kurtosis = .95 phi (Mesokurtic)
Inclusive Mean = 1.97 phi (Medium Sand)
Inclusive Skewness = .09 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_13 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:24:34 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 465.18um = 1.1phi

Surface Weighted Mean:

D[3,2] = 50.96um = 4.29phi

Span :

4.845

Uniformity:

1.51

Specific Surface Area*:

0.118 m²/g

Percentiles:

d(0.05) = 13.66um = 6.19phi

d(0.10) = 33.62um = 4.89phi

d(0.16) = 67.76um = 3.88phi

d(0.50) = 239.55um = 2.06phi

d(0.84) = 995.68um = .01phi

d(0.90) = 1194.36um = -0.26phi

d(0.95) = 1431.05um = -0.52phi

Distribution Properties:

Kurtosis = .22

Skewness = 1.1

Standard Deviation = 467.88 um = 1.1phi

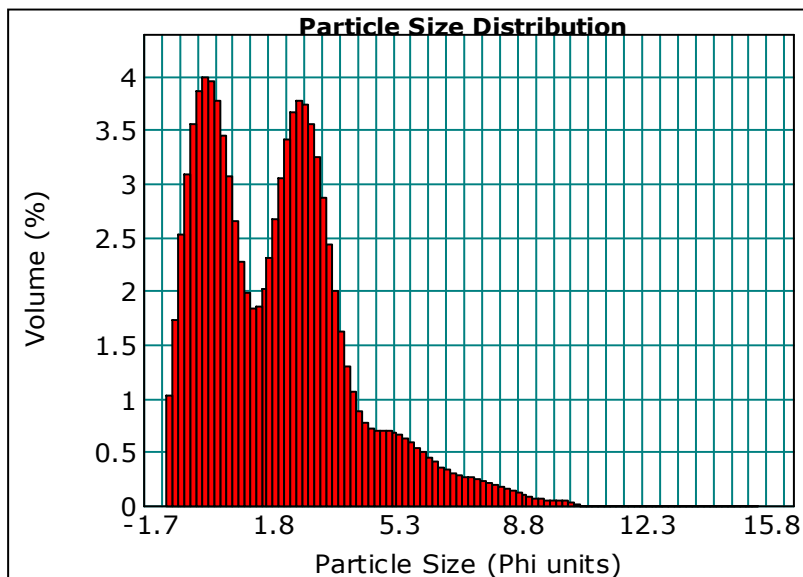
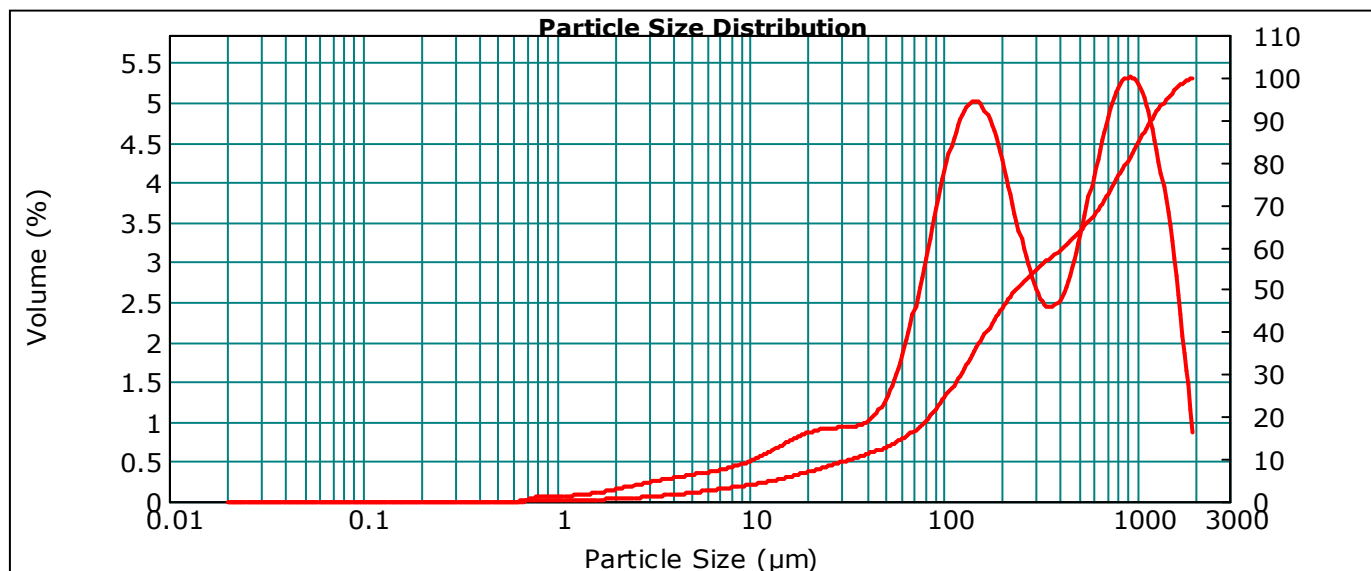
Inclusive SD (Sorting Coeff.) = 1.99 phi (Poorly Sorted)

Inclusive Kurtosis = .95 phi (Mesokurtic)

Inclusive Mean = 1.98 phi (Medium Sand)

Inclusive Skewness = .09 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.29
-0.5	10.56
0.0	11.76
0.5	9.15
1.0	6.08
1.5	6.22
2.0	9.23
2.5	11.21
3.0	9.62
3.5	5.97
4.0	3.18
4.5	2.19
5.0	2.07
5.5	1.87
6.0	1.45
6.5	1.08
7.0	0.86
7.5	

Phi	Volume In %
7.5	0.72
8.0	0.57
8.5	0.40
9.0	0.25
9.5	0.16
10.0	0.12
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:45:12 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 463.53um = 1.11phi

Surface Weighted Mean:
D[3,2] = 58.06um = 4.11phi

Span :
4.858

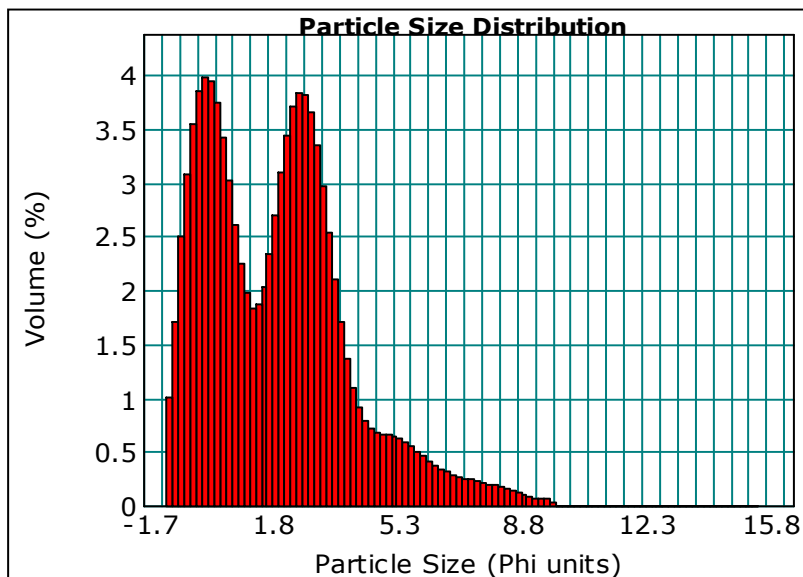
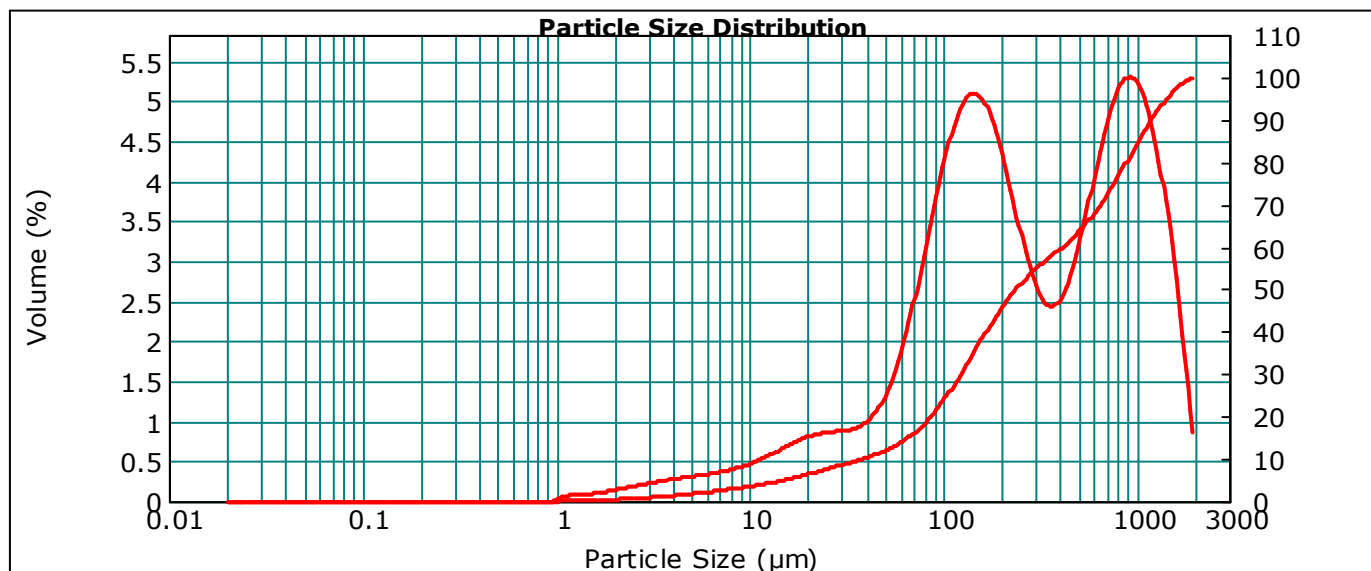
Uniformity:
1.51

Specific Surface Area*:
0.103 m²/g

Percentiles:
d(0.05) = 15.21um = 6.04phi
d(0.10) = 38.08um = 4.71phi
d(0.16) = 70.31um = 3.83phi
d(0.50) = 237.3um = 2.08phi
d(0.84) = 992.59um = .01phi
d(0.90) = 1190.86um = -0.25phi
d(0.95) = 1427.44um = -0.51phi

Distribution Properties:
Kurtosis = .25
Skewness = 1.11
Standard Deviation = 466.24 um = 1.1phi
Inclusive SD (Sorting Coeff.) = 1.95 phi (Poorly Sorted)
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = 1.97 phi (Medium Sand)
Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.22
-0.5	10.52
0.0	11.70
0.5	9.03
1.0	6.04
1.5	6.30
2.0	9.35
2.5	11.40
3.0	9.90
3.5	6.25
4.0	3.31
4.5	2.16
5.0	1.96
5.5	1.75
6.0	1.36
6.5	1.00
7.0	0.80
7.5	

Phi	Volume In %
7.5	0.67
8.0	0.54
8.5	0.38
9.0	0.24
9.5	0.13
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:57:34 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 461.55um = 1.12phi

Surface Weighted Mean:

D[3,2] = 55.4um = 4.17phi

Span :

4.880

Uniformity:

1.51

Specific Surface Area*:

0.108 m²/g

Percentiles:

d(0.05) = 14.72um = 6.09phi

d(0.10) = 36.29um = 4.78phi

d(0.16) = 68.97um = 3.86phi

d(0.50) = 236.63um = 2.08phi

d(0.84) = 990.62um = .01phi

d(0.90) = 1190.99um = -0.25phi

d(0.95) = 1428.87um = -0.51phi

Distribution Properties:

Kurtosis = .28

Skewness = 1.12

Standard Deviation = 466.2 um = 1.1phi

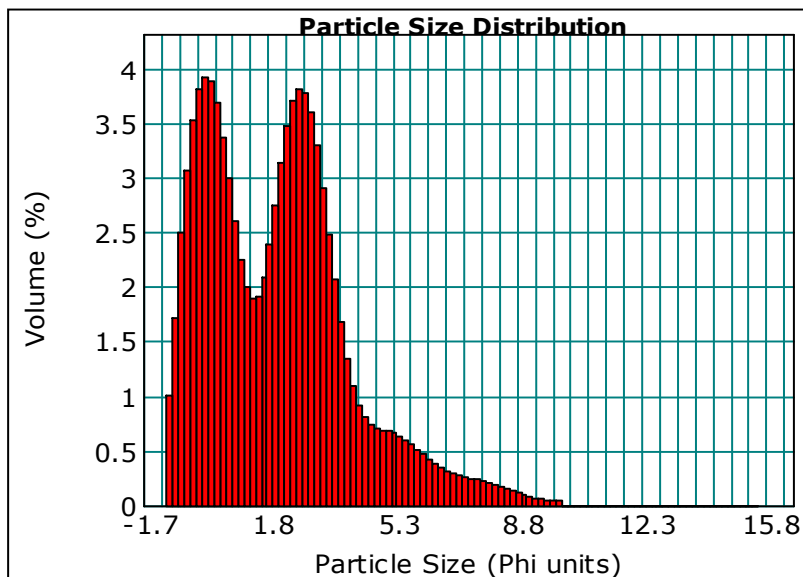
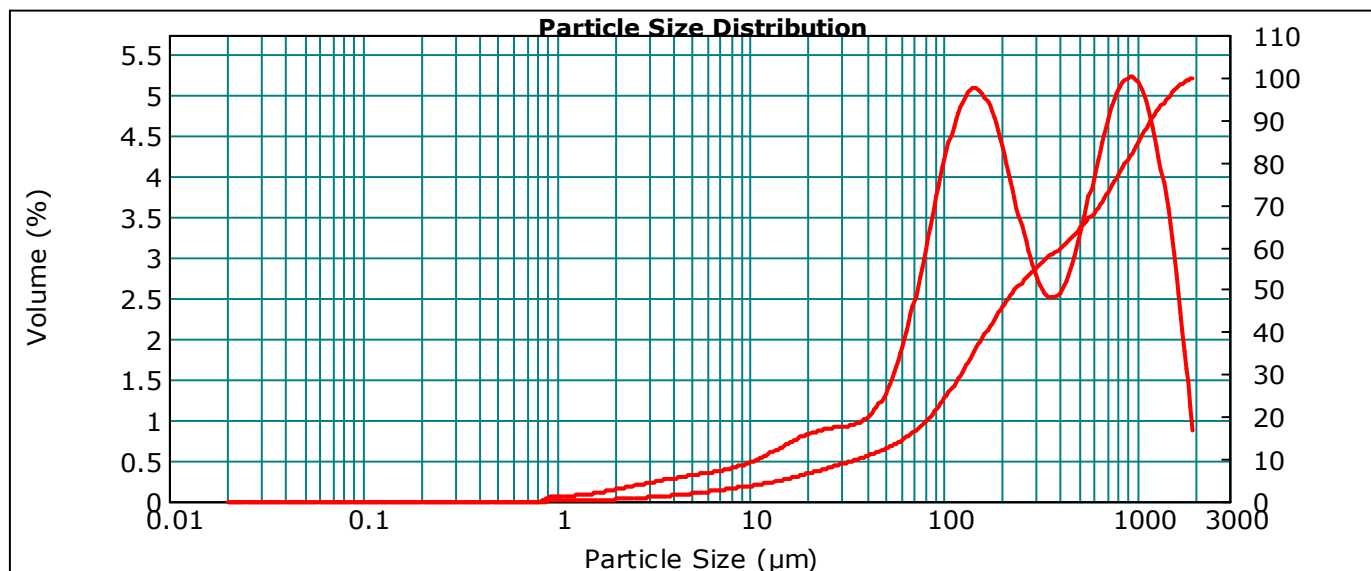
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .95 phi (Mesokurtic)

Inclusive Mean = 1.98 phi (Medium Sand)

Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.25
-0.5	10.43
0.0	11.50
0.5	8.95
1.0	6.14
1.5	6.46
2.0	9.45
2.5	11.34
3.0	9.74
3.5	6.12
4.0	3.30
4.5	2.23
5.0	2.03
5.5	1.79
6.0	1.38
6.5	1.02
7.0	0.82
7.5	

Phi	Volume In %
7.5	0.69
8.0	0.55
8.5	0.39
9.0	0.24
9.5	0.16
10.0	0.03
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:58:41 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 468.5um = 1.09phi

Surface Weighted Mean:

D[3,2] = 57.67um = 4.12phi

Span :

4.871

Uniformity:

1.52

Specific Surface Area*:

0.104 m²/g

Percentiles:

d(0.05) = 14.88um = 6.07phi

d(0.10) = 36.19um = 4.79phi

d(0.16) = 68.99um = 3.86phi

d(0.50) = 239.39um = 2.06phi

d(0.84) = 1004.46um = -0.01phi

d(0.90) = 1202.38um = -0.27phi

d(0.95) = 1437.37um = -0.52phi

Distribution Properties:

Kurtosis = .18

Skewness = 1.09

Standard Deviation = 470.45 um = 1.09phi

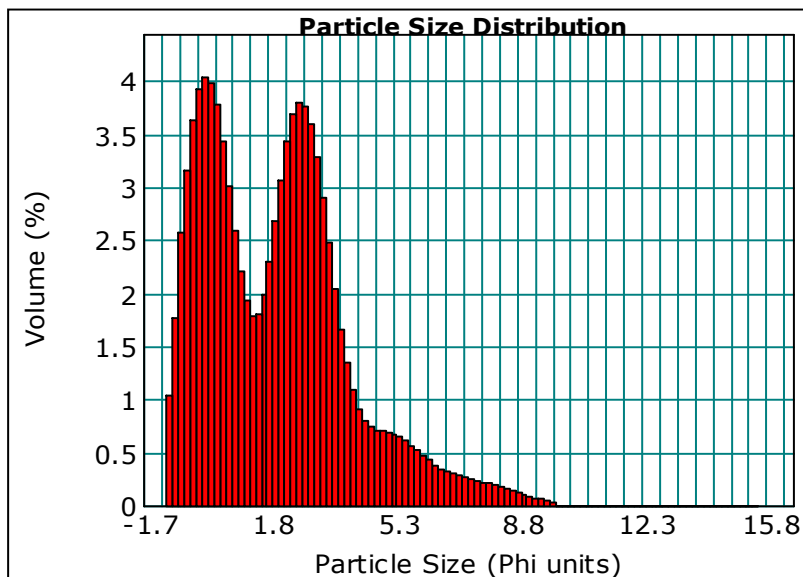
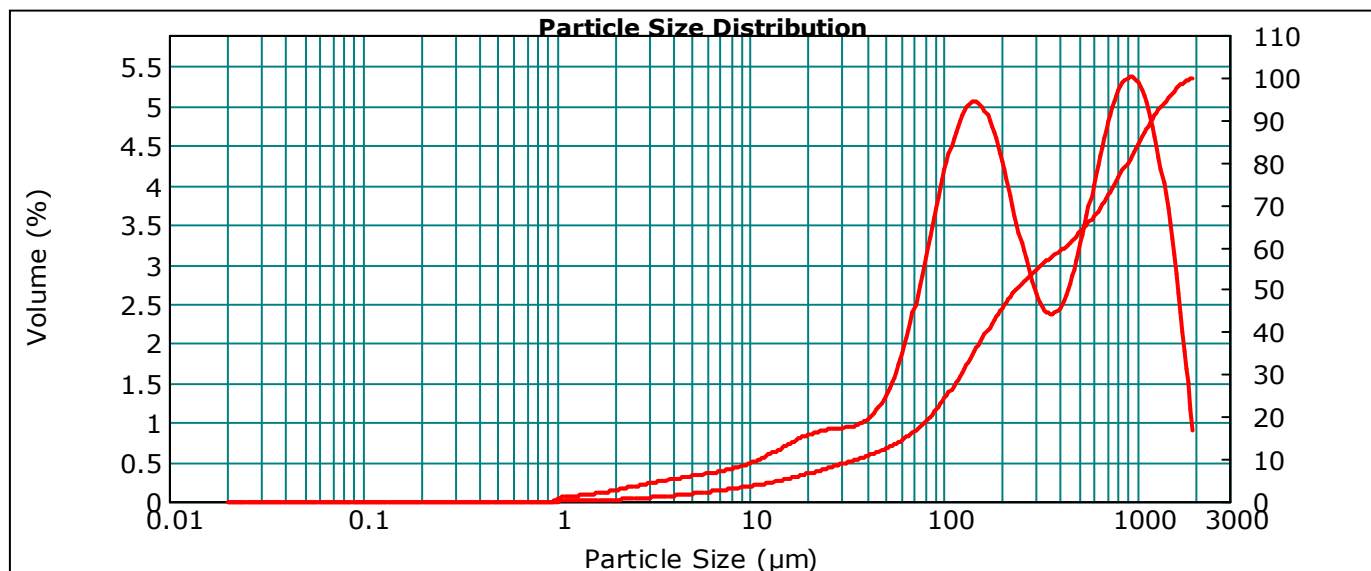
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)

Inclusive Kurtosis = .93 phi (Mesokurtic)

Inclusive Mean = 1.97 phi (Medium Sand)

Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.40
-0.5	10.75
0.0	11.81
0.5	9.01
1.0	5.91
1.5	6.15
2.0	9.26
2.5	11.28
3.0	9.71
3.5	6.08
4.0	3.29
4.5	2.25
5.0	2.06
5.5	1.81
6.0	1.39
6.5	1.03
7.0	0.82
7.5	

Phi	Volume In %
7.5	0.69
8.0	0.54
8.5	0.38
9.0	0.23
9.5	0.13
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_12
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:59:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 490.91um = 1.03phi

Surface Weighted Mean:
D[3,2] = 60.86um = 4.04phi

Span :
4.825

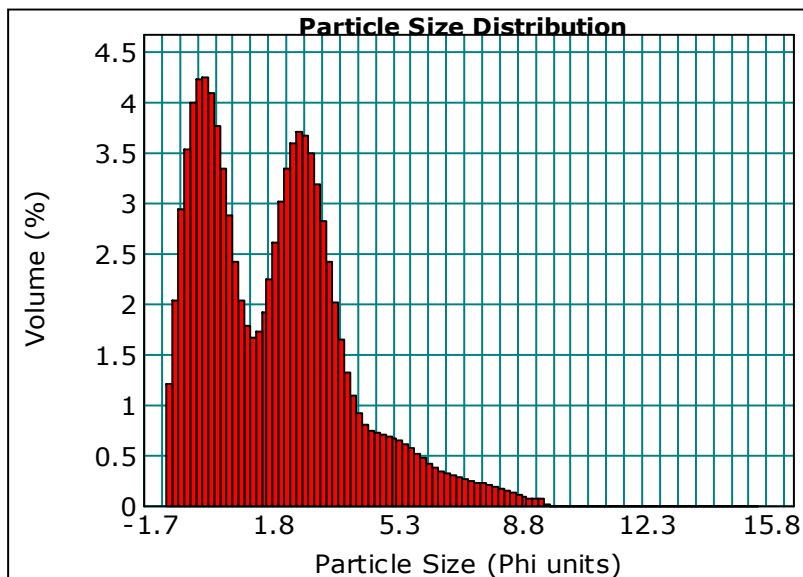
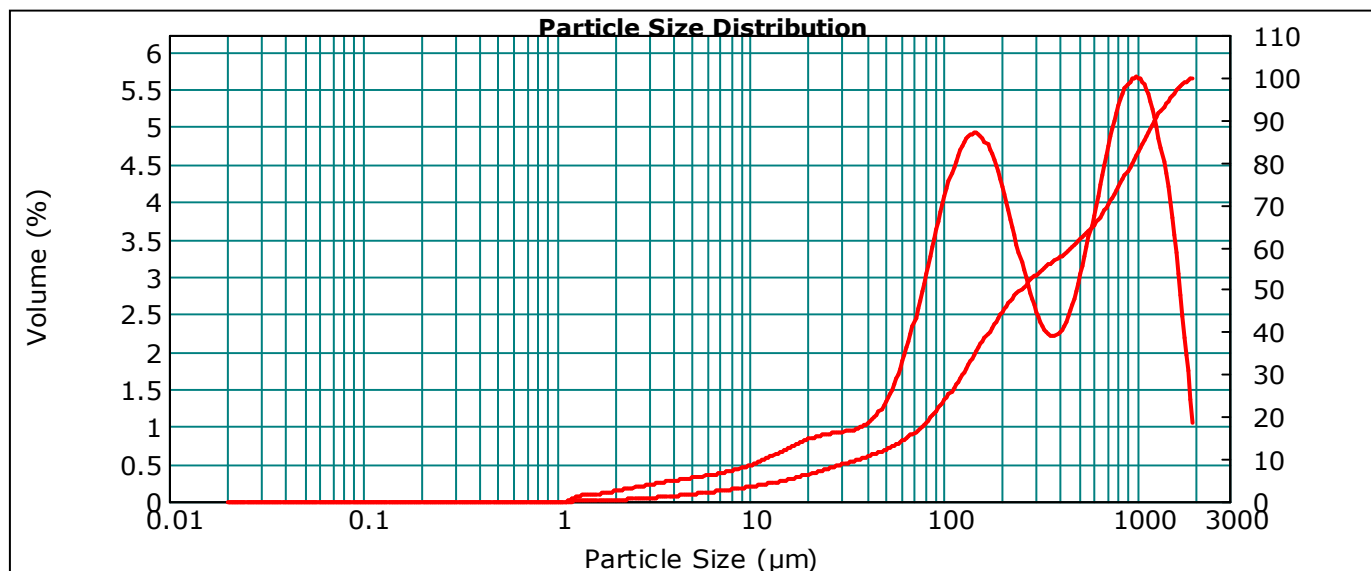
Uniformity:
1.52

Specific Surface Area*:
0.0986 m²/g

Percentiles:
d(0.05) = 15.4um = 6.02phi
d(0.10) = 37.32um = 4.74phi
d(0.16) = 70.01um = 3.84phi
d(0.50) = 251.49um = 1.99phi
d(0.84) = 1055.74um = -0.08phi
d(0.90) = 1250.73um = -0.32phi
d(0.95) = 1477.47um = -0.56phi

Distribution Properties:
Kurtosis = -0.06
Skewness = 1.01
Standard Deviation = 486.71 um = 1.04phi
Inclusive SD (Sorting Coeff.) = 1.98 phi (Poorly Sorted)
Inclusive Kurtosis = .91 phi (Mesokurtic)
Inclusive Mean = 1.92 phi (Medium Sand)
Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	6.20
-0.5	11.81
0.0	12.12
0.5	8.61
1.0	5.46
1.5	5.93
2.0	9.06
2.5	10.98
3.0	9.44
3.5	5.96
4.0	3.28
4.5	2.25
5.0	2.04
5.5	1.79
6.0	1.38
6.5	1.02
7.0	0.81
7.5	

Phi	Volume In %
7.5	0.68
8.0	0.53
8.5	0.37
9.0	0.22
9.5	0.06
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_12 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:57:34 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 473.65um = 1.08phi

Surface Weighted Mean:
D[3,2] = 57.89um = 4.11phi

Span :
4.868

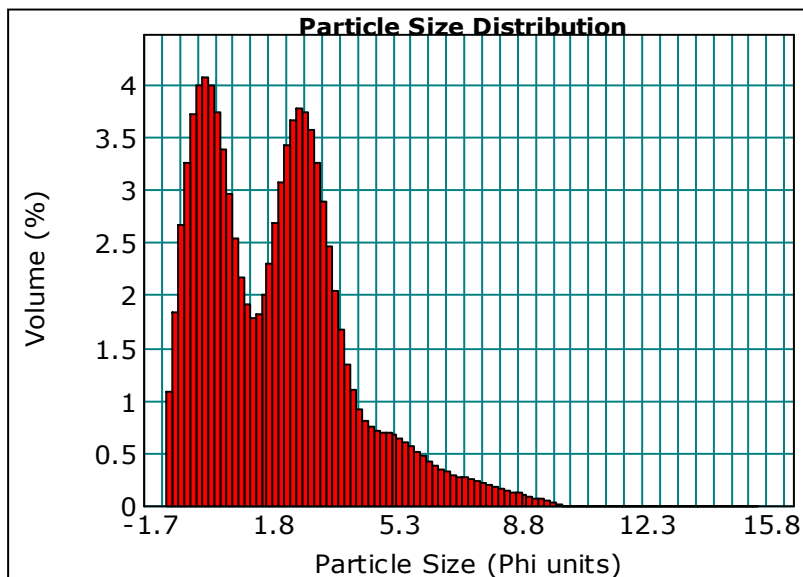
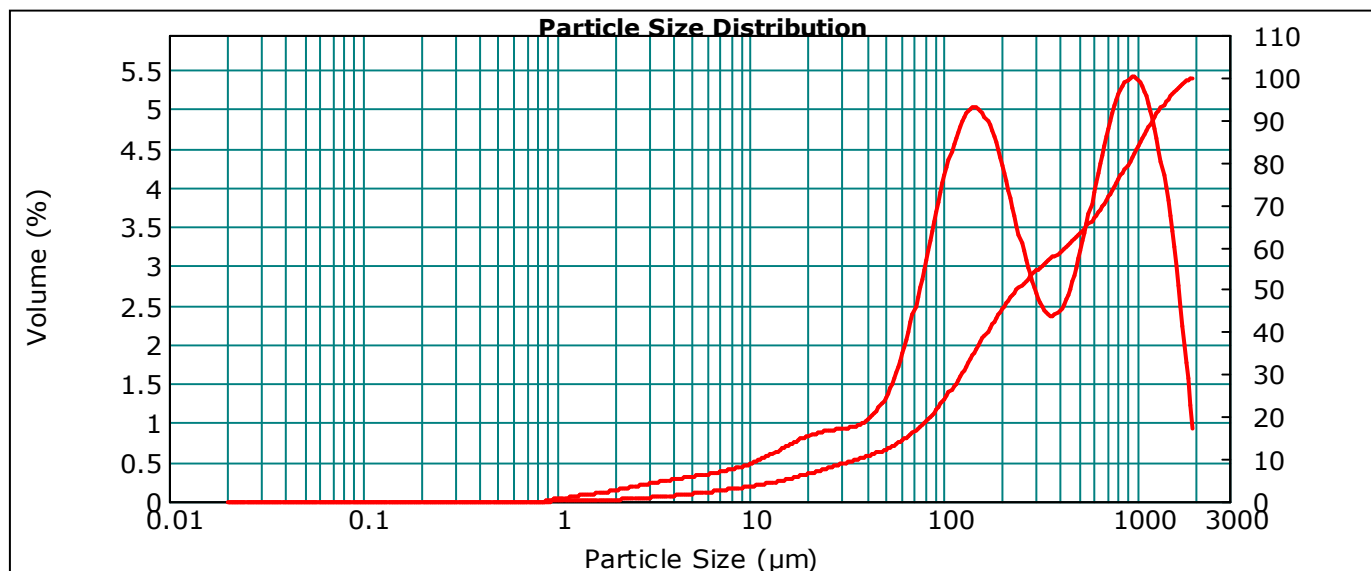
Uniformity:
1.52

Specific Surface Area*:
0.104 m²/g

Percentiles:
d(0.05) = 15um = 6.06phi
d(0.10) = 36.6um = 4.77phi
d(0.16) = 69.32um = 3.85phi
d(0.50) = 242.19um = 2.05phi
d(0.84) = 1017.6um = -0.03phi
d(0.90) = 1215.61um = -0.28phi
d(0.95) = 1448.99um = -0.54phi

Distribution Properties:
Kurtosis = .13
Skewness = 1.07
Standard Deviation = 474.7 um = 1.07phi
Inclusive SD (Sorting Coeff.) = 1.97 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.96 phi (Medium Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	5.62
-0.5	11.00
0.0	11.81
0.5	8.85
1.0	5.84
1.5	6.18
2.0	9.26
2.5	11.20
3.0	9.63
3.5	6.06
4.0	3.29
4.5	2.25
5.0	2.04
5.5	1.79
6.0	1.38
6.5	1.03
7.0	0.82
7.5	

Phi	Volume In %
7.5	0.68
8.0	0.54
8.5	0.38
9.0	0.23
9.5	0.12
10.0	0.01
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:08:58 PM

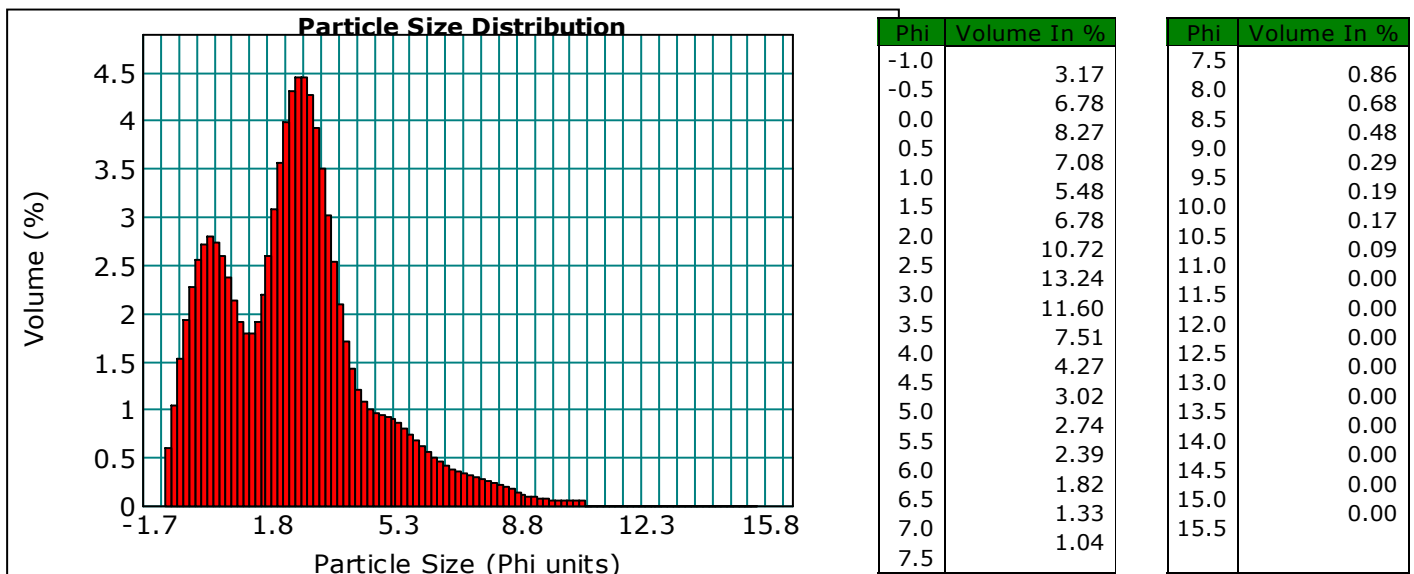
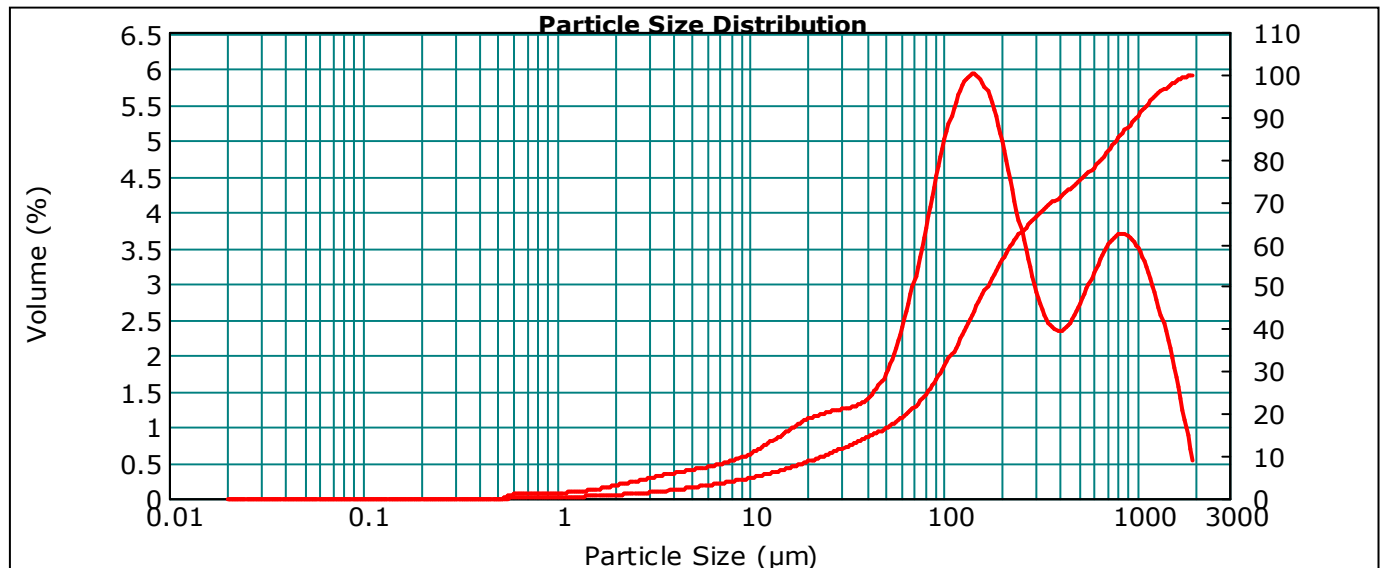
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 352.84um = 1.5phi
Surface Weighted Mean:
D[3,2] = 39.4um = 4.67phi
Span :
5.773
Uniformity:
1.62
Specific Surface Area*:
0.152 m²/g

Percentiles:
d(0.05) = 10.72um = 6.54phi
d(0.10) = 24.06um = 5.38phi
d(0.16) = 48.26um = 4.37phi
d(0.50) = 168.67um = 2.57phi
d(0.84) = 776.58um = .36phi
d(0.90) = 997.84um = 0phi
d(0.95) = 1267.15um = -0.34phi

Distribution Properties:
Kurtosis = 1.91
Skewness = 1.61
Standard Deviation = 408.9 um = 1.29phi
Inclusive SD (Sorting Coeff.) = 2.05 phi (Very Poorly)
Inclusive Kurtosis = 1.07 phi (Mesokurtic)
Inclusive Mean = 2.44 phi (Fine Sand)
Inclusive Skewness = .03 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:10:05 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 368.71um = 1.44phi

Surface Weighted Mean:
D[3,2] = 40.81um = 4.62phi

Span :
5.684

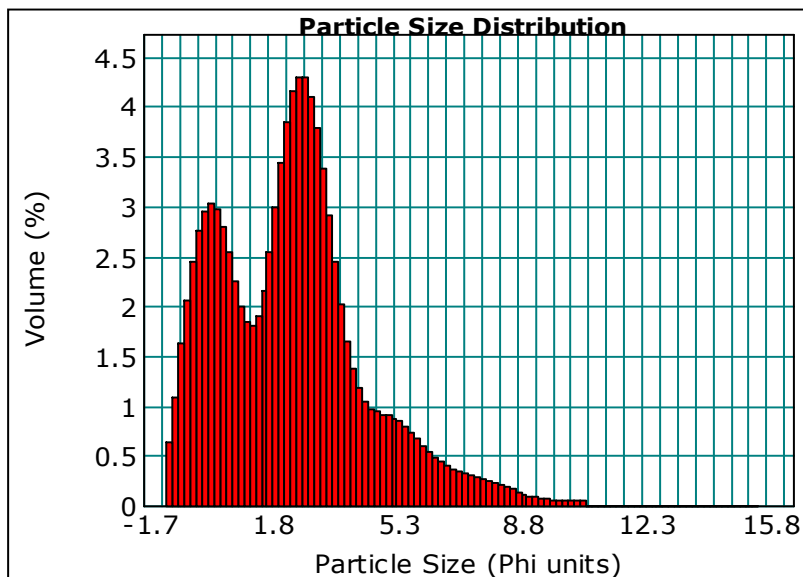
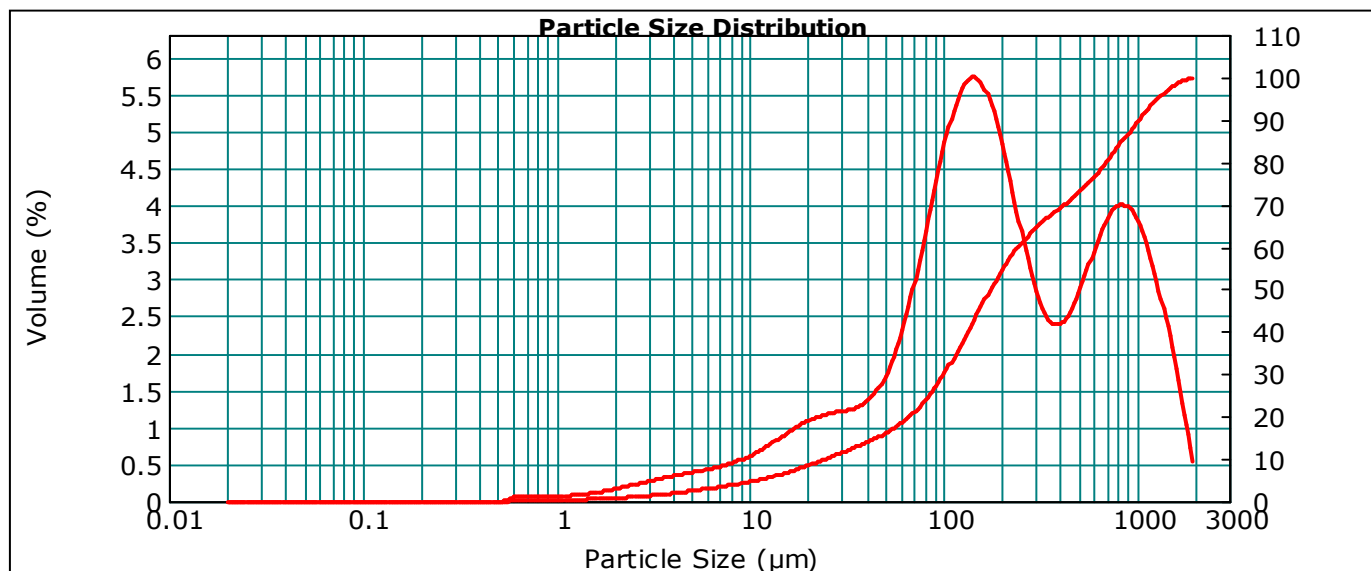
Uniformity:
1.63

Specific Surface Area*:
0.147 m²/g

Percentiles:
d(0.05) = 11.09um = 6.49phi
d(0.10) = 24.83um = 5.33phi
d(0.16) = 50.02um = 4.32phi
d(0.50) = 176.08um = 2.51phi
d(0.84) = 812.48um = .3phi
d(0.90) = 1025.63um = -0.04phi
d(0.95) = 1287.08um = -0.36phi

Distribution Properties:
Kurtosis = 1.55
Skewness = 1.51
Standard Deviation = 417.27 um = 1.26phi
Inclusive SD (Sorting Coeff.) = 2.04 phi (Very Poorly)
Inclusive Kurtosis = 1.03 phi (Mesokurtic)
Inclusive Mean = 2.38 phi (Fine Sand)
Inclusive Skewness = .03 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.35
-0.5	7.27
0.0	8.95
0.5	7.58
1.0	5.63
1.5	6.67
2.0	10.40
2.5	12.79
3.0	11.20
3.5	7.26
4.0	4.14
4.5	2.94
5.0	2.69
5.5	2.35
6.0	2.35
6.5	1.79
7.0	1.31
7.5	1.02

Phi	Volume In %
7.5	0.84
8.0	0.66
8.5	0.46
9.0	0.28
9.5	0.18
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_11
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:11:13 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 393.8um = 1.34phi

Surface Weighted Mean:
D[3,2] = 44.06um = 4.5phi

Span :
5.677

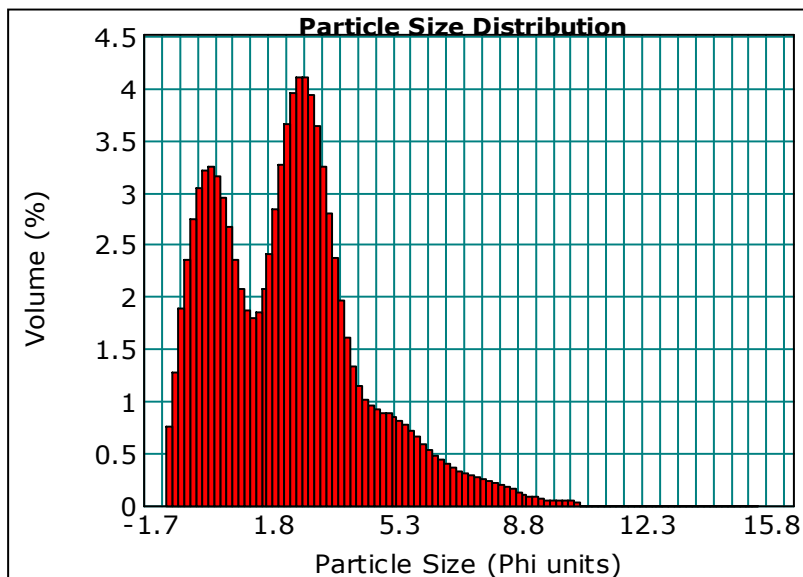
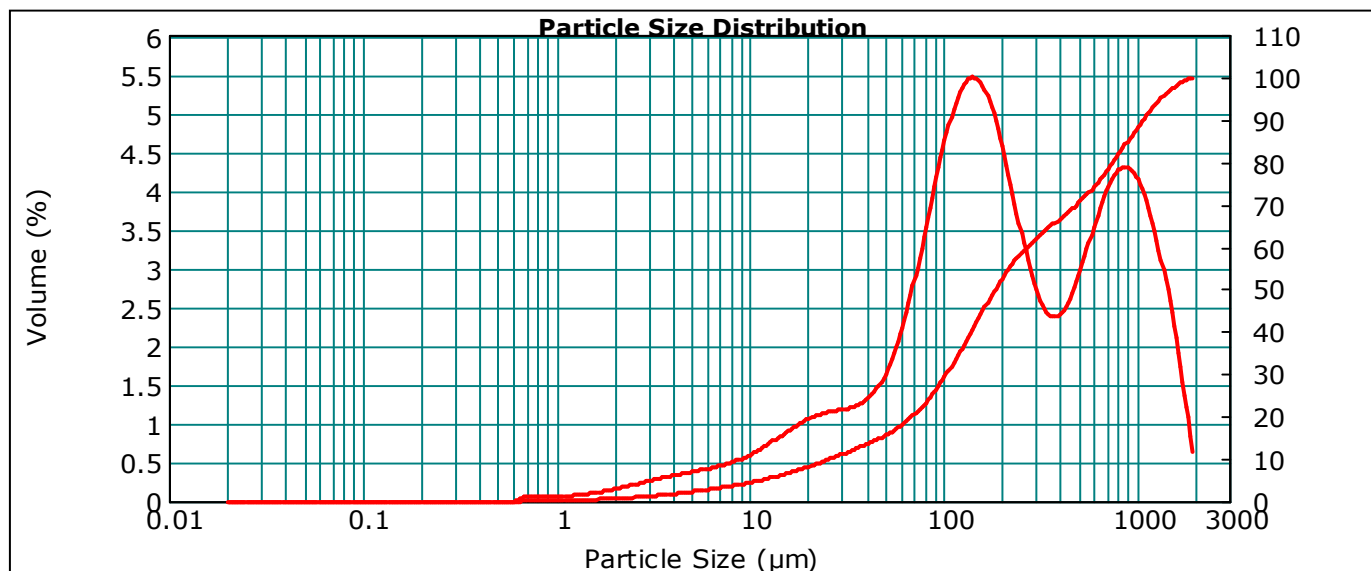
Uniformity:
1.66

Specific Surface Area*:
0.136 m²/g

Percentiles:
d(0.05) = 11.63um = 6.43phi
d(0.10) = 25.93um = 5.27phi
d(0.16) = 52.24um = 4.26phi
d(0.50) = 185.99um = 2.43phi
d(0.84) = 869.82um = .2phi
d(0.90) = 1081.85um = -0.11phi
d(0.95) = 1338.38um = -0.42phi

Distribution Properties:
Kurtosis = 1.09
Skewness = 1.39
Standard Deviation = 435.3 um = 1.2phi
Inclusive SD (Sorting Coeff.) = 2.05 phi (Very Poorly)
Inclusive Kurtosis = .99 phi (Mesokurtic)
Inclusive Mean = 2.3 phi (Fine Sand)
Inclusive Skewness = .04 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.93
-0.5	8.17
0.0	9.62
0.5	7.94
1.0	5.72
1.5	6.41
2.0	9.87
2.5	12.21
3.0	10.76
3.5	7.02
4.0	4.03
4.5	2.87
5.0	2.62
5.5	2.29
6.0	1.75
6.5	1.28
7.0	1.00
7.5	

Phi	Volume In %
7.5	0.82
8.0	0.64
8.5	0.44
9.0	0.26
9.5	0.17
10.0	0.15
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_11 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:08:58 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 371.78um = 1.43phi

Surface Weighted Mean:

D[3,2] = 41.33um = 4.6phi

Span :

5.734

Uniformity:

1.64

Specific Surface Area*:

0.145 m²/g

Percentiles:

d(0.05) = 11.14um = 6.49phi

d(0.10) = 24.92um = 5.33phi

d(0.16) = 50.15um = 4.32phi

d(0.50) = 176.44um = 2.5phi

d(0.84) = 821.14um = .28phi

d(0.90) = 1036.59um = -0.05phi

d(0.95) = 1299.1um = -0.38phi

Distribution Properties:

Kurtosis = 1.49

Skewness = 1.5

Standard Deviation = 420.97 um = 1.25phi

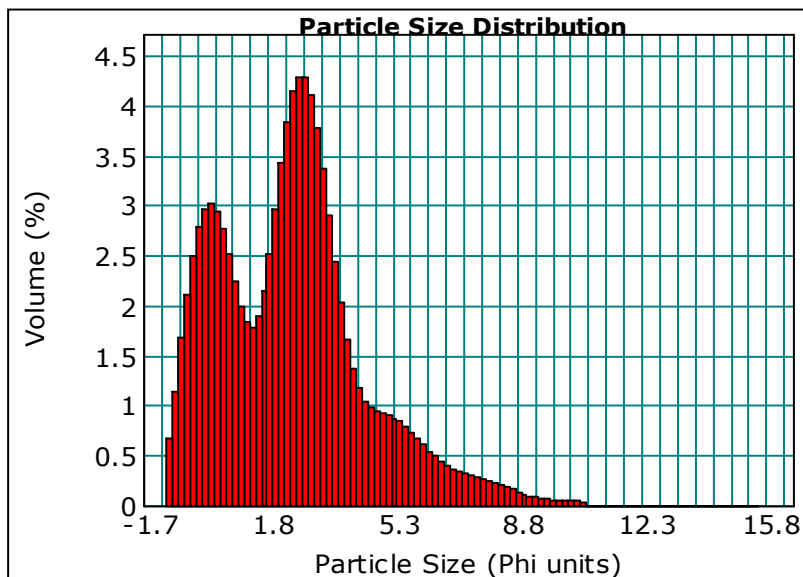
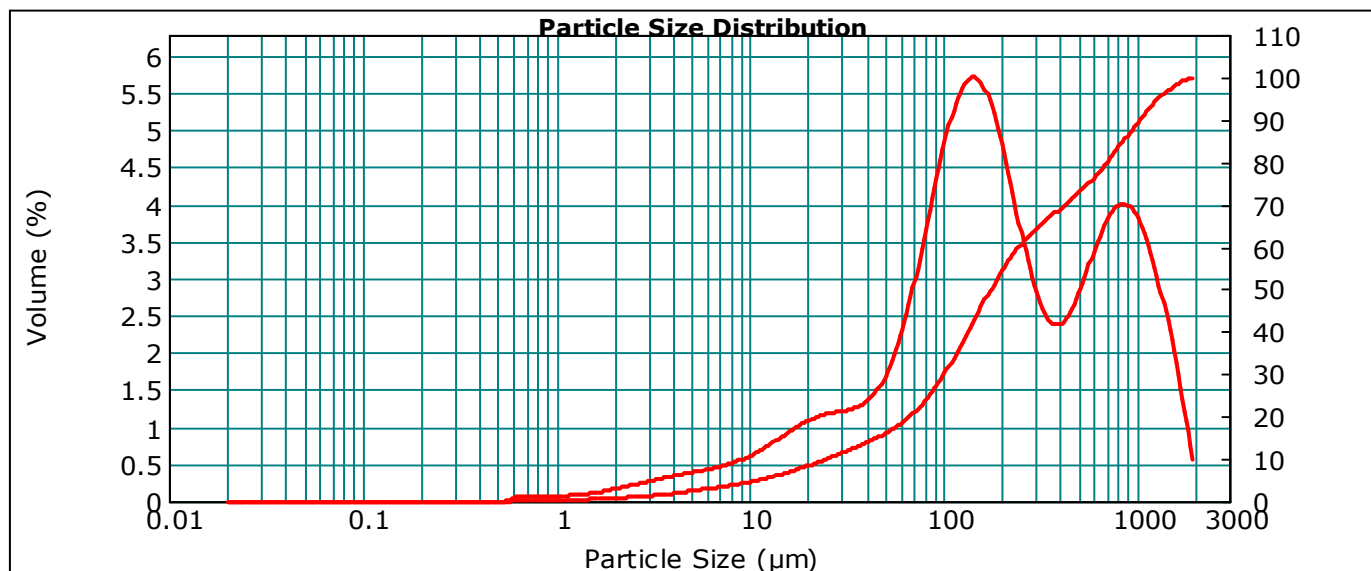
Inclusive SD (Sorting Coeff.) = 2.05 phi (Very Poorly)

Inclusive Kurtosis = 1.03 phi (Mesokurtic)

Inclusive Mean = 2.37 phi (Fine Sand)

Inclusive Skewness = .03 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.49
-0.5	7.41
0.0	8.95
0.5	7.53
1.0	5.61
1.5	6.62
2.0	10.33
2.5	12.75
3.0	11.19
3.5	7.26
4.0	4.15
4.5	2.94
5.0	2.68
5.5	2.34
6.0	1.79
6.5	1.31
7.0	1.02
7.5	

Phi	Volume In %
7.5	0.84
8.0	0.66
8.5	0.46
9.0	0.28
9.5	0.18
10.0	0.16
10.5	0.06
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:19:52 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 382.34um = 1.39phi

Surface Weighted Mean:
D[3,2] = 45.26um = 4.47phi

Span :
6.180

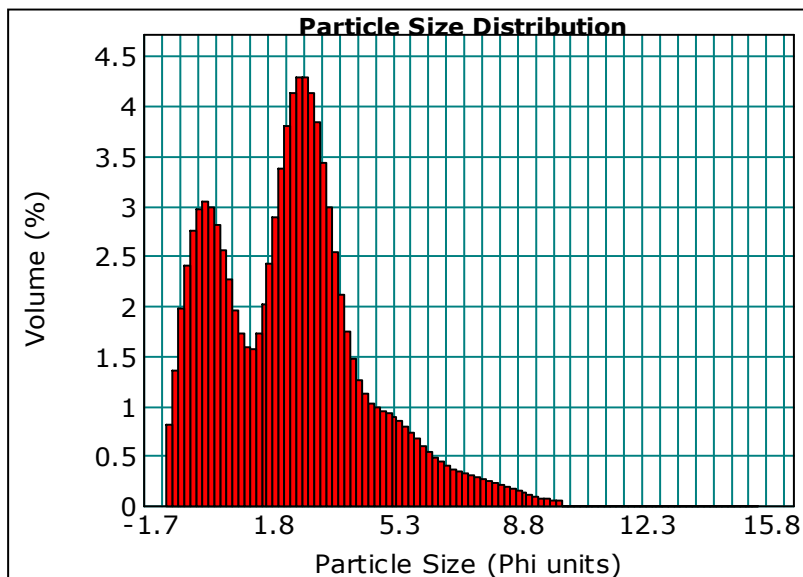
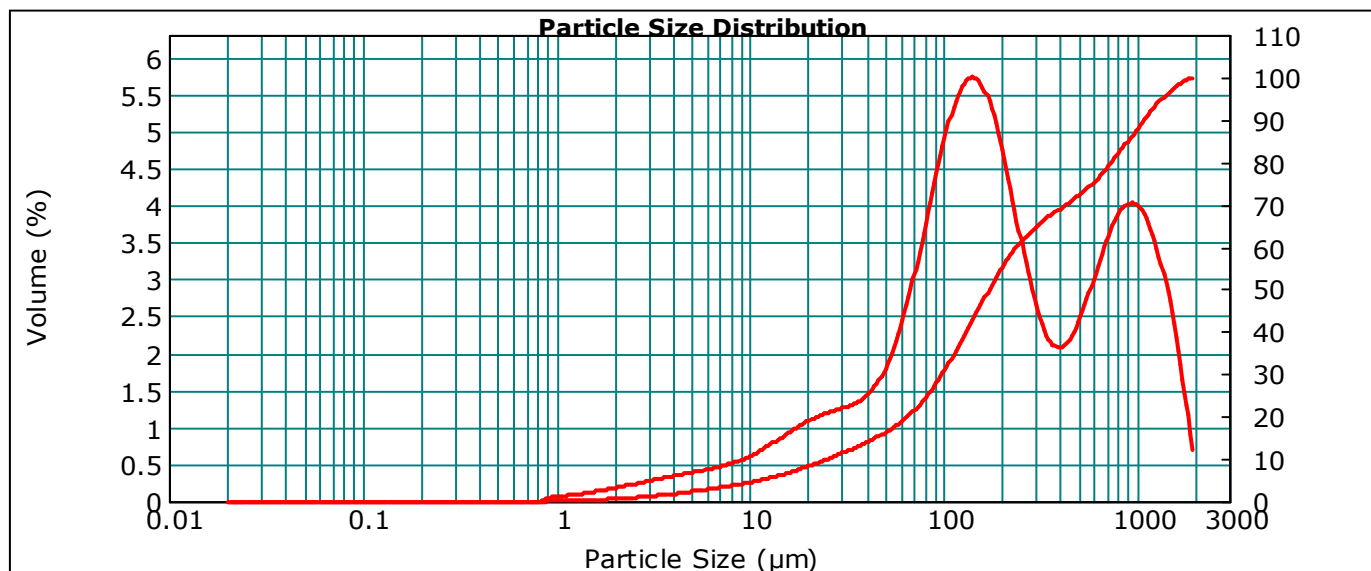
Uniformity:
1.74

Specific Surface Area*:
0.133 m²/g

Percentiles:
d(0.05) = 11.46um = 6.45phi
d(0.10) = 25.34um = 5.3phi
d(0.16) = 49.43um = 4.34phi
d(0.50) = 172.72um = 2.53phi
d(0.84) = 868.72um = .2phi
d(0.90) = 1092.79um = -0.13phi
d(0.95) = 1354.79um = -0.44phi

Distribution Properties:
Kurtosis = 1.24
Skewness = 1.46
Standard Deviation = 439.84 um = 1.18phi
Inclusive SD (Sorting Coeff.) = 2.08 phi (Very Poorly)
Inclusive Kurtosis = 1 phi (Mesokurtic)
Inclusive Mean = 2.36 phi (Fine Sand)
Inclusive Skewness = 0 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	4.15
-0.5	8.15
0.0	8.84
0.5	6.76
1.0	4.88
1.5	6.22
2.0	10.19
2.5	12.76
3.0	11.34
3.5	7.51
4.0	4.40
4.5	3.11
5.0	2.74
5.5	2.34
6.0	1.78
6.5	1.30
7.0	1.02
7.5	

Phi	Volume In %
7.5	0.84
8.0	0.66
8.5	0.48
9.0	0.31
9.5	0.19
10.0	0.04
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:21:00 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 381.77um = 1.39phi

Surface Weighted Mean:
D[3,2] = 40.4um = 4.63phi

Span :
6.071

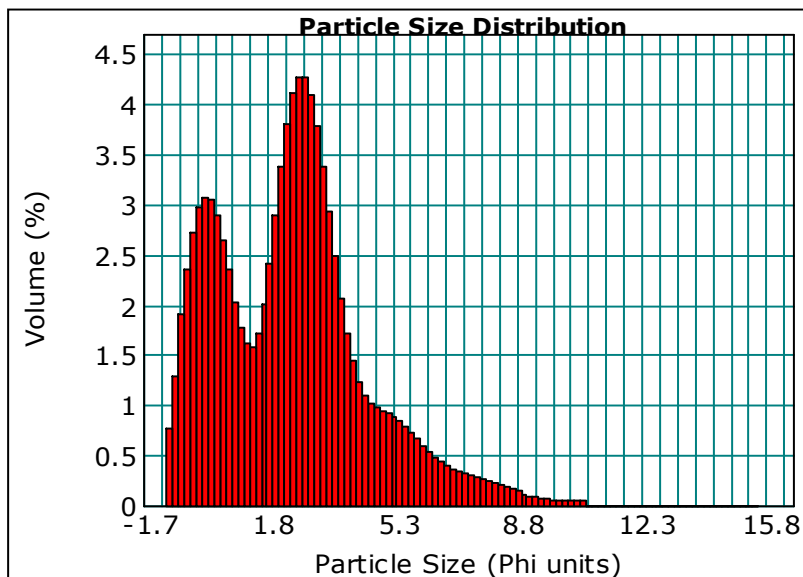
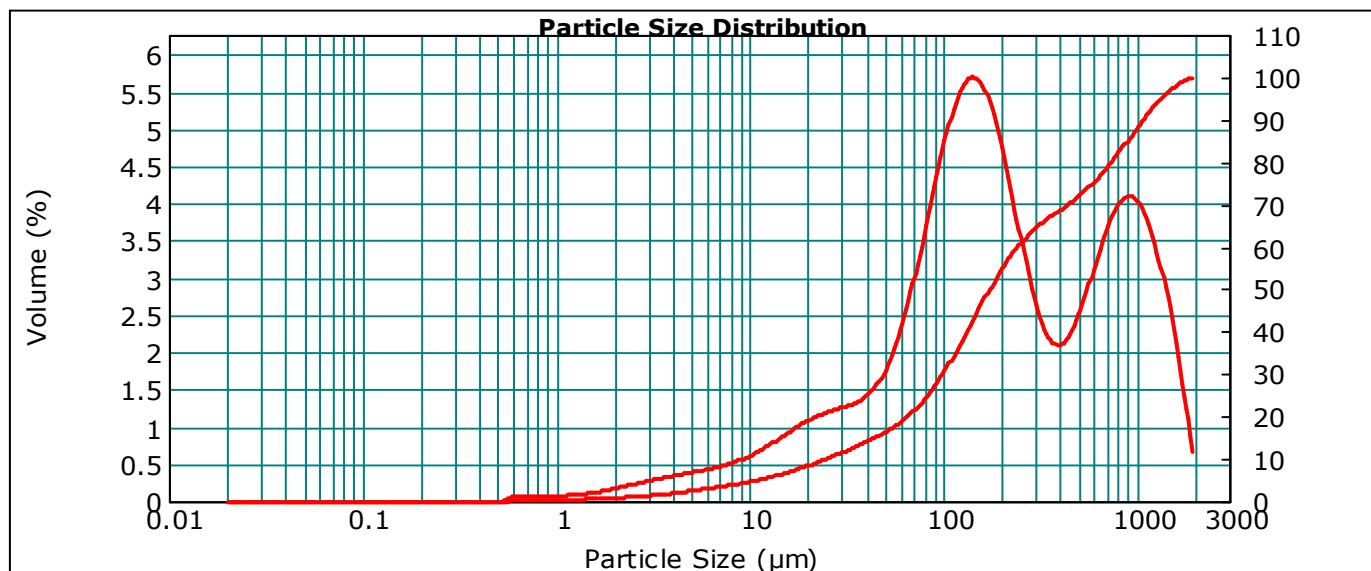
Uniformity:
1.73

Specific Surface Area*:
0.149 m²/g

Percentiles:
d(0.05) = 11.06um = 6.5phi
d(0.10) = 24.84um = 5.33phi
d(0.16) = 48.96um = 4.35phi
d(0.50) = 174.15um = 2.52phi
d(0.84) = 862.66um = .21phi
d(0.90) = 1082.1um = -0.11phi
d(0.95) = 1342.07um = -0.42phi

Distribution Properties:
Kurtosis = 1.24
Skewness = 1.46
Standard Deviation = 436.41 um = 1.2phi
Inclusive SD (Sorting Coeff.) = 2.08 phi (Very Poorly)
Inclusive Kurtosis = 1 phi (Mesokurtic)
Inclusive Mean = 2.36 phi (Fine Sand)
Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.98
-0.5	8.07
0.0	9.04
0.5	7.01
1.0	4.96
1.5	6.22
2.0	10.18
2.5	12.71
3.0	11.20
3.5	7.37
4.0	4.32
4.5	3.08
5.0	2.74
5.5	2.34
6.0	1.78
6.5	1.30
7.0	1.02
7.5	

Phi	Volume In %
7.5	0.84
8.0	0.66
8.5	0.46
9.0	0.28
9.5	0.18
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_10
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:22:07 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 376.28um = 1.41phi

Surface Weighted Mean:
D[3,2] = 39.97um = 4.64phi

Span :
6.146

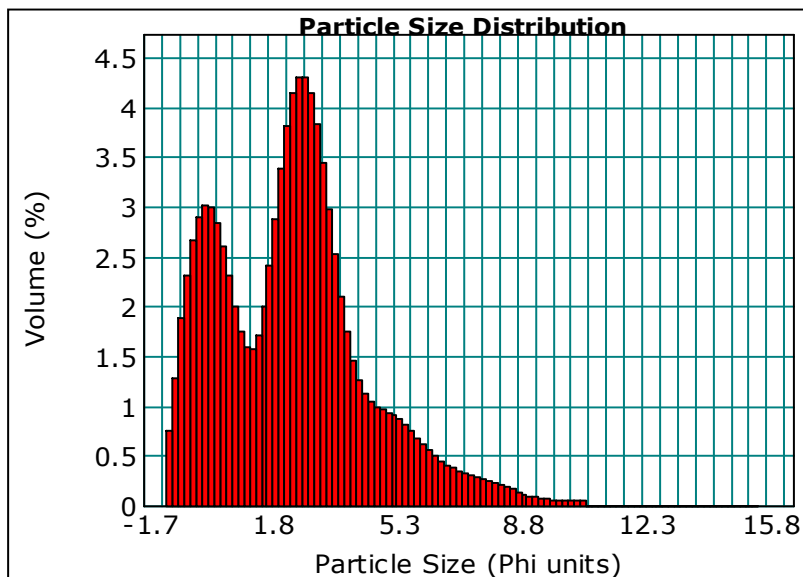
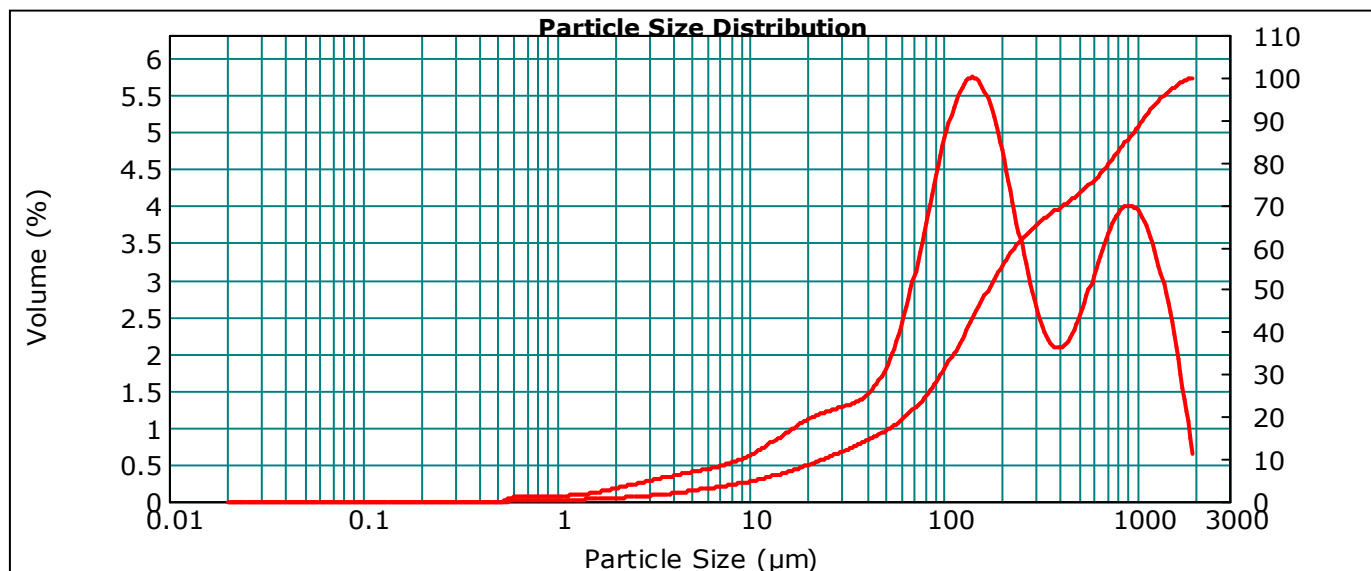
Uniformity:
1.73

Specific Surface Area*:
0.15 m²/g

Percentiles:
d(0.05) = 10.86um = 6.52phi
d(0.10) = 24.25um = 5.37phi
d(0.16) = 47.7um = 4.39phi
d(0.50) = 170.86um = 2.55phi
d(0.84) = 852.24um = .23phi
d(0.90) = 1074.28um = -0.1phi
d(0.95) = 1336.54um = -0.42phi

Distribution Properties:
Kurtosis = 1.33
Skewness = 1.48
Standard Deviation = 434.21 um = 1.2phi
Inclusive SD (Sorting Coeff.) = 2.09 phi (Very Poorly)
Inclusive Kurtosis = 1.01 phi (Mesokurtic)
Inclusive Mean = 2.39 phi (Fine Sand)
Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.92
-0.5	7.91
0.0	8.84
0.5	6.88
1.0	4.90
1.5	6.18
2.0	10.17
2.5	12.79
3.0	11.34
3.5	7.49
4.0	4.39
4.5	3.13
5.0	2.79
5.5	2.39
6.0	1.82
6.5	1.33
7.0	1.04
7.5	

Phi	Volume In %
7.5	0.85
8.0	0.67
8.5	0.46
9.0	0.28
9.5	0.18
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_10 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:19:52 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 380.13um = 1.4phi

Surface Weighted Mean:

D[3,2] = 41.74um = 4.58phi

Span :

6.133

Uniformity:

1.73

Specific Surface Area*:

0.144 m²/g

Percentiles:

d(0.05) = 11.12um = 6.49phi

d(0.10) = 24.8um = 5.33phi

d(0.16) = 48.7um = 4.36phi

d(0.50) = 172.57um = 2.53phi

d(0.84) = 861.19um = .22phi

d(0.90) = 1083.09um = -0.12phi

d(0.95) = 1344.58um = -0.43phi

Distribution Properties:

Kurtosis = 1.27

Skewness = 1.47

Standard Deviation = 436.83 um = 1.19phi

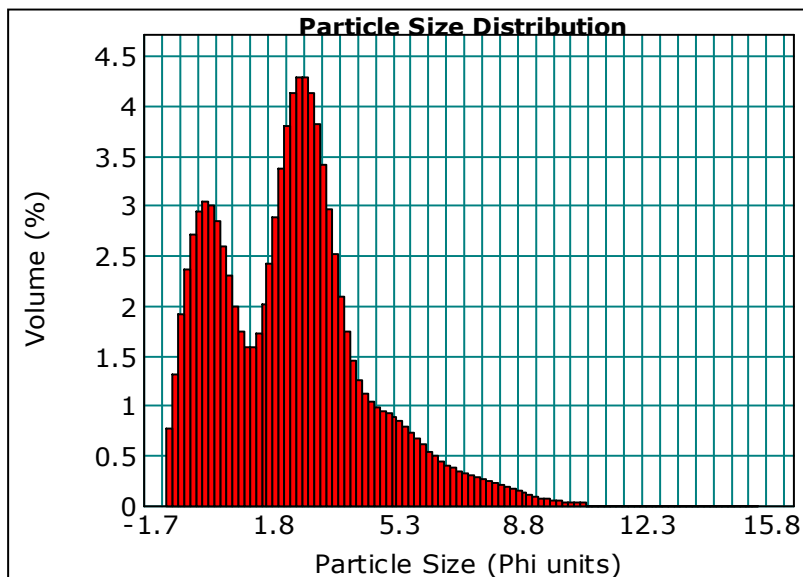
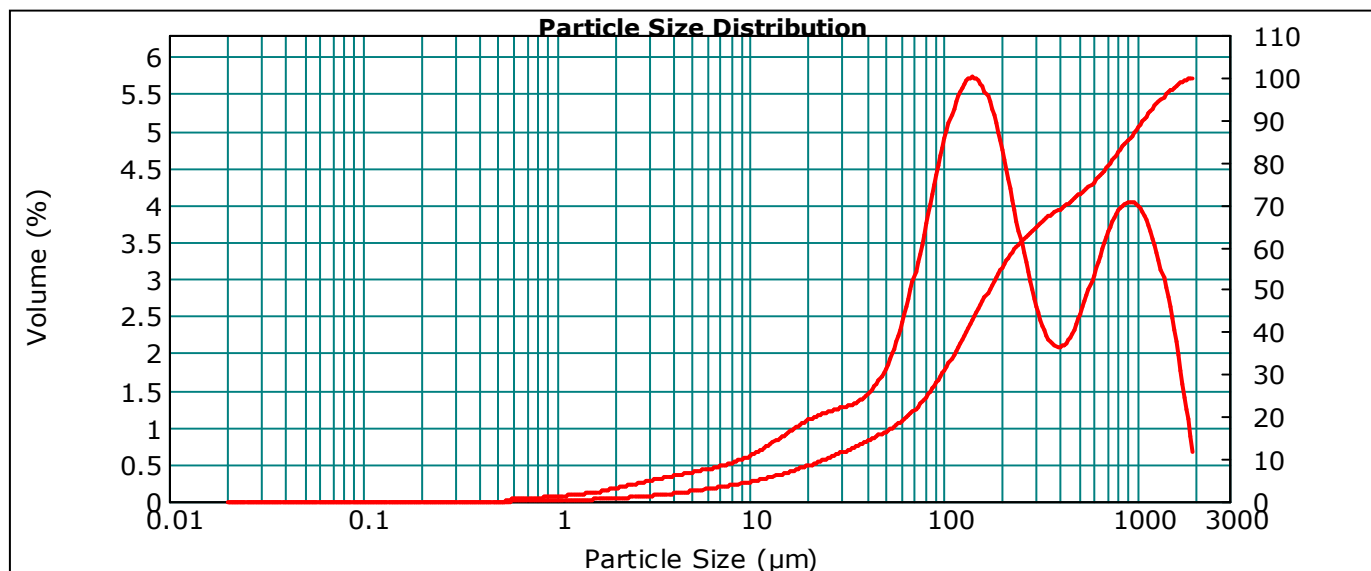
Inclusive SD (Sorting Coeff.) = 2.08 phi (Very Poorly)

Inclusive Kurtosis = 1 phi (Mesokurtic)

Inclusive Mean = 2.37 phi (Fine Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	4.02
-0.5	8.04
0.0	8.91
0.5	6.88
1.0	4.91
1.5	6.21
2.0	10.18
2.5	12.75
3.0	11.30
3.5	7.46
4.0	4.37
4.5	3.11
5.0	2.75
5.5	2.35
6.0	1.79
6.5	1.31
7.0	1.03
7.5	

Phi	Volume In %
7.5	0.84
8.0	0.66
8.5	0.47
9.0	0.29
9.5	0.19
10.0	0.12
10.5	0.06
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:30:24 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 307.55um = 1.7phi

Surface Weighted Mean:

D[3,2] = 32.07um = 4.96phi

Span :

5.895

Uniformity:

1.62

Specific Surface Area*:

0.187 m²/g

Percentiles:

d(0.05) = 8.15um = 6.94phi

d(0.10) = 18.24um = 5.78phi

d(0.16) = 34.24um = 4.87phi

d(0.50) = 148.55um = 2.75phi

d(0.84) = 663.33um = .59phi

d(0.90) = 893.9um = .16phi

d(0.95) = 1175.18um = -.023phi

Distribution Properties:

Kurtosis = 2.98

Skewness = 1.86

Standard Deviation = 379.44 um = 1.4phi

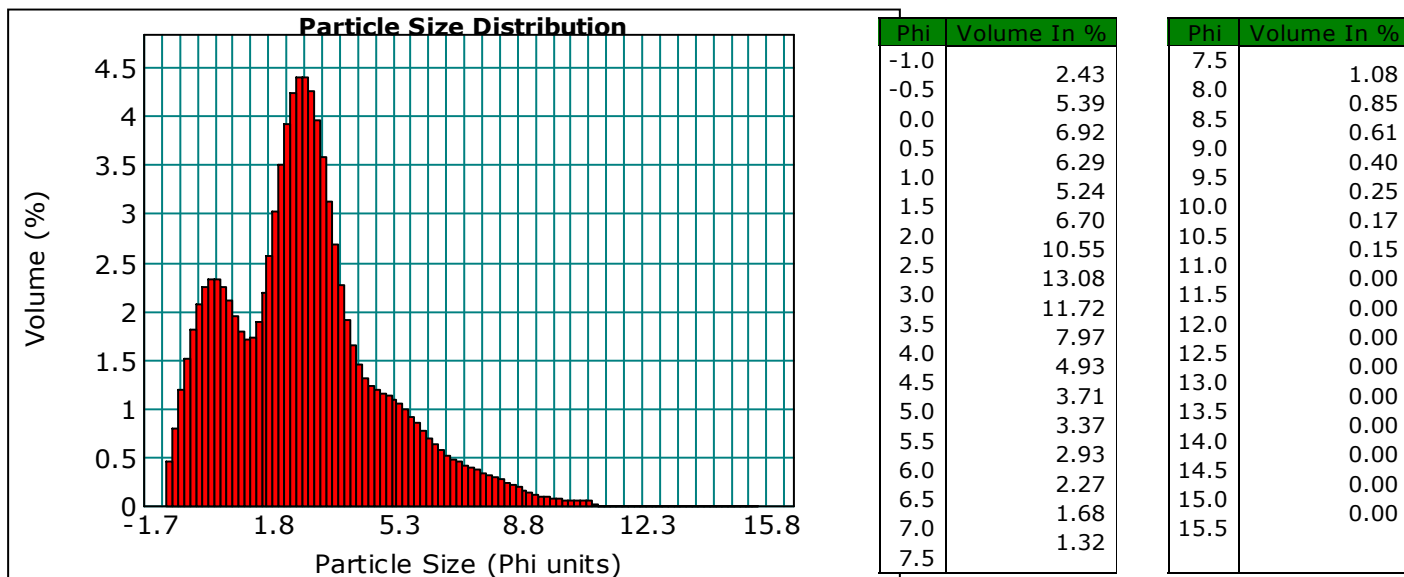
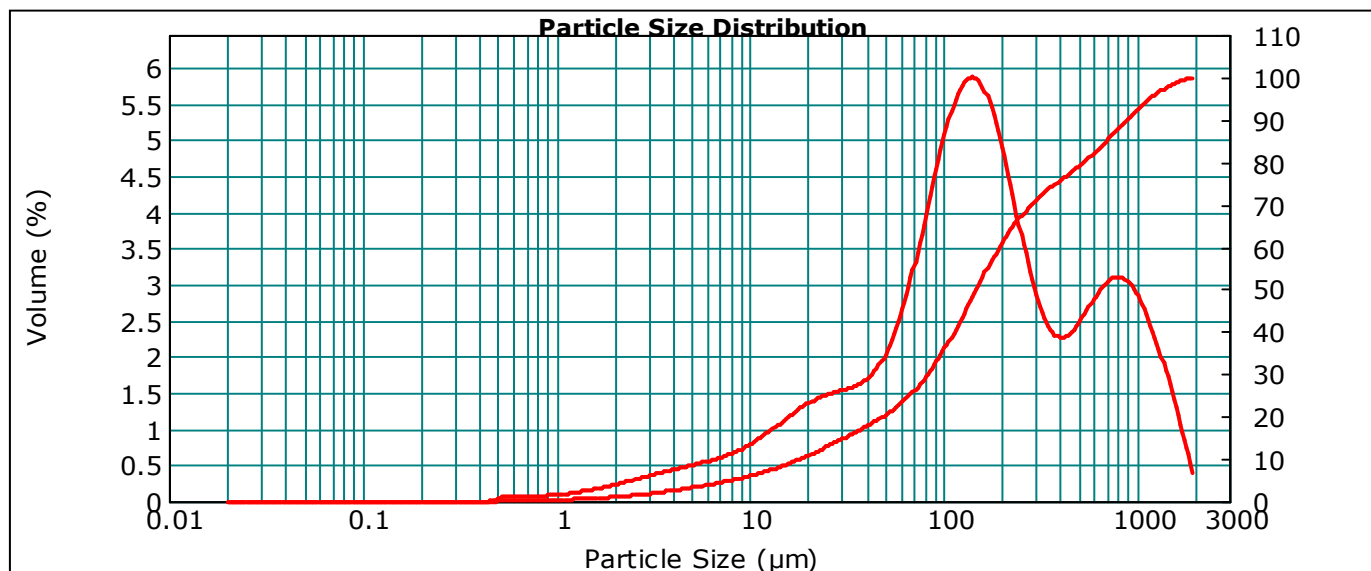
Inclusive SD (Sorting Coeff.) = 2.16 phi (Very Poorly)

Inclusive Kurtosis = 1.17 phi (Leptokurtic)

Inclusive Mean = 2.74 phi (Fine Sand)

Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:31:31 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 321.94um = 1.64phi

Surface Weighted Mean:
D[3,2] = 34.16um = 4.87phi

Span :
5.906

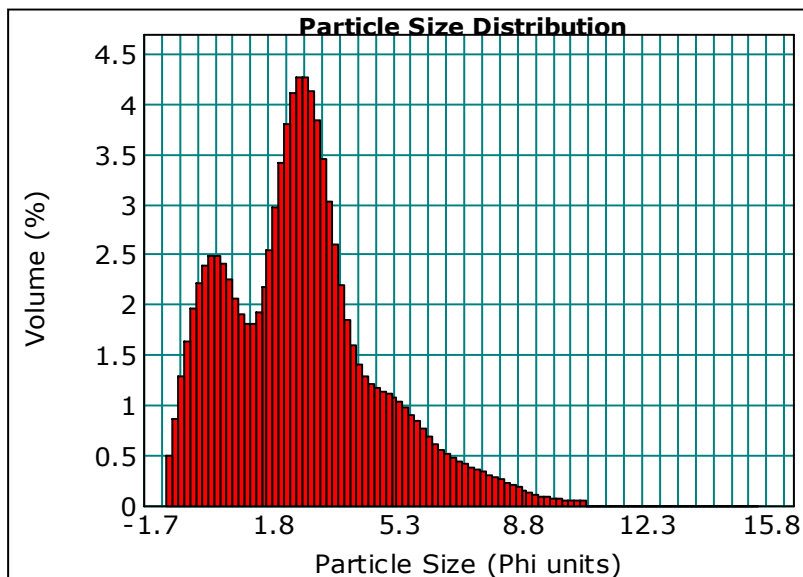
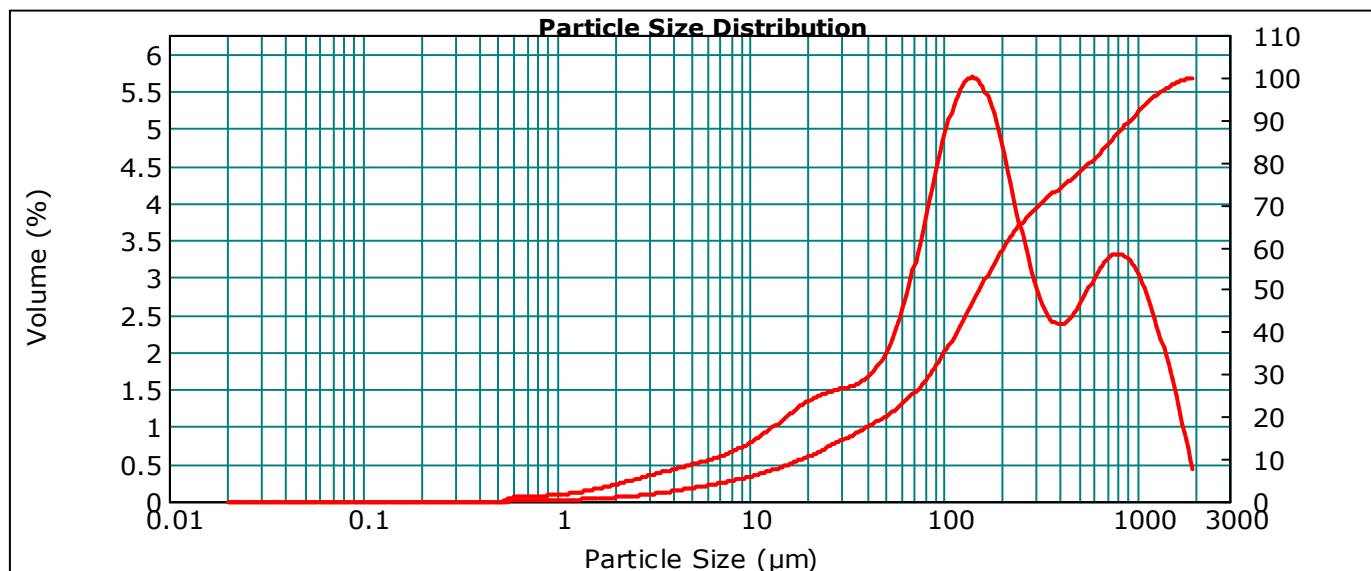
Uniformity:
1.64

Specific Surface Area*:
0.176 m²/g

Percentiles:
d(0.05) = 8.46um = 6.89phi
d(0.10) = 18.7um = 5.74phi
d(0.16) = 35.22um = 4.83phi
d(0.50) = 154.02um = 2.7phi
d(0.84) = 702.44um = .51phi
d(0.90) = 928.31um = .11phi
d(0.95) = 1205.32um = -.027phi

Distribution Properties:
Kurtosis = 2.55
Skewness = 1.76
Standard Deviation = 389.86 um = 1.36phi
Inclusive SD (Sorting Coeff.) = 2.16 phi (Very Poorly)
Inclusive Kurtosis = 1.11 phi (Mesokurtic)
Inclusive Mean = 2.68 phi (Fine Sand)
Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.65
-0.5	5.82
0.0	7.39
0.5	6.70
1.0	5.50
1.5	6.70
2.0	10.28
2.5	12.68
3.0	11.34
3.5	7.71
4.0	4.79
4.5	3.64
5.0	3.33
5.5	2.90
6.0	2.25
6.5	1.67
7.0	1.30
7.5	

Phi	Volume In %
7.5	1.06
8.0	0.83
8.5	0.59
9.0	0.38
9.5	0.24
10.0	0.16
10.5	0.08
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:41:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 311.34um = 1.68phi

Surface Weighted Mean:
D[3,2] = 32.05um = 4.96phi

Span :
6.153

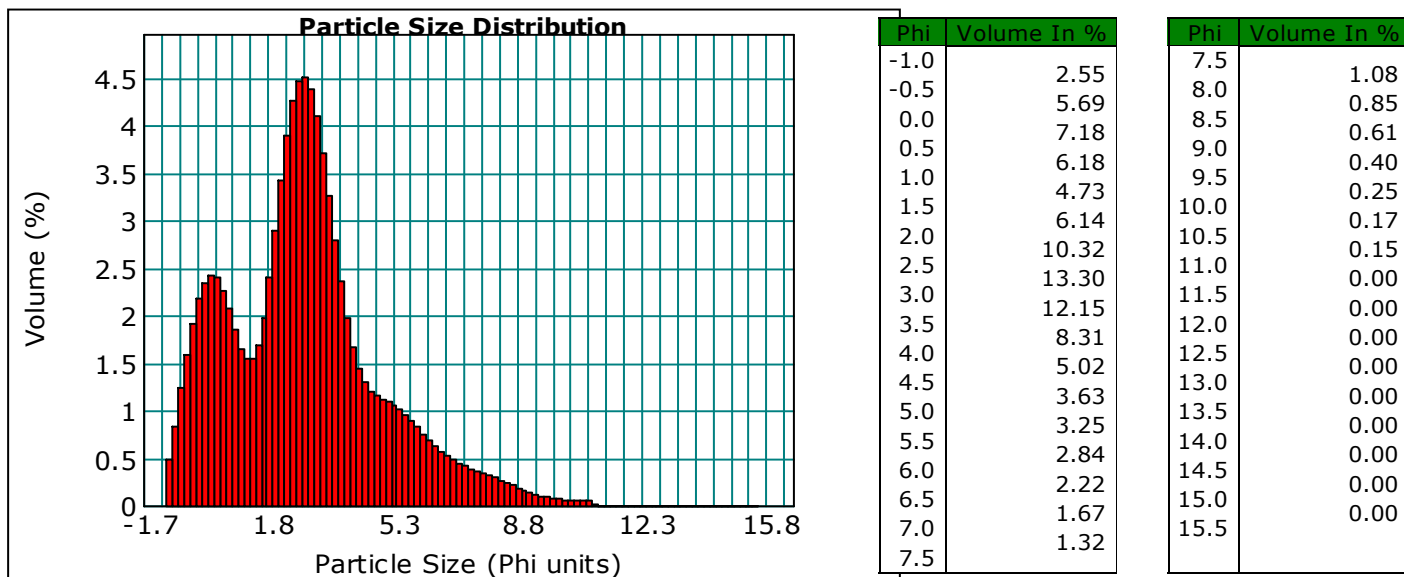
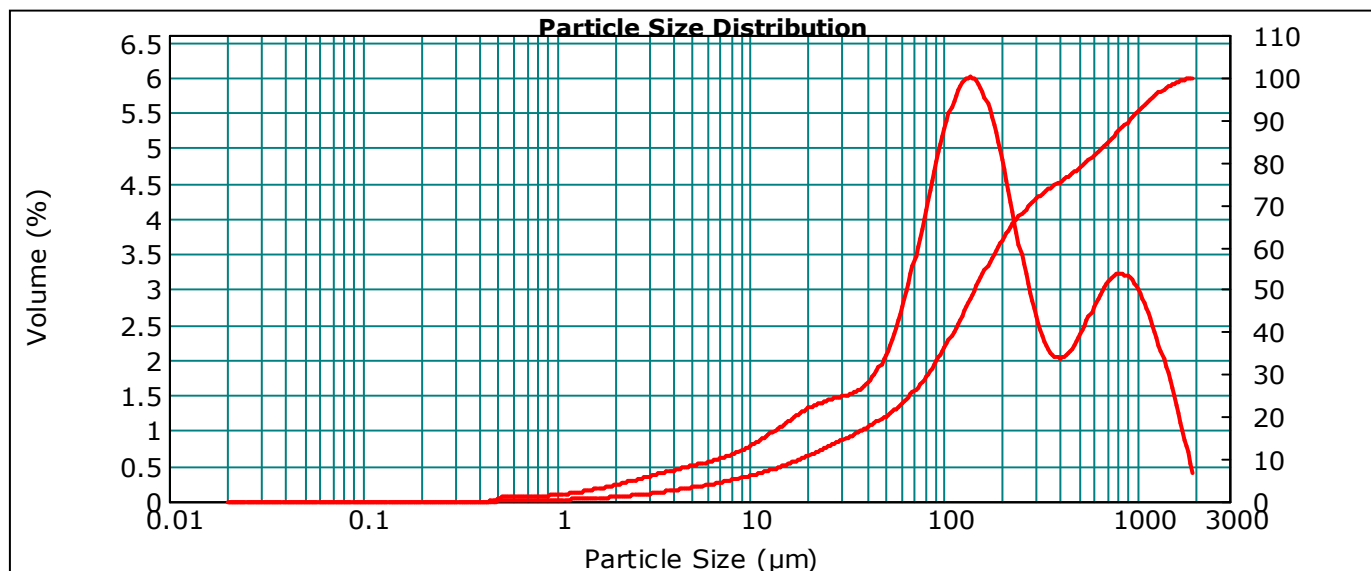
Uniformity:
1.67

Specific Surface Area*:
0.187 m²/g

Percentiles:
d(0.05) = 8.13um = 6.94phi
d(0.10) = 18.42um = 5.76phi
d(0.16) = 35.25um = 4.83phi
d(0.50) = 146.02um = 2.78phi
d(0.84) = 687.01um = .54phi
d(0.90) = 916.92um = .13phi
d(0.95) = 1194.06um = -0.26phi

Distribution Properties:
Kurtosis = 2.76
Skewness = 1.82
Standard Deviation = 386.35 um = 1.37phi
Inclusive SD (Sorting Coeff.) = 2.16 phi (Very Poorly)
Inclusive Kurtosis = 1.17 phi (Leptokurtic)
Inclusive Mean = 2.71 phi (Fine Sand)
Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:42:57 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 323.45um = 1.63phi

Surface Weighted Mean:

D[3,2] = 32.65um = 4.94phi

Span :

6.275

Uniformity:

1.72

Specific Surface Area*:

0.184 m²/g

Percentiles:

d(0.05) = 8.27um = 6.92phi

d(0.10) = 18.59um = 5.75phi

d(0.16) = 35.59um = 4.81phi

d(0.50) = 148.46um = 2.75phi

d(0.84) = 724.83um = .46phi

d(0.90) = 950.22um = .07phi

d(0.95) = 1223.55um = -.02phi

Distribution Properties:

Kurtosis = 2.38

Skewness = 1.74

Standard Deviation = 397.2 um = 1.33phi

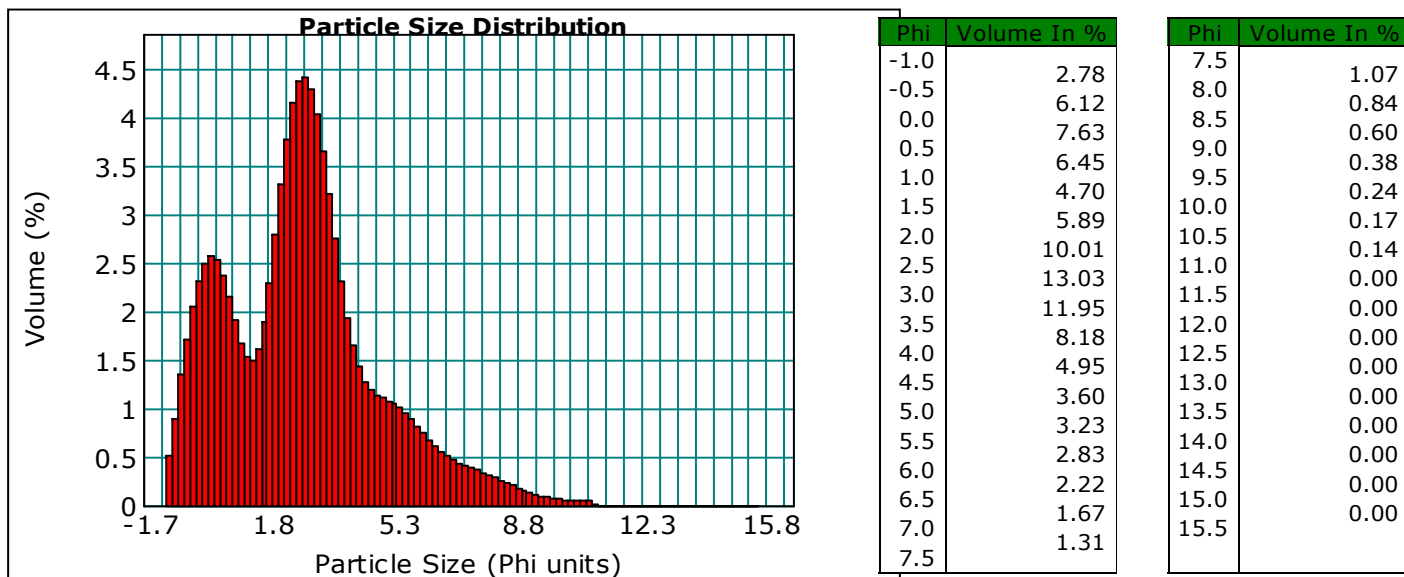
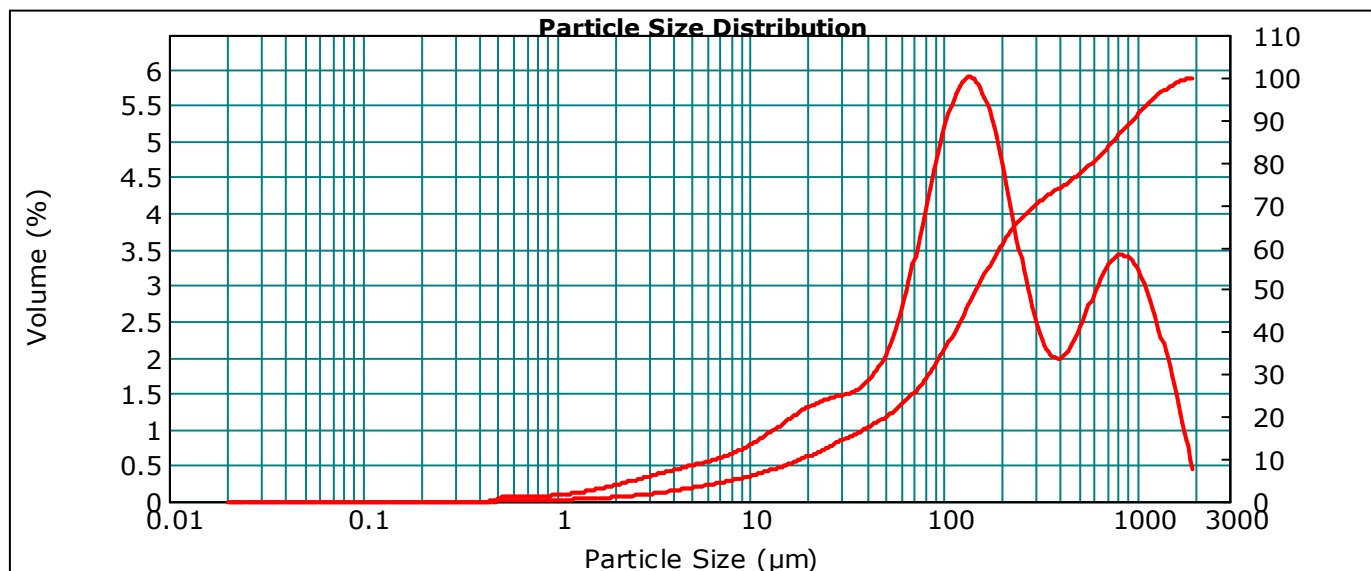
Inclusive SD (Sorting Coeff.) = 2.18 phi (Very Poorly)

Inclusive Kurtosis = 1.11 phi (Mesokurtic)

Inclusive Mean = 2.68 phi (Fine Sand)

Inclusive Skewness = .05 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:44:05 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 298.24um = 1.75phi

Surface Weighted Mean:
D[3,2] = 31.93um = 4.97phi

Span :
5.941

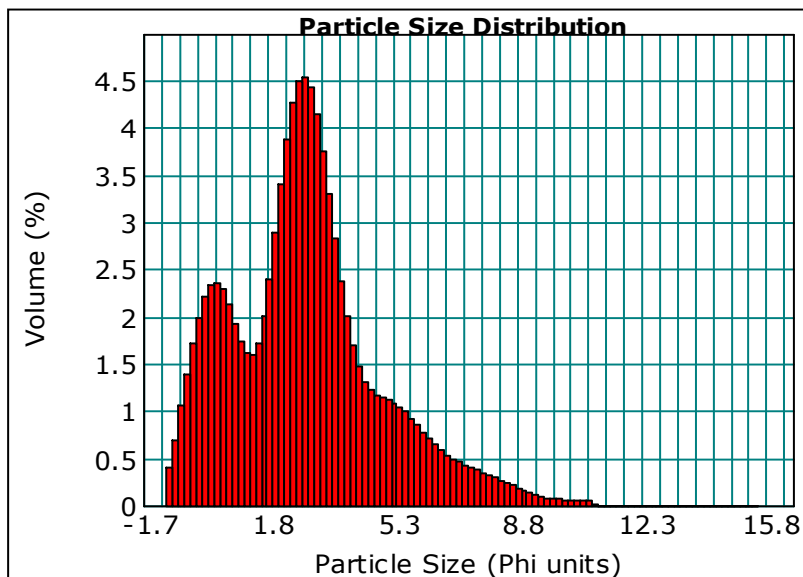
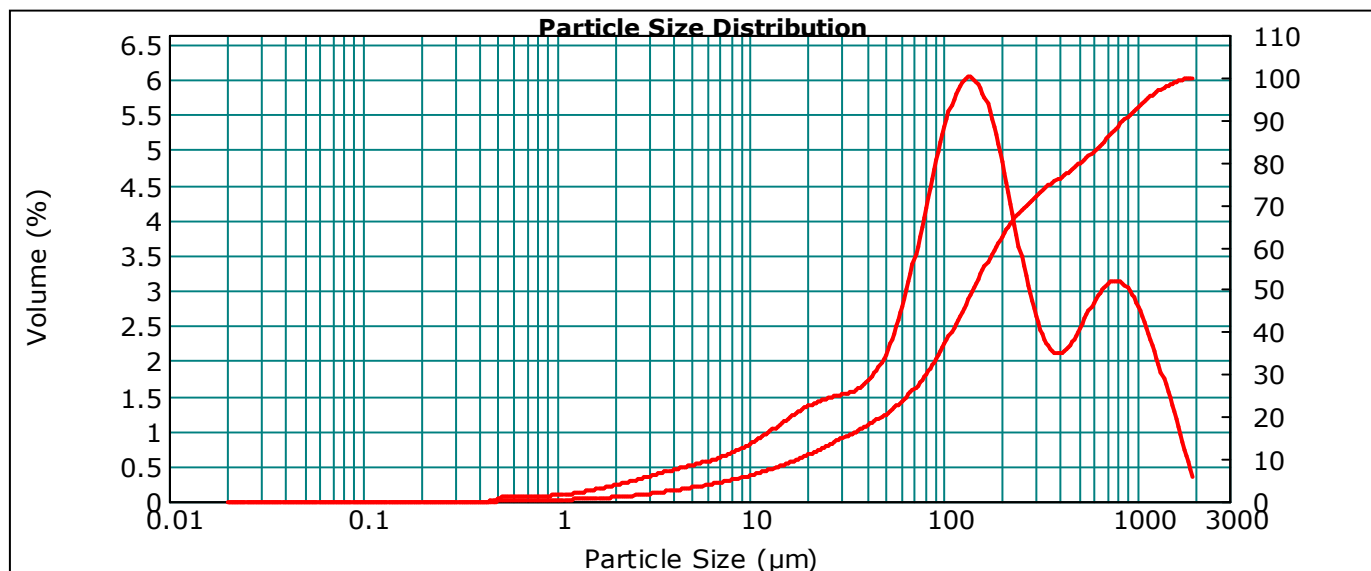
Uniformity:
1.62

Specific Surface Area*:
0.188 m²/g

Percentiles:
d(0.05) = 8.04um = 6.96phi
d(0.10) = 17.89um = 5.8phi
d(0.16) = 33.86um = 4.88phi
d(0.50) = 143.18um = 2.8phi
d(0.84) = 646.06um = .63phi
d(0.90) = 868.55um = .2phi
d(0.95) = 1142.1um = -.19phi

Distribution Properties:
Kurtosis = 3.16
Skewness = 1.89
Standard Deviation = 370.74 um = 1.43phi
Inclusive SD (Sorting Coeff.) = 2.15 phi (Very Poorly)
Inclusive Kurtosis = 1.18 phi (Leptokurtic)
Inclusive Mean = 2.77 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.17
-0.5	5.11
0.0	6.92
0.5	6.34
1.0	4.94
1.5	6.19
2.0	10.31
2.5	13.36
3.0	12.27
3.5	8.40
4.0	5.09
4.5	3.69
5.0	3.34
5.5	2.94
6.0	2.31
6.5	1.73
7.0	1.36
7.5	

Phi	Volume In %
7.5	1.11
8.0	0.86
8.5	0.61
9.0	0.39
9.5	0.24
10.0	0.17
10.5	0.15
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_9 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:41:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 311.01um = 1.68phi

Surface Weighted Mean:
D[3,2] = 32.21um = 4.96phi

Span :
6.132

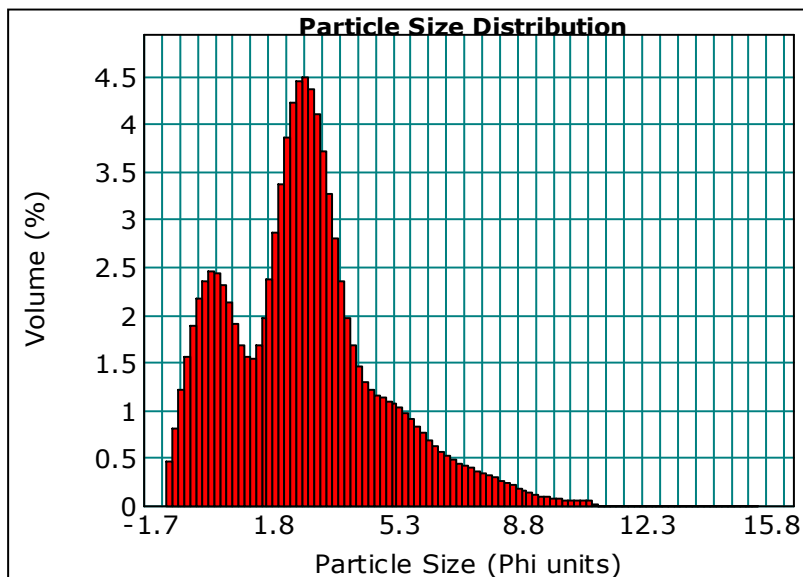
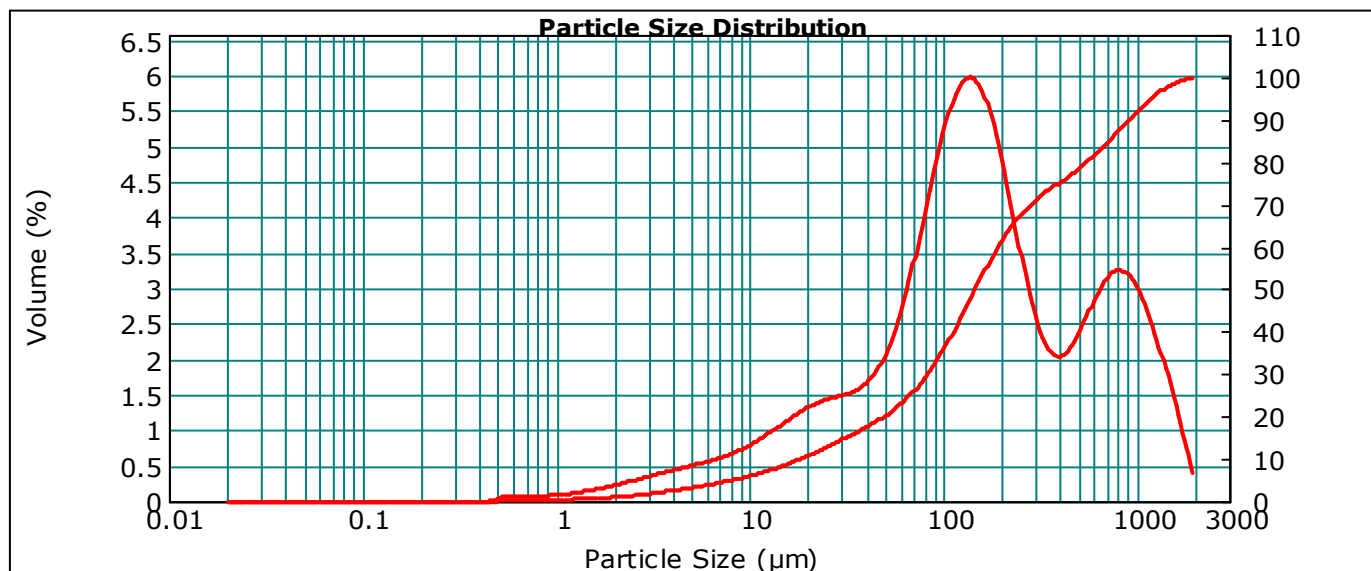
Uniformity:
1.67

Specific Surface Area*:
0.186 m²/g

Percentiles:
d(0.05) = 8.14um = 6.94phi
d(0.10) = 18.3um = 5.77phi
d(0.16) = 34.89um = 4.84phi
d(0.50) = 145.85um = 2.78phi
d(0.84) = 686.22um = .54phi
d(0.90) = 912.58um = .13phi
d(0.95) = 1187.95um = -0.25phi

Distribution Properties:
Kurtosis = 2.76
Skewness = 1.82
Standard Deviation = 385.05 um = 1.38phi
Inclusive SD (Sorting Coeff.) = 2.16 phi (Very Poorly)
Inclusive Kurtosis = 1.15 phi (Leptokurtic)
Inclusive Mean = 2.72 phi (Fine Sand)
Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.50
-0.5	5.64
0.0	7.24
0.5	6.32
1.0	4.79
1.5	6.08
2.0	10.21
2.5	13.23
3.0	12.13
3.5	8.29
4.0	5.02
4.5	3.64
5.0	3.27
5.5	2.87
6.0	2.25
6.5	1.69
7.0	1.33
7.5	

Phi	Volume In %
7.5	1.09
8.0	0.85
8.5	0.60
9.0	0.39
9.5	0.24
10.0	0.17
10.5	0.15
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:54:06 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 290.25um = 1.78phi

Surface Weighted Mean:
D[3,2] = 31.43um = 4.99phi

Span :
6.029

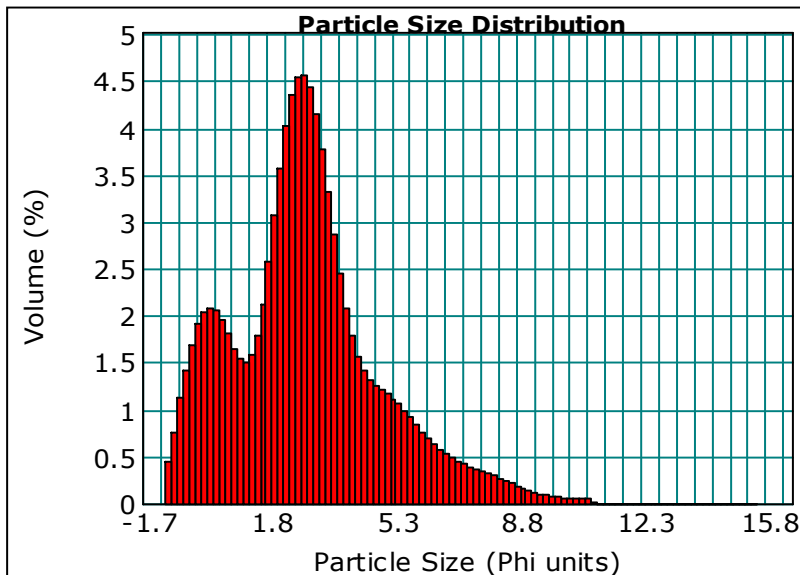
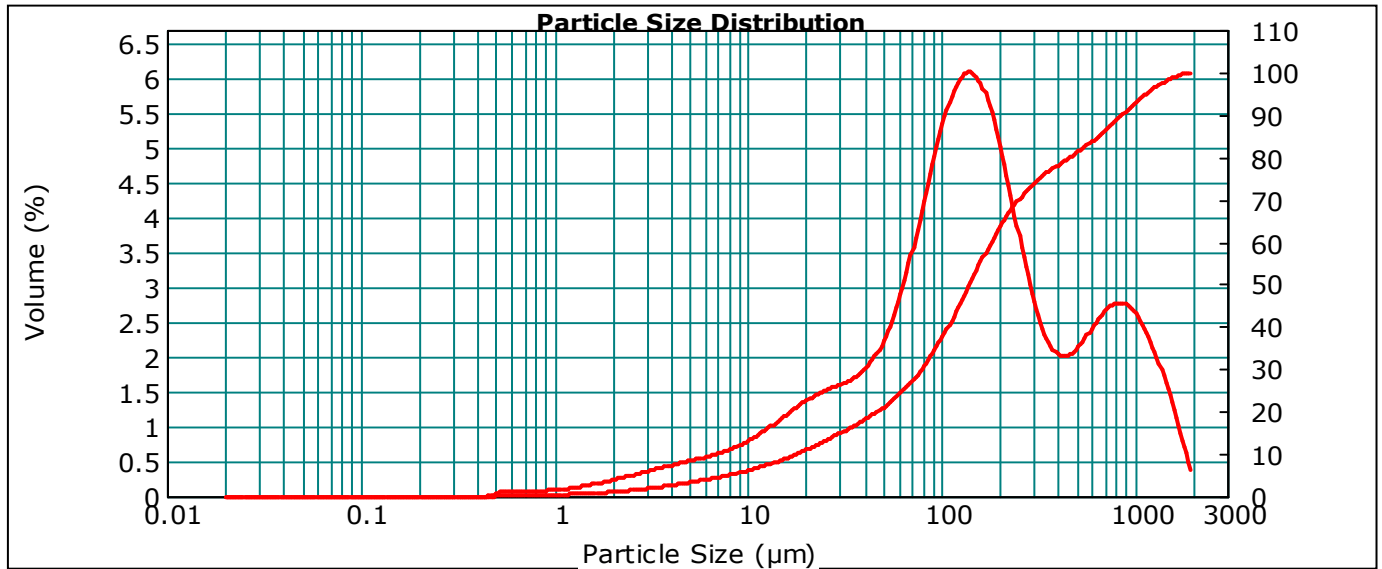
Uniformity:
1.61

Specific Surface Area*:
0.191 m²/g

Percentiles:
d(0.05) = 8um = 6.96phi
d(0.10) = 18.04um = 5.79phi
d(0.16) = 33.44um = 4.9phi
d(0.50) = 140.12um = 2.84phi
d(0.84) = 612.05um = .71phi
d(0.90) = 862.82um = .21phi
d(0.95) = 1156.46um = -.021phi

Distribution Properties:
Kurtosis = 3.58
Skewness = 2
Standard Deviation = 371.17 um = 1.43phi
Inclusive SD (Sorting Coeff.) = 2.14 phi (Very Poorly)
Inclusive Kurtosis = 1.25 phi (Leptokurtic)
Inclusive Mean = 2.82 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.33
-0.5	5.05
0.0	6.20
0.5	5.42
1.0	4.64
1.5	6.57
2.0	10.78
2.5	13.54
3.0	12.30
3.5	8.55
4.0	5.38
4.5	3.96
5.0	3.47
5.5	2.95
6.0	2.27
6.5	1.69
7.0	1.33
7.5	

Phi	Volume In %
7.5	1.10
8.0	0.87
8.5	0.62
9.0	0.40
9.5	0.25
10.0	0.17
10.5	0.15
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:55:13 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 265.8um = 1.91phi

Surface Weighted Mean:

D[3,2] = 30.64um = 5.03phi

Span :

5.631

Uniformity:

1.52

Specific Surface Area*:

0.196 m²/g

Percentiles:

d(0.05) = 7.75um = 7.01phi

d(0.10) = 17.3um = 5.85phi

d(0.16) = 31.57um = 4.99phi

d(0.50) = 134.28um = 2.9phi

d(0.84) = 519.99um = .94phi

d(0.90) = 773.4um = .37phi

d(0.95) = 1073.61um = -.1phi

Distribution Properties:

Kurtosis = 4.68

Skewness = 2.19

Standard Deviation = 345.38 um = 1.53phi

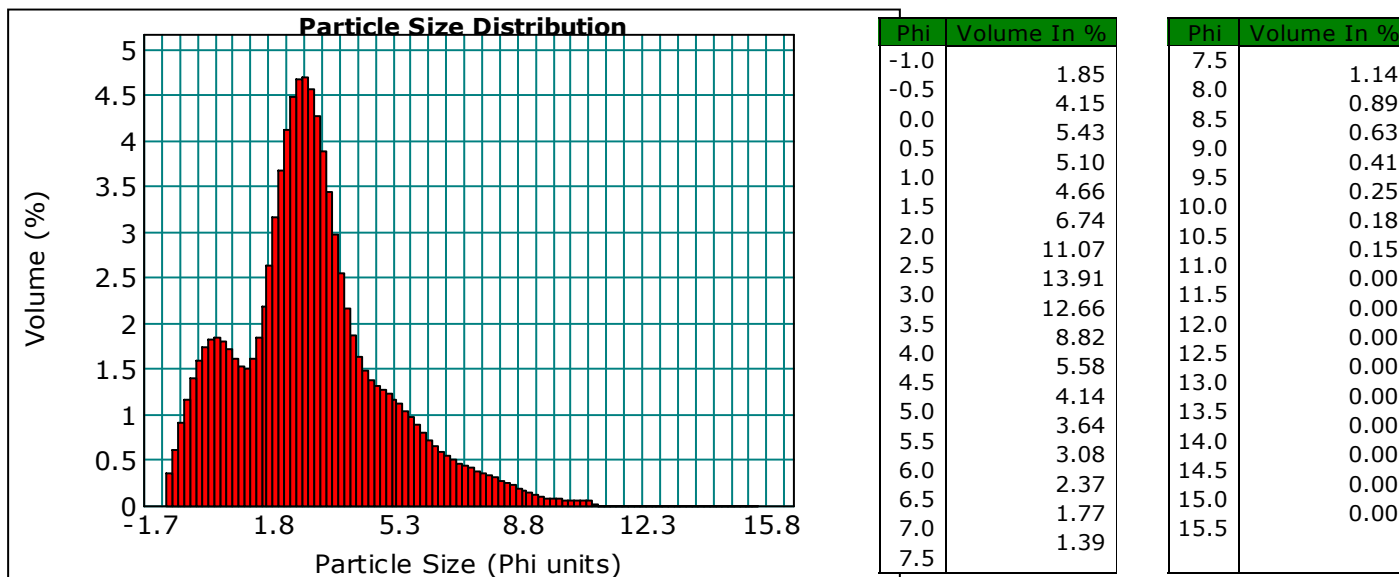
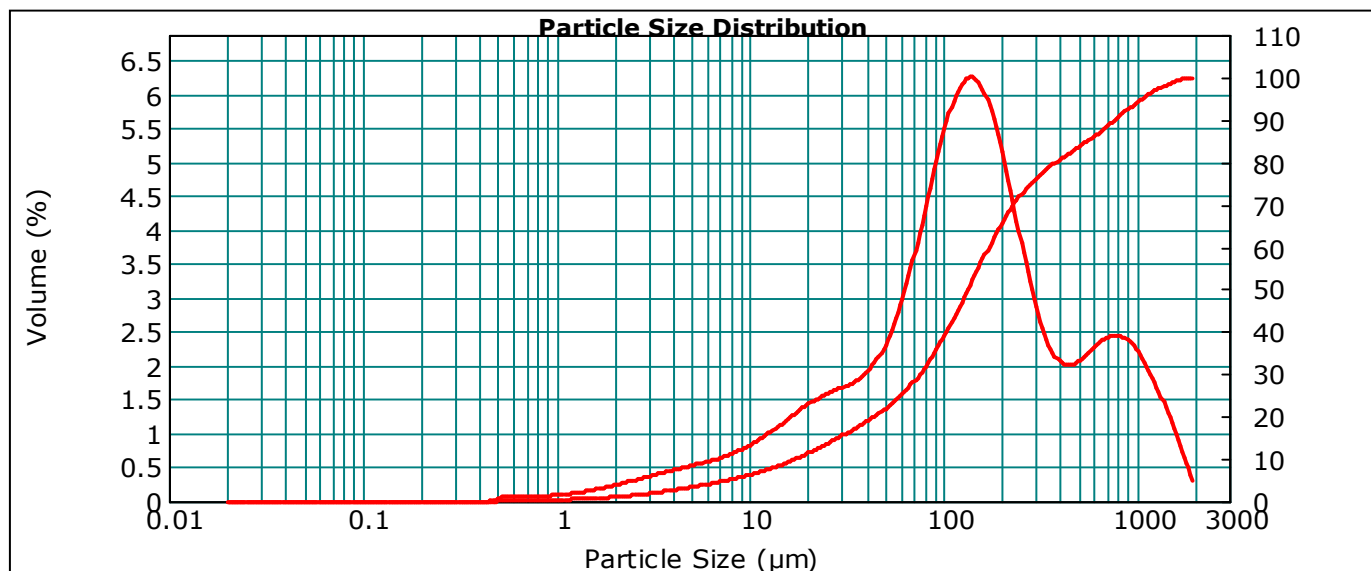
Inclusive SD (Sorting Coeff.) = 2.09 phi (Very Poorly)

Inclusive Kurtosis = 1.31 phi (Leptokurtic)

Inclusive Mean = 2.94 phi (Fine Sand)

Inclusive Skewness = .1 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_8
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:56:21 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 292.52um = 1.77phi

Surface Weighted Mean:
D[3,2] = 33.11um = 4.92phi

Span :
6.089

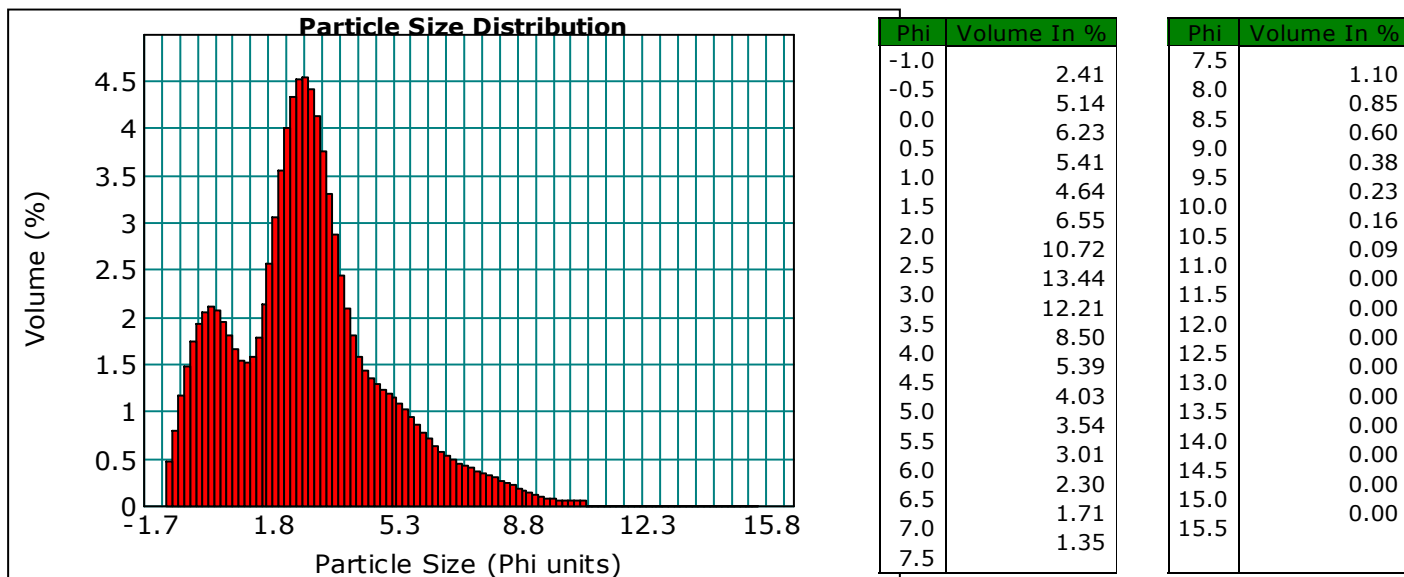
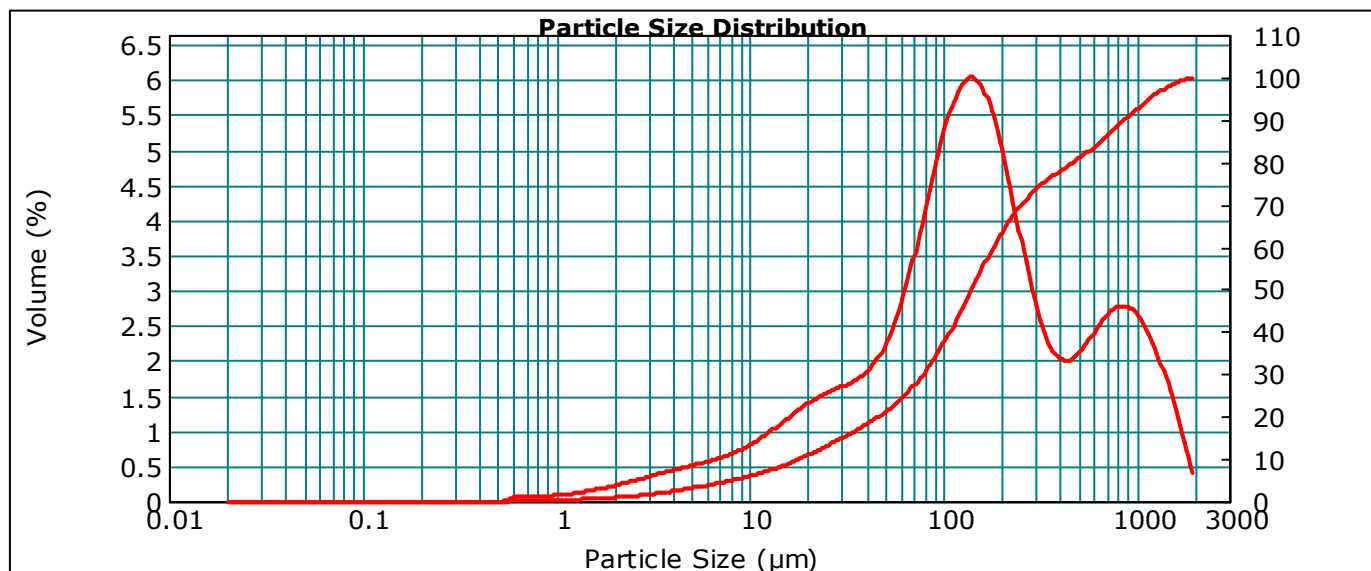
Uniformity:
1.63

Specific Surface Area*:
0.181 m²/g

Percentiles:
d(0.05) = 8.23um = 6.92phi
d(0.10) = 18.14um = 5.78phi
d(0.16) = 33.2um = 4.91phi
d(0.50) = 140.3um = 2.83phi
d(0.84) = 619.67um = .69phi
d(0.90) = 872.37um = .2phi
d(0.95) = 1167.49um = -.022phi

Distribution Properties:
Kurtosis = 3.5
Skewness = 1.99
Standard Deviation = 374.46 um = 1.42phi
Inclusive SD (Sorting Coeff.) = 2.14 phi (Very Poorly)
Inclusive Kurtosis = 1.24 phi (Leptokurtic)
Inclusive Mean = 2.81 phi (Fine Sand)
Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_8 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 3:54:06 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
 alkaline calgon

Vol. Weighted Mean:
 D[4,3] = 282.86um = 1.82phi

Surface Weighted Mean:
 D[3,2] = 31.7um = 4.98phi

Span :
 5.934

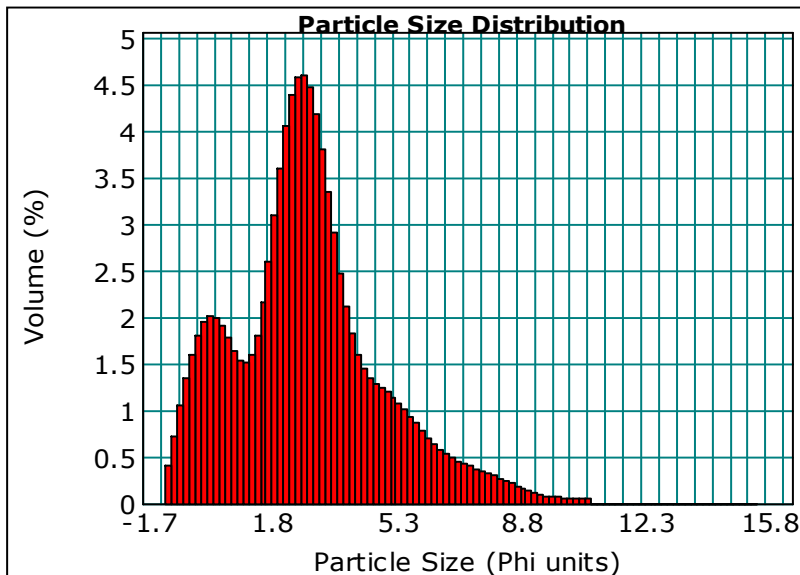
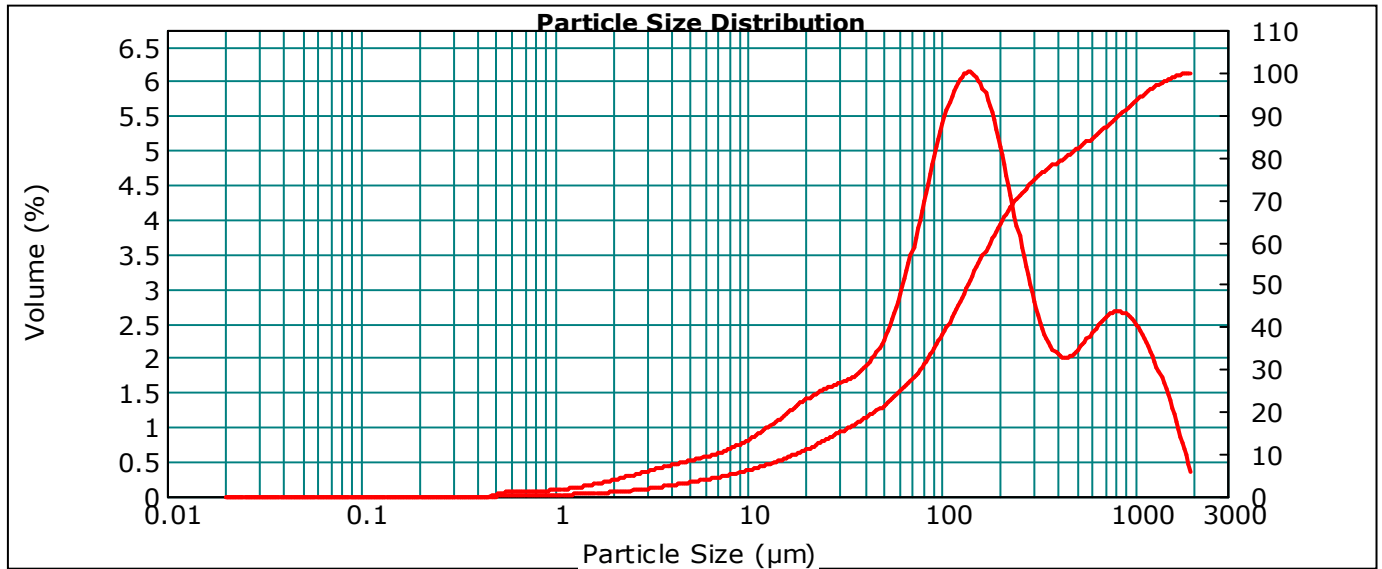
Uniformity:
 1.59

Specific Surface Area*:
 0.189 m²/g

Percentiles:
 d(0.05) = 7.99um = 6.97phi
 d(0.10) = 17.82um = 5.81phi
 d(0.16) = 32.72um = 4.93phi
 d(0.50) = 138.16um = 2.86phi
 d(0.84) = 584.39um = .77phi
 d(0.90) = 837.67um = .26phi
 d(0.95) = 1134.79um = -.18phi

Distribution Properties:
 Kurtosis = 3.89
 Skewness = 2.06
 Standard Deviation = 364.1 um = 1.46phi
 Inclusive SD (Sorting Coeff.) = 2.12 phi (Very Poorly)
 Inclusive Kurtosis = 1.27 phi (Leptokurtic)
 Inclusive Mean = 2.85 phi (Fine Sand)
 Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.20
-0.5	4.78
0.0	5.95
0.5	5.31
1.0	4.64
1.5	6.62
2.0	10.86
2.5	13.63
3.0	12.39
3.5	8.62
4.0	5.45
4.5	4.04
5.0	3.55
5.5	3.01
6.0	2.31
6.5	1.72
7.0	1.36
7.5	

Phi	Volume In %
7.5	1.11
8.0	0.87
8.5	0.62
9.0	0.40
9.5	0.25
10.0	0.17
10.5	0.13
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:05:08 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 326.8um = 1.61phi

Surface Weighted Mean:

D[3,2] = 35.31um = 4.82phi

Span :

5.947

Uniformity:

1.65

Specific Surface Area*:

0.17 m²/g

Percentiles:

d(0.05) = 8.94um = 6.81phi

d(0.10) = 20.18um = 5.63phi

d(0.16) = 38.52um = 4.7phi

d(0.50) = 155.27um = 2.69phi

d(0.84) = 719.29um = .48phi

d(0.90) = 943.54um = .08phi

d(0.95) = 1217.26um = -.028phi

Distribution Properties:

Kurtosis = 2.42

Skewness = 1.73

Standard Deviation = 393.77 um = 1.34phi

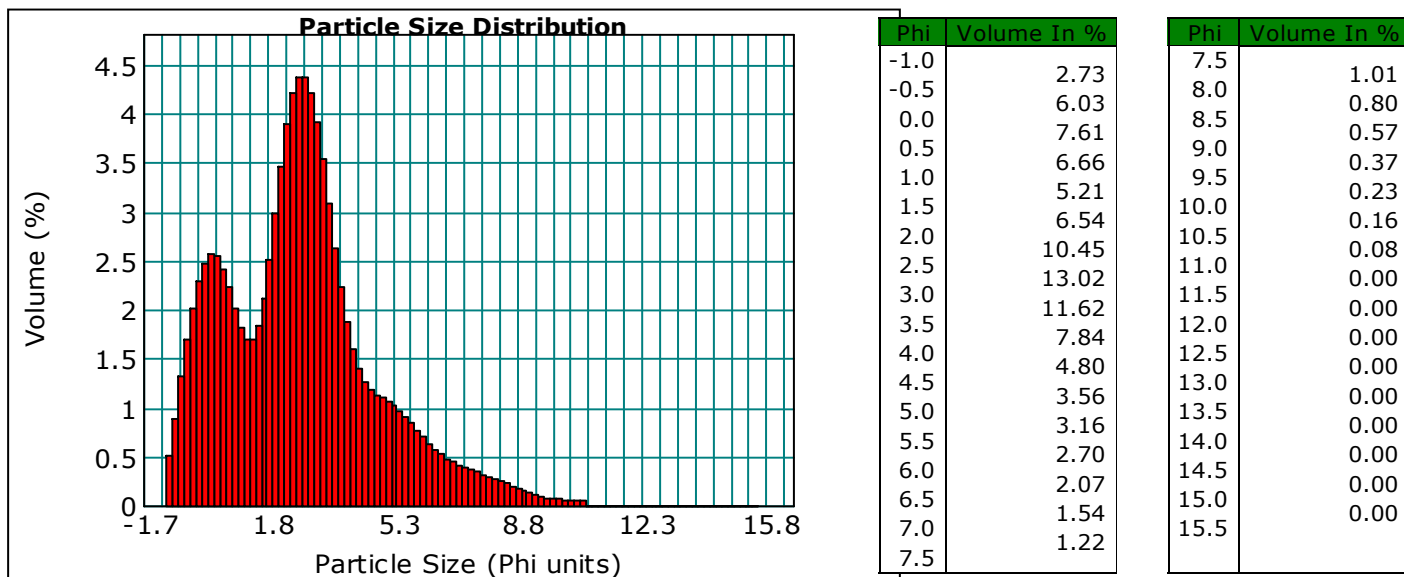
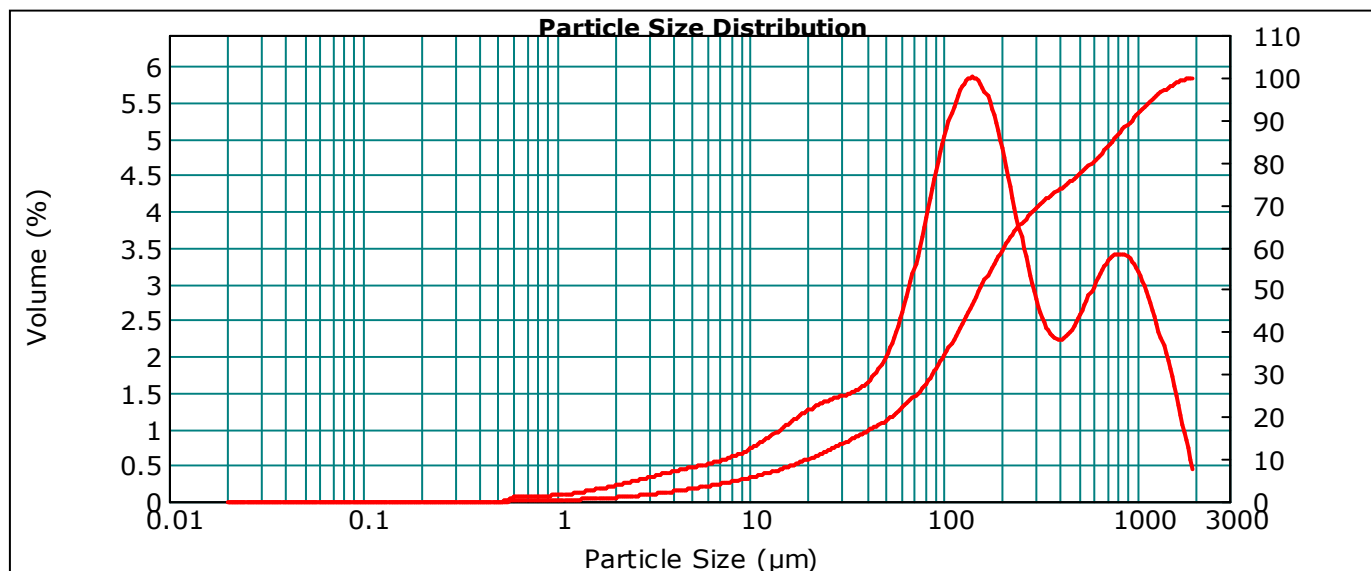
Inclusive SD (Sorting Coeff.) = 2.13 phi (Very Poorly)

Inclusive Kurtosis = 1.11 phi (Leptokurtic)

Inclusive Mean = 2.62 phi (Fine Sand)

Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: Kaoline_USP
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:36:31 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 8.22um = 6.93phi

Surface Weighted Mean:
D[3,2] = 2.97um = 8.4phi

Span :
2.677

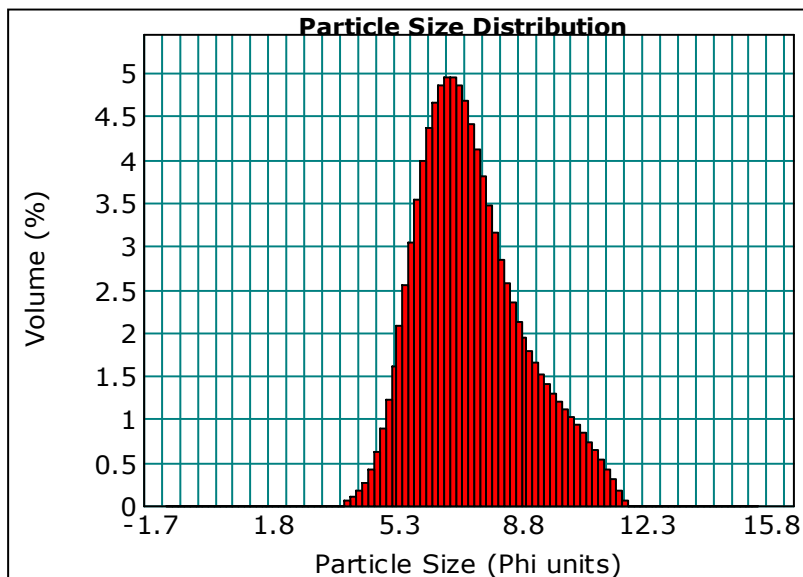
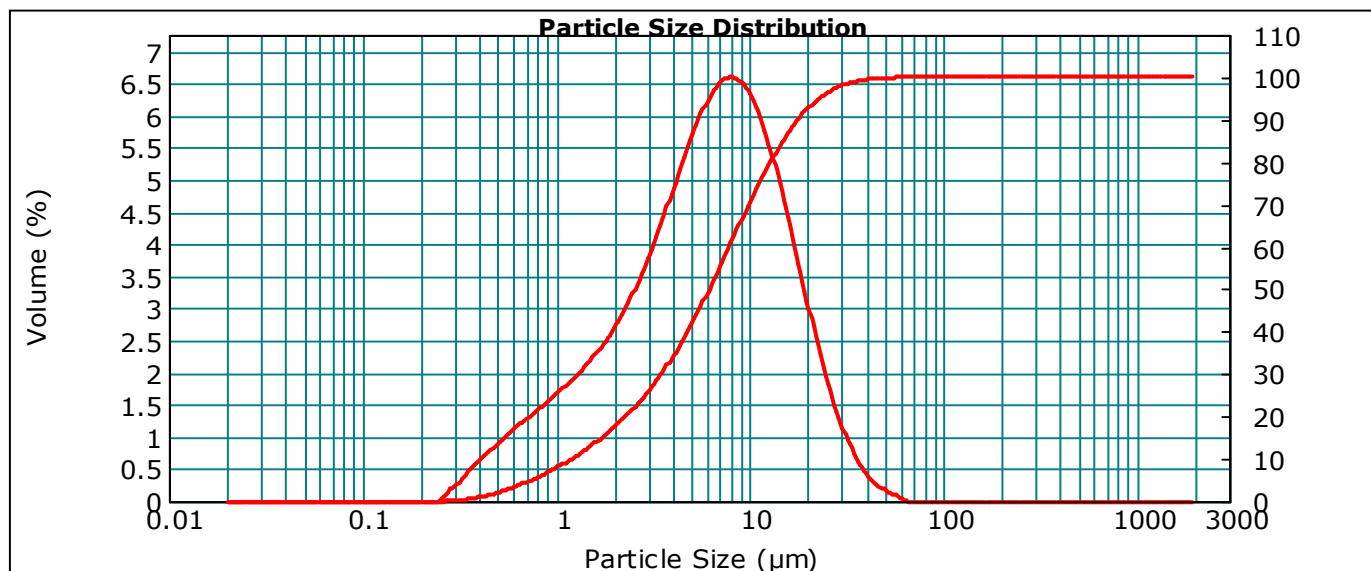
Uniformity:
0.847

Specific Surface Area*:
2.02 m²/g

Percentiles:
d(0.05) = .74um = 10.39phi
d(0.10) = 1.2um = 9.7phi
d(0.16) = 1.83um = 9.09phi
d(0.50) = 6.18um = 7.34phi
d(0.84) = 14.36um = 6.12phi
d(0.90) = 17.74um = 5.82phi
d(0.95) = 22.84um = 5.45phi

Distribution Properties:
Kurtosis = 5.3
Skewness = 1.91
Standard Deviation = 7.46 um = 7.07phi
Inclusive SD (Sorting Coeff.) = 1.49 phi (Poorly Sorted)
Inclusive Kurtosis = 1.04 phi (Mesokurtic)
Inclusive Mean = 7.52 phi (Silt)
Inclusive Skewness = .21 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	0.00
0.0	0.00
0.5	0.00
1.0	0.00
1.5	0.00
2.0	0.00
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.32
5.0	1.34
5.5	3.87
6.0	7.88
6.5	12.07
7.0	14.58
7.5	14.44

Phi	Volume In %
7.5	12.21
8.0	9.31
8.5	6.92
9.0	5.29
9.5	4.16
10.0	3.27
10.5	2.42
11.0	1.48
11.5	0.45
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: Kaoline_USP
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:37:38 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 8.28um = 6.92phi

Surface Weighted Mean:

D[3,2] = 2.97um = 8.39phi

Span :

2.685

Uniformity:

0.855

Specific Surface Area*:

2.02 m²/g

Percentiles:

d(0.05) = .75um = 10.39phi

d(0.10) = 1.2um = 9.7phi

d(0.16) = 1.84um = 9.09phi

d(0.50) = 6.19um = 7.34phi

d(0.84) = 14.41um = 6.12phi

d(0.90) = 17.82um = 5.81phi

d(0.95) = 23.01um = 5.44phi

Distribution Properties:

Kurtosis = 7.17

Skewness = 2.12

Standard Deviation = 7.67 um = 7.03phi

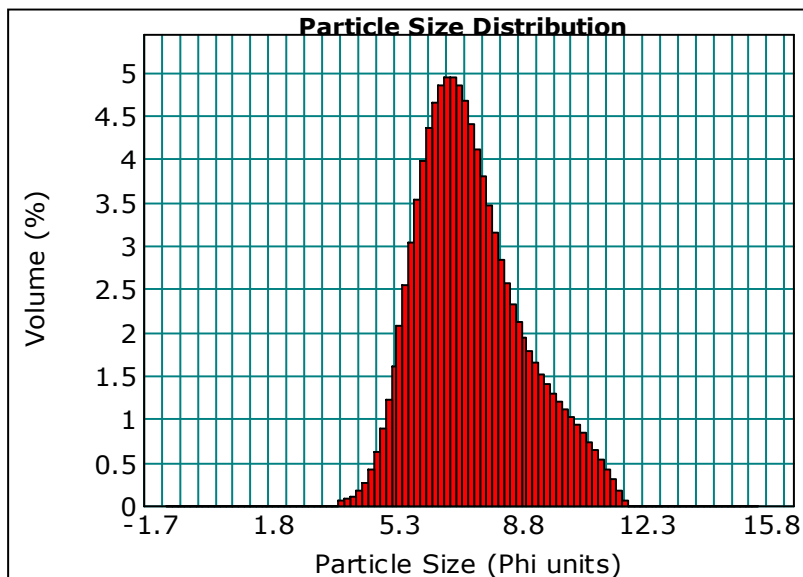
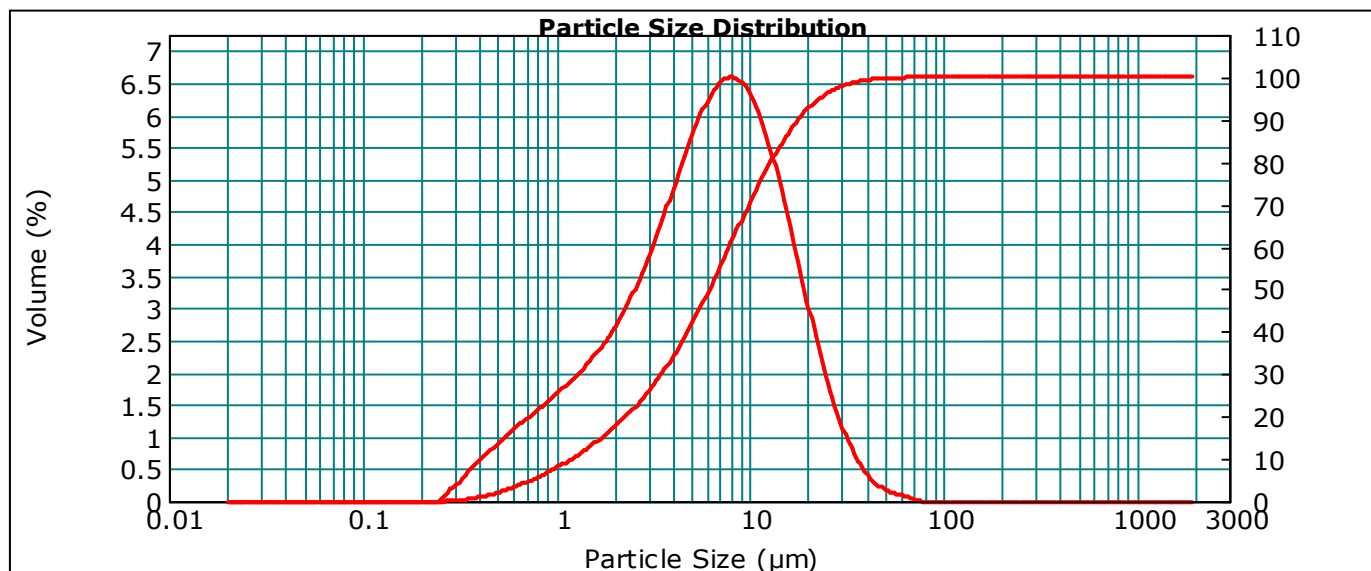
Inclusive SD (Sorting Coeff.) = 1.49 phi (Poorly Sorted)

Inclusive Kurtosis = 1.04 phi (Mesokurtic)

Inclusive Mean = 7.51 phi (Silt)

Inclusive Skewness = .21 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	0.00
0.0	0.00
0.5	0.00
1.0	0.00
1.5	0.00
2.0	0.00
2.5	0.00
3.0	0.00
3.5	0.00
4.0	0.06
4.5	0.37
5.0	1.35
5.5	3.86
6.0	7.88
6.5	12.06
7.0	14.57
7.5	14.42

Phi	Volume In %
7.5	12.19
8.0	9.30
8.5	6.90
9.0	5.28
9.5	4.15
10.0	3.27
10.5	2.42
11.0	1.48
11.5	0.45
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: Kaoline_USP
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:38:46 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 8.79um = 6.83phi

Surface Weighted Mean:
D[3,2] = 3um = 8.38phi

Span :
2.752

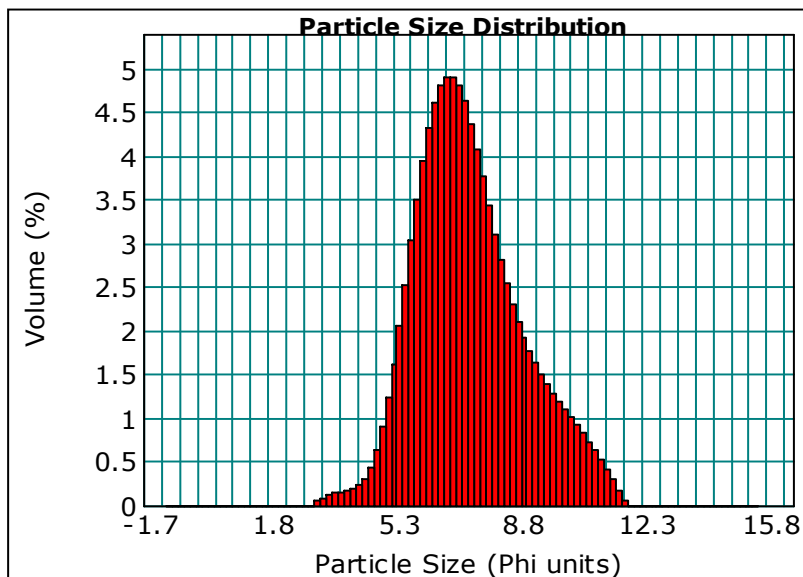
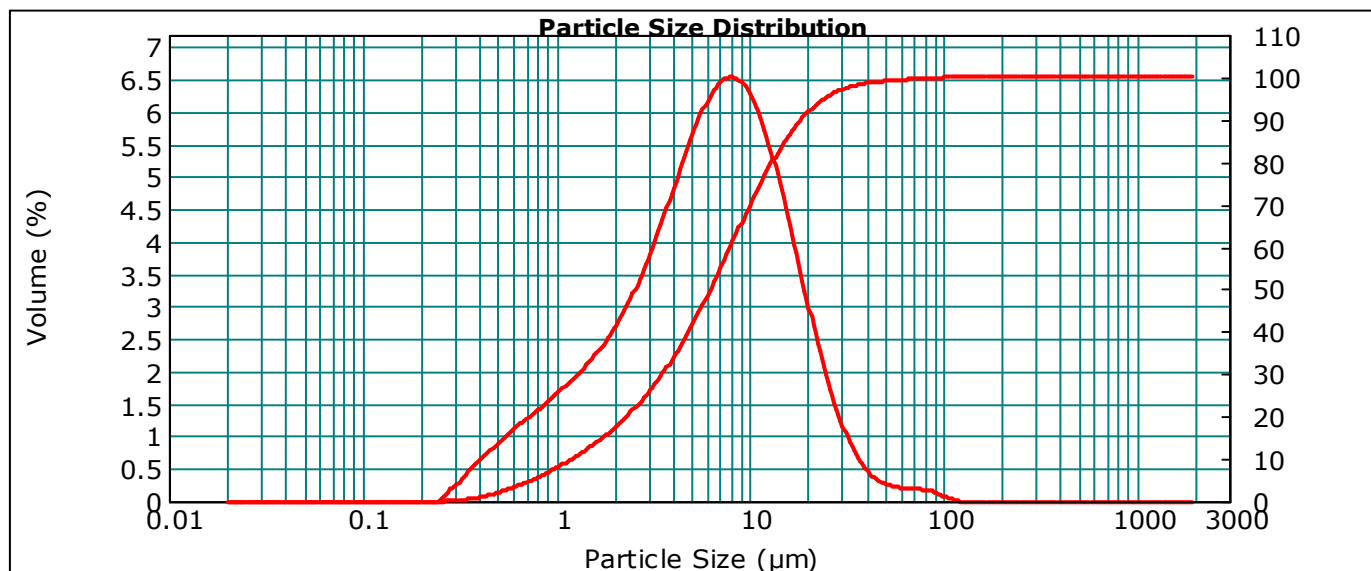
Uniformity:
0.922

Specific Surface Area*:
2 m²/g

Percentiles:
d(0.05) = .75um = 10.38phi
d(0.10) = 1.21um = 9.69phi
d(0.16) = 1.85um = 9.08phi
d(0.50) = 6.26um = 7.32phi
d(0.84) = 14.76um = 6.08phi
d(0.90) = 18.44um = 5.76phi
d(0.95) = 24.39um = 5.36phi

Distribution Properties:
Kurtosis = 22.17
Skewness = 3.66
Standard Deviation = 9.55 um = 6.71phi
Inclusive SD (Sorting Coeff.) = 1.51 phi (Poorly Sorted)
Inclusive Kurtosis = 1.05 phi (Mesokurtic)
Inclusive Mean = 7.49 phi (Silt)
Inclusive Skewness = .2 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	0.00
0.0	0.00
0.5	0.00
1.0	0.00
1.5	0.00
2.0	0.00
2.5	0.00
3.0	0.00
3.5	0.14
4.0	0.41
4.5	0.57
5.0	1.45
5.5	3.88
6.0	7.83
6.5	11.96
7.0	14.44
7.5	14.30

Phi	Volume In %
7.5	12.08
8.0	9.21
8.5	6.84
9.0	5.23
9.5	4.11
10.0	3.24
10.5	2.40
11.0	1.47
11.5	0.45
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: Kaoline_USP - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:36:31 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 8.43um = 6.89phi

Surface Weighted Mean:

D[3,2] = 2.98um = 8.39phi

Span :

2.704

Uniformity:

0.875

Specific Surface Area*:

2.01 m²/g

Percentiles:

d(0.05) = .75um = 10.39phi

d(0.10) = 1.21um = 9.7phi

d(0.16) = 1.84um = 9.09phi

d(0.50) = 6.21um = 7.33phi

d(0.84) = 14.5um = 6.11phi

d(0.90) = 17.99um = 5.8phi

d(0.95) = 23.38um = 5.42phi

Distribution Properties:

Kurtosis = 16.35

Skewness = 2.91

Standard Deviation = 8.29 um = 6.92phi

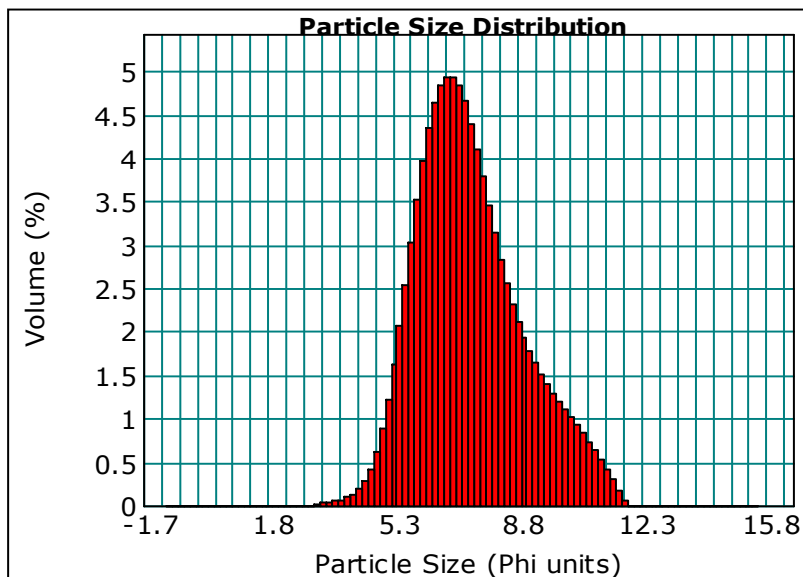
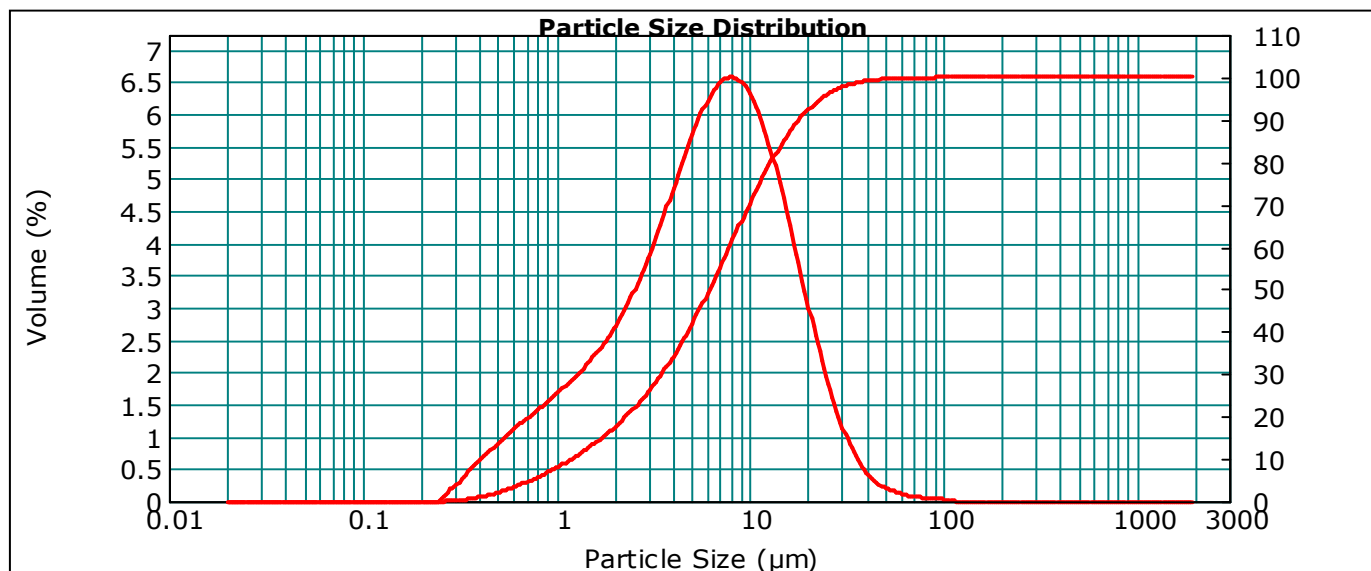
Inclusive SD (Sorting Coeff.) = 1.5 phi (Poorly Sorted)

Inclusive Kurtosis = 1.04 phi (Mesokurtic)

Inclusive Mean = 7.51 phi (Silt)

Inclusive Skewness = .2 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.00
-0.5	0.00
0.0	0.00
0.5	0.00
1.0	0.00
1.5	0.00
2.0	0.00
2.5	0.00
3.0	0.00
3.5	0.05
4.0	0.15
4.5	0.42
5.0	1.38
5.5	3.87
6.0	7.86
6.5	12.03
7.0	14.53
7.5	14.38

Phi	Volume In %
7.5	12.16
8.0	9.27
8.5	6.88
9.0	5.27
9.5	4.14
10.0	3.26
10.5	2.41
11.0	1.48
11.5	0.45
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:46:40 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 334.4um = 1.58phi

Surface Weighted Mean:

D[3,2] = 34.92um = 4.84phi

Span :

6.855

Uniformity:

1.82

Specific Surface Area*:

0.172 m²/g

Percentiles:

d(0.05) = 8.97um = 6.8phi

d(0.10) = 20.12um = 5.64phi

d(0.16) = 37.37um = 4.74phi

d(0.50) = 146.47um = 2.77phi

d(0.84) = 779.27um = .36phi

d(0.90) = 1024.26um = -0.03phi

d(0.95) = 1300.86um = -0.38phi

Distribution Properties:

Kurtosis = 2.08

Skewness = 1.71

Standard Deviation = 420.43 um = 1.25phi

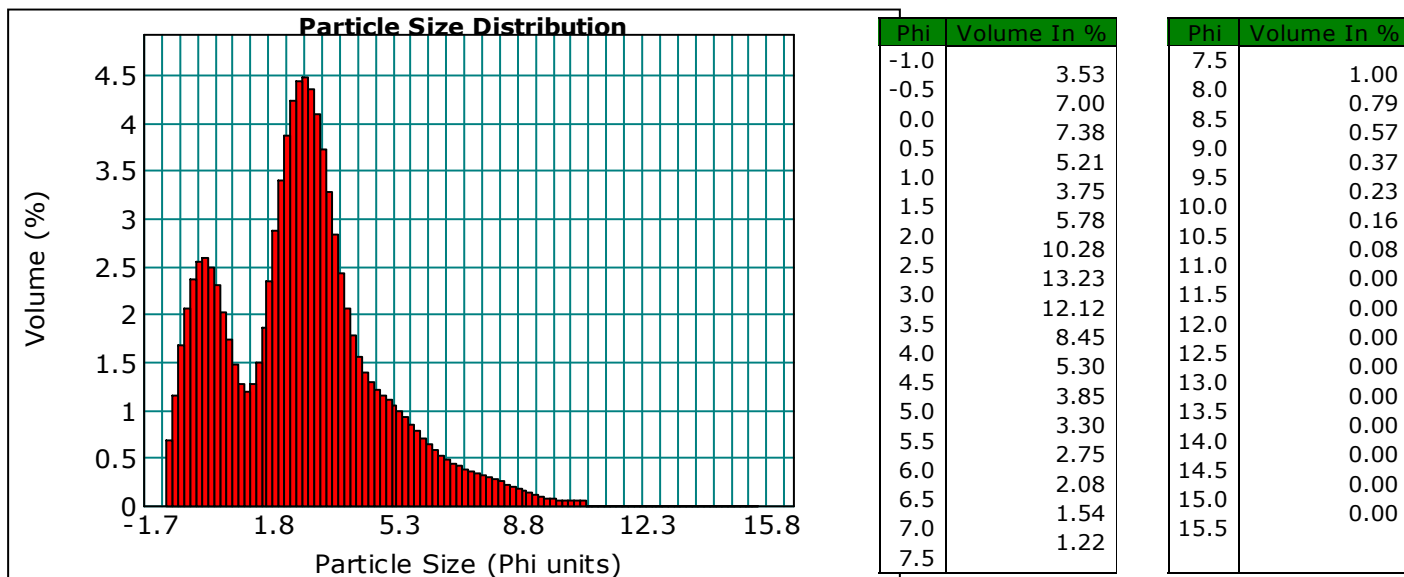
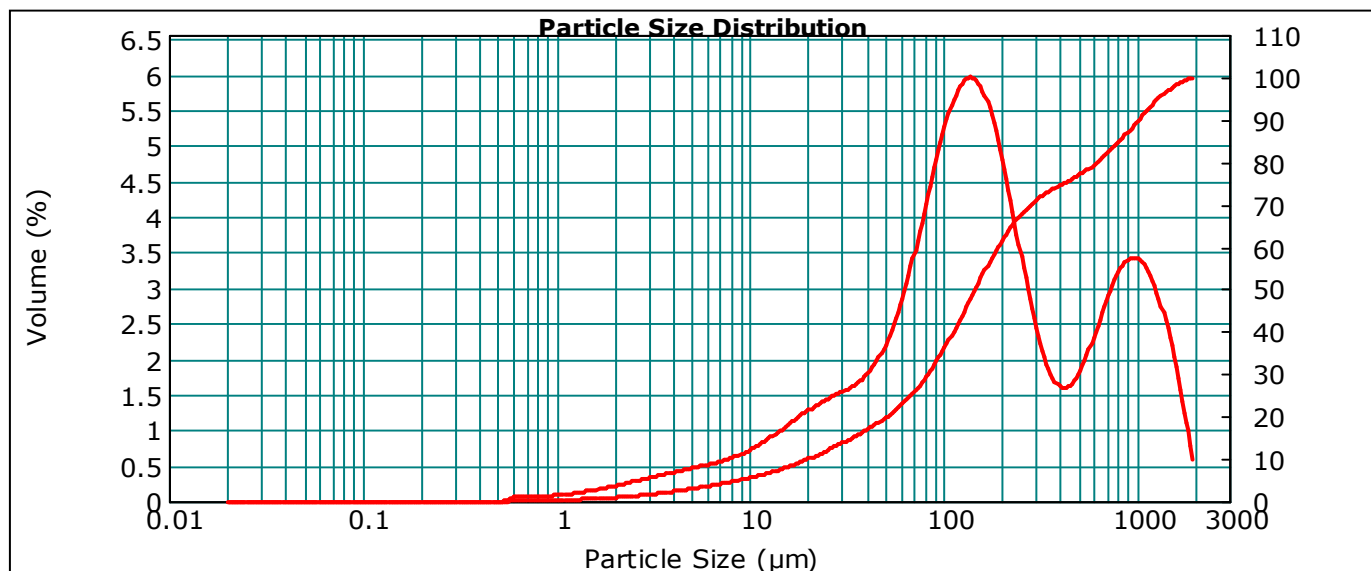
Inclusive SD (Sorting Coeff.) = 2.18 phi (Very Poorly)

Inclusive Kurtosis = 1.12 phi (Leptokurtic)

Inclusive Mean = 2.62 phi (Fine Sand)

Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:47:48 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 337.97um = 1.57phi

Surface Weighted Mean:
D[3,2] = 36.83um = 4.76phi

Span :
6.739

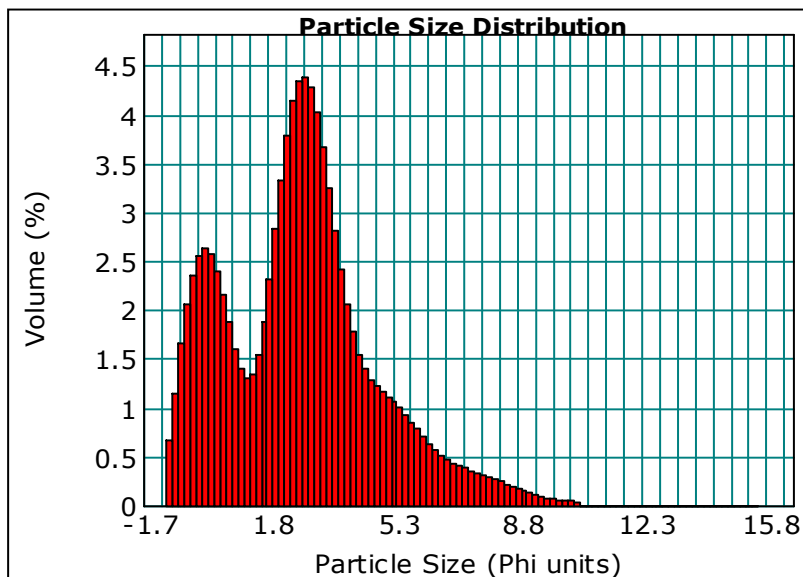
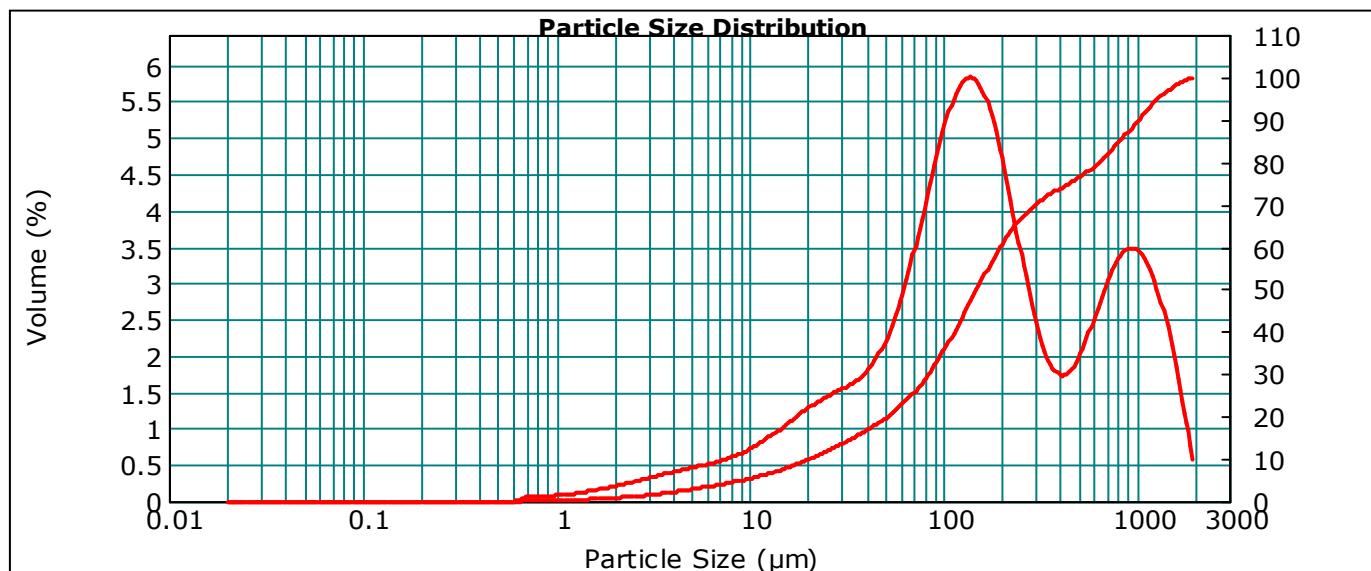
Uniformity:
1.81

Specific Surface Area*:
0.163 m²/g

Percentiles:
d(0.05) = 9.27um = 6.75phi
d(0.10) = 20.47um = 5.61phi
d(0.16) = 37.77um = 4.73phi
d(0.50) = 148.57um = 2.75phi
d(0.84) = 782.29um = .35phi
d(0.90) = 1021.76um = -0.03phi
d(0.95) = 1296.61um = -0.37phi

Distribution Properties:
Kurtosis = 2
Skewness = 1.68
Standard Deviation = 419.78 um = 1.25phi
Inclusive SD (Sorting Coeff.) = 2.17 phi (Very Poorly)
Inclusive Kurtosis = 1.08 phi (Mesokurtic)
Inclusive Mean = 2.61 phi (Fine Sand)
Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	3.48
-0.5	7.00
0.0	7.60
0.5	5.61
1.0	4.05
1.5	5.81
2.0	10.08
2.5	12.94
3.0	11.90
3.5	8.36
4.0	5.31
4.5	3.88
5.0	3.31
5.5	2.75
6.0	2.08
6.5	1.54
7.0	1.21
7.5	

Phi	Volume In %
7.5	0.99
8.0	0.78
8.5	0.56
9.0	0.36
9.5	0.22
10.0	0.16
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_7
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:48:55 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 332.78um = 1.59phi

Surface Weighted Mean:
D[3,2] = 36.56um = 4.77phi

Span :
6.797

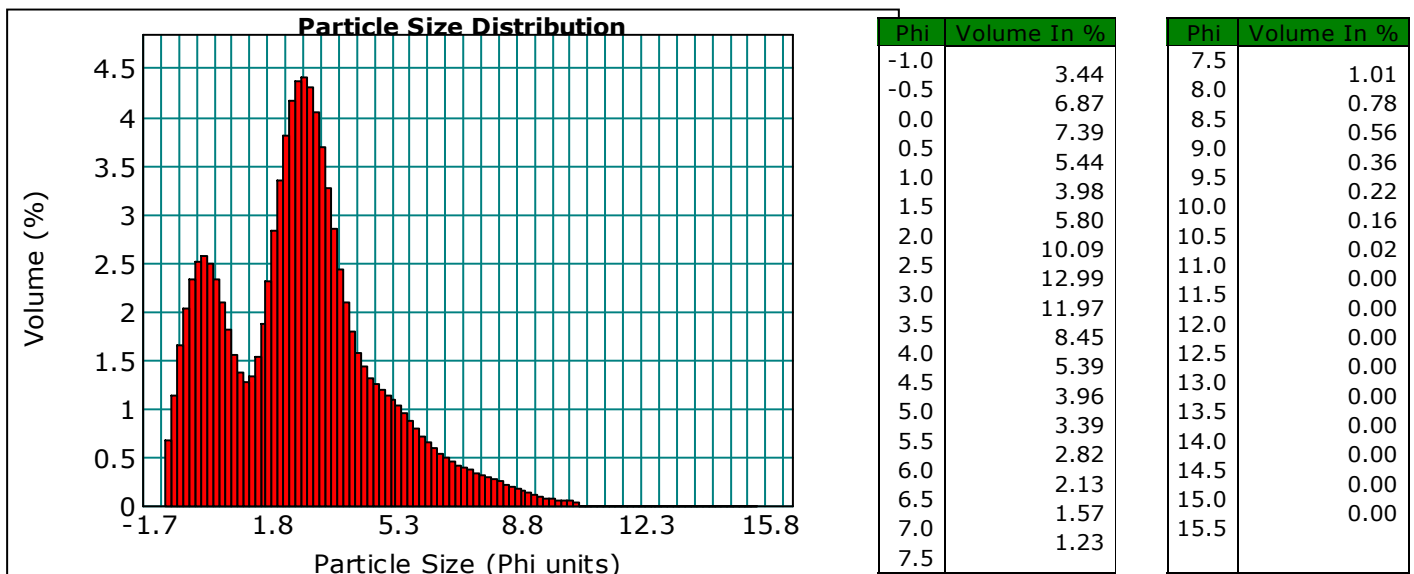
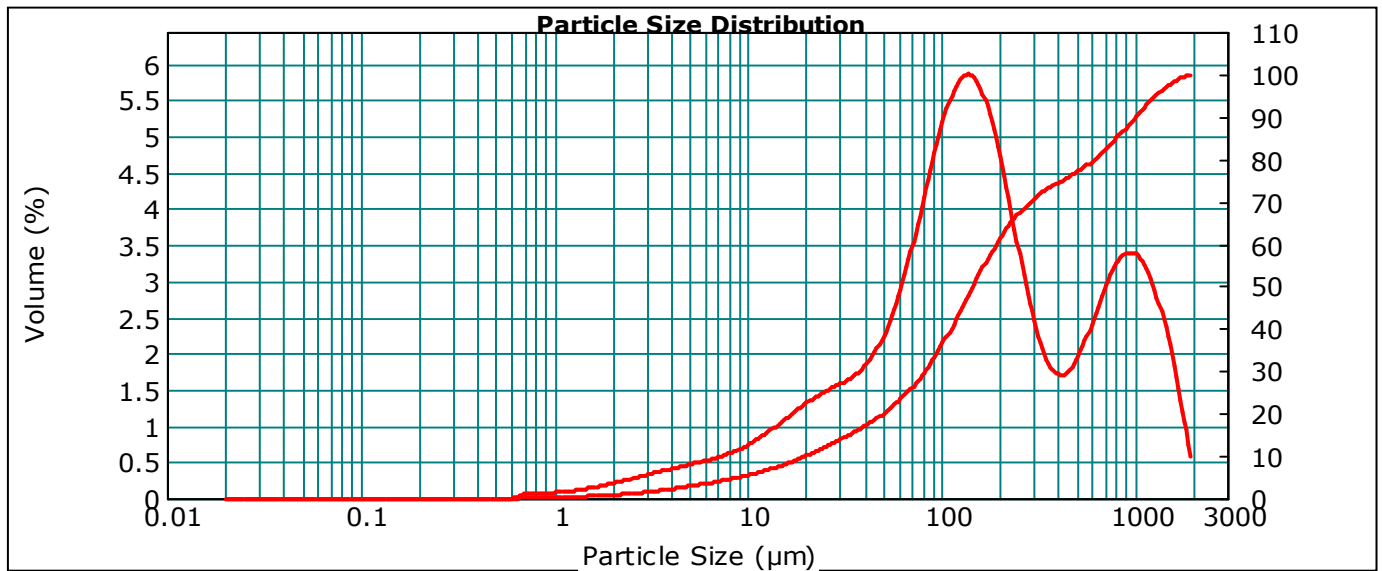
Uniformity:
1.82

Specific Surface Area*:
0.164 m²/g

Percentiles:
d(0.05) = 9.17um = 6.77phi
d(0.10) = 20.08um = 5.64phi
d(0.16) = 36.76um = 4.77phi
d(0.50) = 146.2um = 2.77phi
d(0.84) = 769.88um = .38phi
d(0.90) = 1013.72um = -0.02phi
d(0.95) = 1291.47um = -0.37phi

Distribution Properties:
Kurtosis = 2.13
Skewness = 1.71
Standard Deviation = 417.44 um = 1.26phi
Inclusive SD (Sorting Coeff.) = 2.18 phi (Very Poorly)
Inclusive Kurtosis = 1.1 phi (Mesokurtic)
Inclusive Mean = 2.64 phi (Fine Sand)
Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_7 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:46:40 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 335.05um = 1.58phi

Surface Weighted Mean:
D[3,2] = 36.09um = 4.79phi

Span :
6.798

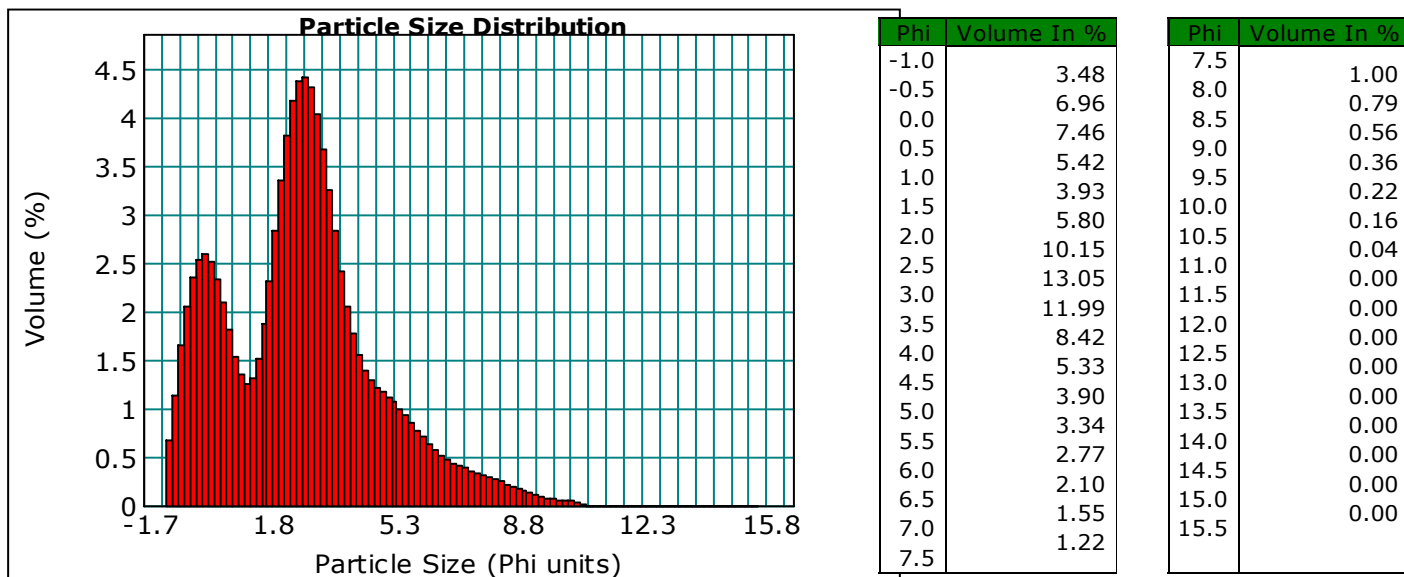
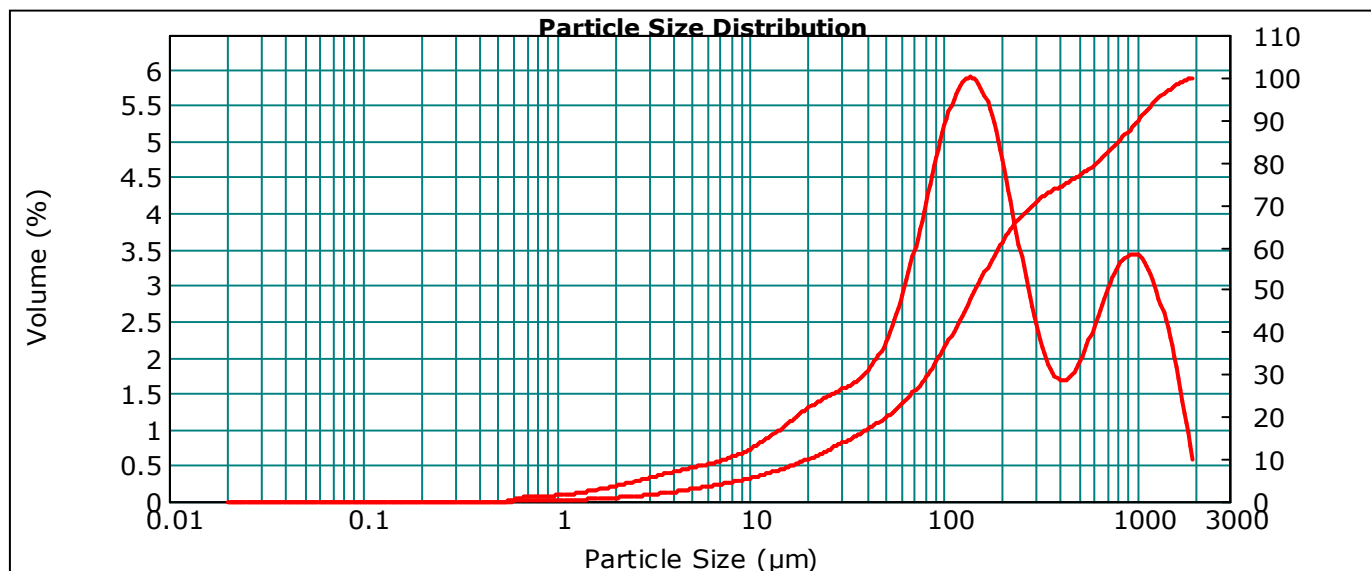
Uniformity:
1.82

Specific Surface Area*:
0.166 m²/g

Percentiles:
d(0.05) = 9.14um = 6.77phi
d(0.10) = 20.22um = 5.63phi
d(0.16) = 37.29um = 4.74phi
d(0.50) = 147.07um = 2.77phi
d(0.84) = 777.17um = .36phi
d(0.90) = 1019.94um = -0.03phi
d(0.95) = 1296.35um = -0.37phi

Distribution Properties:
Kurtosis = 2.07
Skewness = 1.7
Standard Deviation = 419.22 um = 1.25phi
Inclusive SD (Sorting Coeff.) = 2.18 phi (Very Poorly)
Inclusive Kurtosis = 1.1 phi (Mesokurtic)
Inclusive Mean = 2.62 phi (Fine Sand)
Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:56:28 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 301.47um = 1.73phi

Surface Weighted Mean:

D[3,2] = 36.65um = 4.77phi

Span :

5.774

Uniformity:

1.53

Specific Surface Area*:

0.164 m²/g

Percentiles:

d(0.05) = 9.13um = 6.77phi

d(0.10) = 20.58um = 5.6phi

d(0.16) = 39.7um = 4.65phi

d(0.50) = 150.19um = 2.74phi

d(0.84) = 629.81um = .67phi

d(0.90) = 887.81um = .17phi

d(0.95) = 1184.25um = -.24phi

Distribution Properties:

Kurtosis = 3.43

Skewness = 1.97

Standard Deviation = 376.68 um = 1.41phi

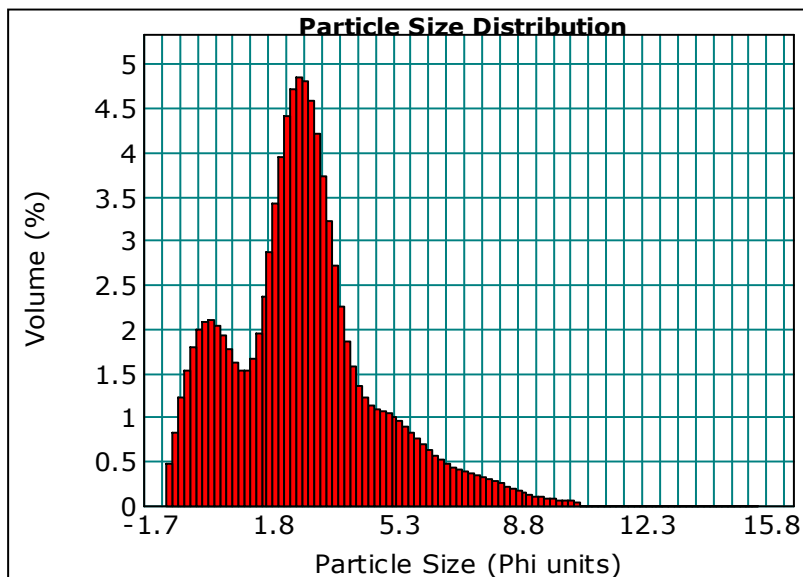
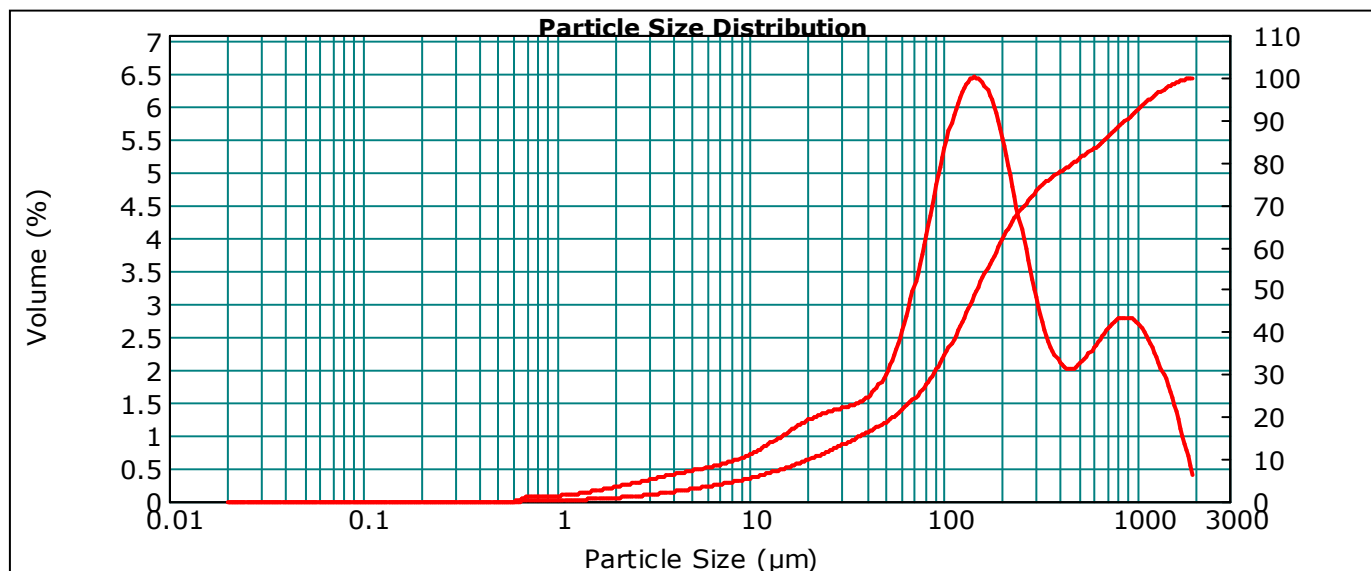
Inclusive SD (Sorting Coeff.) = 2.06 phi (Very Poorly)

Inclusive Kurtosis = 1.31 phi (Leptokurtic)

Inclusive Mean = 2.69 phi (Fine Sand)

Inclusive Skewness = .06 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.53
-0.5	5.31
0.0	6.23
0.5	5.28
1.0	4.74
1.5	7.26
2.0	11.89
2.5	14.42
3.0	12.43
3.5	8.05
4.0	4.71
4.5	3.44
5.0	3.10
5.5	2.66
6.0	2.04
6.5	1.53
7.0	1.21
7.5	

Phi	Volume In %
7.5	1.01
8.0	0.80
8.5	0.57
9.0	0.37
9.5	0.23
10.0	0.16
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:57:36 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 279.95um = 1.84phi

Surface Weighted Mean:
D[3,2] = 35.68um = 4.81phi

Span :
5.550

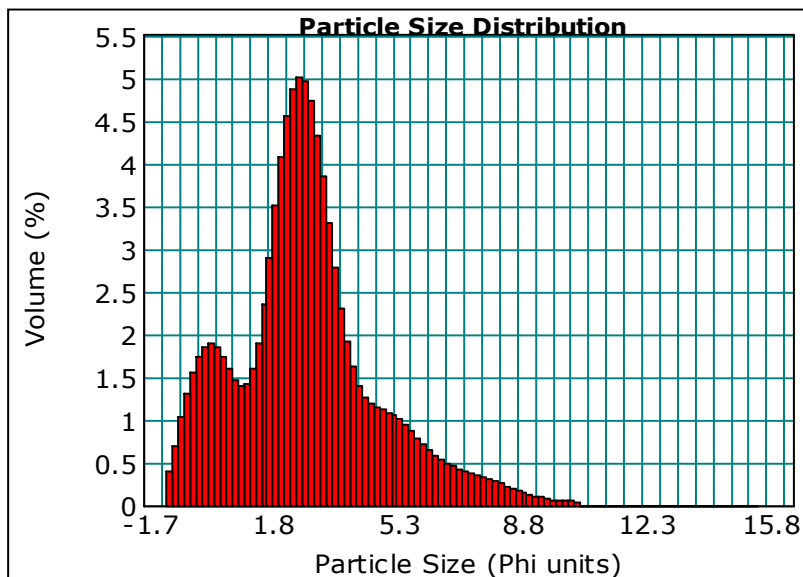
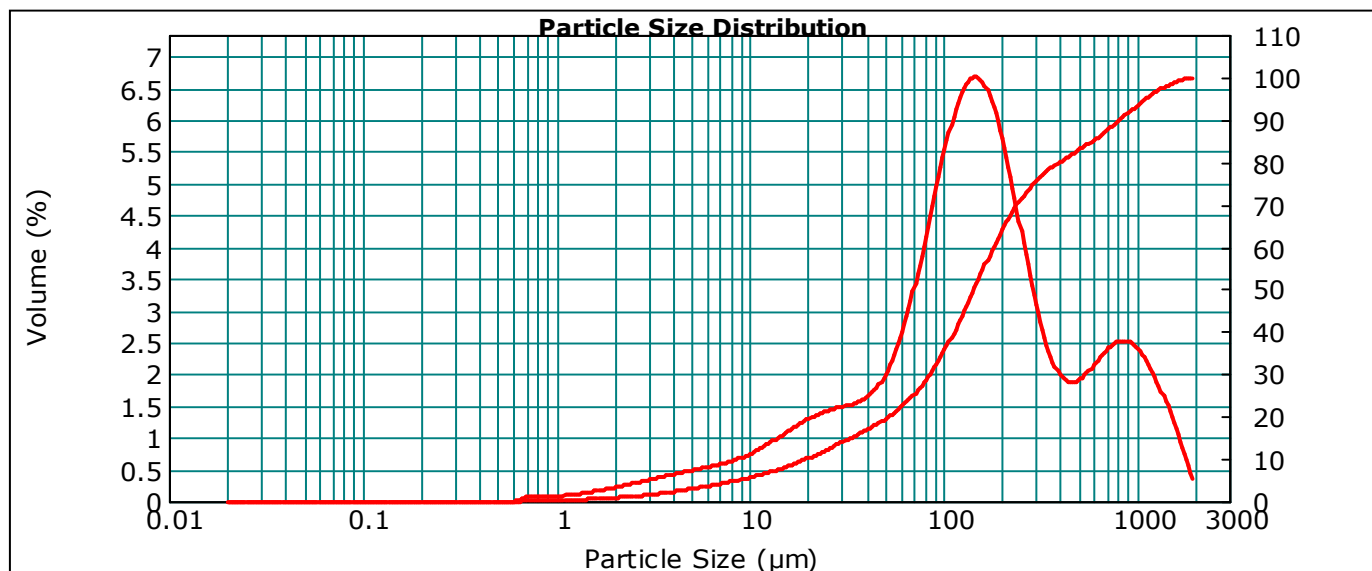
Uniformity:
1.47

Specific Surface Area*:
0.168 m²/g

Percentiles:
d(0.05) = 8.81um = 6.83phi
d(0.10) = 19.64um = 5.67phi
d(0.16) = 37.18um = 4.75phi
d(0.50) = 144.13um = 2.79phi
d(0.84) = 550.55um = .86phi
d(0.90) = 819.51um = .29phi
d(0.95) = 1122.11um = -.17phi

Distribution Properties:
Kurtosis = 4.28
Skewness = 2.14
Standard Deviation = 356.77 um = 1.49phi
Inclusive SD (Sorting Coeff.) = 2.03 phi (Very Poorly)
Inclusive Kurtosis = 1.39 phi (Leptokurtic)
Inclusive Mean = 2.8 phi (Fine Sand)
Inclusive Skewness = .08 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.13
-0.5	4.62
0.0	5.64
0.5	4.83
1.0	4.46
1.5	7.28
2.0	12.26
2.5	14.92
3.0	12.83
3.5	8.29
4.0	4.88
4.5	3.59
5.0	3.25
5.5	2.79
6.0	2.14
6.5	1.60
7.0	1.27
7.5	

Phi	Volume In %
7.5	1.05
8.0	0.82
8.5	0.58
9.0	0.37
9.5	0.23
10.0	0.16
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_6
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:58:43 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 289.22um = 1.79phi

Surface Weighted Mean:
D[3,2] = 36.04um = 4.79phi

Span :
5.773

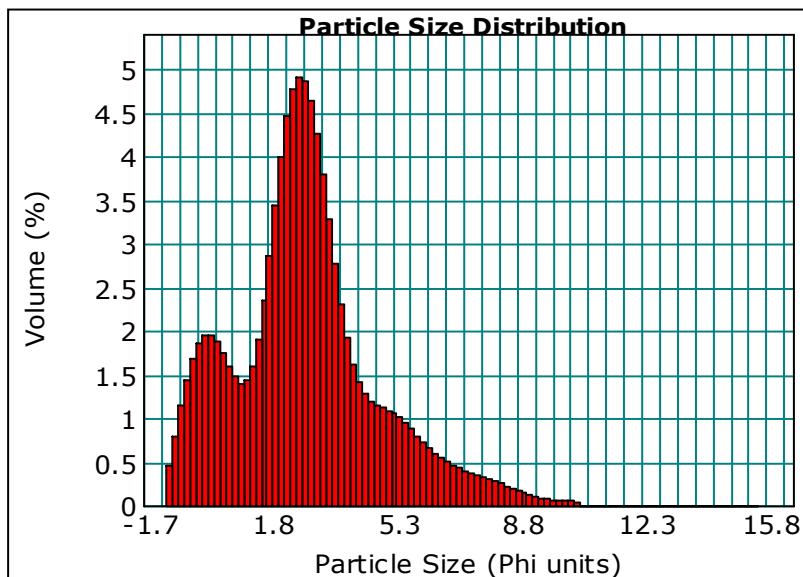
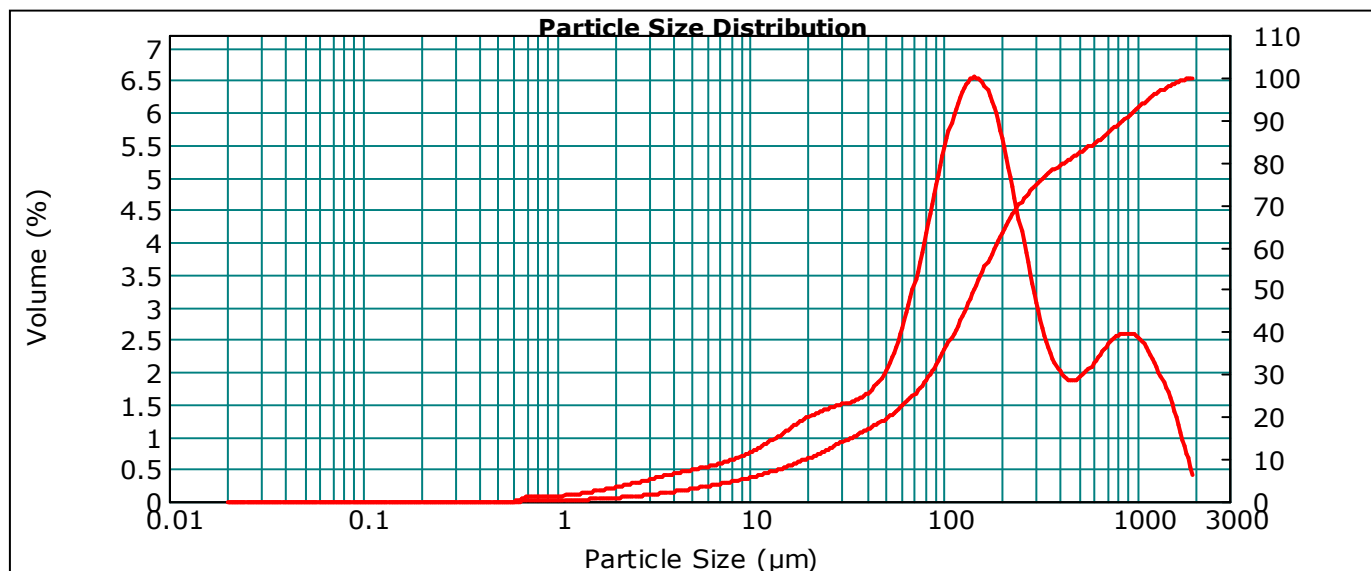
Uniformity:
1.52

Specific Surface Area*:
0.166 m²/g

Percentiles:
d(0.05) = 8.9um = 6.81phi
d(0.10) = 19.69um = 5.67phi
d(0.16) = 37.16um = 4.75phi
d(0.50) = 145.3um = 2.78phi
d(0.84) = 584.92um = .77phi
d(0.90) = 858.53um = .22phi
d(0.95) = 1164.2um = -0.22phi

Distribution Properties:
Kurtosis = 3.88
Skewness = 2.07
Standard Deviation = 369.2 um = 1.44phi
Inclusive SD (Sorting Coeff.) = 2.06 phi (Very Poorly)
Inclusive Kurtosis = 1.36 phi (Leptokurtic)
Inclusive Mean = 2.77 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.41
-0.5	5.01
0.0	5.78
0.5	4.83
1.0	4.47
1.5	7.22
2.0	12.03
2.5	14.62
3.0	12.63
3.5	8.23
4.0	4.89
4.5	3.61
5.0	3.26
5.5	2.80
6.0	2.15
6.5	1.60
7.0	1.27
7.5	

Phi	Volume In %
7.5	1.04
8.0	0.81
8.5	0.57
9.0	0.36
9.5	0.22
10.0	0.16
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_6 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:56:28 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 290.22um = 1.78phi

Surface Weighted Mean:
D[3,2] = 36.12um = 4.79phi

Span :
5.707

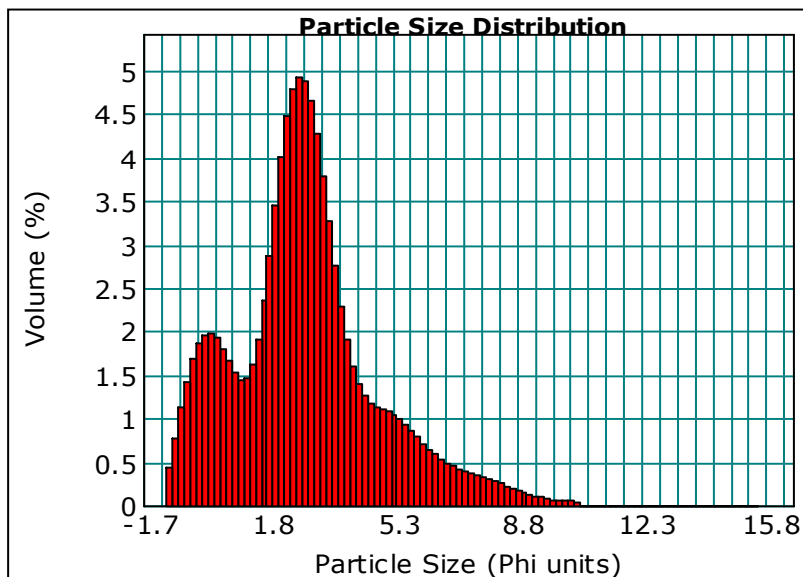
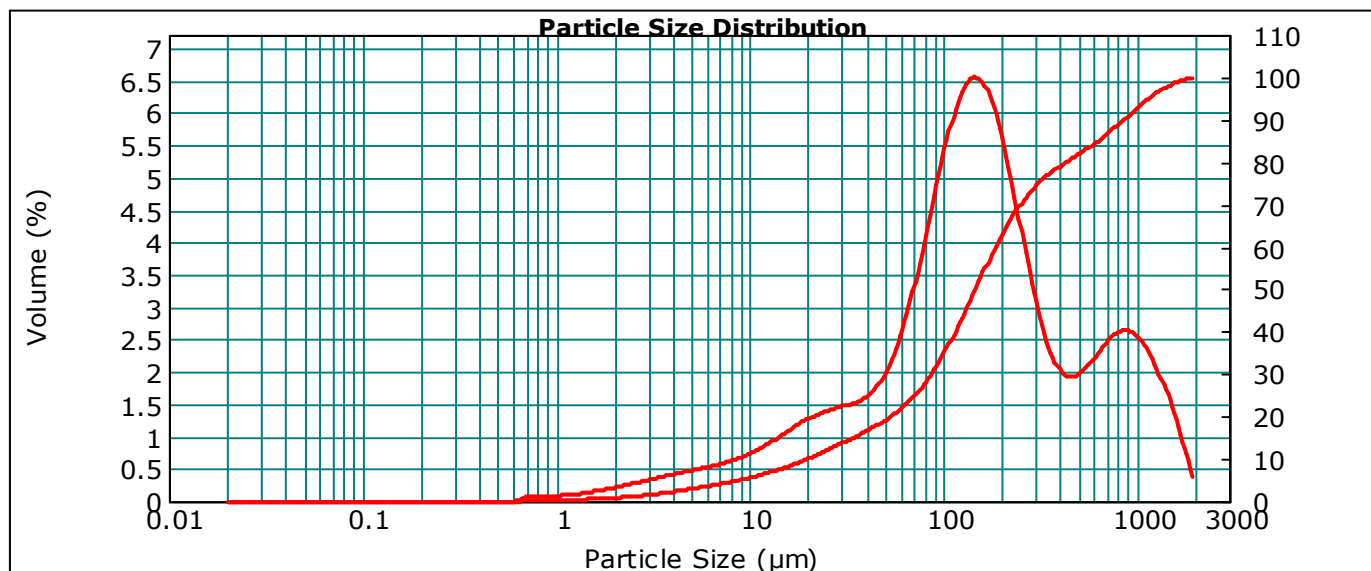
Uniformity:
1.51

Specific Surface Area*:
0.166 m²/g

Percentiles:
d(0.05) = 8.94um = 6.8phi
d(0.10) = 19.96um = 5.65phi
d(0.16) = 37.98um = 4.72phi
d(0.50) = 146.48um = 2.77phi
d(0.84) = 589.48um = .76phi
d(0.90) = 855.96um = .22phi
d(0.95) = 1157.71um = -.021phi

Distribution Properties:
Kurtosis = 3.85
Skewness = 2.06
Standard Deviation = 367.75 um = 1.44phi
Inclusive SD (Sorting Coeff.) = 2.05 phi (Very Poorly)
Inclusive Kurtosis = 1.35 phi (Leptokurtic)
Inclusive Mean = 2.75 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.36
-0.5	4.98
0.0	5.89
0.5	4.98
1.0	4.56
1.5	7.25
2.0	12.06
2.5	14.65
3.0	12.63
3.5	8.19
4.0	4.83
4.5	3.55
5.0	3.20
5.5	2.75
6.0	2.11
6.5	1.57
7.0	1.25
7.5	

Phi	Volume In %
7.5	1.03
8.0	0.81
8.5	0.57
9.0	0.37
9.5	0.23
10.0	0.16
10.5	0.02
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:06:17 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 264.53um = 1.92phi

Surface Weighted Mean:
D[3,2] = 37.01um = 4.76phi

Span :
4.789

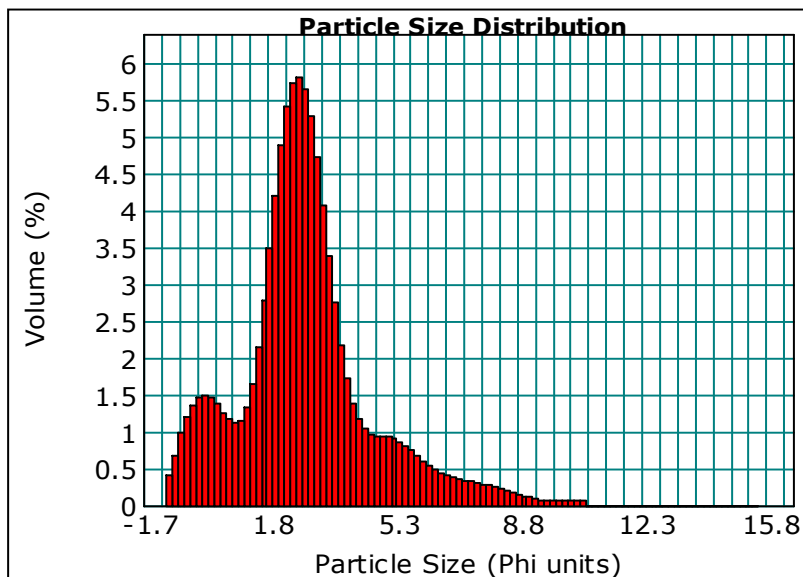
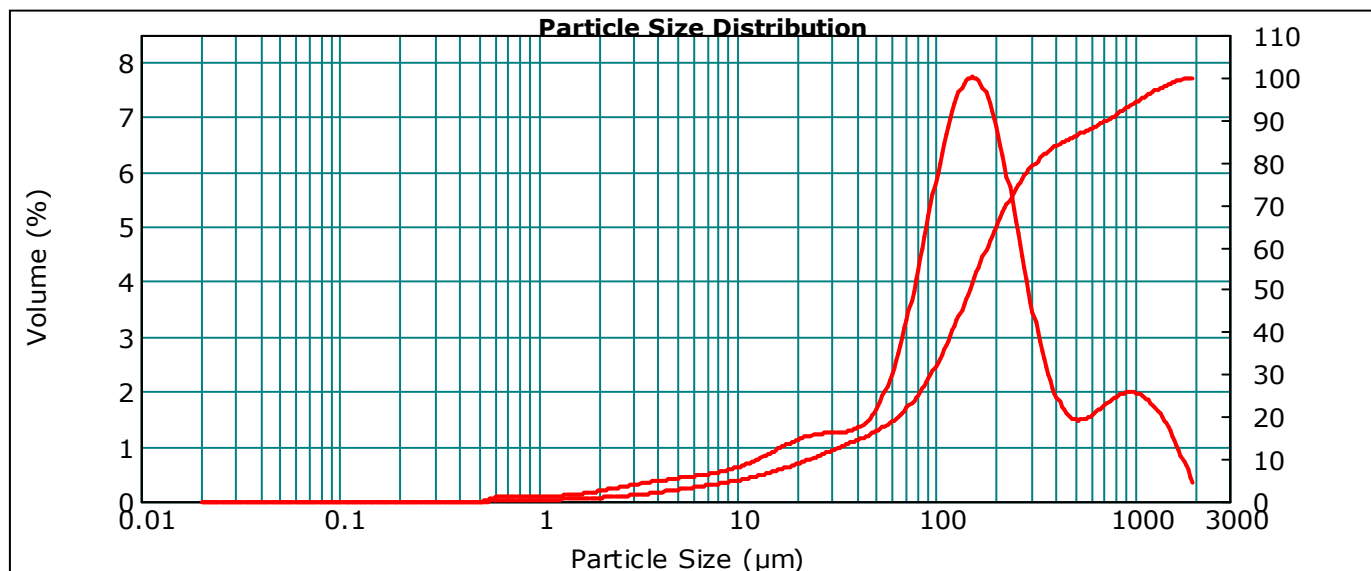
Uniformity:
1.27

Specific Surface Area*:
0.162 m²/g

Percentiles:
d(0.05) = 10.22um = 6.61phi
d(0.10) = 23.52um = 5.41phi
d(0.16) = 47.89um = 4.38phi
d(0.50) = 148.56um = 2.75phi
d(0.84) = 412.43um = 1.28phi
d(0.90) = 734.89um = .44phi
d(0.95) = 1091.65um = -0.13phi

Distribution Properties:
Kurtosis = 5.89
Skewness = 2.44
Standard Deviation = 338.8 um = 1.56phi
Inclusive SD (Sorting Coeff.) = 1.8 phi (Poorly Sorted)
Inclusive Kurtosis = 1.63 phi (Very Leptokurtic)
Inclusive Mean = 2.8 phi (Fine Sand)
Inclusive Skewness = .1 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	2.07
-0.5	4.05
0.0	4.32
0.5	3.53
1.0	4.18
1.5	8.52
2.0	14.63
2.5	17.22
3.0	13.94
3.5	8.14
4.0	4.18
4.5	2.94
5.0	2.76
5.5	2.39
6.0	1.79
6.5	1.32
7.0	1.07
7.5	

Phi	Volume In %
7.5	0.91
8.0	0.72
8.5	0.51
9.0	0.31
9.5	0.21
10.0	0.19
10.5	0.10
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:07:24 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 259.57um = 1.95phi

Surface Weighted Mean:
D[3,2] = 37.03um = 4.76phi

Span :
4.645

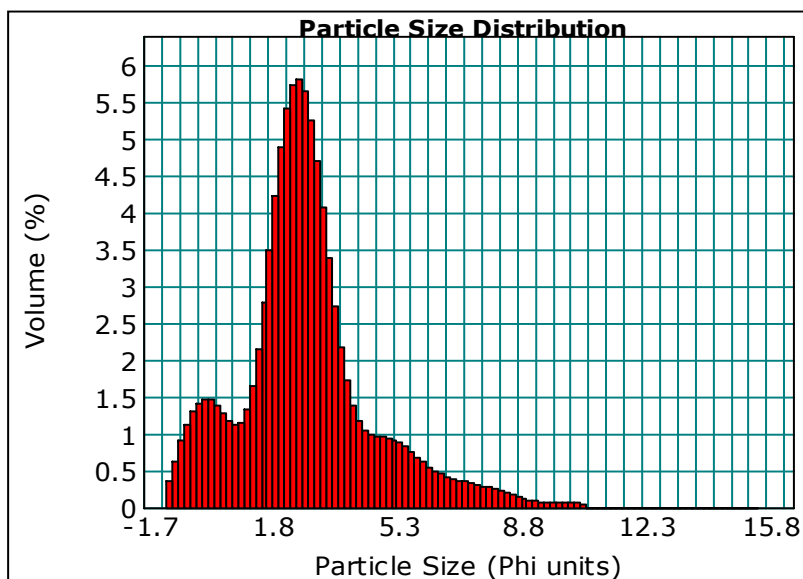
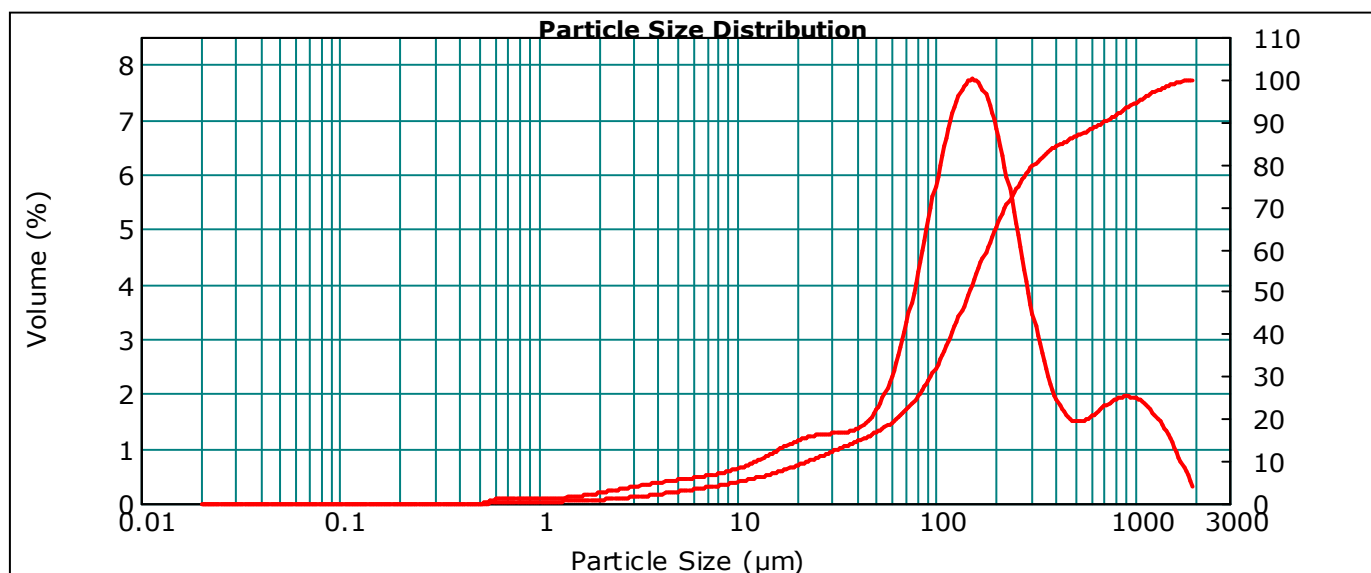
Uniformity:
1.25

Specific Surface Area*:
0.162 m²/g

Percentiles:
d(0.05) = 10.14um = 6.62phi
d(0.10) = 23.15um = 5.43phi
d(0.16) = 46.8um = 4.42phi
d(0.50) = 147.87um = 2.76phi
d(0.84) = 400.86um = 1.32phi
d(0.90) = 710.03um = .49phi
d(0.95) = 1061.64um = -0.09phi

Distribution Properties:
Kurtosis = 6.17
Skewness = 2.47
Standard Deviation = 331.08 um = 1.59phi
Inclusive SD (Sorting Coeff.) = 1.79 phi (Poorly Sorted)
Inclusive Kurtosis = 1.62 phi (Very Leptokurtic)
Inclusive Mean = 2.83 phi (Fine Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.90
-0.5	3.85
0.0	4.30
0.5	4.30
1.0	4.18
1.5	8.54
2.0	14.70
2.5	17.25
3.0	13.92
3.5	8.13
4.0	4.21
4.5	2.99
5.0	2.82
5.5	2.44
6.0	1.83
6.5	1.35
7.0	1.09
7.5	

Phi	Volume In %
7.5	0.92
8.0	0.72
8.5	0.50
9.0	0.31
9.5	0.20
10.0	0.18
10.5	0.10
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_5
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:08:31 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 267.51um = 1.9phi

Surface Weighted Mean:

D[3,2] = 37.93um = 4.72phi

Span :

4.832

Uniformity:

1.28

Specific Surface Area*:

0.158 m²/g

Percentiles:

d(0.05) = 10.42um = 6.58phi

d(0.10) = 23.68um = 5.4phi

d(0.16) = 47.85um = 4.39phi

d(0.50) = 149.86um = 2.74phi

d(0.84) = 437.03um = 1.19phi

d(0.90) = 747.75um = .42phi

d(0.95) = 1083.74um = -.12phi

Distribution Properties:

Kurtosis = 5.6

Skewness = 2.38

Standard Deviation = 338.24 um = 1.56phi

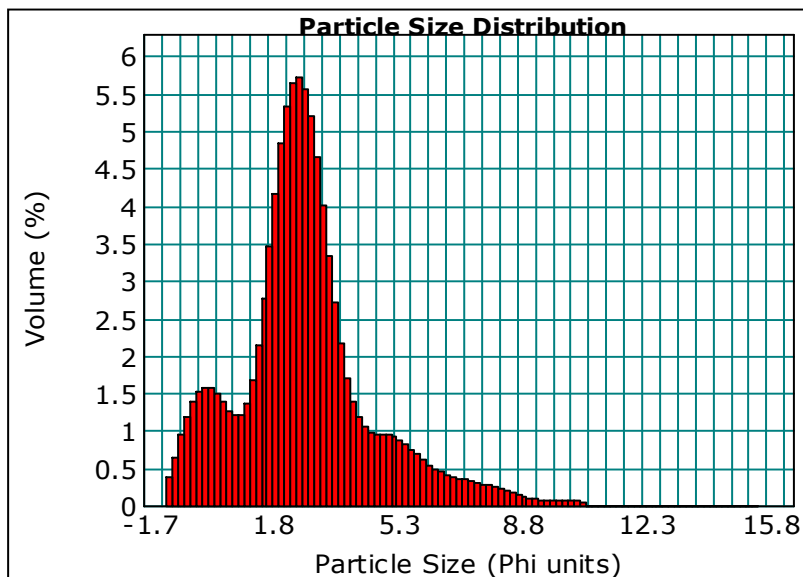
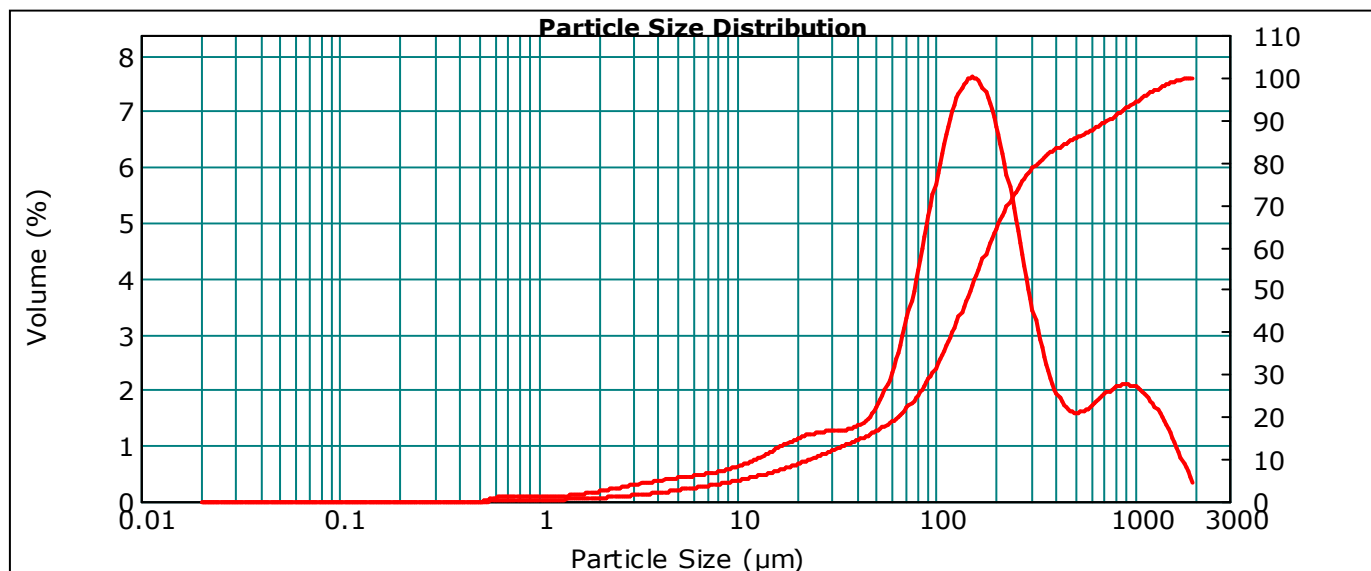
Inclusive SD (Sorting Coeff.) = 1.81 phi (Poorly Sorted)

Inclusive Kurtosis = 1.59 phi (Very Leptokurtic)

Inclusive Mean = 2.77 phi (Fine Sand)

Inclusive Skewness = .09 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.98
-0.5	4.09
0.0	4.64
0.5	3.85
1.0	4.31
1.5	8.49
2.0	14.48
2.5	16.98
3.0	13.72
3.5	8.04
4.0	4.19
4.5	2.97
5.0	2.79
5.5	2.41
6.0	1.80
6.5	1.33
7.0	1.07
7.5	

Phi	Volume In %
7.5	0.90
8.0	0.71
8.5	0.49
9.0	0.30
9.5	0.19
10.0	0.17
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_5 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:06:17 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 263.87um = 1.92phi

Surface Weighted Mean:
D[3,2] = 37.32um = 4.74phi

Span :
4.758

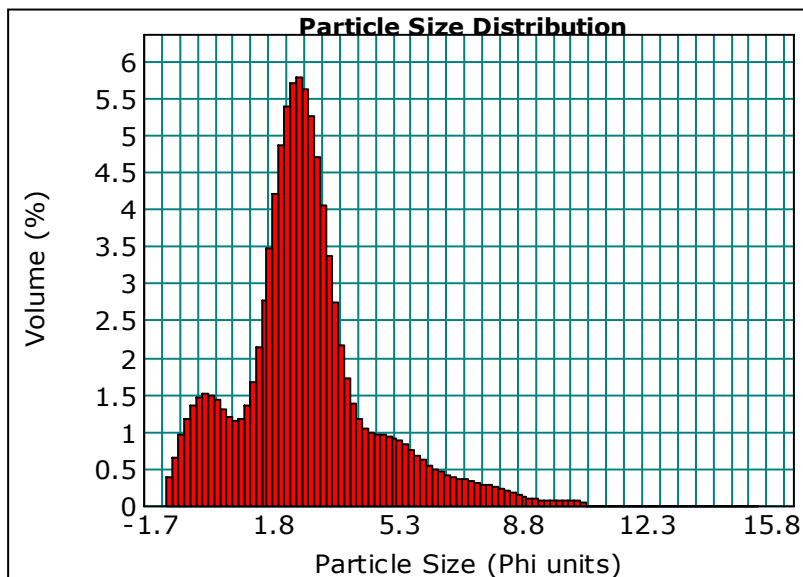
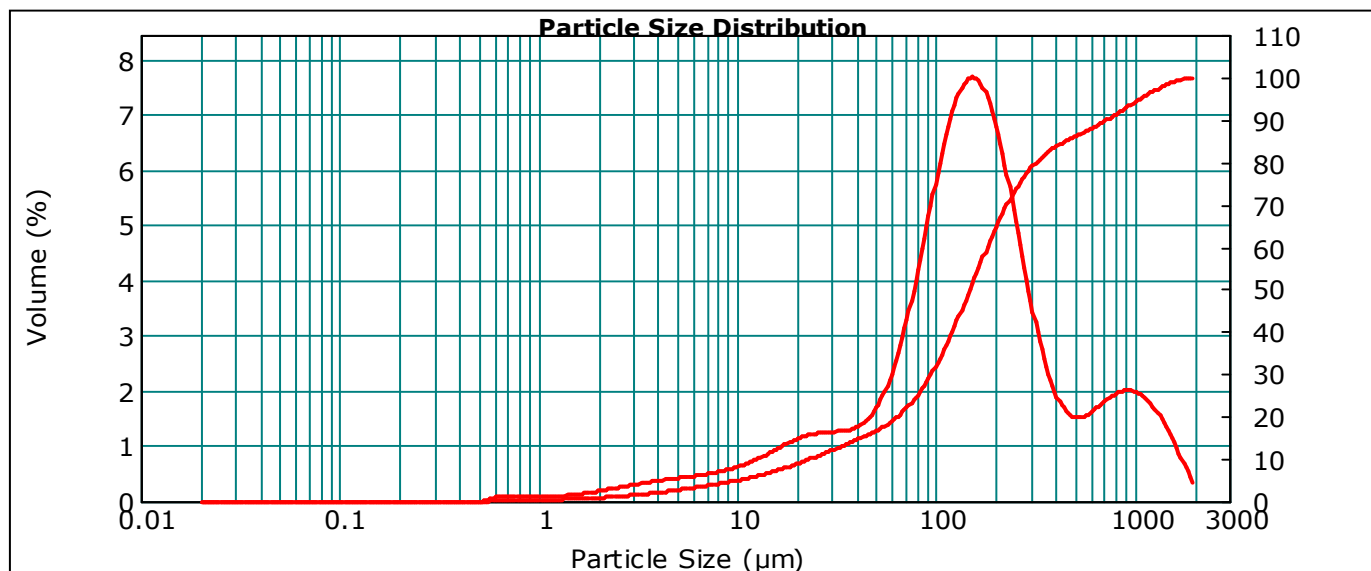
Uniformity:
1.27

Specific Surface Area*:
0.161 m²/g

Percentiles:
d(0.05) = 10.26um = 6.61phi
d(0.10) = 23.45um = 5.41phi
d(0.16) = 47.51um = 4.4phi
d(0.50) = 148.76um = 2.75phi
d(0.84) = 416.27um = 1.26phi
d(0.90) = 731.27um = .45phi
d(0.95) = 1079.18um = -0.11phi

Distribution Properties:
Kurtosis = 5.88
Skewness = 2.43
Standard Deviation = 336.08 um = 1.57phi
Inclusive SD (Sorting Coeff.) = 1.8 phi (Poorly Sorted)
Inclusive Kurtosis = 1.61 phi (Very Leptokurtic)
Inclusive Mean = 2.8 phi (Fine Sand)
Inclusive Skewness = .1 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.98
-0.5	3.99
0.0	4.42
0.5	3.65
1.0	4.22
1.5	8.52
2.0	14.60
2.5	17.15
3.0	13.86
3.5	8.11
4.0	4.20
4.5	2.97
5.0	2.79
5.5	2.41
6.0	1.81
6.5	1.33
7.0	1.08
7.5	

Phi	Volume In %
7.5	0.91
8.0	0.72
8.5	0.50
9.0	0.31
9.5	0.20
10.0	0.18
10.5	0.10
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:16:02 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 230.88um = 2.11phi

Surface Weighted Mean:
D[3,2] = 40.29um = 4.63phi

Span :
3.011

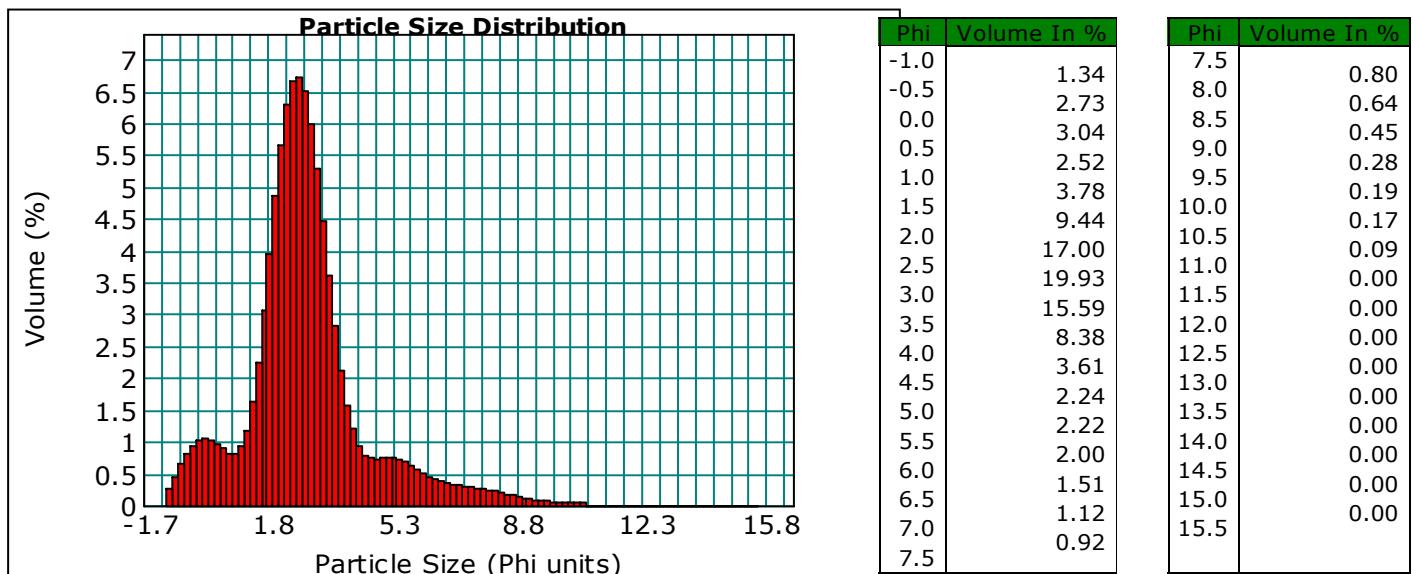
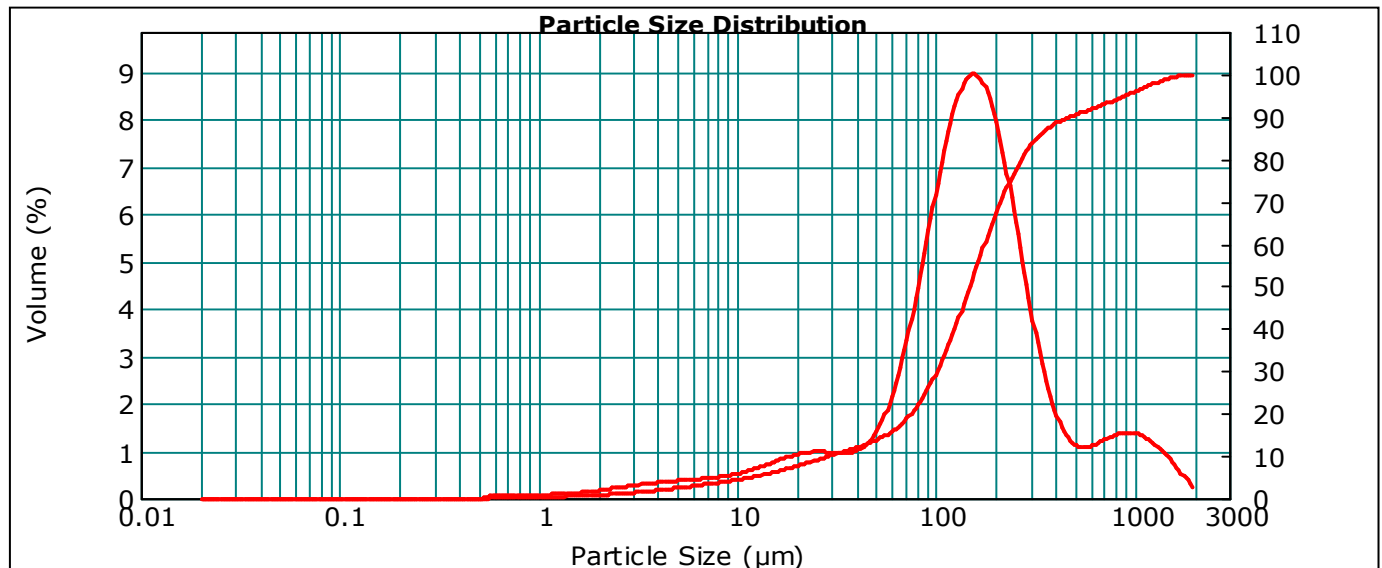
Uniformity:
1.01

Specific Surface Area*:
0.149 m²/g

Percentiles:
d(0.05) = 12.02um = 6.38phi
d(0.10) = 29.33um = 5.09phi
d(0.16) = 61.48um = 4.02phi
d(0.50) = 148.49um = 2.75phi
d(0.84) = 311.21um = 1.68phi
d(0.90) = 476.45um = 1.07phi
d(0.95) = 902.36um = .15phi

Distribution Properties:
Kurtosis = 9.85
Skewness = 2.99
Standard Deviation = 285.66 um = 1.81phi
Inclusive SD (Sorting Coeff.) = 1.53 phi (Poorly Sorted)
Inclusive Kurtosis = 1.81 phi (Very Leptokurtic)
Inclusive Mean = 2.82 phi (Fine Sand)
Inclusive Skewness = .13 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:17:09 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 239.62um = 2.06phi

Surface Weighted Mean:

D[3,2] = 41.4um = 4.59phi

Span :

3.502

Uniformity:

1.04

Specific Surface Area*:

0.145 m²/g

Percentiles:

d(0.05) = 12.34um = 6.34phi

d(0.10) = 30.09um = 5.05phi

d(0.16) = 62.68um = 4phi

d(0.50) = 150.87um = 2.73phi

d(0.84) = 335.75um = 1.57phi

d(0.90) = 558.5um = .84phi

d(0.95) = 923.66um = .11phi

Distribution Properties:

Kurtosis = 8.56

Skewness = 2.8

Standard Deviation = 291.04 um = 1.78phi

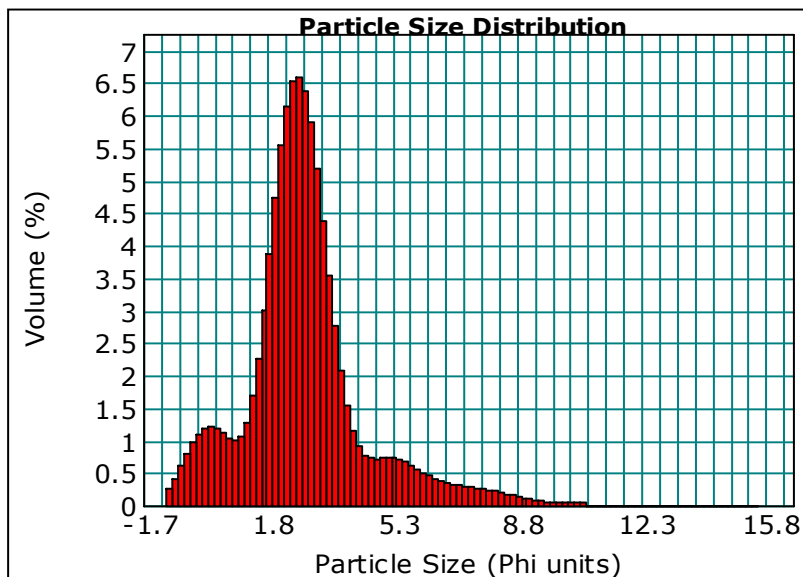
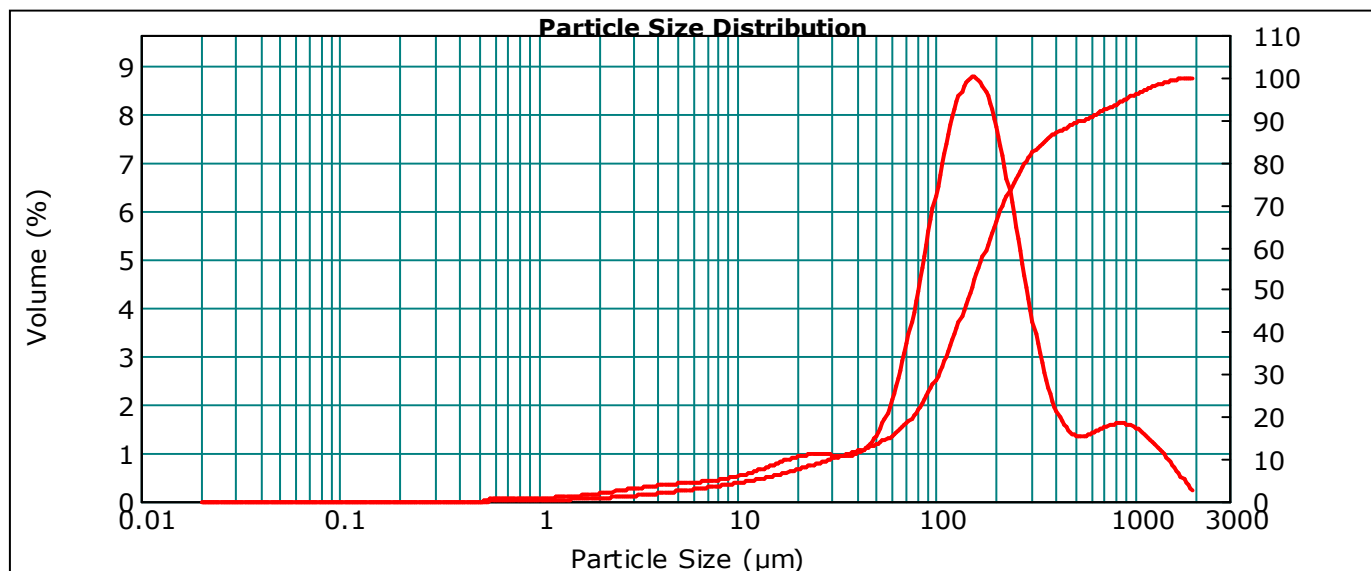
Inclusive SD (Sorting Coeff.) = 1.55 phi (Poorly Sorted)

Inclusive Kurtosis = 1.77 phi (Very Leptokurtic)

Inclusive Mean = 2.77 phi (Fine Sand)

Inclusive Skewness = .1 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.29
-0.5	2.89
0.0	3.60
0.5	3.18
1.0	4.09
1.5	9.29
2.0	16.62
2.5	19.55
3.0	15.32
3.5	8.21
4.0	3.52
4.5	2.20
5.0	2.19
5.5	1.98
6.0	1.49
6.5	1.10
7.0	0.91
7.5	

Phi	Volume In %
7.5	0.79
8.0	0.63
8.5	0.44
9.0	0.27
9.5	0.18
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_4
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:18:16 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 256.11um = 1.97phi

Surface Weighted Mean:
D[3,2] = 42.16um = 4.57phi

Span :
4.191

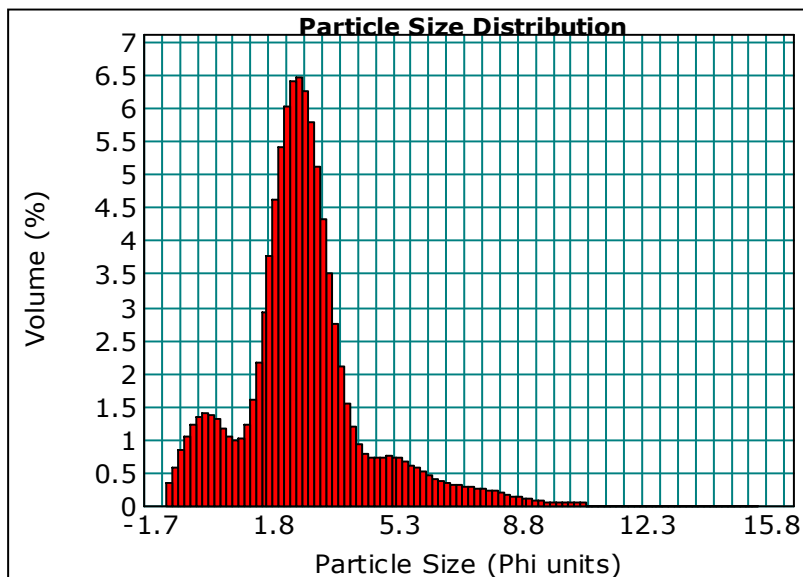
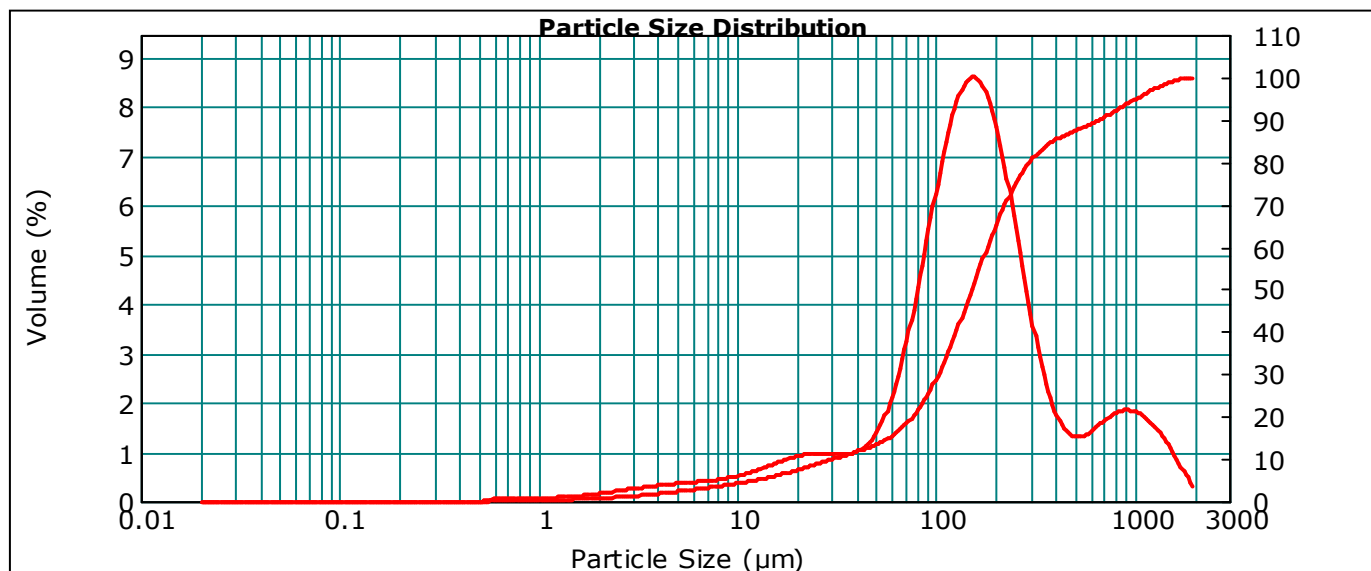
Uniformity:
1.14

Specific Surface Area*:
0.142 m²/g

Percentiles:
d(0.05) = 12.59um = 6.31phi
d(0.10) = 30.49um = 5.04phi
d(0.16) = 62.76um = 3.99phi
d(0.50) = 152.29um = 2.72phi
d(0.84) = 364.38um = 1.46phi
d(0.90) = 668.75um = .58phi
d(0.95) = 1030.09um = -0.04phi

Distribution Properties:
Kurtosis = 6.95
Skewness = 2.6
Standard Deviation = 318.75 um = 1.65phi
Inclusive SD (Sorting Coeff.) = 1.6 phi (Poorly Sorted)
Inclusive Kurtosis = 1.76 phi (Very Leptokurtic)
Inclusive Mean = 2.72 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.74
-0.5	3.62
0.0	4.07
0.5	3.20
1.0	3.85
1.5	8.98
2.0	16.23
2.5	19.15
3.0	15.07
3.5	8.16
4.0	3.56
4.5	2.23
5.0	2.20
5.5	1.97
6.0	1.49
6.5	1.10
7.0	0.90
7.5	

Phi	Volume In %
7.5	0.78
8.0	0.62
8.5	0.43
9.0	0.26
9.5	0.17
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_4 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:16:02 PM

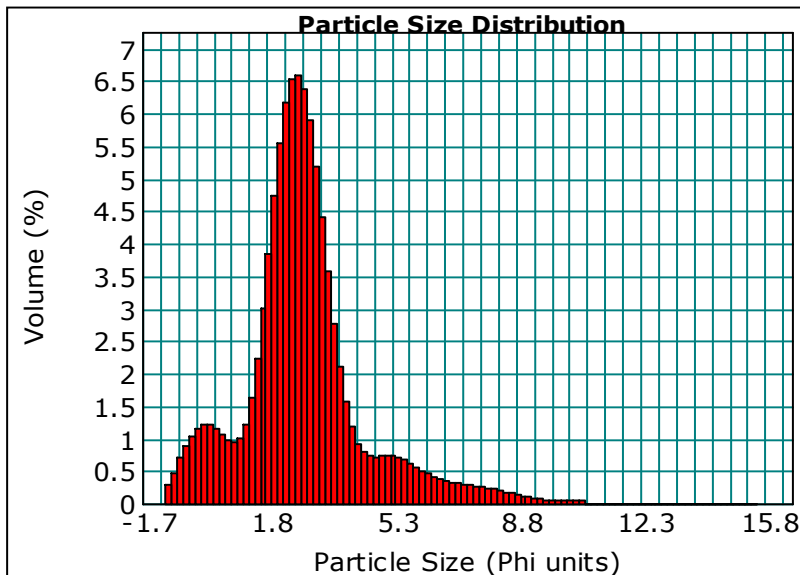
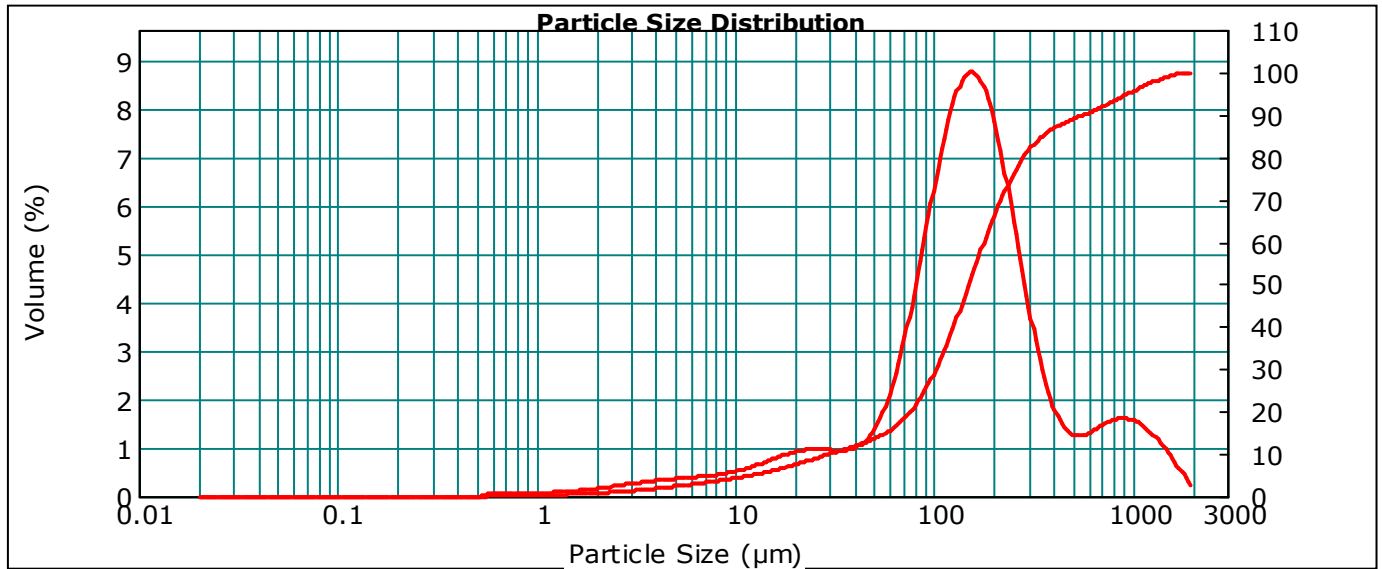
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
 alkaline calgon

Vol. Weighted Mean:
 $D[4,3] = 242.2\mu\text{m} = 2.05\phi$
Surface Weighted Mean:
 $D[3,2] = 41.27\mu\text{m} = 4.6\phi$
Span :
 3.593
Uniformity:
 1.06
Specific Surface Area*:
 0.145 m^2/g

Percentiles:
 $d(0.05) = 12.32\mu\text{m} = 6.34\phi$
 $d(0.10) = 29.97\mu\text{m} = 5.06\phi$
 $d(0.16) = 62.3\mu\text{m} = 4\phi$
 $d(0.50) = 150.51\mu\text{m} = 2.73\phi$
 $d(0.84) = 334.11\mu\text{m} = 1.58\phi$
 $d(0.90) = 570.79\mu\text{m} = .81\phi$
 $d(0.95) = 956.43\mu\text{m} = .06\phi$

Distribution Properties:
 Kurtosis = 8.35
 Skewness = 2.79
 Standard Deviation = $299.02\mu\text{m} = 1.74\phi$
 Inclusive SD (Sorting Coeff.) = 1.56ϕ (Poorly Sorted)
 Inclusive Kurtosis = 1.79ϕ (Very Leptokurtic)
 Inclusive Mean = 2.77ϕ (Fine Sand)
 Inclusive Skewness = .1 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 ($\rho=1$). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.46
-0.5	3.08
0.0	3.57
0.5	2.97
1.0	3.91
1.5	9.23
2.0	16.62
2.5	19.54
3.0	15.33
3.5	8.25
4.0	3.56
4.5	2.22
5.0	2.20
5.5	1.98
6.0	1.50
6.5	1.10
7.0	0.91
7.5	

Phi	Volume In %
7.5	0.79
8.0	0.63
8.5	0.44
9.0	0.27
9.5	0.18
10.0	0.17
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:25:43 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 232.29um = 2.11phi

Surface Weighted Mean:
D[3,2] = 42.79um = 4.55phi

Span :
2.942

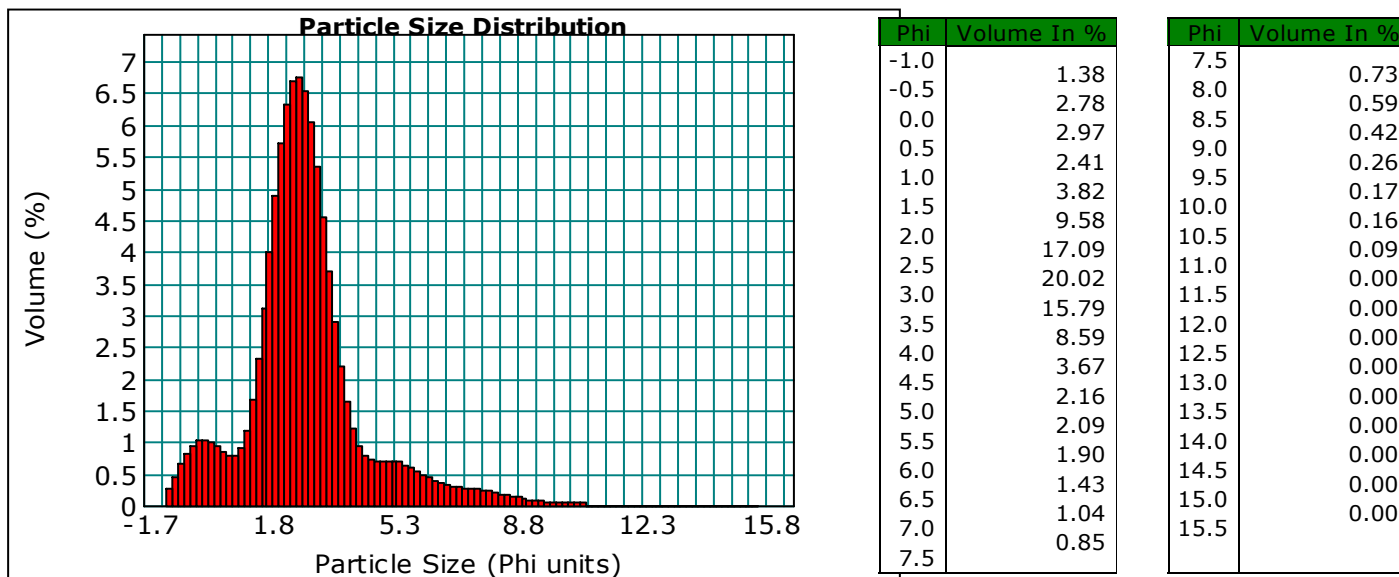
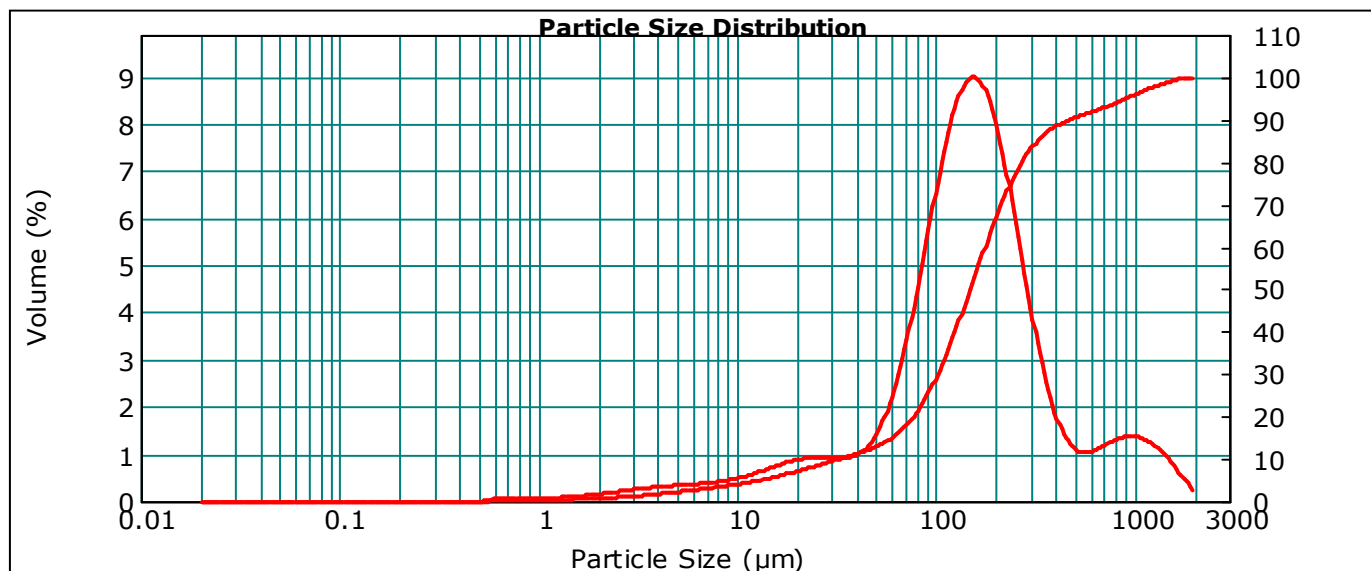
Uniformity:
1

Specific Surface Area*:
0.14 m²/g

Percentiles:
d(0.05) = 13.29um = 6.23phi
d(0.10) = 32.75um = 4.93phi
d(0.16) = 64.23um = 3.96phi
d(0.50) = 149.09um = 2.75phi
d(0.84) = 311.48um = 1.68phi
d(0.90) = 471.42um = 1.08phi
d(0.95) = 911.74um = .13phi

Distribution Properties:
Kurtosis = 9.88
Skewness = 3.01
Standard Deviation = 287.1 um = 1.8phi
Inclusive SD (Sorting Coeff.) = 1.49 phi (Poorly Sorted)
Inclusive Kurtosis = 1.79 phi (Very Leptokurtic)
Inclusive Mean = 2.8 phi (Fine Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:26:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 218.24um = 2.2phi

Surface Weighted Mean:

D[3,2] = 42.27um = 4.56phi

Span :

2.607

Uniformity:

0.93

Specific Surface Area*:

0.142 m²/g

Percentiles:

d(0.05) = 12.98um = 6.27phi

d(0.10) = 31.52um = 4.99phi

d(0.16) = 62.96um = 3.99phi

d(0.50) = 147.17um = 2.76phi

d(0.84) = 298.36um = 1.74phi

d(0.90) = 415.24um = 1.27phi

d(0.95) = 793.69um = .33phi

Distribution Properties:

Kurtosis = 11.75

Skewness = 3.18

Standard Deviation = 260.54 um = 1.94phi

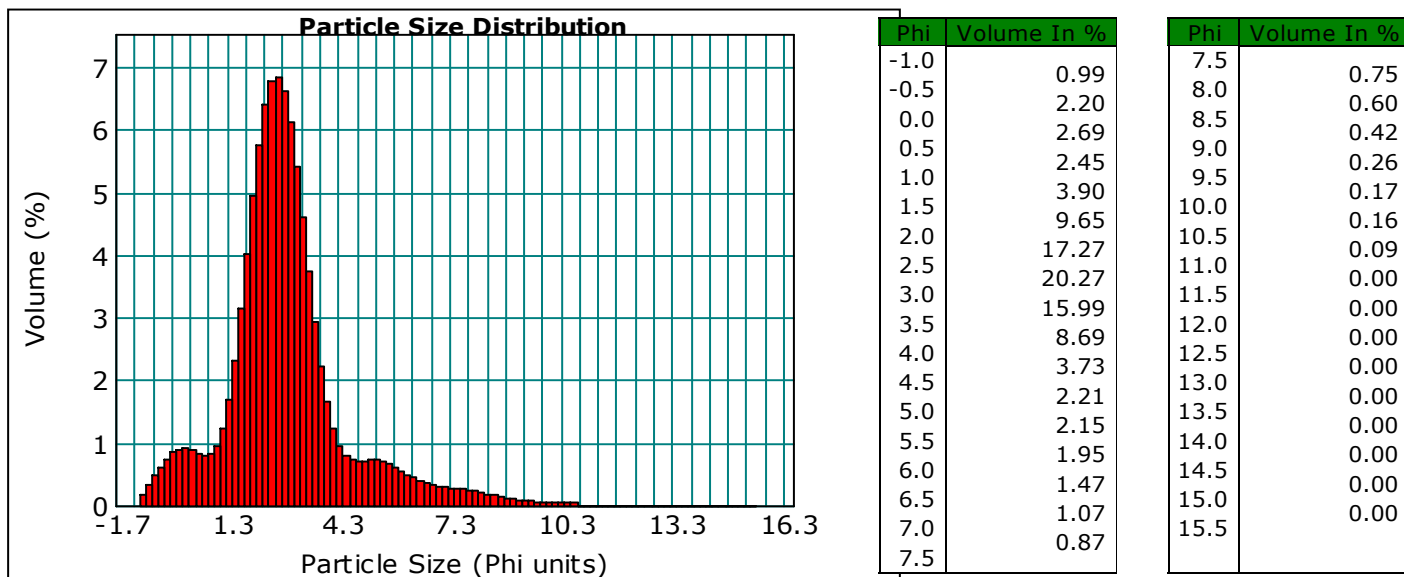
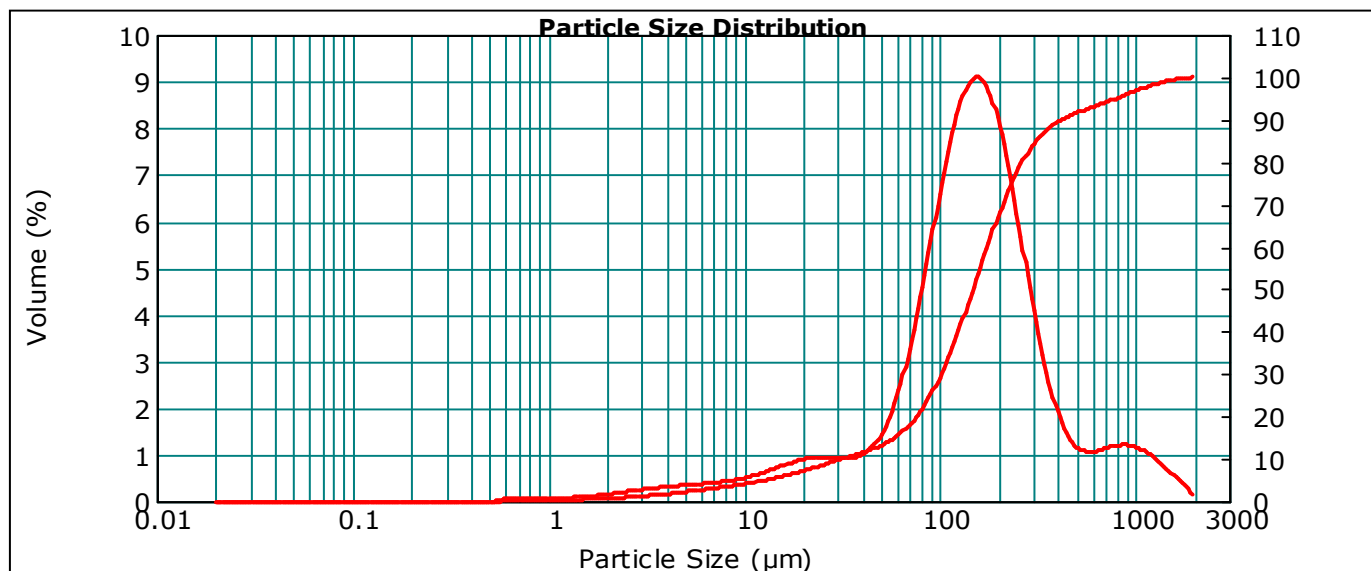
Inclusive SD (Sorting Coeff.) = 1.46 phi (Poorly Sorted)

Inclusive Kurtosis = 1.77 phi (Very Leptokurtic)

Inclusive Mean = 2.83 phi (Fine Sand)

Inclusive Skewness = .14 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_3
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:27:58 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 232.05um = 2.11phi

Surface Weighted Mean:
D[3,2] = 42.45um = 4.56phi

Span :
2.987

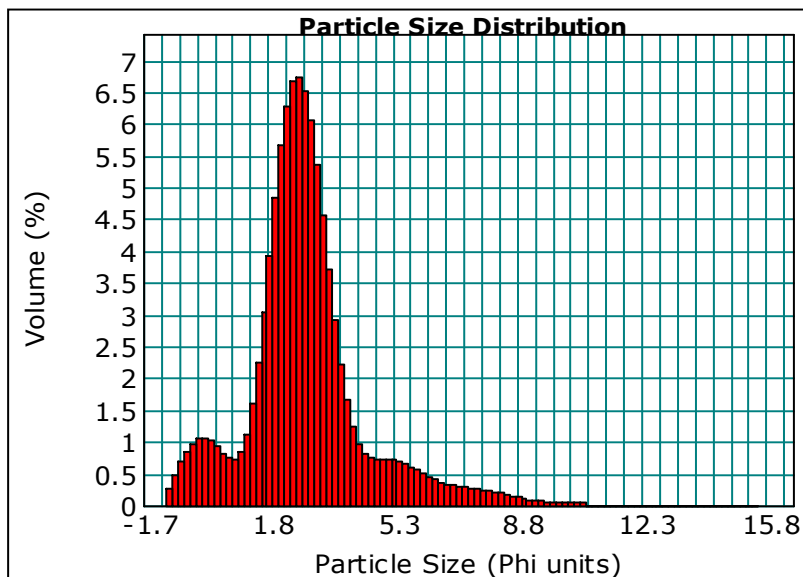
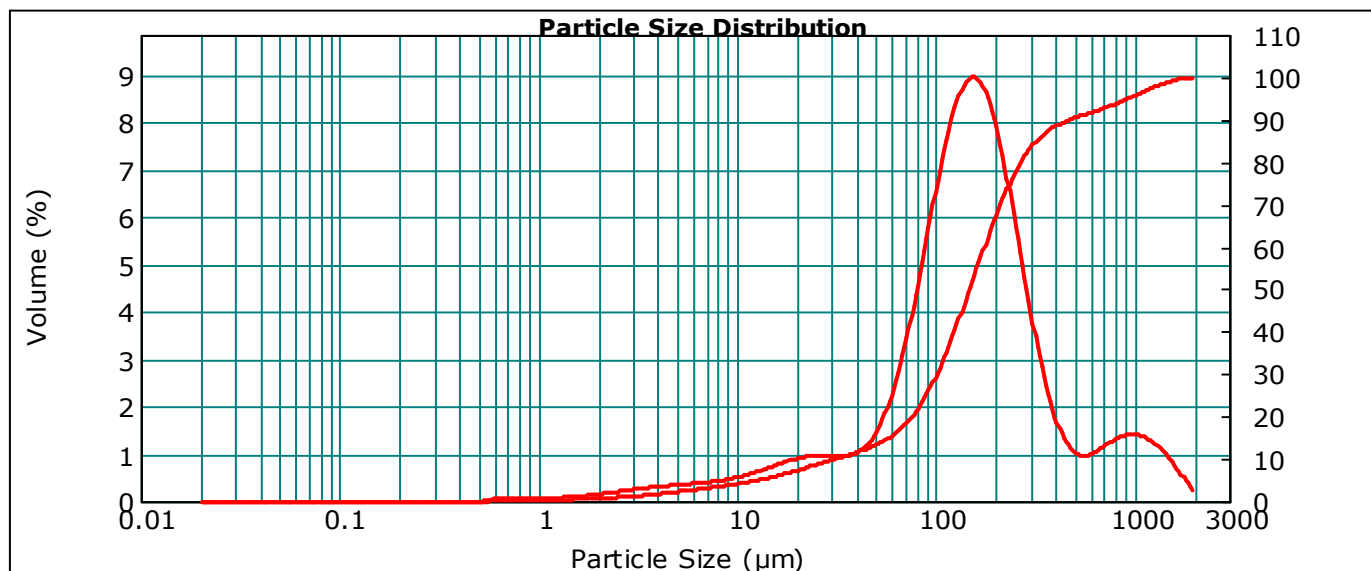
Uniformity:
1.02

Specific Surface Area*:
0.141 m²/g

Percentiles:
d(0.05) = 12.94um = 6.27phi
d(0.10) = 31.36um = 4.99phi
d(0.16) = 62.56um = 4phi
d(0.50) = 147.79um = 2.76phi
d(0.84) = 308.49um = 1.7phi
d(0.90) = 472.82um = 1.08phi
d(0.95) = 926.4um = .11phi

Distribution Properties:
Kurtosis = 9.71
Skewness = 2.99
Standard Deviation = 290.07 um = 1.79phi
Inclusive SD (Sorting Coeff.) = 1.51 phi (Poorly Sorted)
Inclusive Kurtosis = 1.8 phi (Very Leptokurtic)
Inclusive Mean = 2.82 phi (Fine Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.42
-0.5	2.87
0.0	3.02
0.5	2.30
1.0	3.62
1.5	9.38
2.0	16.96
2.5	19.97
3.0	15.81
3.5	8.66
4.0	3.77
4.5	2.24
5.0	2.16
5.5	1.96
6.0	1.48
6.5	1.08
7.0	0.88
7.5	

Phi	Volume In %
7.5	0.75
8.0	0.60
8.5	0.42
9.0	0.25
9.5	0.16
10.0	0.15
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_3 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:25:43 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 227.53um = 2.14phi

Surface Weighted Mean:
D[3,2] = 42.5um = 4.56phi

Span :
2.821

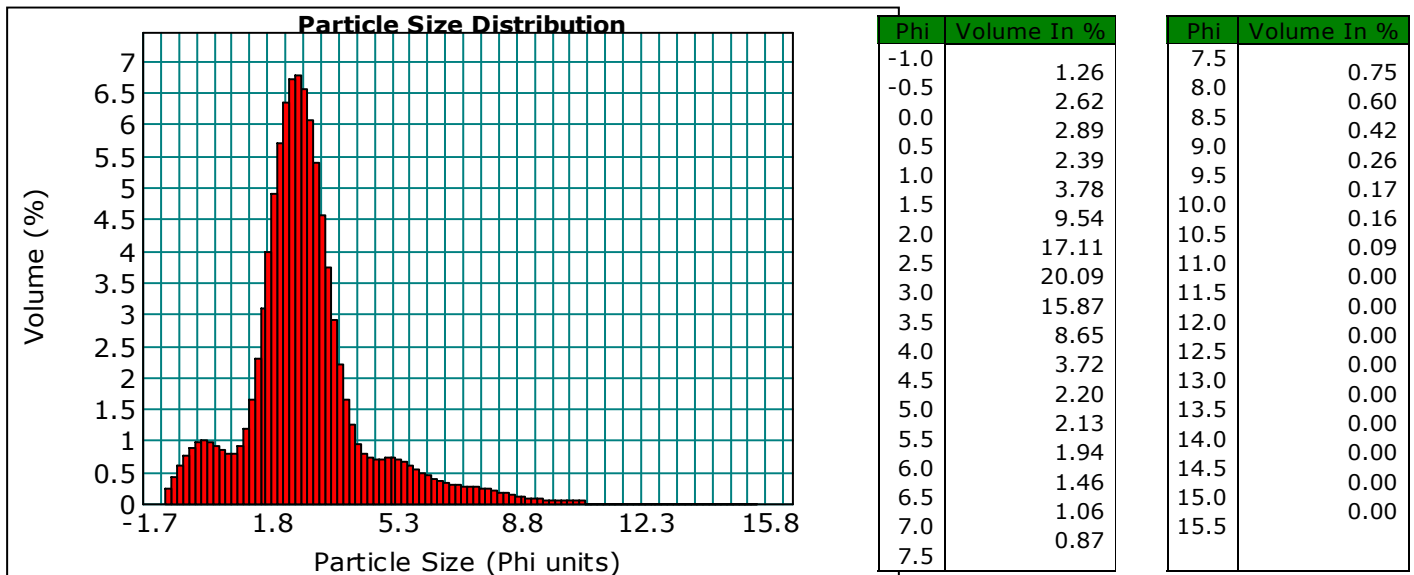
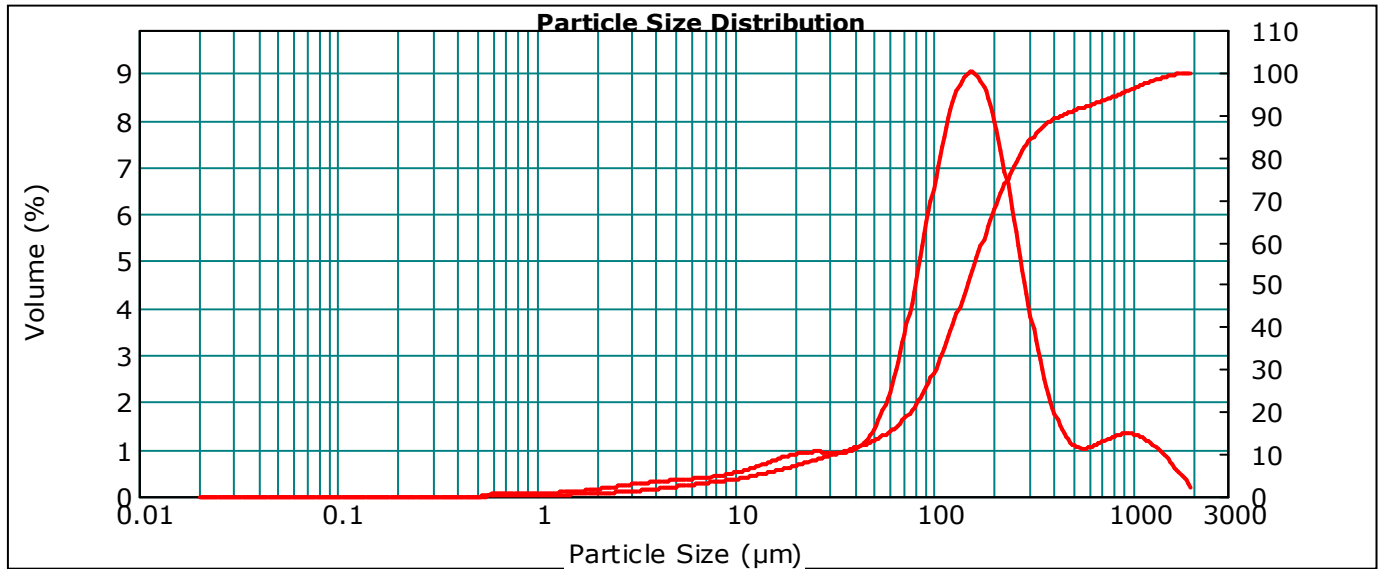
Uniformity:
0.985

Specific Surface Area*:
0.141 m²/g

Percentiles:
d(0.05) = 13.07um = 6.26phi
d(0.10) = 31.86um = 4.97phi
d(0.16) = 63.25um = 3.98phi
d(0.50) = 148.01um = 2.76phi
d(0.84) = 305.91um = 1.71phi
d(0.90) = 449.43um = 1.15phi
d(0.95) = 879.03um = .19phi

Distribution Properties:
Kurtosis = 10.42
Skewness = 3.06
Standard Deviation = 279.63 um = 1.84phi
Inclusive SD (Sorting Coeff.) = 1.49 phi (Poorly Sorted)
Inclusive Kurtosis = 1.79 phi (Very Leptokurtic)
Inclusive Mean = 2.82 phi (Fine Sand)
Inclusive Skewness = .12 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:35:28 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:

D[4,3] = 216.95um = 2.2phi

Surface Weighted Mean:

D[3,2] = 41.64um = 4.59phi

Span :

2.608

Uniformity:

0.969

Specific Surface Area*:

0.144 m²/g

Percentiles:

d(0.05) = 12.91um = 6.28phi

d(0.10) = 31.86um = 4.97phi

d(0.16) = 62.14um = 4.01phi

d(0.50) = 141.81um = 2.82phi

d(0.84) = 284.85um = 1.81phi

d(0.90) = 401.7um = 1.32phi

d(0.95) = 842.23um = .25phi

Distribution Properties:

Kurtosis = 11.6

Skewness = 3.22

Standard Deviation = 271.82 um = 1.88phi

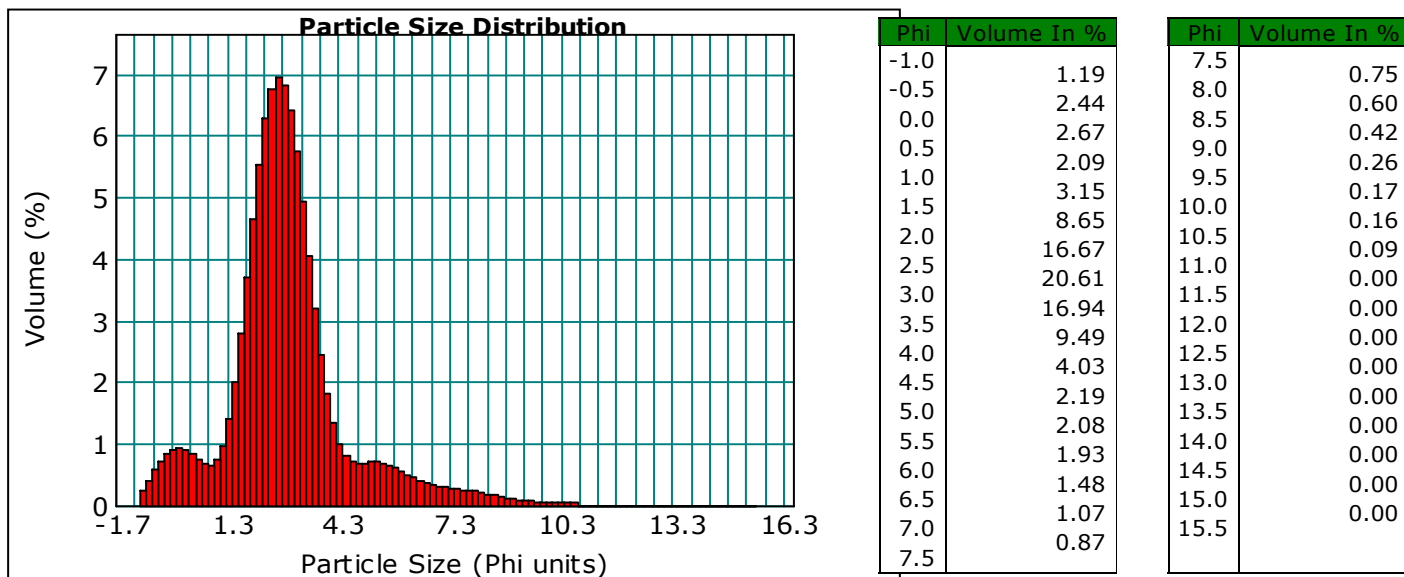
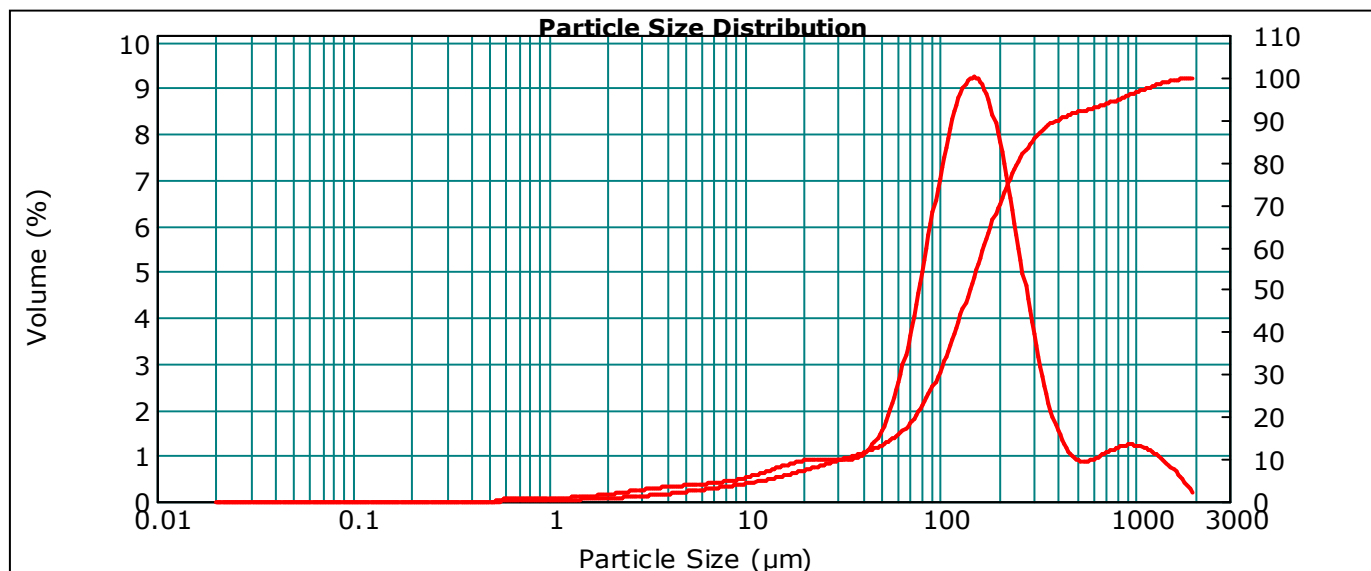
Inclusive SD (Sorting Coeff.) = 1.46 phi (Poorly Sorted)

Inclusive Kurtosis = 1.83 phi (Very Leptokurtic)

Inclusive Mean = 2.88 phi (Fine Sand)

Inclusive Skewness = .12 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:36:35 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 216.72um = 2.21phi

Surface Weighted Mean:
D[3,2] = 41.6um = 4.59phi

Span :
2.524

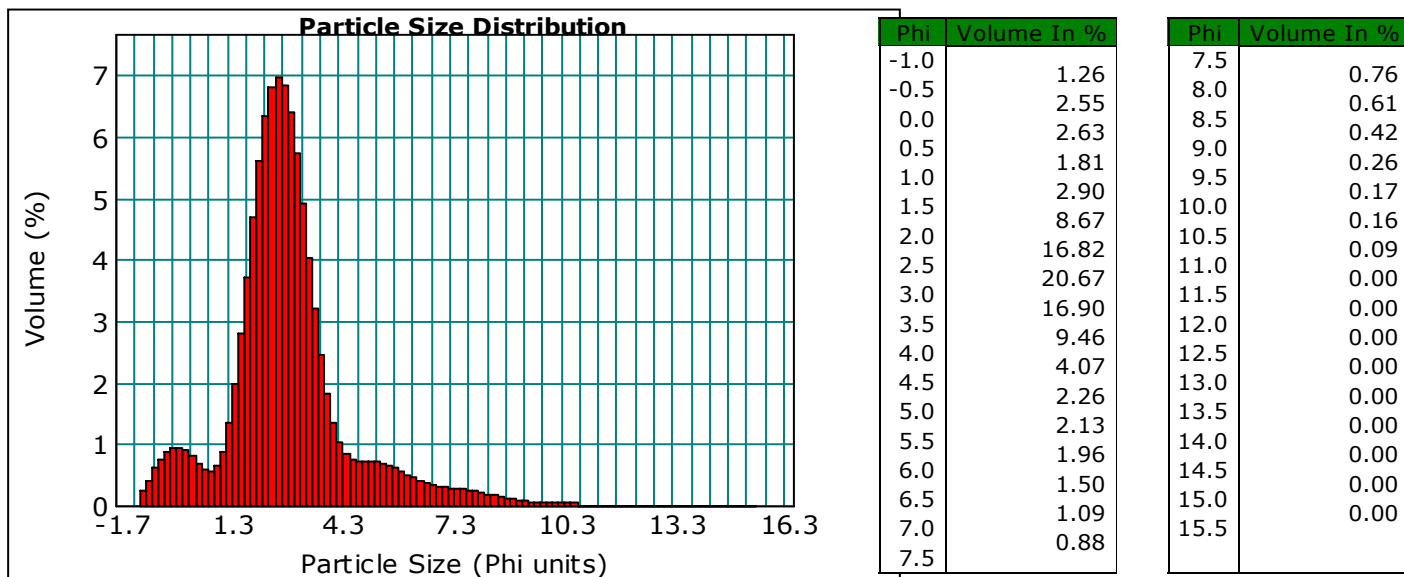
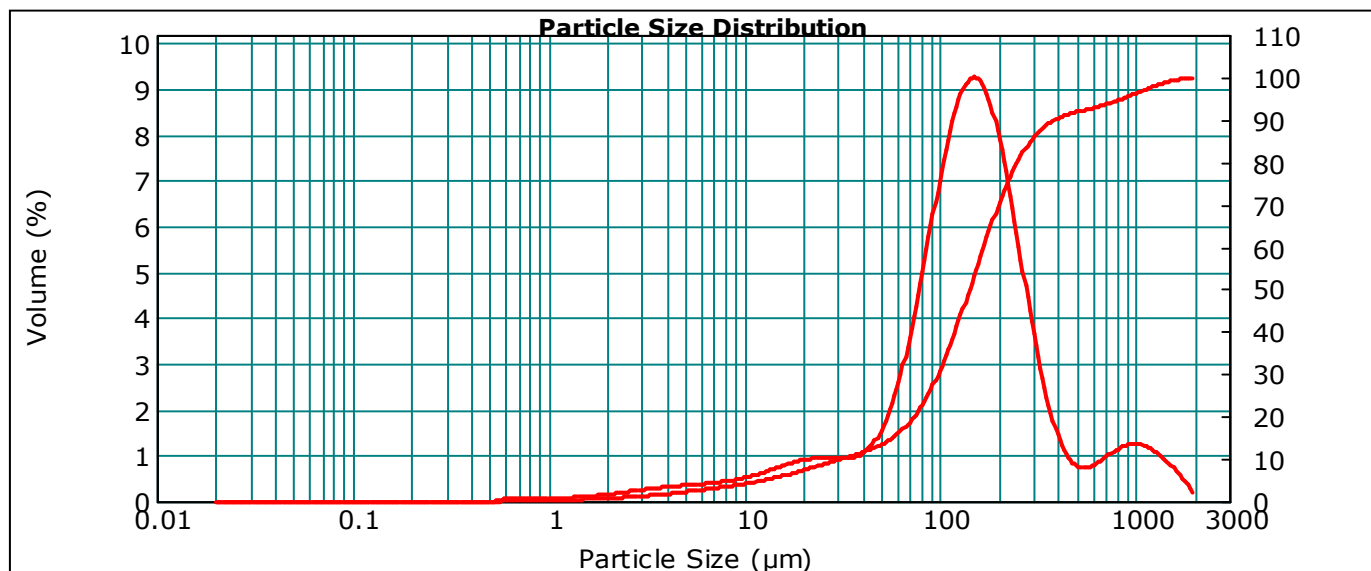
Uniformity:
0.974

Specific Surface Area*:
0.144 m²/g

Percentiles:
d(0.05) = 12.77um = 6.29phi
d(0.10) = 31.15um = 5phi
d(0.16) = 61.27um = 4.03phi
d(0.50) = 141.39um = 2.82phi
d(0.84) = 280.63um = 1.83phi
d(0.90) = 387.95um = 1.37phi
d(0.95) = 862.41um = .21phi

Distribution Properties:
Kurtosis = 11.57
Skewness = 3.24
Standard Deviation = 275.21 um = 1.86phi
Inclusive SD (Sorting Coeff.) = 1.47 phi (Poorly Sorted)
Inclusive Kurtosis = 1.84 phi (Very Leptokurtic)
Inclusive Mean = 2.89 phi (Fine Sand)
Inclusive Skewness = .12 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_2
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:37:42 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 229.24um = 2.13phi

Surface Weighted Mean:
D[3,2] = 42.86um = 4.54phi

Span :
3.160

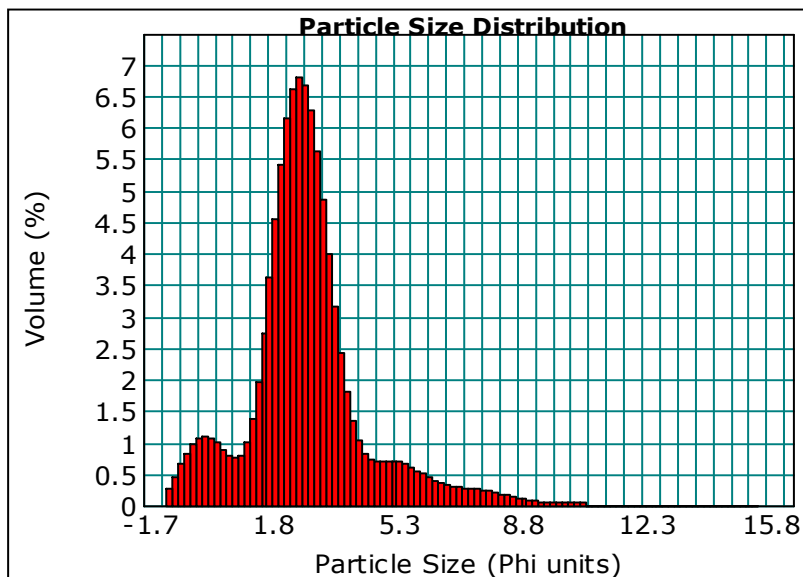
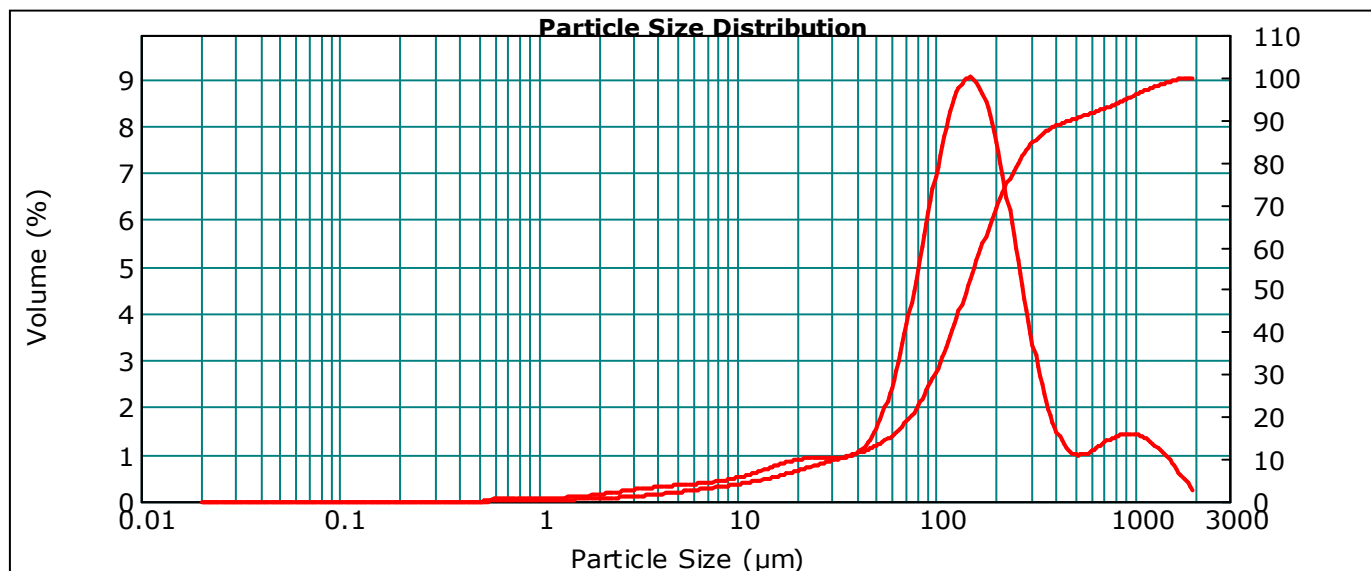
Uniformity:
1.04

Specific Surface Area*:
0.14 m²/g

Percentiles:
d(0.05) = 13.26um = 6.24phi
d(0.10) = 32.49um = 4.94phi
d(0.16) = 62.32um = 4phi
d(0.50) = 143.38um = 2.8phi
d(0.84) = 301.43um = 1.73phi
d(0.90) = 485.57um = 1.04phi
d(0.95) = 924.03um = .11phi

Distribution Properties:
Kurtosis = 9.67
Skewness = 2.99
Standard Deviation = 290.35 um = 1.78phi
Inclusive SD (Sorting Coeff.) = 1.5 phi (Poorly Sorted)
Inclusive Kurtosis = 1.81 phi (Very Leptokurtic)
Inclusive Mean = 2.85 phi (Fine Sand)
Inclusive Skewness = .09 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.40
-0.5	2.86
0.0	3.13
0.5	2.42
1.0	3.23
1.5	8.48
2.0	16.29
2.5	20.16
3.0	16.62
3.5	9.38
4.0	4.05
4.5	2.23
5.0	2.09
5.5	1.92
6.0	1.46
6.5	1.07
7.0	0.86
7.5	

Phi	Volume In %
7.5	0.74
8.0	0.59
8.5	0.40
9.0	0.24
9.5	0.16
10.0	0.15
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_2 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:35:28 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 220.97um = 2.18phi

Surface Weighted Mean:
D[3,2] = 42.02um = 4.57phi

Span :
2.727

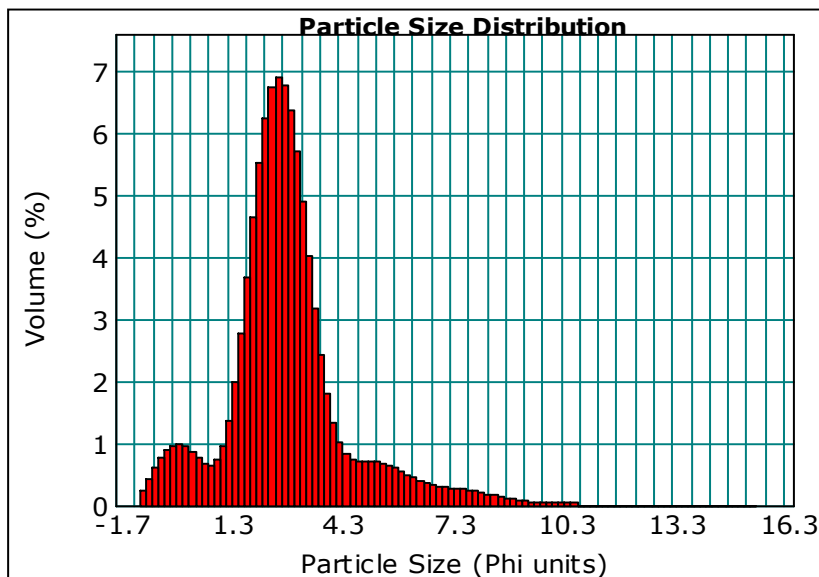
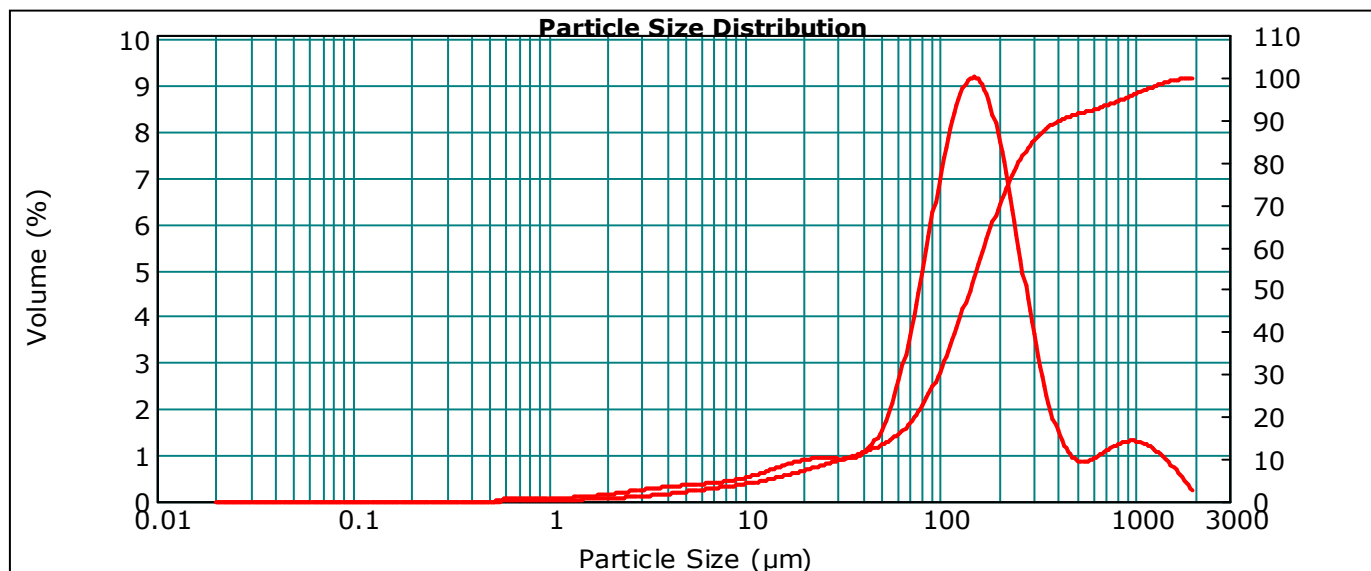
Uniformity:
0.995

Specific Surface Area*:
0.143 m²/g

Percentiles:
d(0.05) = 12.98um = 6.27phi
d(0.10) = 31.83um = 4.97phi
d(0.16) = 61.91um = 4.01phi
d(0.50) = 142.18um = 2.81phi
d(0.84) = 288.4um = 1.79phi
d(0.90) = 419.52um = 1.25phi
d(0.95) = 878.3um = .19phi

Distribution Properties:
Kurtosis = 10.89
Skewness = 3.15
Standard Deviation = 279.31 um = 1.84phi
Inclusive SD (Sorting Coeff.) = 1.48 phi (Poorly Sorted)
Inclusive Kurtosis = 1.83 phi (Very Leptokurtic)
Inclusive Mean = 2.87 phi (Fine Sand)
Inclusive Skewness = .11 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.28
-0.5	2.61
0.0	2.81
0.5	2.11
1.0	3.09
1.5	8.60
2.0	16.59
2.5	20.48
3.0	16.82
3.5	9.44
4.0	4.05
4.5	2.23
5.0	2.10
5.5	1.94
6.0	1.48
6.5	1.08
7.0	0.87
7.5	

Phi	Volume In %
7.5	0.75
8.0	0.60
8.5	0.42
9.0	0.25
9.5	0.17
10.0	0.16
10.5	0.09
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:45:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 172.08um = 2.54phi

Surface Weighted Mean:
D[3,2] = 29.63um = 5.08phi

Span :
2.325

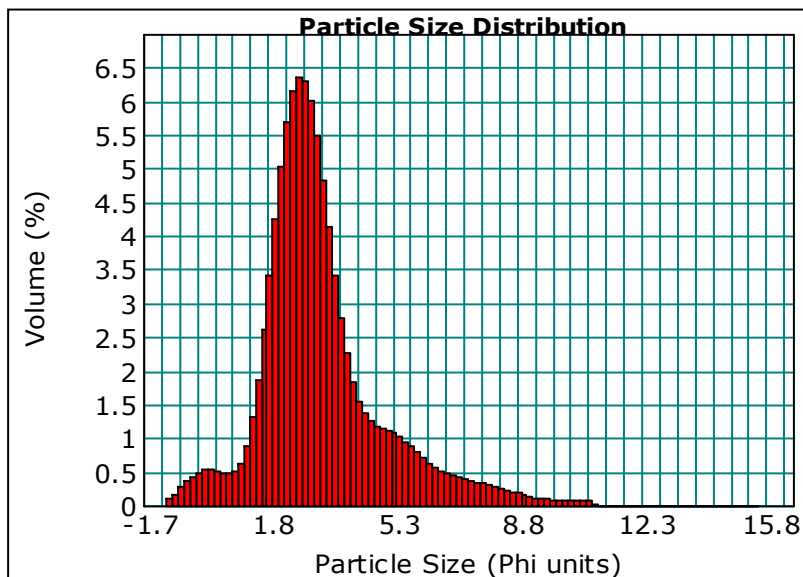
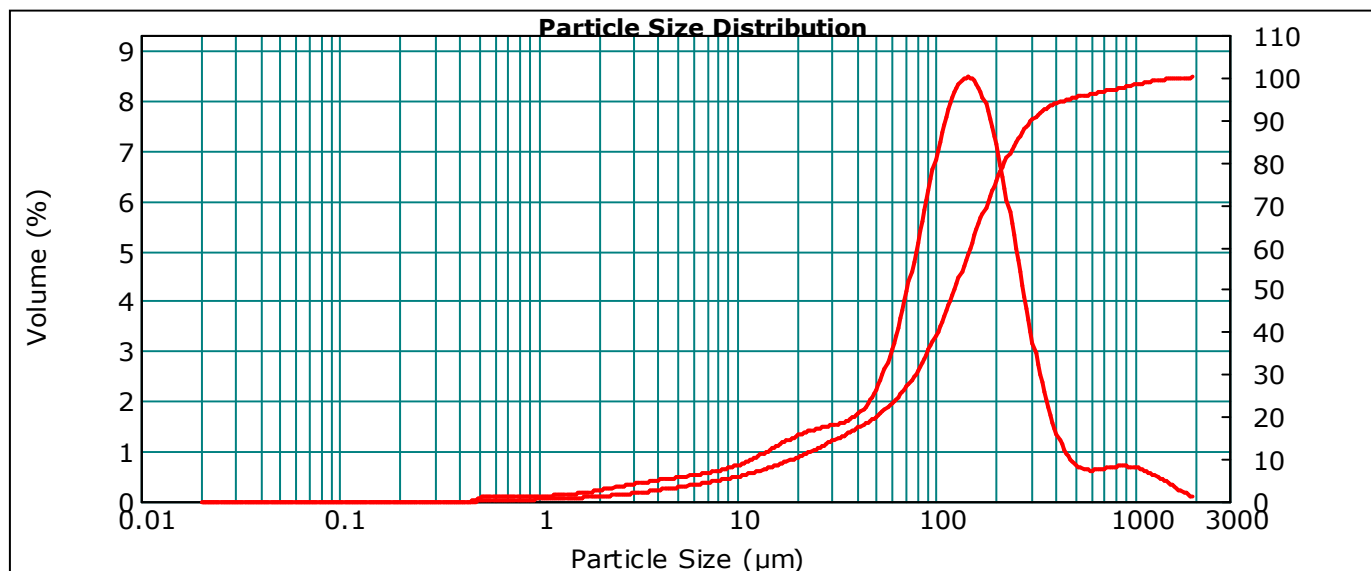
Uniformity:
0.882

Specific Surface Area*:
0.202 m²/g

Percentiles:
d(0.05) = 8.29um = 6.91phi
d(0.10) = 19.23um = 5.7phi
d(0.16) = 36.12um = 4.79phi
d(0.50) = 124.51um = 3.01phi
d(0.84) = 247.84um = 2.01phi
d(0.90) = 308.71um = 1.7phi
d(0.95) = 481.57um = 1.05phi

Distribution Properties:
Kurtosis = 19.29
Skewness = 3.88
Standard Deviation = 210.9 um = 2.25phi
Inclusive SD (Sorting Coeff.) = 1.58 phi (Poorly Sorted)
Inclusive Kurtosis = 1.51 phi (Very Leptokurtic)
Inclusive Mean = 3.27 phi (Very Fine Sand)
Inclusive Skewness = .31 (Strongly Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.54
-0.5	1.27
0.0	1.56
0.5	1.45
1.0	2.87
1.5	8.02
2.0	15.18
2.5	18.90
3.0	16.22
3.5	10.18
4.0	5.54
4.5	3.75
5.0	3.31
5.5	2.81
6.0	2.10
6.5	1.53
7.0	1.22
7.5	

Phi	Volume In %
7.5	1.03
8.0	0.82
8.5	0.59
9.0	0.38
9.5	0.26
10.0	0.25
10.5	0.22
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:46:57 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 178.97um = 2.48phi

Surface Weighted Mean:
D[3,2] = 30.03um = 5.06phi

Span :
2.465

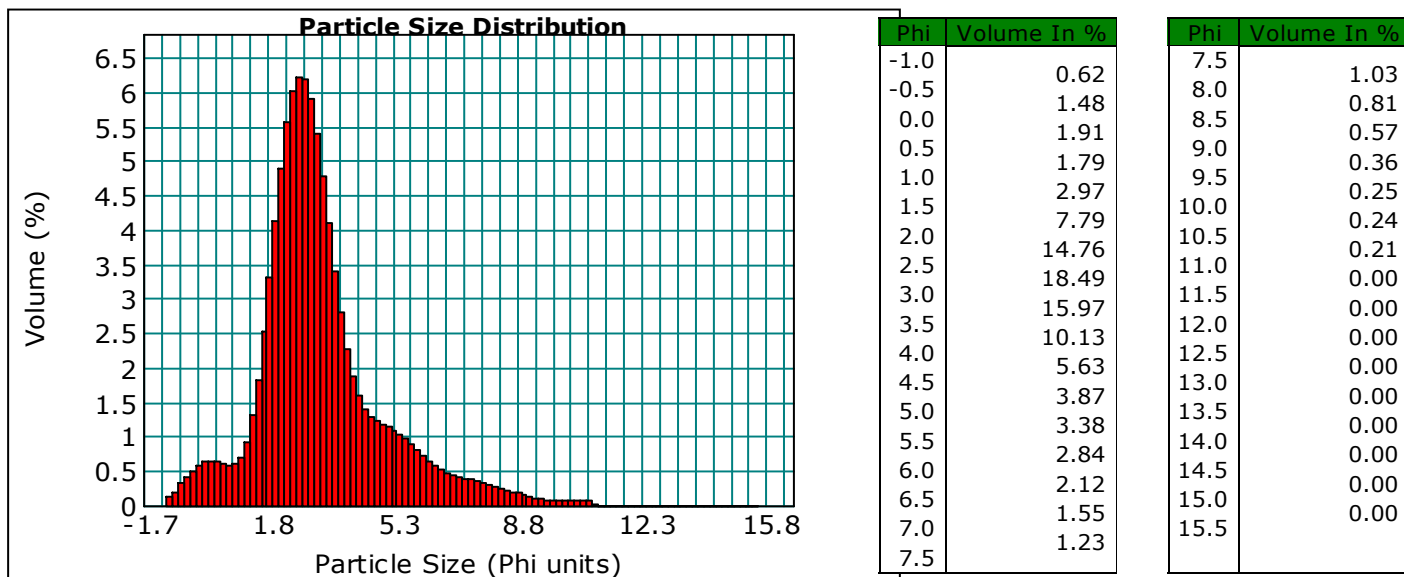
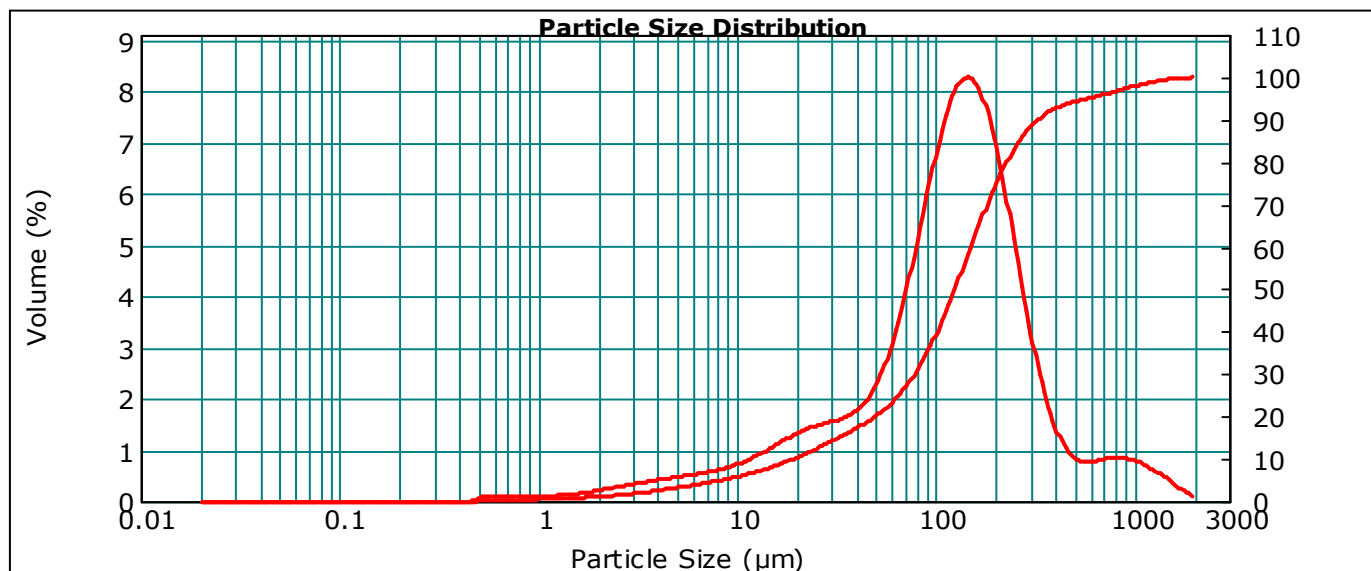
Uniformity:
0.94

Specific Surface Area*:
0.2 m²/g

Percentiles:
d(0.05) = 8.41um = 6.89phi
d(0.10) = 19.25um = 5.7phi
d(0.16) = 35.71um = 4.81phi
d(0.50) = 124.54um = 3.01phi
d(0.84) = 254.4um = 1.97phi
d(0.90) = 326.23um = 1.62phi
d(0.95) = 583.42um = .78phi

Distribution Properties:
Kurtosis = 16.55
Skewness = 3.65
Standard Deviation = 225.06 um = 2.15phi
Inclusive SD (Sorting Coeff.) = 1.64 phi (Poorly Sorted)
Inclusive Kurtosis = 1.54 phi (Very Leptokurtic)
Inclusive Mean = 3.26 phi (Very Fine Sand)
Inclusive Skewness = .27 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.





MASTERSIZER 2000



THE UNIVERSITY OF
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Sample Name: C_1_2_1
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:48:05 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 196.29um = 2.35phi

Surface Weighted Mean:
D[3,2] = 30.56um = 5.03phi

Span :
2.683

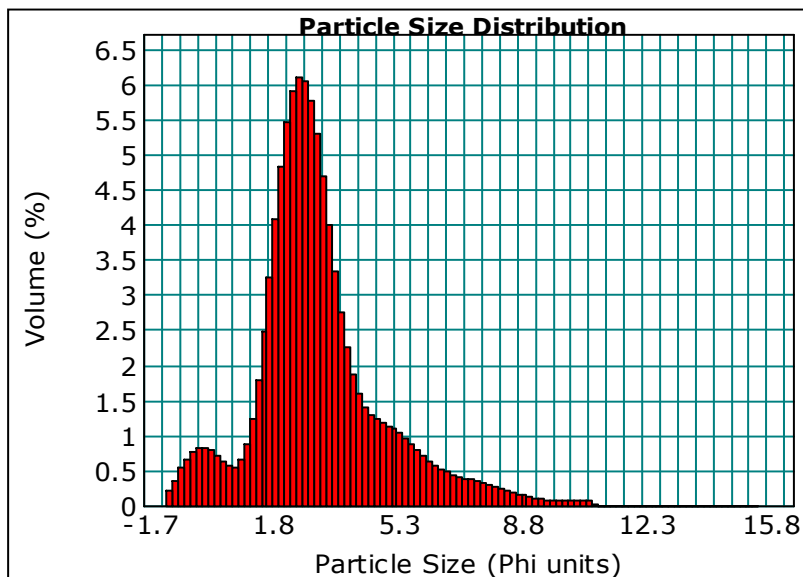
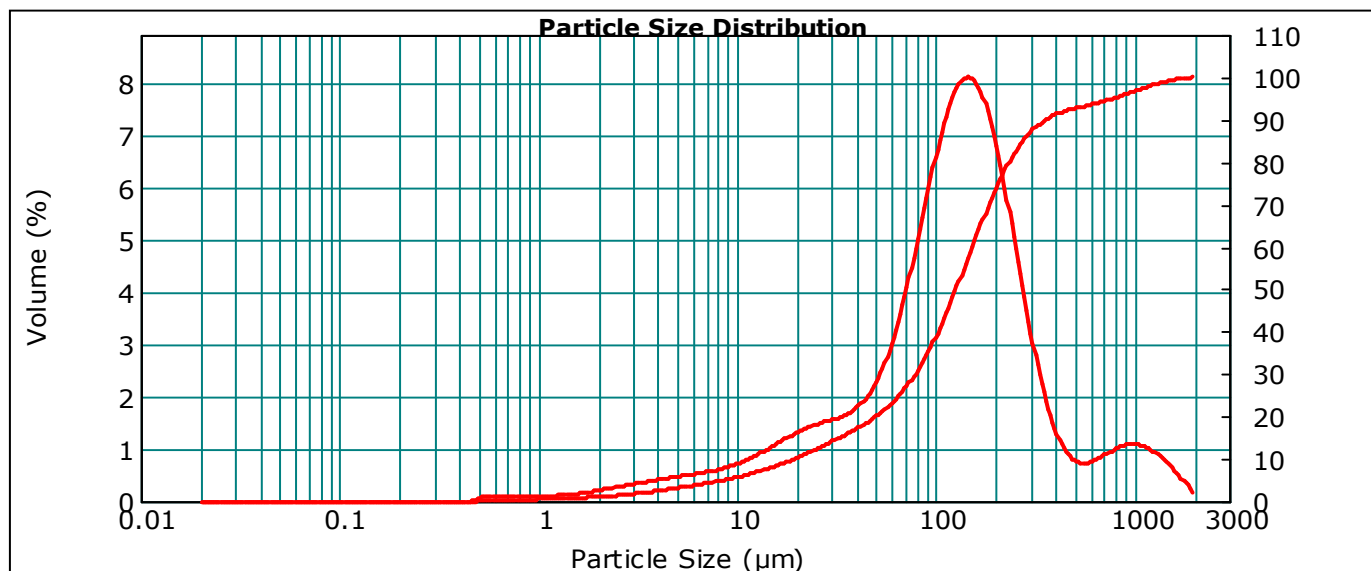
Uniformity:
1.06

Specific Surface Area*:
0.196 m²/g

Percentiles:
d(0.05) = 8.57um = 6.87phi
d(0.10) = 19.53um = 5.68phi
d(0.16) = 36.11um = 4.79phi
d(0.50) = 126.2um = 2.99phi
d(0.84) = 265.43um = 1.91phi
d(0.90) = 358.18um = 1.48phi
d(0.95) = 783.91um = .35phi

Distribution Properties:
Kurtosis = 12.92
Skewness = 3.38
Standard Deviation = 264.83 um = 1.92phi
Inclusive SD (Sorting Coeff.) = 1.71 phi (Poorly Sorted)
Inclusive Kurtosis = 1.61 phi (Very Leptokurtic)
Inclusive Mean = 3.23 phi (Very Fine Sand)
Inclusive Skewness = .22 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	1.10
-0.5	2.22
0.0	2.31
0.5	1.74
1.0	2.79
1.5	7.65
2.0	14.54
2.5	18.13
3.0	15.62
3.5	9.94
4.0	5.60
4.5	3.88
5.0	3.38
5.5	2.82
6.0	2.10
6.5	1.54
7.0	1.22
7.5	

Phi	Volume In %
7.5	1.02
8.0	0.80
8.5	0.56
9.0	0.35
9.5	0.24
10.0	0.23
10.5	0.20
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00



MASTERSIZER 2000



THE UNIVERSITY OF
WESTERN AUSTRALIA

Sample Name: C_1_2_1 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 5:45:50 PM

Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL
alkaline calgon

Vol. Weighted Mean:
D[4,3] = 182.45um = 2.45phi

Surface Weighted Mean:
D[3,2] = 30.07um = 5.06phi

Span :
2.470

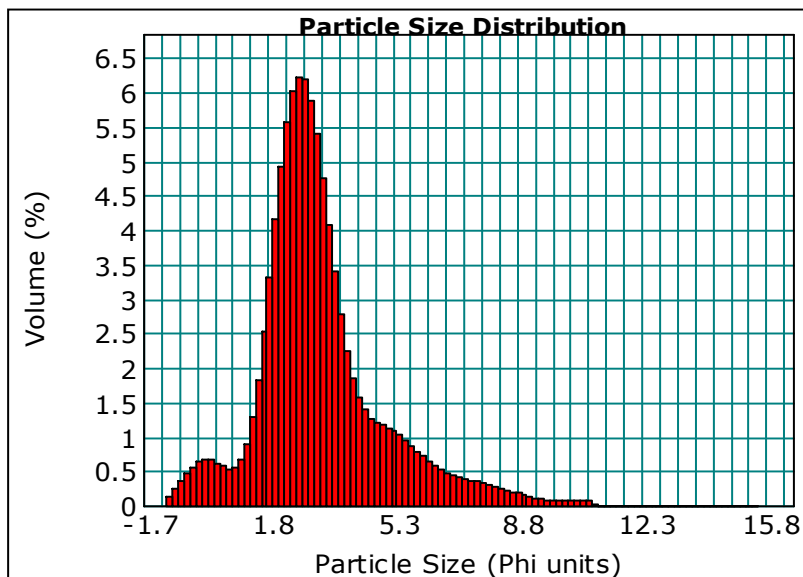
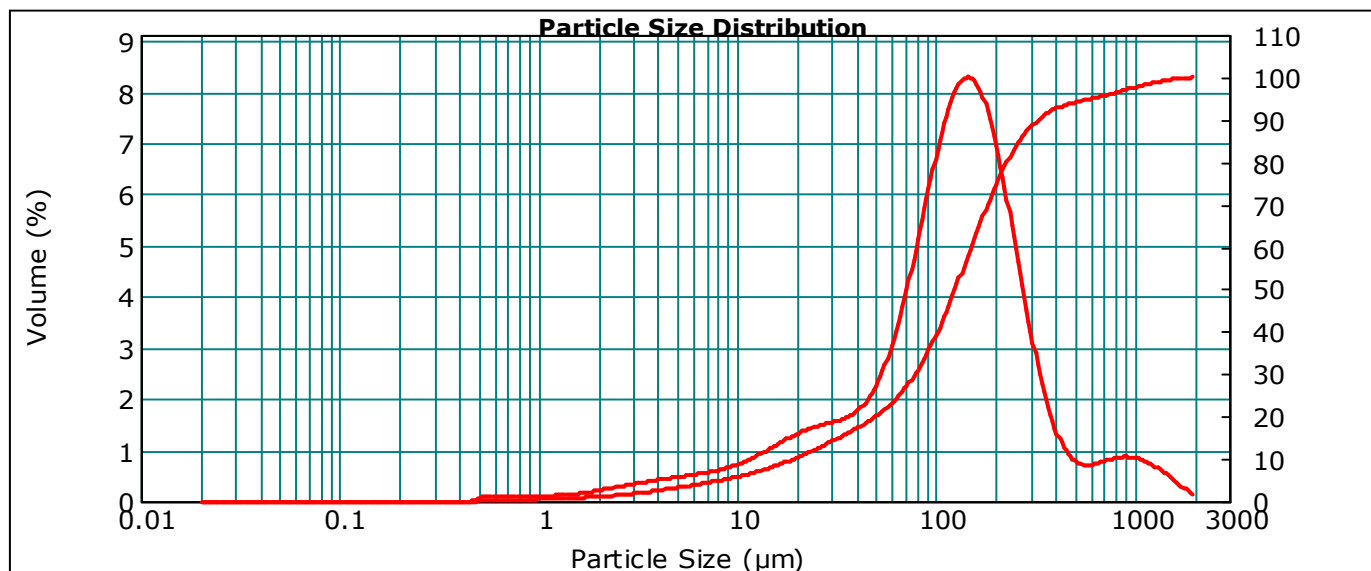
Uniformity:
0.961

Specific Surface Area*:
0.2 m²/g

Percentiles:
d(0.05) = 8.42um = 6.89phi
d(0.10) = 19.34um = 5.69phi
d(0.16) = 35.98um = 4.8phi
d(0.50) = 125.07um = 3phi
d(0.84) = 255.47um = 1.97phi
d(0.90) = 328.26um = 1.61phi
d(0.95) = 617.36um = .7phi

Distribution Properties:
Kurtosis = 16.05
Skewness = 3.65
Standard Deviation = 234.93 um = 2.09phi
Inclusive SD (Sorting Coeff.) = 1.65 phi (Poorly Sorted)
Inclusive Kurtosis = 1.56 phi (Very Leptokurtic)
Inclusive Mean = 3.25 phi (Very Fine Sand)
Inclusive Skewness = .26 (Fine Skewed)

* Specific Surface Area is calculated with particle density = 1 (p=1). For known densities this needs to be recalculated by dividing by actual density.



Phi	Volume In %
-1.0	0.75
-0.5	1.65
0.0	1.93
0.5	1.66
1.0	2.88
1.5	7.82
2.0	14.83
2.5	18.51
3.0	15.94
3.5	10.08
4.0	5.59
4.5	3.83
5.0	3.36
5.5	2.82
6.0	2.10
6.5	1.54
7.0	1.22
7.5	

Phi	Volume In %
7.5	1.02
8.0	0.81
8.5	0.57
9.0	0.36
9.5	0.25
10.0	0.24
10.5	0.21
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00